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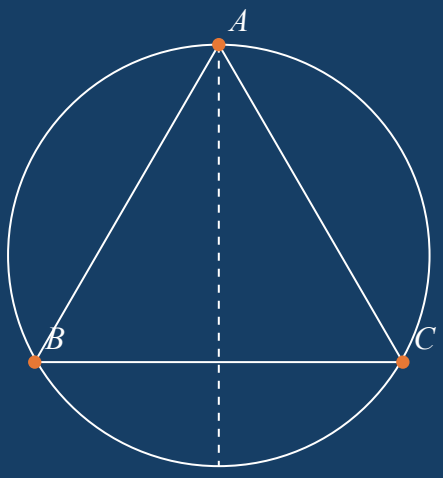
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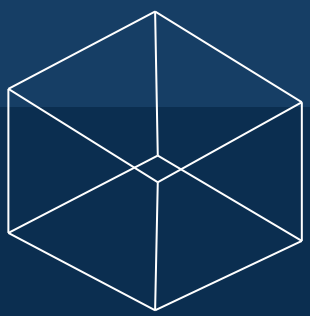
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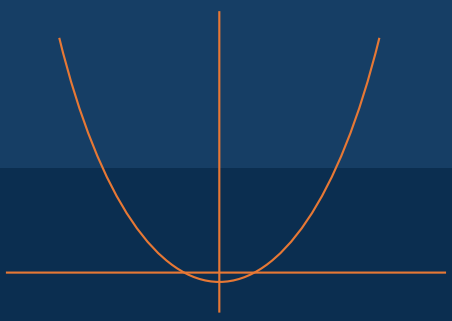
双语知识教材

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2000-2026

810 problems / 44 sections

The Mathematics Atlas

06 CHAPTERS 44 SECTIONS

章节与考点总览

01 Algebra
代数

1.1 Equations & algebraic modeling
方程与代数建模
1.2 Polynomials & Vieta relations
多项式与韦达关系
1.3 Complex numbers & roots of unity
复数与单位根
1.4 Logarithms & exponentials
对数与指数
1.5 Sequences & summation
数列与求和
1.6 Absolute values, floors & piecewise graphs
绝对值、取整与分段图像
1.7 Inequalities & continuous optimization
不等式与连续最值
1.8 Trigonometric algebra & equations
三角恒等式与方程
1.9 Functional constraints & iteration
函数约束与迭代
1.10 Binomial expansions & controlled approximation
二项式展开与误差控制

02 Geometry
几何

2.1 Angles, similarity & parallel lines
角度、相似与平行线
2.2 Areas & area ratios
面积与面积比
2.3 Triangle metrics & trigonometry
三角形度量与三角法
2.4 Circle power & cyclic geometry
圆幂与圆内接几何
2.5 Angle bisectors & triangle centers
角平分线与三角形心
2.6 Tangent circles & common tangents
相切圆与公切线
2.7 Coordinates, vectors & transformations
坐标、向量与变换
2.8 Conics & analytic loci
圆锥曲线与解析轨迹
2.9 Spheres & spatial distances
球与空间距离
2.10 Volumes & solid geometry
体积与立体几何

03 Number Theory
数论

3.1 Prime factorization & valuations
质因数分解与赋值
3.2 Divisibility, GCD & LCM
整除、最大公因数与最小公倍数
3.3 Congruences & Chinese remainders
同余与中国剩余定理
3.4 Orders, modular powers & cycles
乘法阶、模幂与周期
3.5 Diophantine equations & factor pairs
丢番图方程与因子配对
3.6 Number bases, digits & palindromes
进位制、数位与回文
3.7 Rational fractions & repeating decimals
有理分数与循环小数
3.8 Divisor functions & totients
约数函数与欧拉函数

04 Counting & Combinatorics
计数与组合

4.1 Restricted permutations & arrangements
受限排列
4.2 Subsets & binomial identities
子集与二项式恒等式
4.3 Compositions, partitions & generating functions
整数拆分、分拆与生成函数
4.4 Inclusion-exclusion & double counting
容斥与双计数
4.5 Recurrences, lattice paths & tilings
递推、格路与铺砌
4.6 Graphs & geometric counting
图论与几何计数
4.7 Bijections & structural counting
双射与结构计数

05 Probability & Statistics
概率与统计

5.1 Finite probability & combinatorial sampling
有限概率与组合抽样
5.2 Conditional probability & independence
条件概率与独立性
5.3 Geometric probability
几何概率
5.4 Expected values & stopping processes
期望与停止过程
5.5 Distributions & repeated trials
分布与重复试验
5.6 Data & descriptive statistics
数据与描述统计

06 Logic & Strategy
逻辑与策略

6.1 Games, invariants & state strategies
博弈、不变量与状态策略
6.2 Discrete extrema & constructions
离散最值与构造
6.3 Constraint deductions & logical cases
约束推理与逻辑分类

Contents 目录

Algebra
代数

Section 1.1. Equations & algebraic modeling (AL01) 1
方程与代数建模

Section 1.2. Polynomials & Vieta relations (AL02) 21
多项式与韦达关系

Section 1.3. Complex numbers & roots of unity (AL03) 41
复数与单位根

Section 1.4. Logarithms & exponentials (AL04) 60
对数与指数

Section 1.5. Sequences & summation (AL05) 74
数列与求和

Section 1.6. Absolute values, floors & piecewise graphs (AL06) 100
绝对值、取整与分段图像

Section 1.7. Inequalities & continuous optimization (AL07) 124
不等式与连续最值

Section 1.8. Trigonometric algebra & equations (AL08) 164
三角恒等式与方程

Section 1.9. Functional constraints & iteration (AL09) 191
函数约束与迭代

Section 1.10. Binomial expansions & controlled approximation (AL10) 218
二项式展开与误差控制

Geometry
几何

Section 2.1. Angles, similarity & parallel lines (GE01) 224
角度、相似与平行线

Section 2.2. Areas & area ratios (GE02) 241
面积与面积比

Contents 目录

Section 2.3. Triangle metrics & trigonometry (GE03) 266
三角形度量与三角法

Section 2.4. Circle power & cyclic geometry (GE04) 288
圆幂与圆内接几何

Section 2.5. Angle bisectors & triangle centers (GE05) 315
角平分线与三角形心

Section 2.6. Tangent circles & common tangents (GE06) 340
相切圆与公切线

Section 2.7. Coordinates, vectors & transformations (GE07) 362
坐标、向量与变换

Section 2.8. Conics & analytic loci (GE08) 399
圆锥曲线与解析轨迹

Section 2.9. Spheres & spatial distances (GE09) 416
球与空间距离

Section 2.10. Volumes & solid geometry (GE10) 437
体积与立体几何

Number Theory
数论

Section 3.1. Prime factorization & valuations (NT01) 474
质因数分解与赋值

Section 3.2. Divisibility, GCD & LCM (NT02) 492
整除、最大公因数与最小公倍数

Section 3.3. Congruences & Chinese remainders (NT03) 508
同余与中国剩余定理

Section 3.4. Orders, modular powers & cycles (NT04) 528
乘法阶、模幂与周期

Section 3.5. Diophantine equations & factor pairs (NT05) 542
丢番图方程与因子配对

Contents 目录

Section 3.6. Number bases, digits & palindromes (NT06) 562
进位制、数位与回文

Section 3.7. Rational fractions & repeating decimals (NT07) 584
有理分数与循环小数

Section 3.8. Divisor functions & totients (NT08) 594
约数函数与欧拉函数

Counting & Combinatorics
计数与组合

Section 4.1. Restricted permutations & arrangements (CO01) 608
受限排列

Section 4.2. Subsets & binomial identities (CO02) 625
子集与二项式恒等式

Section 4.3. Compositions, partitions & generating functions (CO03) 642
整数拆分、分拆与生成函数

Section 4.4. Inclusion-exclusion & double counting (CO04) 657
容斥与双计数

Section 4.5. Recurrences, lattice paths & tilings (CO05) 670
递推、格路与铺砌

Section 4.6. Graphs & geometric counting (CO06) 691
图论与几何计数

Section 4.7. Bijections & structural counting (CO07) 725
双射与结构计数

Probability & Statistics
概率与统计

Section 5.1. Finite probability & combinatorial sampling (PS01) 743
有限概率与组合抽样

Section 5.2. Conditional probability & independence (PS02) 758
条件概率与独立性

Contents 目录

Section 5.3. Geometric probability (PS03) 771
几何概率

Section 5.4. Expected values & stopping processes (PS04) 783
期望与停止过程

Section 5.5. Distributions & repeated trials (PS05) 800
分布与重复试验

Section 5.6. Data & descriptive statistics (PS06) 818
数据与描述统计

Logic & Strategy
逻辑与策略

Section 6.1. Games, invariants & state strategies (LO01) 826
博弈、不变量与状态策略

Section 6.2. Discrete extrema & constructions (LO02) 841
离散最值与构造

Section 6.3. Constraint deductions & logical cases (LO03) 883
约束推理与逻辑分类

Section 1.1 Equations & algebraic modeling

方程与代数建模

Method 1 / 方法 1 [AL01-M01]

Build equations around the quantity that is conserved or shared. A useful model includes units and domains as well as equations; solving an eliminated equation is only the middle of the argument, followed by reconstruction and feasibility.

围绕守恒量或共同量建立方程。有效模型既包括方程，也包括单位与定义域；求解消元后的方程只在论证中途，之后还须重建与检查可行性。

Write conservation equations for proportions, mixtures and averages

为比例、混合与平均数建立守恒方程

Procedure

1. Choose the conserved amount: people, pure ingredient, total sum, or probability mass.

2. Express each percentage as amount divided by the correct current total; distinguish original and new totals.

3. Eliminate the shared total or scale, then verify nonnegative component amounts and any integer restrictions.

通用步骤

1. 选择守恒量：人数、纯物质量、总和或概率质量。

2. 把每个百分率写成数量除以对应时刻总量，区分原总数与新总数。

3. 消去共同总量或尺度，再核对分量非负与整数限制。

> Example. A 30% solution and a 70% solution are mixed to make 12 liters at 45%. Find the amount of each.

> 例题：将 30% 与 70% 溶液混合成 12 升 45% 溶液，求各用多少。

Section 1.1 Equations & algebraic modeling

方程与代数建模

Method 1 / 方法 1 [AL01-M01] (continued / 续)

Worked solution

1. If x liters are from the first solution, the second contributes $12 - x$ liters.

2. Conserve pure ingredient: $0.3x + 0.7(12 - x) = 0.45 \cdot 12$.

3. Thus $x = 7.5$ and the other amount is 4.5 liters, both nonnegative and totaling twelve.

例题解析

1. 若第一种用 x 升，第二种便用 $12 - x$ 升。

2. 纯物质守恒给出 $0.3x + 0.7(12 - x) = 0.45 \cdot 12$。

3. 得 $x = 7.5$，另一种为 4.5 升，均非负且合计十二升。

Percentages cannot be averaged without weighting by their amounts.

百分率不能脱离其对应数量直接平均。

Practice / 配套练习: AIME-2001-I-Q02 (WB p. 793), AIME-2008-I-Q01 (WB p. 6), AIME-2009-II-Q01 (WB p. 8), AIME-2011-I-Q01 (WB p. 9), AIME-2011-II-Q01 (WB p. 10), AIME-2013-II-Q01 (WB p. 15), AIME-2014-I-Q02 (WB p. 751), AIME-2022-II-Q01 (WB p. 489)

Section 1.1 Equations & algebraic modeling

方程与代数建模

Method 2 / 方法 2 [AL01-M02]

Add work and time over the actual operating stages

按实际阶段累加工作量与时间

Procedure

1. Normalize the whole job and express each rate in a single common time unit.

2. For each stage write work equals rate times time; add rates for simultaneous workers, and count pauses only when they occur.

3. Use the remaining deadline to solve the required final rate; round a workforce upward only after the real threshold is established.

4. Check all stage lengths and final staffing are positive and feasible.

通用步骤

1. 将总工作量归一，并以统一时间单位表达各效率。

2. 逐阶段写工作量等于效率乘时间；同时工作时效率相加，休息只按实际发生次数统计。

3. 用剩余期限求最终所需效率；确定实数门槛后才把人数向上取整。

4. 检查各阶段长度与最终人数均为正且可实现。

> Example. A machine does one third of a job in two hours, then works at half its original rate. How long is the full job?

> 例题：机器先用两小时完成三分之一工作，再以原效率的一半继续，全部工作需要多久？

Section 1.1 Equations & algebraic modeling

方程与代数建模

Method 2 / 方法 2 [AL01-M02] (continued / 续)

Worked solution

1. The original rate is $1/6$ job per hour.

2. The later rate is $1/12$; the remaining $2/3$ job takes $(2/3)/(1/12) = 8$ hours.

3. Total time is $2 + 8 = 10$ hours. Averaging the two rates without duration weights would be incorrect.

例题解析

1. 原效率为每小时 $1/6$ 个工作。

2. 后段效率为 $1/12$，剩余 $2/3$ 工作需 $(2/3)/(1/12) = 8$ 小时。

3. 总时间为 $2 + 8 = 10$ 小时；不按持续时间加权平均效率会出错。

> The last arrival often occurs before the final scheduled rest.

> 最后一次到达通常发生在最后一次休息之前。

Practice / 配套练习: AIME-2004-II-Q05 (WB p. 2), AIME-2007-II-Q04 (WB p. 5), AIME-2008-II-Q02 (WB p. 7), AIME-2013-I-Q01 (WB p. 13), AIME-2014-II-Q01 (WB p. 17), AIME-2013-II-Q07 (WB p. 549)

Section 1.1 Equations & algebraic modeling

方程与代数建模

Method 3 / 方法 3 [AL01-M03]

Use position functions and relative velocity vectors

使用位置函数与相对速度向量

Procedure

1. Choose a common clock and write each position using its actual starting time.

2. For a meeting or midpoint, equate positions or twice the middle position to the sum of the outer positions.

3. For motion in a current, subtract the current velocity vector before applying the given speed norm; subtract squared equations when they share a component.

4. Reject solutions before a departure, after a finish, or on the wrong overtaking branch.

通用步骤

1. 选择共同计时起点，并按各自实际出发时间写位置。

2. 相遇时位置相等，中点条件则为中间位置的两倍等于外侧位置之和。

3. 水流中运动须先减去水流速度向量，再使用给定速率的范数；有共同分量时相减平方方程。

4. 剔除出发前、结束后或错误追赶分支上的解。

> Example. A boat moves across a river at velocity $(3, 4)$ km/h relative to shore; the current is $(1, 0)$. What is its speed relative to water?

> 例题：船相对岸的速度为 $(3, 4)$ 千米每小时，水流为 $(1, 0)$，求船相对水的速率。

Section 1.1 Equations & algebraic modeling

方程与代数建模

Method 3 / 方法 3 [AL01-M03] (continued / 续)

Worked solution

1. Velocities add as vectors: $v_{shore} = v_{water} + v_{current}$.

2. Thus $v_{water} = (3, 4) - (1, 0) = (2, 4)$.

3. The speed is $\sqrt{2^2 + 4^2} = 2\sqrt{5}$, rather than $5 - 1$.

例题解析

1. 速度按向量相加：$v_{岸} = v_{水} + v_{流}$。

2. 故 $v_{水} = (3, 4) - (1, 0) = (2, 4)$。

3. 速率为 $\sqrt{2^2 + 4^2} = 2\sqrt{5}$，而非 $5 - 1$。

> Subtract velocities as vectors, not their magnitudes.

> 速度须以向量相减，不能直接相减其大小。

Practice / 配套练习: AIME-2007-I-Q02 (WB p. 4), AIME-2012-I-Q04 (WB p. 11), AIME-2012-II-Q04 (WB p. 12), AIME-2014-I-Q04 (WB p. 16), AIME-2018-II-Q01 (WB p. 19), AIME-2022-I-Q05 (WB p. 20), AIME-2024-I-Q01 (WB p. 22), AIME-2026-I-Q01 (WB p. 23)

Section 1.1 Equations & algebraic modeling

方程与代数建模

Method 4 / 方法 4 [AL01-M04]

Exploit a shared total before solving a linear system

解线性方程组前先利用共同总量

Procedure

1. Identify repeated sums or boundary totals and give each one a single variable.

2. Add or subtract the equations to isolate the shared total before individual coordinates.

3. In a tiling, match each entire boundary consistently; in a coefficient problem, compare independent coefficients.

4. If integer scale is required, reduce the resulting ratio and check every constructed variable.

通用步骤

1. 找出反复出现的和或边界总长，并为每个总量设置一个变量。

2. 先通过方程相加或相减隔离共同总量，再解单独坐标。

3. 铺砌中须一致匹配整条边界；系数问题则比较独立系数。

4. 若要求整数尺度，约分所得比例，并检查构造中的每个变量。

> Example. Positive x, y, z satisfy $x+y=4$, $y+z=9$, $z+x=7$. Find $x^2+y^2+z^2$.

> 例题：正数 x, y, z 满足 $x+y=4, y+z=9, z+x=7$，求 $x^2+y^2+z^2$。

Worked solution

1. Let $S=x+y+z$. Adding the equations gives $2S=20$, so $S=10$.

Section 1.1 Equations & algebraic modeling

方程与代数建模

Method 4 / 方法 4 [AL01-M04] (continued / 续)

2. Each coordinate is the total minus the opposite pair sum: $x = 10 - 9 = 1, y = 10 - 7 = 3, z = 10 - 4 = 6$.

3. The requested value is $1 + 9 + 36 = 46$; all coordinates are positive and satisfy the data.

例题解析

1. 设 $S = x + y + z$。相加得 $2S = 20$，所以 $S = 10$。

2. 每个坐标等于总和减去另外两项之和：$x = 10 - 9 = 1, y = 10 - 7 = 3, z = 10 - 4 = 6$。

3. 所求为 $1 + 9 + 36 = 46$；各坐标均为正且满足给定条件。

Do not solve for many coordinates when one total closes the system.

若一个总量即可闭合方程组，无需逐个求大量坐标。

Practice / 配套练习: AIME-2000-I-Q04 (WB p. 1), AIME-2000-I-Q10 (WB p. 25), AIME-2001-I-Q11 (WB p. 29), AIME-2004-II-Q06 (WB p. 546), AIME-2008-I-Q03 (WB p. 533), AIME-2008-I-Q13 (WB p. 55), AIME-2009-I-Q14 (WB p. 555)

Section 1.1 Equations & algebraic modeling

方程与代数建模

Method 5 / 方法 5 [AL01-M05]

Replace variables by the symmetric invariants actually requested

以目标真正需要的对称不变量换元

Procedure

1. Introduce a sum s, product p, or common cyclic expression instead of immediately solving each variable.

2. Use $x^2+y^2=s^2-2p$ and $x^3+y^3=s^3-3ps$, or multiply a product-constrained system to eliminate individual variables.

3. Solve only the invariant equations needed for the target; track any division by s, p and check zero branches.

4. If real variables must exist, require the corresponding quadratic discriminant to be nonnegative and reconstruct a valid instance.

通用步骤

1. 设置和 s、积 p 或共同循环表达式，而不立即逐个求变量。

2. 使用 $x^2+y^2=s^2-2p, x^3+y^3=s^3-3ps$，或对带积约束的方程组相乘以消元。

3. 只解目标所需的不变量方程，追踪所有除以 s, p 的操作，并检查零分支。

4. 若要求实变量存在，须对应二次判别式非负，并重建有效实例。

> Example. Real x, y satisfy $x+y=5$ and $xy=3$. Find x^4+y^4 without finding x, y.

> 例题：实数 x, y 满足 $x+y=5, xy=3$，不逐个求 x, y，求 x^4+y^4。

Section 1.1 Equations & algebraic modeling

方程与代数建模

Method 5 / 方法 5 [AL01-M05] (continued / 续)

Worked solution

1. $x^2+y^2=25-6=19$.

2. Square again: $x^4+y^4=(x^2+y^2)^2-2x^2y^2=361-18=343$.

3. The quadratic t^2-5t+3 has discriminant $13>0$, so real values realizing the conditions exist.

例题解析

1. $x^2+y^2=25-6=19$。

2. 再平方得 $x^4+y^4=(x^2+y^2)^2-2x^2y^2=361-18=343$。

3. 二次式 t^2-5t+3 的判别式为 $13>0$，所以存在满足条件的实数。

An invariant equation can have branches that do not correspond to real original variables.

不变量方程可能有不对应实数原变量的分支。

Practice / 配套练习: AIME-2000-I-Q07 (WB p. 24), AIME-2001-I-Q09 (WB p. 247), AIME-2010-I-Q09 (WB p. 26), AIME-2010-II-Q05 (WB p. 105), AIME-2013-I-Q03 (WB p. 14), AIME-2014-I-Q13 (WB p. 400), AIME-2015-II-Q14 (WB p. 31), AIME-2016-II-Q01 (WB p. 139), AIME-2018-II-Q05 (WB p. 64), AIME-2021-II-Q07 (WB p. 28), AIME-2012-II-Q08 (WB p. 75)

Section 1.1 Equations & algebraic modeling

方程与代数建模

Method 6 / 方法 6 [AL01-M06]

Subtract equations to reveal equality branches and complementary pairs

相减方程以揭示相等分支与互补配对

Procedure

1. Subtract two similarly structured equations before expanding the full system.

2. Factor the difference and split all resulting cases, retaining overlaps only once.

3. For pair sums of four numbers, pair complementary edges: each complementary pair has the same total.

4. Reconstruct distinct or real variables to establish that a maximum or a candidate set is attained.

通用步骤

1. 完整展开之前，先相减两个结构相似的方程。

2. 将差因式分解并分类处理所有分支，重叠只保留一次。

3. 四数两两和问题中，将互补边配对，每对具有相同总和。

4. 重建互异或实数变量，证明最大值或候选集合可以实现。

> Example. Classify real triples with $xy + z = yz + x = zx + y$ and $x + y + z = 0$.

> 例题：分类满足 $xy + z = yz + x = zx + y$ 且 $x + y + z = 0$ 的实数三元组。

Section 1.1 Equations & algebraic modeling

方程与代数建模

Method 6 / 方法 6 [AL01-M06] (continued / 续)

Worked solution

1. If all three variables are equal, the sum gives $(0, 0, 0)$. Otherwise relabel an unequal pair as $x \neq z$.

2. Subtracting the first two expressions gives $(y-1)(x-z)=0$, so $y=1$. Equality with the third then gives $(x-1)(z-1)=0$.

3. Thus two coordinates are one and the third is minus two. These permutations and $(0, 0, 0)$ all satisfy the equations.

例题解析

1. 若三变量相等，则由总和得 $(0, 0, 0)$；否则可重新命名，使 $x \neq z$。

2. 前两式相减得 $(y-1)(x-z)=0$，故 $y=1$；再与第三式比较，得 $(x-1)(z-1)=0$。

3. 所以两坐标为一，第三个为负二；这些排列与 $(0, 0, 0)$ 均满足原方程。

A factored difference gives alternative cases, not permission to assume all variables equal.

差的因式分解给出不同分支，不允许直接假设全部变量相等。

Practice / 配套练习: AIME-2017-II-Q05 (WB p. 18), AIME-2018-II-Q12 (WB p. 280), AIME-2023-II-Q04 (WB p. 21)

Section 1.1 Equations & algebraic modeling

方程与代数建模

Method 7 / 方法 7 [AL01-M07]

Isolate radicals and reverse-check every squaring step

隔离根式，并逆向检查每次平方

Procedure

1. Write all radicand and sign restrictions before squaring.

2. Isolate one radical, square, simplify, and repeat only when the isolated side has the required sign.

3. Solve the resulting polynomial equation, then substitute every candidate into the original unsquared equation.

4. For geometric models also check containment and the chosen internal/external tangency branch.

通用步骤

1. 平方之前先写全部被开方式与符号限制。

2. 隔离一个根式后平方并化简；只有隔离侧具有所需符号时才重复。

3. 解所得多项式方程，再把每个候选代回原未平方方程。

4. 几何模型还须检查包含关系与内外切所选分支。

> Example. Solve $\sqrt{x+4}-\sqrt{x}=1$ over the reals.

> 例题：在实数范围解 $\sqrt{x+4}-\sqrt{x}=1$。

Section 1.1 Equations & algebraic modeling

方程与代数建模

Method 7 / 方法 7 [AL01-M07] (continued / 续)

Worked solution

1. The domain is $x \geq 0$. Isolate $\sqrt{x+4} = 1 + \sqrt{x}$, whose right side is positive.

2. Squaring yields $x + 4 = 1 + 2\sqrt{x} + x$, so $\sqrt{x} = 3/2$ and $x = 9/4$.

3. Substitution gives $5/2 - 3/2 = 1$, proving this candidate satisfies the original equation.

例题解析

1. 定义域为 $x \geq 0$。隔离为 $\sqrt{x+4} = 1 + \sqrt{x}$，右侧为正。

2. 平方得 $x + 4 = 1 + 2\sqrt{x} + x$，故 $\sqrt{x} = 3/2, x = 9/4$。

3. 代入得 $5/2 - 3/2 = 1$，证明该候选满足原方程。

A squared equation is generally weaker until the lost sign condition is restored.

恢复丢失的符号条件之前，平方方程一般只是较弱条件。

Practice / 配套练习: AIME-2004-II-Q12 (WB p. 342), AIME-2020-I-Q06 (WB p. 425), AIME-2022-I-Q15 (WB p. 204)

Section 1.1 Equations & algebraic modeling

方程与代数建模

Method 8 / 方法 8 [AL01-M08]

Compare independent radical coefficients consistently

一致比较独立根式的系数

Procedure

1. First confirm that the unknown coefficients are rational, as in an integer-coefficient problem. Choose a radical basis linearly independent over $\mathbb{Q}$, often $1, \sqrt{p}, \sqrt{q}, \sqrt{pq}$.

2. Expand the proposed radical expression and match each basis coefficient separately.

3. Use products of pairwise coefficient equations to recover the required invariant, then check the constant coefficient and original signs.

通用步骤

1. 先确认未知系数为有理数，例如题目要求整数系数。选择在 $\mathbb{Q}$ 上线性无关的根式基，常为 $1, \sqrt{p}, \sqrt{q}, \sqrt{pq}$。

2. 展开待定根式表达式，分别比较每个基的系数。

3. 将两两系数方程相乘以恢复所求不变量，再核对常数系数及原符号。

> Example. If positive integers a, b satisfy $(a\sqrt{2}+b\sqrt{3})^2 = 35+12\sqrt{6}$, find a^2b^2.

> 例题：若正整数 a, b 满足 $(a\sqrt{2}+b\sqrt{3})^2 = 35+12\sqrt{6}$，求 a^2b^2。

Worked solution

1. Expansion gives $2a^2+3b^2+2ab\sqrt{6}$. Independence of $1, \sqrt{6}$ requires $2ab = 12$.

2. Thus $a^2b^2 = 36$. To verify existence, put $a^2 = t, b^2 = 36/t$ in the constant equation.

Section 1.1 Equations & algebraic modeling

方程与代数建模

Method 8 / 方法 8 [AL01-M08] (continued / 续)

3. Then $2t + 108/t = 35$, with roots $t = 4, 27/2$. Only $t = 4$ is an integer square, giving $a = 2, b = 3$ and verifying the constant coefficient.

例题解析

1. 展开得 $2a^2 + 3b^2 + 2ab\sqrt{6}$；由 $1, \sqrt{6}$ 独立，须 $2ab = 12$。

2. 故 $a^2b^2 = 36$。检验存在性时，设 $a^2 = t, b^2 = 36/t$ 代入常数方程。

3. 得 $2t + 108/t = 35$，根为 $t = 4, 27/2$。只有 $t = 4$ 是整数平方，给出 $a = 2, b = 3$，并验证常数系数。

> Matching cross terms does not remove the need to check the constant term.

> 比较交叉项后，仍须核对常数项。

Practice / 配套练习: AIME-2006-I-Q05 (WB p. 3)

Section 1.1 Equations & algebraic modeling

方程与代数建模

Method 9 / 方法 9 [AL01-M09]

Count roots only after restoring domains and real feasibility

恢复定义域与实数可行性后才数根

Procedure

1. Translate the model to a polynomial equation while recording all logarithm, positivity or interval restrictions.

2. Use discriminants, root signs or monotonicity on the permitted interval to count valid roots.

3. A repeated algebraic root counts as one real solution; an excluded boundary or pole counts as none.

通用步骤

1. 将模型转为多项式方程，同时记录全部对数、正性或区间限制。

2. 在允许区间内用判别式、根的符号或单调性统计有效根。

3. 代数重根算一个实数解，被排除的边界或极点不算解。

> Example. For which real k does $\log(x) = \log(k - x)$ have a real solution?

> 例题：实参数 k 取何值时，$\log x = \log(k - x)$ 有实数解?

Worked solution

1. The domain requires $0 < x < k$. Equal logarithms require $x = k - x$.

2. The only candidate is $x = k/2$. It is in the domain exactly when $k > 0$.

3. Therefore there is exactly one real solution for every $k > 0$ and no real solution otherwise.

Section 1.1 Equations & algebraic modeling

方程与代数建模

Method 9 / 方法 9 [AL01-M09] (continued / 续)

例题解析

1. 定义域要求 $0 < x < k$；对数相等要求 $x = k - x$。

2. 唯一候选为 $x = k/2$，它属于定义域当且仅当 $k > 0$。

3. 所以每个 $k > 0$ 恰有一个实数解，其余均无实数解。

Algebraic root count and admissible solution count are different until the domain is checked.

检验定义域之前，代数根数与可接受解数是不同概念。

Practice / 配套练习: AIME-2011-I-Q06 (WB p. 27), AIME-2016-II-Q03 (WB p. 107), AIME-2017-I-Q06 (WB p. 770), AIME-2017-II-Q07 (WB p. 122), AIME-2018-I-Q05 (WB p. 108)

Section 1.1 Equations & algebraic modeling

方程与代数建模

Method 10 / 方法 10 [AL01-M10]

Center symmetric rational expressions and preserve divided-out cases

将对称分式中心化，并保留被除去的分支

Procedure

1. Locate a common midpoint among denominator offsets and substitute $t = x - h$.

2. Pair denominators $t - a, t + a$; their reciprocal sum is $2t/(t^2 - a^2)$.

3. Before dividing by t or another shared factor, test its zero case in the original equation.

4. If the result uses only t^2, set $u = t^2$, solve, then restore both signs and reject original poles.

通用步骤

1. 寻找分母偏移的共同中点，并代换 $t = x - h$。

2. 将分母 $t - a, t + a$ 配对，其倒数和为 $2t/(t^2 - a^2)$。

3. 除以 t 或其他共同因子前，先把零情形代回原方程。

4. 若结果只含 t^2，令 $u = t^2$ 求解，再恢复正负号并剔除原极点。

> Example. Solve $1/(x-1) + 1/(x+1) = x$ over the reals.

> 例题：在实数范围解 $1/(x-1) + 1/(x+1) = x$。

Worked solution

1. The poles $x = \pm 1$ are excluded. Pairing gives $2x/(x^2 - 1) = x$.

Section 1.1 Equations & algebraic modeling

方程与代数建模

Method 10 / 方法 10 [AL01-M10] (continued / 续)

2. Test $x = 0$ before division: it works. For $x \neq 0$, $2/(x^2 - 1) = 1$, hence $x^2 = 3$.

3. The complete solution set is $\{0, \sqrt{3}, -\sqrt{3}\}$, with no pole included.

例题解析

1. 排除极点 $x = \pm 1$，配对得 $2x/(x^2 - 1) = x$。

2. 相除之前检查 $x = 0$，它成立。对 $x \neq 0$，得 $2/(x^2 - 1) = 1$，故 $x^2 = 3$。

3. 完整解集为 $\{0, \sqrt{3}, -\sqrt{3}\}$，均非极点。

Centering removes clutter; it does not permit dropping the zero branch.

中心化可减少繁杂，但不能丢弃零分支。

Practice / 配套练习: AIME-2000-I-Q09 (WB p. 116), AIME-2014-I-Q14 (WB p. 30)

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 1 / 方法 1 [AL02-M01]

A polynomial is a structured object: coefficients encode root symmetries, factorization controls multiplicity, and a degree bound makes identities rigid. Reduce the expression before solving roots, and restore all real-domain restrictions afterward.

多项式具有结构：系数编码根的对称性，因式分解控制重数，次数界使恒等式具有刚性。先约化表达式，再求根，最后恢复全部实数定义域限制。

Recover symmetric root expressions directly from coefficients

直接由系数恢复对称根式

Procedure

1. Normalize the leading coefficient and identify the elementary sums e_1, e_2, e_3.

2. Use $p_2 = e_1^2 - 2e_2$ and $p_3 = e_1p_2 - e_2p_1 + 3e_3$, rather than solving roots individually.

3. For two bases with sum s and product p, their power sums satisfy $u_{n+2} = su_{n+1} - pu_n$.

4. Account explicitly for repeated copies of each root, and check real-root feasibility if the original variables are real.

通用步骤

1. 先归一首项系数，再识别基本对称和 e_1, e_2, e_3。

2. 使用 $p_2 = e_1^2 - 2e_2, p_3 = e_1p_2 - e_2p_1 + 3e_3$，避免逐个解根。

3. 两个底数的和为 s、积为 p 时，其幂和满足 $u_{n+2} = su_{n+1} - pu_n$。

4. 明确处理每个根的重复副本；若原变量为实数，还须检查实根可行性。

> Example. The roots of $x^3 - 2x^2 - 3x + 1$ are r, s, t. Find $r^3 + s^3 + t^3$.

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 1 / 方法 1 [AL02-M01] (continued / 续)

> 例题：x^3-2x^2-3x+1 的根为 r,s,t，求 $r^3+s^3+t^3$。

Worked solution

1. Vieta gives $e_1=2, e_2=-3, e_3=-1$.

2. The square sum is $p_2=2^2-2(-3)=10$.

3. Thus $p_3=e_1p_2-e_2e_1+3e_3=20+6-3=23$.

例题解析

1. 韦达关系给出 $e_1=2, e_2=-3, e_3=-1$。

2. 平方和为 $p_2=2^2-2(-3)=10$。

3. 故 $p_3=e_1p_2-e_2e_1+3e_3=20+6-3=23$。

Vieta signs alternate; divide by the leading coefficient first.

韦达符号交替，须先除首项系数。

If each root z_i is repeated k times, the sum is $k\sum z_i$, while the pair sum is $\binom{k}{2}\sum z_i^2+k^2\sum_{i<j}z_iz_j$. This counts same-root copies separately from different-root copies.

For square roots s_1,s_2,s_3, the product $(s_1+s_2+s_3)(-s_1+s_2+s_3)(s_1-s_2+s_3)(s_1+s_2-s_3)$ equals $4e_2-e_1^2$, where e_i refer to s_1^2,s_2^2,s_3^2. A zero sign sum therefore imposes a symmetric coefficient relation. Still check that other sign sums do not collide.

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 1 / 方法 1 [AL02-M01] (continued / 续)

每个根 z_i 重复 k 次时，总和为 $k\sum z_i$，两两积和为 $\binom{k}{2}\sum z_i^2 + k^2\sum_{i<j} z_i z_j$；须分开同根副本与异根副本。

对平方根 s_1, s_2, s_3，乘积 $(s_1+s_2+s_3)(-s_1+s_2+s_3)(s_1-s_2+s_3)(s_1+s_2-s_3)$ 等于 $4e_2 - e_1^2$，其中 e_i 是 s_1^2, s_2^2, s_3^2 的基本对称式。零符号和因此给出系数关系，但仍须检查其他符号和是否重合。

Practice / 配套练习: AIME-2001-I-Q03 (WB p. 32), AIME-2002-I-Q06 (WB p. 117), AIME-2005-I-Q08 (WB p. 118), AIME-2008-II-Q07 (WB p. 42), AIME-2011-I-Q15 (WB p. 556), AIME-2015-II-Q06 (WB p. 48), AIME-2015-II-Q14 (WB p. 31), AIME-2019-I-Q08 (WB p. 51), AIME-2019-I-Q10 (WB p. 52), AIME-2021-II-Q07 (WB p. 28), AIME-2021-II-Q04 (WB p. 36), AIME-2026-I-Q06 (WB p. 124), AIME-2026-II-Q11 (WB p. 98), AIME-2010-II-Q07 (WB p. 71), AIME-2002-II-Q15 (WB p. 341)

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 2 / 方法 2 [AL02-M02]

Reduce a polynomial at roots and interchange root products

在根处约化多项式并交换根乘积

Procedure

1. Divide $A(x)$ by the polynomial $Q(x)$ defining the roots: at each root only the remainder matters.

2. For a sum, use Vieta or power-sum identities on the lower-degree remainder.

3. For a product, factor one polynomial and interchange the two root products, carefully tracking degree-dependent signs and leading coefficients.

通用步骤

1. 以定义根的多项式 $Q(x)$ 除 $A(x)$；在每个根处只需保留余式。

2. 求和时，对较低次余式使用韦达或幂和恒等式。

3. 求积时，分解一个多项式并交换两组根的乘积，仔细追踪次数决定的符号与首项系数。

> Example. If r, s are the roots of $x^2 - x - 1$, find $r^4 + s^4$ and $(r-2)(s-2)$.

> 例题：若 r, s 是 $x^2 - x - 1$ 的根，求 $r^4 + s^4$ 与 $(r-2)(s-2)$。

Worked solution

1. At either root $x^2 = x + 1$, so $x^4 = 3x + 2$. Hence $r^4 + s^4 = 3(r+s) + 4 = 7$.

2. The monic factorization is $Q(x) = (x - r)(x - s)$.

3. Thus $(r-2)(s-2) = Q(2) = 1$, since two sign reversals cancel.

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 2 / 方法 2 [AL02-M02] (continued / 续)

例题解析

1. 根处 $x^2 = x + 1$，故 $x^4 = 3x + 2$，从而 $r^4 + s^4 = 3(r + s) + 4 = 7$。

2. 首一因式分解为 $Q(x) = (x - r)(x - s)$。

3. 所以 $(r - 2)(s - 2) = Q(2) = 1$，两次反号相消。

At roots of Q, equality modulo Q is exact equality of values.

在 Q 的根处，模 Q 相等就是函数值精确相等。

Practice / 配套练习: AIME-2003-II-Q09 (WB p. 38), AIME-2023-II-Q08 (WB p. 84), AIME-2024-II-Q13 (WB p. 97)

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 3 / 方法 3 [AL02-M03]

Recognize hidden powers, telescoping products and convolutions

识别隐藏幂、望远镜乘积与卷积

Procedure

1. Compare coefficients with $(x-h)^k$, a geometric sum, or a product of two short coefficient lists.

2. Use difference-of-squares or difference-of-cubes identities to telescope products or rationalize a radical.

3. If a multiplier such as $x-1$ is introduced, record its zeros and remove only the roots actually introduced by it.

4. After factoring, check which factors can vanish on the requested real or complex domain.

通用步骤

1. 将系数与 $(x-h)^k$、等比和或两个短系数表的乘积比较。

2. 用平方差或立方差恒等式使乘积望远镜消去，或对根式有理化。

3. 若引入 $x-1$ 等乘子，记录其零点，并只删除由该乘子真正引入的根。

4. 分解后，检查哪些因子能在所求实数或复数定义域内为零。

> Example. Find a polynomial R with $R(0)=1$ such that $R(x)^2=1+2x+3x^2+2x^3+x^4$, and prove uniqueness.

> 例题：求满足 $R(0)=1$ 且 $R(x)^2=1+2x+3x^2+2x^3+x^4$ 的多项式 R，并证明唯一性。

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 3 / 方法 3 [AL02-M03] (continued / 续)

Worked solution

1. Degree comparison forces $R = 1 + ux + vx^2$. The coefficient of x in its square is $2u$, so $u = 1$.

2. The x^2 coefficient is $u^2 + 2v = 3$, so $v = 1$.

3. The candidate $R = 1 + x + x^2$ also gives the required x^3, x^4 coefficients. Both unknown coefficients were forced, proving uniqueness.

例题解析

1. 比较次数得 $R = 1 + ux + vx^2$；平方的 x 系数为 $2u$，故 $u = 1$。

2. x^2 系数为 $u^2 + 2v = 3$，故 $v = 1$。

3. 候选 $R = 1 + x + x^2$ 也给出所需的 x^3, x^4 系数；两个未知系数均被迫确定，故唯一。

Convolution explains a triangular coefficient pattern; inspection alone does not establish multiplicities.

卷积解释三角形系数模式；单凭观察不能确定根的重数。

Practice / 配套练习: AIME-2000-II-Q13 (WB p. 54), AIME-2003-II-Q15 (WB p. 88), AIME-2004-I-Q13 (WB p. 89), AIME-2005-I-Q06 (WB p. 68), AIME-2005-II-Q07 (WB p. 40), AIME-2006-I-Q12 (WB p. 199), AIME-2013-I-Q05 (WB p. 33), AIME-2014-I-Q09 (WB p. 46), AIME-2019-I-Q12 (WB p. 95), AIME-2024-II-Q11 (WB p. 61), AIME-2023-II-Q04 (WB p. 21), AIME-2002-II-Q11 (WB p. 154), AIME-2015-I-Q03 (WB p. 469)

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 4 / 方法 4 [AL02-M04]

Partition preimages and common roots before matching coefficients

比较系数前先划分原像与公共根

Procedure

1. A degree-d polynomial has at most d distinct roots unless it is zero.

2. For a composition or several level values, partition the inputs according to which target root or level they map to.

3. Apply Vieta within each group and compare coefficients that the defining polynomials share.

4. For shared factors, use the degree of the least common multiple, subtract equations to isolate the common root, and verify existence.

通用步骤

1. 非零 d 次多项式至多有 d 个不同根。

2. 对复合式或多个水平值，按输入映到哪个目标根或水平值来分组。

3. 在每组内使用韦达，并比较定义多项式共有的系数。

4. 对公共因子，利用最小公倍多项式的次数，相减方程隔离公共根，并验证存在性。

> Example. Two monic quadratics differ by a nonzero constant. Can they share a root? Can their two root sums differ?

> 例题：两个首一二次多项式相差非零常数，它们能有公共根吗？两组根和能不同吗？

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 4 / 方法 4 [AL02-M04] (continued / 续)

Worked solution

1. At a common root their difference would be zero, contradicting the nonzero constant. Hence no root is shared.

2. The linear coefficients are equal because only the constant changes.

3. Vieta therefore gives equal root sums, even though the two root sets are disjoint.

例题解析

1. 若有公共根，在该根处两式之差为零，与非零常数矛盾，所以不能共根。

2. 因只改变常数项，两式的一次系数相同。

3. 故韦达给出相同根和，尽管两组根集不相交。

Root multiplicity, distinctness and grouping are three separate pieces of information.

重数、互异性与分组是三条不同信息。

Practice / 配套练习: AIME-2007-I-Q08 (WB p. 41), AIME-2014-II-Q05 (WB p. 34), AIME-2015-I-Q10 (WB p. 47), AIME-2020-I-Q11 (WB p. 58), AIME-2020-I-Q14 (WB p. 59), AIME-2020-II-Q11 (WB p. 60)

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 5 / 方法 5 [AL02-M05]

Lower the degree before interpolation or coefficient elimination

插值或系数消元前先降低次数

Procedure

1. Add or subtract polynomials whose leading terms cancel; determine the actual remaining degree.

2. A degree bound plus enough values can determine a polynomial; use a root-count argument to prove uniqueness.

3. For multivariable polynomials, write the coefficient basis and solve the linear vanishing conditions before seeking universal zeros.

通用步骤

1. 将首项可消去的多项式相加或相减，并确定剩余实际次数。

2. 次数界加足够多函数值可确定多项式；以根数论证证明唯一性。

3. 多变量多项式先写系数基，并解线性消零条件，再求全体多项式共有的零点。

> Example. A polynomial P of degree at most three agrees with x^2 at $x = 0, 1, 2, 3$. Find $P(10)$.

> 例题：次数至多为三的多项式 P 在 $x = 0, 1, 2, 3$ 处与 x^2 相等，求 $P(10)$。

Worked solution

1. Form $R(x) = P(x) - x^2$, whose degree is at most three.

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 5 / 方法 5 [AL02-M05] (continued / 续)

2. The four distinct given inputs are roots of R. A nonzero polynomial of degree at most three cannot have four roots.

3. Therefore R is identically zero and $P(10) = 10^2 = 100$.

例题解析

1. 作 $R(x) = P(x) - x^2$，其次数至多为三。

2. 四个不同输入均为 R 的根，非零的至多三次多项式不可能有四个不同根。

3. 故 R 恒为零，$P(10) = 10^2 = 100$。

A canceled leading term changes the degree used in Vieta as well as interpolation.

首项消去既改变插值次数，也改变韦达所使用的次数。

Practice / 配套练习: AIME-2001-I-Q03 (WB p. 32), AIME-2003-II-Q08 (WB p. 143), AIME-2008-I-Q13 (WB p. 55), AIME-2022-I-Q01 (WB p. 37)

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 6 / 方法 6 [AL02-M06]

Use integer roots and coefficient bounds as a finite algebraic count

用整数根与系数界作有限代数计数

Procedure

1. For a monic integer polynomial, rational roots are integers. Factorization into monic rational factors preserves integrality of their coefficients by Gauss's lemma.

2. Convert root sums and products into integer factor pairs or bounded root parameters.

3. When counting exactly one distinct integer solution, distinguish a repeated root from a second root coinciding with an already excluded input.

4. Divide by root permutations only after treating repeated roots; include any unrestricted coefficients as independent factors.

通用步骤

1. 首一整系数多项式的有理根为整数；由高斯引理，其首一有理因子也具有整系数。

2. 把根和与根积转成整数因子对或有界根参数。

3. 恰数一个不同整数解时，区分重根与第二个根恰等于已排除输入。

4. 处理重根后才能除以根的排列数；无约束系数作为独立因子计入。

> Example. Count monic quadratics with nonnegative integer roots and constant term 12.

> 例题：统计根均为非负整数、常数项为 12 的首一二次多项式。

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 6 / 方法 6 [AL02-M06] (continued / 续)

Worked solution

1. Write $(x-r)(x-s)$ with $rs=12$ and $0 \le r \le s$.

2. The unordered positive factor pairs are $(1,12),(2,6),(3,4)$. Zero cannot occur because the product is positive.

3. They give three distinct sums $13, 8, 7$, hence three distinct polynomials.

例题解析

1. 写为 $(x-r)(x-s)$，其中 $rs=12, 0 \le r \le s$。

2. 无序正因子对为 $(1,12),(2,6),(3,4)$；积为正，所以零不能出现。

3. 三对给出不同根和 $13, 8, 7$，所以有三个不同多项式。

Counting root lists is not automatically the same as counting polynomials.

统计根列表并不自动等于统计多项式。

Practice / 配套练习: AIME-2005-II-Q13 (WB p. 553), AIME-2010-II-Q06 (WB p. 44), AIME-2010-II-Q10 (WB p. 713), AIME-2013-I-Q10 (WB p. 76), AIME-2013-II-Q12 (WB p. 56), AIME-2014-I-Q06 (WB p. 45), AIME-2018-I-Q01 (WB p. 35), AIME-2023-I-Q09 (WB p. 53)

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 7 / 方法 7 [AL02-M07]

Use nonnegativity, discriminants and repeated roots at extrema

用非负性、判别式与最值重根

Procedure

1. A real polynomial nonnegative on an open interval has even multiplicity at each interior zero.

2. For a quadratic, use discriminant and sign conditions; for two equal global minima, factor the shifted polynomial by both double roots.

3. Compare coefficients to derive parameter conditions, then prove the roots are distinct, inside the domain and attain the claimed bound.

4. For constrained positive roots, establish compactness first and inspect repeated-root boundary configurations.

通用步骤

1. 实多项式在开区间非负，则每个内部零点的重数为偶数。

2. 二次式用判别式与符号条件；若有两个相等全局最小值，则将平移后的多项式按两个二重根分解。

3. 比较系数得到参数条件，再证明根互异、位于定义域内并达到所求界。

4. 受约束正根的最值，先建立紧致性，再检查重根边界构型。

> Example. A real quadratic Q satisfies $Q(x) \geq x^2$ for every real x and $Q(2) = 4$. Show the form of Q.

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 7 / 方法 7 [AL02-M07] (continued / 续)

> 例题：实二次式 Q 对每个实数 x 满足 $Q(x) \geq x^2$，且 $Q(2) = 4$，求 Q 的形式。

Worked solution

1. The polynomial $D(x) = Q(x) - x^2$ is nonnegative and vanishes at two.

2. A nonzero linear term at that zero would change sign across it. Thus the zero has even multiplicity, or D is identically zero.

3. Therefore $Q(x) = x^2 + c(x-2)^2$ with $c \geq 0$, including $c = 0$. Every such polynomial satisfies the requirements.

例题解析

1. 多项式 $D(x) = Q(x) - x^2$ 非负，且在二处为零。

2. 若零点只有非零一次项，越过它就会变号，所以零点重数须为偶数，或 D 恒为零。

3. 因此 $Q(x) = x^2 + c(x-2)^2$，其中 $c \geq 0$，包括 $c = 0$；每个这种多项式都满足要求。

Repeated-root conditions are necessary at an interior extremum; verify sufficiency and endpoint alternatives.

内部最值的重根条件是必要条件，仍须检验充分性与端点情形。

For positive a, b, c with fixed pair sum and product, an extremum of $a+b+c$ occurs with two variables equal: if the three were distinct positive roots of $t^3 - st^2 + qt - r$, small changes of s would retain three distinct positive roots by continuity, contradicting an extremum. This argument applies only after compactness and positivity exclude escape to the boundary.

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 7 / 方法 7 [AL02-M07] (continued / 续)

To prove four real roots of a quartic, find four disjoint sign-change intervals, counting any boundary root by its nonzero derivative. To prove there are fewer, combine a positive region with the number of derivative zeros on the remaining region.

正数 a, b, c 的两两积和与积固定时，$a+b+c$ 的最值至少有两个变量相等：若三者为 $t^3 - st^2 + qt - r$ 的三个不同正根，连续性使 s 小幅变化后仍保留三个不同正根，与最值矛盾。但此前必须以紧致性及正性排除逃向边界。

证明四次式有四个实根，可找四个不相交变号区间；若根落在区间边界，可用非零导数判断变号。证明不足四根，则可把恒正区域与其余区域的导数零点数结合。

Practice / 配套练习: AIME-2005-II-Q15 (WB p. 419), AIME-2010-I-Q06 (WB p. 43), AIME-2014-I-Q14 (WB p. 30), AIME-2018-II-Q06 (WB p. 50), AIME-2021-I-Q15 (WB p. 420), AIME-2024-I-Q15 (WB p. 189), AIME-2025-I-Q11 (WB p. 180), AIME-2025-II-Q15 (WB p. 191), AIME-2026-II-Q08 (WB p. 318)

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 8 / 方法 8 [AL02-M08]

Exploit polynomial identities through forced roots and finite root orbits

用被迫根与有限根轨道处理多项式恒等式

Procedure

1. Compare degrees and leading coefficients before substituting special inputs.

2. Use zeros of multipliers to force specific polynomial roots, then factor them out as a polynomial identity.

3. If a root is mapped to another root, analyze its finite orbit; a modulus that grows without bound contradicts a finite root set.

4. A polynomial with a nonzero real period is constant, since its finite difference otherwise has degree one less with nonzero leading coefficient.

通用步骤

1. 代入特殊值之前，先比较次数与首项系数。

2. 利用乘子零点强迫特定根，再按多项式恒等式将其提出。

3. 若根被映为另一个根，分析其有限轨道；模长无限增长将与有限根集矛盾。

4. 具有非零实周期的多项式必为常数，否则有限差分将具有低一次的次数与非零首项系数。

> Example. If a nonzero polynomial has every root on or inside the unit circle and constant term equal in modulus to its leading coefficient, prove every root has modulus one.

> 例题：若非零多项式的全部根位于单位圆内或圆上，且常数项与首项系数模长相等，证明所有根的模长均为一。

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 8 / 方法 8 [AL02-M08] (continued / 续)

Worked solution

1. By Vieta, the product of all root moduli is the modulus of the constant-to-leading coefficient ratio, namely one.

2. Every factor is at most one. If any were strictly below one, the product would be strictly below one.

3. Hence all moduli equal one. In particular no root is zero.

例题解析

1. 由韦达，所有根模长之积为常数项与首项系数比的模长，即一。

2. 每个因子均不超过一；若某个严格小于一，乘积就会严格小于一。

3. 故所有模长均为一，特别地不存在零根。

A root-map argument must show the map remains in the same finite root set.

根映射论证须证明映射始终留在同一有限根集中。

Practice / 配套练习: AIME-2007-II-Q14 (WB p. 211), AIME-2016-I-Q11 (WB p. 57)

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 9 / 方法 9 [AL02-M09]

Extract coefficients through evaluations and symmetric selections

通过代值与对称选取提取系数

Procedure

1. The coefficient of x^k in a product is the sum of all selections whose degrees total k.

2. For pair selections use $\sum_{i<j} u_i u_j = [(\sum u_i)^2 - \sum u_i^2]/2$.

3. An evaluation at one sums coefficients. To sum absolute coefficients, first prove a substitution makes all coefficients nonnegative.

通用步骤

1. 乘积中 x^k 系数等于全部所选项次数合计为 k 的选取之和。

2. 两两选取可用 $\sum_{i<j} u_i u_j = [(\sum u_i)^2 - \sum u_i^2]/2$。

3. 代入一可求系数和；若求绝对值和，须先证明某个代换使所有系数非负。

> Example. Find the sum of the absolute coefficients of $(1-x)^3(1+x^2)$.

> 例题：求 $(1-x)^3(1+x^2)$ 的系数绝对值之和。

Worked solution

1. Replacing x by $-x$ gives $(1+x)^3(1+x^2)$, whose coefficients are all nonnegative.

2. The replacement multiplies the degree-j coefficient by $(-1)^j$, so these nonnegative coefficients are the original absolute coefficients.

3. Evaluate at one: the required sum is $2^3 \cdot 2 = 16$.

Section 1.2 Polynomials & Vieta relations

多项式与韦达关系

Method 9 / 方法 9 [AL02-M09] (continued / 续)

例题解析

1. 以 $-x$ 替换 x 得 $(1+x)^3(1+x^2)$，各系数均非负。

2. 该替换把 j 次系数乘 $(-1)^j$，故这些非负系数恰为原系数绝对值。

3. 代入一，所求和为 $2^3 \cdot 2 = 16$。

Evaluating at minus one gives an absolute sum only after a coefficient-sign proof.

只有证明系数符号规律后，代入负一才给出绝对值和。

Practice / 配套练习: AIME-2004-I-Q07 (WB p. 39), AIME-2016-II-Q06 (WB p. 49)

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 1 / 方法 1 [AL03-M01]

Choose coordinates for addition, polar form for multiplication, and geometric loci for a modulus or argument condition. The useful bridge is the complex quotient: its modulus is a length ratio and its argument is a directed angle.

加法用直角坐标，乘法用极形式，模长或辐角条件用几何轨迹。关键桥梁是复商：其模长为长度比，其辐角为有向角。

Separate real and imaginary parts with all sign branches

连同符号分支分离实部与虚部

Procedure

1. Write $z = x + iy$, expand using $i^2 = -1$, and compare real and imaginary parts.

2. Before clearing a complex denominator, exclude its zero. For principal square roots of real quantities, split nonnegative and negative radicands.

3. For product systems, introduce the common product and reconstruct both consistent square-root signs.

4. Check modulus, positivity or integrality restrictions on the resulting coordinates.

通用步骤

1. 写 $z = x + iy$，用 $i^2 = -1$ 展开，再比较实部与虚部。

2. 清除复分母前先排除零；对实数量的主平方根，区分被开方式非负与负。

3. 乘积方程组可设置共同乘积，并恢复两个一致的平方根符号。

4. 检查所得坐标的模长、正性或整数限制。

> Example. Find all complex z with $z + 2\overline{z} = 6 + i$.

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 1 / 方法 1 [AL03-M01] (continued / 续)

> 例题：求满足 $z+2\overline{z}=6+i$ 的全部复数 z。

Worked solution

1. Write $z=x+iy$, so $\overline{z}=x-iy$.

2. The left side is $3x-iy$. Comparing components gives $3x=6, -y=1$.

3. The unique solution is $z=2-i$, and direct substitution gives $6+i$.

例题解析

1. 写 $z=x+iy$，则 $\overline{z}=x-iy$。

2. 左侧为 $3x-iy$，比较分量得 $3x=6, -y=1$。

3. 唯一解为 $z=2-i$，直接代入得 $6+i$。

A complex number is real exactly when its imaginary part vanishes, including the value zero.

复数为实数当且仅当虚部为零，其中包括数值零。

For a Gaussian integer $a+bi$, the norm is a^2+b^2, and its cube has components a^3-3ab^2 and $3a^2b-b^3$. Multiplication by a unit $\pm1, \pm i$ and conjugation change signs or exchange component magnitudes. Before a triangle test, verify both candidate components are positive and use $|X-Y|<L<X+Y$ with proved finite bounds on a, b.

高斯整数 $a+bi$ 的范数为 a^2+b^2，立方的分量为 $a^3-3ab^2, 3a^2b-b^3$。乘单位 $\pm1, \pm i$ 与共轭可改变符号或交换分量绝对值。检验三角形前，须确认候选分量为正，并使用 $|X-Y|<L<X+Y$，且先证明 a, b 的有限界。

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 1 / 方法 1 [AL03-M01] (continued / 续)

Practice / 配套练习: AIME-2007-I-Q03 (WB p. 62), AIME-2009-I-Q02 (WB p. 63), AIME-2012-II-Q08 (WB p. 75), AIME-2016-I-Q07 (WB p. 79), AIME-2018-II-Q05 (WB p. 64), AIME-2019-II-Q08 (WB p. 82), AIME-2023-I-Q15 (WB p. 96)

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 2 / 方法 2 [AL03-M02]

Pair conjugate roots and recover their real polynomial factor

配对共轭根并恢复实系数因子

Procedure

1. A real-coefficient polynomial contains every nonreal root together with its conjugate and equal multiplicity.

2. The pair $u \pm iv$ gives factor $x^2 - 2ux + (u^2 + v^2)$.

3. Use Vieta to connect that factor with the remaining roots. For an integer polynomial, trace and norm of this pair are integral only when the pair is known to form a monic rational quadratic factor; Gauss's lemma then applies.

4. Separate all-real multisets from the one-nonreal-pair case when counting cubics.

通用步骤

1. 实系数多项式的每个非实根都与其共轭以相同重数出现。

2. 根对 $u \pm iv$ 给出因子 $x^2 - 2ux + (u^2 + v^2)$。

3. 用韦达将该因子与其余根联系。整系数多项式中，只有已知该根对构成首一有理二次因子时，才能由高斯引理要求其迹与范数为整数。

4. 统计三次式时，将全实根多重集与一个非实根对的情形分开。

> Example. A monic real cubic has roots $2 + i$ and -3. Its x^2 coefficient is -1. Find the constant term.

> 例题：首一实三次式有根 $2 + i, -3$，其 x^2 系数为 -1，求常数项。

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 2 / 方法 2 [AL03-M02] (continued / 续)

Worked solution

1. The third root must be $2-i$. The root sum is $2+i+2-i-3=1$, consistent with coefficient -1.

2. The conjugate pair has product $(2+i)(2-i)=5$.

3. The three-root product is -15, so the constant term is its negative, 15.

例题解析

1. 第三根必为 $2-i$。根和为 $2+i+2-i-3=1$，与系数 -1 一致。

2. 共轭根对乘积为 $(2+i)(2-i)=5$。

3. 三根乘积为 -15，所以常数项为其相反数 15。

Conjugation fixes the polynomial only when its coefficients are real.

只有系数为实数时，共轭才保持该多项式不变。

Integer coefficients alone do not force each conjugate pair to have integral trace or norm: in x^3-2, the nonreal pair has irrational sum $-\sqrt[3]{2}$. Apply coefficient integrality only after establishing a rational factor, or use the full polynomial coefficient equations.

整系数并不自动使每个共轭根对的迹与范数为整数。例如 x^3-2 的非实根对之和为无理数 $-\sqrt[3]{2}$。须先确定因子系数为有理数，才能使用因子整系数性；否则应使用整个多项式的系数方程。

Practice / 配套练习: AIME-2005-I-Q06 (WB p. 68), AIME-2010-II-Q07 (WB p. 71), AIME-2013-I-Q10 (WB p. 76), AIME-2013-II-Q12 (WB p. 56), AIME-2021-II-Q04 (WB p. 36), AIME-2026-II-Q11 (WB p. 98)

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 3 / 方法 3 [AL03-M03]

Use polar powers and unit-circle angle congruences

使用极形式幂与单位圆辐角同余

Procedure

1. For nonzero z, write $z = re^{i\theta}$; multiplication adds arguments and multiplies moduli.

2. Equality of unit complex numbers is an angle congruence modulo 2π, not an ordinary angle equality.

3. For reality, set the imaginary part to zero or equate a complex expression to its conjugate.

4. Count a half-open complete argument interval and remove overlapping angle families explicitly.

通用步骤

1. 非零 z 写成 $re^{i\theta}$；相乘时辐角相加、模长相乘。

2. 单位复数相等对应模 2π 的辐角同余，而非普通角度相等。

3. 要求实数时，可令虚部为零，或令表达式与其共轭相等。

4. 在半开完整辐角区间计数，并明确消除不同辐角族的重叠。

> Example. How many unit complex numbers satisfy $z^4 \in \mathbb{R}$ but $z^2 \notin \mathbb{R}$?

> 例题：有多少单位复数满足 $z^4 \in \mathbb{R}$ 而 $z^2 \notin \mathbb{R}$?

Worked solution

1. Write $z = e^{i\theta}$ for $0 \le \theta < 2\pi$. Reality of z^4 requires 4θ a multiple of π, giving eight angles.

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 3 / 方法 3 [AL03-M03] (continued / 续)

2. Among them z^2 is real exactly when 2θ is a multiple of π, giving four of those angles.

3. The required difference is $8 - 4 = 4$.

例题解析

1. 写 $z = e^{i\theta}$，其中 $0 \le \theta < 2\pi$。z^4 为实数要求 4θ 为 π 的倍数，给出八角。

2. 其中 z^2 为实数恰当 2θ 为 π 的倍数，共四角。

3. 所求个数为 $8 - 4 = 4$。

A root may have modulus one without being a primitive root of its stated order.

根的模为一，并不表示它一定是该次数的本原单位根。

Practice / 配套练习: AIME-2000-II-Q09 (WB p. 67), AIME-2001-II-Q14 (WB p. 86), AIME-2005-II-Q09 (WB p. 69), AIME-2012-I-Q06 (WB p. 73), AIME-2014-II-Q10 (WB p. 78), AIME-2018-I-Q06 (WB p. 81), AIME-2019-II-Q08 (WB p. 82), AIME-2022-I-Q04 (WB p. 65)

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 4 / 方法 4 [AL03-M04]

Sum rotating displacements with a complex geometric series

以复等比级数累加旋转位移

Procedure

1. Encode the multiplier for one displacement as $q = re^{i\theta}$.

2. For finite motion with $q \neq 1$, use $\sum_{j=0}^{n-1} q^j = (1-q^n)/(1-q)$; if $q = 1$, the sum is n. For a limit verify $|q| < 1$ first.

3. A rotation about c is $z \mapsto c + e^{i\theta}(z-c)$; composing with translations produces an affine map.

4. Compute the final norm only after summing the vectors; for nested disks prove the limit lies in every disk.

通用步骤

1. 将一次位移的倍率编码为 $q = re^{i\theta}$。

2. 有限运动中，$q \neq 1$ 时用 $\sum_{j=0}^{n-1} q^j = (1-q^n)/(1-q)$；$q = 1$ 时和为 n。极限问题先验证 $|q| < 1$。

3. 绕 c 旋转为 $z \mapsto c + e^{i\theta}(z-c)$；与平移复合得到仿射映射。

4. 向量求和后才求最终模长；嵌套圆盘还须证明极限位于每个圆盘内。

> Example. Find the fixed point of $F(z) = iz + 1$ and explain the geometry of all its orbits.

> 例题：求 $F(z) = iz + 1$ 的不动点，并说明全部轨道的几何意义。

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 4 / 方法 4 [AL03-M04] (continued / 续)

Worked solution

1. The fixed point satisfies $c = ic + 1$, so $c = 1/(1-i) = (1+i)/2$.

2. Subtraction gives $F(z) - c = i(z - c)$, a counterclockwise quarter-turn about c.

3. The fixed point has period one. Every other point has exact period four, because $i^k(z - c) = z - c$ first holds at $k = 4$.

例题解析

1. 不动点满足 $c = ic + 1$，故 $c = 1/(1-i) = (1+i)/2$。

2. 相减得 $F(z) - c = i(z - c)$，即绕 c 逆时针四分之一周旋转。

3. 不动点周期为一；其余每点的准确周期为四，因为 $i^k(z - c) = z - c$ 首次在 $k = 4$ 成立。

Adding lengths gives path length; adding complex displacements gives the endpoint.

长度相加得到路程，复位移相加才得到终点。

If nested disks have radii tending to zero and centers converging to B, verify $|B - O_n| \le r_n$ for every n. Any two common points would have distance at most $2r_n \to 0$, proving uniqueness.

嵌套圆盘半径趋零、圆心趋于 B 时，须对每个 n 验证 $|B - O_n| \le r_n$。任意两个公共点距离至多为 $2r_n \to 0$，故公共点唯一。

Practice / 配套练习: AIME-2008-II-Q09 (WB p. 70), AIME-2013-I-Q14 (WB p. 93), AIME-2017-II-Q12 (WB p. 94), AIME-2020-I-Q08 (WB p. 83), AIME-2026-I-Q05 (WB p. 66)

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 5 / 方法 5 [AL03-M05]

Evaluate products over roots of unity

计算单位根上的乘积

Procedure

1. Use $x^n-1=\prod_{k=0}^{n-1}(x-\omega^k)$ when the powers exhaust all nth roots.

2. For nontrivial roots, divide out $x-1$ and evaluate its removable value at one as n.

3. Factor a general polynomial into its roots, interchange the two products and reduce powers modulo the smaller defining polynomial.

4. For chord lengths, take the modulus after the complex product; do not replace a signed algebraic product by a length prematurely.

通用步骤

1. 当各幂穷尽所有 n 次单位根时，使用 $x^n-1=\prod_{k=0}^{n-1}(x-\omega^k)$。

2. 对非平凡单位根，先除去 $x-1$，在一处的可去值为 n。

3. 一般多项式先按其根分解，交换两重乘积，再以较小的定义多项式约化幂。

4. 弦长问题应在复乘积之后取模，不能过早把带符号代数乘积换成长度。

> Example. For a primitive fifth root ω, compute $\prod_{k=1}^{4}(1-\omega^k)$ and the product of distances from 1 to those four roots.

> 例题：对本原五次单位根 ω，计算 $\prod_{k=1}^{4}(1-\omega^k)$，及一到这四根的距离之积。

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 5 / 方法 5 [AL03-M05] (continued / 续)

Worked solution

1. The polynomial of the nontrivial fifth roots is $1+x+x^2+x^3+x^4$.

2. Evaluating at one gives the algebraic product 5.

3. The distance product is the modulus of that product, also 5.

例题解析

1. 非平凡五次单位根的多项式为 $1+x+x^2+x^3+x^4$。

2. 在一处代值得代数乘积 5。

3. 距离乘积为该乘积的模长，也为 5。

If the chosen root is not primitive, its powers may repeat and miss roots.

若所选根非本原根，其幂可能重复并遗漏其他根。

Practice / 配套练习: AIME-2003-II-Q15 (WB p. 88), AIME-2004-I-Q13 (WB p. 89), AIME-2009-II-Q13 (WB p. 91), AIME-2023-II-Q08 (WB p. 84), AIME-2024-II-Q13 (WB p. 97)

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 6 / 方法 6 [AL03-M06]

Translate complex quotients into angles, circles and perpendicularity

将复商转为角、圆与垂直条件

Procedure

1. For nonzero displacements u, v, u/v real means parallel directions; purely imaginary means perpendicular directions.

2. A real cross-ratio equates directed inscribed angles modulo π; for three fixed noncollinear points its locus is their circle.

3. Equal-distance loci are perpendicular bisectors; a modulus equation $|z-a|=R$ is a circle.

4. Exclude quotient poles and zero displacement legs, then use coordinates or a circle center for the requested extremum.

通用步骤

1. 对非零位移 u, v，u/v 为实数表示方向平行，为纯虚数表示方向垂直。

2. 实交比表示有向圆周角模 π 相等；三个固定点不共线时，轨迹为其外接圆。

3. 等距轨迹是垂直平分线，模方程 $|z-a|=R$ 是圆。

4. 排除商的极点与零位移边，再用坐标或圆心求所需最值。

> Example. Find the locus of z for which $(z-1)/(z+1)$ is purely imaginary, including zero as an imaginary value.

> 例题：求使 $(z-1)/(z+1)$ 为纯虚数的 z 的轨迹，其中允许纯虚数取零。

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 6 / 方法 6 [AL03-M06] (continued / 续)

Worked solution

1. The pole $z=-1$ is excluded. Write $z=x+iy$ and multiply numerator and denominator by $x+1-iy$.

2. The real numerator is x^2+y^2-1 and the denominator is positive, so reality of the zero real part requires $x^2+y^2=1$.

3. The locus is the unit circle with -1 deleted; $z=1$ remains because the quotient is zero.

例题解析

1. 排除极点 $z=-1$。写 $z=x+iy$，分子分母乘以 $x+1-iy$。

2. 分子实部为 x^2+y^2-1，分母为正，故实部为零要求 $x^2+y^2=1$。

3. 轨迹为删去 -1 的单位圆；$z=1$ 保留，因为商为零。

Specify whether zero counts as purely imaginary before using it as a perpendicularity test.

用纯虚数判垂直之前，须明确零是否允许；非退化垂直位移不能为零。

Practice / 配套练习: AIME-2012-I-Q14 (WB p. 92), AIME-2017-I-Q10 (WB p. 80), AIME-2019-I-Q12 (WB p. 95), AIME-2025-I-Q08 (WB p. 382)

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 7 / 方法 7 [AL03-M07]

Invert a half-plane and carry unions through the map

反演半平面，并把并集随映射传递

Procedure

1. For $w = 1/z = u + iv$, substitute $z = (u - iv)/(u^2 + v^2)$ with $w \neq 0$.

2. Multiply an inequality only by the positive denominator $u^2 + v^2$, then complete the square.

3. Preserve strictness and boundary exclusions. The image of a union is the union of the images.

4. For a union of symmetric disks, identify which disk reaches farthest on each ray before computing wedge areas.

通用步骤

1. 对 $w = 1/z = u + iv$，代入 $z = (u - iv)/(u^2 + v^2)$，其中 $w \neq 0$。

2. 只以正分母 $u^2 + v^2$ 乘不等式，再配方。

3. 保留严格性与边界排除；并集的像是各像之并。

4. 对对称圆盘并，先确定各射线方向上哪个圆盘伸得最远，再求扇形分区面积。

> Example. Find the image of the half-plane $\Re z > 1$ under $w = 1/z$.

> 例题：求半平面 $\Re z > 1$ 在 $w = 1/z$ 下的像。

Worked solution

1. The inequality becomes $u/(u^2 + v^2) > 1$, with a positive denominator.

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 7 / 方法 7 [AL03-M07] (continued / 续)

2. Rearrange to $(u-1/2)^2+v^2<1/4$.

3. The image is the open disk centered at $1/2$ of radius $1/2$. Its boundary includes zero, so zero is already excluded by strictness.

例题解析

1. 不等式化为 $u/(u^2+v^2)>1$，分母为正。

2. 整理得 $(u-1/2)^2+v^2<1/4$。

3. 像为以 $1/2$ 为圆心、半径 $1/2$ 的开圆盘；零位于其边界，已由严格性排除。

> The map 1/z includes conjugation relative to ordinary radial inversion; use the actual algebra.

> 映射 1/z 相对通常径向反演还含共轭，应使用实际代数表达式。

Practice / 配套练习: AIME-2008-II-Q13 (WB p. 90)

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 8 / 方法 8 [AL03-M08]

Optimize a complex norm or real part and prove equality

优化复模或实部，并证明等号可达

Procedure

1. Factor out a fixed modulus or write $z = R(\cos\theta + i\sin\theta)$.

2. For distance from a fixed point to a circle, align the radius oppositely for a maximum; for a real part use $A\cos\theta + B\sin\theta \le \sqrt{A^2+B^2}$.

3. For independent choices, optimize each summand separately. For an argument range, use tangent rays to the locus.

4. Check the equality point respects half-plane, sign and excluded-point restrictions.

通用步骤

1. 提出固定模长，或写 $z = R(\cos\theta + i\sin\theta)$。

2. 固定点到圆的最远距离使半径方向相反；实部最值用 $A\cos\theta + B\sin\theta \le \sqrt{A^2+B^2}$。

3. 独立选择时分别优化每一项；辐角范围则用轨迹的切射线。

4. 检查等号点符合半平面、符号与排除点限制。

> Example. For $|z| = 1$, find the maximum of $\Re((3+4i)z)$ and a maximizing z.

> 例题：当 $|z| = 1$，求 $\Re((3+4i)z)$ 的最大值及一个达到值的 z。

Worked solution

1. With $z = x + iy$, the objective is $3x - 4y$ and $x^2 + y^2 = 1$.

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 8 / 方法 8 [AL03-M08] (continued / 续)

2. Cauchy bounds it by $\sqrt{3^2+(-4)^2}=5$.

3. Take $z=(3-4i)/5$ to attain five, with modulus one as required.

例题解析

1. 设 $z=x+iy$，目标为 $3x-4y$，且 $x^2+y^2=1$。

2. 柯西不等式给出上界 $\sqrt{3^2+(-4)^2}=5$。

3. 取 $z=(3-4i)/5$ 达到五，其模长确为一。

A norm bound needs the direction of equality, not just the numerical upper bound.

模长估界须给出等号方向，而不只是数值上界。

Practice / 配套练习: AIME-2011-II-Q08 (WB p. 72), AIME-2012-II-Q06 (WB p. 74), AIME-2014-I-Q07 (WB p. 77), AIME-2024-I-Q07 (WB p. 85)

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 9 / 方法 9 [AL03-M09]

Compose fractional maps and control finite polynomial-root orbits

复合分式映射，并控制有限多项式根轨道

Procedure

1. Compose the exact map symbolically, keeping every pole and any excluded intermediate value.

2. If a small iterate is identity, reduce the index only for orbits proved to avoid the poles.

3. For a polynomial root map, a finite invariant root set cannot contain a point whose modulus grows without bound.

4. Combine root-product modulus and conjugation to restrict the final root set, then substitute back into the original identity.

通用步骤

1. 符号复合精确映射，并保留每个极点及被排除中间值。

2. 若某个低次迭代为恒等，只能对已证明避开极点的轨道约化下标。

3. 多项式根映射的有限不变根集不能含模长无限增长的点。

4. 结合根乘积模长与共轭限制最终根集，再代回原恒等式。

> Example. For $F(z)=1/(1-z)$, show the orbit of $z_0=2$ has period three.

> 例题：对 $F(z)=1/(1-z)$，证明 $z_0=2$ 的轨道周期为三。

Section 1.3 Complex numbers & roots of unity

复数与单位根

Method 9 / 方法 9 [AL03-M09] (continued / 续)

Worked solution

1. The successive values are $2, -1, 1/2, 2$.

2. None equals the pole one, so all intermediate maps are defined.

3. The first three values are distinct and the fourth returns to the first, proving exact period three.

例题解析

1. 逐次值为 $2, -1, 1/2, 2$。

2. 各值均不为极点一，因此所有中间映射都有定义。

3. 前三个值互异，第四个返回首值，故准确周期为三。

An algebraic composition formula may cancel a pole that the original step still cannot cross.

复合公式可能约掉某极点，但原逐步映射仍不能经过该点。

Practice / 配套练习: AIME-2002-I-Q12 (WB p. 87), AIME-2007-II-Q14 (WB p. 211)

Section 1.4 Logarithms & exponentials

对数与指数

Method 1 / 方法 1 [AL04-M01]

Take domains seriously, then let logarithms turn multiplication into addition or exponentials turn a shared exponent into one positive variable. Combine identities before solving, and distinguish existence from uniqueness.

先认真处理定义域，再用对数把乘法变为加法，或用指数把共同指数化为一个正变量。求解前先合并恒等式，并区分存在性与唯一性。

Change base and combine powers before solving variables

求变量前先换底并合并幂

Procedure

1. Use $\log_a b = \ln b / \ln a$ with $a > 0, a \neq 1, b > 0$.

2. Use reciprocal logarithms and $c^{\log_c d} = d$ to simplify the target directly.

3. Combine weighted log sums as a single product logarithm; retain positivity when extracting real powers.

通用步骤

1. 在 $a > 0, a \neq 1, b > 0$ 下使用 $\log_a b = \ln b / \ln a$。

2. 使用互反对数及 $c^{\log_c d} = d$，直接化简目标。

3. 把加权对数和合成乘积的对数；取实幂时保留正性。

> Example. Evaluate $(\log_2 9)(\log_3 4)$.

> 例题：计算 $(\log_2 9)(\log_3 4)$。

Section 1.4 Logarithms & exponentials

对数与指数

Method 1 / 方法 1 [AL04-M01] (continued / 续)

Worked solution

1. Extract powers: $\log_2 9 = 2\log_2 3$ and $\log_3 4 = 2\log_3 2$.

2. Change of base gives $(\log_2 3)(\log_3 2) = 1$.

3. The result is 4. All logarithm bases and arguments are valid.

例题解析

1. 提出幂：$\log_2 9 = 2\log_2 3, \log_3 4 = 2\log_3 2$。

2. 换底得 $(\log_2 3)(\log_3 2) = 1$。

3. 结果为 4，各底数与真数均有效。

> A logarithm of a sum does not split into a sum of logarithms.

> 和的对数不能拆为对数之和。

Practice / 配套练习: AIME-2000-II-Q01 (WB p. 99), AIME-2003-I-Q04 (WB p. 100), AIME-2006-II-Q02 (WB p. 102), AIME-2009-II-Q02 (WB p. 103), AIME-2015-II-Q04 (WB p. 262), AIME-2024-I-Q02 (WB p. 113), AIME-2002-II-Q03 (WB p. 529)

Section 1.4 Logarithms & exponentials

对数与指数

Method 2 / 方法 2 [AL04-M02]

Use logarithmic coordinates and a shared value

使用对数坐标与共同值

Procedure

1. Choose one fixed base and replace positive variables by their logarithms.

2. Products and quotients become linear sums and differences; a common variable-base logarithm can be denoted by one shared value.

3. Add or subtract the resulting equations to eliminate repeated coordinates or a repeated exponential term.

4. Reconstruct positive original variables and verify every original base differs from one.

通用步骤

1. 选择统一底数，把正变量换为其对数。

2. 乘除变为线性和差；变底对数的共同值可统一设为一个变量。

3. 将所得方程相加或相减，消去重复坐标或共同指数项。

4. 还原正的原变量，并检查每个原底数不为一。

> Example. Positive x, y satisfy $\log_2(xy) = 5$ and $\log_2(x/y) = 1$. Find x, y.

> 例题：正数 x, y 满足 $\log_2(xy) = 5, \log_2(x/y) = 1$，求 x, y。

Worked solution

1. Put $u = \log_2 x, v = \log_2 y$. The equations become $u + v = 5, u - v = 1$.

Section 1.4 Logarithms & exponentials

对数与指数

Method 2 / 方法 2 [AL04-M02] (continued / 续)

2. Adding and subtracting gives $u = 3, v = 2$.

3. Hence $x = 8, y = 4$; their product and quotient are $32, 2$ as required.

例题解析

1. 令 $u = \log_2 x, v = \log_2 y$，方程化为 $u + v = 5, u - v = 1$。

2. 相加与相减得 $u = 3, v = 2$。

3. 故 $x = 8, y = 4$，积与商分别为 $32, 2$，符合条件。

The logarithmic variables may be negative even though the original variables are positive.

原变量为正，对数变量仍可能为负。

Practice / 配套练习: AIME-2000-I-Q09 (WB p. 116), AIME-2002-I-Q06 (WB p. 117), AIME-2006-I-Q09 (WB p. 474), AIME-2010-II-Q05 (WB p. 105), AIME-2012-I-Q09 (WB p. 120), AIME-2016-II-Q03 (WB p. 107), AIME-2019-I-Q07 (WB p. 477), AIME-2019-II-Q06 (WB p. 123), AIME-2022-II-Q04 (WB p. 111), AIME-2024-II-Q04 (WB p. 114)

Section 1.4 Logarithms & exponentials

对数与指数

Method 3 / 方法 3 [AL04-M03]

Turn an exponential equation into a polynomial on a restricted range

把指数方程转为受限范围内的多项式

Procedure

1. If the exponents share a nonconstant linear expression, put $t = b^{cx}$ with $b > 0, b \neq 1, c \neq 0$. For unrestricted real x its exact range is $t > 0$; intersect this with any original restrictions on x.

2. Solve or analyze the resulting polynomial; discard nonpositive roots before returning to x.

3. Use Vieta for a sum of logarithms or product of original variables, avoiding individual roots when possible.

4. For equations such as $x^y = y^x$, take logarithms or a positive real root only within the original positive domain.

通用步骤

1. 若指数共享非恒定线性表达式，可令 $t = b^{cx}$，其中 $b > 0, b \neq 1, c \neq 0$。当 x 遍历实数时其准确值域为 $t > 0$；若原 x 有限制，还须取相应交集。

2. 求解或分析所得多项式，返回 x 前剔除非正根。

3. 对数和或原变量乘积可用韦达，尽量避免逐个求根。

4. 对 $x^y = y^x$ 等式，只在原正性定义域内取对数或正实根。

> Example. Classify the number of real solutions to $4^x - a2^x + 1 = 0$ as real a varies.

> 例题：按实参数 a 分类 $4^x - a2^x + 1 = 0$ 的实数解个数。

Section 1.4 Logarithms & exponentials

对数与指数

Method 3 / 方法 3 [AL04-M03] (continued / 续)

Worked solution

1. Put $t = 2^x > 0$. The equation is $t^2 - at + 1 = 0$.

2. Real roots require $|a| \geq 2$. Their product is one and sum is a, so they are positive exactly when $a \geq 2$.

3. At $a = 2$ there is one root $t = 1$, giving one real x; at $a > 2$ there are two; for $a < 2$ there are none.

例题解析

1. 令 $t = 2^x > 0$，方程为 $t^2 - at + 1 = 0$。

2. 实根要求 $|a| \geq 2$；根积为一、根和为 a，所以恰在 $a \geq 2$ 时为正。

3. $a = 2$ 时只有根 $t = 1$，对应一个实数 x；$a > 2$ 时有两个；$a < 2$ 时没有。

A real polynomial root may lie outside the range of the exponential substitution.

多项式的实根可能位于指数代换的值域之外。

Practice / 配套练习: AIME-2005-I-Q08 (WB p. 118), AIME-2005-II-Q05 (WB p. 101), AIME-2010-I-Q03 (WB p. 104), AIME-2018-I-Q05 (WB p. 108), AIME-2020-I-Q02 (WB p. 109), AIME-2023-I-Q02 (WB p. 112), AIME-2026-I-Q06 (WB p. 124)

Section 1.4 Logarithms & exponentials

对数与指数

Method 4 / 方法 4 [AL04-M04]

Undo nested logarithms and prove uniqueness monotonically

逐层还原嵌套对数，并用单调性证明唯一

Procedure

1. Work from the outermost equation inward, recording positivity of every inner argument.

2. If the base is unknown, a reciprocal logarithm variable can turn the remaining relation into a scalar equation.

3. Compare monotonicity of the two sides to prove at most one admissible solution; then verify a candidate exactly.

4. For integer base exponents, reduce the result to prime-exponent or bounded divisor conditions.

通用步骤

1. 从最外层方程向内处理，记录每个内层真数的正性。

2. 底数未知时，可用互反对数变量把剩余关系变成单变量方程。

3. 比较两侧单调性，证明至多一个允许解，再精确验证候选。

4. 若底数指数要求为整数，将结果化为质指数或有界约数条件。

> Example. Find the base $b > 1$ for which $\log_b(\log_b 16) = 2$.

> 例题：求使 $\log_b(\log_b 16) = 2$ 的底数 $b > 1$。

Section 1.4 Logarithms & exponentials

对数与指数

Method 4 / 方法 4 [AL04-M04] (continued / 续)

Worked solution

1. Exponentiating the outer logarithm gives $\log_b 16 = b^2$, or $\ln 16 = b^2 \ln b$.

2. For $b > 1$, both positive factors $b^2, \ln b$ strictly increase, so their product strictly increases.

3. The value $b = 2$ works because $4 \ln 2 = \ln 16$. Monotonicity proves it is unique, and the inner logarithm equals four and is positive.

例题解析

1. 还原外层对数得 $\log_b 16 = b^2$，即 $\ln 16 = b^2 \ln b$。

2. 当 $b > 1$，正因子 $b^2, \ln b$ 均严格增加，所以乘积严格增加。

3. $b = 2$ 成立，因为 $4 \ln 2 = \ln 16$；单调性证明唯一，内层对数为正的四。

Checking one guessed base establishes existence; monotonicity establishes uniqueness.

验证猜到的底数只证明存在，单调性才证明唯一。

Practice / 配套练习: AIME-2013-II-Q02 (WB p. 106), AIME-2016-II-Q03 (WB p. 107), AIME-2017-I-Q14 (WB p. 125)

Section 1.4 Logarithms & exponentials

对数与指数

Method 5 / 方法 5 [AL04-M05]

Partition logarithmic floors at exact power thresholds

按精确幂门槛分割对数取整

Procedure

1. Use $\lfloor \log_b t \rfloor = k \iff b^k \le t < b^{k+1}$ for $b > 1$.

2. For a floor in an exponent, partition the original variable into its unit intervals before evaluating its image.

3. Convert the problem into interval lengths, integer counts or a monotone threshold sum; preserve half-open endpoints.

4. At a proposed greatest integer, verify the neighboring integer and invoke monotonicity to exclude all later ones.

通用步骤

1. 当 $b > 1$，使用 $\lfloor \log_b t \rfloor = k \iff b^k \le t < b^{k+1}$。

2. 指数中含取整时，先将原变量按单位区间分块，再求其像。

3. 将题目化为区间长度、整数计数或单调门槛和，并保留半开端点。

4. 拟定最大整数后，检验相邻整数，再用单调性排除其后全部整数。

> Example. Choose x uniformly from $[1, 4]$. Find the probability that $\lfloor \log_2(3x) \rfloor - \lfloor \log_2 x \rfloor = 1$.

> 例题：从 $[1, 4]$ 按长度均匀取 x，求 $\lfloor \log_2(3x) \rfloor - \lfloor \log_2 x \rfloor = 1$ 的概率。

Section 1.4 Logarithms & exponentials

对数与指数

Method 5 / 方法 5 [AL04-M05] (continued / 续)

Worked solution

1. On $[1,2)$ the second floor is zero. The first floor is one precisely on $[1,4/3)$.

2. On $[2,4)$ the second floor is one. The first floor is two precisely on $[2,8/3)$.

3. The favorable lengths total $1/3+2/3=1$. Divide by sample length three to obtain $1/3$; endpoints have zero measure.

例题解析

1. 在 $[1,2)$ 上，第二个取整为零；第一个恰在 $[1,4/3)$ 为一。

2. 在 $[2,4)$ 上，第二个取整为一；第一个恰在 $[2,8/3)$ 为二。

3. 有利长度合计 $1/3+2/3=1$，除以样本长度三得 $1/3$，端点测度为零。

A ceiling-floor difference vanishes at integral logarithms; these are exact powers, not approximate ones.

上取整减下取整在整数对数处为零；这些是精确幂，不能以近似判断。

Practice / 配套练习: AIME-2004-I-Q12 (WB p. 171), AIME-2007-I-Q07 (WB p. 163), AIME-2009-I-Q06 (WB p. 166), AIME-2010-I-Q14 (WB p. 175)

Section 1.4 Logarithms & exponentials

对数与指数

Method 6 / 方法 6 [AL04-M06]

Telescope logarithmic products and exponent progressions

使对数乘积与指数等差式望远镜消去

Procedure

1. Change all logarithms to one base and factor rational exponents before multiplying.

2. Cancel consecutive ratios over the full index range, retaining the first and last factors.

3. For a geometric sequence of powers, its logarithms form an arithmetic sequence; compare a positive sum with its largest term using a rigorous geometric tail bound.

通用步骤

1. 所有对数换至统一底数，并在相乘前分解有理指数。

2. 在完整下标范围内消去相邻比值，保留首尾因子。

3. 幂的等比数列取对数后为等差数列；正项和与最大项之间用严格等比尾和界比较。

> Example. Evaluate $\prod_{k=2}^{7} \log_k(k+1)$.

> 例题：计算 $\prod_{k=2}^{7} \log_k(k+1)$。

Worked solution

1. The kth factor is $\ln(k+1)/\ln k$.

2. All intermediate logarithms cancel, leaving $\ln 8/\ln 2$.

3. The product equals 3. Every base in the range is positive and different from one.

Section 1.4 Logarithms & exponentials

对数与指数

Method 6 / 方法 6 [AL04-M06] (continued / 续)

例题解析

1. 第 k 个因子为 $\ln(k+1)/\ln k$。

2. 所有中间对数相消，留下 $\ln 8/\ln 2$。

3. 乘积等于 3；范围内所有底数均为正且不为一。

Cancel factors across products, never terms across sums.

只能在乘积中约去因子，不能在和式中随意消项。

Practice / 配套练习: AIME-2007-II-Q12 (WB p. 159), AIME-2009-I-Q07 (WB p. 147), AIME-2014-II-Q07 (WB p. 149), AIME-2025-II-Q04 (WB p. 115)

Section 1.4 Logarithms & exponentials

对数与指数

Method 7 / 方法 7 [AL04-M07]

Enforce logarithm domains before inequalities and root counts

先施加对数定义域，再处理不等式与根数

Procedure

1. List positive arguments, positive bases different from one, and any inverse-trigonometric range restrictions.

2. When exponentiating inequalities, distinguish bases above one from bases between zero and one.

3. After a substitution or squaring, test every branch in the original logarithmic domain.

4. Only then count integer parameters or admissible real roots.

通用步骤

1. 列出真数为正、底数为正且不为一，以及反三角函数的范围限制。

2. 指数化不等式时，区分大于一的底数与零一之间的底数。

3. 代换或平方之后，将每个分支代回原对数定义域检验。

4. 之后才统计整数参数或允许实根。

> Example. Find the real domain of $\arcsin(\log_{1/2} x)$.

> 例题：求 $\arcsin(\log_{1/2} x)$ 的实数定义域。

Worked solution

1. The logarithm requires $x > 0$ and its fixed base is valid.

Section 1.4 Logarithms & exponentials

对数与指数

Method 7 / 方法 7 [AL04-M07] (continued / 续)

2. The arcsine requires $-1 \le \log_{1/2} x \le 1$. Since the base is below one, exponentiation reverses the order.

3. Thus $1/2 \le x \le 2$, with both endpoints allowed.

例题解析

1. 对数要求 $x > 0$，固定底数有效。

2. 反正弦要求 $-1 \le \log_{1/2} x \le 1$；底数小于一，指数化会反转顺序。

3. 因此 $1/2 \le x \le 2$，两端点均允许。

The algebraic equation alone cannot certify that its logarithms exist.

单有代数方程不能保证其中的对数存在。

Practice / 配套练习: AIME-2005-II-Q05 (WB p. 101), AIME-2009-II-Q11 (WB p. 174), AIME-2011-I-Q09 (WB p. 119), AIME-2013-I-Q08 (WB p. 121), AIME-2017-II-Q07 (WB p. 122), AIME-2018-I-Q05 (WB p. 108), AIME-2020-I-Q02 (WB p. 109), AIME-2020-II-Q03 (WB p. 110), AIME-2023-I-Q02 (WB p. 112)

Section 1.5 Sequences & summation

数列与求和

Method 1 / 方法 1 [AL05-M01]

Look for a sum structure before calculating terms: arithmetic pairs, geometric blocks, a telescoping difference, or a simple transformed recurrence. Every closed formula needs its index convention, domain and initial values checked.

计算各项之前先找求和结构：等差配对、等比区块、望远镜差分，或简单变换后的递推。每个通项公式都须核对下标约定、定义域与初值。

Pair arithmetic terms and sum polynomial weights

配对等差项并求多项式权重和

Procedure

1. Write $a_j = a_1 + (j-1)d$ and distinguish the number of terms from the final index.

2. Pair terms at complementary indices: their sum is constant. Use $\sum j = n(n+1)/2$, $\sum j^2 = n(n+1)(2n+1)/6$, and $\sum j^3 = [n(n+1)/2]^2$ when needed.

3. For alternating signs or repeated patterns, simplify a full block first, then sum the resulting lower-degree expression.

4. Restore parity, increasing-order and integer-endpoint conditions after the sum is evaluated.

通用步骤

1. 写 $a_j = a_1 + (j-1)d$，并区分项数与末下标。

2. 将互补下标项配对，其和恒定。必要时使用 $\sum j = n(n+1)/2, \sum j^2 = n(n+1)(2n+1)/6, \sum j^3 = [n(n+1)/2]^2$。

3. 对交替符号或重复模式，先化简完整区块，再求所得较低次表达式之和。

4. 求和后恢复奇偶、递增次序与整数端点条件。

Section 1.5 Sequences & summation

数列与求和

Method 1 / 方法 1 [AL05-M01] (continued / 续)

> Example. In an arithmetic sequence, $a_3 + a_8 = 20$. Find $a_1 + a_{10} + a_5 + a_6$.

> 例题：等差数列满足 $a_3 + a_8 = 20$，求 $a_1 + a_{10} + a_5 + a_6$。

Worked solution

1. Every pair whose indices sum to eleven has value $2a_1 + 9d$.

2. Thus both $a_1 + a_{10}$ and $a_5 + a_6$ equal the given pair sum twenty.

3. The requested total is forty, independent of the individual first term and difference.

例题解析

1. 下标和为十一的每对，其值均为 $2a_1 + 9d$。

2. 所以 $a_1 + a_{10}$ 与 $a_5 + a_6$ 均等于给定对和二十。

3. 所求总和为四十，与具体首项及公差无关。

A three-term arithmetic progression is best written as a-d, a, a+d.

三项等差数列宜写成 a-d、a、a+d。

To detect a nonconsecutive four-term progression, write its entries as $u, u + d, u + 2d, u + 3d$ and enumerate the six placements of two fixed entries. Checking only adjacent terms in the original list misses forbidden subsequences.

Section 1.5 Sequences & summation

数列与求和

Method 1 / 方法 1 [AL05-M01] (continued / 续)

Adjacent-sum triangular arrays preserve an arithmetic row. From top entries $a + jd$, after r adjacent-sum steps the entry in position j is $2^r[a + (j + r/2)d]$. The case $r = 0$ is immediate; adding consecutive entries gives exactly the same formula with r replaced by $r + 1$, proving it by induction.

检测非连续出现的四项等差数列，应写成 $u, u + d, u + 2d, u + 3d$，并枚举两个固定项的六种位置；只检查原列表相邻项会漏掉禁用子序列。

相邻求和三角阵会保持每行等差。若顶行是 $a + jd$，经过 r 次相邻求和后，第 j 项为 $2^r[a + (j + r/2)d]$。$r = 0$ 显然成立；将连续两项相加，恰得 r 换为 $r + 1$ 的同式，故归纳得证。

Practice / 配套练习: AIME-2000-I-Q10 (WB p. 25), AIME-2002-II-Q04 (WB p. 236), AIME-2003-I-Q02 (WB p. 237), AIME-2004-I-Q02 (WB p. 128), AIME-2004-I-Q07 (WB p. 39), AIME-2005-I-Q02 (WB p. 600), AIME-2005-I-Q12 (WB p. 609), AIME-2006-I-Q04 (WB p. 465), AIME-2006-I-Q13 (WB p. 480), AIME-2007-I-Q07 (WB p. 163), AIME-2007-I-Q11 (WB p. 172), AIME-2008-II-Q01 (WB p. 130), AIME-2009-II-Q04 (WB p. 535), AIME-2011-II-Q03 (WB p. 131), AIME-2012-I-Q02 (WB p. 133), AIME-2012-II-Q10 (WB p. 168), AIME-2013-I-Q15 (WB p. 513), AIME-2013-II-Q03 (WB p. 135), AIME-2013-II-Q14 (WB p. 514), AIME-2015-I-Q01 (WB p. 136), AIME-2016-I-Q05 (WB p. 138), AIME-2016-II-Q01 (WB p. 139), AIME-2017-I-Q03 (WB p. 517), AIME-2017-I-Q09 (WB p. 501), AIME-2020-II-Q10 (WB p. 503), AIME-2021-I-Q05 (WB p. 538), AIME-2022-I-Q06 (WB p. 666), AIME-2023-I-Q10 (WB p. 170), AIME-2023-II-Q01 (WB p. 140), AIME-2026-II-Q01 (WB p. 141), AIME-2008-I-Q06 (WB p. 145)

Section 1.5 Sequences & summation

数列与求和

Method 2 / 方法 2 [AL05-M02]

Use geometric blocks and the correct dimension scale

使用等比区块与正确维度的缩放

Procedure

1. For n terms use $a(1-r^n)/(1-r)$ when $r \neq 1$; handle $r=1$ separately.

2. An infinite series requires $|r|<1$. Area and volume ratios are the square and cube of a linear similarity ratio.

3. Group consecutive equal-length blocks when partial sums are given; their block sums inherit a geometric ratio.

4. Before adding areas or volumes, verify the pieces have disjoint interiors at every stage used.

通用步骤

1. $r \neq 1$ 时，n 项和用 $a(1-r^n)/(1-r)$，并单列 $r=1$。

2. 无穷级数须 $|r|<1$；面积比与体积比分别为线性相似比的平方与立方。

3. 给定部分和时，可按等长连续区块分组，其区块和继承等比倍率。

4. 面积或体积相加前，须验证所用每阶段的各块内部不相交。

> Example. Inside a unit disk, successive annuli have radii $1, 1/2, 1/4, \ldots$. Color annuli numbered $0, 3, 6, \ldots$ green from the outside. Find the green area fraction.

> 例题：单位圆盘的连续圆环半径为 $1, 1/2, 1/4, \ldots$。从外侧起将编号 $0, 3, 6, \ldots$ 的环涂绿，求绿色面积比例。

Section 1.5 Sequences & summation

数列与求和

Method 2 / 方法 2 [AL05-M02] (continued / 续)

Worked solution

1. Annulus n has area fraction $(1-1/4)(1/4)^n$.

2. The selected annuli form a geometric series with first term $3/4$ and ratio $(1/4)^3 = 1/64$.

3. The fraction is $(3/4)/(1-1/64) = 16/21$; boundaries have zero area and the annuli interiors are disjoint.

例题解析

1. 第 n 环面积比例为 $(1-1/4)(1/4)^n$。

2. 选中圆环形成首项 $3/4$、公比 $(1/4)^3 = 1/64$ 的等比级数。

3. 比例为 $(3/4)/(1-1/64) = 16/21$；边界面积为零，环的内部互不相交。

Equal ratios of lengths and areas are different unless the ratio is zero or one.

长度比与面积比通常不同。

For a replicated geometric construction, multiply the number of pieces by each piece's scaled area or volume to obtain the increment ratio. A face count alone is insufficient if pieces overlap or become hidden. The tetrahedron construction in AIME-2001-II-Q12 is certified only through the three requested stages; no infinite non-overlap claim is used.

复制几何构造时，用块数乘每块缩放后的面积或体积，求增量倍率。若有重叠或隐藏，仅有面数不足以相加。AIME-2001-II-Q12 的四面体构造仅对所求三个阶段完成认证，不使用无限阶段不重叠的结论。

Section 1.5 Sequences & summation

数列与求和

Method 2 / 方法 2 [AL05-M02] (continued / 续)

Practice / 配套练习: AIME-2000-I-Q11 (WB p. 479), AIME-2001-II-Q12 (WB p. 450), AIME-2002-II-Q03 (WB p. 529), AIME-2002-II-Q11 (WB p. 154), AIME-2004-I-Q12 (WB p. 171), AIME-2005-II-Q03 (WB p. 129), AIME-2007-II-Q12 (WB p. 159), AIME-2009-I-Q08 (WB p. 148), AIME-2009-II-Q08 (WB p. 787), AIME-2011-II-Q05 (WB p. 132), AIME-2012-II-Q02 (WB p. 134), AIME-2013-I-Q13 (WB p. 235), AIME-2013-I-Q14 (WB p. 93), AIME-2016-I-Q01 (WB p. 137), AIME-2016-II-Q05 (WB p. 222), AIME-2016-II-Q07 (WB p. 376), AIME-2016-II-Q15 (WB p. 188), AIME-2017-II-Q12 (WB p. 94), AIME-2020-I-Q02 (WB p. 109), AIME-2020-I-Q08 (WB p. 83), AIME-2025-II-Q12 (WB p. 281), AIME-2019-I-Q01 (WB p. 576)

Section 1.5 Sequences & summation

数列与求和

Method 3 / 方法 3 [AL05-M03]

Find the discrete difference that telescopes

寻找能望远镜消去的离散差分

Procedure

1. Try to write the summand as $F(k)-F(k+1)$, possibly after a partial-fraction or trigonometric identity.

2. Write the uncanceled first and last terms explicitly; a shift larger than one leaves several boundary terms.

3. For products, telescope ratios instead of differences. For binomial sums use Pascal's difference identity.

4. If rounding is requested, bound the remaining tail sharply enough to separate it from the rounding boundary.

通用步骤

1. 尝试将每项写成 $F(k)-F(k+1)$，必要时先作部分分式或三角恒等变换。

2. 明确写出未消去的首尾项；位移大于一时会留下多项边界项。

3. 乘积用比值而非差作望远镜；二项式求和使用帕斯卡差分恒等式。

4. 若求舍入结果，须把剩余尾项估计到足以避开舍入边界。

> Example. Find a formula for $\sum_{k=1}^{N} 1/[k(k+1)(k+2)]$.

> 例题：求 $\sum_{k=1}^{N} 1/[k(k+1)(k+2)]$ 的公式。

Section 1.5 Sequences & summation

数列与求和

Method 3 / 方法 3 [AL05-M03] (continued / 续)

Worked solution

1. The summand equals $\frac{1}{2}[1/(k(k+1)) - 1/((k+1)(k+2))]$.

2. Every internal reciprocal pair cancels, leaving $\frac{1}{2}[1/2 - 1/((N+1)(N+2))]$.

3. Thus the sum is $1/4 - 1/[2(N+1)(N+2)]$. Its infinite limit is $1/4$.

例题解析

1. 每项等于 $\frac{1}{2}[1/(k(k+1)) - 1/((k+1)(k+2))]$。

2. 内部倒数对全部消去，留下 $\frac{1}{2}[1/2 - 1/((N+1)(N+2))]$。

3. 故和为 $1/4 - 1/[2(N+1)(N+2)]$，无穷极限为 $1/4$。

Telescope the expression before applying a large-index formula.

应先将表达式望远镜消去，再代大下标。

Pascal gives $\binom{k+1}{r+1} - \binom{k}{r+1} = \binom{k}{r}$; summing proves the hockey-stick identity, with index endpoints retained.

Useful trigonometric differences are $2\sin(h/2)\sin t = \cos(t - h/2) - \cos(t + h/2)$ and $\cot t - \cot(t+h) = \sin h/[\sin t \sin(t+h)]$. Prove nonzero denominators and use one consistent angle unit.

From $|x_{k+1}| = |x_k + c|$, squaring gives $x_{k+1}^2 - x_k^2 = 2cx_k + c^2$. The sum then depends on the terminal square; an integer-parity argument and an explicit attaining path may still be required.

帕斯卡恒等式给出 $\binom{k+1}{r+1} - \binom{k}{r+1} = \binom{k}{r}$；求和即得曲棍球杆恒等式，须保留下标端点。

Section 1.5 Sequences & summation

数列与求和

Method 3 / 方法 3 [AL05-M03] (continued / 续)

常用三角差分为 $2\sin(h/2)\sin t=\cos(t-h/2)-\cos(t+h/2)$ 与 $\cot t-\cot(t+h)=\sin h/[\sin t\sin(t+h)]$。须证明分母非零，并统一角度单位。

由 $|x_{k+1}|=|x_k+c|$，平方得 $x_{k+1}^2-x_k^2=2cx_k+c^2$，从而和只依赖末端平方；之后仍可能需要整数奇偶论证与明确达到路径。

Practice / 配套练习: AIME-2000-II-Q14 (WB p. 590), AIME-2000-II-Q15 (WB p. 198), AIME-2002-I-Q04 (WB p. 127), AIME-2002-II-Q06 (WB p. 142), AIME-2006-I-Q15 (WB p. 156), AIME-2009-I-Q07 (WB p. 147), AIME-2014-II-Q07 (WB p. 149), AIME-2015-II-Q13 (WB p. 203), AIME-2008-II-Q08 (WB p. 194), AIME-2022-II-Q10 (WB p. 641), AIME-2025-II-Q04 (WB p. 115)

Section 1.5 Sequences & summation

数列与求和

Method 4 / 方法 4 [AL05-M04]

Sum weighted geometric terms and regroup positive double series

求加权等比和，并重组正项双级数

Procedure

1. For $|r|<1$, start with $\sum r^k = 1/(1-r)$ and derive weighted sums by a shift-and-subtract argument.

2. For a tail, shift the index first so the weight becomes a constant plus the new index.

3. Nonnegative double series may be regrouped by the total exponent; prove a tail bound before interpreting coefficients as digits.

通用步骤

1. 当 $|r|<1$，从 $\sum r^k = 1/(1-r)$ 出发，以平移相减推导加权和。

2. 尾和先平移下标，使权重变为常数加新下标。

3. 非负双级数可按总指数重组；把系数解释为数位之前须证明尾项界。

> Example. Evaluate $\sum_{k=1}^{\infty} k/3^k$ without differentiating a series.

> 例题：不用级数求导，计算 $\sum_{k=1}^{\infty} k/3^k$。

Worked solution

1. Let $S=\sum_{k\ge 1} kr^k$. Multiplying by r shifts its weights by one.

2. Subtracting gives $(1-r)S=\sum_{k\ge 1} r^k = r/(1-r)$, justified by absolute convergence.

3. So $S=r/(1-r)^2$; at $r=1/3$, the value is $3/4$.

Section 1.5 Sequences & summation

数列与求和

Method 4 / 方法 4 [AL05-M04] (continued / 续)

例题解析

1. 设 $S=\sum_{k\ge 1} kr^k$，乘 r 后其权重平移一位。

2. 相减得 $(1-r)S=\sum_{k\ge 1} r^k = r/(1-r)$，绝对收敛保证操作有效。

3. 故 $S=r/(1-r)^2$；代入 $r=1/3$ 得 $3/4$。

A distant series coefficient can create a carry unless the full scaled tail is controlled.

若未控制完整放大尾和，远处系数也可能造成进位。

Practice / 配套练习: AIME-2009-I-Q08 (WB p. 148), AIME-2026-II-Q07 (WB p. 776), AIME-2026-II-Q09 (WB p. 608)

Section 1.5 Sequences & summation

数列与求和

Method 5 / 方法 5 [AL05-M05]

Solve a linear recurrence through fixed points or characteristic roots

以不动点或特征根求解线性递推

Procedure

1. For $u_{n+1}=ru_n+c$, subtract its fixed point $L=c/(1-r)$ when $r\neq 1$.

2. For a second-order homogeneous recurrence, factor the characteristic quadratic and fit the initial conditions; verify the formula by induction.

3. For large indices modulo a base, prove a return of the entire determining state before reducing the index.

4. For variable initial data, express the late term as a linear combination of those data and apply integer bounds to any collision.

通用步骤

1. 对 $u_{n+1}=ru_n+c$，当 $r\neq 1$ 时减去不动点 $L=c/(1-r)$。

2. 二阶齐次递推可分解特征二次式并代初值；再用归纳验证公式。

3. 求大下标的模值时，须证明整个决定状态返回，才能约化下标。

4. 初值可变时，将后项表示为初值的线性组合，再对重合表示施加整数界。

> Example. Let $u_0=0, u_{n+1}=3u_n+2$. Find u_n.

> 例题：设 $u_0=0, u_{n+1}=3u_n+2$，求 u_n。

Section 1.5 Sequences & summation

数列与求和

Method 5 / 方法 5 [AL05-M05] (continued / 续)

Worked solution

1. The fixed point is -1. Put $v_n = u_n + 1$.

2. Then $v_{n+1} = 3v_n$ and $v_0 = 1$, so $v_n = 3^n$.

3. Hence $u_n = 3^n - 1$, which satisfies both the initial value and recurrence.

例题解析

1. 不动点为 -1，令 $v_n = u_n + 1$。

2. 则 $v_{n+1} = 3v_n, v_0 = 1$，所以 $v_n = 3^n$。

3. 故 $u_n = 3^n - 1$，它满足初值与递推。

A repeated characteristic root needs a factor of n in the second independent solution.

特征根重复时，第二个独立解需要乘以 n。

Expanding a tridiagonal determinant with diagonal a and off-diagonal b gives $D_n = aD_{n-1} - b^2 D_{n-2}$, with $D_0 = 1, D_1 = a$. Derive the sign and the factor b^2 from the minor expansion before solving the recurrence.

For a nontriangle increasing sequence, $a_{j+2} \geq a_j + a_{j+1}$. The least permitted initial values and equality at every step give a sharp Fibonacci-type lower envelope; prove that envelope itself contains no nondegenerate triangle to establish sharpness.

Section 1.5 Sequences & summation

数列与求和

Method 5 / 方法 5 [AL05-M05] (continued / 续)

A higher-order recurrence need not be solved by a quadratic characteristic equation. Combine adjacent recurrences first: if $u_n = \sum_{j=1}^{2r} (-1)^{j+1} u_{n-j}$, substituting the previous recurrence cancels all middle terms and gives $u_n = -u_{n-2r-1}$ once both formulas apply. Thus a period dividing $4r+2$ follows, with the initial transition checked separately.

For tribonacci terms $u_{n+3} = u_{n+2} + u_{n+1} + u_n$, let $T_n = u_{n+2} + u_n$. Direct subtraction gives $T_n - T_{n-1} = 2u_n$. Hence $2\sum_{j=1}^{N} u_j = u_{N+2} + u_N - u_2 - u_0$, where u_0 can be recovered from the first recurrence if necessary. This is a partial-sum identity, not a second-order closed-form claim.

对角元为 a、邻对角元为 b 的三对角行列式展开得 $D_n = aD_{n-1} - b^2 D_{n-2}$，初值 $D_0 = 1, D_1 = a$。解递推前须从余子式展开推导负号与 b^2。

不含三角形的递增数列满足 $a_{j+2} \geq a_j + a_{j+1}$。取最小允许初值并每步取等号，得到准确的 Fibonacci 型下包络；还须证明此包络本身不含非退化三角形，才能说明界准确。

高阶递推未必需要解二次特征方程。先合并相邻递推：若 $u_n = \sum_{j=1}^{2r} (-1)^{j+1} u_{n-j}$，代入前一递推后，中间项全部消去，在两式都适用时得 $u_n = -u_{n-2r-1}$。所以周期整除 $4r+2$，另须检查初始过渡。

对 tribonacci 递推 $u_{n+3} = u_{n+2} + u_{n+1} + u_n$，令 $T_n = u_{n+2} + u_n$，直接相减得 $T_n - T_{n-1} = 2u_n$。故 $2\sum_{j=1}^{N} u_j = u_{N+2} + u_N - u_2 - u_0$；必要时可由首个递推恢复 u_0。这是部分和恒等式，并不声称数列满足二阶通项公式。

Practice / 配套练习: AIME-2001-II-Q03 (WB p. 126), AIME-2001-II-Q05 (WB p. 800), AIME-2002-I-Q08 (WB p. 542), AIME-2003-II-Q13 (WB p. 791), AIME-2006-II-Q11 (WB p. 157), AIME-2011-II-Q11 (WB p. 160), AIME-2018-II-Q02 (WB p. 518), AIME-2023-II-Q13 (WB p. 205)

Section 1.5 Sequences & summation

数列与求和

Method 6 / 方法 6 [AL05-M06]

Reduce constant second differences to an arithmetic first difference

把常二阶差分化为等差一阶差分

Procedure

1. Put $d_n = u_n - u_{n-1}$ when a recurrence contains $u_{n+1} - 2u_n + u_{n-1}$.

2. The second-difference equation gives a simple first-difference law for d_n.

3. Sum the differences and use all initial or boundary values to fix the constants.

通用步骤

1. 递推出现 $u_{n+1} - 2u_n + u_{n-1}$ 时，令 $d_n = u_n - u_{n-1}$。

2. 二阶差分方程转为 d_n 的简单一阶差分关系。

3. 累加差量，并用全部初值或边界值确定常数。

> Example. If $u_0 = 2, u_1 = 5$ and $u_{n+1} - 2u_n + u_{n-1} = 4$, find u_n.

> 例题：若 $u_0 = 2, u_1 = 5$ 且 $u_{n+1} - 2u_n + u_{n-1} = 4$，求 u_n。

Worked solution

1. Let $d_n = u_n - u_{n-1}$. Then $d_1 = 3$ and $d_{n+1} = d_n + 4$.

2. Thus $d_n = 4n - 1$. Sum from one through n.

3. $u_n = 2 + 4n(n+1)/2 - n = 2n^2 + n + 2$.

Section 1.5 Sequences & summation

数列与求和

Method 6 / 方法 6 [AL05-M06] (continued / 续)

例题解析

1. 令 $d_n = u_n - u_{n-1}$，则 $d_1 = 3, d_{n+1} = d_n + 4$。

2. 所以 $d_n = 4n - 1$，从一累加到 n。

3. $u_n = 2 + 4n(n+1)/2 - n = 2n^2 + n + 2$。

Boundary equations can differ from the interior recurrence and must be imposed separately.

边界方程可能不同于内部递推，须单独施加。

Practice / 配套练习: AIME-2003-II-Q08 (WB p. 143), AIME-2016-I-Q13 (WB p. 778)

Section 1.5 Sequences & summation

数列与求和

Method 7 / 方法 7 [AL05-M07]

Linearize nonlinear recurrences with ratios, adjacent products or invariants

用比值、相邻乘积或不变量线性化非线性递推

Procedure

1. Try $r_n = u_n/u_{n-1}$ or $p_n = u_n u_{n-1}$ when the recurrence contains neighboring products or quotients.

2. Clear two successive equations and subtract to reveal a preserved expression if direct linearization is not visible.

3. Solve the transformed arithmetic or geometric recurrence and prove every division remains defined up to the required index.

4. Reconstruct the original quantity rather than assume the transformed relation automatically ensures every original condition.

通用步骤

1. 递推含相邻乘除时，尝试 $r_n = u_n/u_{n-1}$ 或 $p_n = u_n u_{n-1}$。

2. 若未直接出现线性关系，可清除连续两式分母后相减，找保持不变的表达式。

3. 解变换后的等差或等比递推，并证明到所需下标为止每次相除都有定义。

4. 重建原目标，不能假定变换后关系自动保证原全部条件。

> Example. Positive terms satisfy $u_0 = 1, u_1 = 2$ and $u_{n+1}u_n = u_n u_{n-1} + 2$. Find $u_{10}u_{11}$.

> 例题：正项数列满足 $u_0 = 1, u_1 = 2, u_{n+1}u_n = u_n u_{n-1} + 2$，求 $u_{10}u_{11}$。

Section 1.5 Sequences & summation

数列与求和

Method 7 / 方法 7 [AL05-M07] (continued / 续)

Worked solution

1. Put $p_n = u_n u_{n-1}$ for $n \geq 1$. Then $p_1 = 2$ and $p_{n+1} = p_n + 2$.

2. So $p_n = 2n$. Positivity holds inductively since $u_{n+1} = p_{n+1}/u_n > 0$.

3. The desired adjacent product is $p_{11} = 22$.

例题解析

1. 令 $p_n = u_n u_{n-1}$，其中 $n \geq 1$。则 $p_1 = 2, p_{n+1} = p_n + 2$。

2. 故 $p_n = 2n$。因 $u_{n+1} = p_{n+1}/u_n > 0$，正性可归纳保持。

3. 所求相邻乘积为 $p_{11} = 22$。

The transformed sequence may be simple even when the individual original terms are not.

即使原各项复杂，变换后的数列仍可能很简单。

Practice / 配套练习: AIME-2005-II-Q11 (WB p. 155), AIME-2007-I-Q14 (WB p. 158), AIME-2008-II-Q06 (WB p. 146)

Section 1.5 Sequences & summation

数列与求和

Method 8 / 方法 8 [AL05-M08]

Separate absorbing triggers from an invariant growth region

分开吸收触发与不变增长区域

Procedure

1. Identify local configurations that force a zero, a repeated term or another absorbing event.

2. For all other configurations, propose a positive region with a quantitative lower growth or gap bound.

3. Prove the update preserves that region inductively, then count only the exceptional starting configurations.

通用步骤

1. 找出会迫使零、重复项或其他吸收事件的局部构型。

2. 对其他构型，提出具有定量增长下界或间隔界的正区域。

3. 归纳证明更新保持该区域，再只统计例外初始构型。

> Example. Let $u_0 = u_1 = 3$ and $u_{n+2} = u_{n+1}(u_n - 1)$. Prove no term vanishes.

> 例题：设 $u_0 = u_1 = 3, u_{n+2} = u_{n+1}(u_n - 1)$，证明没有一项为零。

Worked solution

1. The initial pair lies in the region $u_n, u_{n+1} \geq 3$.

2. Within this region $u_{n+2} \geq 2u_{n+1} \geq 6$, so the next pair remains in the same region.

3. Induction proves every term is at least three, hence none is zero.

Section 1.5 Sequences & summation

数列与求和

Method 8 / 方法 8 [AL05-M08] (continued / 续)

例题解析

1. 初始对位于 $u_n, u_{n+1} \geq 3$ 区域。

2. 在该区域内，$u_{n+2} \geq 2u_{n+1} \geq 6$，故下一对仍位于同一区域。

3. 归纳得每项至少为三，因而没有零项。

Observed growth for a few steps is insufficient; prove an invariant region.

观察几步增长还不够，须证明一个不变区域。

Practice / 配套练习: AIME-2015-I-Q09 (WB p. 150)

Section 1.5 Sequences & summation

数列与求和

Method 9 / 方法 9 [AL05-M09]

Parametrize alternating arithmetic and geometric triples

参数化交替的等差与等比三元组

Procedure

1. For three geometric positive integers, reduce the ratio and write them as ca^2, cab, cb^2.

2. Use the adjacent arithmetic rule to extend the underlying linear factors by a fixed difference $d = b - a$.

3. Prove the squared-factor and adjacent-product formulas by induction, then apply integer square-divisor or size constraints.

通用步骤

1. 三个正整数成等比时，先约分公比，写成 ca^2, cab, cb^2。

2. 用相邻等差规则，使底层线性因子按固定差 $d = b - a$ 延伸。

3. 归纳证明平方因子与相邻乘积公式，再使用整数平方约数或大小限制。

> Example. For $t_j = a + jd$ with $a, d > 0$, form $b_{2j+1} = t_j^2$ and $b_{2j+2} = t_j t_{j+1}$. Prove that odd-starting triples are geometric and even-starting triples are arithmetic.

> 例题：对 $t_j = a + jd$，其中 $a, d > 0$，作 $b_{2j+1} = t_j^2, b_{2j+2} = t_j t_{j+1}$。证明奇数下标起始三项等比，偶数起始三项等差。

Worked solution

1. The odd-starting triple is $t_j^2, t_j t_{j+1}, t_{j+1}^2$, with common ratio t_{j+1}/t_j.

Section 1.5 Sequences & summation

数列与求和

Method 9 / 方法 9 [AL05-M09] (continued / 续)

2. The even-starting triple is $t_jt_{j+1}, t_{j+1}^2, t_{j+1}t_{j+2}$. Its two differences are dt_{j+1}.

3. Positivity makes every ratio defined; $d > 0$ also makes the constructed sequence strictly increasing.

例题解析

1. 奇数起始三项为 $t_j^2, t_jt_{j+1}, t_{j+1}^2$，公比为 t_{j+1}/t_j。

2. 偶数起始三项为 $t_jt_{j+1}, t_{j+1}^2, t_{j+1}t_{j+2}$，两差均为 dt_{j+1}。

3. 正性使每个公比有定义，$d > 0$ 还使构造的数列严格递增。

A local triple identity needs an induction to describe the entire infinite sequence.

局部三项恒等式需要归纳才能描述整个无限数列。

A single arithmetic-to-geometric transition can be modeled by $a, a+d, a+2d, a+L$, followed by $(a+d)(a+L) = (a+2d)^2$. This is distinct from an infinite alternating rule, but the same positivity and integral-candidate checks remain necessary.

仅一次等差到等比转换可写为 $a, a+d, a+2d, a+L$，再用 $(a+d)(a+L) = (a+2d)^2$。这不同于无限交替规则，但仍须正性与整数候选检查。

Practice / 配套练习: AIME-2003-I-Q08 (WB p. 544), AIME-2004-II-Q09 (WB p. 144), AIME-2009-I-Q01 (WB p. 566), AIME-2016-I-Q10 (WB p. 151), AIME-2016-II-Q09 (WB p. 152)

Section 1.5 Sequences & summation

数列与求和

Method 10 / 方法 10 [AL05-M10]

Prove a rational recurrence period from the full ordered state

从完整有序状态证明有理递推的周期

Procedure

1. For a second-order recurrence the state is (u_n, u_{n+1}), not one term.

2. Compute enough iterates symbolically to return to that state, keeping denominators nonzero.

3. Reduce the requested index by the proved period and evaluate only the remaining terms.

通用步骤

1. 二阶递推的状态是 (u_n, u_{n+1})，而非单项。

2. 符号计算足够多迭代直到状态返回，并保持分母非零。

3. 以已证明周期约化所求下标，只计算剩余项。

> Example. For positive initial values $u_1 = 2, u_2 = 3$ and rule $u_{n+2} = 1/u_n$, find the least period.

> 例题：给定正初值 $u_1 = 2, u_2 = 3$，规则为 $u_{n+2} = 1/u_n$，求最小周期。

Worked solution

1. The terms are $2, 3, 1/2, 1/3, 2, 3, \ldots$.

2. The ordered pair $(2, 3)$ returns after four shifts. All terms are positive, so division is always defined.

Section 1.5 Sequences & summation

数列与求和

Method 10 / 方法 10 [AL05-M10] (continued / 续)

3. The first four terms are distinct, excluding periods one, two and three. The least period is four.

例题解析

1. 各项为 $2, 3, 1/2, 1/3, 2, 3, \ldots$。

2. 有序对 $(2, 3)$ 在四次移动后返回；各项为正，所以除法始终有定义。

3. 前四项互异，排除周期一、二、三，故最小周期为四。

A repeated value may appear before the determining state returns.

决定状态返回之前，单个数值就可能已重复。

For the positive Lyness rule $u_{n+2} = (1 + u_{n+1})/u_n$, an initial pair (a, b) is followed by $(b + 1)/a, (a + b + 1)/(ab), (a + 1)/b, a, b$. This proves a period dividing five; an exact least-period claim needs a separate distinctness check.

对正项 Lyness 规则 $u_{n+2} = (1 + u_{n+1})/u_n$，初始对 (a, b) 后依次为 $(b + 1)/a, (a + b + 1)/(ab), (a + 1)/b, a, b$。这证明周期整除五；若声称准确最小周期，还须另检互异性。

Practice / 配套练习: AIME-2020-II-Q06 (WB p. 153)

Section 1.5 Sequences & summation

数列与求和

Method 11 / 方法 11 [AL05-M11]

Track valid preimages and sum their arithmetic progression

追踪有效原像并求其等差数列之和

Procedure

1. Under $z = |x - c|$, a target z has preimages $c \pm z$ only when $z \geq 0$.

2. Filter invalid targets before taking preimages; then form the union and remove duplicates.

3. If the resulting zeros form an arithmetic progression, prove both endpoints, spacing and complete coverage before using its sum formula.

通用步骤

1. 在 $z = |x - c|$ 下，目标 z 只有非负时才有原像 $c \pm z$。

2. 求原像前先筛掉无效目标，再取并集并去重。

3. 若所得零点为等差数列，须先证明端点、间距与无缺漏覆盖，再用求和公式。

> Example. Find the sum of distinct real x with $|x - 2| \in \{-1, 1, 3\}$.

> 例题：求满足 $|x - 2| \in \{-1, 1, 3\}$ 的不同实数 x 之和。

Worked solution

1. The negative target minus one has no preimage, so retain only one and three.

2. The preimages are 2 ± 1 and 2 ± 3, giving the set $\{-1, 1, 3, 5\}$.

3. This four-term progression has mean two, so its sum is eight.

Section 1.5 Sequences & summation

数列与求和

Method 11 / 方法 11 [AL05-M11] (continued / 续)

例题解析

1. 负目标负一无原像，只保留一与三。

2. 原像为 $2 \pm 1, 2 \pm 3$，得到集合 $\{-1, 1, 3, 5\}$。

3. 此四项等差数列的平均为二，所以和为八。

Negative targets cannot pass through an absolute-value preimage step.

负目标不能通过绝对值原像步骤。

Practice / 配套练习: AIME-2020-II-Q08 (WB p. 209)

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 1 / 方法 1 [AL06-M01]

Floors, ceilings, fractional parts, and absolute values replace one formula by several exact pieces. Write the domain and range of each piece with its open or closed endpoints, then count images or preimages. Separate integer-count arguments from area arguments, where endpoints have zero measure.

取整、上取整、小数部分与绝对值会把一个公式拆成若干精确分段。须写清每段定义域、值域及开闭端点，再计像或原像。整数计数与面积论证应区别处理，后者端点测度为零。

Invert a floor into an exact half-open interval

将取整反解为精确半开区间

Procedure

1. Replace $\lfloor u \rfloor = k$ by $k \le u < k+1$, preserving both endpoint signs.

2. Invert a monotone expression only after fixing its sign and domain.

3. Count integers using floor and ceiling bounds, or count branches whose ranges contain the target.

4. Prove the converse: every counted value or input really lies in its claimed interval.

通用步骤

1. 把 $\lfloor u \rfloor = k$ 换成 $k \le u < k+1$，保留两端开闭。

2. 固定符号与定义域后才反解单调表达式。

3. 用上下取整计算整数，或计算值域包含目标的分支。

4. 证明逆向成立：每个计入的值或输入确实落在对应区间。

> Example. Find all integers x with $\lfloor (2x+1)/5 \rfloor = -2$.

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 1 / 方法 1 [AL06-M01] (continued / 续)

> 例题：求满足 $\lfloor (2x+1)/5 \rfloor = -2$ 的全部整数 x。

Worked solution

1. The floor condition is $-2 \le (2x+1)/5 < -1$.

2. It becomes $-11 \le 2x < -6$, or $-11/2 \le x < -3$.

3. The integer solutions are $-5, -4$. In particular, the upper endpoint -3 is excluded.

例题解析

1. 取整条件为 $-2 \le (2x+1)/5 < -1$。

2. 整理成 $-11 \le 2x < -6$，即 $-11/2 \le x < -3$。

3. 整数解为负五、负四，特别是上端点负三不包含。

Floor means rounding downward, including for negative inputs.

下取整始终向下，负数也一样。

For $A > 0, n > 0$ and positive k, $\lfloor A/n \rfloor = k$ means $A/(k+1) < n \le A/k$. A half-open interval of length at least one always contains an integer. For $x\lfloor x \rfloor$ with $a = \lfloor x \rfloor \ge 1$, the exact range is $[a^2, a^2 + a)$ and each integer in it has inverse $x = N/a$.

For an integer $j \ge 0$, $f(x) = x\{x\}$ on $[j, j+1)$, writing $x = j + t$ gives $f = (j+t)t$, strictly increasing onto $[0, j+1)$. Iterated preimages can therefore be counted by weakly ordered branch indices. Integer endpoints map to zero and must be checked separately.

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 1 / 方法 1 [AL06-M01] (continued / 续)

在 $A>0, n>0, k>0$ 时，$\lfloor A/n \rfloor = k$ 等价于 $A/(k+1) < n \le A/k$。长度至少一的半开区间必含整数。对 $x\lfloor x \rfloor$，若 $a = \lfloor x \rfloor \ge 1$，精确值域为 $[a^2, a^2+a)$，其中每个整数 N 的逆像为 $x = N/a$。

对整数 $j \ge 0$，在 $[j, j+1)$ 上，令 $x = j+t$，则 $x\{x\} = (j+t)t$ 严格递增且值域为 $[0, j+1)$，因此迭代原像可由弱递增分支下标计数。整数端点映到零，须另行检验。

Practice / 配套练习: AIME-2002-II-Q08 (WB p. 162), AIME-2007-II-Q07 (WB p. 164), AIME-2009-I-Q06 (WB p. 166), AIME-2009-II-Q11 (WB p. 174), AIME-2012-II-Q10 (WB p. 168), AIME-2015-II-Q13 (WB p. 203), AIME-2020-II-Q14 (WB p. 178), AIME-2022-II-Q08 (WB p. 504)

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 2 / 方法 2 [AL06-M02]

Group rounding by thresholds and periodic residue classes

按阈值与周期剩余类归组四舍五入

Procedure

1. Write the exact interval producing each rounded value, including any tie convention.

2. If changing the input by a fixed period changes intermediate rounded values by integers, prove a residue-class success pattern.

3. For root rounding or square-free blocks, compare consecutive square gaps with the block width.

4. Count complete blocks and the truncated final block separately.

通用步骤

1. 写出产生每个取整值的精确区间，并说明等距时的规则。

2. 若输入加固定周期使中间取整值增加整数，先证明成功模式按剩余类周期重复。

3. 平方根取整或无平方数区块可比较相邻平方差与区块宽度。

4. 分开计算完整区块与最后截断块。

> Example. For integer n, let $R(n)$ be the nearest integer to $n/3$, and then round $3R(n)$ to an integer. Among $0 \le n \le 20$, which inputs return unchanged?

> 例题：对整数 n，将 $n/3$ 四舍五入为整数 $R(n)$，再取整数 $3R(n)$。零至二十中哪些输入能回到原数?

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 2 / 方法 2 [AL06-M02] (continued / 续)

Worked solution

1. Write $n = 3q + r$, with $r = 0, 1, 2$. The rounded values are $q, q, q + 1$ respectively, with no ties.

2. The returned integers are $3q, 3q, 3q + 3$, so equality holds exactly when $r = 0$.

3. The successful inputs are $0, 3, 6, 9, 12, 15, 18$, seven in total. The residue calculation proves the pattern globally.

例题解析

1. 写 $n = 3q + r$，$r = 0, 1, 2$ 时取整分别为 $q, q, q + 1$，无等距情形。

2. 返回整数分别为 $3q, 3q, 3q + 3$，仅余数零时等于原数。

3. 成功输入为零、三、六、九、十二、十五、十八，共七个；剩余类计算证明了整体周期。

A periodic rounding rule still needs an endpoint count in a finite input interval.

周期取整规则在有限输入范围内仍须处理端点截段。

For positive integers n, k, the nearest integer to $\sqrt{n}$ is k precisely when $(k - 1/2)^2 < n < (k + 1/2)^2$. This becomes $k^2 - k + 1 \le n \le k^2 + k$, with $2k$ inputs. No square-root tie occurs because the squared half-integer endpoints are not integers. The case $n = 0$ gives rounded value zero separately.

A tuple of nondecreasing integer floors stays constant on a consecutive block. To be uniquely identified by it, an integer must have a tuple jump immediately before and after it. This converts singleton floor blocks into divisibility of two consecutive integers.

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 2 / 方法 2 [AL06-M02] (continued / 续)

对正整数 n, k，$\sqrt{n}$ 最近整数为 k 恰要求 $(k-1/2)^2 < n < (k+1/2)^2$，即 $k^2 - k + 1 \le n \le k^2 + k$，共有 $2k$ 个输入。半整数端点的平方非整数，因此无取舍平局；$n = 0$ 时另有取整值零。

若干非降整数取整组成的向量在连续整数区块上恒定。要由该向量唯一确定某整数，它前后必须都发生向量跳跃，可转为相邻整数的整除条件。

Practice / 配套练习: AIME-2007-I-Q05 (WB p. 161), AIME-2007-I-Q11 (WB p. 172), AIME-2008-I-Q07 (WB p. 165), AIME-2013-II-Q06 (WB p. 548), AIME-2022-II-Q08 (WB p. 504)

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 3 / 方法 3 [AL06-M03]

Read logarithmic floors as geometric bands or indicators

把对数取整读成等比区间或指示量

Procedure

1. Translate logarithm thresholds into powers of the base.

2. A floor residue condition selects disjoint bands; sum their lengths with a geometric series.

3. Use $\lceil u \rceil - \lfloor u \rfloor$ as zero at integers and one elsewhere.

4. For a monotone floor sum, certify an extremum by evaluating the last legal input and the next one.

通用步骤

1. 将对数阈值转成底数的幂。

2. 取整剩余类条件选择不交区间带，用几何级数求其长度。

3. 把 $\lceil u \rceil - \lfloor u \rfloor$ 视为整数处零、其他处一的指示量。

4. 单调取整和的最值可由最后合法输入与紧接的下一个输入核验。

> Example. For uniformly chosen $x \in (0, 1]$, find the probability that $\lfloor \log_3(1/x) \rfloor \equiv 1 \pmod 3$.

> 例题：在 $(0, 1]$ 均匀选 x，求 $\lfloor \log_3(1/x) \rfloor$ 模三余一的概率。

Worked solution

1. The permitted floors are $3j + 1$, giving $3^{-3j-2} < x \leq 3^{-3j-1}$ for $j \geq 0$.

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 3 / 方法 3 [AL06-M03] (continued / 续)

2. Each interval length is $(1/3 - 1/9)3^{-3j} = (2/9)27^{-j}$.

3. The total length, hence probability, is $(2/9)/(1 - 1/27) = 3/13$.

例题解析

1. 允许的取整值为 $3j + 1$，对应 $3^{-3j-2} < x \leq 3^{-3j-1}$。

2. 每段长度为 $(2/9)27^{-j}$。

3. 总长度即概率，为 $(2/9)/(1 - 1/27) = 3/13$。

For an integer argument, an integral logarithm may force a special power family; prove which powers are themselves integers.

整数自变量的对数为整数时可能只剩特殊幂族，须证明哪些幂本身为整数。

Practice / 配套练习: AIME-2004-I-Q12 (WB p. 171), AIME-2007-I-Q07 (WB p. 163), AIME-2010-I-Q14 (WB p. 175)

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 4 / 方法 4 [AL06-M04]

Separate a floor sum into its smooth sum and exact residues

将取整和拆成未取整和与精确余数

Procedure

1. Use $\lfloor a/m \rfloor = (a-r)/m$ with the least residue $0 \le r < m$.

2. First bound the total floor error to isolate any unknown integer parameter.

3. Then compute the exact residue correction by full periods and a final partial period.

4. For piecewise-linear area sums, track the fractional part of each unit-strip contribution in the same way.

通用步骤

1. 用 $\lfloor a/m \rfloor = (a-r)/m$，其中最小非负余数为 r。

2. 先估计取整误差总量，隔离可能的未知整数参数。

3. 再按完整周期与尾部截段计算精确余数修正。

4. 分段线性面积和也可同样追踪每个单位条带贡献的小数部分。

> Example. Compute $\sum_{n=1}^{10} \lfloor (3n+2)/4 \rfloor$.

> 例题：计算 $\sum_{n=1}^{10} \lfloor (3n+2)/4 \rfloor$。

Worked solution

1. The unfloored sum is $(3 \cdot 55 + 20)/4 = 185/4$.

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 4 / 方法 4 [AL06-M04] (continued / 续)

2. The residues of $3n+2$ modulo four repeat as $1, 0, 3, 2$. Two full blocks contribute twelve, and the last two terms add one.

3. Subtract the residue total divided by four: $(185-13)/4 = 43$.

例题解析

1. 未取整和为 $(3\cdot 55+20)/4 = 185/4$。

2. $3n+2$ 模四的余数循环为一、零、三、二，两整块贡献十二，最后两项再加一。

3. 扣除余数总量除四，得到 $(185-13)/4 = 43$。

A uniform error bound can prove uniqueness of a parameter before a large sum is evaluated exactly.

统一误差界可以先证明参数唯一，再求大和的精确值。

If N floors each subtract a fractional part in $[0,1)$, their total error lies in $(-N,0]$. If each unit increase of an integer parameter changes the smooth sum in the same direction by more than the permitted window plus this error width, only one parameter can work. For areas under $x\lfloor\sqrt{x}\rfloor$, each integer strip contributes $k(2j+1)/2$; its half-integer parity depends only on whether k is odd.

N 个下取整各减去 $[0,1)$ 内的小数部分，误差总量在 $(-N,0]$。若整数参数每增加一都使未取整和向同一方向变化，且变化量超过允许窗口加误差宽度，就至多一个参数可行。对 $x\lfloor\sqrt{x}\rfloor$ 下的面积，每个整数条带贡献 $k(2j+1)/2$，其半整数奇偶仅取决于 k 是否为奇数。

Practice / 配套练习: AIME-2015-I-Q14 (WB p. 177), AIME-2023-I-Q10 (WB p. 170)

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 5 / 方法 5 [AL06-M05]

Resolve signs and count absolute-value levels

拆分符号并计算绝对值水平线根数

Procedure

1. List all breakpoints where an absolute-value input changes sign.

2. Replace the expression by its exact formula on each interval, retaining its interval restrictions.

3. Construct a level-count table at regular levels, turning levels, zero, and any flat interval.

4. For nested absolute values, translate the outer equation into distinct inner levels and sum their valid root counts.

通用步骤

1. 列出绝对值内部变号的全部分界点。

2. 在各区间用精确公式代替，并保留该区间限制。

3. 对普通高度、转折高度、零高度及平坦段建立根数表。

4. 嵌套绝对值方程先转成不同内部高度，再相加其有效根数。

> Example. Solve $|x-1|+|x+2|=5$, and explain what happens if the right side is three.

> 例题：解 $|x-1|+|x+2|=5$，并说明右端改为三时有何变化。

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 5 / 方法 5 [AL06-M05] (continued / 续)

Worked solution

1. The breakpoints are $-2, 1$. The expression is $-2x - 1$ to the left, three in the middle, and $2x + 1$ to the right.

2. At level five the left and right roots are $-3, 2$, both in their legal intervals; the middle gives none.

3. At level three every $x \in [-2, 1]$ works, so the solution set is an interval rather than a finite root count.

例题解析

1. 分界为负二、一，表达式左段为 $-2x - 1$，中间恒三，右段为 $2x + 1$。

2. 高度五时左右根为负三、二，均在合法段内，中段无解。

3. 高度三时整个 $[-2, 1]$ 都是解，不能按有限根数处理。

Squaring or dropping interval restrictions can turn an illegal branch root into a false solution.

平方或丢掉区间限制，会把无效分支根变成伪解。

An absolute-value preimage $|x - a| = z$ exists only for $z \geq 0$: it gives $a \pm z$, with the two values coinciding at zero. In a two-variable factor-sign region, consider all sign combinations and impose the original plane or ordering constraints before declaring a region bounded.

For an ordered zero-sum tuple with a fixed sum of absolute values, split into total positive and negative mass. An ordered coordinate of rank k cannot carry more than the corresponding mass divided by the number of coordinates forced to share its sign and magnitude bound. The equality tuple must be checked separately.

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 5 / 方法 5 [AL06-M05] (continued / 续)

绝对值原像 $|x-a|=z$ 只在 $z \geq 0$ 时存在，产生 $a \pm z$，零时两值重合。两个变量的乘积符号区域须检查全部符号组合，并施加原平面或次序条件后，才能判断区域有界。

总和为零且绝对值和固定的有序元组，可拆成正质量与负质量。某排名坐标的大小不超过相应质量除以被迫具有同号同量级界的项数；等号元组仍须独立检验。

Practice / 配套练习: AIME-2001-II-Q08 (WB p. 206), AIME-2010-I-Q11 (WB p. 458), AIME-2020-II-Q08 (WB p. 209), AIME-2021-I-Q08 (WB p. 169), AIME-2022-II-Q06 (WB p. 184), AIME-2025-I-Q12 (WB p. 413)

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 6 / 方法 6 [AL06-M06]

Treat floor graphs as cells and strip areas

把取整图形视为单元格与条带面积

Procedure

1. Fix every floor value and translate it into a half-open cell or strip.

2. Within that piece, solve the remaining smooth condition and intersect its range with the cell.

3. For area or probability, sum disjoint piece measures; isolated boundary points have zero measure.

4. For a closed enclosing object, pass to the closure and prove its extremal boundary points are included in the required bound.

通用步骤

1. 固定各个取整值，转成半开单元格或条带。

2. 在分段内解余下光滑条件，并与该格取交。

3. 面积概率按不交分段求和，孤立边界点测度为零。

4. 闭包围对象须包含原集合闭包，并证明极端边界点给出必要界。

> Example. Inside $[0,2)^2$, let R satisfy $\lfloor x \rfloor + \lfloor y \rfloor = 1$. Find its area and the radius of its smallest closed enclosing disk.

> 例题：在 $[0,2)^2$ 中令 R 满足 $\lfloor x \rfloor + \lfloor y \rfloor = 1$，求面积与最小闭包围圆盘半径。

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 6 / 方法 6 [AL06-M06] (continued / 续)

Worked solution

1. The two cells have floor pairs $(0,1),(1,0)$, each of area one, so the area is two.

2. Any closed enclosing disk contains their closures, including opposite points $(0,2)$ and $(2,0)$, whose distance is $2\sqrt{2}$. Thus its radius is at least $\sqrt{2}$.

3. The disk centered at $(1,1)$ of radius $\sqrt{2}$ contains both closed squares, attaining the lower bound.

例题解析

1. 取整对为 $(0,1),(1,0)$ 的两个单元格各面积一，总面积二。

2. 闭包围圆盘须含两格闭包，包括距离 $2\sqrt{2}$ 的对点 $(0,2),(2,0)$，故半径至少 $\sqrt{2}$。

3. 以 $(1,1)$ 为心、半径 $\sqrt{2}$ 的圆盘包含两闭正方形，达到下界。

A boundary excluded from a half-open set can still force the radius of every closed containing disk.

半开集排除的边界仍可能限制所有包含它的闭圆盘半径。

Practice / 配套练习: AIME-2010-I-Q08 (WB p. 167), AIME-2011-II-Q15 (WB p. 176), AIME-2015-I-Q14 (WB p. 177)

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 7 / 方法 7 [AL06-M07]

Filter periodic-piece roots before applying Vieta

先筛周期分段根再使用韦达

Procedure

1. Bound the possible function values, then use the other equation to bound the number of relevant periods.

2. Write one polynomial equation for each rising or falling piece.

3. Keep only roots satisfying that piece domain; count shared endpoints once.

4. Apply Vieta only to a complete retained root pair, and add any single surviving endpoint root separately.

通用步骤

1. 先限制函数值，再用另一方程限制相关周期数量。

2. 对每个上升或下降分段写一个多项式方程。

3. 仅保留符合该分段定义域的根，共用端点只计一次。

4. 只有完整保留的一对根才能直接用韦达求和，单个幸存根须另外相加。

> Example. Let a period-two triangular wave equal x on $[0,1]$ and $2-x$ on $[1,2]$. Find its intersections with $y=x/3$ for $x \geq 0$.

> 例题：周期二的三角波在 $[0,1]$ 为 x、在 $[1,2]$ 为 $2-x$，求其与 $y=x/3$ 在非负横轴上的交点。

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 7 / 方法 7 [AL06-M07] (continued / 续)

Worked solution

1. The wave lies in $[0,1]$, so $0 \le x \le 3$. Only pieces $[0,1], [1,2], [2,3]$ matter.

2. The piece equations are $x = x/3$, $2 - x = x/3$, and $x - 2 = x/3$, giving $x = 0, 3/2, 3$ respectively.

3. All lie in their pieces, so the intersections are $(0,0), (3/2, 1/2), (3,1)$. The endpoint at three is counted once.

例题解析

1. 波值在零与一之间，所以横坐标只须考虑零至三，共三段。

2. 三段方程为 $x = x/3$、$2 - x = x/3$、$x - 2 = x/3$，分别得到 $x = 0, 3/2, 3$。

3. 都位于各自合法段，所以交点为 $(0,0), (3/2, 1/2), (3,1)$，端点三只计一次。

A polynomial may have two roots while its original piece retains only one.

多项式可能有两根，但原分段可能只保留一根。

Practice / 配套练习: AIME-2001-II-Q08 (WB p. 206), AIME-2025-I-Q11 (WB p. 180)

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 8 / 方法 8 [AL06-M08]

Count events in a finite window under stepwise constraints

在阶梯限制下计算有限窗口事件数

Procedure

1. Replace a ceiling-dependent rule by intervals where its integer value is fixed.

2. Optimize the rate or capacity within each interval, preserving whether the endpoint is included.

3. Distinguish a limiting rate from an attainable integer count in a finite closed observation window.

4. For minimal counts, show both a construction and why one fewer item fails the original coverage conditions.

通用步骤

1. 将依赖上取整的规则分成整数值固定的区间。

2. 在每区间内优化速率或容量，并保留端点是否包含。

3. 区分极限速率与有限闭观察窗口中可达到的整数事件数。

4. 最少数量须同时给构造与少一个物体不能满足原覆盖条件的证明。

> Example. Events are exactly fourteen minutes apart. Their phase is free. What is the greatest number occurring in a closed sixty-minute window?

> 例题：事件间隔恰为十四分钟，初相位可自由选择，六十分钟闭窗口中最多发生多少次？

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 8 / 方法 8 [AL06-M08] (continued / 续)

Worked solution

1. If q events occur, the first-to-last span is $(q-1)14$, which must be at most sixty.

2. Thus $q \le \lfloor 60/14 \rfloor + 1 = 5$.

3. Events at minutes $0, 14, 28, 42, 56$ attain five. The floor of the hourly steady rate is four, a different quantity.

例题解析

1. 若发生 q 次，首末跨度为 $(q-1)14$，必须不超过六十。

2. 所以 $q \le \lfloor 60/14 \rfloor + 1 = 5$。

3. 在零、十四、二十八、四十二、五十六分钟发生可达到五次；稳态每小时速率的整数部分为四，是另一数量。

A supremum of rates may still yield an attained integer window count; state the endpoint convention.

速率上确界即使不达到，整数窗口计数仍可能达到，须说明端点约定。

For ordered positive stamp denominations, suppose the already processed stamps have total value S and their subcollections attain every value from zero through S. Adding a stamp of value v extends coverage without a gap exactly when $v \le S+1$. Repeating this criterion gives the necessary low-denomination counts before a high denomination can be used. A final ceiling may give a construction count, but minimality requires showing a collection with one fewer stamp cannot reach the target total.

正邮票面值按序处理，若已处理邮票的总面值为 S，且其子组合能覆盖零至 S，加入面值 v 的一枚邮票能无缺口延伸，当且仅当 $v \le S+1$。反复使用即得到启用较高面值前必须具备的低面值数量。最后的上取整可给构造数量，但最小性还须证明少一枚的总面值无法达到目标。

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 8 / 方法 8 [AL06-M08] (continued / 续)

Practice / 配套练习: AIME-2008-I-Q12 (WB p. 173), AIME-2022-II-Q14 (WB p. 808)

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 9 / 方法 9 [AL06-M09]

Certify intersections of full folded branches

证明完整折叠分支交点计数有效

Procedure

1. Split every absolute and trigonometric fold at all extrema, zeros, and break levels; list each monotone full-range branch.

2. Use the coupled equations to confine every solution and check all branch endpoints separately.

3. Prove one crossing per branch pair: endpoint signs give existence, while orientation or a derivative-product bound gives uniqueness.

4. If a boundary corner replaces a sign change, analyze it locally and certify whether an additional interior root exists.

通用步骤

1. 在所有极值、零点与折叠高度处分开绝对值和三角分段，列全单调满值域分支。

2. 利用联立方程限制全部解，并单独检验所有分支端点。

3. 证明每对分支恰有一次穿越：端点符号给存在性，方向或导数乘积界给唯一性。

4. 若边界角点替代变号，须局部分析是否还有内部根。

> Example. On $[0,1]^2$, solve $y = x^3$ and $x = 2y$. Explain why two increasing functions need not have just one intersection.

> 例题：在单位正方形中解 $y = x^3, x = 2y$，并解释两个递增函数为何不一定只有一个交点。

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 9 / 方法 9 [AL06-M09] (continued / 续)

Worked solution

1. Substitute to get $x = 2x^3$, or $x(1 - 2x^2) = 0$.

2. The two admissible solutions are $(0, 0)$ and $(1/\sqrt{2}, 1/(2\sqrt{2}))$. The negative root lies outside the square.

3. Both graphs are increasing, yet a boundary root and an interior root coexist. At the interior root the derivative product is $(3x^2) \cdot 2 = 3 > 1$, giving a transverse crossing.

例题解析

1. 代入得到 $x = 2x^3$，即 $x(1 - 2x^2) = 0$。

2. 单位方形内解为 $(0, 0)$ 与 $(1/\sqrt{2}, 1/(2\sqrt{2}))$，负根在方形外。

3. 两图都递增，仍可有边界根与内部根。内部根的导数乘积为 $(3x^2) \cdot 2 = 3 > 1$，故为横截穿越。

Counting branch pairs without a uniqueness argument can give an unjustified answer.

没有唯一性证明的分支对相乘，不能保证答案成立。

Crossing lemma: for equally oriented inverse graphs, put $H(y) = G(y) - F^{-1}(y)$. At a root, $H\prime = (F\prime G\prime - 1)/F\prime$. If the derivative product exceeds one at every interior root, the sign of $H\prime$ is fixed on that branch pair. Two roots with that same nonzero crossing direction are impossible. Oppositely oriented graphs are unique directly by monotonicity.

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 9 / 方法 9 [AL06-M09] (continued / 续)

For the coupled trigonometric-fold example, define $h(u)=4\big||u-1/2|-1/4\big|$ on $[0,1]$ and solve $y=F(x)=h(|\sin 2\pi x|)$, $x=G(y)=h(|\cos 3\pi y|)$. The knot values of h at $0,1/4,1/2,3/4,1$ are $1,0,1,0,1$, giving sixteen full-height branches of F and twenty-four full-width branches of G; their only boundary solution is $(1,1)$. The derivative magnitude is $D=96\pi^2|\cos 2\pi x\sin 3\pi y|$. If $|\cos 2\pi x|\le 1/20$, the equations imply $y\ge .99$ and $x\ge 1-18\pi^2(.01)^2>.982$, contradicting that cosine bound. If $D\le 1$, one factor is at most $1/20$, so the sine factor must be small. The equations then force $a=1-x\le .01$, $b=1-y<.01$ and exactly $b=4\sin 2\pi a$, $a=4(1-\cos 3\pi b)$.

Let $K(b)=4(1-\cos 3\pi b)-\arcsin(b/4)/(2\pi)$. On $[0,.01]$, $K''=36\pi^2\cos 3\pi b-b/[128\pi(1-b^2/16)^{3/2}]>0$. Since $K(0)=K(b_*)=0$ for an interior root, strict convexity gives $K'(b_*)>0$, equivalent to $D>1$, a contradiction. Thus the crossing lemma applies. Near zero $K'(0)<0$, while $K(.01)>0$, so the corner branch also has an interior root. This supplies the needed uniqueness and corner certificate for counting the full branch pairs.

穿越引理：同向逆图令 $H(y)=G(y)-F^{-1}(y)$，零点处 $H'=(F'G'-1)/F'$。若每个内部零点导数乘积均大于一，则同一分支对上 H' 在零点具有固定非零符号，不能有两个同向穿越根。反向图则直接由单调性唯一。

联立三角折叠例中，先定义 $h(u)=4\big||u-1/2|-1/4\big|$ $(0\le u\le 1)$，求解 $y=F(x)=h(|\sin 2\pi x|)$、$x=G(y)=h(|\cos 3\pi y|)$。h 在 $0,1/4,1/2,3/4,1$ 的节点值为 $1,0,1,0,1$，故 F 有十六个满高度分支、G 有二十四个满宽度分支，唯一边界解为 $(1,1)$。导数绝对乘积为 $D=96\pi^2|\cos 2\pi x\sin 3\pi y|$。若余弦因子至多二十分之一，方程迫使 $y\ge .99$ 与 $x\ge 1-18\pi^2(.01)^2>.982$，与该余弦界矛盾。若 $D\le 1$，至少一因子不超过二十分之一，所以只能正弦因子小。方程进而迫使 $a=1-x\le .01, b=1-y<.01$，并精确满足 $b=4\sin 2\pi a, a=4(1-\cos 3\pi b)$。

令 $K(b)=4(1-\cos 3\pi b)-\arcsin(b/4)/(2\pi)$。在 $[0,.01]$ 上，其二阶导数 $36\pi^2\cos 3\pi b-b/[128\pi(1-b^2/16)^{3/2}]$ 为正。内部根使 $K(0)=K(b_*)=0$，严格凸性给出 $K'(b_*)>0$，恰等价于 $D>1$，产生矛盾。因此穿越引理适用。零附近 $K'(0)<0$ 而 $K(.01)>0$，故角点分支另有内部根。这补全按满分支对计数所需的唯一性与角点证书。

Section 1.6 Absolute values, floors & piecewise graphs

绝对值、取整与分段图像

Method 9 / 方法 9 [AL06-M09] (continued / 续)

Practice / 配套练习: AIME-2024-I-Q12 (WB p. 179)

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 1 / 方法 1 [AL07-M01]

Every extremum needs a proved bound, the full feasible domain, and a compatible equality or limiting construction. A critical point or an attractive inequality alone does not establish a maximum.

每个最值都需要已证明的界、完整可行域，以及相容的取等或极限构造。只找到驻点或漂亮不等式，并不能证明最大值。

Optimize one variable on its actual interval or residue class

在真实区间或余数类上优化单变量

Procedure

1. Eliminate variables and retain all sign, interval, and integer-congruence restrictions.

2. For a quadratic, complete the square; for a monotone rational expression, compare its derivative or exact difference.

3. An integer optimum uses the nearest permitted lattice points to the vertex, not arbitrary rounding.

4. On an open interval, distinguish an attained value from an infimum or supremum.

通用步骤

1. 消元后保留全部符号、区间及整数同余限制。

2. 二次式配方；单调有理式可比较导数或精确差值。

3. 整数最优应取最接近顶点的允许格点，而不是任意四舍五入。

4. 开区间必须区分实际可取的值与上下确界。

> Example. Maximize $x(30-x)$ for integers $0<x<30$ satisfying $x \equiv 1 \pmod 4$.

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 1 / 方法 1 [AL07-M01] (continued / 续)

> 例题：对满足 $0 < x < 30, x \equiv 1 \pmod 4$ 的整数，求 $x(30-x)$ 最大值。

Worked solution

1. Complete the square: $x(30-x) = 225 - (x-15)^2$.

2. The nearest allowed values are thirteen and seventeen, both distance two from the vertex. Every other allowed integer is farther.

3. Both give $225 - 4 = 221$, so the maximum is 221, attained twice.

例题解析

1. 配方为 $225 - (x-15)^2$。

2. 最近的允许整数为十三和十七，与顶点均差二，其余更远。

3. 两者均给出 221，所以最大值为 221，且有两个取等点。

A congruence restriction can make the two neighboring candidates unequal in value; compare both.

同余限制下，邻近两候选的函数值未必相同，必须都比较。

For a fractional expression on an open feasible interval, prove monotonicity throughout that interval. If the endpoint configuration is degenerate or violates another geometric condition, its limiting value is not attained. To certify the infimum, also construct a sequence of feasible parameters approaching that endpoint.

For an integer probability threshold, clear only positive denominators. Locate the real root between adjacent integers and verify that the rest of the admissible domain lies on the required side.

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 1 / 方法 1 [AL07-M01] (continued / 续)

开可行区间上的分式须在整个区间证明单调性。若端点退化或违反其他几何条件，其极限不可取；要证明下确界，还须给出趋向端点的可行参数序列。

整数概率门槛中，只可清去已知正的分母。将实根定位在相邻整数之间，并证明可行域其余部分均落在正确一侧。

Practice / 配套练习: AIME-2007-II-Q08 (WB p. 803), AIME-2008-I-Q14 (WB p. 186), AIME-2008-II-Q14 (WB p. 277), AIME-2010-II-Q13 (WB p. 764), AIME-2011-I-Q12 (WB p. 765), AIME-2013-I-Q08 (WB p. 121), AIME-2024-I-Q09 (WB p. 416)

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 2 / 方法 2 [AL07-M02]

Complete a product and enforce every AM-GM equality

补成乘积并同时落实所有均值不等式等号

Procedure

1. Group the variables into nonnegative factors whose sum is fixed.

2. Apply AM-GM to each factor group, or $px + q/x \geq 2\sqrt{pq}$ for a reciprocal pair.

3. If the target is a product minus a constrained remainder, impose both equality requirements.

4. Construct original variables satisfying all group sums and side conditions together.

通用步骤

1. 把变量组成和固定的非负因子组。

2. 对因子用均值不等式，倒数对可用 $px + q/x \geq 2\sqrt{pq}$。

3. 若目标为乘积减去受限余量，必须同时满足两处取等。

4. 构造原变量，使所有组和及附加条件共同成立。

> Example. For positive a, b, c, d with sum four, maximize $(a+b)(b+c)(c+d)(d+a)$.

> 例题：正数 a, b, c, d 的和为四，求 $(a+b)(b+c)(c+d)(d+a)$ 的最大值。

Worked solution

1. The opposite factors $a+b, c+d$ sum to four, so their product is at most four. The other opposite factors give the same bound.

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 2 / 方法 2 [AL07-M02] (continued / 续)

2. Therefore the full product is at most sixteen. Equality requires $a+b=c+d=2$ and $b+c=d+a=2$.

3. These are compatible: take $a=c=1/2, b=d=3/2$. Thus sixteen is attained, with a nonconstant equality tuple.

例题解析

1. 相对因子两两和为四，各自乘积至多四。

2. 全积至多十六，取等要求两对因子都等于二。

3. 条件相容，例如 $a=c=1/2, b=d=3/2$，所以十六确可达到，且不必四数全等。

> Equality in one grouped inequality may obstruct equality in another; verify the original tuple.

> 一个分组不等式的取等可能妨碍另一个，须核对原变量。

Practice / 配套练习: AIME-2008-II-Q03 (WB p. 181), AIME-2009-II-Q15 (WB p. 187), AIME-2011-II-Q09 (WB p. 182)

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 3 / 方法 3 [AL07-M03]

Use weighted Cauchy and identify the proportionality

使用加权柯西并确定比例取等

Procedure

1. For positive weights w_i, use $(\sum x_i)^2 \le (\sum x_i^2/w_i)(\sum w_i)$.

2. Equality requires x_i/w_i constant; enforce any positivity and normalization.

3. A linear trigonometric form $A\cos t + B\sin t$ is bounded by $\sqrt{A^2+B^2}$.

4. Check that the proportional vector is permitted by all angular or segment constraints.

通用步骤

1. 正权重下使用加权柯西不等式。

2. 取等须 x_i/w_i 为常数，并满足正性及归一条件。

3. 线性三角式 $A\cos t + B\sin t$ 的上界为 $\sqrt{A^2+B^2}$。

4. 检查比例方向是否符合所有角范围或线段限制。

> Example. For positive $x+y+z=1$, find the minimum of $x^2/2 + y^2/3 + z^2/5$.

> 例题：正数 $x+y+z=1$，求 $x^2/2 + y^2/3 + z^2/5$ 的最小值。

Worked solution

1. Weighted Cauchy gives $1 \le 10(x^2/2 + y^2/3 + z^2/5)$.

2. Equality requires $x/2 = y/3 = z/5$, and the sum condition sets the common ratio to $1/10$.

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 3 / 方法 3 [AL07-M03] (continued / 续)

3. The positive tuple $(1/5, 3/10, 1/2)$ attains $1/10$, so that is the minimum.

例题解析

1. 加权柯西给出目标至少为十分之一。

2. 取等须 $x/2 = y/3 = z/5$，由和为一得公共比为十分之一。

3. 正元组 $(1/5, 3/10, 1/2)$ 确取等，故最小值为十分之一。

An equality constraint already saturating Cauchy can force a unique tuple, even when the question asks for an extremum.

若条件本身已使柯西取等，可能直接迫使唯一元组，即使题目表述为求最值。

Practice / 配套练习: AIME-2016-II-Q15 (WB p. 188), AIME-2017-I-Q15 (WB p. 404), AIME-2024-I-Q07 (WB p. 85)

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 4 / 方法 4 [AL07-M04]

Pair triangle bounds with compatible equality configurations

配对三角不等式并检查相容取等构型

Procedure

1. Group distance sums so each is bounded below by a fixed endpoint distance.

2. For a difference, use the reverse triangle inequality with its collinearity requirement.

3. Check that all required joining segments intersect, or that a rotation construction supplies a common equality point.

4. If using a Fermat point, verify the triangle-angle condition and its position on any additional equality segment.

通用步骤

1. 将距离和分组，使每组由固定端点距离给出下界。

2. 距离差使用反三角不等式，并保留共线取等条件。

3. 检查各连接线段是否相交，或由旋转构造给出共同取等点。

4. 使用费马点时，须检查三角形角条件，以及它是否位于额外的取等线段上。

> Example. For a plane point P, minimize the sum of its distances to $(-3,0),(3,0),(0,-4),(0,4)$.

> 例题：平面点 P 到 $(-3,0),(3,0),(0,-4),(0,4)$ 的距离总和最小是多少?

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 4 / 方法 4 [AL07-M04] (continued / 续)

Worked solution

1. The first opposite pair gives a distance-sum lower bound six, with equality exactly on its horizontal joining segment.

2. The second pair gives a lower bound eight, with equality on its vertical joining segment.

3. The two segments intersect at the origin, so both equalities hold there and the minimum total is fourteen.

例题解析

1. 横向两点的距离和至少六，取等点位于连接它们的水平线段上。

2. 竖向两点的距离和至少八，取等点位于相应竖直线段上。

3. 两线段在原点相交，因此两处同时取等，最小总和为十四。

Two separately sharp lower bounds need not have a common equality point.

两个分别很紧的下界，未必有共同取等点。

A complex modulus obeys the same triangle inequality: for fixed $|w| = R$, the farthest point from a fixed nonzero c is $w = -Rc/|c|$. If w is a power of another variable, verify a permitted root realizes this direction.

For three plane vertices with all angles below 120°, erecting an outward equilateral triangle and rotating one distance by 60° gives the Fermat lower bound. Equality requires three rays meeting at 120°. Additional distance-pair bounds must be attained at that same point.

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 4 / 方法 4 [AL07-M04] (continued / 续)

复数模满足相同三角不等式：固定 $|w| = R$ 时，距非零定点 c 最远的点为 $w = -Rc/|c|$。若 w 是另一变量的幂，还须验证允许的根能实现此方向。

三角形各角小于一百二十度时，在一边外作等边三角形并旋转一段距离六十度，可得费马距离和下界；取等要求三射线夹角一百二十度，额外的距离对下界也须在同一点达到。

Practice / 配套练习: AIME-2008-I-Q10 (WB p. 270), AIME-2012-II-Q06 (WB p. 74), AIME-2022-II-Q12 (WB p. 421), AIME-2025-I-Q14 (WB p. 190)

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 5 / 方法 5 [AL07-M05]

Optimize a fixed-angle area or a fixed-chord height

优化固定角面积或固定弦高

Procedure

1. Express the area as a constant times a sine, its reciprocal, or the perpendicular height above a fixed chord.

2. Identify the exact arc or angle interval supplied by the geometry.

3. Use the allowed sine range or the circular-arc midpoint to prove the bound.

4. Prove the optimizing point is actually reached by the original moving construction.

通用步骤

1. 将面积写为常数乘正弦、除正弦，或固定弦上的垂直高。

2. 确定几何实际给出的弧或角区间。

3. 由允许的正弦范围或圆弧中点证明界。

4. 证明原移动构造确实能到达最优点。

> Example. A circle has radius five and chord $AB = 6$. Find the largest areas of ABP when P lies respectively on the minor and major arcs AB.

> 例题：半径五的圆有弦 $AB = 6$。点 P 分别在劣弧、优弧 AB 上时，三角形面积最大各是多少?

Worked solution

1. The center-to-chord distance is $\sqrt{25 - 9} = 4$.

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 5 / 方法 5 [AL07-M05] (continued / 续)

2. The largest perpendicular heights on the minor and major arcs are respectively $5-4=1$ and $5+4=9$, attained at their arc midpoints.

3. The base is six, so the respective maximum areas are three and twenty-seven. The arc specification changes the answer.

例题解析

1. 圆心到弦的距离为四。

2. 劣弧、优弧上的最大高分别为一、九，均在各自弧中点达到。

3. 底长六，因此最大面积分别为三、二十七；选择哪条弧会改变答案。

A fixed locus bound must include a proof that the original parameter reaches its maximizing locus point.

固定轨迹上的上界，还须证明原参数能到达轨迹的最优点。

Practice / 配套练习: AIME-2009-I-Q15 (WB p. 295), AIME-2013-II-Q10 (WB p. 252), AIME-2018-I-Q13 (WB p. 324)

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 6 / 方法 6 [AL07-M06]

Use even root multiplicities at interior nonnegative contacts

利用内部非负接触点的偶重根

Procedure

1. Subtract the claimed minimum or comparison envelope to obtain a nonnegative polynomial on the relevant interval.

2. Every interior zero has even multiplicity, since an odd multiplicity changes sign.

3. Use the degree and leading coefficient to force a square factorization.

4. Compare coefficients and then verify that the proposed square has the required distinct roots and domain.

通用步骤

1. 减去最小值或比较包络，得到在相应区间非负的多项式。

2. 内部零点的重数为偶数，因为奇重根会变号。

3. 利用次数及首项系数迫使平方因式分解。

4. 比较系数后，验证平方因子具有所需不同根及定义域。

> Example. A monic quartic is nonnegative for every real x and vanishes at $-1, 2$. Find its coefficient of x^2.

> 例题：某首一四次式对所有实数非负，并在负一、二处为零，求二次项系数。

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 6 / 方法 6 [AL07-M06] (continued / 续)

Worked solution

1. Each of the two distinct real zeros has multiplicity at least two.

2. The degree is four and the leading coefficient one, so the polynomial must be $(x+1)^2(x-2)^2=(x^2-x-2)^2$.

3. The x^2 coefficient is $1-4=-3$. The square also verifies global nonnegativity.

例题解析

1. 两个不同实零点的重数均至少二。

2. 由次数四及首一性，必为 $(x+1)^2(x-2)^2=(x^2-x-2)^2$。

3. 二次项系数为 $1-4=-3$，平方形式也验证全域非负。

An endpoint zero need not have even multiplicity; the contact must lie inside the nonnegative domain.

端点零点未必为偶重，接触点必须位于非负定义域内部。

Practice / 配套练习: AIME-2010-I-Q06 (WB p. 43), AIME-2025-II-Q15 (WB p. 191)

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 7 / 方法 7 [AL07-M07]

Bound inscribed shapes by projections and half-planes

用投影及半平面约束内含图形

Procedure

1. For a fixed orientation, compute the width of the shape in every bounding normal direction.

2. A contained disk of radius r must have center-to-boundary signed distances at least r.

3. Combine these linear inequalities to obtain a radius or orientation bound.

4. Exhibit a center and verify every inward half-plane, not just the active pair.

通用步骤

1. 固定朝向后，计算图形在各边法向的投影宽度。

2. 若内含半径 r 的圆盘，圆心到每条边界的有向距离须至少 r。

3. 组合这些线性不等式，得到半径或朝向的界。

4. 给出圆心并检查全部内侧半平面，而不只检查取等的一对。

> Example. Find the largest disk in $0 \le x \le 8, 0 \le y \le 6, x + y \le 10$.

> 例题：求区域 $0 \le x \le 8, 0 \le y \le 6, x + y \le 10$ 内最大圆盘的半径。

Worked solution

1. For center (u, v) and radius r, the lower sides give $u, v \ge r$, while the diagonal side gives $u + v + \sqrt{2}r \le 10$.

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 7 / 方法 7 [AL07-M07] (continued / 续)

2. Thus $(2+\sqrt{2})r \le 10$, or $r \le 5(2-\sqrt{2})$.

3. Take the center (r, r) at this radius. The diagonal constraint is equality, and $2r < 6 < 8$ verifies both upper sides. Hence the bound is attained.

例题解析

1. 圆心 (u, v) 须有 $u, v \ge r$，斜边又要求 $u + v + \sqrt{2}r \le 10$。

2. 故 $r \le 10/(2+\sqrt{2}) = 5(2-\sqrt{2})$。

3. 取圆心为 (r, r)，斜边取等；因 $2r < 6 < 8$，另外两上边界也满足，所以该界可达。

A narrow strip gives an upper bound, but a disk may still fail another side constraint.

最窄带只给上界，圆盘仍可能违反另一条边的限制。

Practice / 配套练习: AIME-2005-I-Q11 (WB p. 185), AIME-2018-I-Q08 (WB p. 378)

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 8 / 方法 8 [AL07-M08]

Find a reachability envelope by a convex hull of disks

用圆盘凸包求可达包络

Procedure

1. First justify a simplified optimal path model from the travel speeds.

2. For each final on-road distance, write the disk of remaining off-road reach.

3. When centers and radii vary linearly, identify the union as a convex hull and derive its common supporting tangent.

4. Take the union over road directions and compute its boundary intersections before the area.

通用步骤

1. 先根据速率证明最优路径可简化。

2. 对每个最后道路距离，写出剩余越野可达圆盘。

3. 圆心和半径线性变化时，将并集识别为凸包并求公共支撑切线。

4. 再对各道路方向取并集，先求边界交点再算面积。

> Example. For $R > 0, 0 < k < 1$, consider disks centered at $(t, 0)$ with radius $R - kt$, where $0 \le t \le R/k$. Describe their union and its upper straight boundary.

> 例题：设 $R > 0, 0 < k < 1$，圆盘圆心为 $(t, 0)$、半径为 $R - kt$，其中 $0 \le t \le R/k$，求并集及上方直边界。

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 8 / 方法 8 [AL07-M08] (continued / 续)

Worked solution

1. Write $\lambda = kt/R$. Each disk is $(1-\lambda)D + \lambda P$, where D is the radius-R disk at the origin and $P = (R/k, 0)$. Thus their union is the convex hull of D and P.

2. The unit normal $n = (k, \sqrt{1-k^2})$ has maximal value $n \cdot (t, 0) + (R - kt) = R$ on every disk.

3. Hence the common upper support is $kx + \sqrt{1-k^2}y = R$. The contact points run from Rn to P, yielding precisely its straight tangent segment.

例题解析

1. 令 $\lambda = kt/R$，各圆盘可写成 $(1-\lambda)D + \lambda P$，其中 D 为原点半径 R 圆盘、$P = (R/k, 0)$，所以并集是它们的凸包。

2. 单位法向 $(k, \sqrt{1-k^2})$ 在每个圆盘上的支撑值均为 $kt + R - kt = R$。

3. 因此上支撑直线为 $kx + \sqrt{1-k^2}y = R$，接触点从 Rn 走到 P，恰给出该直切线段。

Envelope optimization is valid only after the path simplification has been proved.

只有先证明路径简化，包络优化才适用原问题。

Practice / 配套练习: AIME-2000-I-Q13 (WB p. 384)

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 9 / 方法 9 [AL07-M09]

Sum supporting inequalities whose variable parts cancel

相加可抵消变量部分的支撑不等式

Procedure

1. Bound each norm or boundary distance below or above by a linear supporting expression.

2. Choose symmetric directions or weights so the terms involving the unknown point cancel.

3. The remaining constant gives a global bound without solving a derivative system.

4. Check a symmetric point satisfies all supporting equalities and containment conditions.

通用步骤

1. 以线性支撑式对各范数或边界距离估界。

2. 选对称方向或权重，使未知点项相消。

3. 剩余常数给出全局界，不必解导数方程组。

4. 检查对称点同时满足全部支撑取等及容纳条件。

> Example. In space, minimize the sum of distances to the six points $(\pm 1, 0, 0), (0, \pm 1, 0), (0, 0, \pm 1)$.

> 例题：在空间中，求到六个单位坐标轴端点的距离总和最小值。

Worked solution

1. Each vertex vector v has norm one. Cauchy gives $|v - x| \geq (v - x) \cdot v = 1 - x \cdot v$.

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 9 / 方法 9 [AL07-M09] (continued / 续)

2. The six vectors sum to zero, so summing gives a total distance at least six.

3. At $x = 0$, every distance is one. Hence six is the attained global minimum.

例题解析

1. 每个顶点向量为单位向量，由柯西，$|v - x| \geq 1 - x \cdot v$。

2. 六向量和为零，相加得到距离总和至少六。

3. 取原点时每段距离均为一，所以六为可达全局最小值。

Summing symmetric inequalities can be stronger than optimizing them one at a time.

相加对称不等式，可能比逐个优化更强。

For equal-radius vectors v_i summing to zero, $|v_i - x| \geq R - v_i \cdot x/R$ sums to NR. For intersecting cone constraints, use supporting-plane distances and bound radial offsets below by absolute axial coordinates; opposite signs can then be controlled before summing.

等模 R 且和为零的向量，分别满足 $|v_i - x| \geq R - v_i \cdot x/R$，相加得 NR。相交圆锥问题中，可用支撑平面距离，并以轴向坐标绝对值下界径向偏移，处理符号后再相加。

Practice / 配套练习: AIME-2017-II-Q15 (WB p. 405), AIME-2020-II-Q07 (WB p. 426)

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 10 / 方法 10 [AL07-M10]

Restrict the equality case before declaring a geometric maximum

先限制取等构型再宣称几何最大值

Procedure

1. Derive every segment, ray, orientation, and positivity constraint in the chosen parameter.

2. Find the unconstrained equality parameter, then test it against that full feasible interval.

3. If it fails, determine whether the objective is monotone up to an active boundary.

4. Reconstruct the boundary configuration and verify every original distance and containment.

通用步骤

1. 把所有线段、射线、朝向及正性条件写成参数限制。

2. 求无约束取等参数，再对完整可行区间检查。

3. 若不合法，确定目标是否单调至某个约束边界。

4. 重构边界构型，核对全部原距离及容纳关系。

> Example. Maximize $3\cos t + 4\sin t$ on $0 \le t \le \pi/6$.

> 例题：在 $0 \le t \le \pi/6$ 上求 $3\cos t + 4\sin t$ 最大值。

Worked solution

1. The unconstrained Cauchy bound five needs $(\cos t, \sin t) = (3/5, 4/5)$, so $t > \pi/6$ and is infeasible.

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 10 / 方法 10 [AL07-M10] (continued / 续)

2. On the allowed interval, the derivative $-3\sin t + 4\cos t$ is at least $-3/2 + 2\sqrt{3} > 0$.

3. The maximum is therefore at the allowed endpoint, equal to $2 + 3\sqrt{3}/2$, which is strictly below five.

例题解析

1. 无约束柯西上界五需正余弦为 $(3/5, 4/5)$，对应角超过三十度，不可行。

2. 允许区间内，导数至少为 $-3/2 + 2\sqrt{3} > 0$。

3. 因此最大值在右端点，为 $2 + 3\sqrt{3}/2$，严格小于五。

The documented2016 I Q9 source key is an example of an equality configuration violating a side-segment constraint.

已记录的2016 I第九题来源答案差异，正是取等构型违反边线段限制的例子。

Practice / 配套练习: AIME-2016-I-Q09 (WB p. 183)

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 11 / 方法 11 [AL07-M11]

Use saturation of individual coefficient caps

利用各变量上界被同时用满

Procedure

1. Write the equality as a weighted sum with strictly positive weights.

2. Subtract it from the weighted sum of the individual upper bounds.

3. Every deficit is nonnegative, so a zero total forces each deficit to vanish.

4. Recover remaining variables and verify the original complex or polynomial equation.

通用步骤

1. 将等式写成严格正权重的加权和。

2. 从各自上界的加权和中减去已知等式。

3. 每个缺额非负，总缺额为零，就迫使各缺额均为零。

4. 再求其余变量，并验证原复数或多项式方程。

> Example. Reals satisfy $a \le 2, b \le 4, c \le 3$ and $2a + 3b + 5c = 31$. Find abc.

> 例题：实数满足 $a \le 2, b \le 4, c \le 3$ 及 $2a + 3b + 5c = 31$，求 abc。

Worked solution

1. The weighted caps total $2 \cdot 2 + 3 \cdot 4 + 5 \cdot 3 = 31$, equal to the required sum.

2. Thus $2(2 - a) + 3(4 - b) + 5(3 - c) = 0$, a sum of nonnegative terms.

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 11 / 方法 11 [AL07-M11] (continued / 续)

3. Every term is zero, so $(a, b, c) = (2, 4, 3)$ and $abc = 24$.

例题解析

1. 各上界的加权和恰为三十一，与等式相同。

2. 因此 $2(2-a)+3(4-b)+5(3-c)=0$，左边各项非负。

3. 故每项为零，三数为二、四、三，乘积二十四。

Strictly positive weights are essential; a zero weight would leave its variable unconstrained.

权重严格为正很关键，零权重不会限制相应变量。

Practice / 配套练习: AIME-2019-II-Q08 (WB p. 82)

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 12 / 方法 12 [AL07-M12]

Bound sorted coordinates by positive and negative mass

按正负总量限制有序坐标

Procedure

1. Use the ordinary sum and absolute-value sum to determine total positive and negative mass.

2. If $x_k < 0$, at least k entries have magnitude at least $-x_k$; if $x_k > 0$, at least $n-k+1$ entries are at least x_k.

3. Combine the resulting order-statistic bounds with the target coefficients.

4. Construct blocks of equal negative entries, zeros, and equal positive entries to verify simultaneous equality.

通用步骤

1. 由普通和及绝对值和，求正数总量及负数绝对值总量。

2. 若第 k 项为负，前 k 项绝对值均至少为其相反数；若为正，后 $n-k+1$ 项均至少为它。

3. 按目标系数组合所得顺序统计量界。

4. 用等负数块、零块及等正数块构造，验证同时取等。

> Example. Eight ordered reals have sum zero and absolute-value sum two. Maximize $2x_6 - x_2$.

> 例题：八个递增非降实数的和为零、绝对值和为二，求 $2x_6 - x_2$ 的最大值。

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 12 / 方法 12 [AL07-M12] (continued / 续)

Worked solution

1. The total positive mass and total negative magnitude are each one.

2. Therefore $x_2 \geq -1/2$ and $x_6 \leq 1/3$, giving $2x_6 - x_2 \leq 7/6$.

3. The tuple $(-1/2, -1/2, 0, 0, 0, 1/3, 1/3, 1/3)$ satisfies all conditions and attains $7/6$.

例题解析

1. 正数总量及负数绝对值总量均为一。

2. 故 $x_2 \geq -1/2, x_6 \leq 1/3$，目标至多 $7/6$。

3. 取两项负二分之一、三项零、三项三分之一，满足全部条件并取等。

Check that the negative and positive equality blocks do not overlap in the index order.

须检查负、正取等块在下标上没有互相占用。

Practice / 配套练习: AIME-2022-II-Q06 (WB p. 184)

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 13 / 方法 13 [AL07-M13]

Use crossing signs and strict convexity to certify root counts

用穿越符号及严格凸性核实根数

Procedure

1. Split curves into full monotone branches with known endpoint values.

2. Opposite branch orientations give uniqueness by monotonicity; equal orientations need an additional crossing-sign proof.

3. For $y = F(x), x = G(y)$, the derivative of $G(y) - F^{-1}(y)$ at a common point is $(F\prime G\prime - 1)/F\prime$.

4. If a corner root is shared, analyze it separately; strict convexity can prove one further interior crossing rather than justify deleting that branch pair.

通用步骤

1. 将曲线分为端点值明确的完整单调分支。

2. 方向相反时由单调性得唯一交点；同向时必须另证穿越符号。

3. 对互逆形式，在交点处差函数导数为 $(F\prime G\prime - 1)/F\prime$。

4. 共同角点根须单独分析，严格凸性可能证明该分支对还有一个内点交叉，不能直接删去它。

> Example. How many solutions in $[0, 1]$ satisfy $e^{2t} = 1 + 3t$?

> 例题：在 $[0, 1]$ 内，$e^{2t} = 1 + 3t$ 有多少个解?

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 13 / 方法 13 [AL07-M13] (continued / 续)

Worked solution

1. Let $H(t) = e^{2t} - 1 - 3t$. Then $H(0) = 0, H\prime(0) = -1$, and $H\prime\prime(t) = 4e^{2t} > 0$.

2. The derivative increases strictly, so H first decreases and then increases. Since $H(1) = e^2 - 4 > 0$, it crosses zero once after its minimum.

3. There are exactly two roots: the endpoint zero and one interior root. Convexity proves uniqueness of that second crossing.

例题解析

1. 令差函数为 H，则零处函数值零、导数负一，且二阶导数严格正。

2. 导数严格增加，函数先减后增；又在一处为 $e^2 - 4 > 0$，所以最小值之后恰再过零一次。

3. 共有两个根：端点零及一个内点根，第二根的唯一性由凸性保证。

Counting monotone branch pairs is invalid unless intersection existence and uniqueness are both proved.

只有逐对证明交点存在且唯一，才能直接数单调分支对。

If the derivative product has absolute value greater than one at every equally oriented interior crossing, all zeros of the difference have the same strict crossing direction. Two such zeros would force an intervening zero with a contrary or nonstrict crossing, impossible. Endpoint signs then supply existence.

The AL06 folded-branch method for2024 I Q12 supplies the necessary derivative-product certificate, including its separate near-corner function $H(b) = 4(1 - \cos 3\pi b) - \arcsin(b/4)/(2\pi)$ with $H\prime\prime > 0$. This convexity excludes a noncorner weak-derivative crossing and proves that the corner branch pair has an additional interior intersection.

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 13 / 方法 13 [AL07-M13] (continued / 续)

若每个同向内点交叉的导数乘积绝对值都大于一，差函数各零点均有相同严格穿越方向；若有两个，之间必需一个反向或非严格穿越零点，矛盾。再由端点符号证明存在。

AL06中2024 I第十二题的折叠分支方法给出所需导数乘积证明，另用近角点函数 $H(b)=4(1-\cos 3\pi b)-\arcsin(b/4)/(2\pi)$ 的严格凸性排除弱导数非角点根，并证明角点分支对还有一个内部交点。

Practice / 配套练习: AIME-2024-I-Q12 (WB p. 179)

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 14 / 方法 14 [AL07-M14]

Prove existence, then force a repeated-root boundary

先证明最值存在再迫使重根边界

Procedure

1. Bound each positive variable above and away from zero to make the feasible set compact.

2. Represent the variables as roots of a polynomial with the fixed symmetric coefficients.

3. If all roots were distinct and interior, small coefficient perturbations would preserve feasible roots and improve the objective.

4. Therefore test repeated-root cases, compare every positive candidate, and verify the original constraints.

通用步骤

1. 将各正变量同时上界并下界为远离零，使可行集紧致。

2. 把变量看成某个固定对称系数多项式的根。

3. 若根均不同且位于域内，小系数扰动仍保留可行根，并能改进目标。

4. 因此枚举重根情形，比较全部正候选，并验证原限制。

> Example. Show that if $t^3 - st^2 + qt - p$ has three distinct positive roots, then s may be perturbed slightly while retaining three positive roots with the same q, p.

> 例题：若 $t^3 - st^2 + qt - p$ 有三个不同正根，证明可微小改变 s，仍保留三个正根且 q, p 不变。

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 14 / 方法 14 [AL07-M14] (continued / 续)

Worked solution

1. Choose three disjoint closed positive intervals, one around each simple root, with opposite nonzero polynomial signs at the two ends of each interval.

2. At those six endpoints, a sufficiently small perturbation of s changes values only slightly, preserving every sign.

3. The intermediate value theorem gives one positive root in each interval for the perturbed cubic. Degree three makes these exactly its roots, proving the claim and excluding an interior distinct-root extremum of s.

例题解析

1. 在各简单正根周围选三个互不相交的正闭区间，使每个区间两端函数值非零且异号。

2. 足够小地改变 s，只会小幅改变这六处函数值，所有符号保持。

3. 由介值定理，每区间内仍有一正根，三次式恰有这三根。因此成立，也排除了 s 在不同内点根处取得极值。

The repeated-root argument needs compactness or another existence proof before it identifies a maximum.

重根论证之前仍需紧致性或其他存在性证明。

For positive dimensions with $ab+bc+ca=S, abc=V>0$, each $a \geq V/S$ because $bc \leq S$, and $S \geq a(b+c) \geq 2\sqrt{Va}$ gives $a \leq S^2/(4V)$. These closed bounds prove compactness. Setting $b=c=t$ then gives $t^3 - St + 2V = 0$; evaluate all positive roots.

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 14 / 方法 14 [AL07-M14] (continued / 续)

若正尺寸满足 $ab+bc+ca=S, abc=V>0$，由 $bc \le S$ 得 $a \ge V/S$；由 $S \ge a(b+c) \ge 2\sqrt{Va}$ 得 $a \le S^2/(4V)$。这些闭界证明紧致性；再令 $b=c=t$，得 $t^3-St+2V=0$，必须比较全部正根。

Practice / 配套练习: AIME-2024-I-Q15 (WB p. 189)

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 15 / 方法 15 [AL07-M15]

Use strict convexity to distinguish one level intersection from two

用严格凸性区分一次与两次水平相交

Procedure

1. Form a one-parameter membership function for the entire family, not only the given member.

2. Prove strict convexity and divergence at both ends of its parameter interval.

3. A unique level-one membership means the minimum equals one, hence both the value and first derivative equations hold.

4. Check the stationary solution belongs to the stated interior segment.

通用步骤

1. 为整个图形族建立参数成员函数，而不是只写给定成员。

2. 证明严格凸性，并在参数区间两端趋于无穷。

3. 仅有一次值为一的成员，意味着最小值为一，因此联立函数值及导数方程。

4. 检查驻点解确实在所给线段内部。

> Example. For $F(t)=1/t+4/(1-t)$ on $(0,1)$, count solutions of $F(t)=c$ for $c=8,9,10$.

> 例题：对 $(0,1)$ 上 $F(t)=1/t+4/(1-t)$，分别数水平 $c=8,9,10$ 的交点。

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 15 / 方法 15 [AL07-M15] (continued / 续)

Worked solution

1. Cauchy gives $F(t) \geq (1+2)^2 = 9$, with equality at $t = 1/3$. Also $F'' > 0$ and both endpoint limits are infinite.

2. There are no solutions at level eight and exactly one at level nine.

3. At level ten, clearing the positive denominator gives $10t^2 - 7t + 1 = 0$, with roots $1/5, 1/2$. Thus the counts are zero, one, two.

例题解析

1. 柯西给出最小值九，在三分之一处取等；二阶导数正且两端趋于无穷。

2. 因此水平八无解，水平九仅一解。

3. 水平十化为 $10t^2 - 7t + 1 = 0$，根为五分之一、二分之一，所以三种交点数为零、一、二。

The derivative condition alone constructs a candidate envelope; convexity proves uniqueness.

导数条件只给包络候选，严格凸性才证明唯一性。

Practice / 配套练习: AIME-2024-II-Q12 (WB p. 422)

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 16 / 方法 16 [AL07-M16]

Use rearrangement with degree weights and a compatible labeling

以度数权重重排并构造相容标号

Procedure

1. Orient each graph edge from its smaller label to its larger label.

2. The absolute-difference sum becomes a linear label sum with coefficients bounded by vertex degrees and total coefficient zero.

3. Subtract a common center and use rearrangement to pair largest deviations with largest degrees.

4. Construct a bipartite low/high labeling with the needed number of each degree on each side.

通用步骤

1. 把每条边从较小标号定向到较大标号。

2. 绝对差之和成为标号线性和，其系数绝对值受度数限制且系数和为零。

3. 减去公共中心，用重排把最大偏差配给最大度数。

4. 在二分图两色上放低高标号，并核对两色的各度数位置数量。

> Example. Place labels one through five on a five-vertex path. Maximize the sum of differences over its four edges.

> 例题：将一至五放在五顶点路径上，求四条边绝对差总和的最大值。

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 16 / 方法 16 [AL07-M16] (continued / 续)

Worked solution

1. Use center three. There are three degree-two positions and two degree-one positions. The deviations are $2, 2, 1, 1, 0$.

2. Rearrangement bounds the sum by $2(2+2+1)+(1+0)=11$.

3. The path labeling $2, 5, 1, 4, 3$ gives differences $3, 4, 3, 1$, totaling eleven, so the bound is attained.

例题解析

1. 取中心三，三处度二、两处度一，五个偏差为二、二、一、一、零。

2. 重排上界为 $2(2+2+1)+(1+0)=11$。

3. 路径依次标二、五、一、四、三，边差为三、四、三、一，合十一，故取等。

A rearrangement bound may be unattainable unless the graph allows the required sign pattern.

若图不允许所需高低符号模式，重排上界可能不可达到。

Practice / 配套练习: AIME-2026-I-Q11 (WB p. 809)

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 17 / 方法 17 [AL07-M17]

Find extreme directions by tangents to a locus

用轨迹切线找极端方向

Procedure

1. Identify the exact circle or curve traced by the normalized variable.

2. Extreme slopes from an external point occur on tangent lines, provided vertical directions are excluded or separately handled.

3. Use point-to-line distance or a radius-perpendicular triangle for the tangent condition.

4. Choose the correct extremal root and reconstruct a permitted point on the original variable locus.

通用步骤

1. 确定归一变量遍历的准确圆或曲线轨迹。

2. 外点出发的极端斜率位于切线，竖直方向须先排除或另行处理。

3. 用点到直线距离或半径垂直切线的直角三角形写相切条件。

4. 选择正确最值根，并恢复原变量轨迹上的允许点。

> Example. Find the greatest slope y/x at points on $(x-3)^2+(y-4)^2=1$.

> 例题：求圆 $(x-3)^2+(y-4)^2=1$ 上点的斜率 y/x 最大值。

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 17 / 方法 17 [AL07-M17] (continued / 续)

Worked solution

1. The circle has $x \geq 2$, so all slopes are finite. An extremal line $y = mx$ from the origin must be tangent.

2. The distance condition is $(3m-4)^2 = m^2+1$, or $8m^2-24m+15=0$.

3. The two tangent slopes are $(6 \pm \sqrt{6})/4$. The larger is the greatest slope, and its tangency point lies on the circle, proving attainment.

例题解析

1. 圆上 $x \geq 2$，斜率均有限，极端过原点直线须与圆相切。

2. 距离条件化为 $8m^2-24m+15=0$。

3. 两根为 $(6 \pm \sqrt{6})/4$，较大者即最大斜率，其切点确在圆上，故可达。

The origin must lie outside the locus circle for a finite two-tangent argument.

要使用有限的双切线论证，原点必须位于轨迹圆外。

Practice / 配套练习: AIME-2014-I-Q07 (WB p. 77)

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 18 / 方法 18 [AL07-M18]

Find conic-circle tangencies by stationary squared distance

以距离平方驻点求圆锥曲线与圆相切

Procedure

1. Parameterize a smooth point on the given conic.

2. Form squared distance from the prescribed center and differentiate along the conic.

3. At stationarity the radius vector is perpendicular to the tangent; evaluate every such radius.

4. Remove duplicate radii and verify the geometric tangency, including symmetry-related contact points.

通用步骤

1. 参数化给定圆锥曲线的光滑点。

2. 构造到指定圆心的距离平方，并沿曲线求导。

3. 驻点处半径向量垂直切线，计算每种半径。

4. 去重半径，并验证几何相切及对称接触点。

> Example. Find the radii of all circles centered at the origin tangent to $x^2/9 + y^2/4 = 1$.

> 例题：求圆心在原点且与椭圆 $x^2/9 + y^2/4 = 1$ 相切的所有圆半径。

Worked solution

1. Parameterize $(x, y) = (3\cos t, 2\sin t)$. Squared distance is $D = 9\cos^2 t + 4\sin^2 t$.

Section 1.7 Inequalities & continuous optimization

不等式与连续最值

Method 18 / 方法 18 [AL07-M18] (continued / 续)

2. Its derivative is $-10\sin t\cos t$, so stationary points are the four axis endpoints.

3. The distances there are two and three. At each axis endpoint the radial line is normal to the ellipse, so both radii are genuine tangencies and there are no others.

例题解析

1. 参数化后距离平方为 $9\cos^2 t + 4\sin^2 t$。

2. 导数为 $-10\sin t\cos t$，驻点恰为四个坐标轴端点。

3. 相应距离为二及三，且半径均垂直椭圆切线，因此两种均真相切，也无其他半径。

Collect distinct radii rather than counting contact points twice.

应收集不同半径，不能将对称切点重复算成不同半径。

Practice / 配套练习: AIME-2026-II-Q06 (WB p. 418)

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 1 / 方法 1 [AL08-M01]

Trigonometric algebra becomes reliable when each substitution carries its angle range. Keep complementary and supplementary branches, denominator signs, endpoint exclusions, and the attainability of any equality case.

三角代数是否可靠，取决于每次代换是否保留角的范围。补角分支、分母符号、端点排除及等号可达到性都必须保留。

Use a bounded sine-cosine substitution

使用带范围的正余弦代换

Procedure

1. Translate the original domain into allowed signs of sine and cosine.

2. Use $\sin^2 x + \cos^2 x = 1$ to reduce the equation to one bounded variable.

3. Solve the polynomial or inequality, then restore the unsquared signs and any logarithm domain.

4. Use double-angle identities to recover the target rather than solving for the angle itself.

通用步骤

1. 先将原定义域转为正余弦的允许符号。

2. 用平方和恒等式将方程化为一个有界变量。

3. 解多项式或不等式后，恢复未平方符号及对数定义域。

4. 用倍角公式恢复目标，不必求角本身。

> Example. If $0 < x < \pi/4$ and $\sin x \cos x = 1/5$, find $\sin 4x$.

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 1 / 方法 1 [AL08-M01] (continued / 续)

> 例题：若 $0 < x < \pi/4$ 且 $\sin x \cos x = 1/5$，求 $\sin 4x$。

Worked solution

1. The double-angle formula gives $\sin 2x = 2/5$.

2. Since $0 < 2x < \pi/2$, the positive cosine is $\cos 2x = \sqrt{1 - 4/25} = \sqrt{21}/5$.

3. Thus $\sin 4x = 2 \sin 2x \cos 2x = 4\sqrt{21}/25$. The angle interval determined the radical sign.

例题解析

1. 倍角公式给出 $\sin 2x = 2/5$。

2. 由 $0 < 2x < \pi/2$，余弦取正，故 $\cos 2x = \sqrt{21}/5$。

3. 因此 $\sin 4x = 4\sqrt{21}/25$，根式符号由角区间确定。

Squaring removes signs, so every algebraic root must return to the original domain.

平方会丢失符号，所有代数根都须返回原定义域检查。

Practice / 配套练习: AIME-2003-I-Q04 (WB p. 100), AIME-2003-I-Q11 (WB p. 773), AIME-2006-I-Q08 (WB p. 250), AIME-2011-I-Q09 (WB p. 119), AIME-2012-II-Q09 (WB p. 195), AIME-2023-I-Q08 (WB p. 381)

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 2 / 方法 2 [AL08-M02]

List all trigonometric branches over a finite interval

在有限区间列全三角分支

Procedure

1. Use both exact families: $\sin u = \sin v$ gives $u - v = 2k\pi$ or $u + v = (2k+1)\pi$; $\cos u = \cos v$ gives $u = \pm v + 2k\pi$.

2. Intersect each arithmetic angle progression with the stated open or closed interval.

3. Count duplicates by solving the simultaneous progression equations.

4. For a sine level, treat zero, interior levels, and turning-point extrema separately.

通用步骤

1. 列全准确分支：$\sin u = \sin v$ 时，$u - v = 2k\pi$ 或 $u + v = (2k+1)\pi$；$\cos u = \cos v$ 时，$u = \pm v + 2k\pi$。

2. 将每个等差角数列与给定开闭区间相交。

3. 联立两数列方程，检查重复解。

4. 数正弦水平交点时，零、内部水平及转折极值须分别处理。

> Example. How many $x \in [0, 2\pi)$ satisfy $\cos 3x = \sin 2x$?

> 例题：在 $[0, 2\pi)$ 内，有多少个 x 满足 $\cos 3x = \sin 2x$?

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 2 / 方法 2 [AL08-M02] (continued / 续)

Worked solution

1. Write $\sin 2x = \cos(\pi/2 - 2x)$. Equal cosines give $3x = \pi/2 - 2x + 2k\pi$ or $3x = -\pi/2 + 2x + 2k\pi$.

2. The first family is $x = \pi/10 + 2k\pi/5$, with five representatives; the second is $x = -\pi/2 + 2k\pi$, with only $x = 3\pi/2$.

3. That second value is not in the first family, since it would require $k = 7/2$. Hence there are six distinct solutions.

例题解析

1. 将右边写为 $\cos(\pi/2 - 2x)$，余弦相等给出两个差角分支。

2. 第一支为 $x = \pi/10 + 2k\pi/5$，区间内五个；第二支为 $x = -\pi/2 + 2k\pi$，仅有 $3\pi/2$。

3. 后一值若属于第一支，会要求 $k = 7/2$，非整数，因此共有六个不同解。

Equal sine does not mean equal angle; the supplementary family is often decisive.

正弦相等不代表角相等，补角支往往决定答案。

The full rules are $\sin u = \sin v$ iff $u - v = 2k\pi$ or $u + v = (2k+1)\pi$, and $\cos u = \cos v$ iff $u = \pm v + 2k\pi$. For a nested sine, split its inner argument into monotone arcs before counting levels; a zero derivative at an inner extremum can be a tangency rather than two crossings.

For a periodic sign itinerary under angle multiplication, shifting one itinerary period preserves the full condition. If uniqueness is stated, the reduced angle is fixed by that shift; solve its finite congruence candidates and verify every sign in one complete period.

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 2 / 方法 2 [AL08-M02] (continued / 续)

完整规则为：正弦相等当且仅当 $u - v = 2k\pi$ 或 $u + v = (2k+1)\pi$；余弦相等当且仅当 $u = \pm v + 2k\pi$。嵌套正弦数水平交点时，先把内部角分成单调弧；内部极值的零导数可能对应相切，而不是两次穿越。

角倍增下的周期符号轨迹，移动一个信息周期后仍满足完整条件；若已知唯一性，化简后的角须为该移动的不动点。解有限同余候选，再核对一整个符号周期。

Practice / 配套练习: AIME-2002-II-Q10 (WB p. 192), AIME-2018-I-Q06 (WB p. 81), AIME-2019-II-Q10 (WB p. 208), AIME-2024-I-Q12 (WB p. 179), AIME-2025-II-Q09 (WB p. 197)

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 3 / 方法 3 [AL08-M03]

Factor with sum-to-product before dividing

先和差化积再考虑相除

Procedure

1. Pair symmetric angle multiples with sum-to-product identities.

2. Look for a common factor or a cubic identity rather than expanding every term.

3. Keep the zero-factor branches; divide only after handling them separately.

4. Use the triangle angle sum to simplify cyclic expressions when applicable.

通用步骤

1. 将对称倍角用和差化积配对。

2. 寻找公共因子或立方恒等式，不必逐项展开。

3. 保留因子为零的分支，单独处理后才能相除。

4. 涉及三角形循环式时，利用角和进一步化简。

> Example. Solve $\cos x + \cos 3x = \cos 2x$ for $0 < x < \pi$.

> 例题：在 $0 < x < \pi$ 内解 $\cos x + \cos 3x = \cos 2x$。

Worked solution

1. The left side is $2\cos 2x \cos x$, so the equation factors as $\cos 2x(2\cos x - 1) = 0$.

2. The first factor gives $x = \pi/4, 3\pi/4$; the second gives $x = \pi/3$.

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 3 / 方法 3 [AL08-M03] (continued / 续)

3. All three are in the interval and distinct. Dividing at the start by $\cos 2x$ would have lost two roots.

例题解析

1. 左边为 $2\cos 2x\cos x$，所以方程因式分解为 $\cos 2x(2\cos x-1)=0$。

2. 第一因子给出 $\pi/4, 3\pi/4$，第二因子给出 $\pi/3$。

3. 三根均在区间内且互异，若一开始除以 $\cos 2x$，就会漏去两根。

A factorized identity exposes exceptional branches that an early division hides.

因式分解能揭示过早相除所遮住的特殊分支。

For triangle angles, the sine and cosine laws give $\sin^2 A+\sin^2 B-2\sin A\sin B\cos C=\sin^2 C$. Therefore $\cos^2 A+\cos^2 B+2\sin A\sin B\cos C=2-\sin^2 C$. Also $1-\sum\cos 3A=4\sin(3A/2)\sin(3B/2)\sin(3C/2)$.

Products under a fixed angle sum use $2\cos u\cos v=\cos(u-v)+\cos(u+v)$ and $\sin u+\sin v=2\sin((u+v)/2)\cos((u-v)/2)$. A formal maximum of the resulting cosine must still satisfy all original side-segment constraints.

三角形的正余弦定理给出 $\sin^2 A+\sin^2 B-2\sin A\sin B\cos C=\sin^2 C$，故对应余弦平方循环式等于 $2-\sin^2 C$；另有 $1-\sum\cos 3A=4\sin(3A/2)\sin(3B/2)\sin(3C/2)$。

固定角和下的积可用 $2\cos u\cos v=\cos(u-v)+\cos(u+v)$，和可用正弦和差化积。所得余弦的形式最大值，仍须符合原来的边线段容纳条件。

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 3 / 方法 3 [AL08-M03] (continued / 续)

Practice / 配套练习: AIME-2006-I-Q12 (WB p. 199), AIME-2013-II-Q15 (WB p. 200), AIME-2014-II-Q12 (WB p. 201), AIME-2016-I-Q09 (WB p. 183), AIME-2025-II-Q14 (WB p. 256)

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 4 / 方法 4 [AL08-M04]

Add tangents with full angle-branch control

带完整角分支地相加正切

Procedure

1. Track the ranges of the angles before applying tangent addition.

2. Use $\tan(u+v)=(\tan u+\tan v)/(1-\tan u\tan v)$ with separate handling of a zero denominator.

3. For tangent polygons, express each vertex half-angle as $\arctan(r/t)$ and impose their actual total.

4. Filter algebraic roots using monotonicity of the full angle sum, not merely its tangent.

通用步骤

1. 使用正切和角前先跟踪各角范围。

2. 应用和角公式，分母为零时须另行处理。

3. 切圆多边形中，各顶点半角可写为 $\arctan(r/t)$，并保留真实总角。

4. 利用整个角和的单调性筛选代数根，而非只看正切是否相等。

> Example. Find the exact angle $\arctan 2+\arctan 3$.

> 例题：求 $\arctan 2+\arctan 3$ 的精确值。

Worked solution

1. Both angles lie in $(0,\pi/2)$, so the sum lies in $(0,\pi)$.

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 4 / 方法 4 [AL08-M04] (continued / 续)

2. The tangent is $(2+3)/(1-6) = -1$. The negative tangent places the sum in the second quadrant.

3. Therefore the sum is $3\pi/4$, not the principal arctangent $-\pi/4$.

例题解析

1. 两角均在第一象限，和在零与 π 之间。

2. 正切为 $(2+3)/(1-6) = -1$，负号说明角和在第二象限。

3. 因此和为 $3\pi/4$，而不是主值反正切 $-\pi/4$。

Tangent identifies an angle only modulo π; the original angle total chooses the branch.

正切只能确定模 π 的角，须由原角和选择分支。

A convex n-gon has half-angle total $(n-2)\pi/2$. Each positive term $\arctan(r/t_i)$ increases strictly with r, so the radius satisfying this full total is unique. A root of a tangent equation may instead represent another total differing by π.

For a radical sine recurrence, $\sqrt{1-\sin^2\phi} = |\cos\phi|$. Crossing $\pi/2$ changes the sign of the cosine term and can fold an increasing angle sequence into a short cycle.

凸 n 边形半角和为 $(n-2)\pi/2$。每个正项 $\arctan(r/t_i)$ 随 r 严格增加，故满足完整角和的半径唯一；正切方程的某个根可能对应相差 π 的另一个总角。

根式正弦递推中，$\sqrt{1-\sin^2\phi} = |\cos\phi|$；跨越直角会改变余弦项符号，将递增形式角折叠成短循环。

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 4 / 方法 4 [AL08-M04] (continued / 续)

Practice / 配套练习: AIME-2000-II-Q10 (WB p. 331), AIME-2002-II-Q15 (WB p. 341), AIME-2008-I-Q08 (WB p. 193), AIME-2009-II-Q14 (WB p. 213), AIME-2016-II-Q14 (WB p. 433), AIME-2020-II-Q13 (WB p. 347)

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 5 / 方法 5 [AL08-M05]

Use a two-root recurrence for high trigonometric powers

以二根递推处理高次三角幂

Procedure

1. Put $u=\sin^2 x, v=\cos^2 x$, so $u+v=1$ and $p=uv\in[0,1/4]$.

2. The power sums satisfy $S_0=2, S_1=u+v$ and $S_n=(u+v)S_{n-1}-uvS_{n-2}$.

3. Use the allowed product interval to reject extraneous algebraic roots.

4. When irrational coefficients occur, prove the rational/irrational parity pattern by induction before studying last digits.

通用步骤

1. 令 $u=\sin^2 x, v=\cos^2 x$，则和为一，积在零至四分之一之间。

2. 幂和满足 $S_n=(u+v)S_{n-1}-uvS_{n-2}$，初值为二及 $u+v$。

3. 使用乘积允许区间排除多余代数根。

4. 若递推系数含无理数，先归纳证明有理、无理的下标奇偶规律，再研究末位。

> Example. If $\cos 2x=1/3$, find $\sin^8 x+\cos^8 x$ using a power-sum recurrence.

> 例题：若 $\cos 2x=1/3$，用幂和递推求 $\sin^8 x+\cos^8 x$。

Worked solution

1. Here $u+v=1$ and $uv=(1-\cos^2 2x)/4=2/9$.

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 5 / 方法 5 [AL08-M05] (continued / 续)

2. The recurrence gives $S_2 = 1 - 4/9 = 5/9$ and $S_3 = 5/9 - 2/9 = 1/3$.

3. Thus $S_4 = 1/3 - (2/9)(5/9) = 17/81$, the required eighth-power sum.

例题解析

1. 此时 $u+v=1, uv=(1-\cos^2 2x)/4=2/9$。

2. 递推得 $S_2=5/9, S_3=1/3$。

3. 所以 $S_4=1/3-(2/9)(5/9)=17/81$，即所求八次幂和。

An integer subsequence needs an integrality proof before modular arithmetic applies.

只有先证明子序列为整数，才能对它作整数同余。

Practice / 配套练习: AIME-2019-I-Q08 (WB p. 51), AIME-2023-II-Q13 (WB p. 205)

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 6 / 方法 6 [AL08-M06]

Telescope trigonometric sums by endpoint differences

将三角和望远镜化为端点差

Procedure

1. Use product-to-sum or a cotangent difference to express each summand as a discrete difference.

2. Write the finite sum with explicit first and last terms before canceling.

3. For alternating samples, pair supplementary angles and account for unpaired endpoints.

4. Translate the final sign or integer-value condition into a periodic angle interval.

通用步骤

1. 用积化和差或余切差，将每项变成离散差分。

2. 消去之前，明确写出有限和首尾项。

3. 交替采样时配对互补角，并保留未配对端点。

4. 把最终符号或整数值要求转为周期角区间。

> Example. Find a closed form for $\sum_{k=1}^{n} \sin((2k-1)t)$ when $\sin t \neq 0$.

> 例题：当 $\sin t \neq 0$，求 $\sum_{k=1}^{n} \sin((2k-1)t)$ 的闭式。

Worked solution

1. The identity $2\sin t \sin((2k-1)t) = \cos((2k-2)t) - \cos(2kt)$ gives a difference.

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 6 / 方法 6 [AL08-M06] (continued / 续)

2. Summing cancels every interior cosine, leaving $1-\cos 2nt$.

3. Divide by $2\sin t$ and use $1-\cos 2nt = 2\sin^2 nt$. The answer is $\sin^2(nt)/\sin t$.

例题解析

1. 由积化和差，每项乘 $2\sin t$ 后为两个相邻偶倍角余弦之差。

2. 求和时内部余弦全部抵消，只余 $1-\cos 2nt$。

3. 除以 $2\sin t$，再用倍角公式，得闭式 $\sin^2(nt)/\sin t$。

The identity $\cot u - \cot v = \sin(v-u)/(\sin u \sin v)$ is a useful reciprocal-product version.

余切差恒等式是处理正弦乘积倒数的有用版本。

Practice / 配套练习: AIME-2000-II-Q15 (WB p. 198), AIME-2008-II-Q08 (WB p. 194), AIME-2015-II-Q13 (WB p. 203)

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 7 / 方法 7 [AL08-M07]

Pair and double sine products, or use roots of unity

配对倍角正弦积或使用单位根

Procedure

1. Pair complementary or supplementary angles to relate two products.

2. Apply $2\sin x\cos x=\sin 2x$, then match the doubled-angle multiset exactly.

3. Cancel only nonzero products and preserve any central factor equal to one.

4. Alternatively factor a root-of-unity polynomial at one and take magnitudes.

通用步骤

1. 将余角或补角配对，建立两个乘积的关系。

2. 使用正弦倍角公式，准确匹配倍角后的角多重集。

3. 只约去非零乘积，并保留中央等于一的因子。

4. 也可在一处计算单位根多项式的因式分解，再取模。

> Example. Prove $\sin(\pi/5)\sin(2\pi/5)=\sqrt{5}/4$.

> 例题：证明 $\sin(\pi/5)\sin(2\pi/5)=\sqrt{5}/4$。

Worked solution

1. Let $\zeta=e^{2\pi i/5}$. Evaluating $(x^5-1)/(x-1)$ at one gives $\prod_{k=1}^{4}(1-\zeta^k)=5$.

2. Taking magnitudes and using $|1-e^{it}|=2|\sin(t/2)|$ gives $16\prod_{k=1}^{4}\sin(k\pi/5)=5$.

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 7 / 方法 7 [AL08-M07] (continued / 续)

3. Supplementary sines pair equally, so the product is $(\sin(\pi/5)\sin(2\pi/5))^2 = 5/16$. Both factors are positive, giving the stated positive square root.

例题解析

1. 令五次单位根为 ζ，在一处计算 $(x^5-1)/(x-1)$，得四个因子的乘积为五。

2. 取模并用弦长公式，得到 $16\prod_{k=1}^{4}\sin(k\pi/5) = 5$。

3. 补角正弦相等，因此所求积的平方为 $5/16$；两因子均正，故取正根 $\sqrt{5}/4$。

A product identity depends on the entire angle multiset, not only its number of factors.

乘积恒等式依赖整个角多重集，而不只依赖因子数量。

Practice / 配套练习: AIME-2015-I-Q13 (WB p. 202)

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 8 / 方法 8 [AL08-M08]

Convert multiple-angle metric relations into polynomial identities

把倍角度量关系转为多项式恒等式

Procedure

1. Use the geometry to choose one angle and prove its range before algebra.

2. Apply the chord formula $2R\sin(\phi/2)$ or the sine rule to express length ratios.

3. Use double- and triple-angle identities to eliminate the angle in favor of a bounded sine or cosine.

4. Check every root against the geometric ordering and any supplementary-angle choices.

通用步骤

1. 先由几何选择一个角，并证明它的范围。

2. 使用弦长公式或正弦定理表示长度比。

3. 以二倍、三倍角恒等式，将角消为有界正弦或余弦。

4. 按几何顺序及可能的补角选择筛选每个根。

> Example. Prove $\sin 3t/\sin t = 1 + 2\cos 2t$ for $0 < t < \pi$.

> 例题：对 $0 < t < \pi$，证明 $\sin 3t/\sin t = 1 + 2\cos 2t$。

Worked solution

1. Addition gives $\sin 3t = \sin t\cos 2t + \cos t\sin 2t$.

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 8 / 方法 8 [AL08-M08] (continued / 续)

2. Use $\sin 2t = 2\sin t\cos t$ and divide by the positive $\sin t$, obtaining $\cos 2t + 2\cos^2 t$.

3. Since $2\cos^2 t = 1 + \cos 2t$, this equals $1 + 2\cos 2t$, also $4\cos^2 t - 1$.

例题解析

1. 和角公式给出 $\sin 3t = \sin t\cos 2t + \cos t\sin 2t$。

2. 代入二倍角并除以正的 $\sin t$，得 $\cos 2t + 2\cos^2 t$。

3. 再由余弦倍角恒等式，化为 $1 + 2\cos 2t = 4\cos^2 t - 1$。

A chord may correspond to a major consecutive arc; exclude or include that case from the polygon geometry.

一条边可能对应连续的优弧，须由多边形构型决定是否排除。

Practice / 配套练习: AIME-2001-I-Q13 (WB p. 292), AIME-2010-II-Q14 (WB p. 279), AIME-2011-I-Q14 (WB p. 394), AIME-2013-II-Q08 (WB p. 285), AIME-2018-I-Q15 (WB p. 300)

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 9 / 方法 9 [AL08-M09]

Encode geometric angles by dot products and equally turning vectors

用点积及等转角向量表示几何角

Procedure

1. Write each direction as a unit sine-cosine vector, keeping orientation.

2. Use dot products for cosine and determinants for signed sine; a tangent ratio also needs the dot-product sign.

3. For a closed equiangular polygon, the weighted direction vectors sum to zero.

4. Derive exact special-angle cosines from the closure or root-of-unity relation when needed.

通用步骤

1. 将方向写成带定向的单位正余弦向量。

2. 点积给余弦，行列式给有向正弦；求正切还须保留点积符号。

3. 闭合等角多边形的加权边方向向量和为零。

4. 需要时由闭合或单位根关系推导特殊角余弦的精确值。

> Example. Derive exact values for $\cos 72^\circ$ and $\cos 144^\circ$ from a regular pentagon.

> 例题：由正五边形推导 $\cos 72^\circ$ 和 $\cos 144^\circ$ 的精确值。

Worked solution

1. The five equally spaced unit vectors sum to zero. Real parts give $1 + 2\cos 72^\circ + 2\cos 144^\circ = 0$.

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 9 / 方法 9 [AL08-M09] (continued / 续)

2. Put $c = \cos 72^\circ$ and use $\cos 144^\circ = 2c^2 - 1$. This gives $4c^2 + 2c - 1 = 0$.

3. Since 72° is acute, $c = (\sqrt{5} - 1)/4$. Substitution gives $\cos 144^\circ = -(\sqrt{5} + 1)/4$.

例题解析

1. 五个等距单位方向向量和为零，取实部得 $1 + 2\cos 72^\circ + 2\cos 144^\circ = 0$。

2. 令 $c = \cos 72^\circ$，用倍角公式得 $4c^2 + 2c - 1 = 0$。

3. 由七十二度为锐角，取正根 $c = (\sqrt{5} - 1)/4$，继而得 $\cos 144^\circ = -(\sqrt{5} + 1)/4$。

For rolling, ground-frame spin includes rotation of the line of centers as well as relative contact rotation.

滚动的对地自转还包括圆心连线的转动，不能只算相对接触转角。

Practice / 配套练习: AIME-2005-II-Q12 (WB p. 392), AIME-2014-I-Q10 (WB p. 374), AIME-2026-I-Q14 (WB p. 414)

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 10 / 方法 10 [AL08-M10]

Linearize complementary radicals with sine squares

以正弦平方线性化互补根式

Procedure

1. Use the real-radical domain to bound each variable, then put it equal to a scaled sine square.

2. Choose angles in $[0, \pi/2]$ so all square roots have the intended positive signs.

3. The paired radicals become a sine of an angle sum.

4. List both supplementary pair-sum branches, solve their linear angle equations, and reject those outside the allowed range.

通用步骤

1. 由实根式定义域给变量界，再令其等于适当倍数的正弦平方。

2. 选角在第一象限闭区间内，保证所有根式取正确非负符号。

3. 成对互补根式会变成角和的正弦。

4. 列出角和的两种补角支，解线性角方程，并排除超范围分支。

> Example. For $0 \le x, y \le 1$, characterize solutions of $\sqrt{x(1-y)} + \sqrt{y(1-x)} = 1$.

> 例题：对 $0 \le x, y \le 1$，刻画 $\sqrt{x(1-y)} + \sqrt{y(1-x)} = 1$ 的全部解。

Worked solution

1. Put $x = \sin^2 a, y = \sin^2 b$ with $a, b \in [0, \pi/2]$. The left side is $\sin(a+b)$.

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 10 / 方法 10 [AL08-M10] (continued / 续)

2. As $a+b \in [0,\pi]$, equality to one requires $a+b=\pi/2$.

3. Thus $y=\cos^2 a = 1-x$. Conversely every such pair satisfies the original equation, so the complete solution is $x+y=1$ in the given square.

例题解析

1. 令 $x=\sin^2 a, y=\sin^2 b$，角均在零至直角之间，左边为 $\sin(a+b)$。

2. 角和在零至 π 之间，正弦为一只能是角和为直角。

3. 所以 $y=1-x$；反之该关系确满足原方程，故全部解为给定方形内的 $x+y=1$。

> The angle range is what makes the radical identity exact without extra absolute values.

> 角范围保证根式恒等式准确成立，不额外产生绝对值。

Practice / 配套练习: AIME-2022-I-Q15 (WB p. 204)

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 11 / 方法 11 [AL08-M11]

Join sine and cosine series into a complex geometric series

将正余弦级数合成复等比级数

Procedure

1. Express phase shifts through powers of e^{it} or ie^{it}.

2. Combine the two real series as real and imaginary parts of one absolutely convergent geometric series.

3. Sum it exactly, rationalize once, then read both parts.

4. Restore quadrant and denominator conditions after any squared equation.

通用步骤

1. 以 e^{it} 或 ie^{it} 的幂表示相位变化。

2. 将两条实级数合为一个绝对收敛复等比级数的实虚部。

3. 精确求和并统一有理化，再读出两部分。

4. 若后续平方求解，须恢复象限及分母条件。

> Example. Find $\sum_{k=1}^{\infty} 3^{-k}\sin(k\pi/3)$.

> 例题：求 $\sum_{k=1}^{\infty} 3^{-k}\sin(k\pi/3)$。

Worked solution

1. The series is the imaginary part of $\sum_{k\geq 1}(e^{i\pi/3}/3)^k$, whose ratio modulus is $1/3$.

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 11 / 方法 11 [AL08-M11] (continued / 续)

2. For general $|r|<1$, that imaginary part is $r\sin t/(1-2r\cos t+r^2)$, by rationalizing $re^{it}/(1-re^{it})$.

3. With $r=1/3, t=\pi/3$, it is $(\sqrt{3}/6)/(7/9)=3\sqrt{3}/14$.

例题解析

1. 所求为公比 $e^{i\pi/3}/3$ 的复等比级数虚部，公比模小于一。

2. 在公比满足 $|r|<1$ 时，将闭式有理化，一般虚部为 $r\sin t/(1-2r\cos t+r^2)$。

3. 代入 $r=1/3, t=\pi/3$，得 $(\sqrt{3}/6)/(7/9)=3\sqrt{3}/14$。

Absolute convergence permits joining the series; it should be stated before rearranging.

应先说明绝对收敛，再合并或重排级数。

Practice / 配套练习: AIME-2013-I-Q14 (WB p. 93)

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 12 / 方法 12 [AL08-M12]

Check compatibility of simultaneous trigonometric maxima

检查多个三角项能否同时取最大

Procedure

1. If a sum attains the sum of individual upper bounds, every summand must attain its own bound.

2. Translate those equalities to angular congruences and test compatibility using gcd or Bezout.

3. If the bounds are incompatible, reduce the actual expression using an identity and optimize on its true range.

4. Exhibit an angle realizing the final compatible equality case.

通用步骤

1. 和要达到各项上界之和，每项必须同时达到自身上界。

2. 将取等条件转为角同余，并用最大公因数或贝祖检验相容性。

3. 若不相容，利用恒等式在真实范围内优化原表达式。

4. 给出实现最后取等条件的角。

> Example. Find the maximum of $\sin 2x + \cos 4x$ for real x.

> 例题：求实数域上 $\sin 2x + \cos 4x$ 的最大值。

Section 1.8 Trigonometric algebra & equations

三角恒等式与方程

Method 12 / 方法 12 [AL08-M12] (continued / 续)

Worked solution

1. The separate upper bounds sum to two, but $\sin 2x = 1$ forces $\cos 4x = -1$, so they cannot both attain one.

2. Put $u = \sin 2x \in [-1, 1]$. Then the expression is $1 + u - 2u^2 = 9/8 - 2(u - 1/4)^2$.

3. The value $u = 1/4$ is attainable by a real angle, so the exact maximum is $9/8$.

例题解析

1. 各项上界相加为二，但 $\sin 2x = 1$ 会迫使 $\cos 4x = -1$，不能同时为一。

2. 令 $u = \sin 2x \in [-1, 1]$，表达式为 $1 + u - 2u^2 = 9/8 - 2(u - 1/4)^2$。

3. 实角可实现 $u = 1/4$，所以准确最大值为 $9/8$。

An algebraic upper bound is not a maximum until all its equality requirements are compatible.

只有全部取等要求相容，代数上界才能称为最大值。

For positive frequencies $m = gu, n = gv$ with coprime u, v, simultaneous $\sin(mx) = \sin(nx) = 1$ requires $v(4a + 1) = u(4b + 1)$. Bezout shows integer a, b exist exactly when $u \equiv v \pmod 4$, which forces both odd. This is stronger than merely requiring the two frequencies odd.

将正频率写为 $m = gu, n = gv$，其中 u, v 互质。同时正弦为一要求 $v(4a + 1) = u(4b + 1)$。由贝祖，整数 a, b 存在当且仅当 $u \equiv v \pmod 4$，从而两者均奇；仅要求两频率为奇数还不够。

Practice / 配套练习: AIME-2021-I-Q07 (WB p. 196)

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 1 / 方法 1 [AL09-M01]

A functional equation constrains whole orbits, not isolated values. Compose transformations, preserve domains, and use rigidity when a polynomial or an infinite integer orbit cannot support the proposed behavior.

函数方程限制的是完整轨道，而非孤立函数值。应复合变换、保留定义域，并利用多项式或无穷整数轨道无法支持某些行为的刚性。

Compose input symmetries and classify value orbits

复合自变量对称并分类函数值轨道

Procedure

1. Compose the stated input transformations, carrying along any output sign or multiplier.

2. Identify genuine periods once the accumulated multiplier is one; integer periods combine by their gcd.

3. On integer residues, count the remaining reflection orbits, and check whether a sign relation forces a value to zero.

4. Construct a function satisfying every original relation to prove a proposed number of distinct values is attainable.

通用步骤

1. 复合题给自变量变换，同时保留输出的符号或倍数。

2. 累计输出倍数为一时得到真正周期；整数周期可取最大公因数。

3. 在整数余数上数剩余反射轨道，并检查带负号关系是否迫使函数值为零。

4. 构造满足全部原关系的函数，才能证明不同函数值数目可达到。

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 1 / 方法 1 [AL09-M01] (continued / 续)

> Example. A real function obeys $f(-x) = -f(x)$ and $f(2-x) = f(x)$. What is the maximum number of distinct values at integer inputs?

> 例题：实函数满足 $f(-x) = -f(x), f(2-x) = f(x)$，在所有整数输入处最多有多少个不同函数值?

Worked solution

1. Composition gives $f(x+2) = -f(x)$ and hence period four. Oddness gives $f(0) = 0$, so every even integer has value zero.

2. Writing $f(1) = a$, odd integer values alternate $a, -a$. Thus at most the three values $0, a, -a$ occur.

3. The function $f(x) = \sin(\pi x/2)$ satisfies both original identities and realizes $0, 1, -1$, proving the maximum is three.

例题解析

1. 复合得 $f(x+2) = -f(x)$，因此周期为四。奇性给出 $f(0) = 0$，故所有偶整数处值为零。

2. 若 $f(1) = a$，奇整数处交替为 $a, -a$，所以至多有零、a、$-a$ 三值。

3. $f(x) = \sin(\pi x/2)$ 满足两原关系并实现零、一、负一，故最大值为三。

An upper bound on orbit classes still needs a function separating the permitted classes.

轨道分类只给上界，还需函数把允许分开的类别真正区分。

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 1 / 方法 1 [AL09-M01] (continued / 续)

For a period P and reflection $r \mapsto c - r \pmod{P}$, fixed residues solve $2r \equiv c \pmod{P}$. If there are F fixed residues, the number of value orbits is $(P + F)/2$. A cosine centered at $c/2$ separates exactly the reflection pairs when no other relation is present.

周期为 P，剩余反射为 $r \mapsto c - r \pmod{P}$ 时，不动余数满足 $2r \equiv c \pmod{P}$。若有 F 个不动余数，轨道数为 $(P + F)/2$。无其他关系时，以 $c/2$ 为中心的余弦可恰好区分这些反射对。

Practice / 配套练习: AIME-2000-I-Q12 (WB p. 210)

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 2 / 方法 2 [AL09-M02]

Extend a piecewise template by multiplicative scaling

按乘法缩放扩展分段模板

Procedure

1. Use $f(rx) = rf(x)$ to move any positive input into the defining interval $[1, r]$.

2. Write the exact piecewise formula on $[r^k, r^{k+1}]$ and check endpoint consistency.

3. For a target level, compare it with each scaled branch maximum before solving branches.

4. For a least input or a level-set length, combine the branch domains as well as the algebraic solutions.

通用步骤

1. 利用 $f(rx) = rf(x)$，将正输入缩放到定义模板区间 $[1, r]$。

2. 写出 $[r^k, r^{k+1}]$ 上的准确分段式，并检查端点一致。

3. 求给定水平时，先与各缩放分支最大值比较，再解方程。

4. 求最小输入或水平集长度时，必须将分支定义区间与代数解合并。

> Example. For $x > 0$, let $f(2x) = 2f(x)$ and $f(x) = \min(x - 1, 2 - x)$ on $[1, 2]$. Find the length of the set of $x \in [1, 8]$ with $f(x) \geq 1$.

> 例题：正数域函数满足 $f(2x) = 2f(x)$，在 $[1, 2]$ 上为 $\min(x - 1, 2 - x)$，求 $[1, 8]$ 内满足 $f(x) \geq 1$ 的集合长度。

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 2 / 方法 2 [AL09-M02] (continued / 续)

Worked solution

1. On $[1,2]$, the maximum is $1/2$, so there are no qualifying points.

2. On $[2,4]$, $f(x)=\min(x-2,4-x)$ reaches one only at $x=3$, a set of length zero.

3. On $[4,8]$, $f(x)=\min(x-4,8-x)$. The condition is $5\le x\le 7$, so the total length is two.

例题解析

1. 在 $[1,2]$ 上，最大值为二分之一，没有合格点。

2. 在 $[2,4]$ 上，$f(x)=\min(x-2,4-x)$，仅在三处达到一，集合长度为零。

3. 在 $[4,8]$ 上，$f(x)=\min(x-4,8-x)$，条件为五至七，故总长度为二。

A smaller scale may contain no solution even when the branch equation has formal roots.

较小尺度可能根本达不到目标水平，即使形式方程有解。

Practice / 配套练习: AIME-2001-II-Q08 (WB p. 206)

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 3 / 方法 3 [AL09-M03]

Expose finite-order composition and check all poles

找出有限阶复合并检查全部极点

Procedure

1. Compute a few exact compositions of a fractional-linear or affine map.

2. Prove a full composition returns the input, retaining every excluded denominator.

3. Reduce large iteration counts by this order only after proving the candidate orbit avoids the poles.

4. For affine complex maps $z \mapsto \omega z + c$, sum the geometric translation terms rather than tracking each point.

通用步骤

1. 对分式线性或仿射映射计算若干次精确复合。

2. 证明若干次复合回到输入，并保留每个禁用分母。

3. 只有证明候选轨道避开全部极点后，才能按阶约去大迭代次数。

4. 复仿射映射 $z \mapsto \omega z + c$ 应累加等比平移项，不必逐点追踪。

> Example. For $g(x) = 1/(1-x)$, determine all real inputs whose entire forward orbit exists, and the period of each such orbit.

> 例题：对 $g(x) = 1/(1-x)$，求所有完整正向轨道均有定义的实输入，并确定其周期。

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 3 / 方法 3 [AL09-M03] (continued / 续)

Worked solution

1. The pole is one. Its only real preimage is zero, so zero and one must both be excluded.

2. For every other input, the map produces neither zero nor one. Direct composition gives $g^2(x) = (x-1)/x$ and $g^3(x) = x$.

3. A real fixed point would solve $x^2 - x + 1 = 0$, which has negative discriminant. Thus the allowed domain is $\mathbb{R} \setminus \{0, 1\}$ and every orbit has exact period three.

例题解析

1. 极点为一，其唯一实原像为零，因此零、一均须排除。

2. 其余输入的像既不为零也不为一。直接复合得 $g^2(x) = (x-1)/x, g^3(x) = x$。

3. 实不动点须满足 $x^2 - x + 1 = 0$，但判别式为负。因此允许域为 $\mathbb{R} \setminus \{0, 1\}$，每轨道恰为三周期。

A simplified composition can hide intermediate poles; domain checks belong to the unsimplified orbit.

化简复合式可能遮住中途极点，定义域必须按实际轨道检查。

For an affine complex map, $F^N(z) = \omega^N z + c\sum_{j=0}^{N-1}\omega^j$. If $\omega^N = 1$ and $\omega \neq 1$, the translation sum vanishes, proving finite order for the entire map.

复仿射映射满足 $F^N(z) = \omega^N z + c\sum_{j=0}^{N-1}\omega^j$。若 $\omega^N = 1, \omega \neq 1$，则平移和为零，整个映射具有有限阶。

Practice / 配套练习: AIME-2002-I-Q12 (WB p. 87), AIME-2008-II-Q09 (WB p. 70), AIME-2012-II-Q11 (WB p. 214)

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 4 / 方法 4 [AL09-M04]

Use finite root orbits to force polynomial rigidity

用有限根轨道迫使多项式刚性

Procedure

1. Compare leading and constant coefficients of the polynomial identity.

2. Determine how every complex root generates further roots under the induced map.

3. Exclude any root modulus that would grow without bound in a finite root set.

4. Combine the product of root moduli and equality conditions to identify the remaining possible roots.

通用步骤

1. 先比较多项式恒等式的首项及常数项。

2. 确定每个复根在诱导映射下如何产生新根。

3. 根集有限，故排除会不断增大模长的根。

4. 结合根模乘积与等号条件，确定剩余可能根。

> Example. A nonempty finite set R of nonzero complex numbers is closed under squaring and satisfies $\prod_{z\in R}|z|=1$. Show every member is a root of unity.

> 例题：非零复数的非空有限集 R 在平方运算下封闭，且各元素模的乘积为一，证明每个元素都是单位根。

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 4 / 方法 4 [AL09-M04] (continued / 续)

Worked solution

1. If some $|z| > 1$, repeated squaring gives unbounded moduli inside the finite set, impossible. Thus all moduli are at most one.

2. Their product is one and none exceeds one, so every modulus equals one.

3. Finiteness gives $z^{2^j} = z^{2^k}$ for some $j < k$. Since $z \neq 0$, $z^{2^j(2^{k-j}-1)} = 1$, proving the claim.

例题解析

1. 若某元素模大于一，反复平方在有限集中产生无界模，矛盾，因此所有模至多一。

2. 各模乘积为一且均至多一，故每个模均为一。

3. 有限性给出某个 $j < k$ 满足 $z^{2^j} = z^{2^k}$。由 $z \neq 0$，得 $z^{2^j(2^{k-j}-1)} = 1$，即为单位根。

The root map must come from the actual identity; closure under a different transformation gives no information.

根映射必须由原恒等式推出，不能擅自使用另一种封闭变换。

For the map $T(z) = 2z^3 + z$, a modulus $r > 1$ gives $|T(z)| \geq 2r^3 - r > r$, forcing unbounded growth. If a monic polynomial has constant term one, the product of root moduli is one. Hence all root moduli equal one; requiring the image modulus also one gives $|2z^2 + 1| = 1$, forcing $z^2 = -1$.

对 $T(z) = 2z^3 + z$，若模 $r > 1$，则 $|T(z)| \geq 2r^3 - r > r$，会迫使无界增长。首一且常数项为一的多项式，根模乘积为一，所以所有根模均为一；像的模也为一进一步给出 $|2z^2 + 1| = 1$，迫使 $z^2 = -1$。

Practice / 配套练习: AIME-2007-II-Q14 (WB p. 211)

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 5 / 方法 5 [AL09-M05]

Use infinite integer descent to eliminate a nonzero difference

用无穷整数递降排除非零差

Procedure

1. Subtract adjacent cleared recurrences to isolate a sequence of integer differences.

2. Prove a nonzero difference forces the next difference nonzero but smaller in absolute value by a fixed factor.

3. An infinite sequence of positive integer magnitudes cannot decrease this way, so every difference must be zero.

4. Substitute the resulting periodicity back into the original recurrence and optimize only the remaining finite arithmetic choices.

通用步骤

1. 先清分母，再将相邻递推相减，分离整数差序列。

2. 证明非零差会迫使下一差也非零，但绝对值按固定比例缩小。

3. 正整数绝对值不能如此无穷递降，因此每个差都须为零。

4. 将所得周期代回原递推，再对剩余有限算术选择求最值。

> Example. An infinite integer sequence satisfies $b_n = (n+2)b_{n+1}$ for all $n \geq 0$. Prove it is identically zero.

> 例题：无穷整数序列满足 $b_n = (n+2)b_{n+1}$，$n \geq 0$，证明所有项均为零。

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 5 / 方法 5 [AL09-M05] (continued / 续)

Worked solution

1. Iteration gives $b_0 = (k+1)!b_k$ for every $k \geq 1$.

2. If $b_0 \neq 0$, every b_k is a nonzero integer, so $|b_0| \geq (k+1)!$ for every k, impossible.

3. Thus $b_0 = 0$. The nonzero multipliers then force successively $b_1 = b_2 = \cdots = 0$.

例题解析

1. 反复代入得每个 $k \geq 1$ 都有 $b_0 = (k+1)!b_k$。

2. 若 $b_0 \neq 0$，则各 b_k 都是非零整数，迫使 $|b_0| \geq (k+1)!$ 对所有 k 成立，矛盾。

3. 因此 $b_0 = 0$，由各乘数非零，依次推出后续全零。

An inequality allowing the difference to become zero is insufficient; exact propagation of nonzeroness is essential.

若不等式允许差突然变零，就不能这样递降；必须证明非零性被精确传递。

Practice / 配套练习: AIME-2009-I-Q13 (WB p. 212)

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 6 / 方法 6 [AL09-M06]

Normalize a radical recurrence into a folded angle map

将根式递推归一化为折叠角映射

Procedure

1. Remove an exponential scale so the radical becomes $\sqrt{1-u^2}$.

2. Choose a sine parameter $u=\sin\phi$ and write the coefficients as $\cos\theta,\sin\theta$.

3. Use $\sqrt{1-\sin^2\phi}=|\cos\phi|$, splitting at right-angle crossings.

4. Prove the eventual orbit or cycle, then restore the original scale.

通用步骤

1. 先除去指数尺度，使根式变为 $\sqrt{1-u^2}$。

2. 令 $u=\sin\phi$，将系数写成 $\cos\theta,\sin\theta$。

3. 必须使用 $\sqrt{1-\sin^2\phi}=|\cos\phi|$，在跨越直角时分类。

4. 证明最终轨道或循环，再恢复原尺度。

> Example. Let $u_0=0$ and $u_{n+1}=(\sqrt{3}/2)u_n+(1/2)\sqrt{1-u_n^2}$. Describe all terms from u_2 onward.

> 例题：令 $u_0=0,u_{n+1}=(\sqrt{3}/2)u_n+(1/2)\sqrt{1-u_n^2}$，描述从 u_2 起的全部项。

Worked solution

1. The first terms are $u_1=1/2,u_2=\sqrt{3}/2,u_3=1$, using the sine addition formula with angle 30°.

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 6 / 方法 6 [AL09-M06] (continued / 续)

2. At $u = 1$, the next value is $\sqrt{3}/2$. At $u = \sqrt{3}/2$, the positive radical is $1/2$, giving the next value one.

3. Thus the exact two-cycle $\sqrt{3}/2, 1$ starts at u_2. Continuing to increase a formal angle without the absolute cosine would give the wrong later terms.

例题解析

1. 用三十度正弦和角公式，前几项为 $1/2, \sqrt{3}/2, 1$。

2. 当 $u = 1$ 时，下项为 $\sqrt{3}/2$；当 $u = \sqrt{3}/2$ 时，正根式为二分之一，下项为一。

3. 因此从 u_2 起恰交替为 $\sqrt{3}/2, 1$。若忽略余弦绝对值而持续增加形式角，后续会算错。

A nonnegative radical is an absolute value, so angle addition may turn into angle subtraction.

非负根式对应绝对值，因此和角过程可能改为差角。

Practice / 配套练习: AIME-2009-II-Q14 (WB p. 213)

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 7 / 方法 7 [AL09-M07]

Count finite functions by intrinsic image and preimage levels

按内在像集与原像层次计数有限函数

Procedure

1. Translate a composition equality into restrictions on directed functional graphs.

2. Choose intrinsic classes such as fixed points, their direct preimages, and more distant points; the classes must be recoverable from the function.

3. Assign images only to their permitted classes, using independent powers after the classes are chosen.

4. If the image is prescribed exactly, require surjectivity rather than counting all maps into it.

通用步骤

1. 把复合等式转为有向函数图限制。

2. 选择能由函数自身恢复的类别，如不动点、其直接原像和更远点。

3. 选定类别后，只允许把像分配到指定层，再用独立幂计数。

4. 若像集须恰为某集合，必须要求满射，不能把所有映入函数都算入。

> Example. Count functions on a labeled three-element set satisfying $f^2 = f$.

> 例题：求固定标号三元集合上满足 $f^2 = f$ 的函数数。

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 7 / 方法 7 [AL09-M07] (continued / 续)

Worked solution

1. Every image point must be fixed, and every fixed point belongs to the image. Thus choosing the image is exactly choosing the fixed-point set.

2. If it has $k \geq 1$ elements, choose it in $\binom{3}{k}$ ways. Each remaining element maps independently to any of its k points.

3. The count is $\sum_{k=1}^{3} \binom{3}{k} k^{3-k} = 3 + 6 + 1 = 10$. The fixed-point set is intrinsic, so no function is counted twice.

例题解析

1. 每个像点都必须不动，而每个不动点都属于像，因此像集恰为不动点集。

2. 若大小为 $k \geq 1$，有 $\binom{3}{k}$ 种选择，其余点各可独立映入其中任一点。

3. 总数为 $\sum_{k=1}^{3} \binom{3}{k} k^{3-k} = 10$。不动点集由函数唯一确定，故不重复。

A chosen target set is not automatically the exact image set.

指定允许像集并不等于指定准确像集。

If f^2 is constant a, let $S = f^{-1}(a)$. Then $a \in S$, every point of S maps to a, and each outside point maps to $S \setminus \{a\}$. For an n-point domain, a size-k set contributes $n\binom{n-1}{k-1}(k-1)^{n-k}$.

If $f^3 = f^2$, every second image is fixed. Choose i fixed points and j nonfixed direct preimages; all remaining points map to that middle layer. The count for those sizes is $\binom{n}{i}\binom{n-i}{j} i^j j^{n-i-j}$, with $0^0 = 1$ for an empty assignment.

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 7 / 方法 7 [AL09-M07] (continued / 续)

The number of maps onto a specified k-set from an n-set is $\sum_{j=0}^{k}(-1)^j\binom{k}{j}(k-j)^n$. Multiply by the choice of the image set. A second map avoiding that image has $(n-k)^n$ choices.

若 f^2 为常值 a，令 $S=f^{-1}(a)$，则 $a\in S$，S 内各点映到 a，外点各映到 $S\setminus\{a\}$。在 n 元域中，大小 k 的贡献为 $n\binom{n-1}{k-1}(k-1)^{n-k}$。

若 $f^3=f^2$，每个第二像均不动。选 i 个不动点和 j 个非不动直接原像，其余点全映到中层；对应计数为 $\binom{n}{i}\binom{n-i}{j}i^j j^{n-i-j}$，空分配时约定 $0^0=1$。

从 n 元集满射到指定 k 元集的数目为 $\sum_{j=0}^{k}(-1)^j\binom{k}{j}(k-j)^n$，再乘像集选择数。第二个函数若避开该像集，则有 $(n-k)^n$ 种。

Practice / 配套练习: AIME-2013-II-Q11 (WB p. 215), AIME-2014-I-Q12 (WB p. 746), AIME-2018-II-Q10 (WB p. 207)

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 8 / 方法 8 [AL09-M08]

Factor forced roots and apply finite-difference rigidity

分离被迫根并使用有限差分刚性

Procedure

1. Substitute special inputs into a polynomial identity to force explicit roots.

2. Factor those roots before canceling common polynomial factors in the identity.

3. A nonconstant degree-d polynomial has a nonzero first difference of degree $d-1$, so a periodic polynomial is constant.

4. Use normalization to determine the remaining constant and verify the original identity.

通用步骤

1. 向多项式恒等式代入特殊值，找出被迫根。

2. 先分离这些根，再在恒等式中约去公共多项式因子。

3. 非常数 d 次多项式的一阶差分为非零的 $d-1$ 次式，因此周期多项式只能为常数。

4. 由归一条件确定余下常数，并验证原恒等式。

> Example. Find all real polynomials P with $P(x+1)-P(x)=2x+1$.

> 例题：求满足 $P(x+1)-P(x)=2x+1$ 的所有实系数多项式。

Worked solution

1. The polynomial x^2 has the required first difference.

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 8 / 方法 8 [AL09-M08] (continued / 续)

2. Put $Q(x) = P(x) - x^2$. Then $Q(x+1) - Q(x) = 0$, so Q is a periodic polynomial.

3. If Q were nonconstant, its first difference would have nonzero leading coefficient. Hence $Q = C$, and all solutions are $P = x^2 + C$.

例题解析

1. x^2 的一阶差分正好是 $2x + 1$。

2. 令 $Q(x) = P(x) - x^2$，则其差分为零，因此 Q 是周期多项式。

3. 若 Q 非常数，其差分首项系数非零，矛盾。因此 $Q = C$，全部解为 $P = x^2 + C$。

Cancel polynomial factors as identities; do not incorrectly exclude their zeros from the final polynomial.

约去因子是在多项式恒等式层面进行，不能把该因子的零点从最终多项式中排除。

Practice / 配套练习: AIME-2016-I-Q11 (WB p. 57)

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 9 / 方法 9 [AL09-M09]

Periodic information and uniqueness force a fixed residue

周期信息与唯一性迫使固定余数

Procedure

1. Encode the full infinite itinerary by signs, interval labels, or digits.

2. Shift by one full itinerary period and prove the transformed point satisfies the same complete information.

3. Use uniqueness to equate the transformed and original point, giving a fixed-point congruence.

4. Test every finite candidate against the entire repeating itinerary and its domain.

通用步骤

1. 以符号、区间标记或数字编码完整无穷轨迹。

2. 移动一个信息周期，证明变换后的点仍满足同一套完整信息。

3. 利用唯一性令变换后点等于原点，得到不动余数关系。

4. 将每个有限候选代回整个周期轨迹及定义域检查。

> Example. For $F(x)=\{3x\}$ on $[0,1)$, find the point whose labels $\lfloor 3F^n(x)\rfloor$ alternate $0,2,0,2,\ldots$.

> 例题：对 $[0,1)$ 上的 $F(x)=\{3x\}$，求使标记 $\lfloor 3F^n(x)\rfloor$ 交替为零、二、零、二的点。

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 9 / 方法 9 [AL09-M09] (continued / 续)

Worked solution

1. The labels give the ternary expansion $0.\overline{02}_3$. Its first $2k$ digits restrict x to an interval of length 9^{-k}, proving uniqueness.

2. After two shifts the same itinerary remains, so $F^2(x) = x$. Its two-digit integer part is two, hence $9x = 2 + x$ and $x = 1/4$.

3. The orbit is $1/4, 3/4, 1/4, \ldots$, whose labels are exactly zero and two, verifying existence.

例题解析

1. 标记给出三进制展开 $0.\overline{02}_3$，前 $2k$ 位将点限制在长 9^{-k} 的区间内，因此唯一。

2. 两次移位后信息不变，故 $F^2(x) = x$，两位整数部分为二，所以 $9x = 2 + x$，解得四分之一。

3. 轨道为四分之一、四分之三交替，标记恰为零、二，验证存在。

Uniqueness concerns the full infinite itinerary, not a short initial sign table.

唯一性针对完整无穷轨迹，不是有限开头的符号表。

Practice / 配套练习: AIME-2019-II-Q10 (WB p. 208)

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 10 / 方法 10 [AL09-M10]

Propagate preimages branch by branch and prove their pattern

逐分支传播原像并证明其模式

Procedure

1. Determine the exact range of each branch before taking an inverse.

2. For an absolute value retain only nonnegative targets; include both signs and remove coinciding preimages.

3. Use induction to prove an arithmetic zero pattern or nested interval-index rule.

4. Count or sum only after the branch-domain restrictions and duplicates have been settled.

通用步骤

1. 求原像前，先确定每个分支的准确值域。

2. 绝对值只接受非负目标，取正负两支，并去除重合原像。

3. 以归纳证明等差零点模式或嵌套区间下标规则。

4. 只有明确分支定义域及重复后，才能计数或求和。

> Example. On $[0,1]$, let $g(x)=|2x-1|$. Find all zeros of $g\circ g\circ g$.

> 例题：在 $[0,1]$ 上令 $g(x)=|2x-1|$，求三次复合的全部零点。

Worked solution

1. The preimage of zero is the single point $1/2$; the two formal signs coincide there.

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 10 / 方法 10 [AL09-M10] (continued / 续)

2. The preimages of $1/2$ are $1/4, 3/4$. For a target $t \in [0,1]$, its preimages are $(1-t)/2, (1+t)/2$.

3. Applying this to both targets gives $1/8, 3/8, 5/8, 7/8$. They are distinct and remain in the domain, so this is the complete zero set.

例题解析

1. 零的原像只有二分之一，此时形式上的正负分支重合。

2. 二分之一的原像为四分之一、四分之三；一般 $t \in [0,1]$ 的原像为 $(1-t)/2, (1+t)/2$。

3. 对两目标再求原像，得到八分之一、八分之三、八分之五、八分之七，互异且均在域内，故已穷尽。

A negative intermediate zero has no preimage under a nonnegative absolute-value map.

在非负绝对值映射下，负的中间零点没有原像。

For $f(x) = x\{x\}$ on $[j, j+1)$ with $j \geq 0$, write $x = j + t$. The expression $(j+t)t$ is strictly increasing with range $[0, j+1)$. Thus a positive target has exactly one preimage in each admissible unit interval. Repeated inverses produce weakly ordered interval indices, counted by binomial coefficients.

For $F_1(x) = |x-1|, F_n(x) = F_{n-1}(|x-n|)$, the zero set is the step-two progression from $n - T_{n-1}$ to T_n, where $T_n = n(n+1)/2$. Inductively the nonnegative preceding zeros include every value of their parity from zero or one to T_{n-1}; reflecting these as $n \pm z$ produces exactly the stated progression. Its endpoints sum to $2n$ and it has $T_{n-1} + 1$ points.

对 $j \geq 0$ 的区间 $[j, j+1)$，写 $x = j + t$，则 $x\{x\} = (j+t)t$ 严格递增，值域为 $[0, j+1)$。因此每个允许单位区间内，正目标恰有一个原像；反复求逆得到弱递增区间下标，可用组合数计数。

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 10 / 方法 10 [AL09-M10] (continued / 续)

若 $F_1(x)=|x-1|, F_n(x)=F_{n-1}(|x-n|)$，其零点为从 $n-T_{n-1}$ 到 T_n 的公差二数列，其中 $T_n=n(n+1)/2$。归纳时，前一层非负零点含从零或一到 T_{n-1} 的全部同奇偶值；反射为 $n\pm z$ 后恰填满该数列。两端和为 $2n$，点数为 $T_{n-1}+1$。

Practice / 配套练习: AIME-2020-II-Q08 (WB p. 209), AIME-2020-II-Q14 (WB p. 178)

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 11 / 方法 11 [AL09-M11]

Unroll a recursive function to its first reachable terminal input

展开递归直到首个可达终止输入

Procedure

1. Identify the terminal set and the arithmetic progression of inputs visited by the recursion.

2. Prove a terminal point is reachable and choose the first one in that progression, not merely the nearest terminal numerically.

3. Add the cost of each recursive step to the terminal value.

4. For a minimum input, combine the terminal formula with its defining interval and residue constraints.

通用步骤

1. 确定终止集合，以及递归访问的输入等差数列。

2. 证明终止点可达，选择该等差数列中的首个终止点，而非数值最近的任意终止点。

3. 将每次递归的代价加上终止值。

4. 若求最小输入，将终止公式与其定义区间、余数限制合并。

> Example. For positive integers define $H(n) = 0$ if $5 \mid n$, and $H(n) = 1 + H(n+2)$ otherwise. Find $H(7)$.

> 例题：正整数上定义 $H(n)$：若五整除 n 则为零，否则为 $1 + H(n+2)$，求 $H(7)$。

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 11 / 方法 11 [AL09-M11] (continued / 续)

Worked solution

1. The visited inputs are $7+2j$. Because two is invertible modulo five, a multiple of five is reachable.

2. The first nonnegative solution of $7+2j \equiv 0 \pmod 5$ is $j=4$, giving input fifteen.

3. The four preceding calls each add one, so $H(7)=4$. The closer multiple ten is not visited and must not be used.

例题解析

1. 访问输入为 $7+2j$，因二在模五下可逆，必能到达五的倍数。

2. 同余式的首个非负解为 $j=4$，终止输入十五。

3. 此前四次调用各加一，所以结果为四；较近的十不在访问序列中，不能用它。

A square reached by step two must have the input parity; the first square above the input may be skipped.

步长二只能到达与输入同奇偶的平方，紧邻上方的平方可能被跳过。

Let $k=\lceil\sqrt{n}\rceil$. Unit-step recursion first reaches k^2. A step-two recursion first reaches k^2 if $n \equiv k \pmod 2$, otherwise $(k+1)^2$. With cost two per step, its accumulated cost equals terminal square minus n. Always restore $(k-1)^2 < n \le k^2$ before minimizing.

令 $k=\lceil\sqrt{n}\rceil$，步长一首先到 k^2；步长二若 $n \equiv k \pmod 2$ 也到 k^2，否则首先到 $(k+1)^2$。若每步代价为二，累计代价恰为终止平方减去 n。求最小值时必须恢复 $(k-1)^2 < n \le k^2$。

Practice / 配套练习: AIME-2021-II-Q15 (WB p. 216)

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 12 / 方法 12 [AL09-M12]

Conjugate a rational map to a simpler recurrence

将有理映射共轭为较简单递推

Procedure

1. Search for a fractional-linear variable that sends special points or fixed points to simple values.

2. Derive the transformed recurrence exactly and invert the substitution.

3. Track numerator and denominator before and after reduction; use gcd identities to certify the claimed denominator growth.

4. Prove the transformed orbit remains in the interval that gives valid positive original values before taking modular residues.

通用步骤

1. 寻找将特殊点或不动点送到简单数值的分式线性变量。

2. 精确推出变换后的递推，并写出逆变换。

3. 跟踪约分前后的分子分母，用最大公因数关系确认所称分母增长。

4. 取同余前，证明变换轨道留在使原变量为合法正值的区间内。

> Example. Let $x_0 = 3$ and $x_{n+1} = (x_n^2 + 1)/(2x_n)$. Find an exact reduced formula for x_n.

> 例题：令 $x_0 = 3, x_{n+1} = (x_n^2 + 1)/(2x_n)$，求 x_n 的精确既约表达式。

Section 1.9 Functional constraints & iteration

函数约束与迭代

Method 12 / 方法 12 [AL09-M12] (continued / 续)

Worked solution

1. Put $y = (x-1)/(x+1)$. Direct simplification gives $y_{n+1} = y_n^2$ and $y_0 = 1/2$.

2. Thus $y_n = 2^{-2^n}$, always in $(0,1)$, and inversion gives $x_n = (2^{2^n}+1)/(2^{2^n}-1)$.

3. The two numerator/denominator terms differ by two and are both odd. Their gcd is therefore one, so the formula is reduced and positive.

例题解析

1. 令 $y = (x-1)/(x+1)$，直接化简得 $y_{n+1} = y_n^2, y_0 = 1/2$。

2. 因此 $y_n = 2^{-2^n}$ 始终在零一之间，逆变换给出 $x_n = (2^{2^n}+1)/(2^{2^n}-1)$。

3. 分子分母相差二且均奇，最大公因数只能为一，所以该式已既约且为正。

A denominator recurrence is not proved until cancellation has been ruled out.

只有排除约分后，分母递推才算证明。

For a transformed map $y \mapsto y(y-1)$ with $y = c/d$ in lowest terms, the unreduced next fraction is $c(c-d)/d^2$. If $\gcd(c,d) = 1$, both factors in the numerator are coprime to d, so the next denominator is exactly d^2. When inverting a substitution, analyze any new common divisor separately.

若变换映射为 $y \mapsto y(y-1)$，既约 $y = c/d$ 的下一式为 $c(c-d)/d^2$。由 $\gcd(c,d) = 1$，分子两因子都与 d 互质，故下一分母准确为 d^2。逆变换若引入新的公因子，还须另行核查。

Practice / 配套练习: AIME-2025-II-Q13 (WB p. 217)

Section 1.10 Binomial expansions & controlled approximation

二项式展开与误差控制

Method 1 / 方法 1 [AL10-M01]

Extract only the binomial terms the question actually needs. Integer exponents give exact coefficients; fractional exponents require an explicit remainder bound before a decimal or integer part is certified.

只提取题目真正需要的二项式项。整数指数给出精确系数；分数指数则必须先给余项界，才能确定小数位或整数部分。

Compare adjacent coefficients by cancellation

消去公因子比较相邻系数

Procedure

1. In $(ax+b)^n$, write $[x^k]=\binom{n}{k}a^k b^{n-k}$.

2. For consecutive coefficients, divide before expanding: their ratio is $(n-k)a/((k+1)b)$.

3. Keep signs and nonzero conditions; if integer parameters are required, reduce the ratio and apply coprimality.

通用步骤

1. 在 $(ax+b)^n$ 中，$[x^k]=\binom{n}{k}a^k b^{n-k}$。

2. 比较相邻系数时先相除，得到比值 $(n-k)a/((k+1)b)$。

3. 保留符号及非零条件；若参数为整数，约去公因子后使用互质性。

> Example. Find the ratio of the coefficient of x^4 to that of x^3 in $(1-2x)^9$.

> 例题：求 $(1-2x)^9$ 中 x^4 系数与 x^3 系数之比。

Section 1.10 Binomial expansions & controlled approximation

二项式展开与误差控制

Method 1 / 方法 1 [AL10-M01] (continued / 续)

Worked solution

1. The two coefficients are $\binom{9}{4}(-2)^4$ and $\binom{9}{3}(-2)^3$.

2. Their ratio is $((9-3)/4)(-2) = -3$.

3. The negative sign is essential: the even-degree coefficient is positive and the odd-degree coefficient is negative.

例题解析

1. 两系数分别为 $\binom{9}{4}(-2)^4$ 与 $\binom{9}{3}(-2)^3$。

2. 相除得 $((9-3)/4)(-2) = -3$。

3. 负号不可忽略：偶数次系数为正，奇数次系数为负。

A coefficient equality gives a ratio of parameters, not their individual values.

系数相等先给出参数比，不能直接确定各参数。

Practice / 配套练习: AIME-2000-I-Q03 (WB p. 218)

Section 1.10 Binomial expansions & controlled approximation

二项式展开与误差控制

Method 2 / 方法 2 [AL10-M02]

Find the actual degree after leading cancellations

首项抵消后确认实际次数

Procedure

1. Expand from the highest powers downward and combine coefficients before applying a degree-dependent theorem.

2. Continue until the first nonzero coefficient; also extract the next coefficient if Vieta is needed.

3. Use the actual degree and leading coefficient, counting complex roots with multiplicity.

通用步骤

1. 从最高次幂向下展开，合并系数后再使用依赖次数的定理。

2. 直到找到第一个非零系数；若要使用韦达关系，还须提取下一系数。

3. 使用实际次数及首项系数，并按重数计算复根。

> Example. Choose c so $(x+1)^8-(x-1)^8-cx^7$ has degree below seven. Find its actual degree and leading coefficient.

> 例题：选择 c，使 $(x+1)^8-(x-1)^8-cx^7$ 次数小于七，求实际次数及首项系数。

Worked solution

1. The x^8 terms cancel. The x^7 coefficient is $8-(-8)=16$, so choose $c=16$.

2. All even powers cancel in the difference. The next possible power is x^5.

Section 1.10 Binomial expansions & controlled approximation

二项式展开与误差控制

Method 2 / 方法 2 [AL10-M02] (continued / 续)

3. Its coefficient is $2\binom{8}{3} = 112 \neq 0$, so the degree is five and the leading coefficient is 112.

例题解析

1. x^8 项抵消，x^7 系数为 $8-(-8)=16$，故取 $c=16$。

2. 差式中所有偶次幂均抵消，下一可能非零项为 x^5。

3. 其系数为 $2\binom{8}{3} = 112 \neq 0$，所以实际次数为五，首项系数为 112。

Applying Vieta before locating the first surviving term changes the answer.

尚未确定首个保留项就用韦达公式，会使用错误次数。

Practice / 配套练习: AIME-2001-I-Q03 (WB p. 32)

Section 1.10 Binomial expansions & controlled approximation

二项式展开与误差控制

Method 3 / 方法 3 [AL10-M03]

Bound a fractional-power remainder before extracting digits

提取数字前先控制分数幂余项

Procedure

1. Write $f(T+1)=T^r+rT^{r-1}+R$ with T large and positive.

2. Taylor remainder gives $R=f''(\xi)/2$ for some $T<\xi<T+1$, where $f''(t)=r(r-1)t^{r-2}$; use its sign and endpoint bounds.

3. Separate the integer and fractional parts of the retained terms, using modular arithmetic when needed.

4. Compare the full remainder bound with the distance to both neighboring decimal or integer boundaries.

通用步骤

1. 对正的大数 T，写成 $f(T+1)=T^r+rT^{r-1}+R$。

2. 泰勒余项为某个 $T<\xi<T+1$ 处的 $R=f''(\xi)/2$，其中 $f''(t)=r(r-1)t^{r-2}$；用其符号与端点估界。

3. 分离保留项的整数及小数部分，必要时用同余求小数部分。

4. 将全部余项界与相邻两个小数或整数边界的距离比较。

> Example. Find $\lfloor 100(10001)^{3/4}\rfloor$ without decimal approximation.

> 例题：不用小数近似，求 $\lfloor 100(10001)^{3/4}\rfloor$。

Section 1.10 Binomial expansions & controlled approximation

二项式展开与误差控制

Method 3 / 方法 3 [AL10-M03] (continued / 续)

Worked solution

1. Take $T = 10000, r = 3/4$. The first two terms are $1000 + 3/40 = 1000.075$.

2. The second derivative is negative, and $|R| < (3/32)T^{-5/4} = 3/3200000 < 10^{-6}$.

3. Thus $100007.4999 < 100(10001)^{3/4} < 100007.5$. Both bounds have floor 100007, so that is the exact result.

例题解析

1. 取 $T = 10000, r = 3/4$，前两项为 $1000 + 3/40 = 1000.075$。

2. 二阶导数为负，且 $|R| < (3/32)T^{-5/4} = 3/3200000 < 10^{-6}$。

3. 所以 $100007.4999 < 100(10001)^{3/4} < 100007.5$，上下界的整数部分均为 100007，故结果精确确定。

A tiny error is useful only after the nearest digit boundary has been checked.

只有检查了最近的数字边界，“误差很小”才足以确定答案。

The second-order remainder bound follows by integrating f'' twice: $R = \int_T^{T+1} (T + 1 - t) f''(t)\, dt$. The nonnegative weight has integral $1/2$, giving the same endpoint bounds without an infinite-series convergence assumption.

二阶余项界也可由两次积分得到：$R = \int_T^{T+1} (T + 1 - t) f''(t)\, dt$。非负权重的积分为 $1/2$，所以端点界直接给出余项界，不必假定无穷级数收敛。

Practice / 配套练习: AIME-2002-I-Q07 (WB p. 219)

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

Method 1 / 方法 1 [GE01-M01]

Similarity is a statement about ordered correspondence, not a picture's appearance. Prove two angles, name matching vertices and choose a directed scale before transferring lengths, areas or volumes. Parallel cuts and reflections often create the needed triangles.

相似是有序对应关系，不是图形看起来相似。先证明两角、写明对应顶点并选择有向比例，再转移长度、面积或体积；平行截线与反射常能产生所需三角形。

Parallel cuts turn heights into linear scales and areas into squares

平行截线将高度转为线性比、面积转为平方比

Procedure

1. Locate the shared apex and prove the cut triangle is similar by parallel-line angles.

2. Its linear scale equals the ratio of heights measured from that apex.

3. Use the square of scale for area and cube for similar-solid volume; subtract complements only afterward.

4. For equal-area bands, cumulative area determines a square-root length scale, not equal length spacing.

通用步骤

1. 找到公共顶点，由平行线角证明截出三角形相似。

2. 线性相似比等于从该顶点量起的高之比。

3. 面积用比例平方，相似立体体积用立方；之后再扣补集。

4. 等面积条带应由累积面积求平方根长度比例，不是等距截长。

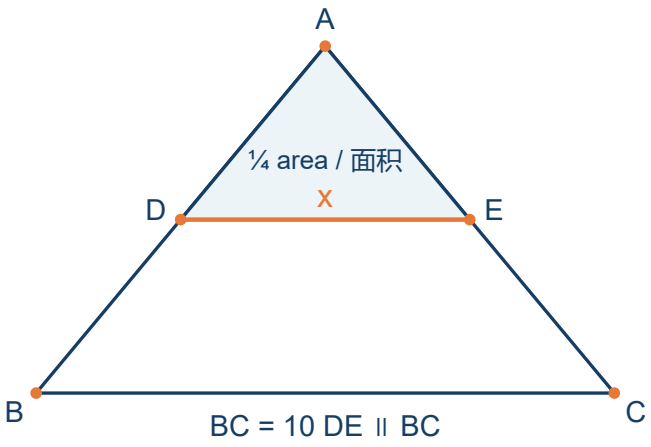
Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

Method 1 / 方法 1 [GE01-M01] (continued / 续)

> Example. A triangle has base ten. A parallel chord cuts off one quarter of its area at the apex. Find the chord length.

> 例题：三角形底边十，一条平行截线在顶端截出原面积的四分之一，求截段长。



Worked solution

1. The small and original triangles are similar, with positive linear scale t.

2. Their area ratio gives $t^2 = 1/4$, so $t = 1/2$.

3. The chord is $10t = 5$. The remaining trapezoid has three quarters of the original area.

例题解析

1. 小三角形与原者相似，正线性比例设为 t。

2. 由面积比 $t^2 = 1/4$ 得 $t = 1/2$。

3. 截段为 $10t = 5$，余梯形面积占原来的四分之三。

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

Method 1 / 方法 1 [GE01-M01] (continued / 续)

Equal-height strips usually have unequal areas.

等高条带通常并不等面积。

Extend trapezoid legs to their common apex. Equal area on the two sides of an intermediate parallel section of length x gives $x^2 = (a^2 + b^2)/2$, where a, b are the two base lengths.

A half-turn about a triangle centroid leaves three exterior corner triangles of linear scale $1/3$, so the union area is $4/3$ of the original.

For a rectangle of width w inscribed above a triangle base of length b and altitude h, similarity gives rectangle height $h(1 - w/b)$.

延长梯形两腰至公共顶点。中间平行截段长 x 将梯形等面积分开时，若底长为 a, b，则 $x^2 = (a^2 + b^2)/2$。

三角形绕重心半周旋转，产生三个线性比例 $1/3$ 的外角三角形，故并集面积是原来的 $4/3$。

三角形底长 b、高 h，内部贴底矩形宽为 w 时，相似给出矩形高 $h(1 - w/b)$。

Practice / 配套练习: AIME-2000-I-Q08 (WB p. 444), AIME-2000-II-Q06 (WB p. 246), AIME-2001-I-Q07 (WB p. 224), AIME-2001-II-Q07 (WB p. 313), AIME-2003-II-Q06 (WB p. 248), AIME-2004-I-Q11 (WB p. 453), AIME-2006-I-Q07 (WB p. 249), AIME-2008-I-Q02 (WB p. 241), AIME-2010-I-Q13 (WB p. 255), AIME-2011-I-Q08 (WB p. 251), AIME-2012-I-Q08 (WB p. 447), AIME-2015-II-Q07 (WB p. 226), AIME-2018-II-Q07 (WB p. 227)

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

Method 2 / 方法 2 [GE01-M02]

Ordered similar triangles transfer ratios and secant products

有序相似三角形转移比例与割线乘积

Procedure

1. Write a named similarity in matching vertex order, justified by two angles.

2. List corresponding sides before choosing a ratio; crossed diagrams can reverse the visual order.

3. Multiply equal ratios to produce a length product when an intermediate side is unknown.

4. For cyclic or tangent configurations, use directed angles first, then verify the positive ray order.

通用步骤

1. 以对应顶点顺序写出相似，并用两个角证明。

2. 选比例前先列对应边；交叉图形可能颠倒视觉次序。

3. 中间边未知时，可相乘比例得到长度乘积。

4. 圆或切线构型先用有向角，再核实正射线点序。

> Example. Triangles ABC, DEF are similar in that order. If $(AB, BC, CA) = (6, 9, 12)$ and $DE = 4$, find EF, FD and their area ratio.

> 例题：ABC, DEF 按所写顺序相似，$(AB, BC, CA) = (6, 9, 12)$ 且 $DE = 4$，求 EF, FD 及面积比。

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

Method 2 / 方法 2 [GE01-M02] (continued / 续)

Worked solution

1. The correspondence is $AB \leftrightarrow DE$, $BC \leftrightarrow EF$, $CA \leftrightarrow FD$.

2. The scale from the first to second is $4/6 = 2/3$, giving $EF = 6, FD = 8$.

3. The second-to-first area ratio is $(2/3)^2 = 4/9$. All three side ratios agree.

例题解析

1. 对应关系为 $AB \leftrightarrow DE, BC \leftrightarrow EF, CA \leftrightarrow FD$。

2. 从前者到后者的比例为 $4/6 = 2/3$，故 $EF = 6, FD = 8$。

3. 后者对前者面积比为 $4/9$，三组边比均一致。

Writing only “the triangles are similar” can hide the crucial correspondence.

只写三角形相似，可能掩盖关键对应关系。

For a right-triangle altitude foot D on hypotenuse AB, similarity gives $AD \cdot AB = AC^2$, $BD \cdot AB = BC^2$ and $CD^2 = AD \cdot BD$.

For an infinite geometric area construction, first identify the retained linear scale and square it; target pieces may require subtracting two similar areas before summing.

直角三角形高的垂足 D 在斜边 AB 上时，相似给出 $AD \cdot AB = AC^2, BD \cdot AB = BC^2, CD^2 = AD \cdot BD$。

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

Method 2 / 方法 2 [GE01-M02] (continued / 续)

无穷等比面积构造应先确定保留的线性比例并平方；每个目标块可能需先作两相似面积之差，再求和。

Practice / 配套练习: AIME-2001-II-Q13 (WB p. 230), AIME-2002-II-Q14 (WB p. 340), AIME-2006-II-Q12 (WB p. 233), AIME-2007-I-Q12 (WB p. 253), AIME-2007-I-Q15 (WB p. 234), AIME-2013-I-Q13 (WB p. 235), AIME-2016-I-Q06 (WB p. 316), AIME-2016-I-Q15 (WB p. 299), AIME-2016-II-Q10 (WB p. 287), AIME-2019-I-Q06 (WB p. 228), AIME-2019-II-Q15 (WB p. 304), AIME-2023-II-Q03 (WB p. 266), AIME-2015-I-Q06 (WB p. 286)

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

Method 3 / 方法 3 [GE01-M03]

Extend parallel sides to reveal a homothety

延长平行边发现位似

Procedure

1. Extend the relevant sides until triangles share an apex or an auxiliary parallelogram appears.

2. Use parallel sides to identify corresponding angles and the homothety center.

3. Keep the scale signed when the image lies across the center; length and area use its absolute value and square.

4. For three parallel-to-side lines, affine or barycentric level equations give the signed scale directly.

通用步骤

1. 延长相关边，使三角形共享顶点或出现辅助平行四边形。

2. 由平行边确定对应角及位似中心。

3. 像在中心另一侧时保留负位似比，长度用绝对值、面积用平方。

4. 三条分别平行于三边的直线，可用仿射或重心坐标水平式直接求有向比例。

> Example. In triangle ABC, choose $D \in AB, E \in AC$ with $AD/AB = AE/AC = 2/5$. Reflect D, E in point A to $D\prime, E\prime$. Find the signed homothety scale from ABC to $AD\prime E\prime$ and their area ratio.

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

Method 3 / 方法 3 [GE01-M03] (continued / 续)

> 例题：三角形中，$D \in AB, E \in AC$ 满足 $AD/AB = AE/AC = 2/5$。将 D, E 关于点 A 作中心反射得 $D\prime, E\prime$。求从 ABC 到 $AD\prime E\prime$ 的有向位似比及面积比。

Worked solution

1. The equal fractions and shared angle give $ADE \sim ABC$, with $DE \parallel BC$ and scale $2/5$.

2. With A as vector origin, $D\prime = -2B/5$ and $E\prime = -2C/5$. Thus the signed homothety scale is $-2/5$: both images lie on the opposite rays.

3. Ordinary area uses the square, so $[AD\prime E\prime]/[ABC] = 4/25$. The negative length scale does not make area negative.

例题解析

1. 两边比例相等且夹角共用，得 $ADE \sim ABC$，且 $DE \parallel BC$，线性比例为 $2/5$。

2. 以 A 为向量原点，则 $D\prime = -2B/5, E\prime = -2C/5$，有向位似比为 $-2/5$，两个像点均在反向射线上。

3. 普通面积取比例平方，故 $[AD\prime E\prime]/[ABC] = 4/25$；负线性比例不会产生负的普通面积。

Parallel-sided triangles may be oppositely oriented.

各边平行的两个三角形可能方向相反。

For barycentric coordinates $\alpha + \beta + \gamma = 1$, lines $\alpha = a, \beta = b, \gamma = c$ form a triangle similar to the reference triangle with absolute ratio $|1 - a - b - c|$.

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

Method 3 / 方法 3 [GE01-M03] (continued / 续)

Common external tangents of unequal circles meet at their external homothety center; center distances have the radius ratio.

If three parallel corner cuts leave a six-equal-sided hexagon inside a triangle of sides a, b, c, its common side s satisfies $1 = s(1/a + 1/b + 1/c)$. On the side of length a, the two removed lengths are as/b and as/c, leaving the middle length s; divide their sum by a.

在重心坐标 $\alpha + \beta + \gamma = 1$ 中，三线 $\alpha = a, \beta = b, \gamma = c$ 所围三角形对参考三角形的绝对相似比为 $|1 - a - b - c|$。

两不等圆公外切线交于外位似中心，到两圆心距离比等于半径比。

若三条平行截角线在边长为 a, b, c 的三角形内留下六边等长的六边形，其公共边长 s 满足 $1 = s(1/a + 1/b + 1/c)$。沿长为 a 的边，两端切去段为 $as/b, as/c$，中央保留 s；将三段之和除以 a 即得。

Practice / 配套练习: AIME-2002-II-Q13 (WB p. 231), AIME-2004-II-Q13 (WB p. 232), AIME-2008-II-Q05 (WB p. 220), AIME-2009-I-Q05 (WB p. 221), AIME-2019-II-Q07 (WB p. 229), AIME-2022-II-Q07 (WB p. 337), AIME-2024-II-Q05 (WB p. 223)

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

Method 4 / 方法 4 [GE01-M04]

Equal lengths and polygon angle sums determine the orientation

等长与多边形角和确定方向

Procedure

1. Mark equal sides and assign the matching base angles in every isosceles triangle.

2. Use the polygon interior sum or exterior turns to determine the remaining angles.

3. An obtuse angle in an isosceles triangle must be its apex; use this to decide which sides are equal.

4. Check positive side portions and convexity before accepting a trigonometric candidate.

通用步骤

1. 标记等边，并给各等腰三角形分配相等底角。

2. 利用多边形内角和或外转角求余角。

3. 等腰三角形中的钝角必为顶角，可据此决定相等的两边。

4. 接受三角方程候选前，检查分段正性及凸性。

> Example. An isosceles triangle has one angle 120°. Determine all its angles and the ratio of its base to an equal side.

> 例题：等腰三角形有一角为 120°，求三角及底边与腰的比。

Worked solution

1. The 120° angle cannot be a base angle, since two such angles exceed 180°.

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

Method 4 / 方法 4 [GE01-M04] (continued / 续)

2. The other two angles are $(180-120)/2=30^\circ$.

3. For equal side a, cosine law gives base squared $2a^2-2a^2\cos 120^\circ=3a^2$, so the ratio is $\sqrt{3}$.

例题解析

1. 120° 不可能是底角，否则两个这样的角超过 180°。

2. 另两角均为 30°。

3. 腰长为 a 时，余弦定理给出底边平方 $3a^2$，故比例为 $\sqrt{3}$。

Do not assign the two equal sides from a sketch alone.

不能仅凭草图决定哪两边相等。

Practice / 配套练习: AIME-2000-I-Q14 (WB p. 272), AIME-2006-II-Q01 (WB p. 240), AIME-2007-II-Q03 (WB p. 259), AIME-2011-II-Q03 (WB p. 131), AIME-2013-I-Q12 (WB p. 398), AIME-2021-II-Q02 (WB p. 244), AIME-2025-II-Q05 (WB p. 282)

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

Method 5 / 方法 5 [GE01-M05]

Bisector reflections expose midpoint lines

角平分线反射显露中点线

Procedure

1. Reflect a vertex across a known internal angle bisector to place it on the other side ray at the same distance.

2. Combine two such reflections with equal-distance or midpoint information.

3. Use the midpoint theorem or an isosceles median to convert the reflected geometry into a parallel segment.

通用步骤

1. 关于已知内角平分线反射顶点，使其落在另一边射线上且到顶点距离不变。

2. 把两次反射与等距或中点信息结合。

3. 用中点定理或等腰中线，将反射构型化为平行线段。

> Example. In triangle ABC, $AB = AC$ and D is the midpoint of BC. Prove AD is both an angle bisector and perpendicular to BC.

> 例题：三角形中 $AB = AC$，D 为 BC 中点，证明 AD 既平分角又垂直于 BC。

Worked solution

1. Triangles ABD, ACD have equal sides $AB = AC$, $BD = DC$ and common AD.

2. SSS congruence gives equal angles at A and equal adjacent angles ADB, ADC.

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

Method 5 / 方法 5 [GE01-M05] (continued / 续)

3. The latter sum to 180°, so each is 90°; hence both conclusions hold.

例题解析

1. ABD, ACD 有 $AB = AC, BD = DC$ 及公共边 AD。

2. SSS 全等给出 A 处两角相等，以及相邻的 ADB, ADC 相等。

3. 后两角和为 180°，故各为 90°，两结论均成立。

An angle bisector becomes a median only after an isosceles relation is proved.

只有证明等腰后，角平分线才能直接视为中线。

Practice / 配套练习: AIME-2011-I-Q04 (WB p. 310), AIME-2022-I-Q03 (WB p. 312)

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

Method 6 / 方法 6 [GE01-M06]

Similar pieces must fit the entire original shapes

相似切块必须容纳于整个原图形

Procedure

1. List all possible side correspondences and cut orientations compatible with the required angles.

2. Express every cut length or altitude in a common linear scale.

3. Enforce the full original side lengths and pairwise non-overlap inequalities, rather than one visible constraint.

4. Construct the equality case to show the best candidate is attainable.

通用步骤

1. 列出符合角条件的全部边对应及切割方向。

2. 把每条截长或高写为共同线性比例。

3. 施加原图全部边长及两两不重叠不等式，不能只检查一个明显限制。

4. 构造等号情形，证明最优候选可达。

> Example. A 3–4–5 right triangle is placed axis-aligned in a 4×5 rectangle. What is the largest scale if its legs must lie along rectangle sides?

> 例题：三、四、五直角三角形两直角边沿长方形边方向放入四乘五矩形，求最大线性比例。

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

Method 6 / 方法 6 [GE01-M06] (continued / 续)

Worked solution

1. With the three-leg along the four-side and four-leg along the five-side, $t \leq \min(4/3, 5/4) = 5/4$.

2. The exchanged orientation requires $t \leq \min(4/4, 5/3) = 1$.

3. Thus the maximum is $5/4$, attained by legs $15/4$ and five inside the rectangle.

例题解析

1. 三长边沿四边、四长边沿五边时，$t \leq \min(4/3, 5/4) = 5/4$。

2. 交换方向时，$t \leq \min(4/4, 5/3) = 1$。

3. 故最大比例为 $5/4$，直角边 $15/4$ 与五确能放入矩形。

A local angle match does not guarantee global containment.

局部角度匹配并不保证整体容纳。

Practice / 配套练习: AIME-2004-I-Q09 (WB p. 225), AIME-2011-I-Q08 (WB p. 251), AIME-2013-I-Q12 (WB p. 398)

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

Method 7 / 方法 7 [GE01-M07]

Iterated right-triangle altitudes have a constant similarity ratio

迭代直角三角形高具有固定相似比

Procedure

1. Identify the fixed acute angle shared by every new right triangle.

2. Use AA similarity to find the linear ratio from one altitude to the next.

3. Compute the first altitude from leg product divided by hypotenuse.

4. Sum the resulting geometric progression, checking the ratio lies strictly between zero and one.

通用步骤

1. 找出各新直角三角形共享的固定锐角。

2. 由 AA 相似求相继高的线性比例。

3. 第一条高由两直角边乘积除以斜边求得。

4. 求所得等比级数和，并检验公比严格介于零与一。

> Example. Let ABC_0 be right at C_0, with $BC_0 = 3, AC_0 = 4$. Let C_1 be the foot from C_0 to AB; for $n \geq 2$, let C_n be the foot from C_{n-1} to line $C_{n-2}B$. Find $\sum_{n\geq 1} C_{n-1}C_n$.

> 例题：直角三角形 ABC_0 在 C_0 为直角，$BC_0 = 3, AC_0 = 4$。C_1 为 C_0 到 AB 的垂足；对 $n \geq 2$，C_n 为 C_{n-1} 到直线 $C_{n-2}B$ 的垂足。求 $\sum_{n\geq 1} C_{n-1}C_n$。

Section 2.1 Angles, similarity & parallel lines

角度、相似与平行线

Method 7 / 方法 7 [GE01-M07] (continued / 续)

Worked solution

1. The first altitude is $3 \cdot 4/5 = 12/5$.

2. The shared-angle similarity ratio is $3/5$, so subsequent altitudes are multiplied by $3/5$.

3. The total is $(12/5)/(1 - 3/5) = 6$, and convergence follows from $3/5 < 1$.

例题解析

1. 第一高为 $3 \cdot 4/5 = 12/5$。

2. 共享角相似比为 $3/5$，以后每条高乘此比例。

3. 总和为 $(12/5)/(1 - 3/5) = 6$，且因公比小于一而收敛。

The first altitude and the ratio come from different parts of the original triangle.

第一高与公比来自原三角形的不同度量关系。

Practice / 配套练习: AIME-2016-II-Q05 (WB p. 222)

Section 2.2 Areas & area ratios

面积与面积比

Method 1 / 方法 1 [GE02-M01]

Area often eliminates an unknown height or point. Use common heights for ratios, squared scales for similarity, determinants for coordinates and sectors for circular pieces. Preserve signed areas during algebra, then assemble the ordinary region without overlap.

面积常能消去未知高度或点。等高用比例，相似用平方比，坐标用行列式，圆块用扇形。代数中保留有向面积，最后再无重叠地拼成普通区域。

Common-height ratios and corner subtraction eliminate shape

等高比例与扣角消去形状

Procedure

1. Triangles with a common altitude have area ratio equal to their base ratio.

2. With a common included angle, area ratio is the product of the two adjacent-side ratios.

3. For an inner triangle, subtract three corner triangles using consistent cyclic side fractions.

4. Only cancel heights after verifying the bases lie on the intended common line.

通用步骤

1. 共高三角形面积比等于底边比。

2. 共夹角三角形面积比等于两邻边比例的乘积。

3. 内部三角形可通过一致的循环边分数扣除三个角三角形。

4. 先确认各底在所需同一直线上，再消去高度。

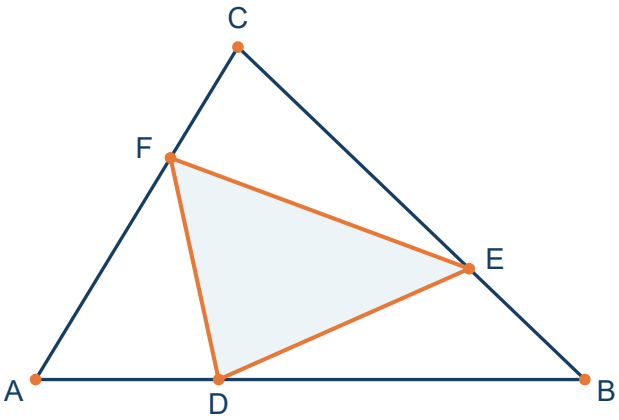
> Example. Points D, E, F lie respectively one third of the way along AB, BC, CA. Find $[DEF]/[ABC]$.

Section 2.2 Areas & area ratios

面积与面积比

Method 1 / 方法 1 [GE02-M01] (continued / 续)

> 例题：D, E, F 分别位于沿 AB, BC, CA 前进三分之一处，求 $[DEF]/[ABC]$。



AD/AB = BE/BC = CF/CA = ⅓

Worked solution

1. At A, the corner triangle has side fractions $1/3$ on AB and $2/3$ on AC, so its area fraction is $2/9$.

2. The other two corner fractions are also $2/9$. Their interiors are disjoint.

3. The inner fraction is $1 - 3(2/9) = 1/3$.

例题解析

1. A 角三角形在 AB, AC 上的边比例为 $1/3, 2/3$，面积占比为 $2/9$。

2. 另两角面积占比同为 $2/9$，且内部不交。

3. 内部比例为 $1 - 3(2/9) = 1/3$。

For cyclic fractions p, q, r, the inner area fraction is $1 - p - q - r + pq + qr + rp$.

Section 2.2 Areas & area ratios

面积与面积比

Method 1 / 方法 1 [GE02-M01] (continued / 续)

循环边分数为 p, q, r 时，内部面积占比为 $1 - p - q - r + pq + qr + rp$。

Practice / 配套练习: AIME-2001-I-Q09 (WB p. 247), AIME-2002-I-Q10 (WB p. 314), AIME-2002-I-Q13 (WB p. 293), AIME-2004-II-Q13 (WB p. 232), AIME-2007-I-Q15 (WB p. 234), AIME-2010-I-Q15 (WB p. 320), AIME-2013-II-Q05 (WB p. 260), AIME-2015-II-Q15 (WB p. 345), AIME-2017-II-Q10 (WB p. 377), AIME-2018-II-Q12 (WB p. 280)

Section 2.2 Areas & area ratios

面积与面积比

Method 2 / 方法 2 [GE02-M02]

Similarity squares linear scales before areas are subtracted

相似先平方线性比再扣面积

Procedure

1. Prove the triangles are similar and determine their positive linear scale.

2. Square that scale for area; keep the whole and complementary pieces separate.

3. For a sequence of parallel bands, compare cumulative area from the same apex.

4. For an infinite construction, identify the area ratio before summing the geometric series.

通用步骤

1. 先证明相似并求正线性比例。

2. 面积取该比例平方，分别记录整体与补块。

3. 平行条带应比较从同一顶点量起的累积面积。

4. 无穷构造先确定面积公比，再求等比级数。

> Example. A triangle has area 72. Two parallel chords lie at one half and one third of its altitude measured from the apex. Find the area of the band between them.

> 例题：一个三角形面积为72。两条平行截线从顶点量起的高度，分别是原高的一半与三分之一。求两截线之间条带的面积。

Worked solution

1. The two apex triangles have linear scales $1/2$ and $1/3$, since all three bases are parallel.

Section 2.2 Areas & area ratios

面积与面积比

Method 2 / 方法 2 [GE02-M02] (continued / 续)

2. Their areas are $72(1/2)^2 = 18$ and $72(1/3)^2 = 8$.

3. The smaller apex triangle is contained in the larger, so the band area is $18 - 8 = 10$, or $5/36$ of the original area.

例题解析

1. 三底平行，所以两个顶端小三角形的线性比例分别为 $1/2, 1/3$。

2. 两面积分别为 $72(1/2)^2 = 18$ 与 $72(1/3)^2 = 8$。

3. 较小顶端三角形包含在较大者内，故条带面积为 $18 - 8 = 10$，即原面积的 $5/36$。

Half the height does not generally mean half the area.

一半高度通常不对应一半面积。

For a trapezoid of parallel bases $0 < a < b$, extend its legs to an apex. A parallel chord x bisects the trapezoid area exactly when $x^2 - a^2 = b^2 - x^2$, hence $x = \sqrt{(a^2 + b^2)/2}$. The positive root lies between the two bases.

两底满足 $0 < a < b$ 的梯形，延长两腰至公共顶点。平行截段 x 平分面积，等价于 $x^2 - a^2 = b^2 - x^2$，所以 $x = \sqrt{(a^2 + b^2)/2}$，正根介于两底之间。

Practice / 配套练习: AIME-2000-II-Q06 (WB p. 246), AIME-2003-II-Q06 (WB p. 248), AIME-2004-I-Q09 (WB p. 225), AIME-2006-I-Q07 (WB p. 249), AIME-2006-II-Q12 (WB p. 233), AIME-2007-I-Q12 (WB p. 253), AIME-2008-I-Q02 (WB p. 241), AIME-2010-I-Q13 (WB p. 255), AIME-2010-II-Q09 (WB p. 372), AIME-2011-I-Q08 (WB p. 251), AIME-2013-I-Q13 (WB p. 235), AIME-2015-I-Q07 (WB p. 375), AIME-2021-II-Q02 (WB p. 244)

Section 2.2 Areas & area ratios

面积与面积比

Method 3 / 方法 3 [GE02-M03]

Determinants measure simple polygons and spatial triangles

行列式度量简单多边形与空间三角形

Procedure

1. List a simple polygon’s vertices in boundary order before using the shoelace formula.

2. Compute twice the signed area as the sum of consecutive cross products, taking absolute value only at the end.

3. For a spatial triangle, use half the norm of the two side-vector cross product.

4. For a self-intersecting boundary, split its bounded regions and add ordinary areas instead of taking one global shoelace absolute value.

通用步骤

1. 用鞋带公式前，按边界顺序列出简单多边形顶点。

2. 相邻点叉积和为两倍有向面积，最后才取绝对值。

3. 空间三角形用两边向量叉积长度的一半。

4. 自交边界应分割各有界区域并加普通面积，不能对全局鞋带结果直接取绝对值。

> Example. Find the area of the concave polygon $(0,0),(4,0),(4,1),(1,1),(1,3),(0,3)$.

> 例题：求凹多边形 $(0,0),(4,0),(4,1),(1,1),(1,3),(0,3)$ 的面积。

Worked solution

1. The consecutive cross products are $0,4,3,2,3,0$, summing to twelve.

Section 2.2 Areas & area ratios

面积与面积比

Method 3 / 方法 3 [GE02-M03] (continued / 续)

2. The area is $|12|/2 = 6$.

3. As a check, subtract a 3×2 notch from a 4×3 rectangle: $12 - 6 = 6$.

例题解析

1. 相邻叉积依次为 $0, 4, 3, 2, 3, 0$，总和十二。

2. 面积为 $|12|/2 = 6$。

3. 也可从四乘三矩形扣三乘二缺口，得到 $12 - 6 = 6$。

The signed area of a crossed polygon may cancel bounded lobes.

交叉多边形的有向面积可能把部分有界叶片相互抵消。

For lattice points on $ax + y = N$, triangle area with the origin is $N|i - j|/2$ after parameterizing points by their integer x coordinates. Integrality can therefore become a parity condition.

An affine transformation multiplies every signed area by the same determinant. Normalize a parallelogram to the unit square when only area ratios are needed.

将 $ax + y = N$ 上格点按整数横坐标编号后，与原点组成的面积为 $N|i - j|/2$，面积整性因而可能转化为奇偶条件。

仿射变换把全部有向面积乘同一个行列式。只求面积比时，可把平行四边形规范为单位正方形。

Section 2.2 Areas & area ratios

面积与面积比

Method 3 / 方法 3 [GE02-M03] (continued / 续)

Practice / 配套练习: AIME-2000-I-Q13 (WB p. 384), AIME-2003-II-Q14 (WB p. 389), AIME-2005-I-Q10 (WB p. 368), AIME-2009-I-Q11 (WB p. 254), AIME-2013-I-Q07 (WB p. 373), AIME-2013-II-Q13 (WB p. 399), AIME-2014-I-Q13 (WB p. 400), AIME-2015-I-Q04 (WB p. 357), AIME-2017-II-Q03 (WB p. 767), AIME-2017-II-Q10 (WB p. 377), AIME-2018-II-Q04 (WB p. 358), AIME-2018-II-Q09 (WB p. 379), AIME-2019-II-Q01 (WB p. 359), AIME-2025-I-Q02 (WB p. 245), AIME-2025-I-Q12 (WB p. 413), AIME-2025-II-Q01 (WB p. 364), AIME-2026-I-Q10 (WB p. 383), AIME-2026-II-Q03 (WB p. 365)

Section 2.2 Areas & area ratios

面积与面积比

Method 4 / 方法 4 [GE02-M04]

Sectors, segments and annuli assemble circular area

扇形、弓形及圆环拼成圆面积

Procedure

1. A circular segment is a sector minus its central triangle; specify the minor or complementary part.

2. Annulus area is $\pi(R^2 - r^2)$, so consecutive radius squares often telescope or pair.

3. For a symmetric disk union, assign each angular wedge to the disk reaching farthest along its rays.

4. Confirm all subtractions or unions are disjoint except at boundaries.

通用步骤

1. 圆弓形为扇形减圆心三角形，先明确劣弓形或补集。

2. 圆环面积为 $\pi(R^2 - r^2)$，连续半径平方常可配对或消项。

3. 对称圆盘并集可按射线伸得最远的圆盘分配角扇区。

4. 核实各扣块或拼块除边界外互不重叠。

> Example. A chord in a radius-two circle is distance one from its center. Find the smaller circular segment area.

> 例题：半径二的圆中，一弦距圆心为一，求较小弓形面积。

Section 2.2 Areas & area ratios

面积与面积比

Method 4 / 方法 4 [GE02-M04] (continued / 续)

Worked solution

1. The half central angle has cosine $1/2$, so the full angle is 120°.

2. The sector area is $4\pi/3$, and its isosceles central triangle has area $\frac{1}{2}(2)(2)\sin 120^\circ = \sqrt{3}$.

3. The smaller segment area is $4\pi/3 - \sqrt{3}$.

例题解析

1. 中心半角余弦为 $1/2$，所以中心角为 120°。

2. 扇形面积为 $4\pi/3$，圆心等腰三角形面积为 $\frac{1}{2}(2)(2)\sin 120^\circ = \sqrt{3}$。

3. 小弓形面积为 $4\pi/3 - \sqrt{3}$。

The sum of disk areas counts overlaps repeatedly.

直接相加各圆盘面积会重复计算重叠。

Six unit disks centered at sixth roots of unity have a union radial boundary $2\cos\theta$ on each 60° wedge. Each wedge area is $\pi/3 + \sqrt{3}/2$, obtained from a 120° unit sector and two triangles.

六个以六次单位根为圆心的单位圆盘，其并集在每个 60° 扇区内的径向边界为 $2\cos\theta$。每区面积为 $\pi/3 + \sqrt{3}/2$，可分为一个 120° 单位扇形与两个三角形。

Practice / 配套练习: AIME-2003-I-Q02 (WB p. 237), AIME-2004-I-Q04 (WB p. 238), AIME-2004-II-Q01 (WB p. 239), AIME-2005-I-Q01 (WB p. 329), AIME-2008-II-Q13 (WB p. 90), AIME-2010-I-Q13 (WB p. 255), AIME-2015-I-Q15 (WB p. 459), AIME-2026-I-Q03 (WB p. 424)

Section 2.2 Areas & area ratios

面积与面积比

Method 5 / 方法 5 [GE02-M05]

Regular polygons and hinged shapes reduce to triangle pieces

正多边形与铰接图形化为三角形块

Procedure

1. Decompose a regular polygon, rhombus or trapezoid into triangles with known angles or common heights.

2. Use $ab\sin\theta/2$ for each piece and count cell rings before multiplying a cell area.

3. For a hinged deformation recover the perpendicular height from the rigid bar lengths.

4. If the shape varies, state its full angle interval so strict endpoints and attainable areas are distinguished.

通用步骤

1. 将正多边形、菱形或梯形分成已知角或共高三角形。

2. 每块用 $ab\sin\theta/2$，蜂窝环先计格数再乘单位格面积。

3. 铰接形变先由固定杆长求垂直高度。

4. 活动构型应列完整角区间，区分严格端点与可达面积。

> Example. A rhombus has side two and acute angle $0<\theta<90^\circ$. Which positive integer areas are attainable?

> 例题：菱形边长二，锐角满足 $0<\theta<90^\circ$，哪些正整数面积可达?

Section 2.2 Areas & area ratios

面积与面积比

Method 5 / 方法 5 [GE02-M05] (continued / 续)

Worked solution

1. Its area is $4\sin\theta$. Since sine increases continuously from zero to one on this open interval, area ranges over $(0, 4)$.

2. The positive integers in that interval are one, two and three.

3. Each is attained by $\theta = \arcsin(K/4)$; area four requires the excluded right angle.

例题解析

1. 面积为 $4\sin\theta$。正弦在该开区间连续严格增，从而面积遍历 $(0, 4)$。

2. 其中正整数为一、二、三。

3. 各值均由 $\theta = \arcsin(K/4)$ 实现，面积四需要被排除的直角。

An endpoint square may be excluded even when approached arbitrarily closely.

即使能任意接近正方形，端点构型仍可能被排除。

A regular unit-side hexagon has area $3\sqrt{3}/2$. A cluster's jth ring contains $6j$ cells; sum rings before subtracting an outer paved layer.

In a symmetric five-rhombus assembly with fixed outer area C and outer meeting angle α, a straight-angle condition can make the central area $2C\cos\alpha$. Check the entire nondegenerate angle range before counting integer areas.

An equilateral triangle has area $\sqrt{3}s^2/4$. A trapezoid with parallel bases a, b and perpendicular height h has area $(a+b)h/2$; do not use a slanted leg as height.

Section 2.2 Areas & area ratios

面积与面积比

Method 5 / 方法 5 [GE02-M05] (continued / 续)

边长一的正六边形面积为 $3\sqrt{3}/2$。蜂窝簇第 j 环有 $6j$ 格，应先求环格总数，再扣最外层。

对称五菱形拼图中，若外块面积固定为 C、外块相接角为 α，平角条件可能使中央面积为 $2C\cos\alpha$；计整数面积前须检查全部非退化角范围。

等边三角形面积为 $\sqrt{3}s^2/4$。两平行底为 a, b、垂直高为 h 的梯形面积为 $(a+b)h/2$，不能以斜腰代替高度。

Practice / 配套练习: AIME-2002-II-Q04 (WB p. 236), AIME-2006-I-Q08 (WB p. 250), AIME-2006-II-Q01 (WB p. 240), AIME-2012-I-Q13 (WB p. 396), AIME-2013-I-Q03 (WB p. 14), AIME-2013-I-Q12 (WB p. 398), AIME-2014-II-Q03 (WB p. 242), AIME-2014-II-Q10 (WB p. 78), AIME-2019-I-Q03 (WB p. 243), AIME-2021-I-Q09 (WB p. 380), AIME-2023-II-Q09 (WB p. 288), AIME-2025-I-Q06 (WB p. 290), AIME-2025-II-Q06 (WB p. 339), AIME-2025-II-Q14 (WB p. 256)

Section 2.2 Areas & area ratios

面积与面积比

Method 6 / 方法 6 [GE02-M06]

Choose between sine area, Heron and inradius area

在正弦面积、海伦与内切圆面积中选择

Procedure

1. Use $K = ab\sin C/2$ when two sides and their included angle are known.

2. Use Heron for three sides, or $K = rs$ when inradius and semiperimeter are available.

3. Equate formulas to eliminate an unknown angle, radius or scale.

4. A fixed sine usually has acute and obtuse angle branches; test both against the original conditions.

通用步骤

1. 两边及夹角已知时用 $K = ab\sin C/2$。

2. 三边已知用海伦，内切圆半径与半周长已知用 $K = rs$。

3. 联立公式消去未知角、半径或尺度。

4. 正弦值固定通常有锐角与钝角两分支，应按原条件逐一检验。

> Example. Two triangle sides are three and five and the area is six. Find both possible third-side lengths.

> 例题：三角形两边为三与五，面积为六，求第三边的两种可能长度。

Worked solution

1. The included angle satisfies $6 = (15/2)\sin C$, so $\sin C = 4/5$.

Section 2.2 Areas & area ratios

面积与面积比

Method 6 / 方法 6 [GE02-M06] (continued / 续)

2. The two possible cosines are $3/5, -3/5$. Cosine law gives third-side squares $34 - 30(3/5) = 16$ and $34 + 18 = 52$.

3. The possible lengths are 4 and $2\sqrt{13}$, both satisfying the strict triangle inequalities with three and five.

例题解析

1. 夹角满足 $6 = (15/2)\sin C$，故 $\sin C = 4/5$。

2. 余弦可为 $3/5, -3/5$，余弦定理给出第三边平方为 $16, 52$。

3. 所以第三边为 4 或 $2\sqrt{13}$，两者均与三、五满足严格三角不等式。

Equal area and two fixed sides do not necessarily give congruent triangles.

同面积及两固定边，不一定确定全等三角形。

Practice / 配套练习: AIME-2005-I-Q15 (WB p. 294), AIME-2006-II-Q15 (WB p. 275), AIME-2009-I-Q12 (WB p. 319), AIME-2010-II-Q12 (WB p. 278), AIME-2020-II-Q13 (WB p. 347), AIME-2021-II-Q05 (WB p. 265), AIME-2024-I-Q14 (WB p. 462), AIME-2024-II-Q10 (WB p. 317)

Section 2.2 Areas & area ratios

面积与面积比

Method 7 / 方法 7 [GE02-M07]

Area differences are linear normal projections

面积差是法向线性投影

Procedure

1. Orient a fixed base and write its signed triangle area as half base length times signed perpendicular distance.

2. When a circular arc is fixed, its region area differs from the central sector by this linear projection term.

3. Add normal-vector identities to eliminate an unknown point.

4. For an interior point of an equilateral triangle, the three positive side distances sum to its altitude.

通用步骤

1. 给固定底边定向，将有向三角面积写为底长一半乘有向垂距。

2. 固定圆弧时，相应区域面积与圆心扇形的差就是这个线性投影项。

3. 把法向量恒等式相加以消去未知点。

4. 等边三角形内部点到三边的正距离和等于三角形高。

> Example. An equilateral triangle has altitude six. An interior point has distances one and two to two sides. Find its distance to the third side.

> 例题：等边三角形高六，内部点到其中两边距离为一、二，求到第三边的距离。

Section 2.2 Areas & area ratios

面积与面积比

Method 7 / 方法 7 [GE02-M07] (continued / 续)

Worked solution

1. Join the point to all three vertices. The three smaller triangles have a common base length equal to the equilateral side a.

2. Area addition gives $a(d_1 + d_2 + d_3)/2 = a \cdot 6/2$.

3. Hence $d_3 = 6 - 1 - 2 = 3$. Positivity agrees with the interior-point condition.

例题解析

1. 连点与三顶点，三个小三角形可分别以等边长 a 为底。

2. 面积相加给出 $a(d_1 + d_2 + d_3)/2 = a \cdot 6/2$。

3. 故 $d_3 = 3$，其为正，符合内部点条件。

Outside the triangle, use signed distances rather than three positive distances.

点在三角形外时，应使用有向距离，而不是三个正距离。

For a regular polygon in a circle, let u_j be the outward unit normal toward side j and a its side length. A fixed-arc region at P has area equal to its central sector minus $(a/2)P \cdot u_j$. Linear relations among normals give exact relations among region areas.

圆内正多边形第 j 边的外法向单位向量为 u_j、边长为 a 时，点 P 对应固定弧区域面积等于圆心扇形面积减 $(a/2)P \cdot u_j$。法向量的线性关系因而给出区域面积的精确关系。

Practice / 配套练习: AIME-2019-II-Q13 (WB p. 407), AIME-2023-I-Q12 (WB p. 411)

Section 2.2 Areas & area ratios

面积与面积比

Method 8 / 方法 8 [GE02-M08]

Diagonals encode quadrilateral area and angle

对角线编码四边形面积及夹角

Procedure

1. For convex $ABCD$, orient the full diagonals as $u = \overrightarrow{AC}$ and $v = \overrightarrow{BD}$; then $K = |u \times v|/2$.

2. The side-square difference satisfies $BC^2 + DA^2 - AB^2 - CD^2 = 2u \cdot v$.

3. Divide cross and dot magnitudes to obtain the acute diagonal-angle tangent.

4. In a cyclic quadrilateral, Ptolemy and Brahmagupta supply the diagonal product and area separately.

通用步骤

1. 对凸四边形 $ABCD$，将整条对角线定向为 $u = \overrightarrow{AC}, v = \overrightarrow{BD}$，则 $K = |u \times v|/2$。

2. 边长平方差满足 $BC^2 + DA^2 - AB^2 - CD^2 = 2u \cdot v$。

3. 以叉积、点积绝对值之比得到对角线锐夹角的正切。

4. 共圆时，托勒密与婆罗摩笈多可分别提供对角线乘积及面积。

> Example. A convex quadrilateral has diagonals six and eight meeting at 30°. Find its area and the magnitude of their dot product.

> 例题：凸四边形对角线长六、八，夹角为 30°，求面积及点积绝对值。

Section 2.2 Areas & area ratios

面积与面积比

Method 8 / 方法 8 [GE02-M08] (continued / 续)

Worked solution

1. The diagonal cross-product magnitude is $6 \cdot 8 \sin 30^\circ = 24$.

2. Area is half that, 12.

3. The dot-product magnitude is $48 \cos 30^\circ = 24\sqrt{3}$, checking the angle tangent $24/(24\sqrt{3}) = 1/\sqrt{3}$.

例题解析

1. 对角向量叉积长度为 $6 \cdot 8 \sin 30^\circ = 24$。

2. 面积为其一半，即十二。

3. 点积绝对值为 $24\sqrt{3}$，两者之比 $1/\sqrt{3}$ 核实夹角。

Use full diagonals, not just the two pieces next to their intersection.

应使用整条对角线，不是交点旁的两个分段。

Practice / 配套练习: AIME-2018-I-Q15 (WB p. 300), AIME-2018-II-Q12 (WB p. 280), AIME-2021-I-Q11 (WB p. 306), AIME-2021-II-Q12 (WB p. 410)

Section 2.2 Areas & area ratios

面积与面积比

Method 9 / 方法 9 [GE02-M09]

Projection factors convert planar strips into spatial section areas

投影系数将平面条带化为空间截面积

Procedure

1. Locate every section vertex or curved boundary before projecting.

2. Compute the projected polygon or disk-strip area exactly.

3. For section plane $z = px + qy + c$, multiply projected area by $\sqrt{1 + p^2 + q^2}$; equivalently divide by the normal-angle cosine.

通用步骤

1. 投影前先定位全部截面顶点或曲边。

2. 精确求投影多边形或圆盘条带面积。

3. 平面 $z = px + qy + c$ 的截面积等于投影面积乘 $\sqrt{1 + p^2 + q^2}$，即除以法向夹角余弦。

> Example. Find the area of the patch $z = 2x$ above $0 \le x \le 3, 0 \le y \le 2$.

> 例题：求平面 $z = 2x$ 在 $0 \le x \le 3, 0 \le y \le 2$ 上方截片的面积。

Worked solution

1. The horizontal projection is a three-by-two rectangle, area six.

2. The plane has $p = 2, q = 0$, so the enlargement factor is $\sqrt{1 + 4} = \sqrt{5}$.

3. The patch area is $6\sqrt{5}$, also checked by adjacent edge vectors $(3, 0, 6), (0, 2, 0)$.

Section 2.2 Areas & area ratios

面积与面积比

Method 9 / 方法 9 [GE02-M09] (continued / 续)

例题解析

1. 水平投影为三乘二矩形，面积六。

2. 平面有 $p = 2, q = 0$，放大系数为 $\sqrt{5}$。

3. 截片面积为 $6\sqrt{5}$，亦可由相邻边向量 $(3, 0, 6), (0, 2, 0)$ 核实。

A finite cylinder may project to a clipped strip rather than a full disk.

有限圆柱截面可能投影为裁剪条带，而非整圆盘。

Practice / 配套练习: AIME-2007-I-Q13 (WB p. 456), AIME-2015-I-Q15 (WB p. 459)

Section 2.2 Areas & area ratios

面积与面积比

Method 10 / 方法 10 [GE02-M10]

Area bounds require a matching geometric construction

面积界必须配套几何构造

Procedure

1. Express the target as a constant times a triangle area or sine of one angle.

2. Use the area bound for fixed adjacent sides, or a containment comparison of altitudes.

3. Identify the exact equality angle and prove it lies in the original feasible family.

通用步骤

1. 把目标写成某三角面积或一个角正弦的常数倍。

2. 对固定邻边使用面积界，或用容纳关系比较高。

3. 明确等号角，并证明原可行构型能达到它。

> Example. Points P, Q lie on concentric circles of radii two and three. Find the greatest area of triangle OPQ.

> 例题：P, Q 分别位于圆心为 O、半径二与三的同心圆上，求 OPQ 最大面积。

Worked solution

1. Area is $\frac{1}{2}(2)(3)|\sin\angle POQ| = 3|\sin\angle POQ|$.

2. This is at most three.

3. Choose perpendicular radii OP, OQ; both points remain on their circles, so area three is attainable.

Section 2.2 Areas & area ratios

面积与面积比

Method 10 / 方法 10 [GE02-M10] (continued / 续)

例题解析

1. 面积为 $3|\sin\angle POQ|$。

2. 因此面积至多三。

3. 取两半径垂直，两点仍位于各自圆上，故面积三确能达到。

An equality angle may be excluded by a segment or non-overlap condition.

等号角可能被线段或不重叠条件排除。

Practice / 配套练习: AIME-2006-I-Q08 (WB p. 250), AIME-2011-I-Q08 (WB p. 251), AIME-2013-II-Q10 (WB p. 252)

Section 2.2 Areas & area ratios

面积与面积比

Method 11 / 方法 11 [GE02-M11]

Central symmetry pairs equal area on opposite sides of a line

中心对称把直线两侧面积配对

Procedure

1. Identify pieces exchanged by a half-turn about a point on the proposed bisector.

2. The half-turn maps one open side of the line to the other and preserves area.

3. Handle remaining uncut pieces by equal-area pairing, then add the disjoint contributions.

4. For overlap polygons, intersect every boundary half-plane before measuring the result.

通用步骤

1. 找出关于拟议平分线上一点半周旋转后互换的部分。

2. 半周旋转交换直线两开半平面并保持面积。

3. 其余未切块按等面积配对，再相加互不重叠的贡献。

4. 重叠多边形须先相交全部边界半平面，再测量所得区域。

> Example. Two unit disks are centered at $(-1,0),(1,0)$. Show every line through the origin bisects the area of their union.

> 例题：两个单位圆盘圆心为 $(-1,0),(1,0)$，证明任何过原点直线都平分它们并集的面积。

Worked solution

1. A half-turn about the origin exchanges the two disks and preserves their union.

Section 2.2 Areas & area ratios

面积与面积比

Method 11 / 方法 11 [GE02-M11] (continued / 续)

2. For any line through the origin, the same half-turn exchanges its two sides and preserves area.

3. The line itself has area zero, so each side receives half the union area, namely π.

例题解析

1. 关于原点半周旋转交换两圆盘，并保持并集。

2. 对任一过原点直线，同一旋转交换两侧且保持面积。

3. 直线本身面积零，所以两侧各占并集一半，即 π。

A line can bisect a pair’s total area without bisecting each component.

一条线可以平分两个部件的总面积，而不分别平分每个部件。

Practice / 配套练习: AIME-2006-I-Q10 (WB p. 369), AIME-2021-I-Q02 (WB p. 361)

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

Method 1 / 方法 1 [GE03-M01]

Triangle metrics connect geometry to exact algebra. Choose a right section, the sine rule, the cosine rule or an area identity according to the known data. Every algebraic solution must still satisfy triangle inequalities, angle ranges and the stated side-segment conditions.

三角形度量把几何转为精确代数。依据已知条件选择直角截面、正弦定理、余弦定理或面积恒等式。所有代数解仍须满足三角不等式、角范围及题目规定的边上线段限制。

Right sections and special triangles recover perpendicular heights

直角截面与特殊三角形恢复垂直高度

Procedure

1. Drop a perpendicular or choose a symmetry-axis cross section.

2. Use Pythagoras, the $1:1:\sqrt{2}$ or $1:\sqrt{3}:2$ triangle, and scaled Pythagorean triples.

3. Keep a slant length separate from its perpendicular projection.

4. If two distances share an unknown height, subtract their squared equations before solving.

通用步骤

1. 作垂线或选择对称轴截面。

2. 使用勾股、$1:1:\sqrt{2}$ 或 $1:\sqrt{3}:2$ 比例，以及缩放勾股数组。

3. 把斜长与垂直投影分开。

4. 若两个距离共用未知高度，先相减其平方方程。

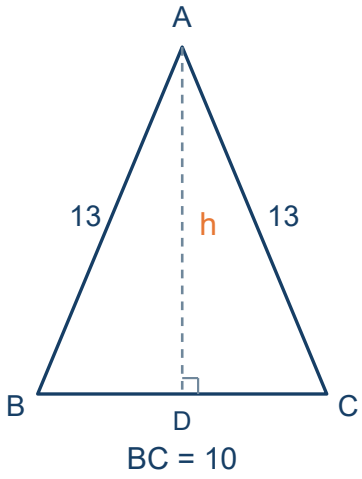
> Example. An isosceles triangle has equal sides thirteen and base ten. Find its altitude and area.

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

Method 1 / 方法 1 [GE03-M01] (continued / 续)

> 例题：等腰三角形腰十三、底十，求高与面积。



Worked solution

1. The apex altitude bisects the base into two lengths five.

2. Pythagoras gives height $\sqrt{13^2 - 5^2} = 12$.

3. The area is $10 \cdot 12/2 = 60$; the two right sections are scaled 5–12–13 triangles.

例题解析

1. 顶角高把底边平分为两个五。

2. 勾股得高 $\sqrt{13^2 - 5^2} = 12$。

3. 面积为 $10 \cdot 12/2 = 60$，两截块均为五、十二、十三直角三角形。

A three-dimensional height must be perpendicular to the base plane.

空间高度必须垂直于底面。

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

Method 1 / 方法 1 [GE03-M01] (continued / 续)

Equal chords have center-to-chord distance $\sqrt{R^2 - c^2/4}$. A sphere section uses the same right triangle with sphere radius and section radius.

In a tilted square or cube section, establish the relevant projection or diagonal factor first; equal spatial edge lengths do not imply equal vertical heights.

An isosceles triangle with equal sides r and base angle α has base $2r\cos\alpha$. In a right-angle wedge, establish the angular order of two internal equal-length rays before subtracting their equilateral60-degree separation.

长度为 c 的弦距圆心为 $\sqrt{R^2 - c^2/4}$。球截面使用球半径与截圆半径组成的同类直角三角形。

倾斜正方形或立方体截面中，应先确定投影或对角线系数；空间棱等长并不意味着竖直高度相等。

腰为 r、底角为 α 的等腰三角形底边为 $2r\cos\alpha$。在直角域中，先明确两条等长内部射线的次序，再减去等边三角形的六十度夹角。

Practice / 配套练习: AIME-2001-I-Q04 (WB p. 257), AIME-2001-I-Q05 (WB p. 415), AIME-2002-I-Q02 (WB p. 328), AIME-2003-I-Q07 (WB p. 543), AIME-2003-II-Q11 (WB p. 388), AIME-2004-II-Q07 (WB p. 367), AIME-2006-I-Q01 (WB p. 258), AIME-2006-II-Q06 (WB p. 370), AIME-2007-II-Q03 (WB p. 259), AIME-2007-II-Q09 (WB p. 315), AIME-2008-I-Q05 (WB p. 440), AIME-2008-I-Q15 (WB p. 457), AIME-2008-II-Q05 (WB p. 220), AIME-2012-I-Q14 (WB p. 92), AIME-2012-II-Q04 (WB p. 12), AIME-2012-II-Q05 (WB p. 441), AIME-2014-I-Q01 (WB p. 261), AIME-2014-II-Q03 (WB p. 242), AIME-2014-II-Q11 (WB p. 401), AIME-2015-I-Q15 (WB p. 459), AIME-2015-II-Q04 (WB p. 262), AIME-2015-II-Q09 (WB p. 448), AIME-2016-I-Q04 (WB p. 442), AIME-2017-I-Q04 (WB p. 443), AIME-2019-I-Q03 (WB p. 243), AIME-2019-I-Q06 (WB p. 228), AIME-2019-II-Q15 (WB p. 304), AIME-2021-II-Q10 (WB p. 427), AIME-2024-II-Q15 (WB p. 706), AIME-2025-I-Q06 (WB p. 290), AIME-2025-II-Q14 (WB p. 256)

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

Method 2 / 方法 2 [GE03-M02]

Sine and cosine laws eliminate a shared side or angle

正弦与余弦定理消去公共边或角

Procedure

1. Name the same angle or diagonal consistently across all subtriangles.

2. Use $a/\sin A = b/\sin B$ for side–angle pairs and $a^2 = b^2 + c^2 - 2bc\cos A$ for included-angle data.

3. Subtract two cosine-law equations to cancel the common diagonal; factor only after retaining any excluded equal-side case.

4. Check the original side order, positive lengths and angle interval after solving.

通用步骤

1. 在各子三角形中一致命名公共角或对角线。

2. 边角对使用 $a/\sin A = b/\sin B$，夹角资料使用 $a^2 = b^2 + c^2 - 2bc\cos A$。

3. 两余弦定理相减可消公共对角线；因式分解除法前保留被排除的等边情形。

4. 求解后检查原边次序、正长及角区间。

> Example. Two triangles share a diagonal. Their adjacent side pairs are $(5, u)$ and $(5, v)$ with the same included angle θ, where $u \neq v$ and $u + v = 6$. Find $\cos\theta$.

> 例题：两个三角形共用一条对角线，邻边分别为 $(5, u), (5, v)$，夹角同为 θ，且 $u \neq v, u + v = 6$，求 $\cos\theta$。

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

Method 2 / 方法 2 [GE03-M02] (continued / 续)

Worked solution

1. Equal diagonal squares give $25 + u^2 - 10u\cos\theta = 25 + v^2 - 10v\cos\theta$.

2. Factor to $(u - v)(u + v - 10\cos\theta) = 0$. Since $u \neq v$, the second factor is zero.

3. Thus $\cos\theta = 6/10 = 3/5$, a valid acute-angle cosine.

例题解析

1. 对角线平方相等给出 $25 + u^2 - 10u\cos\theta = 25 + v^2 - 10v\cos\theta$。

2. 分解得 $(u - v)(u + v - 10\cos\theta) = 0$，由 $u \neq v$ 可令第二因子为零。

3. 故 $\cos\theta = 3/5$，为合法锐角余弦。

Routine quadratic roots may encode a point on a line extension rather than a side segment.

普通二次方程根可能对应延长线上的点，而不是边内点。

If consecutive radial lengths have a fixed product from equal triangle areas, the radii alternate between two values. A common included angle then makes the intervening polygon edges equal by cosine law.

A coordinate angle tangent is cross product divided by dot product. Before equating tangents, retain the dot-product sign to distinguish an acute angle from its supplement.

若由等三角面积得相邻径长乘积固定，则径长交替取两值；再由共同夹角与余弦定理，可使中间多边形边等长。

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

Method 2 / 方法 2 [GE03-M02] (continued / 续)

坐标夹角正切为叉积除以点积。相等正切之前，应保留点积符号，区分锐角与其补角。

Practice / 配套练习: AIME-2000-I-Q14 (WB p. 272), AIME-2003-I-Q10 (WB p. 267), AIME-2003-I-Q12 (WB p. 273), AIME-2003-II-Q07 (WB p. 268), AIME-2004-II-Q11 (WB p. 455), AIME-2004-II-Q13 (WB p. 232), AIME-2005-I-Q07 (WB p. 269), AIME-2006-II-Q12 (WB p. 233), AIME-2007-I-Q12 (WB p. 253), AIME-2008-II-Q14 (WB p. 277), AIME-2010-I-Q15 (WB p. 320), AIME-2010-II-Q14 (WB p. 279), AIME-2011-II-Q10 (WB p. 284), AIME-2012-I-Q12 (WB p. 321), AIME-2012-I-Q13 (WB p. 396), AIME-2012-II-Q13 (WB p. 397), AIME-2013-I-Q09 (WB p. 271), AIME-2013-II-Q05 (WB p. 260), AIME-2014-II-Q11 (WB p. 401), AIME-2014-II-Q12 (WB p. 201), AIME-2014-II-Q14 (WB p. 322), AIME-2018-I-Q04 (WB p. 263), AIME-2018-II-Q12 (WB p. 280), AIME-2019-I-Q13 (WB p. 302), AIME-2022-I-Q11 (WB p. 308), AIME-2022-I-Q14 (WB p. 326), AIME-2022-II-Q15 (WB p. 309), AIME-2023-I-Q05 (WB p. 362), AIME-2023-II-Q03 (WB p. 266), AIME-2023-II-Q12 (WB p. 412), AIME-2024-I-Q10 (WB p. 289), AIME-2025-II-Q12 (WB p. 281)

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

Method 3 / 方法 3 [GE03-M03]

Heron and radius identities connect three sides to area

海伦与半径恒等式连接三边及面积

Procedure

1. Compute $s=(a+b+c)/2$ and check all three factors $s-a, s-b, s-c$ are positive.

2. Use $K^2=s(s-a)(s-b)(s-c)$, $r=K/s$ and $R=abc/(4K)$.

3. When side lengths depend on a scale, factor out its square from area before imposing integrality.

4. For a triangle with altitude sums given by radicals, verify the altitude feet are inside the sides before using positive base pieces.

通用步骤

1. 求 $s=(a+b+c)/2$，并检验 $s-a, s-b, s-c$ 均正。

2. 使用 $K^2=s(s-a)(s-b)(s-c), r=K/s, R=abc/(4K)$。

3. 边长含尺度时，先从面积提出尺度平方，再施加整性。

4. 根式给出高线底边分段时，应确认垂足在边内，才能使用正分段相加。

> Example. Find the area, inradius and circumradius of a triangle of sides $4, 13, 15$.

> 例题：求边长四、十三、十五的三角形面积、内切圆及外接圆半径。

Worked solution

1. The semiperimeter is sixteen, so $K^2=16\cdot 12\cdot 3\cdot 1=576$ and $K=24$.

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

Method 3 / 方法 3 [GE03-M03] (continued / 续)

2. The inradius is $r = 24/16 = 3/2$.

3. The circumradius is $R = 4 \cdot 13 \cdot 15/(4 \cdot 24) = 65/8$.

例题解析

1. 半周长十六，$K^2 = 16 \cdot 12 \cdot 3 \cdot 1 = 576$，故面积二十四。

2. 内切圆半径 $r = 24/16 = 3/2$。

3. 外接圆半径 $R = 4 \cdot 13 \cdot 15/(4 \cdot 24) = 65/8$。

Heron’s zero factor signals a degenerate triangle, not an allowed zero area.

海伦零因子表示退化三角形，不是可接受的零面积解。

An identity $x = \sqrt{y^2 - h^2} + \sqrt{z^2 - h^2}$ describes the two nonnegative pieces of side x made by an internal altitude h. It implies strict triangle inequalities, but the constructed side-ratio triangle must be checked acute to make all such altitude feet internal.

$x = \sqrt{y^2 - h^2} + \sqrt{z^2 - h^2}$ 表示高 h 在边 x 上分出的两个非负段。它给出严格三角不等式，但若三个式子都如此解释，仍须检验所构造边比三角形为锐角，使各垂足均在边内。

Practice / 配套练习: AIME-2000-II-Q12 (WB p. 430), AIME-2005-I-Q15 (WB p. 294), AIME-2006-II-Q15 (WB p. 275), AIME-2007-II-Q15 (WB p. 343), AIME-2010-II-Q12 (WB p. 278), AIME-2015-II-Q07 (WB p. 226), AIME-2017-I-Q04 (WB p. 443), AIME-2024-II-Q10 (WB p. 317), AIME-2022-II-Q15 (WB p. 309)

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

Method 4 / 方法 4 [GE03-M04]

Sine-area ratios compare cevians and cyclic chords

正弦面积比比较劈线与圆内弦

Procedure

1. Write the two subtriangle areas with one common vertex angle or common base altitude.

2. Cancel shared side factors and express the remaining ratio using angle sines.

3. Use isogonal or cyclic-angle equality to cancel sine factors, then convert the ratio to the required side division.

通用步骤

1. 以共同顶角或共同底高表示两个子三角形面积。

2. 约去共同边因子，把剩余比写为夹角正弦比。

3. 用等角共轭或圆周角相等约去正弦，再转为目标线段分比。

> Example. For $D \in BC$ in a triangle with $AB = AC$, suppose $\sin\angle BAD$ is twice $\sin\angle DAC$. Find $BD : DC$.

> 例题：三角形中 $AB = AC, D \in BC$，且 $\sin\angle BAD$ 为 $\sin\angle DAC$ 的两倍，求 $BD : DC$。

Worked solution

1. The common altitude from A gives $BD/DC = [ABD]/[ACD]$.

2. The sine-area formulas give this ratio as $(AB/AC)\sin\angle BAD/\sin\angle DAC$.

3. The first factor is one and the sine ratio is two, so $BD : DC = 2 : 1$.

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

Method 4 / 方法 4 [GE03-M04] (continued / 续)

例题解析

1. 从 A 的共同高给出 $BD/DC = [ABD]/[ACD]$。

2. 正弦面积式使其为 $(AB/AC)\sin\angle BAD/\sin\angle DAC$。

3. 第一因子为一，正弦比为二，所以 $BD : DC = 2 : 1$。

Supplementary angles have equal sine and can support the same sine-rule ratio.

互补角的正弦相等，因此也可给出相同正弦定理比例。

Practice / 配套练习: AIME-2005-II-Q14 (WB p. 274), AIME-2018-II-Q14 (WB p. 301), AIME-2009-II-Q15 (WB p. 187), AIME-2020-II-Q15 (WB p. 305), AIME-2013-II-Q05 (WB p. 260)

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

Method 5 / 方法 5 [GE03-M05]

Half-angle right triangles locate tangent centers

半角直角三角形定位相切圆心

Procedure

1. A circle tangent to two sides has center on their angle bisector.

2. Use $r = d\sin(A/2)$ for center-to-vertex distance d, and tangent length $r\cot(A/2)$.

3. Recover half-angle values from $\sin^2(A/2) = (1-\cos A)/2$ or $\tan(A/2) = \sin A/(1+\cos A)$.

4. Check the chosen sector and positive tangent portions.

通用步骤

1. 与两边相切的圆心位于角平分线上。

2. 圆心到顶点距离 d 满足 $r = d\sin(A/2)$，切线长为 $r\cot(A/2)$。

3. 用 $\sin^2(A/2) = (1-\cos A)/2$ 或 $\tan(A/2) = \sin A/(1+\cos A)$ 求半角。

4. 检查所选角域及切线分段正性。

> Example. A radius-two circle is tangent inside a 60° angle. Find its center-to-vertex distance and each vertex tangent length.

> 例题：半径二的圆在 60° 角内与两边相切，求圆心到顶点距离及每条顶点切线长。

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

Method 5 / 方法 5 [GE03-M05] (continued / 续)

Worked solution

1. The bisector makes 30° with either side, and the contact radius is perpendicular to the side.

2. Thus center distance is $2/\sin 30^\circ = 4$.

3. The tangent length is $2\cot 30^\circ = 2\sqrt{3}$, also checked by $4^2 - 2^2 = 12$.

例题解析

1. 平分线与边夹 30°，切点半径垂直边。

2. 故圆心距为 $2/\sin 30^\circ = 4$。

3. 切线长为 $2\cot 30^\circ = 2\sqrt{3}$，亦由 $4^2 - 2^2 = 12$ 核实。

> The inward ray matters; the opposite angle-bisector ray describes a different sector.

> 向内射线很重要，反向平分射线对应另一角域。

Practice / 配套练习: AIME-2007-I-Q09 (WB p. 334), AIME-2007-II-Q15 (WB p. 343), AIME-2008-II-Q11 (WB p. 344), AIME-2009-II-Q10 (WB p. 371), AIME-2018-I-Q13 (WB p. 324), AIME-2020-I-Q13 (WB p. 325), AIME-2022-I-Q11 (WB p. 308), AIME-2024-I-Q08 (WB p. 338), AIME-2024-II-Q10 (WB p. 317)

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

Method 6 / 方法 6 [GE03-M06]

Triangle angle sums control trigonometric branches

三角形角和控制三角函数分支

Procedure

1. Use $A+B+C=\pi$ to replace sums or multiple angles by one variable.

2. Record whether each angle must be acute, obtuse or inside a specified subtriangle.

3. After a sine or polynomial equation is solved, test every candidate against positive side portions and the angle interval.

4. Two sine-area branches can give different noncongruent triangles; decide which satisfy the required angle type.

通用步骤

1. 利用 $A+B+C=\pi$，把和角或多角化为一个变量。

2. 记录各角应为锐角、钝角，或属于哪个子三角形。

3. 解正弦或多项式方程后，用正线段及角区间检验每个候选。

4. 同一正弦面积可能给出两种不全等三角形，应分别检查所需角型。

> Example. Two sides are two and three, and their included angle has sine $1/2$. Find both third-side possibilities.

> 例题：两边为二、三，夹角正弦为 $1/2$，求第三边的两种可能值。

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

Method 6 / 方法 6 [GE03-M06] (continued / 续)

Worked solution

1. The included angle lies in $(0, 180^\circ)$, so it is 30° or 150°.

2. Cosine law gives $c^2 = 13 - 12(\pm\sqrt{3}/2) = 13 \mp 6\sqrt{3}$.

3. Both squared values are positive and arise from valid included angles, giving $c = \sqrt{13 - 6\sqrt{3}}$ or $\sqrt{13 + 6\sqrt{3}}$.

例题解析

1. 夹角在 $(0, 180^\circ)$ 内，故为 30° 或 150°。

2. 余弦定理给出 $c^2 = 13 \mp 6\sqrt{3}$。

3. 两平方值均正且对应合法夹角，故第三边为 $\sqrt{13 - 6\sqrt{3}}$ 或 $\sqrt{13 + 6\sqrt{3}}$。

A squared trigonometric equation loses the sign of a cosine.

三角方程平方会丢失余弦符号。

Practice / 配套练习: AIME-2000-I-Q14 (WB p. 272), AIME-2010-II-Q14 (WB p. 279), AIME-2013-II-Q15 (WB p. 200), AIME-2014-II-Q12 (WB p. 201), AIME-2021-II-Q05 (WB p. 265)

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

Method 7 / 方法 7 [GE03-M07]

Geometric feasibility can sharpen a formal metric bound

几何可行性可收紧形式度量界

Procedure

1. Write strict triangle inequalities and side-segment coordinate bounds before optimizing.

2. Find the formal equality condition of the chosen inequality.

3. If it violates a segment constraint, optimize on the actual feasible endpoint and construct that boundary case.

4. For acute or obtuse requirements, compare the largest side square with the sum of the other two squares.

通用步骤

1. 优化前写出严格三角不等式及边上线段坐标限制。

2. 求所用不等式形式上的等号条件。

3. 若违反线段限制，应在实际可行端点优化，并构造该边界情形。

4. 锐钝角条件可通过最长边平方与另两边平方和比较。

> Example. A right triangle has hypotenuse one and legs $x, y > 0$, with $x \leq 1/2$. Maximize $x + y$.

> 例题：直角三角形斜边一，直角边 $x, y > 0$ 且 $x \leq 1/2$，求 $x + y$ 最大值。

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

Method 7 / 方法 7 [GE03-M07] (continued / 续)

Worked solution

1. The unrestricted bound is $x+y \le \sqrt{2}$, but equality requires $x=y=1/\sqrt{2}$, violating $x \le 1/2$.

2. On $0 < x \le 1/2$, write $y=\sqrt{1-x^2}$. The derivative of the sum is $1 - x/\sqrt{1-x^2} > 0$.

3. The maximum is at $x=1/2, y=\sqrt{3}/2$, giving $(1+\sqrt{3})/2$, an allowed triangle.

例题解析

1. 无约束界为 $x+y \le \sqrt{2}$，等号要求 $x=y=1/\sqrt{2}$，违反限制。

2. 在 $0 < x \le 1/2$ 上写 $y=\sqrt{1-x^2}$，和的导数 $1 - x/\sqrt{1-x^2} > 0$。

3. 故最大在 $x=1/2, y=\sqrt{3}/2$，值为 $(1+\sqrt{3})/2$，确为合法三角形。

A historical keyed bound can fail attainment when a stated segment is silently extended to a line.

若把题目线段悄悄延长为直线，历史参考界也可能无法达到。

For lengths $a,b,c>0$, triangle existence is $|a-b| < c < a+b$. In integer or Gaussian-integer searches this exact test can give a sharp finite bound before enumerating candidates.

正长 a,b,c 成三角形当且仅当 $|a-b| < c < a+b$。整数或高斯整数搜索中，此精确条件可在枚举前给出严格有限界。

Practice / 配套练习: AIME-2006-II-Q02 (WB p. 102), AIME-2008-I-Q10 (WB p. 270), AIME-2016-I-Q09 (WB p. 183), AIME-2017-I-Q15 (WB p. 404), AIME-2021-II-Q05 (WB p. 265), AIME-2023-I-Q15 (WB p. 96), AIME-2023-II-Q12 (WB p. 412)

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

Method 8 / 方法 8 [GE03-M08]

Equal-side chains propagate base angles

等边链逐次传递底角

Procedure

1. Assign a variable to one apex angle and mark the base-angle equalities of each isosceles subtriangle.

2. Use straight-angle supplements when a point lies inside a side.

3. Close the chain with the original triangle angle sum and reject a candidate if a point leaves its segment.

通用步骤

1. 给一个顶角设变量，并标记各等腰子三角形底角相等。

2. 点位于边内时，使用共线平角的补角关系。

3. 用原三角形角和闭合等式，若点离开线段则排除该候选。

> Example. In isosceles ABC, $AB = AC$ and $A = 30^\circ$. An interior point $D \in AC$ satisfies $BD = BC$. Find $\angle ABD$.

> 例题：等腰三角形 $AB = AC, A = 30^\circ$，边内点 $D \in AC$ 满足 $BD = BC$，求 $\angle ABD$。

Worked solution

1. The original base angles are $B = C = 75^\circ$.

2. In BDC, equal sides $BD = BC$ make the angles at D, C equal to 75°, so $\angle DBC = 30^\circ$.

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

Method 8 / 方法 8 [GE03-M08] (continued / 续)

3. Subtract within the original angle at B: $\angle ABD = 75 - 30 = 45^\circ$.

例题解析

1. 原底角均为 75°。

2. BDC 中 $BD = BC$，所以 D, C 两角均 75°，得 $\angle DBC = 30^\circ$。

3. 在原 B 角内相减，得 $\angle ABD = 45^\circ$。

> The direction of a side ray decides whether angles add or subtract.

> 边射线的方向决定角度应相加还是相减。

Practice / 配套练习: AIME-2020-I-Q01 (WB p. 264)

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

Method 9 / 方法 9 [GE03-M09]

A sixty-degree rotation proves the Fermat distance bound

六十度旋转证明费马距离和下界

Procedure

1. Erect an outward equilateral triangle ABB' on one side of triangle ABC.

2. Rotate X with B by 60° about A to X', B'; then $XX' = AX$ and $X'B' = XB$.

3. Triangle inequality along C, X, X', B' gives $AX + BX + CX \geq CB'$.

4. When all original angles are below 120°, the Fermat point with three 120° rays attains the bound; verify any simultaneous external constraint separately.

通用步骤

1. 在三角形 ABC 的一边外作等边三角形 ABB'。

2. 绕 A 旋转 60°，使 X, B 映为 X', B'，则 $XX' = AX, X'B' = XB$。

3. 对折线 C, X, X', B' 用三角不等式，得 $AX + BX + CX \geq CB'$。

4. 原三角均小于 120° 时，三射线夹角均 120° 的费马点取等；其他同时限制须另行检验。

> Example. Find the least sum of distances from a plane point to the vertices of an equilateral triangle of side two.

> 例题：求平面一点到边长二的等边三角形三顶点距离和的最小值。

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

Method 9 / 方法 9 [GE03-M09] (continued / 续)

Worked solution

1. An outward equilateral construction makes $CB\prime$ an altitude across two equilateral triangles, length $2\sqrt{3}$. The rotation argument bounds every distance sum by this value.

2. At the common centroid, each vertex distance is $2/\sqrt{3}$.

3. The sum is $3(2/\sqrt{3}) = 2\sqrt{3}$, so the bound is attained.

例题解析

1. 外作等边构造后，$CB\prime$ 是两个等边三角形高的和，长 $2\sqrt{3}$；旋转论证给出距离和下界。

2. 在重心处，到每顶点距离为 $2/\sqrt{3}$。

3. 总和为 $2\sqrt{3}$，下界确可达到。

If a vertex angle is at least 120°, the minimum is at that vertex instead of an interior Fermat point.

若某顶角至少 120°，最小值在该顶点而非内部费马点。

Practice / 配套练习: AIME-2025-I-Q14 (WB p. 190)

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

Method 10 / 方法 10 [GE03-M10]

No-slip rolling separates center sweep from spin

无滑动滚动区分圆心扫角与自转

Procedure

1. In an axial section, use center distance $R+r$ while one circle rolls over a fixed external circle.

2. The center sweep θ has arc length $(R+r)\theta$, equaling $R\phi$ for spin ϕ.

3. Use the ground-contact right triangle to find the transition angles and horizontal displacement.

4. Subtract obstacle spin from the prescribed total spin before converting the remaining ground roll into distance.

通用步骤

1. 轴截面中，动圆绕固定圆从外侧滚动时，圆心距为 $R+r$。

2. 圆心扫角 θ 的弧长为 $(R+r)\theta$，等于自转弧长 $R\phi$。

3. 由地面接触直角三角形求转换角及水平位移。

4. 从规定总自转扣除绕障碍的自转，再将剩余地面转动化为路程。

> Example. A radius-three circle rolls around a fixed radius-one circle. Its center sweeps 90°. Find its spin angle.

> 例题：半径三的圆绕固定半径一的圆从外侧无滑动滚动，圆心扫过 90°，求自转角。

Section 2.3 Triangle metrics & trigonometry

三角形度量与三角法

Method 10 / 方法 10 [GE03-M10] (continued / 续)

Worked solution

1. The center-path radius is four, so the arc length is $4(\pi/2) = 2\pi$.

2. This equals moving-radius three times spin angle ϕ.

3. Thus $\phi = 2\pi/3 = 120^\circ$, distinct from the 90° center sweep.

例题解析

1. 圆心轨迹半径四，弧长为 2π。

2. 其等于动圆半径三乘自转角 ϕ。

3. 所以 $\phi = 2\pi/3 = 120^\circ$，不同于圆心扫角。

Angles in the no-slip length equation must be in radians.

无滑动弧长式中的角必须使用弧度。

Practice / 配套练习: AIME-2007-II-Q11 (WB p. 276)

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 1 / 方法 1 [GE04-M01]

Circle geometry has three complementary languages: arc angles, point powers and circle equations. Choose one that preserves the given order of points. Directed-angle identities are powerful, but an ordinary angle or positive segment length still needs a branch check.

圆几何可用弧角、点幂及圆方程三种互补语言。选择能保留已知点序的方法。有向角恒等式虽然有力，但普通角及正线段长仍须筛选分支。

Chord lengths and arc angles share a common radius

弦长与弧角共享同一半径

Procedure

1. A chord subtending minor central angle θ has length $2R\sin(\theta/2)$.

2. An inscribed angle measures half the arc not containing its vertex; identify that arc from cyclic order.

3. For repeated chord types, add the actual consecutive central angles and exclude a hidden major-arc branch.

4. Use multiple-angle identities only after fixing those angle ranges.

通用步骤

1. 劣中心角为 θ 的弦长为 $2R\sin(\theta/2)$。

2. 圆周角等于不含其顶点的对应弧一半，应由圆上次序明确该弧。

3. 多次出现同类弦时，累加实际相邻中心角，并排除隐藏的优弧分支。

4. 确定角范围后，再用倍角或多角恒等式。

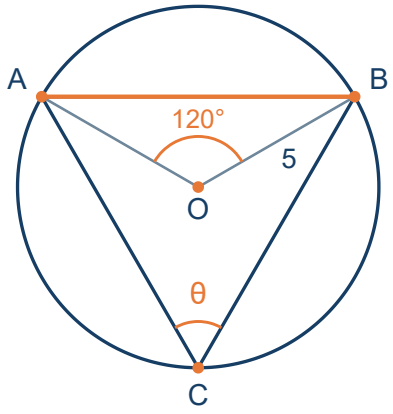
Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 1 / 方法 1 [GE04-M01] (continued / 续)

> Example. In a radius-5 circle a chord subtends 120°. Find its length and the inscribed angle subtending its minor arc.

> 例题：半径五的圆内，一弦所对劣中心角为 120°，求弦长及所对劣弧的圆周角。



Worked solution

1. The half central angle is 60°, so the chord is $10\sin 60^\circ = 5\sqrt{3}$.

2. An inscribed vertex outside this minor arc sees the 120° arc and has angle 60°.

3. A vertex on the minor arc instead sees the 240° complement and has angle 120°; the vertex location is essential.

例题解析

1. 中心半角为 60°，所以弦长为 $10\sin 60^\circ = 5\sqrt{3}$。

2. 劣弧外的圆周顶点所见对应弧为 120°，圆周角为 60°。

3. 若顶点在该劣弧上，则对应补弧为 240°，圆周角为 120°；顶点位置不可省略。

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 1 / 方法 1 [GE04-M01] (continued / 续)

Equal chords determine equal minor arcs, but a consecutive arc could be the major complement.

等弦对应相等劣弧，但相邻顶点之间的实际弧可能是优弧补段。

A circular segment area is its sector area minus the center–chord triangle. Choose the smaller or complementary segment explicitly before forming a ratio.

圆弓形面积等于扇形减去圆心弦三角形；求比例前先明确使用小弓形还是补集。

Practice / 配套练习: AIME-2001-I-Q13 (WB p. 292), AIME-2004-II-Q01 (WB p. 239), AIME-2013-II-Q08 (WB p. 285), AIME-2015-I-Q06 (WB p. 286), AIME-2018-I-Q15 (WB p. 300), AIME-2025-II-Q05 (WB p. 282)

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 2 / 方法 2 [GE04-M02]

Power multiplies directed secant distances

点幂使用有向割线距离乘积

Procedure

1. For a point outside a circle, $PT^2 = PA \cdot PB$ on a secant with near point A and far point B.

2. For intersecting chords at an interior point, the two positive endpoint-distance products are equal.

3. Converse equal products can create an auxiliary cyclic quadrilateral, provided all required points are distinct.

4. Use equal tangents to transfer lengths, then return to the actual segment order.

通用步骤

1. 圆外点满足 $PT^2 = PA \cdot PB$，其中 A 为近交点、B 为远交点。

2. 圆内弦交点到两端点的正距离乘积，在两弦上相等。

3. 反过来，等乘积可构造辅助共圆四边形，但须确认所需点互异。

4. 用等切线转移长度，再还原实际线段次序。

> Example. A secant from P meets a circle after distances $4, 9$. Another meets it first after distance three. Find its far-intersection distance and the tangent length.

> 例题：从 P 的一条割线在距离四、九处交圆，另一割线近交距为三，求远交距及切线长。

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 2 / 方法 2 [GE04-M02] (continued / 续)

Worked solution

1. The power of P is $4 \cdot 9 = 36$.

2. If the second far distance is u, then $3u = 36$, giving $u = 12 > 3$.

3. The tangent length is $\sqrt{36} = 6$. Both positive-distance interpretations agree with P being outside.

例题解析

1. P 的点幂为 $4 \cdot 9 = 36$。

2. 第二条远交距 u 满足 $3u = 36$，所以 $u = 12 > 3$。

3. 切线长为 $\sqrt{36} = 6$，两组正距离均符合圆外位置。

Near distance times chord length is not the power; use near distance times full far distance.

近交距乘弦长不是点幂，应乘到远交点的全距。

Practice / 配套练习: AIME-2002-I-Q13 (WB p. 293), AIME-2005-I-Q15 (WB p. 294), AIME-2009-I-Q12 (WB p. 319), AIME-2016-I-Q06 (WB p. 316), AIME-2016-II-Q10 (WB p. 287), AIME-2019-II-Q15 (WB p. 304), AIME-2022-I-Q11 (WB p. 308), AIME-2024-I-Q10 (WB p. 289), AIME-2025-I-Q06 (WB p. 290), AIME-2026-II-Q10 (WB p. 291)

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 3 / 方法 3 [GE04-M03]

Right angles identify diameter circles and antipodal pairs

直角识别直径圆与对径点

Procedure

1. A right angle subtending two fixed points places its vertex on their diameter circle.

2. The midpoint of a right triangle's hypotenuse is its circumcenter.

3. Use an antipodal point to replace a cyclic angle by a perpendicular or a line intersection.

4. An inscribed rectangle has two opposite-vertex pairs that are diameters.

通用步骤

1. 两固定点所张直角的顶点位于以两点连线为直径的圆上。

2. 直角三角形斜边中点即外心。

3. 利用对径点把圆内角换成垂直关系或直线求交。

4. 圆内接矩形的两组对顶点均互为对径点。

> Example. Points P, Q lie on the circle with diameter $AB = 10$ and subtend angle $PAQ = 30^\circ$. Find PQ.

> 例题：P, Q 在直径 $AB = 10$ 的圆上，且 $\angle PAQ = 30^\circ$，求 PQ。

Worked solution

1. The radius is five. The arc PQ not containing A has measure 60°.

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 3 / 方法 3 [GE04-M03] (continued / 续)

2. The corresponding chord is $2 \cdot 5 \sin 30^\circ = 5$.

3. Thus $PQ = 5$; the same chord value follows if the minor central angle is taken directly as 60°.

例题解析

1. 圆半径为五，不含 A 的对应弧 PQ 为 60°。

2. 弦长为 $2 \cdot 5 \sin 30^\circ = 5$。

3. 所以 $PQ = 5$，亦可直接用 60° 劣中心角核实。

A midpoint locus needs a converse construction, not just one right-angle implication.

中点轨迹还需反向构造，不能只证一侧的直角推论。

Practice / 配套练习: AIME-2004-I-Q04 (WB p. 238), AIME-2011-II-Q10 (WB p. 284), AIME-2012-II-Q15 (WB p. 297), AIME-2014-I-Q15 (WB p. 298), AIME-2017-I-Q08 (WB p. 771), AIME-2020-II-Q15 (WB p. 305), AIME-2023-II-Q07 (WB p. 626), AIME-2025-I-Q14 (WB p. 190)

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 4 / 方法 4 [GE04-M04]

Circle subtraction gives radical axes and their concurrence

两圆方程相减给出根轴及共线关系

Procedure

1. For circles with distinct centers, subtract their equations to cancel $x^2 + y^2$; the equal-power locus is the resulting radical-axis line, containing any common chord.

2. A point of equal powers lies on this radical axis even when its tangency segments are easier to compute than coordinates.

3. Three pairwise radical axes concur or are parallel, yielding auxiliary intersections and tangent-midpoint alignments.

4. Track the full ray order before turning directed power products into positive length products.

通用步骤

1. 对圆心不同的两圆，相减方程消去 $x^2 + y^2$；等点幂轨迹是所得根轴直线，若有公共弦则包含它。

2. 一点对两圆点幂相等便在根轴上；有时切线段比坐标更易求。

3. 三个两两根轴共点或平行，可得到辅助交点及公切线中点的共线关系。

4. 将有向点幂积转为正长度积之前，明确完整射线点序。

> Example. Find the common chord line of $x^2 + y^2 = 25$ and $(x-4)^2 + y^2 = 9$.

> 例题：求 $x^2 + y^2 = 25$ 与 $(x-4)^2 + y^2 = 9$ 的公共弦直线。

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 4 / 方法 4 [GE04-M04] (continued / 续)

Worked solution

1. Expand the second equation as $x^2 + y^2 - 8x + 7 = 0$. The first is $x^2 + y^2 - 25 = 0$.

2. Subtract to get $-8x + 32 = 0$, so the radical axis is $x = 4$.

3. On that line $y^2 = 9$, giving distinct common points $(4, 3), (4, -3)$ and confirming a genuine chord.

例题解析

1. 第二式展开为 $x^2 + y^2 - 8x + 7 = 0$，第一式为 $x^2 + y^2 - 25 = 0$。

2. 相减得 $-8x + 32 = 0$，根轴为 $x = 4$。

3. 代入得 $y^2 = 9$，交点为 $(4, 3), (4, -3)$，确认是真正公共弦。

The radical axis is a line of equal powers, not necessarily a line of equal distances to centers.

根轴是等点幂直线，不一定是到两圆心等距直线。

The power difference has gradient magnitude $2O_1O_2$. Therefore its magnitude at P equals $2O_1O_2$ times distance from P to the radical axis. Along a center line, signed distances interpolate affinely.

The midpoint of a common external tangent segment has equal squared tangent powers, hence lies on the radical axis. Combine this with near/far intersection order before using a difference of squares.

点幂差的梯度长度为 $2O_1O_2$，所以一点幂差绝对值等于其到根轴距离乘 $2O_1O_2$。沿圆心直线，有向距离按仿射比例变化。

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 4 / 方法 4 [GE04-M04] (continued / 续)

公外切线段中点的两切线幂相等，故位于根轴上。使用平方差之前，须结合近远交点的次序。

Practice / 配套练习: AIME-2005-II-Q08 (WB p. 332), AIME-2010-II-Q15 (WB p. 296), AIME-2016-I-Q15 (WB p. 299), AIME-2019-I-Q13 (WB p. 302), AIME-2019-I-Q15 (WB p. 303), AIME-2019-II-Q11 (WB p. 406), AIME-2021-I-Q13 (WB p. 348), AIME-2023-II-Q09 (WB p. 288)

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 5 / 方法 5 [GE04-M05]

Circle equations linearize symmetric squared-distance sums

圆方程线性化对称距离平方和

Procedure

1. Choose a circle center or a distinguished vertex as origin.

2. For a point on a known circle, replace $x^2 + y^2$ by the circle equation's affine expression.

3. For circumcenter vector o with a vertex at zero, use $2o \cdot a = |a|^2$ for every other vertex vector.

4. A tangent can be written with a unit normal, giving exact projection coordinates and a bounded parameter range.

通用步骤

1. 以圆心或特殊顶点为原点。

2. 点在已知圆上时，将 $x^2 + y^2$ 替换为圆方程给出的仿射表达式。

3. 若一顶点在原点、外心向量为 o，其余顶点向量 a 满足 $2o \cdot a = |a|^2$。

4. 用单位法向量表示切线，可得到精确投影坐标及有界参数范围。

> Example. A point $P = (x, y)$ lies on $x^2 + y^2 = 4y$. Show $PA^2 + PB^2$ is affine in y when $A = (-1, 0), B = (1, 0)$.

> 例题：点 $P = (x, y)$ 在 $x^2 + y^2 = 4y$ 上，$A = (-1, 0), B = (1, 0)$，将 $PA^2 + PB^2$ 化为关于 y 的仿射式。

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 5 / 方法 5 [GE04-M05] (continued / 续)

Worked solution

1. Expanding gives $PA^2 + PB^2 = 2(x^2 + y^2) + 2$.

2. The circle relation replaces $x^2 + y^2$ by $4y$.

3. Thus the sum equals $8y + 2$. This linearization avoids solving both point coordinates.

例题解析

1. 展开得 $PA^2 + PB^2 = 2(x^2 + y^2) + 2$。

2. 圆方程给出 $x^2 + y^2 = 4y$。

3. 所以平方和为 $8y + 2$，无须分别解出两坐标。

Use the circle equation only for points already proved to lie on that circle.

只有已经证明位于该圆上的点，才能代入其圆方程。

Practice / 配套练习: AIME-2001-II-Q06 (WB p. 283), AIME-2008-I-Q14 (WB p. 186), AIME-2015-II-Q11 (WB p. 402), AIME-2020-II-Q15 (WB p. 305), AIME-2023-II-Q12 (WB p. 412), AIME-2024-I-Q05 (WB p. 363)

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 6 / 方法 6 [GE04-M06]

Directed circle angles still need ordinary-angle branch checks

圆内有向角仍需普通角分支检验

Procedure

1. Transfer angles through a shared chord or the tangent–chord theorem modulo 180°.

2. Use the converse cyclic-angle criterion to create useful auxiliary circles.

3. Recover the actual rays from convexity, cyclic point order, acute-angle bounds or explicit coordinates.

4. If a complex cross ratio is real, interpret equality of directed arguments as a generalized-circle condition and check excluded points.

通用步骤

1. 通过同弦或切线弦定理传递模 180° 的角。

2. 用共圆角判定的逆命题构造辅助圆。

3. 利用凸性、圆上点序、锐角界或明确坐标恢复实际射线。

4. 复交比为实时，将有向辐角相等解释为广义圆条件，并检查无定义点。

> Example. Points A, B, C, D are cyclic. If $\angle ABC = 70^\circ$ and B, D lie on opposite sides of AC, find $\angle ADC$.

> 例题：A, B, C, D 共圆，$\angle ABC = 70^\circ$，且 B, D 位于弦 AC 两侧，求 $\angle ADC$。

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 6 / 方法 6 [GE04-M06] (continued / 续)

Worked solution

1. The vertices B, D see the two complementary arcs of AC.

2. Their ordinary inscribed angles sum to 180°, so $\angle ADC = 110^\circ$.

3. Directed equality modulo 180° remains true, but replacing it by ordinary-angle equality would incorrectly give seventy.

例题解析

1. B, D 分别看到 AC 的两条互补弧。

2. 两普通圆周角互补，所以 $\angle ADC = 110^\circ$。

3. 有向角模 180° 的相等仍成立，但不能误写为普通角也相等。

A supplementary branch can satisfy every squared length equation.

补角分支可能满足所有距离平方方程。

For noncollinear fixed complex points z_1, z_2, z_3, reality of $[(z_3 - z_1)/(z_2 - z_1)][(z - z_2)/(z - z_3)]$ gives their circumcircle, excluding denominator-zero points. An ordinary angle must still be chosen from the actual rays.

对不共线定复点 z_1, z_2, z_3，$[(z_3 - z_1)/(z_2 - z_1)][(z - z_2)/(z - z_3)]$ 为实数时，轨迹为其外接圆，须排除分母为零的点。普通角仍需按实际射线筛选。

Practice / 配套练习: AIME-2012-II-Q15 (WB p. 297), AIME-2014-II-Q14 (WB p. 322), AIME-2015-I-Q06 (WB p. 286), AIME-2016-I-Q15 (WB p. 299), AIME-2017-I-Q10 (WB p. 80), AIME-2019-I-Q13 (WB p. 302), AIME-2021-II-Q14 (WB p. 307)

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 7 / 方法 7 [GE04-M07]

A near incircle tangent gives a reciprocal side identity

内切圆近切线给出边长倒数恒等式

Procedure

1. Let the incircle touch AB, AC, BC at Z, W, X. Let Y be the first circle intersection on ray AX, and let its tangent cut side segments AB, AC at P, Q. Set $\ell = AZ = AW, p = AP, q = AQ$.

2. Equal tangents give $PY = \ell - p$ and $BX = AB - \ell$. The common angle at A and supplementary tangent–chord angles give the sine-rule ratio $PY/p = BX/AB$.

3. Rearrange to $\ell/p + \ell/AB = 2$. Repeating on the other side gives $\ell/q + \ell/AC = 2$.

4. Check the near-intersection order and positive cut segments. An arbitrary tangent to the near incircle arc does not satisfy these reciprocal laws without A, Y, X collinear.

通用步骤

1. 设内切圆切 AB, AC, BC 于 Z, W, X。令 Y 为射线 AX 先遇到的圆交点，Y 处切线交边线段 AB, AC 于 P, Q；记 $\ell = AZ = AW, p = AP, q = AQ$。

2. 等切线给出 $PY = \ell - p, BX = AB - \ell$。由共同的 A 角及互补的弦切角，正弦定理得 $PY/p = BX/AB$。

3. 整理得 $\ell/p + \ell/AB = 2$，另一边同理得 $\ell/q + \ell/AC = 2$。

4. 检查近交点次序及各截段为正。若缺少 A, Y, X 共线条件，任意近弧切线并不满足这些倒数关系。

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 7 / 方法 7 [GE04-M07] (continued / 续)

> Example. The incircle of ABC touches BC at X. Let Y be the first intersection of ray AX with the incircle, and let the tangent at Y meet AB, AC at P, Q inside the sides. Prove $AP = AQ$ if and only if $AB = AC$.

> 例题：三角形内切圆切 BC 于 X。射线 AX 先交圆于 Y，Y 处切线交边内于 $P \in AB, Q \in AC$。证明 $AP = AQ$ 当且仅当 $AB = AC$。

Worked solution

1. Write $p = AP, q = AQ$ and let ℓ be the common vertex tangent length. The tangent–chord and sine-rule calculation gives $1/p + 1/AB = 2/\ell$.

2. The same argument on the AC side gives $1/q + 1/AC = 2/\ell$. Subtracting yields $1/p - 1/q = 1/AC - 1/AB$.

3. All four lengths are positive. Thus the left side vanishes exactly when $p = q$, and the right side exactly when $AB = AC$, proving both directions.

例题解析

1. 记 $p = AP, q = AQ$，公共顶点切线长为 ℓ。由弦切角与正弦定理得到 $1/p + 1/AB = 2/\ell$。

2. 在 AC 一侧同理得到 $1/q + 1/AC = 2/\ell$。相减得 $1/p - 1/q = 1/AC - 1/AB$。

3. 四个长度均为正，所以左边为零等价于 $p = q$，右边为零等价于 $AB = AC$，两方向均得证。

The far incircle tangent does not have the same segment signs.

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 7 / 方法 7 [GE04-M07] (continued / 续)

内切圆远切线的线段符号并不相同。

Practice / 配套练习: AIME-2018-II-Q14 (WB p. 301)

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 8 / 方法 8 [GE04-M08]

Cyclic intercept ratios are products of chord ratios

圆内截比等于弦比乘积

Procedure

1. For a cyclic quadrilateral, split triangles at a chord intersection and use common-altitude area ratios.

2. Apply the sine rule to the corresponding equal inscribed angles; products of chord lengths replace unknown sines.

3. Convert a division ratio t on a unit segment to position $t/(1+t)$ before subtracting positions.

4. Check that the free point traverses the complete positive parameter range needed for equality.

通用步骤

1. 对圆内接四边形，在弦交点分三角形并使用等高面积比。

2. 对相等圆周角使用正弦定理，用弦长乘积替换未知正弦。

3. 单位线段上分比为 t 时，其位置是 $t/(1+t)$，先换位置再相减。

4. 检查活动点是否遍历等号所需的全部正参数范围。

> Example. On a unit diameter, two chord-intersection points have ratios from the left endpoint x and $2x$, with $x>0$. Find their largest separation.

> 例题：在单位直径上，两弦交点从左端量起的分比分别为 $x, 2x$，其中 $x>0$，求两点最大间距。

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 8 / 方法 8 [GE04-M08] (continued / 续)

Worked solution

1. The positions are $x/(1+x)$ and $2x/(1+2x)$, so their difference is $x/((1+x)(1+2x))$.

2. Its reciprocal is $1/x+3+2x \geq 3+2\sqrt{2}$, with equality at $x=1/\sqrt{2}$.

3. The largest separation is $1/(3+2\sqrt{2}) = 3-2\sqrt{2}$.

例题解析

1. 两位置为 $x/(1+x)$ 与 $2x/(1+2x)$，差为 $x/((1+x)(1+2x))$。

2. 倒数为 $1/x+3+2x \geq 3+2\sqrt{2}$，在 $x=1/\sqrt{2}$ 取等。

3. 故最大间距为 $3-2\sqrt{2}$。

For diameter endpoints M, N and cyclic B, C, $MP/PN = MB \cdot MC/(NB \cdot NC)$ at $P = BC \cap MN$.

直径端点为 M, N、圆上点为 B, C 时，$P = BC \cap MN$ 满足 $MP/PN = MB \cdot MC/(NB \cdot NC)$。

Practice / 配套练习: AIME-2009-II-Q15 (WB p. 187)

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 9 / 方法 9 [GE04-M09]

Cyclic quadrilaterals combine Ptolemy, area and circumradius

圆内接四边形结合托勒密、面积与外接半径

Procedure

1. For consecutive sides a, b, c, d, use Ptolemy: the diagonal product equals $ac + bd$.

2. Brahmagupta gives $K^2 = (s-a)(s-b)(s-c)(s-d)$.

3. If two pieces share a circumcircle, equate $16K^2R^2 = (ab+cd)(ac+bd)(ad+bc)$.

4. Reject algebraic roots violating positive side lengths, convexity or the cyclic point order.

通用步骤

1. 连续四边为 a, b, c, d 时，托勒密给出对角线乘积为 $ac + bd$。

2. 婆罗摩笈多公式为 $K^2 = (s-a)(s-b)(s-c)(s-d)$。

3. 两块共外接圆时，可比较 $16K^2R^2 = (ab+cd)(ac+bd)(ad+bc)$。

4. 排除违反正边长、凸性或圆上次序的代数根。

> Example. A cyclic quadrilateral has consecutive sides $3, 4, 3, 4$. Find its area and diagonal product.

> 例题：圆内接四边形连续边长为三、四、三、四，求面积及对角线乘积。

Worked solution

1. The semiperimeter is seven, so $K^2 = 4 \cdot 3 \cdot 4 \cdot 3 = 144$ and $K = 12$.

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 9 / 方法 9 [GE04-M09] (continued / 续)

2. Ptolemy gives diagonal product $3 \cdot 3 + 4 \cdot 4 = 25$.

3. It is a rectangle with diagonals five; the direct 3×4 area checks both formulas.

例题解析

1. 半周长为七，所以 $K^2 = 4 \cdot 3 \cdot 4 \cdot 3 = 144$，面积为十二。

2. 托勒密给出对角线乘积 $3 \cdot 3 + 4 \cdot 4 = 25$。

3. 该构型为三乘四矩形、对角线五，直接计算核实两式。

Brahmagupta’s short formula needs cyclicity; a general quadrilateral has an extra angle term.

婆罗摩笈多简式需要共圆，一般四边形还含额外角度项。

Cross-projection onto two intersecting diagonals is reflection across an angle bisector followed by scale $\cos\theta$. Hence all projected side lengths scale by that factor, while $K = \frac{1}{2} d_1 d_2 \sin\theta$ determines it from Ptolemy and area.

向两相交对角线作交叉投影，等价于关于角平分线反射再按 $\cos\theta$ 缩放。投影边长均按此比例变化，而 $K = \frac{1}{2} d_1 d_2 \sin\theta$ 可结合托勒密与面积求该比例。

Practice / 配套练习: AIME-2021-I-Q11 (WB p. 306), AIME-2022-II-Q15 (WB p. 309)

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 10 / 方法 10 [GE04-M10]

Arc and semicircle structure controls circular counting

圆弧及半圆结构控制圆周计数

Procedure

1. An inscribed obtuse triangle has all three vertices in one open semicircle; choose a unique first vertex when counting.

2. Distinct circles intersect in at most two points; coincident boundary circles give no proper strictly separated subset. Thus a disk selects one consecutive run of vertices on a circle, unless empty or full.

3. A random chord misses a fixed inscribed convex polygon exactly when both endpoints lie in the same open boundary arc between consecutive vertices.

通用步骤

1. 圆内接钝角三角形的三顶点位于同一开半圆；计数时可指定唯一首顶点。

2. 不同圆至多交于两点；边界圆重合时不能严格分离非空真子集。因此圆盘在圆周顶点中选出的子集应为连续一段，或空集、全集。

3. 随机弦不交固定圆内接凸多边形，当且仅当两端点位于同一对相邻顶点之间的开圆弧。

> Example. A fixed inscribed equilateral triangle is given. Two independent uniform circumference points form a chord. Find the probability the chord meets the triangle.

> 例题：给定圆内接等边三角形，从圆周独立均匀取两点连弦，求弦与三角形相交的概率。

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 10 / 方法 10 [GE04-M10] (continued / 续)

Worked solution

1. The three complementary boundary arcs each have one third of the circumference.

2. The chord misses the triangle precisely when both endpoints belong to the same one of these arcs, apart from zero-probability endpoint cases.

3. The miss probability is $3(1/3)^2 = 1/3$, so the intersection probability is $2/3$.

例题解析

1. 三条相邻顶点间的边界圆弧各占圆周三分之一。

2. 除概率零的端点情况外，弦不交三角形恰在两端位于同一此类圆弧时发生。

3. 不交概率为 $3(1/3)^2 = 1/3$，故相交概率为 $2/3$。

An open semicircle excludes antipodal right-angle cases.

开半圆不包含对径点对应的直角情形。

Practice / 配套练习: AIME-2011-I-Q10 (WB p. 739), AIME-2014-I-Q05 (WB p. 693), AIME-2017-I-Q06 (WB p. 770)

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 11 / 方法 11 [GE04-M11]

The orthocenter reflects a circumcircle across a side

垂心把外接圆关于一边反射

Procedure

1. Assume the orthocenter H and B, C are distinct. With the circumcenter at the vector origin, use $h = a + b + c$.

2. The circle through H, B, C is the reflection of the original circumcircle across BC.

3. Its center is $b + c = h - a$; the radius toward H is parallel to a.

4. Use chord midpoint normals or a tangent perpendicular to that radius to recover lengths and area.

通用步骤

1. 先假设垂心 H 与 B, C 互不重合。以外心为向量原点，使用 $h = a + b + c$。

2. 过 H, B, C 的圆是原外接圆关于 BC 的反射。

3. 其圆心为 $b + c = h - a$，指向 H 的半径平行于 a。

4. 利用弦中点法向或垂直该半径的切线，求长度与面积。

> Example. On the unit circle take $A = (0, 1), B = (-1, 0), C = (1, 0)$. Find the center of the circle through the orthocenter H and B, C.

> 例题：单位圆上取 $A = (0, 1), B = (-1, 0), C = (1, 0)$，求过垂心 H 与 B, C 的圆心。

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 11 / 方法 11 [GE04-M11] (continued / 续)

Worked solution

1. Triangle ABC is right at A, so $H = A = (0, 1)$; the vector identity also gives $a + b + c = a$.

2. The side BC is the x axis, so reflection preserves the original circle in this special case.

3. The new center is $b + c = (0, 0)$, and its radius is one.

例题解析

1. 三角形在 A 为直角，故 $H = A = (0, 1)$，向量恒等式也给出 $a + b + c = a$。

2. BC 为 x 轴，在此特殊情况下反射保持原圆不变。

3. 新圆心为 $b + c = (0, 0)$，半径为一。

The reflected circle need not have the same center as the original in a general acute triangle.

一般锐角三角形中，反射后的圆心不等于原圆心。

Practice / 配套练习: AIME-2020-I-Q15 (WB p. 408)

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 12 / 方法 12 [GE04-M12]

A fixed inscribed angle gives a circular arc with an attainable height bound

固定圆周角给出圆弧及可达高度界

Procedure

1. Use cyclic angle relations to show a moving point sees a fixed chord under a fixed ordinary angle.

2. This places it on a specified arc; height above the chord is greatest at the arc midpoint.

3. Prove the midpoint is attainable using an explicit configuration or a continuous parameter with distinct endpoint limits.

通用步骤

1. 由圆周角关系证明活动点看固定弦的普通角恒定。

2. 因此活动点位于指定圆弧，距弦高度在弧中点最大。

3. 用明确构造或具有不同端点极限的连续参数，证明弧中点确实可达。

> Example. Point P lies on the minor circular arc below a chord $BC = 2$ and satisfies $\angle BPC = 120^\circ$. Find the largest triangle area.

> 例题：点 P 位于弦 $BC = 2$ 下方的劣圆弧上，且 $\angle BPC = 120^\circ$，求三角形面积最大值。

Worked solution

1. At the maximum $PB = PC = x$. Cosine law gives $4 = 2x^2 - 2x^2 \cos 120^\circ = 3x^2$.

Section 2.4 Circle power & cyclic geometry

圆幂与圆内接几何

Method 12 / 方法 12 [GE04-M12] (continued / 续)

2. The corresponding height is $x\cos 60^\circ = 1/\sqrt{3}$, reached at the arc midpoint.

3. The area is $2(1/\sqrt{3})/2 = 1/\sqrt{3}$.

例题解析

1. 最大时 $PB = PC = x$，余弦定理给出 $4 = 3x^2$。

2. 相应高为 $x\cos 60^\circ = 1/\sqrt{3}$，在弧中点取到。

3. 面积为 $2(1/\sqrt{3})/2 = 1/\sqrt{3}$。

A locus bound is insufficient if the moving configuration reaches only part of the arc.

若原构型只能到达圆弧的一部分，仅有轨迹上的上界还不充分。

Practice / 配套练习: AIME-2009-I-Q15 (WB p. 295), AIME-2013-II-Q10 (WB p. 252)

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 1 / 方法 1 [GE05-M01]

Triangle centers are useful because they preserve exact ratios, equal distances and half-angle relations. Choose the appropriate center identity, retain internal/external bisector branches, and connect lengths to area only after the point order is established.

三角形中心可提供精确比例、等距及半角关系。应选择合适的中心恒等式，保留内外角平分线分支，并先明确点序再联系长度与面积。

Angle-bisector ratios persist in nested triangles

角平分线比例在嵌套三角形中仍成立

Procedure

1. For the internal bisector AD, write $BD/DC = AB/AC$ with positive segment lengths.

2. If a second segment crosses the same angle, apply the theorem inside the smaller triangle as well.

3. For a needed length use $AD^2 = AB \cdot AC - BD \cdot DC$, or Stewart's theorem for a general cevian.

4. Restore the signs and triangle inequalities after eliminating or squaring.

通用步骤

1. 对内角平分线 AD，写出正线段比例 $BD/DC = AB/AC$。

2. 若另一条线段横跨同一角，在小三角形内也可使用同一定理。

3. 需要长度时用 $AD^2 = AB \cdot AC - BD \cdot DC$，一般劈线则用斯图尔特定理。

4. 消元或平方后补查符号与三角不等式。

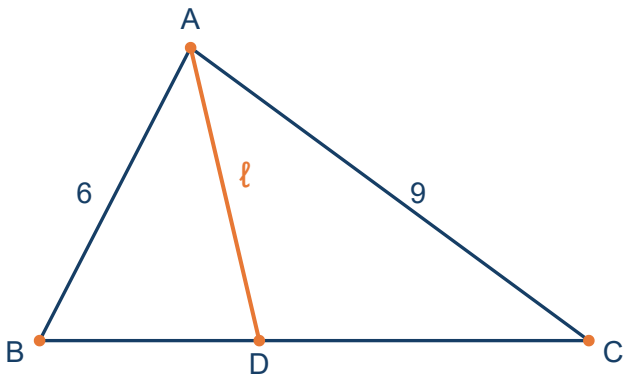
Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 1 / 方法 1 [GE05-M01] (continued / 续)

> Example. A triangle has $AB = 6, AC = 9, BC = 10$. Find the length of the internal bisector from A.

> 例题：三角形满足 $AB = 6, AC = 9, BC = 10$，求从 A 出发的内角平分线长度。



BC = 10 ∠BAD = ∠DAC

Worked solution

1. The ratio $BD/DC = 2/3$ gives $BD = 4, DC = 6$.

2. The bisector identity gives $AD^2 = 6 \cdot 9 - 4 \cdot 6 = 30$.

3. Thus $AD = \sqrt{30}$. The original sides satisfy all strict triangle inequalities, so the internal construction is valid.

例题解析

1. 由 $BD/DC = 2/3$ 得 $BD = 4, DC = 6$。

2. 角平分线恒等式给出 $AD^2 = 6 \cdot 9 - 4 \cdot 6 = 30$。

3. 故 $AD = \sqrt{30}$；原三边满足严格三角不等式，内部构造合法。

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 1 / 方法 1 [GE05-M01] (continued / 续)

An angle bisector is not generally a median.

角平分线一般不是中线。

Stewart's theorem for $BD = m, DC = n, BC = m + n$, $AB = c, AC = b, AD = d$ is $b^2m + c^2n = (m + n)(d^2 + mn)$. Internal-bisector substitution yields the short identity above.

A bisected subangle can create an isosceles auxiliary triangle. Determine its other two angles before assuming the bisector splits the original side in half.

若 $BD = m, DC = n, BC = m + n, AB = c, AC = b, AD = d$，斯图尔特定理为 $b^2m + c^2n = (m + n)(d^2 + mn)$；代入内角平分线比例即可得到上面的简式。

被平分的子角可能产生等腰辅助三角形，应先求另外两角，不能直接假设原边被平分。

Practice / 配套练习: AIME-2001-I-Q04 (WB p. 257), AIME-2002-I-Q10 (WB p. 314), AIME-2009-I-Q05 (WB p. 221), AIME-2010-I-Q15 (WB p. 320), AIME-2010-II-Q15 (WB p. 296), AIME-2012-I-Q12 (WB p. 321), AIME-2026-II-Q10 (WB p. 291)

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 2 / 方法 2 [GE05-M02]

Balance masses to locate cevian intersections

平衡质量定位劈线交点

Procedure

1. Assign positive endpoint masses inversely proportional to the side-division lengths.

2. Scale the assignments so shared vertices carry the same mass.

3. Add endpoint masses when replacing a side by its division point, then read a new ratio inversely.

通用步骤

1. 端点质量按分段长度的反比分配。

2. 统一缩放，使共用顶点质量一致。

3. 将两端质量相加作为分点质量，再按反比读取新的线段比例。

> Example. In triangle ABC, $D \in BC$ has $BD : DC = 2 : 3$, and $E \in CA$ has $CE : EA = 1 : 2$. If $P = AD \cap BE$, find $AP : PD$.

> 例题：三角形中 $BD : DC = 2 : 3, CE : EA = 1 : 2$，且 $P = AD \cap BE$，求 $AP : PD$。

Worked solution

1. Take masses $m_B = 3, m_C = 2$ to balance D. The ratio at E requires $m_A = 1$.

2. The mass at D is $m_B + m_C = 5$.

3. On AD, the balance relation is $AP/PD = m_D/m_A = 5$, so the ratio is $5 : 1$.

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 2 / 方法 2 [GE05-M02] (continued / 续)

例题解析

1. 由 D 取质量 $m_B = 3, m_C = 2$；E 的比例再要求 $m_A = 1$。

2. D 的质量为 $m_B + m_C = 5$。

3. 沿 AD，平衡关系给出 $AP/PD = m_D/m_A = 5$，故为 $5:1$。

> Masses add at a side point; segment lengths do not.

> 分点处相加的是质量，不是线段长度。

Practice / 配套练习: AIME-2002-II-Q13 (WB p. 231), AIME-2011-II-Q04 (WB p. 311)

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 3 / 方法 3 [GE05-M03]

Reflect across a bisector to create an isosceles midpoint

关于角平分线反射构造等腰中点

Procedure

1. Reflection across an internal angle bisector exchanges the two side rays.

2. Reflect a known endpoint onto the other side at the same distance from the vertex.

3. Transfer equal distances to the reflected points, then use perpendicular bisectors, midpoint theorems or parallel-line isosceles triangles.

通用步骤

1. 关于内角平分线反射会交换两条边射线。

2. 把已知端点映到另一边上距顶点相等的位置。

3. 将等距条件转移到反射点，再用垂直平分线、中点定理或平行线等腰三角形。

> Example. Angle A is bisected by line AM. Point B lies on one ray with $AB = 3$. Its reflection is $B\prime$. If M lies on AM and $MB = 4$, find $MB\prime$.

> 例题：直线 AM 平分角 A，一边上 $AB = 3$，将 B 反射得 $B\prime$。若 M 在平分线上且 $MB = 4$，求 $MB\prime$。

Worked solution

1. The reflection fixes A, M and sends B onto the other ray with $AB\prime = 3$.

2. Reflections preserve all distances, so $MB\prime = MB = 4$.

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 3 / 方法 3 [GE05-M03] (continued / 续)

3. Thus M lies on the perpendicular bisector of $BB\prime$, a useful linear condition when several bisectors meet a midpoint.

例题解析

1. 反射固定 A, M，并把 B 映到另一射线上，$AB\prime = 3$。

2. 反射保持距离，故 $MB\prime = MB = 4$。

3. 因此 M 在 $BB\prime$ 的垂直平分线上；多个角平分线通过中点时，这可给出线性条件。

Reflection preserves a ray's distance from the vertex, not its original coordinate.

反射保持到顶点的距离，不保持原坐标。

In a trapezoid, parallel bases often turn an angle-bisector intercept triangle into an isosceles triangle. The second bisector then becomes its median, placing the intersection at a midpoint.

To align coordinates with a bisector, place adjacent vertices at $(c\cos t, c\sin t), (a\cos t, -a\sin t)$. Locate the axis intercept by the bisector theorem before intersecting a perpendicular cut.

梯形中，平行底边常使角平分线截出的三角形成为等腰；另一条平分线于是成为中线，使交点成为中点。

若把平分线作为坐标轴，可将邻接两顶点写为 $(c\cos t, c\sin t), (a\cos t, -a\sin t)$。先用平分线定理定位轴上截点，再求垂直截线。

Practice / 配套练习: AIME-2003-I-Q15 (WB p. 387), AIME-2011-I-Q04 (WB p. 310), AIME-2022-I-Q03 (WB p. 312), AIME-2022-II-Q11 (WB p. 327)

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 4 / 方法 4 [GE05-M04]

Cancel area in inradius, altitude and semiperimeter equations

在内切圆、三角高及半周长式中消去面积

Procedure

1. Write $K = rs$ and $K = ah/2$ for the same triangle.

2. Ratios such as $r/h = a/(2s)$ eliminate area without computing radicals.

3. For equal inradii in adjacent triangles, replace radius equality by area divided by semiperimeter equality.

4. If tangent lengths are used, combine $K = rs$ with Heron and verify the circle is an incircle.

通用步骤

1. 对同一三角形写 $K = rs$ 与 $K = ah/2$。

2. 用 $r/h = a/(2s)$ 等比例消去面积，无须先计算根式。

3. 相邻三角形内切圆相等时，改写为面积除以半周长相等。

4. 若使用切线长，可联立 $K = rs$ 与海伦公式，并确认所用圆确为内切圆。

> Example. A triangle has sides $5, 6, 7$. Through its incenter draw a line parallel to the side of length six. Find the resulting interior parallel chord.

> 例题：三角形三边为 $5, 6, 7$，过内心作平行于六长边的直线，求内部平行截段长。

Worked solution

1. The semiperimeter is nine. If altitude to the six-side is h, then $r/h = 6/(2 \cdot 9) = 1/3$.

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 4 / 方法 4 [GE05-M04] (continued / 续)

2. The upper similar triangle has height $h - r = 2h/3$.

3. Its base chord is therefore $(2/3)6 = 4$.

例题解析

1. 半周长为九。以六长边为底的高为 h，则 $r/h = 6/(2 \cdot 9) = 1/3$。

2. 上方相似小三角形高为 $h - r = 2h/3$。

3. 所以平行底截段长为 $(2/3)6 = 4$。

Use the semiperimeter of the correct subtriangle.

应使用对应子三角形的半周长。

Practice / 配套练习: AIME-2001-I-Q07 (WB p. 224), AIME-2001-II-Q07 (WB p. 313), AIME-2009-I-Q12 (WB p. 319), AIME-2010-I-Q15 (WB p. 320), AIME-2005-I-Q15 (WB p. 294)

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 5 / 方法 5 [GE05-M05]

Tangent lengths are semiperimeter differences

切线长等于半周长之差

Procedure

1. For sides a, b, c and semiperimeter s, the two tangent lengths from A equal $s - a$.

2. Use these values to locate incircle contact points on a common side.

3. Connect a vertex tangent length to the half-angle by $r = (s - a)\tan(A/2)$.

4. When chords or a second tangent meet a triangle side, retain the actual segment order before adding lengths.

通用步骤

1. 三边为 a, b, c、半周长为 s 时，从 A 出发的两切线长为 $s - a$。

2. 利用这些长度定位公共边上的内切圆切点。

3. 由 $r = (s - a)\tan(A/2)$ 将顶点切线长联系到半角。

4. 弦或另一切线与边相交时，先明确线段次序，再作长度相加。

> Example. The incircle of a 5–12–13 right triangle touches the hypotenuse. Find the two hypotenuse pieces.

> 例题：五、十二、十三直角三角形的内切圆与斜边相切，求斜边两分段长。

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 5 / 方法 5 [GE05-M05] (continued / 续)

Worked solution

1. The semiperimeter is fifteen. At the endpoints of the hypotenuse the opposite side lengths are twelve and five.

2. The tangent pieces are therefore $15-12=3$ and $15-5=10$.

3. They sum to thirteen, checking the hypotenuse length. Equal tangents transfer the same values onto the adjacent legs.

例题解析

1. 半周长为十五，斜边两端对应的对边长为十二与五。

2. 所以切线分段为 $15-12=3$ 与 $15-5=10$。

3. 两段和为十三，核实斜边长度；等切线可把相同数值转移至邻接直角边。

Tangent-length addition differs from secant-power multiplication.

切线段相加与割线点幂相乘不可混用。

A circle row along a triangle base is controlled by the same half-angle cotangents as its incircle. Compare two rows to recover the common base coefficient without solving either base angle.

三角形底边圆链由与内切圆相同的两个半角余切控制。比较两种圆链可求共同底边系数，无需分别求两个底角。

Practice / 配套练习: AIME-2007-II-Q09 (WB p. 315), AIME-2018-II-Q14 (WB p. 301), AIME-2022-I-Q11 (WB p. 308), AIME-2024-I-Q08 (WB p. 338)

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 6 / 方法 6 [GE05-M06]

Circumradius converts a side into a sine

外接圆半径把边长转化为正弦

Procedure

1. Use $a = 2R\sin A$ or $R = abc/(4K)$ to connect a center radius with triangle metrics.

2. For an equidistant apex over a triangle plane, its projection is the base circumcenter.

3. Retain the correct half-angle and diagonal choices in a rhombus, and verify the circumcenter location when a filled region is involved.

通用步骤

1. 使用 $a = 2R\sin A$ 或 $R = abc/(4K)$ 联系圆心半径与三角形度量。

2. 到三角形三个顶点等距的空间顶点，其投影为底面外心。

3. 菱形中保留正确半角与对角线选择；涉及填满区域时核实外心位置。

> Example. A triangle has sides $13, 14, 15$. Find its circumradius.

> 例题：三角形三边为 $13, 14, 15$，求外接圆半径。

Worked solution

1. The semiperimeter is twenty-one, so Heron gives $K = \sqrt{21 \cdot 8 \cdot 7 \cdot 6} = 84$.

2. Use $R = abc/(4K) = 13 \cdot 14 \cdot 15/(4 \cdot 84)$.

3. This simplifies to $R = 65/8$. Since $15^2 < 13^2 + 14^2$, the triangle is acute and its circumcenter is interior.

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 6 / 方法 6 [GE05-M06] (continued / 续)

例题解析

1. 半周长为二十一，海伦公式给出 $K=\sqrt{21\cdot 8\cdot 7\cdot 6}=84$。

2. 使用 $R=abc/(4K)=13\cdot 14\cdot 15/(4\cdot 84)$。

3. 化简为 $R=65/8$；由于 $15^2<13^2+14^2$，三角形为锐角，外心在内部。

A circumradius is not generally the distance from the centroid to a vertex.

外接圆半径一般不是重心到顶点的距离。

Practice / 配套练习: AIME-2000-II-Q12 (WB p. 430), AIME-2003-II-Q07 (WB p. 268), AIME-2017-I-Q04 (WB p. 443)

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 7 / 方法 7 [GE05-M07]

Incenter distances are half-angle right triangles

内心距离来自半角直角三角形

Procedure

1. Drop a radius to a side: $AI = r/\sin(A/2)$ and the adjacent tangent length is $r\cot(A/2)$.

2. The internal bisectors meet with $\angle BIC = 90^\circ + A/2$.

3. In an isosceles triangle, put the center on its symmetry axis and use one half-triangle.

4. For excircles use $r_a = K/(s-a)$; preserve external-center directions.

通用步骤

1. 向边作半径，得 $AI = r/\sin(A/2)$，邻接切线长为 $r\cot(A/2)$。

2. 内角平分线满足 $\angle BIC = 90^\circ + A/2$。

3. 等腰三角形中把内心放在对称轴上，使用半个三角形。

4. 旁切圆使用 $r_a = K/(s-a)$，并保留外部圆心方向。

> Example. In a triangle with inradius two and angle $A = 60^\circ$, find AI and $\angle BIC$.

> 例题：三角形内切圆半径二，$A = 60^\circ$，求 AI 与 $\angle BIC$。

Worked solution

1. At a contact point, the radius is perpendicular to the side and the angle at A is 30°.

2. Therefore $AI = 2/\sin 30^\circ = 4$.

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 7 / 方法 7 [GE05-M07] (continued / 续)

3. The bisector-angle identity gives $\angle BIC = 90^\circ + 30^\circ = 120^\circ$.

例题解析

1. 切点处半径垂直边，A 处夹角为 30°。

2. 所以 $AI = 2/\sin 30^\circ = 4$。

3. 平分线交角恒等式给出 $\angle BIC = 120^\circ$。

An angle between unoriented bisector lines may be supplementary to the angle between their inward rays.

无向角平分线的夹角，可能是向内射线交角的补角。

Three equal corner circles have centers homothetic to the vertices about the incenter, with ratio $(r - x)/r$. In an isosceles triangle with base a, equal sides b and altitude h, use $r = ah/(a + 2b)$, base exradius $ah/(2b - a)$ and other exradii h.

三个等半径顶角圆的圆心，是三顶点以内心为中心、比例 $(r - x)/r$ 的位似像。等腰三角形底边为 a、腰为 b、高为 h 时，可用 $r = ah/(a + 2b)$、底边旁切圆半径 $ah/(2b - a)$、另两旁切圆半径 h。

Practice / 配套练习: AIME-2007-II-Q15 (WB p. 343), AIME-2009-I-Q15 (WB p. 295), AIME-2009-II-Q10 (WB p. 371), AIME-2015-I-Q11 (WB p. 323), AIME-2018-I-Q13 (WB p. 324), AIME-2019-I-Q11 (WB p. 346), AIME-2026-II-Q08 (WB p. 318)

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 8 / 方法 8 [GE05-M08]

Euler’s center identity links inradius and circumradius

欧拉圆心恒等式联系内外接半径

Procedure

1. For incenter I and circumcenter O, use $OI^2 = R^2 - 2Rr$.

2. Combine a perpendicular-center condition with a right triangle involving a vertex on the circumcircle.

3. Recover a half-angle from $AI = r/\sin(A/2)$, then use $K = rs = bc\sin A/2$.

通用步骤

1. 对内心 I、外心 O，使用 $OI^2 = R^2 - 2Rr$。

2. 将圆心垂直条件与包含圆上顶点的直角三角形结合。

3. 由 $AI = r/\sin(A/2)$ 求半角，再用 $K = rs = bc\sin A/2$。

> Example. A triangle has $R = 5, r = 2$ and $IA \perp OI$. Find AI.

> 例题：三角形有 $R = 5, r = 2$ 且 $IA \perp OI$，求 AI。

Worked solution

1. Euler gives $OI^2 = 25 - 20 = 5$.

2. In right triangle OIA, $AI^2 = OA^2 - OI^2 = 25 - 5 = 20$.

3. Thus $AI = 2\sqrt{5}$ and $\sin(A/2) = 2/(2\sqrt{5}) = 1/\sqrt{5}$, consistent with a positive angle.

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 8 / 方法 8 [GE05-M08] (continued / 续)

例题解析

1. 欧拉恒等式给出 $OI^2 = 25 - 20 = 5$。

2. 直角三角形 OIA 中，$AI^2 = OA^2 - OI^2 = 20$。

3. 所以 $AI = 2\sqrt{5}$，且 $\sin(A/2) = 1/\sqrt{5}$，符合正角条件。

Write the right triangle with its actual right-angle vertex.

必须按实际直角顶点写勾股关系。

Practice / 配套练习: AIME-2024-II-Q10 (WB p. 317)

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 9 / 方法 9 [GE05-M09]

The bisector’s arc midpoint is equidistant from the incenter and endpoints

角平分线弧中点到内心及弧端点等距

Procedure

1. The internal bisector from A meets the circumcircle at the midpoint M of the arc BC not containing A.

2. Angle chasing gives $MB = MC = MI$ for the incenter I.

3. Use these equal distances with secant power, similar triangles or an antipodal point.

4. Keep the order of the bisector’s circle, side and incenter intersections explicit.

通用步骤

1. 从 A 出发的内角平分线交外接圆于不含 A 的弧 BC 中点 M。

2. 追角可得 $MB = MC = MI$，其中 I 为内心。

3. 将等距与割线点幂、相似三角形或对径点结合。

4. 明确角平分线上圆交点、边交点及内心的次序。

> Example. An internal bisector from C meets side AB at L, then its circumcircle at M. The order is C, I, L, M, with $LI = 1, LM = 2$. Find MC if $MA^2 = LM \cdot MC$.

> 例题：从 C 的内角平分线交 AB 于 L、再交外接圆于 M，点序为 C, I, L, M，且 $LI = 1, LM = 2$。已知 $MA^2 = LM \cdot MC$，求 MC。

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 9 / 方法 9 [GE05-M09] (continued / 续)

Worked solution

1. By the arc-midpoint lemma, $MA = MI$. The given order gives $MI = LI + LM = 3$.

2. The secant relation becomes $9 = 2MC$.

3. Thus $MC = 9/2$, and $CI = MC - MI = 3/2 > 0$ checks the stated order.

例题解析

1. 弧中点引理给出 $MA = MI$，由点序得 $MI = LI + LM = 3$。

2. 割线关系成为 $9 = 2MC$。

3. 所以 $MC = 9/2$，且 $CI = MC - MI = 3/2 > 0$，核实点序。

An antipode turns a cyclic right angle into a line relation; choose the correct arc midpoint first.

对径点可将圆内直角转为直线关系，但须先选对弧中点。

To prove $MA = MI$ in the version with bisector from C and arc midpoint M of AB, let half-angles at A, C be u, v. Both $\angle MAI$ and $\angle MIA$ equal $u + v$, making triangle AMI isosceles.

对从 C 平分至弧 AB 中点 M 的版本，设 A, C 半角为 u, v。则 $\angle MAI$ 与 $\angle MIA$ 均为 $u + v$，故 AMI 等腰，得到 $MA = MI$。

Practice / 配套练习: AIME-2012-II-Q15 (WB p. 297), AIME-2014-II-Q14 (WB p. 322), AIME-2016-I-Q06 (WB p. 316), AIME-2020-I-Q13 (WB p. 325)

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 10 / 方法 10 [GE05-M10]

Centroid, circumcenter and nine-point circle have exact coordinate roles

重心、外心与九点圆具有精确坐标角色

Procedure

1. The centroid is the average of the three vertex vectors and divides each median in ratio $2:1$.

2. Find a circumcenter from perpendicular-bisector equations, and use symmetry to identify it with the centroid only in special triangles.

3. The side midpoints and altitude feet lie on the nine-point circle; convert its arcs using inscribed angles.

4. When right-angle circles transfer directed angles, finish with a ray and supplement check.

通用步骤

1. 重心为三个顶点向量的平均，按 $2:1$ 分每条中线。

2. 外心由垂直平分线方程确定，只有特殊三角形才能凭对称性认定其与重心相同。

3. 三边中点及三条高的垂足共九点圆，可用圆周角换算弧。

4. 利用直角圆传递有向角后，最后仍须检验射线与补角分支。

> Example. Triangle vertices are $(0,0),(6,0),(0,6)$. Find its centroid and circumcenter.

> 例题：三角形顶点为 $(0,0),(6,0),(0,6)$，求重心与外心。

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 10 / 方法 10 [GE05-M10] (continued / 续)

Worked solution

1. The average vertex vector gives centroid $G = (2, 2)$.

2. The two axis legs have perpendicular bisectors $x = 3, y = 3$, so circumcenter $O = (3, 3)$.

3. They are distinct. As a right-triangle check, O is the midpoint of the hypotenuse.

例题解析

1. 顶点平均给出重心 $G = (2, 2)$。

2. 两条轴上直角边的垂直平分线为 $x = 3, y = 3$，所以外心 $O = (3, 3)$。

3. 两中心不同；直角三角形的斜边中点为外心，也核实了结果。

Directed-angle equality modulo 180° does not determine the ordinary angle without ray information.

模 180° 的有向角相等，不能在缺少射线信息时直接确定普通角。

Practice / 配套练习: AIME-2002-I-Q13 (WB p. 293), AIME-2011-II-Q13 (WB p. 395), AIME-2013-II-Q04 (WB p. 356), AIME-2021-II-Q14 (WB p. 307), AIME-2025-II-Q05 (WB p. 282)

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 11 / 方法 11 [GE05-M11]

Isogonal cevians give a sine-area product relation

等角劈线给出正弦面积乘积关系

Procedure

1. For cevians from one vertex, write each pair of adjacent subtriangle areas using a common pair of side lengths and angle sines.

2. Equal reflected angles exchange the sine factors, causing the product of opposite area ratios to simplify.

3. Use the remaining side ratios or area totals to solve, retaining which cevian is internal.

通用步骤

1. 对同顶点劈线，用共同侧边长度及夹角正弦表示相邻子三角形面积。

2. 关于角平分线对称的角交换正弦因子，使两组面积比乘积化简。

3. 再用剩余边比或面积总和求解，并保留内部劈线位置。

> Example. In triangle ABC, $AB = 2AC$. Rays AD, AE inside angle A satisfy $\angle BAD = \angle EAC$. Show $(BD/DC)(BE/EC) = 4$.

> 例题：三角形中 $AB = 2AC$，内部射线 AD, AE 满足 $\angle BAD = \angle EAC$。证明 $(BD/DC)(BE/EC) = 4$。

Worked solution

1. By common-altitude area ratios, $BD/DC = [ABD]/[ACD] = (AB/AC)\sin\angle BAD/\sin\angle DAC$.

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 11 / 方法 11 [GE05-M11] (continued / 续)

2. Similarly $BE/EC = (AB/AC)\sin\angle BAE/\sin\angle EAC$. Isogonality makes $\angle BAE = \angle DAC$.

3. Multiplication cancels both sine ratios, giving $(AB/AC)^2 = 4$.

例题解析

1. 由等高面积比，$BD/DC = (AB/AC)\sin\angle BAD/\sin\angle DAC$。

2. 同理 $BE/EC = (AB/AC)\sin\angle BAE/\sin\angle EAC$，而等角关系使 $\angle BAE = \angle DAC$。

3. 相乘后正弦比相消，得到 $(AB/AC)^2 = 4$。

Two equal-looking angles must be located at the same vertex before applying isogonality.

应用等角共轭前，应确认所比较的角位于同一顶点。

Practice / 配套练习: AIME-2005-II-Q14 (WB p. 274)

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 12 / 方法 12 [GE05-M12]

A midpoint perimeter bisector is parallel to the opposite angle bisector

过边中点的周长平分线平行于对角平分线

Procedure

1. Equal polygon perimeters after a cut mean the two original boundary routes between cut endpoints have equal length.

2. Through a side midpoint, construct the second endpoint using the difference of the adjacent side lengths.

3. Write its displacement as a scalar multiple of the sum of the two unit side directions at the opposite vertex.

4. The sum points along the internal angle bisector; convert the acute angle between two such lines with a supplement check.

通用步骤

1. 切开后的两多边形周长相等，等价于切点之间原边界两条路线等长。

2. 从边中点出发，利用相邻边长之差确定第二切点。

3. 将该切线向量写成对角顶点两边单位方向向量之和的倍数。

4. 单位向量和沿内角平分线，最后换算两条此类直线的锐夹角并检验补角。

> Example. Triangle ABC has $CA = 5, CB = 7$. Let M be the midpoint of AB, and $X \in CB$ with $CX = 1$. Prove MX bisects perimeter and is parallel to the C-bisector.

Section 2.5 Angle bisectors & triangle centers

角平分线与三角形心

Method 12 / 方法 12 [GE05-M12] (continued / 续)

> 例题：三角形中 $CA=5, CB=7$，M 为 AB 中点，$X \in CB$ 且 $CX=1$。证明 MX 平分周长并平行于 C 的角平分线。

Worked solution

1. Boundary routes $M-A-C-X$ and $M-B-X$ have lengths $AB/2+5+1$ and $AB/2+6$, hence equal.

2. With unit vectors u, v from C toward A, B, $M=(5u+7v)/2$ and $X=v$.

3. Thus $M-X=(5/2)(u+v)$, parallel to the internal bisector at C. Adding the common cut MX preserves equality of the two new perimeters.

例题解析

1. 边界路线 $M-A-C-X$ 与 $M-B-X$ 的长度分别为 $AB/2+5+1$ 与 $AB/2+6$，相等。

2. 以从 C 指向 A, B 的单位向量为 u, v，则 $M=(5u+7v)/2, X=v$。

3. 所以 $M-X=(5/2)(u+v)$，平行于 C 的内角平分线；两边都加入共同切段 MX 后，新周长仍相等。

Perimeter bisection and area bisection are different conditions.

平分周长与平分面积是不同条件。

Practice / 配套练习: AIME-2022-I-Q14 (WB p. 326)

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 1 / 方法 1 [GE06-M01]

Tangency has an orientation: external circles have center distance $r_1 + r_2$, internally tangent circles have $|r_1 - r_2|$. Identify the correct center ray, side of each tangent and radius sign before eliminating variables.

相切具有方向：外切圆心距为 $r_1 + r_2$，内切为 $|r_1 - r_2|$。消元之前先确定圆心射线、切线两侧及半径正性。

Center-distance equations need ray and containment checks

圆心距离方程需要射线与容纳检验

Procedure

1. Use symmetry or angle bisectors to locate the unknown center with as few variables as possible.

2. Write each tangency as a radius sum or difference, keeping absolute values until the center order is known.

3. Solve and restore all unsquared signs, positive radii, segment contact positions and containment inequalities.

通用步骤

1. 利用对称或角平分线，以尽可能少的变量定位未知圆心。

2. 将各相切写为半径和或差；圆心次序未明时保留绝对值。

3. 求解后补查未平方符号、正半径、线段切点位置及容纳不等式。

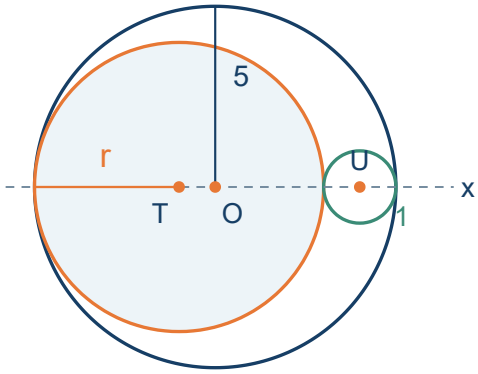
> Example. A radius-5 circle is centered at $(0, 0)$ and contains a radius-1 circle centered at $(4, 0)$. A third circle has center on the negative x axis, is internally tangent to the large circle and externally tangent to the unit circle. Find its radius.

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 1 / 方法 1 [GE06-M01] (continued / 续)

例题：圆心 $(0,0)$、半径 5 的圆内有圆心 $(4,0)$、半径 1 的圆。第三圆圆心在负 x 轴，与大圆内切、与单位圆外切，求其半径。



O = (0,0) U = (4,0) T is on −x / T 在负半轴

Worked solution

1. Write its center as $(-t,0)$ with $t>0$ and its radius as $r>0$. Internal tangency gives $t+r=5$.

2. External tangency gives $4+t=r+1$. Together these yield $t=1, r=4$.

3. The circle is contained in the large one because center distance plus radius is $1+4=5$; its center distance to the unit circle is five, equal to the radius sum.

例题解析

1. 设圆心为 $(-t,0)$，其中 $t>0$，半径 $r>0$。内切给出 $t+r=5$。

2. 外切给出 $4+t=r+1$，联立得 $t=1, r=4$。

3. 圆心距加半径为 $1+4=5$，确在大圆内；到单位圆的圆心距为五，等于半径和。

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 1 / 方法 1 [GE06-M01] (continued / 续)

A squared distance equation cannot distinguish internal from external tangency by itself.

仅凭距离平方方程无法区分内切与外切。

If a center lies on a vertex angle bisector at distance d from the vertex, its radius is $d\sin(A/2)$. When the circumcenter lies on the same line, preserve both branches of $|R-d|=R-r$.

In a symmetric isosceles triangle, the incenter and excenters have explicit axial coordinates. Internal tangency may require the new circle to contain an excircle rather than lie inside it; decide from center distance before choosing the radius difference.

圆心在顶角平分线上，距顶点为 d 时，半径为 $d\sin(A/2)$。若外心也在此线上，须保留 $|R-d|=R-r$ 的两个分支。

等腰三角形的内心、旁心可取明确轴向坐标。内切时，新圆可能包含旁切圆，而非位于旁切圆内；应先比较圆心距再选半径差。

Practice / 配套练习: AIME-2004-II-Q12 (WB p. 342), AIME-2008-II-Q11 (WB p. 344), AIME-2009-II-Q05 (WB p. 330), AIME-2014-II-Q08 (WB p. 335), AIME-2019-I-Q11 (WB p. 346), AIME-2022-I-Q08 (WB p. 336), AIME-2025-II-Q06 (WB p. 339)

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 2 / 方法 2 [GE06-M02]

Equal-circle arrays are polygons of centers

等圆排列转化为圆心多边形

Procedure

1. Replace each tangent circle by its center, with neighbor distance twice the radius.

2. For staggered rows use equilateral-center triangles; for a ring use a regular-center polygon.

3. Add radius margins to obtain enclosing dimensions, then subtract disk areas only after verifying disjoint interiors.

通用步骤

1. 将相切圆换成圆心，相邻圆心距为半径两倍。

2. 交错行使用等边圆心三角形，环形排列使用正圆心多边形。

3. 求外框尺寸时加入半径边距；扣圆盘面积前须确认内部互不重叠。

> Example. Six unit circles surround a central unit circle, all neighbors tangent. Find the radius of the smallest concentric enclosing circle.

> 例题：六个单位圆围绕一个中心单位圆，相邻圆均相切，求同心最小外包圆半径。

Worked solution

1. Each outer center is distance two from the central center.

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 2 / 方法 2 [GE06-M02] (continued / 续)

2. Neighboring outer centers also have separation two, so they form a regular hexagon of circumradius two.

3. The farthest point of each outer disk is one more unit away. The enclosing radius is 3, and it is attained on each outward radial line.

例题解析

1. 每个外圆圆心距中心圆心为二。

2. 相邻外圆心间距也为二，所以形成外接圆半径二的正六边形。

3. 每个外圆盘最远点再向外一单位，故外包半径为 3，在各向外半径上取到。

In a regular n-circle ring, $2r = 2d\sin(\pi/n)$ relates circle radius r and center-ring radius d.

正 n 圆环中，圆半径 r 与圆心环半径 d 满足 $2r = 2d\sin(\pi/n)$。

Practice / 配套练习: AIME-2002-I-Q02 (WB p. 328), AIME-2005-I-Q01 (WB p. 329)

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 3 / 方法 3 [GE06-M03]

Homothety controls common tangents and opposite circle chords

位似控制公切线与相反方向圆弦

Procedure

1. For unequal radii, common external tangents meet at the external similarity center, dividing center distances in the radius ratio. Equal-radius external tangents are parallel and have no finite similarity center.

2. Common internal tangents pass through the internal similarity center; include its directed location on the center segment.

3. At an external circle-contact point, a negative homothety maps a secant chord to the other circle; use signed distances when areas are compared.

通用步骤

1. 半径不等时，两条公外切线交于外位似中心，到两圆心距离比等于半径比；半径相等时公外切线平行，不存在有限外位似中心。

2. 公内切线经过内位似中心，应保留其在圆心线段上的有向位置。

3. 两圆外切点处，负位似将一圆割弦映至另一圆；比较面积时须使用有向距离。

> Example. Externally tangent circles have radii $1, 4$. Find the distance from the smaller center to the intersection of their common external tangents.

> 例题：两外切圆半径为 $1, 4$，求小圆圆心到两条公外切线交点的距离。

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 3 / 方法 3 [GE06-M03] (continued / 续)

Worked solution

1. The centers are five units apart. The external homothety center lies beyond the smaller circle.

2. If its distance to the smaller center is d, its distance to the larger center is $d+5$, and $(d+5)/d=4$.

3. Thus $d=5/3$. The tangent half-angle satisfies $\sin\theta=1/d=3/5$, consistent for both circles.

例题解析

1. 两圆心距为五，外位似中心在小圆外侧。

2. 设到小圆心距离为 d，则到大圆心为 $d+5$，比例给出 $(d+5)/d=4$。

3. 所以 $d=5/3$；切线半角满足 $\sin\theta=1/d=3/5$，对两圆均一致。

An external-contact homothety has negative scale, so corresponding chord points lie on opposite rays.

以外切点为中心的位似比为负，对应弦点位于相反射线。

If chord lengths from the contact point scale by R/r, equality of two triangle areas forces the opposite inverse ratio between their vertex-to-secant distances. A same-side condition determines whether those distances have the same signed ratio.

若从切点量起的弦长按 R/r 缩放，两个三角形面积相等便要求相应顶点到割线距离满足逆比例。同侧条件决定其有向比例是否同号。

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 3 / 方法 3 [GE06-M03] (continued / 续)

Practice / 配套练习: AIME-2006-II-Q09 (WB p. 333), AIME-2015-II-Q15 (WB p. 345), AIME-2022-II-Q07 (WB p. 337)

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 4 / 方法 4 [GE06-M04]

Half-angle tangent lengths solve tangential polygons and circle chains

半角切线长解外切多边形与圆链

Procedure

1. Equal tangents give the two side portions belonging to each polygon vertex.

2. For tangent length t and inradius r, write $\tan(A/2) = r/t$; the half-angle sum is $(n-2)\pi/2$.

3. Use tangent identities with branch checks; the sum of positive arctangents increases strictly with r.

4. For a row of equal circles along a triangle base, the end-center offsets are $r\cot(B/2), r\cot(C/2)$ and neighbor gaps are $2r$.

通用步骤

1. 由同点等切线长求各多边形顶点所占两段。

2. 切线长为 t、内切圆半径为 r 时，$\tan(A/2) = r/t$，半角总和为 $(n-2)\pi/2$。

3. 使用正切恒等式并检验分支；正反正切之和随 r 严格递增。

4. 三角形底边圆链的两端圆心偏移为 $r\cot(B/2), r\cot(C/2)$，相邻圆心间距为 $2r$。

> Example. A triangle has side lengths $8, 10, 12$. Use half-angle tangent lengths to find its inradius.

> 例题：三角形三边为 $8, 10, 12$，用半角切线长求内切圆半径。

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 4 / 方法 4 [GE06-M04] (continued / 续)

Worked solution

1. The semiperimeter is fifteen, so the three vertex tangent lengths are $3, 5, 7$.

2. The three half-angles sum to $\pi/2$. The tangent-sum denominator therefore vanishes: $1 - r^2(1/15 + 1/21 + 1/35) = 0$.

3. The parenthesis is $1/7$, so $r = \sqrt{7}$. The positive half-angle sum cannot be $3\pi/2$, since each half-angle is strictly below $\pi/2$.

例题解析

1. 半周长为十五，三个顶点切线长为 $3, 5, 7$。

2. 三个半角和为 $\pi/2$，因此正切和公式分母为零：$1 - r^2(1/15 + 1/21 + 1/35) = 0$。

3. 括号等于 $1/7$，故 $r = \sqrt{7}$。每个半角严格小于 $\pi/2$，所以正半角和不可能为 $3\pi/2$。

A polynomial produced by a tangent identity can retain a wrong angle-sum branch.

正切恒等式产生的多项式可能保留错误的角度和分支。

An N-circle row of radius r along base a satisfies $a = r[\cot(B/2) + \cot(C/2)] + 2r(N - 1)$. The incircle corresponds to $N = 1$. For an odd-sided tangential polygon, the linear side equations uniquely recover all vertex tangent lengths.

底边长为 a，半径 r 的 N 圆链满足 $a = r[\cot(B/2) + \cot(C/2)] + 2r(N - 1)$；内切圆对应 $N = 1$。奇数边外切多边形的边长线性方程可唯一确定全部顶点切线长。

Practice / 配套练习: AIME-2000-II-Q10 (WB p. 331), AIME-2002-II-Q14 (WB p. 340), AIME-2007-I-Q09 (WB p. 334), AIME-2020-II-Q13 (WB p. 347), AIME-2024-I-Q08 (WB p. 338)

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 5 / 方法 5 [GE06-M05]

Eliminate an unknown tangent radius to obtain an ellipse

消去未知切圆半径得到椭圆

Procedure

1. For one internal and one external tangency write $PF_1 = R - r, PF_2 = s + r$.

2. The radius cancels to $PF_1 + PF_2 = R + s$, identifying the center ellipse.

3. Check $r = R - PF_1 > 0$ and then impose a line-intersection discriminant if slope or direction is optimized.

通用步骤

1. 一内切一外切时，写为 $PF_1 = R - r, PF_2 = s + r$。

2. 消去半径得 $PF_1 + PF_2 = R + s$，即圆心椭圆。

3. 检验 $r = R - PF_1 > 0$；若优化斜率或方向，再用直线与椭圆的判别式。

> Example. Circles centered at $(-2, 0), (2, 0)$ have radii $7, 1$. Describe centers of circles internally tangent to the first and externally tangent to the second.

> 例题：两圆圆心为 $(-2, 0), (2, 0)$，半径为 $7, 1$，求与前者内切、与后者外切的圆之圆心轨迹。

Worked solution

1. The center distances sum to eight, so the ellipse has $a = 4, c = 2, b^2 = 12$.

2. Its equation is $x^2/16 + y^2/12 = 1$. Distance to the first focus ranges from two to six.

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 5 / 方法 5 [GE06-M05] (continued / 续)

3. Thus $r = 7 - PF_1$ ranges from one to five and is always positive; every ellipse point gives the required internal/external tangencies.

例题解析

1. 圆心到两焦点距离和为八，所以 $a = 4, c = 2, b^2 = 12$。

2. 椭圆方程为 $x^2/16 + y^2/12 = 1$，到第一焦点距离介于二与六。

3. 因此 $r = 7 - PF_1$ 介于一与五且始终正，椭圆各点都满足所需内切与外切。

The ellipse equation is necessary; the recovered radius makes it sufficient.

椭圆方程只先给出必要性，恢复的半径条件才保证充分性。

Practice / 配套练习: AIME-2005-II-Q15 (WB p. 419)

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 6 / 方法 6 [GE06-M06]

Power differences measure signed distance from the radical axis

圆幂差度量到根轴的有向距离

Procedure

1. Expand the powers of a point in both circles; squared point coordinates cancel.

2. The resulting affine function vanishes on the common chord line and has gradient magnitude twice the center distance.

3. Use a chord angle or center-line ratio to express the point's distance from that line, retaining its side.

通用步骤

1. 展开一点对两圆的点幂，点坐标平方项会消去。

2. 所得仿射函数在公共弦直线上为零，其梯度长度是圆心距的两倍。

3. 用弦所对角或圆心线比例表示到该直线的距离，并保留所在侧。

> Example. Two intersecting circles have radii $5, 3$. A radius-2 circle is externally tangent to both, and its center is distance one from their radical axis. Find the two original centers' distance.

> 例题：两相交圆半径为 $5, 3$，另一个半径 2 的圆与两者外切，其圆心到原两圆根轴距离为一。求原两圆心距。

Worked solution

1. The new center has powers $(5+2)^2 - 5^2 = 24$ and $(3+2)^2 - 3^2 = 16$.

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 6 / 方法 6 [GE06-M06] (continued / 续)

2. Their difference is eight. If original center distance is d, its magnitude is also $2d \cdot 1$.

3. Hence $d = 4$, consistent with the intersection condition $|5 - 3| < d < 5 + 3$.

例题解析

1. 新圆心的两圆幂分别为 $(5+2)^2 - 5^2 = 24$ 与 $(3+2)^2 - 3^2 = 16$。

2. 差为八；若原圆心距为 d，幂差绝对值又为 $2d \cdot 1$。

3. 故 $d = 4$，并满足相交条件 $|5 - 3| < d < 5 + 3$。

Use a signed affine distance when comparing points on opposite sides of the chord line.

比较公共弦两侧的点时，应使用有向仿射距离。

Along the line of the two centers, signed distance to the radical axis is affine. A prescribed division point therefore has the same weighted combination of the two center distances; convert its half-chord length by $\sqrt{R^2 - d^2}$.

沿两圆心直线，到根轴的有向距离为仿射函数。按比例分点的距离是两圆心距离的同权线性组合，再由 $\sqrt{R^2 - d^2}$ 求半弦长。

Practice / 配套练习: AIME-2005-II-Q08 (WB p. 332), AIME-2021-I-Q13 (WB p. 348)

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 7 / 方法 7 [GE06-M07]

Offset corner circles are an incenter homothety

内移角圆组成以内心为中心的位似

Procedure

1. A circle of radius x tangent to a vertex's two sides has its center on the vertex-to-incenter line.

2. For corner circles inside the triangle with $0 < x < r$, the three centers are images of the vertices under scale $(r - x)/r$ about the incenter, where r is the triangle inradius.

3. A fourth equal circle tangent to all three has its center at their circumcenter; impose circumradius $2x$ and verify the fourth disk stays inside.

通用步骤

1. 半径 x 的圆与某顶点两邻边相切时，圆心在顶点到内心的直线上。

2. 若三个角圆均在三角形内且 $0 < x < r$，圆心是三顶点以内心为中心、比例 $(r - x)/r$ 的位似像，其中 r 为原内切圆半径。

3. 第四等圆与三圆均外切时，其圆心为该圆心三角形外心，故外接半径为 $2x$；最后核实第四圆也在内部。

> Example. In an equilateral triangle of inradius one, three equal circles inside the triangle are each tangent to the two sides at a different vertex. A fourth circle of the same radius lies inside and is externally tangent to all three. Find the common radius.

> 例题：内切圆半径为一的等边三角形内，有三个等圆分别与三个顶点的两邻边相切。第四个同半径圆也在三角形内，并与前三圆均外切，求公共半径。

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 7 / 方法 7 [GE06-M07] (continued / 续)

Worked solution

1. The original circumradius is two. Corner-center homothety gives circumradius $2(1-x)$.

2. Tangency to the fourth equal circle requires $2(1-x)=2x$, hence $x=1/2$.

3. The fourth center is the triangle center, distance one from every side, so its radius-$1/2$ disk lies inside; the corner circles also fit.

例题解析

1. 原外接圆半径为二，角圆心位似给出外接半径 $2(1-x)$。

2. 第四等圆外切要求 $2(1-x)=2x$，所以 $x=1/2$。

3. 第四圆心为原中心，距三边均一，故半径 $1/2$ 的圆在内部；三个角圆也可容纳。

Equal center distances alone do not prove that the fourth disk fits inside the triangle.

到三个圆心等距本身，并不能证明第四圆盘位于三角形内。

Practice / 配套练习: AIME-2007-II-Q15 (WB p. 343)

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 8 / 方法 8 [GE06-M08]

Rolling and nested tangency use radius sums or differences

滚动与嵌套相切使用半径和或差

Procedure

1. For a circle rolling externally around a fixed circle, the moving center follows radius $R + r$.

2. For fixed-circle radius R, moving radius r, ground-frame spin ϕ, and center sweep θ, no slip gives $r\,d\phi = (R + r)\,d\theta$.

3. For nested internal tangency, center displacement has length $R - r$; if radii shrink to zero, the nested disks have at most one common point.

通用步骤

1. 圆绕固定圆从外侧滚动时，动圆心轨迹半径为 $R + r$。

2. 固定圆半径为 R、动圆半径为 r、对地自转角为 ϕ、圆心公转角为 θ 时，无滑动给出 $r\,d\phi = (R + r)\,d\theta$。

3. 嵌套内切时圆心位移长为 $R - r$；若半径趋零，嵌套圆盘至多有一个公共点。

> Example. A radius-2 circle rolls without slipping around a fixed radius-1 circle. During a 90° sweep of the center, how much does the moving circle spin?

> 例题：半径 2 的圆沿半径 1 的固定圆从外侧无滑动滚动，圆心转过 90° 时动圆自转多少度?

Worked solution

1. The moving center follows a circle of radius three, so its path length is $3\pi/2$.

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 8 / 方法 8 [GE06-M08] (continued / 续)

2. No slip makes this length equal to moving radius two times spin angle in radians.

3. The spin is $3\pi/4 = 135^\circ$.

例题解析

1. 动圆心轨迹半径为三，路程为 $3\pi/2$。

2. 无滑动使该路程等于动圆半径二乘以弧度自转角。

3. 自转角为 $3\pi/4 = 135^\circ$。

Center revolution and circle spin are different angles.

圆心公转角与圆本身自转角不同。

For a nested sequence with radii r^n and successive radial directions rotating by a fixed unit complex factor u, center increments form a geometric series proportional to $(ru)^n$. After computing the limit, verify its distance to every stage center is less than that stage radius.

嵌套圆半径为 r^n、相继半径方向乘固定单位复数 u 时，圆心增量形成与 $(ru)^n$ 成比例的等比级数。求极限后，仍需检验它到每阶段圆心的距离小于该阶段半径。

Practice / 配套练习: AIME-2007-II-Q11 (WB p. 276), AIME-2017-II-Q12 (WB p. 94)

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 9 / 方法 9 [GE06-M09]

A rotational contact circle follows an axial similarity

旋转接触圆由轴截面相似确定

Procedure

1. Cut the rotational surfaces through their common axis.

2. Internal and external circle tangency give center distance $R-r$ and $R+r$, respectively.

3. The contact point lies on the center line; scale its distance from the axis by the corresponding radius ratio.

通用步骤

1. 作通过旋转曲面公共轴的截面。

2. 截圆内切、外切分别给出圆心距 $R-r, R+r$。

3. 切点在连心线上，其到轴距离按相应半径比缩放。

> Example. A radius-1 generating circle has center distance three from a rotation axis. It is externally tangent to an axial sphere of radius five. Find the contact-circle radius.

> 例题：半径一的母圆圆心距旋转轴为三，并与轴上圆心、半径五的球外切，求接触圆半径。

Worked solution

1. The two section centers are distance $5+1=6$ apart.

2. The contact point lies $5/6$ of the way from the sphere center to the generating-circle center.

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 9 / 方法 9 [GE06-M09] (continued / 续)

3. Its distance from the axis is $(5/6)3 = 5/2$, the contact-circle radius. The center placement exists because six exceeds three.

例题解析

1. 截面两圆心距为 $5 + 1 = 6$。

2. 切点从球心出发沿连心线走全长的 $5/6$。

3. 到轴距离为 $(5/6)3 = 5/2$；因六大于三，圆心轴向摆放确实存在。

> Use center distances in the section, not three-dimensional surface distances.

> 应使用截面中的圆心距，不能使用三维曲面距离。

Practice / 配套练习: AIME-2024-II-Q08 (WB p. 429)

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 10 / 方法 10 [GE06-M10]

An incidence quadratic combines two angle-bisector circles

过定点的二次方程联系两个角平分线圆

Procedure

1. Intersecting circles tangent to the same two lines occupy one common angular sector.

2. Parameterize their centers by (x, tx) and radii tx; a shared point gives one quadratic whose roots are the two center parameters.

3. Use Vieta to connect the radius product to t^2, choose the sector sign, and double the angle to recover the second tangent line.

通用步骤

1. 相交且同时与两直线相切的圆处于同一角域。

2. 把圆心写为 (x, tx)、半径为 tx；共同定点给出同一二次方程，两根即两圆心参数。

3. 用韦达定理把半径积联系到 t^2，筛选角域符号，再用倍角求另一切线。

> Example. Two circles through $(1, 1)$ have radius product $1/2$. Both are tangent to the x axis and a positive-slope line through the origin. Find that slope.

> 例题：两圆均过 $(1, 1)$，半径积为 $1/2$，并均与 x 轴及过原点的一条正斜率直线相切，求该斜率。

Worked solution

1. The common point above the axis places both circles above it. Their centers (x, tx) satisfy $(1 - x)^2 + (1 - tx)^2 = t^2x^2$, or $x^2 - 2(1 + t)x + 2 = 0$.

Section 2.6 Tangent circles & common tangents

相切圆与公切线

Method 10 / 方法 10 [GE06-M10] (continued / 续)

2. Vieta gives $x_1x_2 = 2$, so the radius product is $2t^2 = 1/2$. The obtuse upper sector would have $t < -1$, impossible; hence $t = 1/2$.

3. The second tangent has slope $2t/(1 - t^2) = 4/3$. The center parameters are one and two, giving positive radii $1/2, 1$ and two genuinely intersecting circles.

例题解析

1. 共同点在轴上方，使两圆均位于上方。圆心 (x, tx) 满足 $(1 - x)^2 + (1 - tx)^2 = t^2x^2$，即 $x^2 - 2(1 + t)x + 2 = 0$。

2. 韦达给出 $x_1x_2 = 2$，故半径积为 $2t^2 = 1/2$。上方钝角域会有 $t < -1$，不可能，所以 $t = 1/2$。

3. 另一切线斜率为 $2t/(1 - t^2) = 4/3$。圆心参数为一与二，半径为 $1/2, 1$，确形成两个相交圆。

The sign of the angle-bisector slope carries the sector information.

角平分线斜率的符号包含角域信息。

Practice / 配套练习: AIME-2002-II-Q15 (WB p. 341)

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 1 / 方法 1 [GE07-M01]

Coordinates succeed when they encode the strongest geometry with the fewest variables. Use affine coordinates for incidence and ratios, orthonormal coordinates for lengths and angles, and signed normal equations for sides and containment. A final coordinate solution must reproduce every segment and orientation restriction.

坐标法应以最少变量编码最强几何条件。关联与比例适用仿射坐标，长度与角度适用标准正交坐标，两侧及容纳适用有向法向方程。最后须用坐标还原全部线段及方向限制。

Parameterize lines, midpoints and affine side divisions

参数化直线、中点及仿射分点

Procedure

1. Normalize a convenient triangle, rectangle or parallelogram, preserving only the metric information actually needed.

2. Write a point on a line as $P + tv$ and a side point as $(1-t)A + tB$ with $0 \le t \le 1$.

3. Midpoint and centroid conditions are linear vector averages.

4. Solve the linear incidence equations before any squared-distance equation.

通用步骤

1. 规范方便的三角形、矩形或平行四边形，仅保留实际需要的度量信息。

2. 直线点写为 $P + tv$，边上点写为 $(1-t)A + tB$ 并保留 $0 \le t \le 1$。

3. 中点与重心条件是向量线性平均。

4. 先解线性关联，再使用距离平方方程。

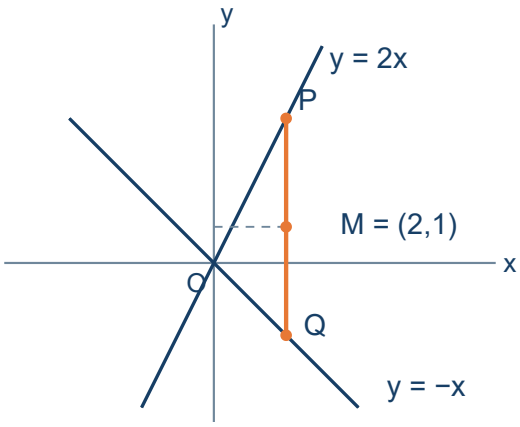
Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 1 / 方法 1 [GE07-M01] (continued / 续)

> Example. Point P lies on $y = 2x$, point Q on $y = -x$, and their midpoint is $(2, 1)$. Find PQ.

> 例题：P 在 $y = 2x$ 上，Q 在 $y = -x$ 上，中点为 $(2, 1)$，求 PQ。



Worked solution

1. Write $P = (t, 2t), Q = (u, -u)$. Midpoint equations are $t + u = 4, 2t - u = 2$.

2. They give $t = u = 2$, hence $P = (2, 4), Q = (2, -2)$.

3. The segment is vertical of length six; its midpoint is indeed $(2, 1)$.

例题解析

1. 设 $P = (t, 2t), Q = (u, -u)$，中点方程为 $t + u = 4, 2t - u = 2$。

2. 解得 $t = u = 2$，所以 $P = (2, 4), Q = (2, -2)$。

3. 线段竖直，长为六，中点也确为 $(2, 1)$。

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 1 / 方法 1 [GE07-M01] (continued / 续)

An affine normalization preserves ratios and area ratios, but not arbitrary lengths or angles.

仿射规范保持比例与面积比，不保持一般长度或角度。

With barycentric coordinates $\alpha+\beta+\gamma=1$, a line parallel to the side opposite A fixes α. Three fixed coordinate levels produce a signed similar triangle with scale $1-a-b-c$.

Align an angle bisector with the x-axis to place adjacent vertices at $(c\cos t, c\sin t), (a\cos t, -a\sin t)$; this keeps the bisector intercept and perpendicular cuts linear.

在重心坐标 $\alpha+\beta+\gamma=1$ 下，平行于 A 对边的直线固定 α。三条固定水平线形成有向比例为 $1-a-b-c$ 的相似三角形。

以角平分线为横轴，可把两邻顶点写为 $(c\cos t, c\sin t), (a\cos t, -a\sin t)$，使平分线截点及垂直截线保持线性。

Practice / 配套练习: AIME-2001-II-Q04 (WB p. 350), AIME-2003-I-Q15 (WB p. 387), AIME-2003-II-Q11 (WB p. 388), AIME-2005-I-Q10 (WB p. 368), AIME-2006-II-Q06 (WB p. 370), AIME-2009-I-Q04 (WB p. 351), AIME-2011-I-Q02 (WB p. 353), AIME-2011-II-Q02 (WB p. 355), AIME-2011-II-Q04 (WB p. 311), AIME-2013-II-Q13 (WB p. 399), AIME-2014-II-Q11 (WB p. 401), AIME-2017-II-Q10 (WB p. 377), AIME-2019-II-Q07 (WB p. 229), AIME-2020-II-Q12 (WB p. 409), AIME-2022-I-Q03 (WB p. 312)

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 2 / 方法 2 [GE07-M02]

Equal distances linearize centers and radical axes

等距线性化圆心与根轴

Procedure

1. Expand squared distances to two fixed points and subtract; the quadratic point terms cancel.

2. Find circumcenters from two perpendicular-bisector equations, or circle centers from symmetry and tangency normals.

3. Subtract two circle equations to obtain their radical axis before solving a remaining quadratic.

4. Check positive radius, the distinct second intersection and the required side or ray.

通用步骤

1. 展开到两定点的距离平方并相减，点的二次项会消去。

2. 由两垂直平分线求外心，或由对称及相切法向定位圆心。

3. 两圆方程先相减求根轴，再解剩余二次式。

4. 检查正半径、不同第二交点及所需边或射线。

> Example. Find the circle through $(0,0),(6,0),(0,8)$.

> 例题：求经过 $(0,0),(6,0),(0,8)$ 的圆。

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 2 / 方法 2 [GE07-M02] (continued / 续)

Worked solution

1. Let its center be (u, v). Equality of the first two squared distances gives $12u = 36$, so $u = 3$.

2. Equality with the third gives $16v = 64$, so $v = 4$.

3. The radius is $\sqrt{3^2 + 4^2} = 5$, hence $(x-3)^2 + (y-4)^2 = 25$.

例题解析

1. 设圆心 (u, v)，前两点等距给出 $12u = 36$，故 $u = 3$。

2. 与第三点等距给出 $16v = 64$，故 $v = 4$。

3. 半径为五，所以圆方程为 $(x-3)^2 + (y-4)^2 = 25$。

A line of equal powers differs from a perpendicular bisector when the radii differ.

半径不等时，等点幂直线不同于垂直平分线。

Practice / 配套练习: AIME-2001-II-Q06 (WB p. 283), AIME-2001-II-Q07 (WB p. 313), AIME-2002-I-Q15 (WB p. 452), AIME-2005-II-Q08 (WB p. 332), AIME-2006-II-Q09 (WB p. 333), AIME-2008-II-Q11 (WB p. 344), AIME-2009-II-Q05 (WB p. 330), AIME-2009-II-Q10 (WB p. 371), AIME-2011-II-Q13 (WB p. 395), AIME-2012-II-Q15 (WB p. 297), AIME-2014-I-Q15 (WB p. 298), AIME-2014-II-Q08 (WB p. 335), AIME-2017-I-Q10 (WB p. 80), AIME-2019-I-Q13 (WB p. 302), AIME-2019-I-Q15 (WB p. 303), AIME-2019-II-Q11 (WB p. 406), AIME-2020-II-Q15 (WB p. 305), AIME-2022-I-Q08 (WB p. 336), AIME-2023-II-Q09 (WB p. 288), AIME-2024-I-Q05 (WB p. 363), AIME-2025-I-Q08 (WB p. 382), AIME-2025-II-Q01 (WB p. 364), AIME-2025-II-Q06 (WB p. 339)

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 3 / 方法 3 [GE07-M03]

Dot and cross products encode metric constraints

点积与叉积编码度量约束

Procedure

1. Use $u \cdot v = |u||v| \cos\theta$ for angles and perpendicularity.

2. Use the planar determinant or spatial cross product for signed area and orientation.

3. Divide cross by dot only after checking the dot sign and whether the requested angle is acute.

4. Expand squared lengths to expose cancellations, closure identities or Gram determinants.

通用步骤

1. 用 $u \cdot v = |u||v| \cos\theta$ 处理夹角及垂直。

2. 用平面行列式或空间叉积处理有向面积与方向。

3. 叉积除点积之前，检查点积符号及所求是否锐角。

4. 展开距离平方，以显露消项、闭合恒等式或格拉姆行列式。

> Example. For $u = (1, 2), v = (3, 1)$, find their acute angle and the parallelogram area.

> 例题：给定 $u = (1, 2), v = (3, 1)$，求两向量锐夹角及平行四边形面积。

Worked solution

1. The dot product is five and the determinant is $1 - 6 = -5$. Both vectors are nonzero and the dot is positive.

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 3 / 方法 3 [GE07-M03] (continued / 续)

2. Thus $\tan\theta = |-5|/5 = 1$, so $\theta = 45^\circ$.

3. The parallelogram area is the determinant magnitude five.

例题解析

1. 点积为五，行列式为 $1-6=-5$；两向量非零且点积正。

2. 故 $\tan\theta = 1$，锐角为 45°。

3. 平行四边形面积为行列式绝对值五。

An absolute cross-to-dot ratio does not distinguish an acute angle from its supplement; retain the dot sign.

叉积与点积的绝对值比不能区分锐角及其补角，应保留点积符号。

A closed polygon has side-vector sum zero; squaring this equation relates pairwise side products weighted by direction cosines. In an equiangular pentagon only the72- and144-degree cosine classes remain.

For vectors v_i of equal positive length R and sum zero, $|v_i - x| \geq R - v_i \cdot x/R$ sums to $\sum |v_i - x| \geq nR$, attained at $x = 0$.

For generating vectors u, v, w, the Gram determinant $\det(u_i \cdot u_j)$ equals squared parallelepiped volume. Face diagonals determine pairwise norms and dot products through $|u \pm v|^2$.

闭合多边形的边向量和为零，将其平方可联系按方向余弦加权的两两边积。等角五边形只剩七十二度与一百四十四度两类余弦。

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 3 / 方法 3 [GE07-M03] (continued / 续)

若 v_i 均长 $R>0$ 且向量和为零，则 $|v_i-x|\geq R-v_i\cdot x/R$ 相加得到 $\sum|v_i-x|\geq nR$，在 $x=0$ 取等。

生成向量 u,v,w 的格拉姆行列式 $\det(u_i\cdot u_j)$ 等于平行六面体体积平方。面对角线可由 $|u\pm v|^2$ 求范数及点积。

Practice / 配套练习: AIME-2000-II-Q08 (WB p. 366), AIME-2002-II-Q02 (WB p. 436), AIME-2003-II-Q14 (WB p. 389), AIME-2005-II-Q12 (WB p. 392), AIME-2009-II-Q03 (WB p. 352), AIME-2011-II-Q13 (WB p. 395), AIME-2012-I-Q14 (WB p. 92), AIME-2013-I-Q07 (WB p. 373), AIME-2016-I-Q09 (WB p. 183), AIME-2017-II-Q15 (WB p. 405), AIME-2021-I-Q06 (WB p. 449), AIME-2021-II-Q12 (WB p. 410), AIME-2022-I-Q05 (WB p. 20), AIME-2023-I-Q05 (WB p. 362), AIME-2023-I-Q12 (WB p. 411), AIME-2023-I-Q13 (WB p. 460), AIME-2023-II-Q12 (WB p. 412), AIME-2026-I-Q14 (WB p. 414)

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 4 / 方法 4 [GE07-M04]

Unit normals give signed distances and projections

单位法向量给出有向距离与投影

Procedure

1. For line $ax+by=c$, use signed distance $(ax+by-c)/\sqrt{a^2+b^2}$.

2. The perpendicular foot is $P-[(n\cdot P-c)/|n|^2]n$.

3. Use the sign to identify a half-plane and retain the normalization when the line direction varies.

4. The spatial plane formula is identical with a three-component normal.

通用步骤

1. 直线 $ax+by=c$ 的有向距离为 $(ax+by-c)/\sqrt{a^2+b^2}$。

2. 垂足为 $P-[(n\cdot P-c)/|n|^2]n$。

3. 利用正负识别半平面，直线方向变化时仍须保留归一化。

4. 空间平面公式相同，只需使用三分量法向量。

> Example. Find the distance and perpendicular foot from $(4,3)$ to $3x+4y=10$.

> 例题：求点 $(4,3)$ 到直线 $3x+4y=10$ 的距离及垂足。

Worked solution

1. The normal is $(3,4)$ of length five, and the residual is $12+12-10=14$. Thus distance is $14/5$.

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 4 / 方法 4 [GE07-M04] (continued / 续)

2. The foot is $(4,3)-(14/25)(3,4)=(58/25,19/25)$.

3. Substitution gives $3(58/25)+4(19/25)=10$, checking incidence.

例题解析

1. 法向量为 $(3,4)$、长五，残量为十四，故距离为 $14/5$。

2. 垂足为 $(4,3)-(14/25)(3,4)=(58/25,19/25)$。

3. 代入 $3x+4y=10$ 成立，核实垂足在线上。

Distance to a segment needs an endpoint check if the foot leaves the segment.

垂足若离开线段，求线段距离还须比较端点。

Practice / 配套练习: AIME-2001-I-Q12 (WB p. 431), AIME-2004-I-Q10 (WB p. 769), AIME-2005-I-Q14 (WB p. 391), AIME-2006-I-Q14 (WB p. 393), AIME-2008-I-Q14 (WB p. 186), AIME-2011-I-Q03 (WB p. 354), AIME-2013-I-Q12 (WB p. 398), AIME-2015-II-Q11 (WB p. 402), AIME-2015-II-Q15 (WB p. 345), AIME-2016-I-Q14 (WB p. 403), AIME-2018-I-Q08 (WB p. 378), AIME-2019-II-Q13 (WB p. 407), AIME-2021-I-Q09 (WB p. 380), AIME-2023-I-Q08 (WB p. 381), AIME-2026-I-Q12 (WB p. 434), AIME-2026-II-Q12 (WB p. 435)

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 5 / 方法 5 [GE07-M05]

Rotations and reflections preserve lengths but change orientation

旋转与反射保持长度但改变方向

Procedure

1. Use $R_\theta(x,y)=(x\cos\theta-y\sin\theta, x\sin\theta+y\cos\theta)$.

2. A rotation about C maps P to $C+R_\theta(P-C)$; compare edge vectors to recover its angle first.

3. For squares use a signed quarter-turn, and for equilateral apices use the two signed60-degree turns.

4. A fold crease is the perpendicular bisector of a point and its image; repeated folds also reverse layer order.

通用步骤

1. 使用 $R_\theta(x,y)=(x\cos\theta-y\sin\theta, x\sin\theta+y\cos\theta)$。

2. 绕 C 旋转为 $C+R_\theta(P-C)$，先比较边向量确定角度。

3. 正方形用带符号四分之一转，等边顶点用正负六十度转。

4. 折痕是点与其像的垂直平分线，重复折叠还会颠倒层序。

> Example. A 90° counterclockwise rotation sends $(0,0)$ to $(3,1)$. Find its center and the image of $(1,0)$.

> 例题：一次逆时针九十度旋转将 $(0,0)$ 映为 $(3,1)$，求旋转中心及 $(1,0)$ 的像。

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 5 / 方法 5 [GE07-M05] (continued / 续)

Worked solution

1. For center (u, v), the origin image is $(u+v, v-u) = (3, 1)$.

2. Solve to get $(u, v) = (1, 2)$.

3. The image of $(1, 0)$ is $(1, 2) + R_{90}(0, -2) = (3, 2)$.

例题解析

1. 若中心为 (u, v)，原点像为 $(u+v, v-u) = (3, 1)$。

2. 解得中心 $(1, 2)$。

3. 点 $(1, 0)$ 的像为 $(1, 2) + R_{90}(0, -2) = (3, 2)$。

Both signed rotations often solve length equations; use the specified quadrant or side of a line.

正负旋转常都满足长度方程，应按象限或直线两侧筛选。

Folding the right half of an L-cell strip of thickness H over the left maps a right-half cell position p to $L+1-p$ and bottom-up layer b to $2H+1-b$; left-half data remain unchanged before updating L, H.

For a rolling marker, compute the moving center's orbital position and then add the marker's ground-frame spin offset. A full spin restores the marker direction, not the center position.

将长 L 格、厚 H 层纸条右半折到左半上方时，右半格位置 p 变为 $L+1-p$，自底层号 b 变为 $2H+1-b$；左半数据不变，然后更新 L, H。

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 5 / 方法 5 [GE07-M05] (continued / 续)

滚动标记点先算圆心公转位置，再加地面参照下的自转偏移。自转整周恢复的是标记方向，不是圆心位置。

Practice / 配套练习: AIME-2000-I-Q02 (WB p. 349), AIME-2004-II-Q07 (WB p. 367), AIME-2004-II-Q15 (WB p. 390), AIME-2011-I-Q14 (WB p. 394), AIME-2012-I-Q13 (WB p. 396), AIME-2012-II-Q13 (WB p. 397), AIME-2013-I-Q09 (WB p. 271), AIME-2013-II-Q04 (WB p. 356), AIME-2014-I-Q10 (WB p. 374), AIME-2015-I-Q04 (WB p. 357), AIME-2015-I-Q07 (WB p. 375), AIME-2019-II-Q01 (WB p. 359), AIME-2020-II-Q04 (WB p. 360), AIME-2022-II-Q11 (WB p. 327), AIME-2025-I-Q02 (WB p. 245), AIME-2025-I-Q09 (WB p. 417), AIME-2025-I-Q14 (WB p. 190), AIME-2026-I-Q05 (WB p. 66), AIME-2026-I-Q10 (WB p. 383)

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 6 / 方法 6 [GE07-M06]

Centroid averages expose a common homothety

重心平均显露共同位似

Procedure

1. Write midpoint and centroid vectors as exact averages.

2. Rearrange all target points into $T_i = O + t(V_i - O)$ with one center and scale.

3. Use scale $|t|$ for length, t^2 for planar area and $|t|^3$ for volume.

4. For a centrally symmetric family, pair opposite points to prove equal area or a center bound.

通用步骤

1. 把中点及重心向量写为精确平均。

2. 将各目标点整理为同一中心、同一比例的 $T_i = O + t(V_i - O)$。

3. 长度用 $|t|$，面积用 t^2，体积用 $|t|^3$。

4. 中心对称点族可配对相反点，证明等面积或圆心界。

> Example. Let P be inside a unit square. The centroids of its four triangles with the square sides form a quadrilateral. Find its area.

> 例题：P 为单位正方形内部点，P 与四条边分别组成三角形，其四重心组成四边形，求面积。

Worked solution

1. If M_i is a side midpoint, its fan-triangle centroid is $T_i = (P + 2M_i)/3$.

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 6 / 方法 6 [GE07-M06] (continued / 续)

2. Thus the centroid quadrilateral is a homothety of the midpoint quadrilateral with linear scale $2/3$.

3. The midpoint quadrilateral has area $1/2$, so the result is $(4/9)(1/2) = 2/9$, independent of P.

例题解析

1. 若 M_i 为边中点，则对应三角形重心为 $T_i = (P + 2M_i)/3$。

2. 所以重心四边形是中点四边形按 $2/3$ 缩放的位似像。

3. 中点四边形面积为 $1/2$，故目标面积为 $2/9$，与 P 无关。

Writing one common center is stronger than checking separate side parallels.

写出共同位似中心，比逐一验证边平行更有力。

With circumcenter as origin, $h = a + b + c$ is the orthocenter vector. Reflection of the circumcircle across BC has center $b + c = h - a$.

以外心为原点时，垂心向量为 $h = a + b + c$，外接圆关于 BC 反射后的圆心为 $b + c = h - a$。

Practice / 配套练习: AIME-2003-II-Q04 (WB p. 438), AIME-2005-II-Q10 (WB p. 446), AIME-2006-I-Q10 (WB p. 369), AIME-2010-I-Q08 (WB p. 167), AIME-2018-II-Q09 (WB p. 379), AIME-2019-II-Q01 (WB p. 359), AIME-2020-I-Q15 (WB p. 408), AIME-2022-I-Q03 (WB p. 312), AIME-2026-II-Q12 (WB p. 435)

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 7 / 方法 7 [GE07-M07]

Half-plane clipping describes overlaps and nearest-point regions

半平面截割描述重叠及最近点区域

Procedure

1. Convert each boundary into a signed linear inequality; compare squared distances to turn nearest-point conditions into perpendicular-bisector half-planes.

2. Intersect every inequality and find feasible pairwise boundary intersections.

3. For a slope inequality, split by the sign of its horizontal denominator before multiplying.

4. Use the resulting actual polygon for area, not the unbounded supporting lines.

通用步骤

1. 将边界转为有向线性不等式；相减距离平方可把最近点条件转为垂直平分线半平面。

2. 相交全部不等式，求可行的两两边界交点。

3. 斜率不等式先按横向分母符号分区，再相乘。

4. 用实际截割多边形求面积，不能使用无界支撑直线。

> Example. Inside the unit square, find the area closer to $(0,0)$ than to either $(1,0)$ or $(0,1)$.

> 例题：单位正方形内，求比 $(1,0),(0,1)$ 都更靠近原点的区域面积。

Worked solution

1. Comparison with $(1,0)$ gives $x^2+y^2<(x-1)^2+y^2$, or $x<1/2$.

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 7 / 方法 7 [GE07-M07] (continued / 续)

2. Comparison with $(0,1)$ gives $y < 1/2$.

3. The intersection is a half-by-half square, area $1/4$. Strict boundaries have area zero.

例题解析

1. 与 $(1,0)$ 比较距离平方，得 $x < 1/2$。

2. 与 $(0,1)$ 比较，得 $y < 1/2$。

3. 交集为二分之一乘二分之一的正方形，面积 $1/4$；严格边界面积为零。

A factorized inequality can define several sign regions; identify the bounded one before measuring.

因式分解不等式可能给出多个符号区域，应先辨认有界部分。

For a constant-speed road followed by slower off-road travel, moving-center endpoint disks form a reachable envelope. First prove an optimal route can be reduced to the permitted road/straight-segment form, then derive the common tangent boundaries.

For a bounded lattice-cell union enclosed by a closed disk, pass to cell closures and check all polygon corners; convexity of the disk then covers each whole cell.

先沿高速道路、再慢速离路行驶时，移动圆心的终点圆盘组成可达包络。先证明最优路径可简化为允许的道路段加直线段，再求公切线边界。

闭圆盘包围有限格单元并集时，可先取各格闭包再检查全部顶点；圆盘凸性保证整格被包含。

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 7 / 方法 7 [GE07-M07] (continued / 续)

Practice / 配套练习: AIME-2000-I-Q13 (WB p. 384), AIME-2004-I-Q10 (WB p. 769), AIME-2010-I-Q11 (WB p. 458), AIME-2012-I-Q08 (WB p. 447), AIME-2013-I-Q12 (WB p. 398), AIME-2017-II-Q03 (WB p. 767), AIME-2020-II-Q02 (WB p. 768), AIME-2021-I-Q02 (WB p. 361), AIME-2025-I-Q12 (WB p. 413)

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 8 / 方法 8 [GE07-M08]

Supporting widths enforce full containment

支撑宽度施加整体容纳限制

Procedure

1. For a unit normal n, compute extreme projections $\min n \cdot v, \max n \cdot v$ of the entire figure.

2. Their difference is its width in that direction, giving strip or square-fit bounds.

3. For an inscribed disk, require the center's signed distance from every supporting side to be at least its radius.

4. After a formal optimum is found, check every side-segment condition and construct a feasible equality center.

通用步骤

1. 给定单位法向 n，求整个图形投影的最小及最大值。

2. 两者差为该方向宽度，给出条带或正方形容纳界。

3. 对内接圆，要求圆心到每条支撑边的向内有向距离至少为半径。

4. 求形式最优后，检查全部边上线段限制并构造可行取等圆心。

> Example. A unit square is rotated 45° about its center. Find the side of the largest concentric axis-parallel square inside it.

> 例题：单位正方形绕中心转四十五度，求其内部同心、边平行于坐标轴的最大正方形边长。

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 8 / 方法 8 [GE07-M08] (continued / 续)

Worked solution

1. If the inner side is t, its projection width onto a rotated-side normal is $t(\cos 45^\circ + \sin 45^\circ) = t\sqrt{2}$.

2. This must be at most the outer square's width one.

3. Thus $t \le 1/\sqrt{2}$, and the centered square of that side touches all four outer sides, attaining equality.

例题解析

1. 内方边长为 t 时，在外方边法向的投影宽为 $t\sqrt{2}$。

2. 它不能超过外方该方向宽一。

3. 故 $t \le 1/\sqrt{2}$，同心取此边长可同时接触四边，达到等号。

A line-intersection construction can lie beyond a side even when its equations are correct.

直线求交方程即使正确，构造点仍可能落在边的延长线上。

A side-t square rotated by θ has axis widths $t(|\cos\theta| + |\sin\theta|)$. For two concentric nested squares, apply the same support factor at both successive contacts.

For a symmetric semicircle, width in a nonparallel direction depends on both a straight-diameter endpoint and the curved support point. Derive the support function rather than treating the semicircle as a full circle.

边长 t 的正方形转角 θ 后，两坐标方向宽均为 $t(|\cos\theta| + |\sin\theta|)$；同心嵌套方形的相邻接触可连续应用同一支撑系数。

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 8 / 方法 8 [GE07-M08] (continued / 续)

半圆在非平行方向的宽度同时取决于直径端点与曲线支撑点，应推导支撑函数，不能把半圆当整圆。

Practice / 配套练习: AIME-2005-I-Q11 (WB p. 185), AIME-2005-I-Q14 (WB p. 391), AIME-2010-I-Q08 (WB p. 167), AIME-2013-I-Q12 (WB p. 398), AIME-2016-I-Q09 (WB p. 183), AIME-2016-II-Q07 (WB p. 376), AIME-2017-I-Q15 (WB p. 404), AIME-2018-I-Q08 (WB p. 378)

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 9 / 方法 9 [GE07-M09]

Primitive lattice directions and unfolded reflection rays

本原格点方向与反射展开射线

Procedure

1. Divide integer direction components by their GCD to obtain a primitive direction.

2. Unfold specular reflections so the entire path becomes one straight ray in repeated boxes.

3. A vertex return means every unfolded coordinate is a box-side multiple; solve the simultaneous divisibility conditions.

4. For a thickened lattice object, test its full linear-form range and correct shared endpoints when repeating a primitive segment.

通用步骤

1. 整数方向分量除以最大公因数，得到本原方向。

2. 展开镜面反射，使整个路径成为重复长方体中的一条直射线。

3. 回到顶点等价于展开各坐标均为边长倍数，应联立整除条件。

4. 厚格点物体应检查整个线性型范围，重复本原线段时还须修正共端点。

> Example. A ray in the cube $[0,2]^3$ starts at the origin in direction $(2,1,1)$ and reflects at every face. Find the distance to its first later vertex.

> 例题：光线在立方体 $[0,2]^3$ 中从原点沿 $(2,1,1)$ 方向发出并在各面反射，求首次再到顶点的路程。

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 9 / 方法 9 [GE07-M09] (continued / 续)

Worked solution

1. The unfolded ray is $(2t, t, t)$ with $t > 0$. A folded vertex requires all three coordinates divisible by two.

2. The first coordinate requires integral t, and the other two then require even t. The first is $t = 2$.

3. Distance is $2\sqrt{2^2 + 1^2 + 1^2} = 2\sqrt{6}$.

例题解析

1. 展开射线为 $(2t, t, t)$，折回顶点要求三坐标都为二的倍数。

2. 第一坐标要求 t 为整数，另两坐标进一步要求偶数，最小为二。

3. 路程为 $2\sqrt{6}$。

For radius-ε circles, use Euclidean normal length; for axis-aligned squares use the sum of absolute normal components.

半径 ε 圆使用法向量欧氏长度，轴平行正方形使用法向分量绝对值和。

A line with primitive direction v has successive lattice points separated by v. In the grid box $\{0, \ldots, N-1\}^d$, where N is a positive integer, containing k grid points requires $(k-1)|v_i| \le N - 1$ for every component. Normalize one endpoint to avoid counting a line repeatedly.

For square centers (u, v) near line $ax + by = 0$, a side-2ε square intersects the line when $|au + bv| \le \varepsilon(|a| + |b|)$. A radius-$\varepsilon$ circle instead uses $\varepsilon\sqrt{a^2 + b^2}$. Check segment endpoints separately.

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 9 / 方法 9 [GE07-M09] (continued / 续)

本原方向为 v 的直线，相继格点间差为 v。在格点盒 $\{0,\dots,N-1\}^d$ 中（N 为正整数），若直线含 k 个格点，则各分量满足 $(k-1)|v_i| \le N-1$；规范一个端点可避免重复计线。

格点中心 (u,v) 附近的边长 2ε 轴平行方形与 $ax+by=0$ 相交，当 $|au+bv| \le \varepsilon(|a|+|b|)$；半径 ε 圆则使用 $\varepsilon\sqrt{a^2+b^2}$，线段端点还须单独检查。

Practice / 配套练习: AIME-2000-II-Q11 (WB p. 385), AIME-2002-I-Q11 (WB p. 386), AIME-2016-I-Q14 (WB p. 403), AIME-2017-II-Q14 (WB p. 704), AIME-2020-II-Q12 (WB p. 409)

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 10 / 方法 10 [GE07-M10]

Complex inversion maps half-planes to disks

复数倒数将半平面映为圆盘

Procedure

1. Put $w = 1/z = u + iv$ and express z through u, v.

2. Translate the original half-plane inequality and multiply only by the positive quantity $u^2 + v^2$.

3. Complete squares to identify the image disk; preserve unions, intersections and excluded zero points.

通用步骤

1. 令 $w = 1/z = u + iv$，用 u, v 表示 z。

2. 翻译原半平面不等式，只乘正数 $u^2 + v^2$。

3. 配方识别像圆盘，保留并集、交集及零点排除条件。

> Example. Find the image of $\Re z > 1$ under $w = 1/z$.

> 例题：求半平面 $\Re z > 1$ 在 $w = 1/z$ 下的像。

Worked solution

1. Since $z = (u - iv)/(u^2 + v^2)$, the condition is $u/(u^2 + v^2) > 1$.

2. Thus $u^2 + v^2 - u < 0$, or $(u - 1/2)^2 + v^2 < 1/4$.

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 10 / 方法 10 [GE07-M10] (continued / 续)

3. The image is the open disk centered at $1/2$ of radius $1/2$; zero is on its boundary and is automatically excluded.

例题解析

1. 由 $z = (u - iv)/(u^2 + v^2)$，条件为 $u/(u^2 + v^2) > 1$。

2. 所以 $u^2 + v^2 - u < 0$，即 $(u - 1/2)^2 + v^2 < 1/4$。

3. 像为圆心 $1/2$、半径 $1/2$ 的开圆盘，零在边界上而自动被排除。

The exterior of a convex polygon is a union of exterior half-planes, so its inversion is a union of the corresponding disks.

凸多边形外部是各外侧半平面的并，故其倒数像为相应圆盘的并。

Practice / 配套练习: AIME-2008-II-Q13 (WB p. 90)

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 11 / 方法 11 [GE07-M11]

Develop a cylindrical path and preserve tangent pitch

展开圆柱路径并保持相切坡度

Procedure

1. Unroll the cylindrical surface into a plane with horizontal coordinate equal to circumferential arc length.

2. The shortest surface path is a straight segment in an appropriate winding lift.

3. For a taut rope leaving tangentially, extend that line's slope into its free portion and use the same length-to-horizontal-projection ratio.

通用步骤

1. 展开圆柱面，水平坐标取圆周弧长。

2. 适当绕行提升中的直线段给出最短表面路程。

3. 绷紧绳索相切离开时，其自由段延续展开直线坡度，并使用相同实际长度与水平投影比。

> Example. On a radius-one cylinder, two points are on opposite generators with height difference two. Find their shortest surface distance.

> 例题：半径一的圆柱上，两点位于相对母线、高度相差二，求最短表面距离。

Worked solution

1. The unfolded horizontal separation is $\pi + 2k\pi$ for integer winding number k.

2. The smallest absolute separation is π, attained for the two opposite winding choices.

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 11 / 方法 11 [GE07-M11] (continued / 续)

3. Pythagoras gives distance $\sqrt{\pi^2+4}$.

例题解析

1. 展开水平距离为 $\pi+2k\pi$，其中 k 为整数绕数。

2. 最小绝对水平距离为 π，对应两种相反绕向。

3. 由勾股，最短距离为 $\sqrt{\pi^2+4}$。

The free tangent length in horizontal projection is $\sqrt{d^2-R^2}$ for endpoint axis distance d.

端点距轴 d 时，自由切线水平投影长为 $\sqrt{d^2-R^2}$。

Practice / 配套练习: AIME-2004-I-Q14 (WB p. 454)

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 12 / 方法 12 [GE07-M12]

Orthonormal spatial coordinates separate height, plane and solid

标准正交空间坐标分离高度、平面及立体

Procedure

1. Choose three perpendicular solid-edge directions as a basis; encode the vertical direction by a unit vector.

2. Height differences along equal orthogonal edges have squared sum equal to the edge length squared.

3. Write cutting faces as plane equations and compute actual segment intersections, not just line intersections.

4. Use point–plane distances, determinants or projected-area factors according to the requested quantity.

通用步骤

1. 以三条垂直立体棱方向为基，用单位向量表示竖直方向。

2. 等长正交棱的高度差平方和等于棱长平方。

3. 把截面写为平面方程，并求实际棱线段交点，不仅是直线交点。

4. 按目标选择点面距离、行列式或投影面积系数。

> Example. A cube has a lowest vertex at height zero. Its three neighboring vertices have heights two,three,six. Find its volume.

> 例题：立方体最低顶点高度零，三个相邻顶点高度分别二、三、六，求体积。

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 12 / 方法 12 [GE07-M12] (continued / 续)

Worked solution

1. Let the edge length be s. The three vertical direction components are $2/s, 3/s, 6/s$.

2. Their squared sum is one, so $s^2 = 4 + 9 + 36 = 49$ and $s = 7$.

3. The volume is 343. All three components are positive, so the nominated vertex is indeed lowest.

例题解析

1. 设棱长 s，竖直单位向量三个分量为 $2/s, 3/s, 6/s$。

2. 平方和为一，所以 $s^2 = 49, s = 7$。

3. 体积为 343；三分量皆正，给定顶点确为最低点。

Plane incidence and a filled face's nearest point are different constraints.

平面关联与到填满面的最近点，是不同约束。

Common same-side tangent planes to spheres resting on $z = 0$ give $ux + vy - c = (1 - h)\rho$ for a unit normal (u, v, h). Excluding the repeated plane $h = 1$ makes radius an affine function of contact position; interpolate it from three noncollinear contacts and verify unit-normal feasibility.

球与 $z = 0$ 相切、又与同侧第二平面相切时，单位法向量 (u, v, h) 给出 $ux + vy - c = (1 - h)\rho$。排除重复平面 $h = 1$ 后，半径成为接触位置的仿射函数，可由三个不共线接触点插值，并需检验单位法向可行性。

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 12 / 方法 12 [GE07-M12] (continued / 续)

Practice / 配套练习: AIME-2001-II-Q15 (WB p. 451), AIME-2002-I-Q15 (WB p. 452), AIME-2006-I-Q14 (WB p. 393), AIME-2007-I-Q13 (WB p. 456), AIME-2008-I-Q15 (WB p. 457), AIME-2011-I-Q13 (WB p. 432), AIME-2012-I-Q08 (WB p. 447), AIME-2013-I-Q07 (WB p. 373), AIME-2017-II-Q15 (WB p. 405), AIME-2023-I-Q13 (WB p. 460), AIME-2023-II-Q14 (WB p. 461), AIME-2026-I-Q12 (WB p. 434), AIME-2026-II-Q12 (WB p. 435)

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 13 / 方法 13 [GE07-M13]

Coordinate area uses the boundary's true order

坐标面积使用真实边界次序

Procedure

1. Compute intersection points exactly and arrange them in actual boundary order.

2. Use a planar determinant or shoelace sum for a simple polygon, and a cross-product norm for a spatial triangle.

3. For a crossed outline, partition all bounded lobes before adding positive areas.

4. Normalize by an affine map only when the common determinant factor cancels from an area ratio.

通用步骤

1. 精确求交点，并按实际边界次序排列。

2. 简单多边形用行列式或鞋带和，空间三角形用叉积长度。

3. 交叉轮廓需分割全部有界叶片，再加正面积。

4. 只有共同面积行列式因子可约去时，才作仿射规范。

> Example. Find the area of triangle $(1,1),(5,2),(2,4)$.

> 例题：求三角形 $(1,1),(5,2),(2,4)$ 的面积。

Worked solution

1. From the first vertex, the two side vectors are $(4,1),(1,3)$.

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 13 / 方法 13 [GE07-M13] (continued / 续)

2. Their determinant is $4 \cdot 3 - 1 \cdot 1 = 11$.

3. The triangle area is $|11|/2 = 11/2$.

例题解析

1. 从第一顶点出发，两边向量为 $(4, 1), (1, 3)$。

2. 行列式为 $4 \cdot 3 - 1 \cdot 1 = 11$。

3. 三角形面积为 $11/2$。

A global absolute shoelace value is an algebraic area for a self-intersecting polygon.

自交多边形的全局鞋带绝对值仍是代数面积。

Practice / 配套练习: AIME-2005-I-Q10 (WB p. 368), AIME-2010-II-Q09 (WB p. 372), AIME-2013-I-Q07 (WB p. 373), AIME-2013-II-Q13 (WB p. 399), AIME-2014-I-Q13 (WB p. 400), AIME-2015-I-Q04 (WB p. 357), AIME-2018-II-Q04 (WB p. 358), AIME-2018-II-Q09 (WB p. 379), AIME-2019-II-Q01 (WB p. 359), AIME-2025-I-Q02 (WB p. 245), AIME-2025-I-Q12 (WB p. 413), AIME-2026-I-Q10 (WB p. 383), AIME-2026-II-Q03 (WB p. 365)

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 14 / 方法 14 [GE07-M14]

Velocity vectors distinguish the moving and ground frames

速度向量区分运动介质与地面参照系

Procedure

1. Write ground velocity as medium-relative velocity plus the medium's velocity.

2. Use the target displacement divided by time for the ground-frame components.

3. Apply speed norms in the correct frame, and subtract squared equations to eliminate the shared transverse component.

通用步骤

1. 地面速度等于相对介质速度加介质速度。

2. 由目标位移除以时间写出地面分量。

3. 在正确参照系下使用速率范数，并相减平方式以消去共同横向分量。

> Example. A river is twelve units wide and flows east at one unit per minute. A swimmer of water-relative speed $\sqrt{5}$ travels directly north in the ground frame. Find the crossing time.

> 例题：河宽十二，水流向东速度一单位每分，泳者相对水速为 $\sqrt{5}$，地面轨迹正北，求过河时间。

Worked solution

1. To cancel east current, the swimmer's water-relative horizontal component must be -1.

2. If its north component is $v > 0$, then $1 + v^2 = 5$, so $v = 2$.

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 14 / 方法 14 [GE07-M14] (continued / 续)

3. The ground north speed remains two, giving time $12/2 = 6$ minutes.

例题解析

1. 抵消向东水流要求相对水的横向分量为 -1。

2. 若向北分量为 $v > 0$，则 $1 + v^2 = 5$，所以 $v = 2$。

3. 地面向北速度仍为二，故过河时间为六分钟。

The magnitude of a sum of velocities is not generally the sum of their magnitudes.

速度向量和的长度一般不等于速率之和。

Practice / 配套练习: AIME-2022-I-Q05 (WB p. 20)

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 15 / 方法 15 [GE07-M15]

Oblique unit-vector bases retain a Gram metric

斜交单位向量基保留格拉姆度量

Procedure

1. Choose linearly independent unit side directions u, v and record $u \cdot v = c$, with $|c| < 1$.

2. A point $xu + yv$ has norm square $x^2 + y^2 + 2cxy$; incidence stays affine but Euclidean lengths use this quadratic form.

3. The vector $u + v$ points along the internal bisector because its dot products with the two unit sides are equal.

4. Subtract circle equations sharing this metric form to obtain a linear radical axis.

通用步骤

1. 取线性无关的单位边方向 u, v，记录 $u \cdot v = c$，其中 $|c| < 1$。

2. 点 $xu + yv$ 的范数平方为 $x^2 + y^2 + 2cxy$；关联保持仿射，但欧氏长度必须使用此二次型。

3. $u + v$ 对两单位边的点积相同，因此指向内角平分线。

4. 同一度量下的两圆方程相减，仍得到线性根轴。

> Example. Unit vectors u, v meet at 60°. Find $|2u + v|$ and a unit vector along their internal bisector.

> 例题：单位向量 u, v 夹角六十度，求 $|2u + v|$ 及其内角平分线单位方向。

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 15 / 方法 15 [GE07-M15] (continued / 续)

Worked solution

1. The Gram coefficient is $u \cdot v = 1/2$. Thus $|2u+v|^2 = 4+1+4(1/2) = 7$.

2. Also $|u+v|^2 = 2+2(1/2) = 3$, so its unit direction is $(u+v)/\sqrt{3}$.

3. The direction has equal dot product $\sqrt{3}/2$ with u, v, confirming that each angle is 30°.

例题解析

1. 格拉姆系数为 $1/2$，所以 $|2u+v|^2 = 4+1+4(1/2) = 7$。

2. 又 $|u+v|^2 = 3$，故单位方向为 $(u+v)/\sqrt{3}$。

3. 该方向与 u, v 点积均为 $\sqrt{3}/2$，核实两角各三十度。

Using x^2+y^2 in an oblique basis silently changes the geometry.

斜交坐标中若使用 x^2+y^2，便悄悄改变了几何度量。

For a midpoint M of side AB, with $C = 0, A = bu, B = av$, subtracting $\min(a,b)(u+v)/2$ from M locates the opposite perimeter-bisector endpoint on the longer adjacent side. The two boundary routes then have equal length; this proves the cleaver direction, rather than assuming it.

若 M 为 AB 中点，取 $C = 0, A = bu, B = av$，从 M 减去 $\min(a,b)(u+v)/2$，便得到较长邻边上的周长平分线另一端点。两条边界路线长度相等，从而证明劈周长线方向，而非直接假设该结论。

Practice / 配套练习: AIME-2010-II-Q15 (WB p. 296), AIME-2022-I-Q14 (WB p. 326)

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

Method 1 / 方法 1 [GE08-M01]

A conic equation is most useful after its geometry has selected the variables. Distinguish intersection, focal-distance optimization, tangency and envelope questions; each needs its own existence and branch check.

先由几何关系选择变量，圆锥曲线方程才最有效。求交、焦距优化、相切与包络问题应分别处理，并检验存在性与分支。

Exploit symmetry before intersecting an ellipse

求椭圆交点前先利用对称性

Procedure

1. Place the symmetry axis on a coordinate axis and write paired points as $(u, v), (-u, v)$.

2. Translate a metric condition, such as an equilateral altitude, into one linear relation between u, v.

3. Substitute into the conic, reject collapsed configurations, and recover the requested length.

通用步骤

1. 把对称轴放在坐标轴上，将成对的点写为 $(u, v), (-u, v)$。

2. 将等边三角形高等度量条件转为 u, v 的一个线性关系。

3. 代入圆锥曲线，排除退化构型，再求目标长度。

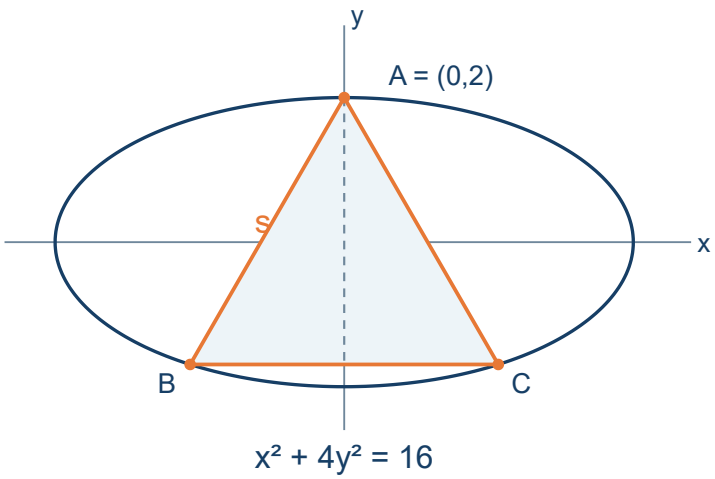
> Example. An equilateral triangle has top vertex $(0, 2)$ on $x^2 + 4y^2 = 16$ and altitude on the y axis. Find its squared side length.

> 例题：等边三角形的顶点 $(0, 2)$ 在椭圆 $x^2 + 4y^2 = 16$ 上，一条高在 y 轴上。求边长平方。

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

Method 1 / 方法 1 [GE08-M01] (continued / 续)



Worked solution

1. The other vertices are $(u, v), (-u, v)$, where $u > 0$ and $v = 2 - \sqrt{3}u$.

2. Substitution gives $u^2 + 4(2 - \sqrt{3}u)^2 = 16$, hence $u(13u - 16\sqrt{3}) = 0$.

3. Reject $u = 0$. The side is $2u = 32\sqrt{3}/13$, whose square is $3072/169$.

例题解析

1. 另两顶点为 $(u, v), (-u, v)$，其中 $u > 0, v = 2 - \sqrt{3}u$。

2. 代入得 $u^2 + 4(2 - \sqrt{3}u)^2 = 16$，即 $u(13u - 16\sqrt{3}) = 0$。

3. 排除 $u = 0$，边长为 $32\sqrt{3}/13$，平方为 $3072/169$。

A zero algebraic root can represent a collapsed triangle.

代数零根可能对应退化三角形。

Practice / 配套练习: AIME-2001-I-Q05 (WB p. 415)

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

Method 2 / 方法 2 [GE08-M02]

Combine conics, then certify the number of intersections

线性组合圆锥曲线后严格核实交点数

Procedure

1. Rewrite both conics with all terms on one side and look for a linear combination having equal x^2, y^2 coefficients.

2. Complete squares to identify the circle containing every common point.

3. Independently eliminate one variable and certify distinct real roots with disjoint sign intervals or derivative monotonicity.

4. Check a boundary root separately: a zero value alone does not certify distinct crossings.

通用步骤

1. 把两圆锥曲线都移项，寻找使 x^2, y^2 系数相同的线性组合。

2. 配方，确定包含全部公共点的圆。

3. 另行消元，用互不重叠的变号区间或导数单调性核实不同实根数。

4. 边界根须单独检验；仅有函数值为零不能证明相邻区间仍有另一根。

> Example. Find a circle through all intersections of $y = x^2 - 2$ and $x = y^2 - 2$, and determine the intersection count.

> 例题：求经过 $y = x^2 - 2$ 与 $x = y^2 - 2$ 全部交点的圆，并确定交点数。

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

Method 2 / 方法 2 [GE08-M02] (continued / 续)

Worked solution

1. Adding gives $x^2+y^2-x-y=4$, or $(x-1/2)^2+(y-1/2)^2=9/2$.

2. Subtracting gives $(x-y)(x+y+1)=0$. On $x=y$, the equation is $x^2-x-2=0$, giving $x=2,-1$.

3. On $x+y=-1$, substitute $y=-1-x$ into $y=x^2-2$: $x^2+x-1=0$, giving two distinct roots. Neither equals its paired y, so these are two further points.

4. There are four distinct real intersections, all on the identified circle.

例题解析

1. 相加得 $x^2+y^2-x-y=4$，即 $(x-1/2)^2+(y-1/2)^2=9/2$。

2. 相减得 $(x-y)(x+y+1)=0$。当 $x=y$ 时，$x^2-x-2=0$ 给出 $x=2,-1$。

3. 当 $x+y=-1$ 时，代入得 $x^2+x-1=0$，有两个不同根；它们均不等于对应的 y，所以另给两点。

4. 因此共有四个不同实交点，且均位于所求圆上。

A containing circle does not by itself guarantee four conic intersections.

找到公共圆并不能自动保证有四个圆锥曲线交点。

For a quartic with positive leading coefficient, four alternating sign intervals certify exactly four roots. If a boundary sample is zero, a nonzero derivative can supply the nearby sign change.

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

Method 2 / 方法 2 [GE08-M02] (continued / 续)

首项系数为正的四次式，四个交错变号区间可核实恰有四根。若边界样点恰为零，可利用非零导数补出附近的符号变化。

Practice / 配套练习: AIME-2021-I-Q15 (WB p. 420)

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

Method 3 / 方法 3 [GE08-M03]

A tangent-circle center traces an ellipse

相切圆的圆心形成椭圆轨迹

Procedure

1. Turn internal/external tangency into center-distance equations involving the unknown radius.

2. Eliminate that radius; a constant sum of two distances identifies an ellipse.

3. Before optimizing, verify every proposed ellipse point gives a positive radius and the required containment.

4. Substitute a variable line and require its quadratic discriminant to be nonnegative; an endpoint slope occurs at tangency.

通用步骤

1. 把内切、外切写为包含未知半径的圆心距离方程。

2. 消去半径；到两定点距离和为常数时得到椭圆。

3. 优化前先验证椭圆点给出的半径为正且满足容纳关系。

4. 代入可变直线，要求二次方程判别式非负；极端斜率常在相切时出现。

> Example. A circle lies inside the radius-7 circle centered at $(-3,0)$ and is externally tangent to the radius-3 circle centered at $(3,0)$. Show its center lies on an ellipse and decide whether every point of that ellipse is admissible.

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

Method 3 / 方法 3 [GE08-M03] (continued / 续)

> 例题：某圆在圆心 $(-3,0)$、半径 7 的圆内并与之内切，又与圆心 $(3,0)$、半径 3 的圆外切。求其圆心轨迹，并判断轨迹是否全部可行。

Worked solution

1. For center P and radius t, write $PF_1 = 7 - t$ and $PF_2 = 3 + t$. Thus $PF_1 + PF_2 = 10$.

2. The foci are six units apart, so the ellipse has $a = 5, c = 3, b = 4$, equation $x^2/25 + y^2/16 = 1$.

3. On this ellipse, PF_1 ranges from 2 to 8. But $t = 7 - PF_1 > 0$ requires $PF_1 < 7$, so part of the algebraic ellipse is inadmissible.

例题解析

1. 设圆心为 P、半径为 t，有 $PF_1 = 7 - t, PF_2 = 3 + t$，所以距离和为 10。

2. 焦距为六，故 $a = 5, c = 3, b = 4$，椭圆方程为 $x^2/25 + y^2/16 = 1$。

3. 椭圆上 PF_1 介于 2 与 8，而 $t = 7 - PF_1 > 0$ 要求 $PF_1 < 7$，所以代数椭圆并非全部可行。

Eliminating the radius can add inadmissible points; restore its sign.

消去半径会加入不可行点，必须补回正性条件。

Practice / 配套练习: AIME-2005-II-Q15 (WB p. 419)

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

Method 4 / 方法 4 [GE08-M04]

Pair focal distances so triangle bounds attain equality together

配对焦距使三角不等式同时取等

Procedure

1. Express each ellipse parameter as half its sum of focal distances.

2. Try both pairings of the four foci and apply triangle inequalities through the common point.

3. Choose a pairing whose two joining segments actually intersect.

4. At that intersection, recompute the parameters and check strict ellipse restrictions.

通用步骤

1. 将每个椭圆参数写成两焦距之和的一半。

2. 尝试四焦点的两种配对，对经过公共点的折线使用三角不等式。

3. 选择两条焦点连线确实相交的配对。

4. 在交点重新计算参数，并检验椭圆参数的严格限制。

> Example. Ellipse families have foci $(-2,0),(2,0)$ and $(4,2),(4,4)$, with distance sums $2a, 2b$. Find the least possible $a+b$ at a common point.

> 例题：两族椭圆焦点分别为 $(-2,0),(2,0)$ 与 $(4,2),(4,4)$，焦距和为 $2a, 2b$。求有公共点时 $a+b$ 的最小值。

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

Method 4 / 方法 4 [GE08-M04] (continued / 续)

Worked solution

1. Pair $(-2,0)$ with $(4,2)$ and $(2,0)$ with $(4,4)$. The joining lengths are $2\sqrt{10}, 2\sqrt{5}$, so $a+b \geq \sqrt{10}+\sqrt{5}$.

2. The segments intersect at $P=(14/5, 8/5)$, found from $y=(x+2)/3$ and $y=2(x-2)$. The point lies strictly inside both segments.

3. At P, both distance inequalities are equalities. The point is on neither focal axis, so the two ellipses are nondegenerate, attaining $\sqrt{10}+\sqrt{5}$.

例题解析

1. 把 $(-2,0)$ 与 $(4,2)$、$(2,0)$ 与 $(4,4)$ 配对，连线长为 $2\sqrt{10}, 2\sqrt{5}$，所以 $a+b \geq \sqrt{10}+\sqrt{5}$。

2. 由两直线 $y=(x+2)/3, y=2(x-2)$，交点为 $P=(14/5, 8/5)$，严格位于两线段内部。

3. 在 P，两处三角不等式同时取等。此点不在任一焦轴上，两椭圆均非退化，故达到 $\sqrt{10}+\sqrt{5}$。

A lower bound from the wrong pairing may have incompatible equality cases.

错误配对所得下界，其等号条件可能互不相容。

Practice / 配套练习: AIME-2022-II-Q12 (WB p. 421)

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

Method 5 / 方法 5 [GE08-M05]

Perpendicular hyperbola radii give an open feasible interval

双曲线垂直径向量给出开可行区间

Procedure

1. Write opposite vertices as $P, -P, Q, -Q$ and use perpendicularity of rhombus diagonals.

2. Substitute a slope into the conic and require a positive finite squared coordinate.

3. Apply the same existence condition to the perpendicular slope.

4. Optimize only on their common open interval and distinguish an infimum from an attained minimum.

通用步骤

1. 把对角顶点写为 $P, -P, Q, -Q$，并使用菱形对角线垂直。

2. 将斜率代入曲线，要求坐标平方为正且有限。

3. 对垂直斜率使用同样的存在条件。

4. 只在公共开区间上优化，区分下确界与可取到的最小值。

> Example. A centered rhombus has vertices on $x^2/3 - y^2/4 = 1$. Find the infimum of a squared diagonal.

> 例题：中心在原点的菱形顶点位于 $x^2/3 - y^2/4 = 1$，求一条对角线平方的下确界。

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

Method 5 / 方法 5 [GE08-M05] (continued / 续)

Worked solution

1. Put $P=(x,mx)$. Then $x^2=12/(4-3m^2)$, and the perpendicular line exists only when $3/4<m^2<4/3$.

2. The diagonal square is $4|P|^2=48(1+m^2)/(4-3m^2)$, strictly increasing in m^2.

3. At the lower endpoint the limit is 48. Every feasible value is greater than 48, and slopes approaching the endpoint attain values arbitrarily close to it.

例题解析

1. 令 $P=(x,mx)$，则 $x^2=12/(4-3m^2)$；垂直直线也有交点要求 $3/4<m^2<4/3$。

2. 对角线平方为 $48(1+m^2)/(4-3m^2)$，随 m^2 严格递增。

3. 趋向下端点时极限为 48。所有可行值均大于 48，且可任意接近它。

The endpoint fails because the other diagonal becomes asymptotic, not because the objective is undefined there.

端点不可行是因为另一对角线成为渐近方向，并非目标表达式在该处无定义。

Practice / 配套练习: AIME-2024-I-Q09 (WB p. 416)

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

Method 6 / 方法 6 [GE08-M06]

A one-segment point is an envelope tangency

仅属于一条线段的点对应包络相切

Procedure

1. Parameterize unit-axis intercepts by $(\cos\theta, \sin\theta)$.

2. For a positive point, form $F(\theta) = x\sec\theta + y\csc\theta$; membership is $F = 1$.

3. Use strict convexity and endpoint divergence to show exactly one segment means the minimum equals one.

4. Solve $F = 1, F\prime = 0$ at the specified segment parameter.

通用步骤

1. 用 $(\cos\theta, \sin\theta)$ 参数化单位线段的轴截距。

2. 对正坐标点构造 $F(\theta) = x\sec\theta + y\csc\theta$，在线段上等价于 $F = 1$。

3. 由严格凸性和端点趋于无穷，证明仅有一条线段等价于最小值恰为一。

4. 在指定线段参数处联立 $F = 1, F\prime = 0$。

> Example. On the unit segment joining $(\sqrt{3}/2, 0)$ and $(0, 1/2)$, find the positive point lying on no other unit segment joining the axes.

> 例题：在连接 $(\sqrt{3}/2, 0)$ 与 $(0, 1/2)$ 的单位线段上，求不在其他同类单位截轴线段上的正坐标点。

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

Method 6 / 方法 6 [GE08-M06] (continued / 续)

Worked solution

1. The parameter is $\theta = \pi/6$. The equations are $2x/\sqrt{3} + 2y = 1$ and $(2/3)x - 2\sqrt{3}y = 0$.

2. The derivative equation gives $x = 3\sqrt{3}y$. Substitution yields $y = 1/8, x = 3\sqrt{3}/8$.

3. For positive x, y, F is strictly convex and diverges at both endpoints. Its stationary value is one, so this point belongs to exactly one unit segment.

例题解析

1. 参数为 $\theta = \pi/6$，两式为 $2x/\sqrt{3} + 2y = 1$ 与 $(2/3)x - 2\sqrt{3}y = 0$。

2. 导数式给出 $x = 3\sqrt{3}y$，代入得 $y = 1/8, x = 3\sqrt{3}/8$。

3. 正坐标下 F 严格凸且两端趋于无穷；其驻点值为一，因此该点只属于一条单位线段。

A formal envelope equation needs a uniqueness proof; convexity supplies it.

形式上的包络方程仍需唯一性证明，严格凸性可提供此证明。

Practice / 配套练习: AIME-2024-II-Q12 (WB p. 422)

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

Method 7 / 方法 7 [GE08-M07]

Use reflection symmetry to construct a rotated-conic intersection

利用反射对称构造旋转圆锥曲线交点

Procedure

1. Identify a reflection preserving the original conic.

2. Seek a point whose inverse rotation is exactly that reflection.

3. The resulting linear relation intersects the original conic in explicit candidates.

4. Check the requested quadrant and use stated uniqueness only after verifying the constructed point.

通用步骤

1. 找出保持原圆锥曲线不变的反射。

2. 寻找逆旋转恰等于该反射的点。

3. 所得线性关系与原曲线相交，给出明确候选。

4. 检验所需象限；须先验证构造点，再使用题给唯一性。

> Example. Construct a fourth-quadrant common point of $y = x^2 - 1$ and its 90° counterclockwise rotation.

> 例题：构造 $y = x^2 - 1$ 与其绕原点逆时针旋转 90° 后曲线的一个第四象限公共点。

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

Method 7 / 方法 7 [GE08-M07] (continued / 续)

Worked solution

1. The inverse rotation is $(x, y) \mapsto (y, -x)$, and the original parabola is preserved by $(x, y) \mapsto (-x, y)$.

2. Equating these images gives $y = -x$. With $y = x^2 - 1$, the positive root is $x = (\sqrt{5} - 1)/2$.

3. The point $((\sqrt{5} - 1)/2, (1 - \sqrt{5})/2)$ lies in the fourth quadrant, and its inverse rotation is its reflection on the original parabola, so it belongs to both curves.

例题解析

1. 逆旋转为 $(x, y) \mapsto (y, -x)$，原抛物线关于 $(x, y) \mapsto (-x, y)$ 不变。

2. 令两像相等，得 $y = -x$。与 $y = x^2 - 1$ 联立，正根为 $x = (\sqrt{5} - 1)/2$。

3. 所得点 $((\sqrt{5} - 1)/2, (1 - \sqrt{5})/2)$ 在第四象限；逆旋转像是原曲线上的反射点，所以它在两曲线上。

This construction proves existence; uniqueness needs a separate argument unless explicitly given.

此构造证明存在；若题目未给唯一性，还须另证。

Practice / 配套练习: AIME-2025-I-Q09 (WB p. 417)

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

Method 8 / 方法 8 [GE08-M08]

Tangency radii are stationary distances to a conic

相切半径对应到圆锥曲线的驻定距离

Procedure

1. Parameterize a smooth point on the conic.

2. Form squared distance from the prescribed circle center.

3. Differentiate: a stationary distance means the radius vector is normal to the conic tangent.

4. Evaluate every stationary distance, remove duplicate radii, and verify actual tangency.

通用步骤

1. 参数化圆锥曲线上的光滑点。

2. 构造到指定圆心的距离平方。

3. 求导：距离驻定等价于半径向量垂直于曲线切线。

4. 计算全部驻定距离，去掉重复半径，并核实实际相切。

> Example. Find all circle radii centered at $(0,3)$ tangent to $y=x^2$.

> 例题：求圆心 $(0,3)$ 且与抛物线 $y=x^2$ 相切的所有圆半径。

Worked solution

1. At (t,t^2), squared distance is $D(t)=t^2+(t^2-3)^2$.

2. Its derivative is $D\prime(t)=2t(2t^2-5)$, giving $t=0$ or $t^2=5/2$.

Section 2.8 Conics & analytic loci

圆锥曲线与解析轨迹

Method 8 / 方法 8 [GE08-M08] (continued / 续)

3. These give $D = 9$ and $D = 11/4$. Thus the distinct radii are 3 and $\sqrt{11}/2$; in each case the radius is perpendicular to the tangent.

例题解析

1. 在点 (t, t^2)，距离平方为 $D(t) = t^2 + (t^2 - 3)^2$。

2. 导数为 $D\prime(t) = 2t(2t^2 - 5)$，所以 $t = 0$ 或 $t^2 = 5/2$。

3. 对应距离平方为 $9, 11/4$，故不同半径为 $3, \sqrt{11}/2$；两种情况下半径均垂直于切线。

> Tangency at one point does not require the circle to avoid the conic everywhere else.

> 在一点相切，并不要求圆与曲线在其他位置没有交点。

Practice / 配套练习: AIME-2026-II-Q06 (WB p. 418)

Section 2.9 Spheres & spatial distances

球与空间距离

Method 1 / 方法 1 [GE09-M01]

Spatial distance problems become manageable when the projection, symmetry axis or supporting plane is chosen first. A sphere center is not the same as a contact point; retain signed heights and verify containment after solving equal-distance equations.

空间距离题应先选择投影、对称轴或支撑平面。球心不是切点；解等距方程时保留有向高度，并在求解后核实容纳条件。

Sphere sections separate vertical and planar distances

球截面分离竖直距离与平面距离

Procedure

1. For sphere radius R and section radius a, write the center height as $h = \pm\sqrt{R^2 - a^2}$.

2. Decompose a center-to-center squared distance as planar distance squared plus the squared height difference.

3. When section radii are equal, subtract height squares to obtain a difference-of-squares equation.

4. After any squaring, restore the chosen side of the plane and the signs of all radicals.

通用步骤

1. 球半径为 R、截圆半径为 a 时，球心高度为 $h = \pm\sqrt{R^2 - a^2}$。

2. 把球心距平方分成平面投影距离平方与高度差平方。

3. 截圆半径相等时，相减高度平方，得到平方差方程。

4. 若曾平方，须补回球心所在半空间及各根式符号条件。

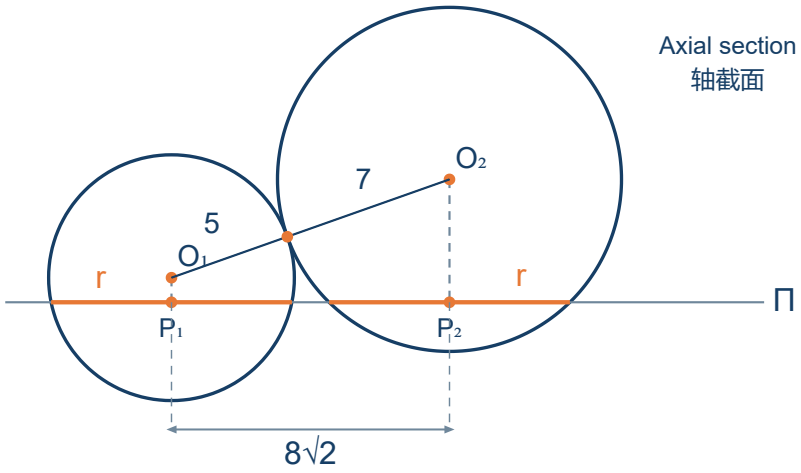
Section 2.9 Spheres & spatial distances

球与空间距离

Method 1 / 方法 1 [GE09-M01] (continued / 续)

Example. Two externally tangent spheres of radii $5, 7$ are cut into equal-radius circles by one plane. Their centers are on the same side, and their projected centers are distance $8\sqrt{2}$ apart. Find the common section radius.

例题：半径 $5, 7$ 的两外切球被同一平面截得等半径圆，球心位于平面同侧，投影圆心距为 $8\sqrt{2}$。求截圆半径。



Worked solution

1. Let positive heights be $u < v$. Since the sphere-center distance is 12, $(v-u)^2 = 144 - 128 = 16$, so $v - u = 4$.

2. Equal section radii give $v^2 - u^2 = 49 - 25 = 24$, hence $v + u = 6$ and $(u, v) = (1, 5)$.

3. The section radius squared is $25 - 1 = 24$, so it is $2\sqrt{6}$. Both heights are positive as required.

例题解析

1. 设正高度为 $u < v$。球心距为 12，故 $(v-u)^2 = 144 - 128 = 16$，得到 $v - u = 4$。

Section 2.9 Spheres & spatial distances

球与空间距离

Method 1 / 方法 1 [GE09-M01] (continued / 续)

2. 截圆半径相同给出 $v^2-u^2=49-25=24$，所以 $v+u=6$，即 $(u,v)=(1,5)$。

3. 截圆半径平方为 $25-1=24$，所以半径为 $2\sqrt{6}$；两高度均正，符合要求。

Use the difference of heights, not the difference of sphere radii.

应使用高度差，不能直接使用球半径差。

Practice / 配套练习: AIME-2020-I-Q06 (WB p. 425), AIME-2022-I-Q10 (WB p. 428)

Section 2.9 Spheres & spatial distances

球与空间距离

Method 2 / 方法 2 [GE09-M02]

Containment is a center-distance or diameter bound

容纳条件是球心距界或直径界

Procedure

1. A radius-r ball fits inside a radius-R ball precisely when the center distance is at most $R - r$.

2. For a ball resting on a plane, its center height is r; project the containment inequality to the contact plane.

3. A box needs sphere radius at least half its space diagonal because opposite vertices must both fit.

4. The box center attains that radius; for a family of boxes, maximize the diagonal separately.

通用步骤

1. 半径 r 的球能放入半径 R 的球，当且仅当球心距不超过 $R - r$。

2. 球在平面上相切时，球心高度为 r；把容纳不等式投影到接触平面。

3. 长方体需要的球半径至少是体对角线一半，因为两个对顶点均须被容纳。

4. 以长方体中心为球心可达到该半径；长方体族另需最大化体对角线。

> Example. A radius-2 ball rests on the equatorial disk of a radius-5 hemisphere and must be contained in it. What fraction of the disk allows its contact point?

> 例题：半径 2 的球在半径 5 的上半球底圆盘上相切且完全被容纳，允许接触点占底圆盘面积的几分之几?

Section 2.9 Spheres & spatial distances

球与空间距离

Method 2 / 方法 2 [GE09-M02] (continued / 续)

Worked solution

1. For contact distance d from the disk center, the small center is distance $\sqrt{d^2+4}$ from the hemisphere center.

2. Containment requires $\sqrt{d^2+4}+2 \le 5$, so $d^2 \le 5$. The small ball is automatically above the bottom plane.

3. The permitted disk has radius $\sqrt{5}$, giving area fraction $5/25 = 1/5$.

例题解析

1. 设接触点距底面圆心为 d，小球心距半球心为 $\sqrt{d^2+4}$。

2. 容纳要求 $\sqrt{d^2+4}+2 \le 5$，即 $d^2 \le 5$；小球同时自动位于底面上方。

3. 允许圆盘半径为 $\sqrt{5}$，面积占比为 $5/25 = 1/5$。

The containing sphere can be positioned separately for each box; its radius is the common worst-case requirement.

对每个长方体可分别摆放容球；共同要求的是最坏情形半径。

Practice / 配套练习: AIME-2024-I-Q15 (WB p. 189), AIME-2026-I-Q03 (WB p. 424)

Section 2.9 Spheres & spatial distances

球与空间距离

Method 3 / 方法 3 [GE09-M03]

Use normalized distances to supporting planes

使用归一化支撑平面距离

Procedure

1. Write each face or support as $ax+by+cz=d$ and divide signed distance by $\sqrt{a^2+b^2+c^2}$.

2. Choose an interior sign convention before equating face distances to a radius.

3. Exploit coordinate-plane tangency or reflection symmetry to reduce center variables.

4. For a largest insphere, prove a bound using supporting planes and then verify one center attains all inequalities.

通用步骤

1. 将每个面或支撑写为 $ax+by+cz=d$，有向距离须除以 $\sqrt{a^2+b^2+c^2}$。

2. 令各面距离等于半径前，先固定内部的正负号约定。

3. 利用坐标面相切或反射对称减少球心变量。

4. 求最大内球时，先由支撑平面证明界，再核实某球心满足全部不等式并取等。

> Example. Find the inradius of the tetrahedron in the first octant bounded by $x+y+z=6$.

> 例题：求第一卦限内由 $x+y+z=6$ 与三个坐标面围成的四面体内切球半径。

Section 2.9 Spheres & spatial distances

球与空间距离

Method 3 / 方法 3 [GE09-M03] (continued / 续)

Worked solution

1. Tangency to the coordinate planes gives center (r, r, r) with $r > 0$.

2. The distance to the fourth face is $(6 - 3r)/\sqrt{3}$, so $6 - 3r = r\sqrt{3}$.

3. Hence $r = 6/(3 + \sqrt{3}) = 3 - \sqrt{3}$. Its center is interior and the four positive face distances all equal r.

例题解析

1. 与三个坐标面相切使球心为 (r, r, r)，其中 $r > 0$。

2. 到第四面的距离为 $(6 - 3r)/\sqrt{3}$，故 $6 - 3r = r\sqrt{3}$。

3. 所以 $r = 3 - \sqrt{3}$。球心在内部，到四面的正距离均为 r。

Plane coefficients alone are not distances: normalization matters.

平面系数不是距离，法向量长度的归一化不可省略。

For a solid cone described by $\alpha u + \beta\rho \le \gamma$, where u is the axial coordinate, ρ the radial coordinate and $\beta > 0$, every lateral supporting plane stays at least a ball radius away precisely when $\sqrt{\alpha^2 + \beta^2}\, r \le \gamma - \alpha u - \beta\rho$ at its center. Two perpendicular cones give coupled absolute-value bounds; add them and check the base-plane distances too.

A rigid tripod can be kept fixed while its three contact points define a new support plane. The height is the distance of the apex to that plane, avoiding a full spatial rotation.

Section 2.9 Spheres & spatial distances

球与空间距离

Method 3 / 方法 3 [GE09-M03] (continued / 续)

若实心圆锥由 $\alpha u + \beta\rho \le \gamma$ 描述，其中 u 为轴向坐标、ρ 为径向坐标且 $\beta > 0$，则各侧面支撑平面到球心距离均不小于球半径，等价于球心满足 $\sqrt{\alpha^2 + \beta^2}\, r \le \gamma - \alpha u - \beta\rho$。两个垂直圆锥给出含绝对值的耦合界，可相加求界，但仍需检验底面距离。

刚性三脚架可保持自身坐标不变，由三个接触点确定新支撑平面；高度即顶点到该平面距离，无需求整体空间旋转。

Practice / 配套练习: AIME-2001-I-Q12 (WB p. 431), AIME-2006-I-Q14 (WB p. 393), AIME-2020-II-Q07 (WB p. 426), AIME-2026-II-Q12 (WB p. 435)

Section 2.9 Spheres & spatial distances

球与空间距离

Method 4 / 方法 4 [GE09-M04]

Symmetry locates an axial circumsphere center

对称性确定轴上的外接球心

Procedure

1. Project a sphere center to a plane through three sphere points; the projection is their planar circumcenter.

2. For an axial pyramid or symmetric tetrahedron, place the center at signed height z on the symmetry axis.

3. Equate squared distances to an apex and a base vertex; quadratic terms usually cancel.

4. When asked for distance to a filled triangle, verify the perpendicular foot is inside that triangle.

通用步骤

1. 球心向三个球面点所在平面投影，垂足即三点的平面外心。

2. 对轴对称棱锥或对称四面体，把球心放在对称轴有向高度 z 处。

3. 令到顶点和底面顶点的距离平方相等，二次项通常会消去。

4. 若求到填满三角形的距离，必须确认垂足在三角形内部。

> Example. A right square pyramid has base edge 4 and height 3. Find the radius of its circumsphere.

> 例题：正四棱锥底边长 4、高 3，求外接球半径。

Section 2.9 Spheres & spatial distances

球与空间距离

Method 4 / 方法 4 [GE09-M04] (continued / 续)

Worked solution

1. Put the base center at height zero and let the sphere center have signed height z. A base vertex is planar distance $2\sqrt{2}$ from the axis.

2. Equate $z^2 + 8 = (3 - z)^2$, obtaining $z = 1/6$.

3. The radius is $3 - z = 17/6$. Equal distances to all four base vertices follow by symmetry.

例题解析

1. 以底面中心为零高度，球心有向高度为 z；底顶点到轴的平面距离为 $2\sqrt{2}$。

2. 由 $z^2 + 8 = (3 - z)^2$ 得 $z = 1/6$。

3. 半径为 $3 - z = 17/6$，对称性保证到另三个底顶点距离相同。

The center can lie below a base plane; do not assume $z > 0$ without proof.

球心可能位于底面下方，不可未经证明假设 $z > 0$。

For a triangle of circumradius R on a sphere of radius L, its plane distance is $\sqrt{L^2 - R^2}$. This equals distance to the filled triangle only when its circumcenter lies inside, for example when the triangle is acute.

若球面上三角形外接圆半径为 R、球半径为 L，球心到其平面的距离为 $\sqrt{L^2 - R^2}$。只有外心在三角形内部时，才等于到填满三角形的距离；锐角三角形即满足此条件。

Practice / 配套练习: AIME-2000-II-Q12 (WB p. 430), AIME-2022-II-Q03 (WB p. 423), AIME-2026-II-Q12 (WB p. 435)

Section 2.9 Spheres & spatial distances

球与空间距离

Method 5 / 方法 5 [GE09-M05]

Orthonormal edge projections have a fixed square sum

正交单位边投影具有固定平方和

Procedure

1. Use three mutually perpendicular edge directions as an orthonormal basis.

2. The components of a unit vertical vector in that basis have squared sum one.

3. Multiply by edge length squared to relate three height differences.

4. Choose the root that gives all required projection signs and the specified nearest or farthest vertex.

通用步骤

1. 以三条相互垂直的棱方向作为标准正交基。

2. 竖直单位向量在此基中的三个分量平方和为一。

3. 乘以棱长平方，得到三个高度差的关系。

4. 根据投影正负及最近、最远顶点要求筛选代数根。

> Example. A side-3 cube has a lowest vertex at height $h > 0$, and its three adjacent vertices all at height 2. Find h.

> 例题：棱长 3 的立方体最低顶点高度为 $h > 0$，三个相邻顶点高度都为 2，求 h。

Worked solution

1. The three edge height projections are all $2 - h > 0$. Their squared sum must equal 3^2.

Section 2.9 Spheres & spatial distances

球与空间距离

Method 5 / 方法 5 [GE09-M05] (continued / 续)

2. Thus $3(2-h)^2 = 9$, giving $h = 2 \pm \sqrt{3}$.

3. The plus sign makes adjacent vertices lower, contradicting the premise. Hence $h = 2 - \sqrt{3} > 0$.

例题解析

1. 三条棱的高度投影均为正数 $2-h$，平方和必须为 3^2。

2. 所以 $3(2-h)^2 = 9$，得 $h = 2 \pm \sqrt{3}$。

3. 正号会使相邻顶点更低，故只能取 $h = 2 - \sqrt{3} > 0$。

The fixed square sum comes from orthogonality, not equal inclination.

固定平方和来自正交性，不依赖三条棱倾角相同。

Practice / 配套练习: AIME-2011-I-Q13 (WB p. 432)

Section 2.9 Spheres & spatial distances

球与空间距离

Method 6 / 方法 6 [GE09-M06]

A perpendicular section converts dihedrals into tangent-center triangles

垂直截面将二面角化为切球圆心三角形

Procedure

1. Cut perpendicularly to the intersection line of the planes; their dihedral becomes a planar angle.

2. A sphere tangent to both planes has its center on an angle bisector, transverse distance $r / \sin\theta$ for half-angle θ.

3. Keep the along-line coordinate separate and use tangency center distances in the resulting right triangle.

4. For axial sphere points, identify an actual diameter before using a right-angle or altitude-product identity.

通用步骤

1. 作垂直于两平面交线的截面，将二面角转为平面角。

2. 同时与两面相切的球心在角平分线上，半角为 θ 时横向距为 $r / \sin\theta$。

3. 交线方向坐标另行保留，在所得直角三角形中使用切球心距。

4. 对轴上球面点，先证实为直径端点，再使用直角或高线乘积关系。

> Example. Two perpendicular planes meet along a line. Two spheres of radius 2 lie in the same sector, tangent to both planes and each other. What is the separation of their centers along the intersection line?

Section 2.9 Spheres & spatial distances

球与空间距离

Method 6 / 方法 6 [GE09-M06] (continued / 续)

> 例题：两垂直平面相交于一直线，两个半径 2 的球位于同一角域，同时与两平面及彼此外切。求两球心沿交线方向的距离。

Worked solution

1. In the perpendicular cross section, both centers have coordinates $(2, 2)$ relative to the planes, so their transverse projections coincide.

2. External tangency makes their full center distance $2 + 2 = 4$.

3. All of that distance is along the intersection line, so the requested separation is 4.

例题解析

1. 在垂直截面内，两球心相对两面的坐标均为 $(2, 2)$，所以横向投影相同。

2. 外切使球心距为 $2 + 2 = 4$。

3. 该距离全部沿交线方向，故所求为 4。

The two-plane cross section does not capture separation parallel to their intersection line.

两平面的垂直截面不会显示沿交线方向的间距。

For unequal radii R, r in the same sector, the transverse center separation is $|R - r| / \sin\theta$. If a small sphere touches two equal spheres whose centers differ only along the intersection line, it lies halfway between them along that line.

Section 2.9 Spheres & spatial distances

球与空间距离

Method 6 / 方法 6 [GE09-M06] (continued / 续)

A sphere through three noncollinear triangle vertices has its center on their perpendicular circumcenter axis. Two distinct sphere points on that axis are diameter endpoints. A triangle vertex then supplies a right triangle and, when its projection lies between the endpoints, the altitude-product relation.

同一角域内半径为 R, r 的球心横向间距为 $|R-r|/\sin\theta$。若小球同时与两个仅沿交线分离的等球相切，则小球在交线方向上位于两者正中。

经过三角形三个不共线顶点的球，其球心在外心垂轴上。该轴上两个不同球面点必为直径端点。再用一个三角形顶点构造直角三角形；若其投影位于两端点之间，还可使用高线乘积关系。

Practice / 配套练习: AIME-2016-II-Q14 (WB p. 433), AIME-2021-II-Q10 (WB p. 427)

Section 2.9 Spheres & spatial distances

球与空间距离

Method 7 / 方法 7 [GE09-M07]

Prism distances split into horizontal chord and height classes

棱柱距离分成水平弦长与高度类别

Procedure

1. Classify pairs by their horizontal separation in the base polygon.

2. For vertices in different parallel bases, add the fixed height square to the horizontal distance square.

3. Test each possible equality of two sides, including equality with a base side rather than only two cross-base sides.

4. Count disjoint vertex classes and multiply by symmetry only when it preserves the class.

通用步骤

1. 按底面多边形中的水平距离对点对分类。

2. 不同平行底面的顶点距离平方，等于水平距离平方加固定高度平方。

3. 检查各种两边相等条件，既包括两条跨底边相等，也包括跨底边与底边相等。

4. 对互斥顶点类别计数，仅在对称保持类别时使用对称倍数。

> Example. A right square prism has unit base edge and height $\sqrt{2}$. Choose a diagonal pair in its bottom base. How many top vertices form an isosceles triangle with that pair?

> 例题：正四棱柱底边为一、高为 $\sqrt{2}$。固定底面一对对角顶点，顶面有多少顶点能与之组成等腰三角形?

Section 2.9 Spheres & spatial distances

球与空间距离

Method 7 / 方法 7 [GE09-M07] (continued / 续)

Worked solution

1. The bottom diagonal has length $\sqrt{2}$. A top vertex above either endpoint has cross distances $\sqrt{2}, 2$, so it forms an isosceles triangle by matching the bottom diagonal.

2. Each of the other two top vertices has horizontal distance one to both bottom endpoints, hence both cross distances are $\sqrt{3}$.

3. All four top vertices qualify. The two mechanisms are disjoint and neither may be omitted.

例题解析

1. 底面对角线长为 $\sqrt{2}$。正上方对应两个端点的顶点，其跨底距离为 $\sqrt{2}, 2$，通过底边与跨底边相等构成等腰。

2. 另外两个顶点到两底端点的水平距离均为一，跨底距离均为 $\sqrt{3}$。

3. 四个顶点全部符合；两种机制互斥且不能漏掉任一种。

Equal-height symmetry is only one way to make an isosceles spatial triangle.

等高对称不是空间等腰三角形的唯一来源。

Practice / 配套练习: AIME-2018-I-Q07 (WB p. 699)

Section 2.9 Spheres & spatial distances

球与空间距离

Method 8 / 方法 8 [GE09-M08]

An axial torus section gives the contact-circle radius

圆环面轴截面给出接触圆半径

Procedure

1. In the coaxial configuration, the sphere center lies on the torus rotation axis. A plane through that axis cuts the torus into its generating circles; analyze either one.

2. For sphere radius R and tube radius r, use center distance $R - r$ for internal tangency (requiring $R > r$), or $R + r$ for external tangency.

3. The contact point lies on the center line, so its distance to the axis scales by $R/(R \mp r)$.

4. Check that the required center distance is at least the major torus radius, so the axial placement exists.

通用步骤

1. 在同轴构型中，球心位于圆环旋转轴上。经过该轴的平面将圆环面截成母圆，分析其中一个即可。

2. 球半径为 R、管半径为 r 时，内切圆心距为 $R - r$（要求 $R > r$），外切为 $R + r$。

3. 切点在连心线上，到轴距离按 $R/(R \mp r)$ 缩放。

4. 检查所需圆心距不小于圆环主半径，保证轴向摆放存在。

> Example. A torus has major radius 3 and tube radius 1. Find its contact-circle radius when internally tangent to a sphere of radius 5.

> 例题：圆环面主半径为 3、管半径为 1，与半径 5 的球从内部相切，求接触圆半径。

Section 2.9 Spheres & spatial distances

球与空间距离

Method 8 / 方法 8 [GE09-M08] (continued / 续)

Worked solution

1. The generating-circle center is distance $5-1=4$ from the sphere center and distance 3 from the axis.

2. An axial displacement $\sqrt{4^2-3^2}=\sqrt{7}$ exists, so this placement is feasible.

3. The contact point is $5/4$ times as far from the axis as the generating-circle center. The contact-circle radius is $15/4$.

例题解析

1. 母圆圆心距球心为 $5-1=4$，距旋转轴为 3。

2. 轴向位移为 $\sqrt{4^2-3^2}=\sqrt{7}$，所以摆放可行。

3. 切点到轴距离是母圆心的 $5/4$ 倍，故接触圆半径为 $15/4$。

The contact circle is not generally the torus midline circle.

接触圆通常不是圆环的中心圆。

Practice / 配套练习: AIME-2024-II-Q08 (WB p. 429)

Section 2.9 Spheres & spatial distances

球与空间距离

Method 9 / 方法 9 [GE09-M09]

Two common tangent planes make sphere radii affine

两个公共切平面使球半径成为仿射函数

Procedure

1. Put one tangent plane at $z = 0$, giving center (x, y, ρ) for radius ρ.

2. Write the second plane with unit normal (u, v, h) and a consistent signed-distance equation.

3. Its tangency equation is linear in x, y, ρ; for a distinct plane solve $\rho = \alpha x + \beta y + \gamma$.

4. Interpolate this affine function from three noncollinear contact points and verify a unit normal exists.

通用步骤

1. 把一个切平面设为 $z = 0$，半径 ρ 的球心即 (x, y, ρ)。

2. 以单位法向量 (u, v, h) 表示第二平面，保持有向距离符号一致。

3. 切触式对 x, y, ρ 线性；平面不同便可解为 $\rho = \alpha x + \beta y + \gamma$。

4. 用三个不共线接触点插值仿射函数，并验证存在单位法向量。

> Example. Spheres tangent above $z = 0$ have radii $1, 2, 3$ at contact points $(0, 0), (2, 0), (0, 2)$. They share a second tangent plane on the same side. Find the radius at contact point $(1, 1)$ that shares this plane.

> 例题：在 $z = 0$ 上方相切的球，在接触点 $(0, 0), (2, 0), (0, 2)$ 处半径分别为 $1, 2, 3$，并与另一个平面同侧相切。求接触点 $(1, 1)$ 处与该平面也相切的球半径。

Section 2.9 Spheres & spatial distances

球与空间距离

Method 9 / 方法 9 [GE09-M09] (continued / 续)

Worked solution

1. The interpolating radius function is $g(x, y) = 1 + x/2 + y$. Therefore the desired radius is $g(1, 1) = 5/2$.

2. To verify the plane, set $u = (1 - h)/2, v = 1 - h$; the unit-normal equation gives $h = 1$ or $h = 1/9$.

3. Take the distinct-plane value $h = 1/9$, yielding $(u, v, h) = (4, 8, 1)/9$ and plane $4x + 8y + z = -8$. Its signed distance from each center is exactly its radius.

例题解析

1. 插值得半径函数 $g(x, y) = 1 + x/2 + y$，因此所求为 $5/2$。

2. 验证平面时，取 $u = (1 - h)/2, v = 1 - h$；单位法向条件给出 $h = 1$ 或 $h = 1/9$。

3. 取不同平面对应的 $h = 1/9$，得单位法向量 $(4, 8, 1)/9$ 及平面 $4x + 8y + z = -8$。各球心到该面的有向距离恰为其半径。

The $h = 1$ solution reproduces the first plane and must be excluded.

$h = 1$ 对应原来的第一个平面，必须排除。

Practice / 配套练习: AIME-2026-I-Q12 (WB p. 434)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 1 / 方法 1 [GE10-M01]

Choose the solid model before computing: similarity controls scale, a perpendicular section controls height, and coordinates control clipping. Surface area, volume and surface path length require different decompositions. Every additive calculation must respect overlap and newly created faces.

先确定立体模型，再计算：相似控制缩放，垂直截面控制高度，坐标控制截割。表面积、体积与表面路径需使用不同分解；相加之前必须核实重叠及新产生的面。

Similarity has different area and volume exponents

相似的面积与体积具有不同指数

Procedure

1. Identify a common linear scale t between corresponding solids.

2. Use area scale t^2 and volume scale $|t|^3$; for a complementary piece subtract from the whole.

3. Distinguish inherited paint from a newly cut face, and check $0 < t < 1$ for a true subdivision.

通用步骤

1. 确定对应立体的线性相似比 t。

2. 面积比为 t^2，体积比为 $|t|^3$；补块须从整体相减。

3. 区分原有油漆与新切面，真正分割须满足 $0 < t < 1$。

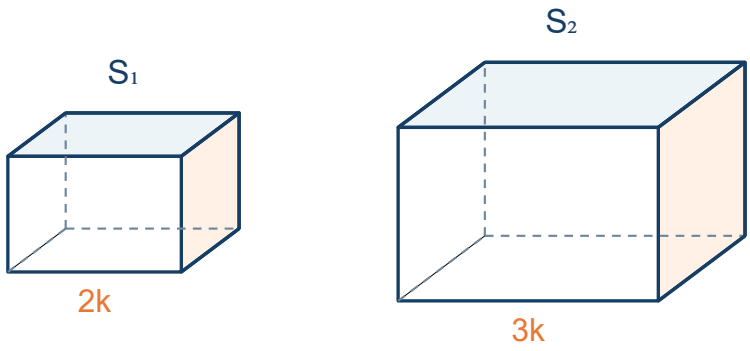
> Example. Two similar closed boxes have corresponding edge lengths in ratio $2:3$. Their total surface area is 130. Find each surface area and the ratio of their volumes.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 1 / 方法 1 [GE10-M01] (continued / 续)

> 例题：两个相似的封闭长方体，对应棱长比为 $2:3$，表面积之和为 130。求各自表面积及体积比。



S₁ + S₂ = 130 edge ratio / 棱长比 = 2 : 3

Worked solution

1. Area scales as the square of the linear ratio, so write the two surface areas as $4c$ and $9c$.

2. The total gives $13c = 130$, hence the areas are 40 and 90.

3. Volume scales as the cube, so the volume ratio is $8:27$. Similar boxes realizing these areas exist by uniformly scaling any fixed box.

例题解析

1. 面积按线性比的平方缩放，故把两表面积写成 $4c, 9c$。

2. 总和给出 $13c = 130$，所以表面积分别为 $40, 90$。

3. 体积按线性比的立方缩放，故体积比为 $8:27$。将任一固定长方体整体缩放，即可实现这些表面积。

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 1 / 方法 1 [GE10-M01] (continued / 续)

A fresh cut surface carries no original paint.

新切面没有原来的油漆。

For a tetrahedron with centroid G, the opposite face centroid is $G-(A-G)/3$; the four face centroids form a tetrahedron of volume $1/27$ of the original.

四面体重心为 G 时，顶点 A 对面重心为 $G-(A-G)/3$；四个面重心形成的四面体体积为原来的 $1/27$。

Practice / 配套练习: AIME-2000-I-Q08 (WB p. 444), AIME-2003-II-Q04 (WB p. 438), AIME-2004-I-Q11 (WB p. 453)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 2 / 方法 2 [GE10-M02]

Recursive face counts require non-overlap certification

递归面数需要不重叠证明

Procedure

1. Track the number of exposed faces and their common linear scale at each stage.

2. Multiply the number of newly attached copies by the cube of their scale, forming an exact increment sequence.

3. Before summing, prove distinct attachments have disjoint interiors and the counted faces remain exposed.

通用步骤

1. 逐阶段记录外露面数及其共同线性尺度。

2. 新附加副本数乘以尺度立方，得到精确体积增量数列。

3. 相加之前，先证明各附加块内部不交，所计表面仍然外露。

> Example. Attach one outward square pyramid of height $1/2$ to each face of a unit cube, using that entire face as its base. Find the new volume and justify adding the six pyramid volumes.

> 例题：在单位立方体每个面外侧，各接一个以整面为底、高为 $1/2$ 的正四棱锥。求新体积，并说明为何可将六个棱锥的体积相加。

Worked solution

1. Each pyramid has base area one and volume $1 \cdot (1/2)/3 = 1/6$.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 2 / 方法 2 [GE10-M02] (continued / 续)

2. The pyramid on $x = 1$ lies in $x \geq 1$, with its other coordinates in $[0, 1]$; the corresponding statement holds at every face. Distinct outward pyramids therefore have disjoint interiors and meet the cube only on its boundary.

3. The total is $1 + 6(1/6) = 2$. The geometric separation, rather than the number of faces alone, justifies the sum.

例题解析

1. 每个棱锥底面积为一，体积为 $1 \cdot (1/2)/3 = 1/6$。

2. $x = 1$ 面上的棱锥满足 $x \geq 1$，另两坐标均在 $[0, 1]$ 内；其余面同理。不同外接棱锥的内部互不相交，且只在边界接触原立方体。

3. 总体积为 $1 + 6(1/6) = 2$。能相加的依据是几何隔离，而非仅仅知道面数。

A plausible face recursion can fail after self-intersections; do not extend it without proof.

发生自交后，貌似合理的面数递推也可能失效，不能无证明推广。

For convex polyhedra, separating-axis tests use face normals and cross products of edge directions. A disjoint projection interval proves interior separation; exact rational coordinates prevent numerical ambiguity at boundary contacts. For repeated attachments, also test final boundary faces for hidden positive-area overlap.

凸多面体的分离轴可取各面法向及两棱方向叉积。投影区间内部不交即可证明内部隔离；使用精确有理坐标可避免边界接触的浮点歧义。递归附加还需检验最终表面是否存在隐藏的正面积重叠。

Practice / 配套练习: AIME-2001-II-Q12 (WB p. 450)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 3 / 方法 3 [GE10-M03]

Coordinate planes recover solid geometry and exposed surfaces

坐标平面重建立体几何与外露表面

Procedure

1. Place known faces in coordinate planes and translate planarity into linear equations.

2. Use equal-distance subtraction or a linear combination of squared distances to cancel unknown coordinates.

3. For a drilled solid, split surface area into retained exterior faces and new tunnel-wall polygons; count both separately.

通用步骤

1. 将已知面放在坐标面内，把共面条件写成线性方程。

2. 通过等距相减或距离平方线性组合消去未知坐标。

3. 对钻孔立体，把表面积分为保留外表面及新隧道壁多边形，分别计数。

> Example. In a cube with vertex $A=(0,0,0)$, let B, C, D be its three neighbors and G the opposite vertex. If $PB^2+PC^2+PD^2-PG^2=50$, find PA.

> 例题：立方体顶点 $A=(0,0,0)$，B, C, D 为三个邻点，G 为对顶点。若 $PB^2+PC^2+PD^2-PG^2=50$，求 PA。

Worked solution

1. Write neighbors as $(s,0,0), (0,s,0), (0,0,s)$, opposite vertex (s,s,s), and $P=(x,y,z)$.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 3 / 方法 3 [GE10-M03] (continued / 续)

2. Expanding the four squared distances cancels all linear terms and all s^2 terms, leaving $2(x^2+y^2+z^2)=50$.

3. Therefore $PA^2=25$ and $PA=5$, independently of cube side length.

例题解析

1. 把三邻点写为 $(s,0,0),(0,s,0),(0,0,s)$，对顶点为 (s,s,s)，并设 $P=(x,y,z)$。

2. 展开四个距离平方，线性项与全部 s^2 项消去，得 $2(x^2+y^2+z^2)=50$。

3. 因此 $PA=5$，与立方体棱长无关。

Distance to a filled face equals distance to its plane only when the perpendicular foot lies in the face.

只有垂足在面内时，到填满面的距离才等于到所在平面的距离。

A tunnel bounded by planes parallel to a cube diagonal can meet each cube face in different polygons. Solve each plane–face intersection before subtracting exterior openings or adding wall areas.

If a tetrahedron has two faces with exceptional edge length $2a$ and two with exceptional length $2b$, the exceptional edges must be opposite: adjacent exceptional edges would belong to one face together. With four equal cross edges L, use vertices $(\pm a,0,h),(0,\pm b,0)$, where $h^2=L^2-a^2-b^2$. This differs from a tetrahedron whose three opposite-edge pairs are equal.

由平行于立方体体对角线的平面围成的隧道，可能在不同立方体面上截出不同多边形。必须先求各平面与面的交线，才能扣除外部开口并加入隧道壁面积。

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 3 / 方法 3 [GE10-M03] (continued / 续)

若四面体两面含特殊棱长 $2a$，另两面含特殊棱长 $2b$，两特殊棱必为对棱；若相邻，它们会同时出现在某一面。其余四棱均为 L 时，可取顶点 $(\pm a, 0, h), (0, \pm b, 0)$，其中 $h^2 = L^2 - a^2 - b^2$。此模型不同于三组对棱各自相等的四面体。

Practice / 配套练习: AIME-2001-II-Q15 (WB p. 451), AIME-2002-I-Q15 (WB p. 452), AIME-2021-I-Q06 (WB p. 449), AIME-2026-II-Q12 (WB p. 435)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 4 / 方法 4 [GE10-M04]

Cube-vertex distances have only three types

立方体顶点距离只有三类

Procedure

1. Normalize the cube side to s; vertex-pair squared distances are $s^2, 2s^2, 3s^2$.

2. Use adjacency and opposite-vertex constraints to reject impossible triples of distance types.

3. Count triangles by their squared-side patterns and check their counts against $\binom{8}{3}$.

通用步骤

1. 设立方体棱长为 s，顶点间距离平方仅为 $s^2, 2s^2, 3s^2$。

2. 利用邻接与唯一对顶点关系排除不可能的三边类别。

3. 按边长平方型统计三角形，并与 $\binom{8}{3}$ 核对总数。

> Example. How many equilateral triangles have all three vertices among the vertices of a cube? What fraction of the cube-vertex triples do they form?

> 例题：立方体顶点中有多少组三点能构成等边三角形？它们占全部顶点三元组的比例是多少？

Worked solution

1. Normalize vertices to $\{0,1\}^3$. A pairwise equal-distance triple cannot use cube edges or space diagonals, so all its squared side lengths must be two.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 4 / 方法 4 [GE10-M04] (continued / 续)

2. Partition the eight vertices by the parity of their coordinate sum. Each class contains four vertices; any two distinct members differ in exactly two coordinates.

3. Each parity class therefore forms a regular tetrahedron, contributing $\binom{4}{3} = 4$ equilateral triangles. There are eight altogether.

4. There are $\binom{8}{3} = 56$ vertex triples, so their fraction is $8/56 = 1/7$.

例题解析

1. 将顶点归一为 $\{0,1\}^3$。两两等距三点不能以棱或体对角线相连，所以各边平方只能为二。

2. 按坐标和的奇偶把八顶点分为两类，每类四点；同类不同点恰有两个坐标不同。

3. 所以每类形成正四面体，贡献 $\binom{4}{3} = 4$ 个等边三角形，共八个。

4. 全部顶点三元组有 $\binom{8}{3} = 56$ 个，所占比例为 $8/56 = 1/7$。

An equilateral vertex triple is not a face triangle.

顶点组成的等边三角形不在某一个正方形面内。

Practice / 配套练习: AIME-2002-II-Q02 (WB p. 436), AIME-2003-I-Q06 (WB p. 445)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 5 / 方法 5 [GE10-M05]

Rounded boxes decompose by the nearest feature

圆角长方体按最近特征分解

Procedure

1. Separate the original box, face slabs, edge quarter-cylinders and corner sphere octants.

2. Group congruent fractional pieces before evaluating their volumes.

3. Use the nearest-feature partition to justify disjoint interiors and complete coverage.

通用步骤

1. 分成原长方体、面上薄层、棱旁四分之一圆柱及顶点八分之一球。

2. 先合并全等的分数块，再计算体积。

3. 按最近特征划分可证明各块内部不交且完整覆盖。

> Example. A solid box has side lengths $a, b, c > 0$. A coating consists of all exterior points at distance at most $r > 0$ from the box. Derive the coating volume as a polynomial in r.

> 例题：实心长方体三边为 $a, b, c > 0$。涂层由长方体外部距其不超过 $r > 0$ 的点组成。把涂层体积写成关于 r 的多项式。

Worked solution

1. The six face slabs have total volume $2(ab + ac + bc)r$; the original box is excluded from the coating.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 5 / 方法 5 [GE10-M05] (continued / 续)

2. The four edges parallel to a side of length a contribute four quarter-cylinders, totaling $\pi a r^2$. The three directions together give $\pi(a+b+c)r^2$.

3. The eight corner octants combine into one sphere of volume $4\pi r^3/3$.

4. The coating volume is $2(ab+ac+bc)r+\pi(a+b+c)r^2+4\pi r^3/3$. Assigning each exterior point to its nearest box face, edge or vertex proves coverage and disjoint interiors.

例题解析

1. 六个面上的薄层总体积为 $2(ab+ac+bc)r$；原长方体不属于涂层。

2. 平行于长度 a 的四条棱各贡献四分之一圆柱，共为 $\pi a r^2$；三个方向合计 $\pi(a+b+c)r^2$。

3. 八个角部八分之一球合成一个球，体积为 $4\pi r^3/3$。

4. 涂层体积为 $2(ab+ac+bc)r+\pi(a+b+c)r^2+4\pi r^3/3$。按外部点的最近面、棱或顶点分类，可证明完整覆盖及内部互不相交。

Adding twice the radius to every box dimension replaces curved edge and corner pieces by rectangular ones and gives the wrong volume.

直接把各边长加上两倍半径，会错误地把曲面角部换成方角。

Practice / 配套练习: AIME-2003-I-Q05 (WB p. 437)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 6 / 方法 6 [GE10-M06]

An affine height averages to its value at the base centroid

仿射高度的平均值等于底面重心处高度

Procedure

1. Write the upper-minus-lower plane separation as an affine height $h(x, y)$.

2. On a centrally symmetric base, pair opposite points: their heights sum to twice the central height.

3. Multiply base area by this average height, checking the height is nonnegative throughout.

通用步骤

1. 将上下平面之差写为仿射高度 $h(x, y)$。

2. 在中心对称底面上配对相反点，其高度和为中心高度的两倍。

3. 底面积乘平均高度，并检查高度在整个底面非负。

> Example. Above the disk $x^2 + y^2 \le 4$, a solid lies between $z = 0$ and $z = 3 + x$. Find its volume.

> 例题：立体位于圆盘 $x^2 + y^2 \le 4$ 上方、$z = 0$ 与 $z = 3 + x$ 之间，求体积。

Worked solution

1. The height is nonnegative because $-2 \le x \le 2$, so $1 \le 3 + x \le 5$.

2. Opposite disk points have heights $3 + x$ and $3 - x$, averaging to 3.

3. The disk area is 4π, so volume is 12π.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 6 / 方法 6 [GE10-M06] (continued / 续)

例题解析

1. 由于 $-2 \le x \le 2$，高度介于 1 与 5，始终非负。

2. 相反圆盘点的高度为 $3+x, 3-x$，平均值为 3。

3. 底面积为 4π，所以体积为 12π。

If height changes sign, the desired solid needs a separate region split.

若高度变号，所求立体必须另行分区。

Practice / 配套练习: AIME-2003-II-Q05 (WB p. 439)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 7 / 方法 7 [GE10-M07]

Visibility and paint depend on a cube cell’s boundary position

可见性与油漆取决于单位格的边界位置

Procedure

1. Encode whether a cell touches each designated outer face.

2. Count hidden cells with shifted dimensions, such as $(p-1)(q-1)(r-1)$ for three adjacent visible faces.

3. For exposed or unpainted face pairs, classify corner, edge, face-interior and solid-interior positions before counting orientations.

通用步骤

1. 记录单位格是否接触各指定外表面。

2. 用平移后的尺寸计隐藏格，例如三个相邻可见面对应 $(p-1)(q-1)(r-1)$。

3. 计外露或未染面组合时，先按角、棱、面内及体内位置分类，再计朝向。

> Example. A $3\times4\times5$ block of unit cubes is viewed with three adjacent faces visible. How many cubes touch none of those faces?

> 例题：从三个相邻面观察 $3\times4\times5$ 单位立方体块，有多少小立方体完全不接触这些面?

Worked solution

1. In each coordinate direction exclude the first layer touching the corresponding visible face.

2. The remaining index choices are $2,3,4$ independently.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 7 / 方法 7 [GE10-M07] (continued / 续)

3. Thus the hidden count is $2 \cdot 3 \cdot 4 = 24$. This is different from removing all six outer layers.

例题解析

1. 每个坐标方向排除与相应可见面相接的首层。

2. 三个方向独立剩余 $2, 3, 4$ 个指标。

3. 所以隐藏格数为 24；这不同于移除全部六个外表面层。

Visible from three faces does not mean exposed on any of six faces.

从三个面可见，不等于是否接触六个表面中的任意一个。

Two adjacent marked cube faces meet along one of twelve equally likely edges. To keep both inward, count unordered inward-face pairs minus opposite-face pairs. For a corner, edge position and face center these counts are $\binom{3}{2} = 3$, $\binom{4}{2} - 1 = 5$ and $\binom{5}{2} - 2 = 8$, respectively, each out of twelve.

立方体两个相邻标记面交于十二条等可能棱之一。要使两面均朝内，可计向内面的无序二元组，再扣除对面组合。大立方体角格、棱格、面心格的合法数分别为 $\binom{3}{2} = 3$, $\binom{4}{2} - 1 = 5$, $\binom{5}{2} - 2 = 8$，总朝向类别均为十二。

Practice / 配套练习: AIME-2004-II-Q03 (WB p. 531), AIME-2005-I-Q09 (WB p. 737)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 8 / 方法 8 [GE10-M08]

Develop curved surfaces before measuring paths or rolling

测量路径或滚动前先展开曲面

Procedure

1. A cylinder develops to a rectangle; a cone of slant height ℓ and base radius r develops to angle $2\pi r/\ell$.

2. Taut geodesics become straight lines in a valid development; compare angular lifts or winding classes.

3. For rolling, equate no-slip arc lengths and then recover the desired height or length.

4. At a tangent departure from a cylinder, preserve the developed line's pitch in the free segment.

通用步骤

1. 圆柱展开为矩形；母线长 ℓ、底半径 r 的圆锥展开角为 $2\pi r/\ell$。

2. 绷紧最短路径在合法展开中成为直线，并须比较绕行或角度提升类别。

3. 对无滑动滚动令对应弧长相等，再恢复目标高度或长度。

4. 从圆柱相切离开的自由线段，须保持展开直线的坡度。

> Example. A cylinder has circumference 8 and height 3. Find the shortest path on its lateral surface from a point on its bottom rim to the diametrically opposite point on its top rim. How many shortest paths are there?

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 8 / 方法 8 [GE10-M08] (continued / 续)

例题：圆柱周长为 8、高为 3。从下底圆周一点沿侧面到上底圆周上与其对径的位置，求最短距离及最短路径条数。

Worked solution

1. Unroll the lateral surface into its periodic strip. Put the start at $(0,0)$; the endpoint lifts are $(4+8k,3)$ for integers k.

2. A straight lifted path has squared length $(4+8k)^2+9$. The least absolute horizontal displacement is four, attained exactly at $k=0,-1$.

3. The minimum distance is therefore 5. Both straight segments stay between the two rim lines, giving two shortest paths, one around each side of the cylinder.

例题解析

1. 把侧面展开为周期条带。起点取 $(0,0)$，终点各提升为 $(4+8k,3)$，其中 k 为整数。

2. 直线提升的长度平方为 $(4+8k)^2+9$。最小水平位移绝对值为四，且仅在 $k=0,-1$ 达到。

3. 所以最短距离为 5；两条直线均在上下底边之间，投回圆柱后得到分别绕两侧的两条最短路径。

A planar angle between generators is not generally their developed angle.

母线的空间角通常不等于展开后的角。

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 8 / 方法 8 [GE10-M08] (continued / 续)

For a rolling cone completing one circuit about its fixed vertex, the contact travel is $2\pi\ell$, equal to the number of axial spins times $2\pi r$. For a cylinder rope, the free tangent's horizontal length is $\sqrt{d^2-R^2}$; use the common length-to-horizontal-projection ratio.

圆锥绕固定顶点滚完一圈，接触路程为 $2\pi\ell$，等于轴向自转圈数乘 $2\pi r$。圆柱绳索自由切线的水平长度为 $\sqrt{d^2-R^2}$，再用共同的实际长度与水平投影比。

Practice / 配套练习: AIME-2004-I-Q14 (WB p. 454), AIME-2004-II-Q11 (WB p. 455), AIME-2008-I-Q05 (WB p. 440)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 9 / 方法 9 [GE10-M09]

Find the perpendicular height before using a solid-volume formula

套用立体体积公式前先求垂直高度

Procedure

1. Distinguish edge length, face slant height and perpendicular solid height.

2. Use a right section, base circumradius or explicit center coordinates to recover that height.

3. Apply $V = Kh/3$ for a pyramid, or a dimension product for a box; only then simplify or optimize.

通用步骤

1. 区分棱长、面上斜高及立体的垂直高度。

2. 用直角截面、底面外接圆半径或明确圆心坐标求垂直高。

3. 棱锥用 $V = Kh/3$，长方体用三维乘积；最后再化简或优化。

> Example. A tetrahedron has a right-triangle base with legs $6, 8$. Each edge from its apex to a base vertex has length 13. Find its volume.

> 例题：四面体底面为两直角边长 $6, 8$ 的直角三角形，顶点到三个底面顶点的距离均为 13。求体积。

Worked solution

1. The base hypotenuse is ten, so its circumcenter is the hypotenuse midpoint and its circumradius is five. Its area is $6 \cdot 8/2 = 24$.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 9 / 方法 9 [GE10-M09] (continued / 续)

2. Equal apex distances put its perpendicular projection at that circumcenter. The vertical height is $\sqrt{13^2-5^2}=12$.

3. Thus the volume is $24\cdot 12/3=96$. Placing the apex twelve units above the circumcenter realizes all three required lateral lengths.

例题解析

1. 底面斜边为十，外心为斜边中点，外接圆半径为五，面积为 $6\cdot 8/2=24$。

2. 顶点到三底面点等距，故其垂足在外心；垂直高为 $\sqrt{13^2-5^2}=12$。

3. 体积为 $24\cdot 12/3=96$。将顶点放在外心正上方十二处，确能实现三条长为十三的侧棱。

Equal lateral edges put the apex over the base circumcenter, not necessarily its centroid.

侧棱相等使顶点在底面外心上方，不一定在重心上方。

For the regular octahedron with axial vertices $(\pm1,0,0),(0,\pm1,0),(0,0,\pm1)$, two square pyramids give volume $4/3$. Its face centers are $(\pm1/3,\pm1/3,\pm1/3)$, a cube of side $2/3$.

If box cuts preserve a fixed sum of the remaining dimensions, AM–GM bounds the volume; confirm the equal-dimension candidate uses an allowed integral number of cuts.

轴上顶点为 $(\pm1,0,0),(0,\pm1,0),(0,0,\pm1)$ 的正八面体，可由两个正四棱锥求得体积 $4/3$。各面重心为 $(\pm1/3,\pm1/3,\pm1/3)$，形成边长 $2/3$ 的立方体。

若切割长方体后剩余三维和固定，可由均值不等式限制体积；还须确认等边长候选对应合法的整数切片数。

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 9 / 方法 9 [GE10-M09] (continued / 续)

Practice / 配套练习: AIME-2005-II-Q10 (WB p. 446), AIME-2008-II-Q03 (WB p. 181), AIME-2012-II-Q05 (WB p. 441), AIME-2017-I-Q04 (WB p. 443), AIME-2022-II-Q03 (WB p. 423)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 10 / 方法 10 [GE10-M10]

Planar section area is projected area divided by a cosine

平面截面积等于投影面积除以余弦

Procedure

1. Find the actual section polygon by intersecting the cutting plane with solid edges or curved boundaries.

2. Project to a convenient coordinate plane, retaining exactly the allowed projected region.

3. If the section plane is $z = px + qy + c$, multiply projected area by $\sqrt{1 + p^2 + q^2}$.

通用步骤

1. 将截平面与立体各棱或曲边相交，确定实际截面多边形。

2. 向方便的坐标面投影，精确保留合法投影区域。

3. 若截平面为 $z = px + qy + c$，则投影面积乘 $\sqrt{1 + p^2 + q^2}$。

> Example. A plane $z = x + y$ cuts the vertical prism over the unit square, with enough height to contain the entire patch. Find the patch area.

> 例题：平面 $z = x + y$ 截过以单位正方形为底的直棱柱，棱柱高度足以包含整片截面，求截片面积。

Worked solution

1. The four section vertices are $(0, 0, 0), (1, 0, 1), (1, 1, 2), (0, 1, 1)$.

2. Their horizontal projection is the unit square of area one. Here $p = q = 1$.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 10 / 方法 10 [GE10-M10] (continued / 续)

3. The section area is $1\sqrt{1+1+1}=\sqrt{3}$. The same value follows from the cross product of adjacent section edges.

例题解析

1. 四个截面顶点为 $(0,0,0),(1,0,1),(1,1,2),(0,1,1)$。

2. 水平投影为面积一的单位正方形，且 $p=q=1$。

3. 实际面积为 $\sqrt{1+1+1}=\sqrt{3}$，亦可由相邻截面边叉积核实。

An infinite-plane ellipse is not always the cylinder’s bounded section: top and bottom caps may truncate it.

无限平面与圆柱面的椭圆，不一定就是有限圆柱截面；上下底面可能将它截断。

A tilted plane through a cylinder center can project to a disk strip. Compute the strip by subtracting circular caps, then apply the tilt factor; do not replace the strip by a full ellipse.

过圆柱中心的斜平面可能投影为圆盘条带，应先扣除圆弓形计算条带，再乘倾斜系数，不可直接当成完整椭圆。

Practice / 配套练习: AIME-2007-I-Q13 (WB p. 456), AIME-2015-I-Q15 (WB p. 459)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 11 / 方法 11 [GE10-M11]

Fold rigid panels by matching their projected seam points

通过拼缝投影相合求刚性侧板折叠

Procedure

1. Choose coordinates in the original sheet and locate the hinge lines and cut vertices.

2. After folding, distances to a hinge line remain fixed; symmetry determines which seam points must coincide.

3. Split the invariant panel distance into horizontal and vertical components and solve for height.

通用步骤

1. 在原纸面取坐标，定位折线与切口顶点。

2. 折叠后到铰链直线的距离保持不变，对称性确定必须重合的拼缝点。

3. 把不变的侧板距离分成水平与竖直分量，求高度。

> Example. Two perpendicular hinge lines are $x=3, y=3$. Points $(2,0)$ and $(0,2)$ on their respective symmetric panels are folded up and joined above $(2,2)$. Find the common height.

> 例题：两条垂直折线为 $x=3, y=3$。对称侧板上的点 $(2,0),(0,2)$ 折起后在 $(2,2)$ 正上方拼合，求高度。

Worked solution

1. The point $(2,0)$ is distance three from its hinge $y=3$. Its perpendicular hinge foot is $(2,3)$.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 11 / 方法 11 [GE10-M11] (continued / 续)

2. After joining, its horizontal separation from that foot is one. Rigid folding keeps the total separation three.

3. Thus $h^2 + 1^2 = 3^2$, giving $h = 2\sqrt{2}$. The other panel gives the same height by reflection.

例题解析

1. 点 $(2, 0)$ 到折线 $y = 3$ 的距离为三，垂足为 $(2, 3)$。

2. 拼合后其到垂足的水平距离为一，刚性折叠使总距离仍为三。

3. 故 $h^2 + 1^2 = 3^2$，得 $h = 2\sqrt{2}$；另一侧板由对称给出同高。

Folding a tray does not imply its walls become vertical.

托盘折起后，其侧壁不一定竖直。

Practice / 配套练习: AIME-2008-I-Q15 (WB p. 457)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 12 / 方法 12 [GE10-M12]

Revolve a triangle as two cones sharing a base

将三角形旋转分成共底双圆锥

Procedure

1. Resolve all planar inequalities to locate the triangular region and the rotation axis.

2. Project the off-axis vertex perpendicularly and check whether its foot lies on the base segment.

3. If the foot is internal, the two cones share radius equal to that perpendicular distance; their heights sum to the base length.

通用步骤

1. 先解平面不等式，确定三角形区域及旋转轴。

2. 从轴外顶点作垂足，并检查垂足是否在底边线段内。

3. 若垂足在内部，两圆锥共底半径为该垂距，其高之和为底边长。

> Example. Revolve triangle $A=(-2,0), B=(4,0), C=(1,3)$ about the x axis. Find its volume.

> 例题：将三角形 $A=(-2,0), B=(4,0), C=(1,3)$ 绕 x 轴旋转，求体积。

Worked solution

1. The perpendicular foot of C is $(1,0)$, inside AB. Both cone radii are three.

2. The left and right heights are three and three, summing to six.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 12 / 方法 12 [GE10-M12] (continued / 续)

3. Total volume is $\pi \cdot 3^2(3+3)/3 = 18\pi$. Their interiors meet only at the common base.

例题解析

1. C 的垂足为 $(1,0)$，位于 AB 内，两圆锥半径均为三。

2. 左右高分别为三，总高为六。

3. 总体积为 $\pi \cdot 3^2(3+3)/3 = 18\pi$；两块仅在共底边界接触。

If the foot is outside, adding two cones double-counts; a difference model is required.

若垂足在外，相加会重复计算，应改用圆锥差。

Practice / 配套练习: AIME-2010-I-Q11 (WB p. 458)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 13 / 方法 13 [GE10-M13]

Clip a first-octant tetrahedron by box constraints

用长方体限制截割第一卦限四面体

Procedure

1. Write the cutting plane in edge coordinates and choose the desired half-space.

2. Use the first-octant simplex volume $abc/6$ from its three intercepts.

3. Subtract portions beyond the box faces, using translated intercepts and inclusion–exclusion when needed.

4. For a tilted immersed cube, derive the height functional from orthonormal edge projections before clipping.

通用步骤

1. 在棱坐标中写出截平面，并选择目标半空间。

2. 由三轴截距 a, b, c 使用第一卦限四面体体积 $abc/6$。

3. 扣除超出长方体各面的部分，使用平移截距，必要时作容斥。

4. 倾斜浸水立方体应先由正交棱投影求出高度函数，再进行截割。

> Example. Find the volume in $[0,2]^3$ satisfying $x+y+z<3$.

> 例题：求 $[0,2]^3$ 内满足 $x+y+z<3$ 的部分体积。

Worked solution

1. The full first-octant tetrahedron has intercepts three, so volume $27/6$.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 13 / 方法 13 [GE10-M13] (continued / 续)

2. Beyond each of $x = 2, y = 2, z = 2$, translate that coordinate by two; the excluded simplex has intercepts one and volume $1/6$.

3. No point can exceed two coordinates of two while their sum is below three, so there are no overlapping exclusions. Volume is $27/6 - 3/6 = 4$.

例题解析

1. 完整第一卦限四面体三截距均为三，体积为 $27/6$。

2. 超出 $x = 2, y = 2, z = 2$ 的每块，分别平移相应坐标二，得到截距均一、体积 $1/6$ 的四面体。

3. 坐标和小于三时，不可能有两个坐标同时大于二，因此排除块不重叠，体积为 4。

Check every coordinate ceiling; not all three truncations necessarily occur.

各坐标上界须分别检查，不一定三个方向都发生截割。

For a cube with a body diagonal vertical, a near-corner horizontal section has equal edge intercepts and is equilateral. Verify those intercepts are below the edge length and that the submerged depth is less than the container depth.

立方体体对角线竖直时，靠近角的水平截面具有相等棱截距，为等边三角形。须检查截距小于棱长，并检查浸没深度小于容器深度。

Practice / 配套练习: AIME-2012-I-Q08 (WB p. 447), AIME-2015-II-Q09 (WB p. 448), AIME-2023-II-Q14 (WB p. 461)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 14 / 方法 14 [GE10-M14]

Dihedral angles live in a perpendicular section

二面角应在垂直截面中测量

Procedure

1. Locate the line where the two planes meet.

2. Choose one line in each plane perpendicular to that common line at the same point.

3. The angle between these section lines is the dihedral angle; use planar trigonometry to recover spatial height.

通用步骤

1. 找出两平面的公共交线。

2. 在交线同一点，分别于两平面内作垂直交线的直线。

3. 这两截线夹角才是二面角，再用平面三角法求空间高度。

> Example. A triangular base has altitude 4 from vertex A to side UV. A point W is directly h above A. If planes AUV, WUV form 60°, find h.

> 例题：三角形底面中 A 到边 UV 的高为四，点 W 位于 A 正上方高 h 处。若平面 AUV, WUV 的二面角为 60°，求 h。

Worked solution

1. Let M be the perpendicular foot from A to UV. Since AW is normal to the base, WM is also perpendicular to UV.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 14 / 方法 14 [GE10-M14] (continued / 续)

2. The required section angle is $\angle AMW = 60^\circ$. Triangle AMW is right at A.

3. Thus $h = AM\tan 60^\circ = 4\sqrt{3}$.

例题解析

1. 令 M 为 A 到 UV 的垂足；因 AW 垂直底面，WM 也垂直 UV。

2. 所需截面角为 $\angle AMW = 60^\circ$，而 AMW 在 A 为直角。

3. 所以 $h = 4\tan 60^\circ = 4\sqrt{3}$。

Two arbitrary lines in the planes do not measure their dihedral.

分别取两平面内任意两条线，不能测得二面角。

Practice / 配套练习: AIME-2016-I-Q04 (WB p. 442), AIME-2016-II-Q14 (WB p. 433)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 15 / 方法 15 [GE10-M15]

Face diagonals determine a Gram volume

面对角线确定格拉姆体积

Procedure

1. Choose generating edge vectors u, v, w and use $|u \pm v|^2$ to recover edge norms and dot products.

2. Form their Gram matrix, whose determinant equals squared parallelepiped volume.

3. Keep all feasible dot-product sign choices; the sign of the triple product of pairwise dots can change volume.

通用步骤

1. 取生成棱向量 u, v, w，用 $|u \pm v|^2$ 求棱范数与点积。

2. 组成格拉姆矩阵，其行列式等于平行六面体体积平方。

3. 保留全部可行点积符号组合；三个两两点积之积的符号可能改变体积。

> Example. Unit vectors u, v, w satisfy $u \cdot v = u \cdot w = v \cdot w = 1/2$. Find the parallelepiped volume.

> 例题：单位向量 u, v, w 的两两点积均为 $1/2$，求所生成平行六面体体积。

Worked solution

1. The Gram matrix has diagonal entries one and off-diagonal entries $1/2$.

2. Its determinant is $1 - 3(1/2)^2 + 2(1/2)^3 = 1/2$.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 15 / 方法 15 [GE10-M15] (continued / 续)

3. The volume is $1/\sqrt{2}$. Positive determinant together with positive leading principal minors confirms a nondegenerate realization.

例题解析

1. 格拉姆矩阵对角元均一，非对角元均 $1/2$。

2. 行列式为 $1-3(1/2)^2+2(1/2)^3=1/2$。

3. 体积为 $1/\sqrt{2}$；各顺序主子式均正，确认存在非退化空间实现。

An algebraic dot-product pattern needs a positive semidefinite Gram matrix to be geometrically feasible.

代数点积模式需有半正定格拉姆矩阵，才能在几何上实现。

Practice / 配套练习: AIME-2023-I-Q13 (WB p. 460)

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 16 / 方法 16 [GE10-M16]

Opposite equal edges embed a tetrahedron in a box

对棱相等可把四面体嵌入长方体

Procedure

1. Use alternate vertices of a box so opposite tetrahedron edges are diagonals of corresponding box faces.

2. Solve the three pair-sum equations for squared box dimensions and require positivity.

3. Compute volume as one third of box volume; compute face area and use $3V = rS$ for an insphere.

通用步骤

1. 取长方体相间顶点，使四面体对棱成为相应长方体面的对角线。

2. 解三组平方和方程求长方体边长平方，并要求均为正。

3. 四面体体积为长方体的三分之一；求各面面积后，用 $3V = rS$ 求内切球。

> Example. Could a tetrahedron have its three opposite-edge pairs of lengths $\sqrt{2}, \sqrt{3}, \sqrt{6}$? Each proposed face is a valid triangle; determine whether the spatial solid exists.

> 例题：四面体三组对棱长度能否分别为 $\sqrt{2}, \sqrt{3}, \sqrt{6}$？虽然每个拟定面都满足三角不等式，请判断空间立体是否存在。

Worked solution

1. The strongest face inequality is $\sqrt{2} + \sqrt{3} > \sqrt{6}$, so planar face validity alone gives no obstruction.

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 16 / 方法 16 [GE10-M16] (continued / 续)

2. For an opposite-edge-equal tetrahedron, box dimensions would satisfy $a^2+b^2=2, a^2+c^2=3, b^2+c^2=6$.

3. The first two equations minus the third give $2a^2=-1$, impossible for a real box dimension. Thus the tetrahedron does not exist, despite the valid individual faces.

例题解析

1. 最强的面三角不等式为 $\sqrt{2}+\sqrt{3}>\sqrt{6}$，所以单独检查各面无法排除。

2. 若对棱相等四面体存在，长方体边长应满足 $a^2+b^2=2, a^2+c^2=3, b^2+c^2=6$。

3. 前两式相加再减第三式得 $2a^2=-1$，不可能对应实数边长。因此各面虽合法，空间四面体仍不存在。

If exceptional edge lengths occur in pairs of faces, first establish which edges are opposite.

若特殊棱长通过各面条件给出，应先确定哪些棱互为对棱。

For face squared sides A, B, C, $16K^2=2(AB+BC+CA)-(A^2+B^2+C^2)$. When all four faces are congruent, total surface area is $4K$.

The box representation is a theorem, not an extra assumption. Put the tetrahedron centroid at zero and let u, v, w be the midpoints of the three edges from one vertex. The four vertex vectors are $u+v+w, u-v-w, -u+v-w, -u-v+w$. Equality of each opposite-edge pair gives $|v+w|=|v-w|$ and its cyclic counterparts, so u, v, w are pairwise perpendicular. Their doubled lengths are the box dimensions; nondegeneracy requires all three to be positive.

若一面的三边平方为 A, B, C，则 $16K^2=2(AB+BC+CA)-(A^2+B^2+C^2)$；四面全等时总表面积为 $4K$。

Section 2.10 Volumes & solid geometry

体积与立体几何

Method 16 / 方法 16 [GE10-M16] (continued / 续)

长方体表示是定理而非额外假设。把四面体重心放在原点，令 u, v, w 为同一顶点引出的三条棱的中点向量，则四顶点向量为 $u+v+w, u-v-w, -u+v-w, -u-v+w$。每组对棱相等给出 $|v+w| = |v-w|$ 及循环同式，所以 u, v, w 两两垂直。其长度的两倍即长方体三边，非退化要求三者均正。

Practice / 配套练习: AIME-2024-I-Q14 (WB p. 462)

Section 3.1 Prime factorization & valuations

质因数分解与赋值

Method 1 / 方法 1 [NT01-M01]

Prime exponents turn multiplicative conditions into independent arithmetic constraints. Use exact valuation rules for divisibility, parity for powers, and a certified bound before any finite search.

质指数可把乘法条件化为独立的算术限制：整除用精确赋值，幂次用指数奇偶，有限搜索之前先证明界。

Compare prime exponents, then round the required threshold

比较质指数，再向上取整门槛

Procedure

1. Factor the fixed integer and write each unknown prime exponent explicitly.

2. Use $A \mid B \iff v_p(A) \le v_p(B)$ for every prime; a kth power has every exponent divisible by k.

3. For $D \mid x^k$, require $v_p(x) \ge \lceil v_p(D)/k \rceil$. Keep any extra positivity or size bounds.

4. Build a candidate from these exponents and verify all original conditions, including minimality.

通用步骤

1. 分解固定整数，并明确写出每个未知质指数。

2. 对每个质数使用 $A \mid B \iff v_p(A) \le v_p(B)$；$k$ 次幂的每个指数均为 k 的倍数。

3. 由 $D \mid x^k$ 得 $v_p(x) \ge \lceil v_p(D)/k \rceil$，并保留正性或大小限制。

4. 由这些指数构造候选，并验证原条件及最小性。

> Example. Find the least positive integer u such that $72u$ is a cube and u is a square.

Section 3.1 Prime factorization & valuations

质因数分解与赋值

Method 1 / 方法 1 [NT01-M01] (continued / 续)

> 例题：求最小正整数 u，使 $72u$ 是立方数且 u 是平方数。

Worked solution

1. Write $72 = 2^3 3^2$ and $u = 2^a 3^b$; introducing any other prime cannot reduce the minimum.

2. Squareness requires even a, b. Cubeness requires $a + 3 \equiv 0$ and $b + 2 \equiv 0 \pmod 3$.

3. The smallest choices are $a = 0, b = 4$, giving $u = 81$. Indeed $72 \cdot 81 = 18^3$ and $81 = 9^2$.

例题解析

1. 写成 $72 = 2^3 3^2, u = 2^a 3^b$；增加其他质数不会使最小值更小。

2. 平方条件要求 a, b 为偶数；立方条件要求 $a + 3 \equiv 0, b + 2 \equiv 0 \pmod 3$。

3. 最小选择为 $a = 0, b = 4$，故 $u = 81$；确有 $72 \cdot 81 = 18^3$ 且 $81 = 9^2$。

Exponent conditions are simultaneous; satisfying one prime is insufficient.

各质数的指数条件必须同时满足。

If an exponent itself occurs inside the integer, a finite check needs a growth proof. For $c \geq 2$ and $n \geq 1$, the ratio of $c^{n+1}/(n+1)$ to c^n/n is $cn/(n+1) \geq 1$. Thus once $c^n \geq Kn$ holds, it persists at all later indices. In a product of independent prime powers, bound the other factors below before applying this comparison to each exponent.

若指数本身也出现在整数表达式中，有限检验必须有增长证明。对 $c \geq 2, n \geq 1$，$c^{n+1}/(n+1)$ 与 c^n/n 的比为 $cn/(n+1) \geq 1$，所以 $c^n \geq Kn$ 一旦成立，此后各下标均成立。多个独立质数幂相乘时，先给其余因子建立下界，再逐指数使用此比较。

Section 3.1 Prime factorization & valuations

质因数分解与赋值

Method 1 / 方法 1 [NT01-M01] (continued / 续)

Practice / 配套练习: AIME-2002-II-Q05 (WB p. 464), AIME-2007-I-Q01 (WB p. 467), AIME-2016-I-Q10 (WB p. 151), AIME-2016-II-Q07 (WB p. 376), AIME-2018-II-Q07 (WB p. 227), AIME-2020-I-Q10 (WB p. 478), AIME-2020-II-Q01 (WB p. 471), AIME-2026-II-Q10 (WB p. 291), AIME-2002-II-Q07 (WB p. 497), AIME-2015-I-Q13 (WB p. 202)

Section 3.1 Prime factorization & valuations

质因数分解与赋值

Method 2 / 方法 2 [NT01-M02]

Count valuation layers in factorials and weighted products

按赋值层统计阶乘与加权乘积

Procedure

1. Use $v_p(n!) = \sum_{j\ge 1}\lfloor n/p^j\rfloor$; stop when $p^j > n$.

2. For a product with repeated factors, count how often each factor occurs before summing its valuation.

3. For a factorial quotient, subtract denominator valuations prime by prime; for trailing zeros take the smaller of the two- and five-adic valuations.

4. If a sum has different powers of one prime in its denominators, test whether the largest denominator term is uniquely odd after clearing denominators.

通用步骤

1. 使用 $v_p(n!) = \sum_{j\ge 1}\lfloor n/p^j\rfloor$，当 $p^j > n$ 时停止。

2. 若乘积中因子重复出现，先统计每个因子的出现次数，再累加赋值。

3. 阶乘商按质数相减分母赋值；末尾零数取二与五赋值的较小者。

4. 若和式各分母含同一质数的不同幂，清分母后检查最大分母对应项是否唯一不被该质数整除。

> Example. Nine labeled objects are partitioned into three unlabeled groups of three. What is the exponent of 3 in the number of partitions?

> 例题：把九个有标签物品分为三个无标签三元组，分法总数中的质因数 3 的指数是多少？

Section 3.1 Prime factorization & valuations

质因数分解与赋值

Method 2 / 方法 2 [NT01-M02] (continued / 续)

Worked solution

1. The count is $N = 9!/((3!)^3 3!)$. The last $3!$ removes the ordering of the three groups.

2. Legendre gives $v_3(9!) = 3 + 1 = 4$ and $v_3(3!) = 1$.

3. Hence $v_3(N) = 4 - 3 - 1 = 0$. In fact $N = 280$, consistent with the valuation.

例题解析

1. 总数为 $N = 9!/((3!)^3 3!)$；最后的 $3!$ 消去三个组的次序。

2. 勒让德公式给出 $v_3(9!) = 3 + 1 = 4$，而 $v_3(3!) = 1$。

3. 所以 $v_3(N) = 4 - 3 - 1 = 0$；实际 $N = 280$，与赋值一致。

A multiple of p^j contributes at every layer up to j.

p^j 的倍数会在前 j 层各贡献一次。

Practice / 配套练习: AIME-2006-I-Q04 (WB p. 465), AIME-2006-II-Q03 (WB p. 466), AIME-2009-II-Q07 (WB p. 475), AIME-2010-II-Q03 (WB p. 468), AIME-2020-I-Q07 (WB p. 639), AIME-2023-I-Q04 (WB p. 473), AIME-2025-I-Q10 (WB p. 627)

Section 3.1 Prime factorization & valuations

质因数分解与赋值

Method 3 / 方法 3 [NT01-M03]

Use exponent parity and squarefree parts

使用指数奇偶与无平方因子部分

Procedure

1. Write $n = st^2$ with squarefree s by separating the odd exponent of each prime.

2. A product is square exactly when each total prime exponent is even; encode these parities as bits if many factors are chosen.

3. For rational square roots, reduce the fraction first: numerator and denominator must each be squares.

4. Count or bound each allowed parity pattern and confirm its realization.

通用步骤

1. 把每个质数指数的奇数部分分离，写成 $n = st^2$，其中 s 无平方因子。

2. 乘积为平方数当且仅当各质指数总和均为偶数；多次选择时可把奇偶编码为二元位。

3. 有理数平方根问题先约分，既约分数的分子与分母必须各为平方数。

4. 对允许的奇偶模式逐个计数或估界，并检验可实现性。

> Example. What is the least positive rational number r such that $18r^2$ is an integer square?

> 例题：求最小正有理数 r，使 $18r^2$ 为整数平方。

Section 3.1 Prime factorization & valuations

质因数分解与赋值

Method 3 / 方法 3 [NT01-M03] (continued / 续)

Worked solution

1. If $18r^2 = k^2$ for a positive integer k, then $\sqrt{18} = k/r$ would be rational.

2. But $18 = 2 \cdot 3^2$ has odd exponent of 2, so $\sqrt{18} = 3\sqrt{2}$ is irrational.

3. There is no such positive rational r. Multiplying by a rational square cannot change the parity of a prime exponent.

例题解析

1. 若 $18r^2 = k^2$，其中 k 为正整数，则 $\sqrt{18} = k/r$ 必为有理数。

2. 然而 $18 = 2 \cdot 3^2$ 的质因数 2 指数为奇数，故 $\sqrt{18} = 3\sqrt{2}$ 无理。

3. 不存在这样的正有理数 r；乘有理平方数不会改变质指数的奇偶。

A parity obstruction can prove impossibility before any enumeration.

赋值奇偶障碍可在枚举之前证明不可能。

Practice / 配套练习: AIME-2019-II-Q04 (WB p. 470)

Section 3.1 Prime factorization & valuations

质因数分解与赋值

Method 4 / 方法 4 [NT01-M04]

Allocate prime exponents independently

独立分配各质数的指数

Procedure

1. For $N = \prod p^{e_p}$, encode each factor by its exponent for each prime.

2. Apply the local rule: bounded intervals for divisors, weak compositions for ordered products, or one-sided allocation for coprimality.

3. Multiply independent prime contributions. If factors are unordered, remove repeated-factor exceptions before dividing by permutations.

4. For sums rather than counts, factor the sum into finite geometric series; signed exponents also encode reduced rational factors.

通用步骤

1. 对 $N = \prod p^{e_p}$，用各质数的指数编码每个因子。

2. 按局部规则计数：约数对应有界区间，有序乘积对应弱拆分，互质对应单侧分配。

3. 独立质数的贡献相乘；若因子无序，先处理重复因子例外，再除排列数。

4. 若求和而非计数，则将和分解为有限等比级数；带符号指数还可编码既约有理因子。

> Example. How many divisors d of 360 satisfy $\gcd(d, 360/d) = 1$? Also find their sum.

> 例题：有多少个 360 的约数 d 满足 $\gcd(d, 360/d) = 1$？并求这些约数之和。

Section 3.1 Prime factorization & valuations

质因数分解与赋值

Method 4 / 方法 4 [NT01-M04] (continued / 续)

Worked solution

1. Factor $360 = 2^3 3^2 5$. Coprimality forces the entire power of each prime into exactly one side.

2. Each of the three prime powers has two independent destinations, so there are $2^3 = 8$ divisors.

3. Their sum is $(1+8)(1+9)(1+5) = 540$, by expanding one choice from each factor.

例题解析

1. 分解 $360 = 2^3 3^2 5$；互质要求每个质数的整块幂只放到一侧。

2. 三块质数幂各有两种独立归属，故共有 $2^3 = 8$ 个约数。

3. 从各因子各选一项展开，约数和为 $(1+8)(1+9)(1+5) = 540$。

Prime choices are independent; ordering of whole factors is a separate issue.

质指数的独立性与整个因子的有序性是不同问题。

Practice / 配套练习: AIME-2000-I-Q11 (WB p. 479), AIME-2000-II-Q02 (WB p. 528), AIME-2000-II-Q04 (WB p. 599), AIME-2010-II-Q10 (WB p. 713), AIME-2016-II-Q08 (WB p. 476), AIME-2019-I-Q09 (WB p. 606), AIME-2019-II-Q09 (WB p. 607), AIME-2020-I-Q09 (WB p. 741), AIME-2024-II-Q14 (WB p. 593)

Section 3.1 Prime factorization & valuations

质因数分解与赋值

Method 5 / 方法 5 [NT01-M05]

Force prime structure and prove finite prime bounds

强迫质因数结构并证明有限质数界

Procedure

1. Factor an identity or exclude small prime divisors by residue tables.

2. Use primality to force one factor to be one, or use the least permitted primes to bound a product.

3. Prove that the remaining candidate list is exhaustive before testing it.

4. Certify each claimed prime by trial division only through its square root, then substitute into the original identity.

通用步骤

1. 分解恒等式，或用剩余表排除小质因数。

2. 利用质性迫使某因子为一，或用最小允许质数为乘积建立下界。

3. 测试候选之前先证明剩余列表穷尽。

4. 对声称为质数的候选只需试除至其平方根，再代回原恒等式。

> Example. An integer has exactly three prime factors counted with multiplicity, each at least 7, and is not a cube. Find its smallest possible value.

> 例题：某整数恰有三个质因数（计重），各不小于 7，且它不是立方数。求其最小值。

Section 3.1 Prime factorization & valuations

质因数分解与赋值

Method 5 / 方法 5 [NT01-M05] (continued / 续)

Worked solution

1. The unrestricted least product is 7^3, which is forbidden because it is a cube.

2. At least one prime must therefore be at least 11, the next prime after 7. The product is at least $7^2 \cdot 11$.

3. The value $539 = 7^2 \cdot 11$ satisfies every condition. It is therefore the minimum.

例题解析

1. 无限制最小乘积为 7^3，但它是立方数而被排除。

2. 至少一个质因数须不小于 7 后的下一质数 11，故乘积至少为 $7^2 \cdot 11$。

3. $539 = 7^2 \cdot 11$ 满足全部条件，因此为最小值。

A prime search needs a certified bound, not a long unchecked list.

搜索质数须有已证明的界，不能只列一长串未核验候选。

Practice / 配套练习: AIME-2000-I-Q01 (WB p. 463), AIME-2015-I-Q03 (WB p. 469), AIME-2016-I-Q12 (WB p. 482), AIME-2023-I-Q15 (WB p. 96), AIME-2026-I-Q04 (WB p. 540), AIME-2013-I-Q11 (WB p. 512), AIME-2004-I-Q15 (WB p. 678)

Section 3.1 Prime factorization & valuations

质因数分解与赋值

Method 6 / 方法 6 [NT01-M06]

Use a valuation to distinguish values and control cancellation

用赋值区分数值并控制消去

Procedure

1. Choose a prime for which one factor is always a unit. Its valuation isolates the remaining exponent.

2. If two expressions are equal, compare valuations first, then divide out the common prime power.

3. Use the remaining odd or unit part to recover the other parameters, proving no duplicate counting.

4. For an extremal power representation, translate the condition into exponent inequalities before a finite search.

通用步骤

1. 选择使某个因子恒不被整除的质数，其赋值便可隔离另一指数。

2. 若两个表达式相等，先比较赋值，再除去共同质数幂。

3. 由剩下的奇数部分或单位部分恢复其他参数，证明计数无重复。

4. 幂表示最值题应先把条件转为指数不等式，再进行有限搜索。

> Example. Each positive integer is written as $2^a u$ with u odd. Prove that numbers with different pairs (a, u) cannot coincide, and count such numbers below 17 with $a \geq 2$.

Section 3.1 Prime factorization & valuations

质因数分解与赋值

Method 6 / 方法 6 [NT01-M06] (continued / 续)

例题：将正整数写成 $2^a u$，其中 u 为奇数。证明不同 (a, u) 不会表示同一数，并统计小于 17 且 $a \geq 2$ 的这些整数。

Worked solution

1. If $2^a u = 2^b v$ and $a < b$, dividing by 2^a would make the odd number u even. Hence $a = b$.

2. Division now gives $u = v$, proving uniqueness.

3. For $a = 2$, the odd parts are $1, 3$; for $a = 3, 4$, only 1 works. The four values are $4, 12, 8, 16$.

例题解析

1. 若 $2^a u = 2^b v$ 且 $a < b$，约去 2^a 将使奇数 u 变为偶数，矛盾，所以 $a = b$。

2. 再相除得到 $u = v$，唯一性得证。

3. $a = 2$ 时奇数部分为 $1, 3$；$a = 3, 4$ 时只可为 1。共四个值 $4, 12, 8, 16$。

The valuation proves uniqueness before a parameter count is summed.

应先用赋值证明唯一性，再累加参数计数。

Practice / 配套练习: AIME-2009-I-Q08 (WB p. 148), AIME-2021-I-Q03 (WB p. 472), AIME-2021-II-Q09 (WB p. 520)

Section 3.1 Prime factorization & valuations

质因数分解与赋值

Method 7 / 方法 7 [NT01-M07]

Translate logarithmic and divisor constraints into exponent systems

把对数与约数条件转为指数方程组

Procedure

1. Exponentiate a logarithmic identity before assigning prime exponents.

2. Use $v_p(\gcd(x,y)) = \min(a,b)$ and $v_p(\operatorname{lcm}(x,y)) = \max(a,b)$.

3. Split the order cases for these minima and maxima; reject negative or nonintegral exponent solutions.

4. Reconstruct the numbers or count bounded lattice solutions in the exponent variables.

通用步骤

1. 先把对数恒等式指数化，再设置质指数。

2. 使用 $v_p(\gcd(x,y)) = \min(a,b)$ 与 $v_p(\operatorname{lcm}(x,y)) = \max(a,b)$。

3. 按最小值与最大值的大小关系分类，剔除负数或非整数指数解。

4. 重建原数，或统计指数变量中的有界整点解。

> Example. Positive integers x, y satisfy $x\gcd(x,y) = 2^8$ and $y/\gcd(x,y) = 2^2$. Find all ordered pairs.

> 例题：正整数 x, y 满足 $x\gcd(x,y) = 2^8$ 且 $y/\gcd(x,y) = 2^2$，求全部有序对。

Section 3.1 Prime factorization & valuations

质因数分解与赋值

Method 7 / 方法 7 [NT01-M07] (continued / 续)

Worked solution

1. No odd prime divides x. Write $x = 2^a$ and $g = \gcd(x, y) = 2^c$; the second equation gives $y = 2^{c+2}$.

2. The conditions become $a + c = 8$ and $c = \min(a, c + 2)$. Since $c < c + 2$, the latter forces $a = c$.

3. Thus $a = c = 4$, giving $(x, y) = (16, 64)$. Direct substitution verifies both equations.

例题解析

1. 没有奇质数可整除 x。写 $x = 2^a, g = \gcd(x, y) = 2^c$，第二式给出 $y = 2^{c+2}$。

2. 条件化为 $a + c = 8, c = \min(a, c + 2)$。因 $c < c + 2$，后者迫使 $a = c$。

3. 所以 $a = c = 4$，得 $(x, y) = (16, 64)$，代回可验证两式。

A prime absent from one side of a product identity must be absent from every positive integral factor.

乘积一侧不存在的质因数，不会出现在另一侧任一正整数因子中。

Practice / 配套练习: AIME-2006-I-Q09 (WB p. 474), AIME-2006-I-Q13 (WB p. 480), AIME-2006-II-Q14 (WB p. 481), AIME-2013-II-Q02 (WB p. 106), AIME-2019-I-Q07 (WB p. 477)

Section 3.1 Prime factorization & valuations

质因数分解与赋值

Method 8 / 方法 8 [NT01-M08]

Lift odd-prime valuations and split multiplicative costs

提升奇质数赋值与拆分乘法代价

Procedure

1. For exponential differences, first find a power for which the two bases agree modulo the prime.

2. For odd p with $p \mid A-B$ and $p \nmid AB$, use $v_p(A^k-B^k)=v_p(A-B)+v_p(k)$. Check these hypotheses each time.

3. In a multiplicative construction, compare the cost of a composite factor with the costs of its smaller factors.

4. Combine local necessities, then construct an object attaining all of them simultaneously.

通用步骤

1. 对指数差，先找使两底数模该质数相等的幂次。

2. 若奇质数 $p \mid A-B$ 且 $p \nmid AB$，使用 $v_p(A^k-B^k)=v_p(A-B)+v_p(k)$，每次核对前提。

3. 在乘法构造中，比较合数因子的代价与其较小因子拆分后的总代价。

4. 合并局部必要条件，再构造同时达到所有条件的对象。

> Example. Find the least positive k such that $3^4 \mid 10^k-1$.

> 例题：求使 $3^4 \mid 10^k-1$ 的最小正整数 k。

Section 3.1 Prime factorization & valuations

质因数分解与赋值

Method 8 / 方法 8 [NT01-M08] (continued / 续)

Worked solution

1. Here $3 \mid 10-1$ and $3 \nmid 10$, so the odd-prime lifting hypotheses hold.

2. Therefore $v_3(10^k-1) = v_3(9) + v_3(k) = 2 + v_3(k)$.

3. The exact requirement is $9 \mid k$, whose least positive solution is $k = 9$.

例题解析

1. 这里 $3 \mid 10-1$ 且 $3 \nmid 10$，满足奇质数赋值提升条件。

2. 因此 $v_3(10^k-1) = v_3(9) + v_3(k) = 2 + v_3(k)$。

3. 准确要求为 $9 \mid k$，最小正解为 $k = 9$。

The lifting formula for odd primes is not a general substitute for the special $p = 2$ formula.

奇质数赋值提升公式不能直接替代 $p = 2$ 的特殊公式。

For odd p, write $A = B + pc$ and expand. If $p \nmid k$, the first binomial term has the least valuation; replacing k by pk raises the valuation by one. Factoring $k = p^e u$ proves the displayed lifting rule. At $p = 3$ for cubes, this also shows that two unit inputs congruent modulo 3^6 have cubes congruent modulo 3^7, while the induced map on the relevant residue representatives is injective.

For product $N = \prod f_i$ with cost $\sum (f_i - 1)$, splitting gh into $g, h > 1$ reduces cost by $(g-1)(h-1) > 0$. Thus a minimum splits completely into prime factors. This is a cost theorem only after the combinatorial construction has proved that the factors act independently.

Section 3.1 Prime factorization & valuations

质因数分解与赋值

Method 8 / 方法 8 [NT01-M08] (continued / 续)

A prime-index update can encode binary carries: if a state contains each prime at most once, removing all primes before the first missing prime and inserting that prime implements an increment of its binary indicator. Prove squarefreeness and this invariant by induction before decoding an index.

对奇质数 p，写 $A = B + pc$ 后展开；当 $p \nmid k$ 时，首个二项式项赋值最低，将 k 换为 pk 则赋值增加一。写 $k = p^e u$ 即得所述公式。对 $p = 3$ 的立方，这还表明模 3^6 相等的单位输入之立方模 3^7 相等，而相关代表元所诱导的立方映射为单射。

若乘积 $N = \prod f_i$ 的代价为 $\sum (f_i - 1)$，将 gh 拆成 $g, h > 1$ 可减少 $(g-1)(h-1) > 0$。故最小代价必须完全拆成质因数；但只有先证明组合构造的各因子独立，这一代价定理才能适用。

质数索引更新可编码二进制进位：若每个质数最多出现一次，删除首个缺失质数前的全部质数并加入该质数，就使其二元指示数加一。解码下标之前须归纳证明无平方因子性与此不变量。

Practice / 配套练习: AIME-2020-I-Q12 (WB p. 524), AIME-2025-I-Q15 (WB p. 527), AIME-2026-II-Q13 (WB p. 718), AIME-2014-II-Q15 (WB p. 592)

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 1 / 方法 1 [NT02-M01]

Divisibility becomes simpler after a reversible change of variables. Reduce ratios, subtract multiples, and use the gcd to isolate the true scale or period; count only after these structural constraints are fixed.

整除在可逆换元后往往更简单：约分比例、减去倍数，并用最大公因数隔离真正尺度或周期；确定这些结构限制后再计数。

Use Euclidean reduction before listing divisors

先作欧几里得化简，再列约数

Procedure

1. Replace $A \mid B$ by $A \mid (B - qA)$ with a useful integer q.

2. For equal remainders, the modulus divides every pairwise difference. For $n + c \mid P(n)$, substitute $n = -c$ into the remainder.

3. List only divisors meeting the original positivity, digit or size restrictions.

4. Recover the original variables and check the remainder lies between zero and the modulus.

通用步骤

1. 以合适整数 q 将 $A \mid B$ 改写为 $A \mid (B - qA)$。

2. 余数相等时，模数整除所有两两差；对 $n + c \mid P(n)$，以 $n = -c$ 求余数。

3. 只列满足原正性、数位或大小限制的约数。

4. 还原原变量，并检查余数介于零与模数之间。

> Example. Find all positive integers n such that $n + 3$ divides $n^2 + 2n + 8$.

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 1 / 方法 1 [NT02-M01] (continued / 续)

> 例题：求全部满足 $n+3 \mid n^2+2n+8$ 的正整数 n。

Worked solution

1. Divide: $n^2+2n+8=(n+3)(n-1)+11$.

2. The condition is exactly $n+3 \mid 11$. Positivity requires $n+3 \geq 4$.

3. Only divisor 11 qualifies, giving $n=8$. Substitution gives 88, divisible by 11.

例题解析

1. 相除得 $n^2+2n+8=(n+3)(n-1)+11$。

2. 条件恰为 $n+3 \mid 11$，正性又要求 $n+3 \geq 4$。

3. 只有约数 11 合格，得 $n=8$；代入为 88，确可被 11 整除。

A remainder reduction is an equivalence; preserve both directions.

求余化简须保留充要关系。

Practice / 配套练习: AIME-2001-I-Q01 (WB p. 558), AIME-2006-I-Q03 (WB p. 564), AIME-2017-I-Q02 (WB p. 486), AIME-2020-I-Q04 (WB p. 488), AIME-2025-I-Q01 (WB p. 582), AIME-2025-II-Q02 (WB p. 490)

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 2 / 方法 2 [NT02-M02]

Reduce ratios and use coprimality to fix integer scale

约分比例并用互质性确定整数尺度

Procedure

1. Write every rational ratio in lowest terms before assigning an integer scale.

2. Use Euclid's lemma: if $\gcd(a,b)=1$ and $a \mid bc$, then $a \mid c$.

3. For equal fractions or a reduced numerator-denominator pair, resolve the whole common scale, not just one variable.

4. Check that the resulting counts or digits are feasible, since integrality alone may not suffice.

通用步骤

1. 设置整数尺度前，先把所有比例约至最简。

2. 使用欧几里得引理：若 $\gcd(a,b)=1$ 且 $a \mid bc$，则 $a \mid c$。

3. 分数相等或既约分子分母题，应确定整个公因尺度，不能只确定一个变量。

4. 检验所得人数或数位是否可实现，因为整数性未必充分。

> Example. A class has boys:girls $=4:7$. After five boys join, the ratio is $3:4$. Find the original class size.

> 例题：某班男女人数比为 $4:7$，加入五名男生后变为 $3:4$。求原班人数。

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 2 / 方法 2 [NT02-M02] (continued / 续)

Worked solution

1. Since $4, 7$ are coprime, write the original counts as $4t, 7t$ with positive integer t.

2. The new ratio gives $4(4t+5) = 3(7t)$, so $t = 4$.

3. The original size is $11t = 44$. The final counts $21, 28$ indeed have ratio $3:4$.

例题解析

1. 因 $4, 7$ 互质，原人数可写成 $4t, 7t$，其中 t 为正整数。

2. 新比例给出 $4(4t+5) = 3(7t)$，故 $t = 4$。

3. 原班共有 $11t = 44$ 人；最后人数 $21, 28$ 之比确为 $3:4$。

A reduced ratio makes the scale integral; an unreduced one may not.

最简整数比才保证对应尺度为整数。

Practice / 配套练习: AIME-2000-I-Q03 (WB p. 218), AIME-2012-I-Q06 (WB p. 73), AIME-2013-I-Q08 (WB p. 121), AIME-2015-II-Q01 (WB p. 485), AIME-2019-I-Q07 (WB p. 477), AIME-2022-II-Q01 (WB p. 489), AIME-2023-II-Q05 (WB p. 595)

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 3 / 方法 3 [NT02-M03]

Parametrize multiples and common-divisor constraints

参数化倍数与公约数限制

Procedure

1. If $a \mid b, c$, write $b = au, c = av$ before using a sum or product condition.

2. Combine repeated divisibility conditions with gcd or lcm. Distinguish a divisor bound from a multiple bound.

3. For a fixed parameter, count remaining positive solutions explicitly, or list bounded factor pairs.

4. Prove the parameter list complete and construct the extremal or counted objects.

通用步骤

1. 若 $a \mid b, c$，先写成 $b = au, c = av$，再代入和或积条件。

2. 用最大公因数或最小公倍数合并重复整除条件，区分约数界与倍数界。

3. 固定参数后，明确统计剩余正整数解，或列出有界因子对。

4. 证明参数列表完整，并构造相应最值对象或被统计对象。

> Example. Count positive ordered pairs (x, y) such that $x \mid y$ and $x + y = 24$.

> 例题：统计满足 $x \mid y$ 且 $x + y = 24$ 的正整数有序对 (x, y)。

Worked solution

1. Write $y = kx$ with $k \geq 1$. Then $x(k + 1) = 24$.

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 3 / 方法 3 [NT02-M03] (continued / 续)

2. Thus x can be any divisor of 24 no greater than 12: $1, 2, 3, 4, 6, 8, 12$.

3. Each determines the unique positive $y = 24 - x$ and satisfies divisibility. There are seven pairs.

例题解析

1. 写 $y = kx$，其中 $k \geq 1$，则 $x(k+1) = 24$。

2. 故 x 为 24 的不大于 12 的约数：$1, 2, 3, 4, 6, 8, 12$。

3. 每个 x 都确定唯一正整数 $y = 24 - x$ 并满足整除，共七对。

A parameter count must respect strict increase, positivity and ordering.

参数计数须同时尊重严格递增、正性与有序性。

Practice / 配套练习: AIME-2007-II-Q02 (WB p. 484), AIME-2007-II-Q07 (WB p. 164), AIME-2010-II-Q01 (WB p. 567), AIME-2010-II-Q06 (WB p. 44), AIME-2013-I-Q11 (WB p. 512), AIME-2014-I-Q06 (WB p. 45), AIME-2016-I-Q05 (WB p. 138), AIME-2016-II-Q04 (WB p. 659), AIME-2019-II-Q03 (WB p. 487), AIME-2026-II-Q01 (WB p. 141)

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 4 / 方法 4 [NT02-M04]

Use the gcd to find periods, orbit lengths and primitive segments

用最大公因数求周期、轨道长度与本原格段

Procedure

1. For motion modulo n by a fixed step a, the first return solves $n \mid ka$.

2. Dividing by $g = \gcd(a, n)$ gives orbit length n/g and g separate cycles.

3. For lattice displacement (u, v), divide by $\gcd(u, v)$ to get a primitive step; for reflections, compose them to get translations.

4. For unfolding or multiple modular coordinates, solve all return divisibilities simultaneously; correct shared endpoints when repeating a segment.

通用步骤

1. 模 n 每次走固定步长 a 时，首次返回满足 $n \mid ka$。

2. 约去 $g = \gcd(a, n)$，轨道长度为 n/g，共有 g 个循环。

3. 格点位移 (u, v) 除以 $\gcd(u, v)$ 得本原步长；反射则先复合成平移。

4. 展开反射或处理多个模坐标时，同时求解全部返回整除条件；重复格段须修正公共端点。

> Example. A token moves 8 positions each turn around 30 labeled positions. How many cycles does this rule form, and how long is each cycle?

> 例题：棋子在 30 个有标签位置组成的环上，每次前进 8 格。该规则形成多少个循环？每个多长？

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 4 / 方法 4 [NT02-M04] (continued / 续)

Worked solution

1. The return condition is $30 \mid 8k$. Since $\gcd(30, 8) = 2$, it is equivalent to $15 \mid k$.

2. Each cycle therefore contains 15 distinct positions before returning.

3. The 30 positions split into $30/15 = 2$ cycles, namely the two parity classes.

例题解析

1. 返回条件为 $30 \mid 8k$。因 $\gcd(30, 8) = 2$，等价于 $15 \mid k$。

2. 故每个循环返回前经过 15 个不同位置。

3. 30 个位置分为 $30/15 = 2$ 个循环，恰为奇偶两个类。

Multiplication of residue labels by a is a permutation iff gcd(a,n)=1. Addition by a is always a permutation and has one full orbit iff gcd(a,n)=1.

剩余类标签乘 a 为置换当且仅当 gcd(a,n)=1；加 a 总是置换，而只有 gcd(a,n)=1 时才形成单一满轨道。

Bézout gives $ua + vb = \gcd(a, b)$. Hence two integer periods a, b force their gcd to be a period. For reflected function values, retain the residual reflection on the resulting residue classes; period reduction alone does not count the final classes.

The segment from $(0, 0)$ to positive lattice point (m, n) crosses $m + n - \gcd(m, n)$ open unit-cell interiors. Start with one cell, add the $m - 1$ vertical and $n - 1$ horizontal grid crossings, and subtract the $\gcd(m, n) - 1$ interior lattice crossings counted in both directions. A simultaneous crossing enters only one new diagonal cell.

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 4 / 方法 4 [NT02-M04] (continued / 续)

裴蜀关系 $ua + vb = \gcd(a, b)$ 表明两个整数周期 a, b 会迫使其最大公因数也为周期。反射函数题在约化剩余类后仍须保留剩余反射，单有周期约化并未统计最终等价类。

从 $(0, 0)$ 到正整点 (m, n) 的线段穿过 $m + n - \gcd(m, n)$ 个单位格内部。先计首格，再加 $m - 1$ 次竖直与 $n - 1$ 次水平格线穿越，减去被两方向重复计算的 $\gcd(m, n) - 1$ 个内部整点穿越；同时穿越时只进入一个新的对角格。

Practice / 配套练习: AIME-2000-I-Q12 (WB p. 210), AIME-2002-I-Q11 (WB p. 386), AIME-2012-I-Q15 (WB p. 510), AIME-2016-I-Q14 (WB p. 403), AIME-2021-I-Q07 (WB p. 196), AIME-2023-I-Q14 (WB p. 705), AIME-2025-II-Q11 (WB p. 707), AIME-2007-II-Q05 (WB p. 692)

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 5 / 方法 5 [NT02-M05]

Track a gcd invariant through reduced-fraction updates

在既约分数更新中追踪最大公因数不变量

Procedure

1. Write the numerator and denominator update before reduction.

2. Find a linear combination preserved by that update, and express the reduction gcd through it and another changing coordinate.

3. After division by the gcd, update both auxiliary coordinates; list the exact indices at which cancellation occurs.

4. Prove no later return is possible when the invariant becomes a unit.

通用步骤

1. 先写约分之前的分子分母更新。

2. 寻找更新保持的线性组合，并用它与另一变化坐标表达约分公因数。

3. 除以公因数后，同时更新两个辅助坐标，并精确记录发生约分的下标。

4. 当不变量变成单位时，证明以后不会再次返回。

> Example. Reduce m/n after replacing it by $(m+2)/(n+3)$. If $d = n - m$ and $I = 3m - 2n$, express the next reduced d, I in terms of the cancellation gcd.

> 例题：将 m/n 更新为 $(m+2)/(n+3)$ 后约分。设 $d = n - m, I = 3m - 2n$，用约分公因数表达新 d, I。

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 5 / 方法 5 [NT02-M05] (continued / 续)

Worked solution

1. Before reduction the difference is $d+1$, while $3(m+2)-2(n+3)=I$.

2. The gcd is $g=\gcd(m+2,n+3)=\gcd(m+2,d+1)=\gcd(I,d+1)$, because $I=(m+2)-2(d+1)$.

3. Consequently $d_{new}=(d+1)/g$ and $I_{new}=I/g$. If $I=1$, every future gcd is one and d increases forever.

例题解析

1. 约分前差为 $d+1$，而 $3(m+2)-2(n+3)=I$ 保持不变。

2. 公因数为 $g=\gcd(m+2,n+3)=\gcd(m+2,d+1)=\gcd(I,d+1)$，因 $I=(m+2)-2(d+1)$。

3. 因此 $d_{新}=(d+1)/g, I_{新}=I/g$。若 $I=1$，以后公因数恒为一，而 d 永远递增。

Reduction changes the preserved linear combination by division; it is not literally constant across reduced states.

约分会把线性组合也除以公因数，它并非在既约状态间字面保持常数。

Practice / 配套练习: AIME-2021-I-Q10 (WB p. 491)

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 6 / 方法 6 [NT02-M06]

Combine divisor sets and divide a chain by its first term

组合约数集合，并用首项约化整除链

Procedure

1. The common divisors of A, B are exactly the divisors of $\gcd(A, B)$; an LCM target requires each full prime-power exponent to appear.

2. For subsets, count missing-exponent events and apply inclusion-exclusion.

3. For an increasing divisibility chain beginning at d, divide every later term by d to obtain a smaller-sum instance.

4. Keep base cases and empty/singleton objects explicit so a recurrence is exhaustive.

通用步骤

1. A, B 的公约数恰为 $\gcd(A, B)$ 的约数；达到某最小公倍数须让每个完整质数幂指数出现。

2. 对子集，统计缺少完整指数的事件并用容斥。

3. 对首项为 d 的递增整除链，将以后各项除以 d 得总和更小的问题。

4. 明确递推初值、空对象与单元素对象，保证穷尽。

> Example. Choose a subset of the six divisors of 12. How many subsets have LCM exactly 12?

> 例题：从 12 的六个正约数中选子集，有多少个子集的最小公倍数恰为 12?

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 6 / 方法 6 [NT02-M06] (continued / 续)

Worked solution

1. The divisors are $1, 2, 3, 4, 6, 12$. A successful subset needs a divisor carrying 2^2 and one carrying 3.

2. There are 2^6 subsets; 2^4 miss the full two-exponent, 2^3 miss the three-exponent, and 2^2 miss both.

3. Inclusion-exclusion gives $64 - 16 - 8 + 4 = 44$. The empty subset is automatically excluded by both missing conditions.

例题解析

1. 约数为 $1, 2, 3, 4, 6, 12$。合格子集需有携带 2^2 的约数及携带 3 的约数。

2. 全部为 2^6 个子集；缺完整二指数有 2^4 个，缺三指数有 2^3 个，两者均缺有 2^2 个。

3. 容斥得 $64 - 16 - 8 + 4 = 44$；空集已由两个缺失条件自动排除。

A recurrence requires a reversible first-step decomposition, not only a numerical pattern.

递推须来自可逆的首步分解，不能只凭数值规律。

For public-information candidates involving common multiples, first list all admissible bounded runs using the divisibility conditions. A person's uncertainty is a statement about how many surviving candidates contain their value; repeat the candidate elimination after each public announcement. The gcd/LCM arithmetic creates the initial list, while the elimination itself belongs to LO03.

对涉及公倍数的公开信息候选，先用整除条件列出全部有界连续区间。某人不能确定，表示含有其数值的存活候选不止一个；每次公开声明后须重新筛选候选。最大公因数与最小公倍数算术产生初始列表，筛选本身属于 LO03。

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 6 / 方法 6 [NT02-M06] (continued / 续)

Practice / 配套练习: AIME-2019-II-Q12 (WB p. 686), AIME-2025-II-Q07 (WB p. 492), AIME-2021-II-Q11 (WB p. 810)

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 7 / 方法 7 [NT02-M07]

Bound a factorial divisor before testing a quotient

检验阶乘商前先确定阶乘约数的界

Procedure

1. Factor the expression around one convenient factorial.

2. Use monotonicity of $n!$ to rule out all sufficiently large indices at once.

3. Test the few remaining indices by exact integer quotients.

通用步骤

1. 围绕方便的某个阶乘分解表达式。

2. 利用 $n!$ 的单调性一次排除所有足够大的下标。

3. 用精确整数商检验剩下的少量下标。

> Example. Find the largest n such that $n! \mid 10! + 9!$.

> 例题：求使 $n! \mid 10! + 9!$ 的最大 n。

Worked solution

1. Write $10! + 9! = 11 \cdot 9!$, so $9!$ divides the expression.

2. But division by $10!$ gives $11/10$, which is not an integer.

3. For $n \geq 11$, $n! \geq 11! = 110 \cdot 9! > 11 \cdot 9!$. Thus no larger index works, and the answer is 9.

Section 3.2 Divisibility, GCD & LCM

整除、最大公因数与最小公倍数

Method 7 / 方法 7 [NT02-M07] (continued / 续)

例题解析

1. 写成 $10! + 9! = 11 \cdot 9!$，故 $9!$ 可整除。

2. 除以 $10!$ 得 $11/10$，不是整数。

3. 当 $n \geq 11$，有 $n! \geq 11! = 110 \cdot 9! > 11 \cdot 9!$，所以更大下标均不可能，答案为 9。

Magnitude is a useful first filter; divisibility still needs an exact quotient at the boundary.

大小是有效的初筛，但边界处仍须检验精确整除。

Practice / 配套练习: AIME-2003-I-Q01 (WB p. 483)

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

Method 1 / 方法 1 [NT03-M01]

Choose a modulus for a reason: it should simplify a coefficient, isolate a prime power, or expose a short pattern. Keep necessity and sufficiency separate, especially when clearing denominators or converting a modular invariant into an existence claim.

选模数须有目的：化简系数、隔离质数幂，或揭示短模式。尤其在清分母与把模不变量转为存在结论时，须分清必要性与充分性。

Solve a linear congruence and count its bounded representatives

解线性同余并统计有界代表元

Procedure

1. Write the condition as $ax \equiv b \pmod{m}$. Let $g = \gcd(a, m)$; solvability requires $g \mid b$.

2. Divide the entire congruence by g, including the modulus, then invert the now coprime coefficient.

3. Write $x = r + kt$ for the resulting modulus t and convert the original interval into exact integer bounds on k.

通用步骤

1. 写成 $ax \equiv b \pmod{m}$。设 $g = \gcd(a, m)$；可解须 $g \mid b$。

2. 将同余式整体除以 g，包括模数，再对已互质的系数求逆。

3. 对所得模数 t 写 $x = r + kt$，把原区间转为 k 的精确整数界。

> Example. How many integers $1 \le x \le 80$ satisfy $6x \equiv 9 \pmod{15}$?

> 例题：满足 $6x \equiv 9 \pmod{15}$ 的整数 $1 \le x \le 80$ 有多少个?

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

Method 1 / 方法 1 [NT03-M01] (continued / 续)

Worked solution

1. Divide by 3: $2x \equiv 3 \pmod 5$.

2. The inverse of 2 is 3 modulo 5, so $x \equiv 4 \pmod 5$.

3. The values are $4, 9, \ldots, 79$, numbering 16.

例题解析

1. 整体除以 3，得 $2x \equiv 3 \pmod 5$。

2. 模 5 下 2 的逆元为 3，故 $x \equiv 4 \pmod 5$。

3. 数值为 $4, 9, \ldots, 79$，共 16 个。

Canceling a nonunit coefficient changes the modulus.

约去非单位系数时，模数也必须改变。

To prove a modular motion criterion sufficient, track the state inside a fixed representative interval, or construct moves generating every allowed lattice point. A congruence invariant gives only necessity until such a construction is supplied.

证明模运动判据充分时，须把状态限制在固定代表区间中追踪，或构造能生成每个允许格点的动作。未给构造之前，同余不变量只提供必要性。

Practice / 配套练习: AIME-2001-I-Q11 (WB p. 29), AIME-2007-I-Q04 (WB p. 493), AIME-2008-I-Q06 (WB p. 145), AIME-2009-I-Q11 (WB p. 254), AIME-2011-II-Q03 (WB p. 131), AIME-2012-I-Q04 (WB p. 11), AIME-2012-II-Q01 (WB p. 537), AIME-2013-I-Q15 (WB p. 513), AIME-2018-I-Q10 (WB p. 715), AIME-2022-I-Q04 (WB p. 65), AIME-2026-II-Q03 (WB p. 365)

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

Method 2 / 方法 2 [NT03-M02]

Reduce a polynomial remainder and preserve noninvertible information

化简多项式余数并保留不可逆信息

Procedure

1. Reduce every power or coefficient modulo the proposed divisor before expanding a large expression.

2. For a variable divisor $n+c$, substitute $n \equiv -c$. Clear denominators only when their invertibility is known.

3. If multiplication by a denominator produced only a necessary condition, test each surviving candidate in the original expression.

通用步骤

1. 在展开大式之前，先对预定除数化简每个幂或系数。

2. 可变除数 $n+c$ 下代入 $n \equiv -c$；清分母时须核对可逆性。

3. 若乘分母后只得到必要条件，须把每个存活候选代回原式。

> Example. Find positive n for which the remainder of $n(n+1)/2$ modulo $n+2$ is 1.

> 例题：求正整数 n，使 $n(n+1)/2$ 除以 $n+2$ 余 1。

Worked solution

1. Put $d = n+2$. Since $n(n+1) \equiv 2 \pmod{d}$, twice the desired congruence holds for every d. It does not by itself prove the original congruence.

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

Method 2 / 方法 2 [NT03-M02] (continued / 续)

2. Compute exactly: $n(n+1)/2 - 1 = d(d-3)/2$. This is divisible by d exactly when $d-3$ is even.

3. Hence d and n are odd. Every positive odd n works, including $n = 1$.

例题解析

1. 令 $d = n+2$。因 $n(n+1) \equiv 2 \pmod{d}$，目标同余的两倍对所有 d 都成立，但这不足以证明原同余。

2. 精确计算得 $n(n+1)/2 - 1 = d(d-3)/2$；它被 d 整除当且仅当 $d-3$ 为偶数。

3. 因此 d, n 均为奇数；每个正奇数 n 都满足，包括 $n = 1$。

A multiplied congruence may lose information when the multiplier is not coprime to the modulus.

乘数不与模数互质时，放大后的同余可能丢失信息。

Practice / 配套练习: AIME-2010-I-Q02 (WB p. 494), AIME-2017-I-Q02 (WB p. 486), AIME-2017-II-Q08 (WB p. 502), AIME-2020-II-Q10 (WB p. 503), AIME-2025-II-Q02 (WB p. 490)

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

Method 3 / 方法 3 [NT03-M03]

Split a composite modulus into prime powers and recombine

把合数模拆为质数幂再合并

Procedure

1. Factor the modulus into pairwise coprime prime powers.

2. Solve each local condition. If $p^e \mid uv$ with $p \nmid \gcd(u, v)$, the whole power must divide one factor.

3. Combine every compatible local choice by the Chinese remainder theorem, then apply the original range and leading-digit restrictions.

4. For counting, count complete periods and handle the final partial block explicitly.

通用步骤

1. 把模数分成两两互质的质数幂。

2. 分别求解局部条件。若 $p^e \mid uv$ 且 $p \nmid \gcd(u, v)$，整块幂须归于一个因子。

3. 以中国剩余定理合并每组相容局部选择，再施加原范围与首位限制。

4. 计数时先统计完整周期，再明确处理最后不完整区块。

> Example. Find every residue $x \pmod{40}$ satisfying $x^2 = x$.

> 例题：求满足 $x^2 = x$ 的全部模 40 剩余类。

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

Method 3 / 方法 3 [NT03-M03] (continued / 续)

Worked solution

1. Factor $40 = 8 \cdot 5$. Consecutive factors $x, x-1$ are coprime, so x is 0 or 1 modulo each of $8, 5$.

2. The local choices $(0,0), (1,1), (0,1), (1,0)$ give residues $0, 1, 16, 25$ modulo 40.

3. All four square to themselves modulo 40, and CRT proves no other residues exist.

例题解析

1. 分解 $40 = 8 \cdot 5$。连续因子 $x, x-1$ 互质，故 x 模 $8, 5$ 分别只能为 0 或 1。

2. 局部选择 $(0,0), (1,1), (0,1), (1,0)$ 分别给出模 40 的 $0, 1, 16, 25$。

3. 四者平方均模 40 等于自身，且中国剩余定理证明无其他解。

Allocate prime powers, not merely individual primes, to coprime factors.

互质因子间应分配整块质数幂，不能只分配单个质数。

When a polynomial coefficient is a nonnegative solution count, congruences modulo complementary primes can force offsets in every variable. Substitute those offsets first; a common residual coefficient may then reduce the count to stars and bars (CO03).

若多项式系数转化为非负整数解计数，按互补质数取模可强制各变量的偏移。先代入偏移，剩余公共系数可能使计数化为隔板法（CO03）。

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

Method 3 / 方法 3 [NT03-M03] (continued / 续)

Practice / 配套练习: AIME-2002-II-Q07 (WB p. 497), AIME-2008-II-Q08 (WB p. 194), AIME-2011-II-Q14 (WB p. 509), AIME-2012-I-Q10 (WB p. 499), AIME-2012-II-Q12 (WB p. 511), AIME-2013-I-Q11 (WB p. 512), AIME-2014-I-Q08 (WB p. 500), AIME-2016-II-Q11 (WB p. 611), AIME-2017-I-Q09 (WB p. 501), AIME-2022-II-Q08 (WB p. 504), AIME-2022-II-Q13 (WB p. 656), AIME-2023-I-Q07 (WB p. 505), AIME-2024-II-Q14 (WB p. 593)

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

Method 4 / 方法 4 [NT03-M04]

Filter bounded arithmetic and geometry by small residues

用小模数筛选有界算术与几何条件

Procedure

1. Choose a modulus for which powers, digits, parity or angles have few possible residues.

2. Tabulate the possibilities and translate them into restrictions on the original variables.

3. Use the size bounds to shorten the surviving list; then check both the arithmetic and the geometric feasibility.

通用步骤

1. 选使幂、数位、奇偶或角度只出现少数剩余的模数。

2. 列出可能剩余，并转为原变量限制。

3. 用大小界缩短候选表，再同时检查算术与几何可实现性。

> Example. Can a triangle with integer sides have side squares summing to a number congruent to 7 modulo 8?

> 例题：整数边长三角形的三边平方和能否模 8 余 7?

Worked solution

1. A square is $0, 1$, or 4 modulo 8.

2. Three such residues cannot sum to 7: with zero, one, two or three odd squares the sums have residues among $0, 1, 2, 3, 4, 5, 6$.

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

Method 4 / 方法 4 [NT03-M04] (continued / 续)

3. There are no such integer side lengths even before the triangle inequality is considered.

例题解析

1. 平方数模 8 只能为 $0, 1, 4$。

2. 三个这样的剩余无法和为 7：按奇平方数的个数分类，所得剩余只在 $0, 1, 2, 3, 4, 5, 6$ 中。

3. 尚未考虑三角不等式，就已排除所有这样的整数边长。

A residue obstruction proves impossibility; a permitted residue does not prove existence.

剩余障碍能证明不可能；允许的剩余却不能单独证明存在。

Practice / 配套练习: AIME-2001-I-Q10 (WB p. 734), AIME-2003-II-Q02 (WB p. 562), AIME-2004-I-Q01 (WB p. 563), AIME-2005-II-Q09 (WB p. 69), AIME-2008-II-Q15 (WB p. 554), AIME-2012-I-Q01 (WB p. 568), AIME-2012-I-Q15 (WB p. 510), AIME-2013-I-Q02 (WB p. 570), AIME-2014-II-Q04 (WB p. 594), AIME-2015-II-Q03 (WB p. 571), AIME-2016-I-Q12 (WB p. 482), AIME-2016-I-Q14 (WB p. 403), AIME-2016-II-Q09 (WB p. 152), AIME-2022-I-Q02 (WB p. 579), AIME-2022-II-Q05 (WB p. 695), AIME-2024-II-Q07 (WB p. 506)

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

Method 5 / 方法 5 [NT03-M05]

Separate floor and rounding blocks from their residue corrections

把取整与舍入区块和剩余修正分开

Procedure

1. Write $\lfloor A/m \rfloor = (A-r)/m$ with r the least nonnegative residue.

2. Sum the smooth part first and tabulate residue corrections over a full period.

3. For uniqueness or maximum remainders, identify the exact integer interval on which a quotient tuple stays constant.

4. Include boundary jumps and compare against integers outside the requested counting range when uniqueness is global.

通用步骤

1. 写 $\lfloor A/m \rfloor = (A-r)/m$，其中 r 为最小非负剩余。

2. 先求无取整部分之和，再按完整周期列出剩余修正。

3. 对唯一性或最大余数问题，找出商元组保持不变的精确整数区间。

4. 计入边界跳变；若唯一性是全局性质，还须与所统计范围外的整数比较。

> Example. Calculate $\sum_{n=1}^{12} \lfloor n^2/3 \rfloor$.

> 例题：计算 $\sum_{n=1}^{12} \lfloor n^2/3 \rfloor$。

Worked solution

1. The square residue modulo 3 is zero for multiples of 3 and one otherwise.

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

Method 5 / 方法 5 [NT03-M05] (continued / 续)

2. There are eight nonmultiples of 3 in the range, while $\sum n^2 = 650$.

3. Thus the floor sum is $(650 - 8)/3 = 214$.

例题解析

1. 平方模 3 在 3 的倍数处为零，其余为一。

2. 范围内有八个非 3 的倍数，而 $\sum n^2 = 650$。

3. 所以取整和为 $(650 - 8)/3 = 214$。

Strict and weak interval endpoints can change an entire residue class.

区间端点的严格或非严格形式可能改变整个剩余类。

The remainder $n \bmod k$ is bounded by both $k - 1$ and $n - qk$ once a quotient lower bound q is known. Compare these intersecting bounds near their crossing, then handle exact-divisibility exceptions separately.

若已知商至少为 q，余数 $n \bmod k$ 同时不超过 $k - 1$ 与 $n - qk$。在两个界的交会附近比较，并单独处理恰好整除的例外。

Practice / 配套练习: AIME-2007-I-Q05 (WB p. 161), AIME-2013-II-Q14 (WB p. 514), AIME-2015-I-Q14 (WB p. 177), AIME-2022-II-Q08 (WB p. 504), AIME-2023-I-Q10 (WB p. 170)

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

Method 6 / 方法 6 [NT03-M06]

Reduce a counting problem to residue patterns

将计数问题化为剩余模式

Procedure

1. Replace each label by its residue, retaining how many original labels realize each class.

2. Use cyclic symmetry, subtraction of adjacent equations, or a reduced polynomial to find all permitted residue patterns.

3. Count the residue patterns first, then restore within-class label permutations and remove symmetry exactly once.

4. For a coefficient sum in residue classes, work modulo both the prime and $x^m - 1$; explain why the reduced coefficient has the intended meaning.

通用步骤

1. 以剩余替换标签，并保留每类原标签的数量。

2. 用循环对称、相邻方程相减或约化多项式求全部允许剩余模式。

3. 先数剩余模式，再恢复类内标签排列，且只消除一次对称。

4. 对按指数剩余类分组的系数和，同时模质数与 $x^m - 1$ 运算，并说明约化系数的含义。

> Example. How many subsets of $\{1, 2, 3, 4\}$ have sum divisible by 3?

> 例题：$\{1, 2, 3, 4\}$ 中有多少子集的元素和可被 3 整除?

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

Method 6 / 方法 6 [NT03-M06] (continued / 续)

Worked solution

1. The element 3 may be included or excluded freely. For $\{1,2,4\}$ the residue polynomial is $(1+x)^2(1+x^2)$.

2. Expand and reduce using $x^3=1$: $1+2x+2x^2+2x^3+x^4=3+3x+2x^2$.

3. The constant coefficient is 3; the free choice of 3 doubles it to 6.

例题解析

1. 元素 3 可自由选或不选。对 $\{1,2,4\}$，剩余生成式为 $(1+x)^2(1+x^2)$。

2. 展开并用 $x^3=1$ 约化：$1+2x+2x^2+2x^3+x^4=3+3x+2x^2$。

3. 常数系数为 3，元素 3 的自由选择使其加倍为 6。

Residue classes may have different multiplicities; restore the actual labels after the residue count.

剩余类的重数未必相同；统计模式后须恢复实际标签。

Over a prime field, $\binom{p}{j}\equiv 0 \pmod p$ for $0<j<p$, so $(1+x)^p=1+x^p$. In a quotient with $x^{p-1}=1$, this becomes $1+x$. Reducing a large exponent by this identity is valid as a polynomial identity in that quotient; it does not assert equality of unreduced integer coefficients.

The seven triples $\{i,i+1,i+3\}$ modulo seven form the Fano-plane lines: every pair occurs in exactly one triple. To use this incidence structure, classify zero positions first, then count the sign products on lines avoiding them. CO06 supplies the incidence and small-graph enumeration; modular reduction alone does not finish that count.

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

Method 6 / 方法 6 [NT03-M06] (continued / 续)

A necessary condition modulo $m-1$ for a sum of powers can become sufficient by repeatedly replacing one m^j by m copies of m^{j-1}; each split adds exactly $m-1$ summands. Construct the required number of splits explicitly.

在质数域中，$0<j<p$ 时 $\binom{p}{j}\equiv 0 \pmod p$，故 $(1+x)^p=1+x^p$。若再模 $x^{p-1}-1$，它变为 $1+x$。以此约化大指数是在商环中的多项式恒等式，并非未约化整数系数相等。

模七的七个三元组 $\{i,i+1,i+3\}$ 构成 Fano 平面直线，每一对点恰在一个三元组中。使用此关联结构时，先按零位置分类，再统计避开零点的直线上符号乘积。CO06 提供关联与小图枚举；仅有模约化并未完成该计数。

幂和模 $m-1$ 的必要条件，可通过反复把一个 m^j 拆为 m 个 m^{j-1} 证明充分；每次拆分使项数恰增加 $m-1$。须明确构造所需拆分次数。

Practice / 配套练习: AIME-2007-II-Q13 (WB p. 643), AIME-2011-I-Q05 (WB p. 618), AIME-2011-I-Q07 (WB p. 498), AIME-2018-I-Q12 (WB p. 748), AIME-2021-II-Q03 (WB p. 623), AIME-2023-II-Q10 (WB p. 716), AIME-2025-I-Q05 (WB p. 495), AIME-2026-I-Q08 (WB p. 507), AIME-2026-I-Q13 (WB p. 646), AIME-2026-II-Q15 (WB p. 708)

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

Method 7 / 方法 7 [NT03-M07]

Combine power cycles with local lifting and CRT

结合模幂周期、局部提升与中国剩余定理

Procedure

1. Handle nonunits modulo a prime power separately from invertible bases.

2. Use the multiplicative order or a proved exponent cycle to reduce exponents, not the modulus itself.

3. For a root modulo p, expand $f(a+pt)$ modulo p^2 and solve $f(a)/p + f\prime(a)t \equiv 0 \pmod{p}$.

4. Recombine the resulting residues, and prove the least representative or least exponent really is minimal.

通用步骤

1. 把质数幂模下不可逆的底数与可逆底数分别处理。

2. 用乘法阶或已证明的指数周期约化指数，不能直接用模数约化。

3. 对模 p 的根，模 p^2 展开 $f(a+pt)$，求解 $f(a)/p + f\prime(a)t \equiv 0 \pmod{p}$。

4. 合并所得剩余，并证明最小代表元或最小指数确为最小。

> Example. Lift the root $x \equiv 2 \pmod{7}$ of $x^2 - 4$ to a root modulo 49.

> 例题：将 $x^2 - 4$ 的模 7 根 $x \equiv 2 \pmod{7}$ 提升为模 49 的根。

Worked solution

1. Write $x = 2 + 7t$. Then $x^2 - 4 \equiv 28t \pmod{49}$.

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

Method 7 / 方法 7 [NT03-M07] (continued / 续)

2. Division by 7 gives $4t \equiv 0 \pmod 7$, so $t \equiv 0 \pmod 7$.

3. The unique lift in this residue class is $x \equiv 2 \pmod{49}$. The other root -2 belongs to a different initial class.

例题解析

1. 写 $x = 2 + 7t$，则 $x^2 - 4 \equiv 28t \pmod{49}$。

2. 除以 7 得 $4t \equiv 0 \pmod 7$，故 $t \equiv 0 \pmod 7$。

3. 该初始剩余类中唯一提升为 $x \equiv 2 \pmod{49}$；另一根 -2 属于不同初始类。

Unique lifting from modulo p to modulo p^2 requires a nonzero derivative modulo p. If the derivative vanishes, the linear condition gives either no lift or all p lifts in that initial root class.

从模 p 到模 p^2 的唯一提升要求导数模 p 非零。若导数为零，线性条件会给出无提升，或该初始根类中的全部 p 个提升。

Practice / 配套练习: AIME-2001-II-Q14 (WB p. 86), AIME-2009-I-Q07 (WB p. 147), AIME-2011-I-Q11 (WB p. 521), AIME-2017-I-Q14 (WB p. 125), AIME-2018-I-Q06 (WB p. 81), AIME-2018-I-Q11 (WB p. 522), AIME-2019-I-Q14 (WB p. 523), AIME-2019-II-Q10 (WB p. 208), AIME-2021-II-Q13 (WB p. 516), AIME-2023-II-Q15 (WB p. 525), AIME-2024-I-Q13 (WB p. 526), AIME-2025-II-Q13 (WB p. 217)

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

Method 8 / 方法 8 [NT03-M08]

Use least representable amounts in each residue class

使用各剩余类的最小可表示金额

Procedure

1. Fix a denomination m and find the least representable amount M_r in every residue class modulo m. If a class has no representable amount, infinitely many positive amounts are missing.

2. Adding m represents every larger amount in that class. Its largest missing amount is therefore $M_r - m$.

3. Prove each proposed minimum by considering all smaller counts of the other denominations, then take the maximum over residues.

通用步骤

1. 固定面额 m，求模 m 每个剩余类中的最小可表示金额 M_r；若某类根本不可表示，则有无穷多个正整数金额无法表示。

2. 不断加 m 可表示该类所有更大金额，故该类最大缺失金额为 $M_r - m$。

3. 考虑其他面额的所有较少张数，证明所列最小值，再对剩余类取最大。

> Example. With unlimited 4- and 7-unit stamps, find the largest positive amount that cannot be made.

> 例题：有无限张面值 4 与 7 的邮票，求无法表示的最大正整数金额。

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

Method 8 / 方法 8 [NT03-M08] (continued / 续)

Worked solution

1. Modulo 4, the smallest representable amounts for residues $0, 1, 2, 3$ are $0, 21, 14, 7$.

2. These use respectively $0, 3, 2, 1$ seven-unit stamps; fewer sevens cannot have the required residue.

3. The largest missing amount is $\max M_r - 4 = 17$. Each amount above it is reached by adding fours within its residue class.

例题解析

1. 模 4 的剩余 $0, 1, 2, 3$ 中，最小可表示金额分别为 $0, 21, 14, 7$。

2. 它们分别使用 $0, 3, 2, 1$ 张七元票，更少七元票无法具有相应剩余。

3. 最大缺失金额为 $\max M_r - 4 = 17$；每个更大金额都可在其剩余类内加四得到。

A residue minimum must be proved minimal; finding one representation gives only an upper bound.

剩余类最小值必须证明其最小性；找到一种表示只给出上界。

To compare a greedy coin choice with an optimum, use exchange moves to bound the number of small coins in an optimum. This can restrict the number of large coins to two neighboring possibilities. Compare their exact counts for each residual amount; discard any rival requiring a negative coin count.

比较贪心取币与最优取币时，先用交换论证限制最优方案的小面额枚数；这可把大面额枚数限制为相邻两种。按剩余金额比较确切枚数，并排除需要负枚数的对手方案。

Practice / 配套练习: AIME-2019-II-Q14 (WB p. 515), AIME-2025-II-Q08 (WB p. 805)

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

Method 9 / 方法 9 [NT03-M09]

Prove a residue invariant is sufficient with a construction

用构造证明剩余不变量的充分性

Procedure

1. Identify the residues forced by every permitted move or every retained condition.

2. Prove that no forbidden residue can occur, using induction or subtraction of neighboring conditions.

3. Give an explicit move, partition or construction for every permitted residue; then count the bounded representatives.

通用步骤

1. 确定每个允许动作或保留条件强制的剩余。

2. 用归纳或相邻条件相减，证明禁用剩余不可能出现。

3. 对每个允许剩余给出明确动作、分划或构造，最后统计有界代表元。

> Example. A game permits removing 1 or 2 tokens, with the last move winning. Prove the losing pile sizes for the player to move.

> 例题：游戏每次可拿走 1 或 2 枚，拿走最后一枚者胜。证明轮到行动者时的必败堆数。

Worked solution

1. Zero is losing. From any positive multiple of 3, either move reaches a number congruent to 1 or 2.

Section 3.3 Congruences & Chinese remainders

同余与中国剩余定理

Method 9 / 方法 9 [NT03-M09] (continued / 续)

2. From residue 1 remove one; from residue 2 remove two. Both moves reach a smaller multiple of 3.

3. Induction now proves that exactly the multiples of 3 are losing.

例题解析

1. 零为必败态。从任意正的 3 倍数出发，两种动作均到达余 1 或 2 的数。

2. 余 1 时拿一枚，余 2 时拿两枚，均到达更小的 3 倍数。

3. 因此归纳证明必败态恰为 3 的倍数。

> An invariant alone rarely proves reachability or a winning strategy.

> 单有不变量通常不能证明可达性或必胜策略。

Practice / 配套练习: AIME-2002-I-Q09 (WB p. 496), AIME-2002-I-Q14 (WB p. 508), AIME-2006-I-Q15 (WB p. 156), AIME-2007-II-Q08 (WB p. 803), AIME-2012-I-Q11 (WB p. 799), AIME-2024-I-Q03 (WB p. 797)

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

Method 1 / 方法 1 [NT04-M01]

A residue cycle must be proved at the correct state level. Multiplicative order handles units; prime-power valuations handle nonunits and lifting. Separate these roles before combining them by CRT.

剩余周期须在正确的状态层面证明。乘法阶处理单位，质数幂赋值处理不可逆部分与提升；先区分这些作用，再用中国剩余定理合并。

Certify the multiplicative order before reducing exponents

约化指数前先确认乘法阶

Procedure

1. For $\gcd(a,m)=1$, define the order as the least positive d with $a^d \equiv 1 \pmod{m}$.

2. Find a candidate exponent from a known group bound or direct calculation. For each prime $q \mid d$, verify $a^{d/q} \not\equiv 1$ to certify exactness.

3. Then $a^n = 1$ precisely when $d \mid n$; a prescribed minus-one power also constrains the two-part of d.

4. For prime moduli, use $d \mid p-1$ to restrict prime candidates, and verify the surviving prime.

通用步骤

1. 当 $\gcd(a,m)=1$，乘法阶为使 $a^d \equiv 1 \pmod{m}$ 的最小正整数 d。

2. 由群阶界或直接计算找候选指数。对每个质因数 $q \mid d$，检验 $a^{d/q} \not\equiv 1$，以确认准确阶数。

3. 此后 $a^n = 1$ 恰当 $d \mid n$；指定幂为负一还会限制 d 的二因子部分。

4. 质数模下以 $d \mid p-1$ 限制质数候选，再验证存活质数。

> Example. Find the order of 3 modulo 13 and decide whether $13 \mid 3^{100}+1$.

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

Method 1 / 方法 1 [NT04-M01] (continued / 续)

> 例题：求 3 模 13 的乘法阶，并判断 $13 \mid 3^{100}+1$ 是否成立。

Worked solution

1. $3^3 = 27 \equiv 1 \pmod{13}$, while $3 \not\equiv 1$, so the order is exactly 3.

2. Since $100 \equiv 1 \pmod{3}$, one has $3^{100} \equiv 3 \pmod{13}$.

3. The expression is congruent to 4, so divisibility fails. A power of odd order cannot equal -1 modulo an odd prime.

例题解析

1. $3^3 = 27 \equiv 1 \pmod{13}$，而 $3 \not\equiv 1$，故阶恰为 3。

2. 因 $100 \equiv 1 \pmod{3}$，有 $3^{100} \equiv 3 \pmod{13}$。

3. 原式余 4，故不能整除；在奇质数模下，奇数阶的幂也不可能等于 -1。

Euler gives an exponent that works; it need not be the least one.

欧拉定理给出可行指数，但未必是最小指数。

If $a^m \equiv -1 \pmod{p}$ for odd prime p, then the order divides $2m$ but not m. Therefore its two-adic valuation is exactly $v_2(m)+1$. Conversely, factor identities such as $x^{2t}-1$ and $x^{2t+1}+1$ can construct a common divisor; an order obstruction by itself proves only necessity.

若奇质数 p 下 $a^m \equiv -1$，阶整除 $2m$ 却不整除 m，故其二赋值恰为 $v_2(m)+1$。反过来，可用 $x^{2t}-1$ 与 $x^{2t+1}+1$ 等分解恒等式构造公约数；单有阶的障碍只证明必要性。

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

Method 1 / 方法 1 [NT04-M01] (continued / 续)

Practice / 配套练习: AIME-2001-II-Q10 (WB p. 519), AIME-2002-I-Q07 (WB p. 219), AIME-2004-II-Q10 (WB p. 736), AIME-2019-I-Q14 (WB p. 523), AIME-2021-II-Q09 (WB p. 520)

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

Method 2 / 方法 2 [NT04-M02]

Prove a cycle by returning to the full determining state

回到完整决定状态来证明周期

Procedure

1. Identify how much information determines the next term: one residue, a pair, or a longer consecutive tuple.

2. Compute successive full states until the initial or a later state repeats. A repeated output alone does not prove a cycle.

3. Separate any nonperiodic prefix, then count complete cycles and the final partial cycle.

4. For a polynomial sequence, prove a period algebraically by showing its difference at shifted indices vanishes modulo the base.

通用步骤

1. 确定下一项由多少信息决定：一个剩余、一对数，或更长连续元组。

2. 计算完整状态直到初始或后续状态重复；单个输出重复不能证明周期。

3. 分开非周期前缀，再数完整周期与最后不完整周期。

4. 多项式数列应通过证明平移下标后的差模底数为零来确认周期。

> Example. Let $u_0 = 0, u_1 = 1$ and $u_{n+2} \equiv u_{n+1} + u_n \pmod 2$. Find u_{100}.

> 例题：设 $u_0 = 0, u_1 = 1, u_{n+2} \equiv u_{n+1} + u_n \pmod 2$，求 u_{100}。

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

Method 2 / 方法 2 [NT04-M02] (continued / 续)

Worked solution

1. The determining pairs are $(0,1),(1,1),(1,0),(0,1)$.

2. The first pair returns after three updates, so the terms repeat $0,1,1$.

3. Since $100 \equiv 1 \pmod 3$, the answer is $u_{100} = 1$.

例题解析

1. 决定状态对依次为 $(0,1),(1,1),(1,0),(0,1)$。

2. 初始状态对三次更新后返回，故数列重复 $0,1,1$。

3. 因 $100 \equiv 1 \pmod 3$，故 $u_{100} = 1$。

A recurrence of order k normally needs k consecutive residues as its state.

k 阶递推通常需要连续 k 个剩余组成状态。

Practice / 配套练习: AIME-2017-I-Q03 (WB p. 517), AIME-2018-II-Q02 (WB p. 518), AIME-2023-II-Q13 (WB p. 205), AIME-2023-II-Q15 (WB p. 525)

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

Method 3 / 方法 3 [NT04-M03]

Split nonunit prefixes and use CRT for a composite modulus

分离不可逆前缀，并对合数模使用中国剩余定理

Procedure

1. Factor the modulus into prime powers. If the base shares a prime factor, first find when its power becomes zero locally.

2. For the remaining unit parts, reduce exponents by a certified order or Euler period.

3. Combine local residues by CRT. For a sum of distinct residues, look for a fixed-point-free pairing such as $r \mapsto m - r$.

通用步骤

1. 将模数分为质数幂；底数若含相同质因数，先求其幂何时在该局部变为零。

2. 其余可逆部分，以确认的阶或欧拉周期约化指数。

3. 用中国剩余定理合并局部剩余。对不同剩余之和，可寻找 $r \mapsto m - r$ 等无不动点配对。

> Example. Find $6^{20} \pmod{40}$.

> 例题：求 6^{20} 模 40 的余数。

Worked solution

1. Modulo 8, every power from the third onward is zero, so $6^{20} \equiv 0$.

2. Modulo 5, the base is one, so $6^{20} \equiv 1$.

3. The unique residue modulo 40 that is zero modulo 8 and one modulo 5 is 16.

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

Method 3 / 方法 3 [NT04-M03] (continued / 续)

例题解析

1. 模 8 时，从三次幂起均为零，故 $6^{20} \equiv 0$。

2. 模 5 时底数为一，故 $6^{20} \equiv 1$。

3. 模 40 唯一满足模 8 余零、模 5 余一的剩余为 16。

Do not apply Euler to a base that is not coprime to the local modulus.

底数不与局部模数互质时不能使用欧拉定理。

For a power tower a^{b^N}, first find an exponent period h for the outer unit base and only then reduce b^N modulo h. If b is not coprime to h, split h into prime powers and handle its zero-producing local prefix before using any order.

对幂塔 a^{b^N}，先求外层单位底数的指数周期 h，再将 b^N 模 h 约化。若 b 不与 h 互质，须将 h 分为质数幂，先处理产生零的局部前缀，再使用乘法阶。

Practice / 配套练习: AIME-2011-I-Q11 (WB p. 521), AIME-2017-I-Q14 (WB p. 125), AIME-2025-II-Q13 (WB p. 217)

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

Method 4 / 方法 4 [NT04-M04]

Lift an order or a root to a prime-square modulus

把阶或根提升到质数平方模

Procedure

1. Find the root or exact order modulo the prime first.

2. For an order d, expand $(a^d)^t = (1+pc)^t$ modulo p^2; if $p \nmid c$, the order multiplies by p.

3. For a root a, write $a+pt$ and solve the resulting linear derivative congruence; inspect every initial root.

4. For higher powers use an explicitly justified lifting valuation, and keep singular derivative cases separate.

通用步骤

1. 先求模该质数的根或准确乘法阶。

2. 若模质数阶为 d，将 $(a^d)^t = (1+pc)^t$ 模 p^2 展开；若 $p \nmid c$，则阶乘以 p。

3. 对根 a 写成 $a+pt$，求所得导数线性同余，并检查每个初始根。

4. 更高次质数幂须使用已证明的赋值提升，并单列导数为零的奇异情形。

> Example. Find the order of 2 modulo 25.

> 例题：求 2 模 25 的乘法阶。

Worked solution

1. The order modulo 5 is 4, since $2^2 = -1$ modulo 5. Any order modulo 25 is a multiple of 4.

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

Method 4 / 方法 4 [NT04-M04] (continued / 续)

2. $2^{4t} = 16^t = (1+15)^t \equiv 1 + 15t \pmod{25}$.

3. This is one precisely when $5 \mid t$, proving the least exponent is 20.

例题解析

1. 模 5 的阶为 4，因 $2^2 = -1$；所以模 25 的阶必为 4 的倍数。

2. $2^{4t} = 16^t = (1+15)^t \equiv 1 + 15t \pmod{25}$。

3. 它等于一当且仅当 $5 \mid t$，故最小指数为 20。

A root may lift uniquely even though its order calculation uses a different argument.

根的唯一提升与阶的计算可能需要不同论证。

For odd p, $p \mid A - B$ and $p \nmid AB$ imply $v_p(A^t - B^t) = v_p(A - B) + v_p(t)$. The binomial expansion proves the rule first for $p \nmid t$, then for each additional factor of p.

Cubing on units modulo powers of 3 is singular: for units a, b with $a \equiv b \pmod 3$, $v_3(a^3 - b^3) = v_3(a - b) + 1$. Thus representatives modulo 3^k map injectively by cubing into the unit cube classes modulo 3^{k+1}, precisely the classes congruent to $\pm 1 \pmod 9$. The image has $2 \cdot 3^{k-1}$ elements, equal to the number of inputs.

对奇质数 p，若 $p \mid A - B, p \nmid AB$，则 $v_p(A^t - B^t) = v_p(A - B) + v_p(t)$。先用二项式展开证明 $p \nmid t$ 的情况，再逐次加入 p 因子即得。

模 3 幂的单位立方映射是奇异情形：若单位 a, b 满足 $a \equiv b \pmod 3$，则 $v_3(a^3 - b^3) = v_3(a - b) + 1$。故模 3^k 代表元的立方单射到模 3^{k+1} 的单位立方类，后者恰为模 9 余 ± 1 的类；其元素数 $2 \cdot 3^{k-1}$ 与输入数相同。

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

Method 4 / 方法 4 [NT04-M04] (continued / 续)

Practice / 配套练习: AIME-2018-I-Q11 (WB p. 522), AIME-2020-I-Q12 (WB p. 524), AIME-2024-I-Q13 (WB p. 526), AIME-2025-I-Q15 (WB p. 527)

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

Method 5 / 方法 5 [NT04-M05]

Refine an exponent fixed point one prime-power layer at a time

逐层细化指数不动点的剩余

Procedure

1. Split the target congruence into coprime prime powers and obtain a first constraint on the exponent itself.

2. Use that constraint to reduce the power modulo a small prime power, obtaining a stronger residue class for the exponent.

3. Repeat only while the established exponent period justifies the next refinement, then recombine and verify the original equation.

通用步骤

1. 将目标同余拆为互质质数幂，并先得到指数本身的初步限制。

2. 用该限制约化小质数幂模下的幂，得到指数更强的剩余类。

3. 只有已知指数周期允许时才继续细化，最后合并并验证原方程。

> Example. Find all residues $n \pmod{20}$ satisfying $n \equiv 3^n \pmod{10}$.

> 例题：求满足 $n \equiv 3^n \pmod{10}$ 的全部模 20 剩余类。

Worked solution

1. The right side is odd, so n is odd. Powers of 3 have period four modulo 10.

2. If $n \equiv 1 \pmod{4}$, the equation requires $n \equiv 3 \pmod{10}$, giving $n \equiv 13 \pmod{20}$.

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

Method 5 / 方法 5 [NT04-M05] (continued / 续)

3. If $n \equiv 3 \pmod 4$, it requires $n \equiv 7 \pmod{10}$, giving $n \equiv 7 \pmod{20}$. Both classes work, and the parity split exhausts all cases.

例题解析

1. 右侧为奇数，故 n 为奇数；3 的幂模 10 的周期为四。

2. 若 $n \equiv 1 \pmod 4$，原式要求 $n \equiv 3 \pmod{10}$，得 $n \equiv 13 \pmod{20}$。

3. 若 $n \equiv 3 \pmod 4$，则须 $n \equiv 7 \pmod{10}$，得 $n \equiv 7 \pmod{20}$。两类均成立，奇偶分类已穷尽。

Each stronger exponent class must follow from the previous one; guessing a long cycle is not a proof of minimality.

更强的指数剩余类须由前一步推出，猜测长周期不能证明最小性。

Practice / 配套练习: AIME-2021-II-Q13 (WB p. 516)

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

Method 6 / 方法 6 [NT04-M06]

Make a universal geometric sum vanish in every local case

使全称等比和在每种局部情形均消失

Procedure

1. Use multiplicativity to reduce a divisor-sum condition to $1+a+\cdots+a^N$ for prime-power factors.

2. Modulo a prime, separate $a=0$, $a=1$, and $a\neq 0,1$; the last case permits division by $a-1$.

3. For a least universal exponent, prove necessity with actual admissible inputs attaining the required residues or orders.

通用步骤

1. 用积性把约数和条件约为质数幂因子的 $1+a+\cdots+a^N$。

2. 模质数分别处理 $a=0, a=1, a\neq 0,1$；最后一种才可除以 $a-1$。

3. 求最小全称指数时，须用确实可取且达到所需剩余或阶的输入证明必要性。

> Example. Find the least positive N such that $a+a^2+\cdots+a^N\equiv 0 \pmod 5$ for every integer a.

> 例题：求最小正整数 N，使每个整数 a 都满足 $a+a^2+\cdots+a^N\equiv 0 \pmod 5$。

Worked solution

1. At $a=1$, necessity gives $5\mid N$. At $a=2$, the geometric formula gives $2^N=1$, so $4\mid N$.

Section 3.4 Orders, modular powers & cycles

乘法阶、模幂与周期

Method 6 / 方法 6 [NT04-M06] (continued / 续)

2. Thus $20 \mid N$. For $N = 20$, $a = 0$ is automatic and $a = 1$ gives a multiple of 5.

3. For the other nonzero residues, $a^4 = 1$ and $a - 1$ is invertible, so $a(a^{20} - 1)/(a - 1) = 0$. The minimum is 20.

例题解析

1. $a = 1$ 给出 $5 \mid N$；$a = 2$ 时等比公式给出 $2^N = 1$，故 $4 \mid N$。

2. 故 $20 \mid N$。当 $N = 20$，$a = 0$ 自动成立，$a = 1$ 得 5 的倍数。

3. 其余非零剩余满足 $a^4 = 1$，且 $a - 1$ 可逆，故 $a(a^{20} - 1)/(a - 1) = 0$；最小值为 20。

A universal assertion needs all residue cases, including the one where the geometric denominator vanishes.

全称结论须覆盖所有剩余情形，包括等比公式分母为零者。

Practice / 配套练习: AIME-2021-I-Q14 (WB p. 612)

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

Method 1 / 方法 1 [NT05-M01]

Integrality is a source of structure: it turns a continuous range into factor pairs, residue classes or a finite list. Every search below begins with a bound and ends with a reconstruction of the original object.

整数性本身提供结构：它把连续范围化为因子对、剩余类或有限列表。这里每次搜索都以界开始，以重建原对象结束。

Parametrize a linear integer system with its domain bounds

以定义域界参数化线性整数方程组

Procedure

1. Eliminate variables over the rationals, but retain all integer and positivity restrictions.

2. Solve a remaining congruence and write every solution using one integer parameter.

3. Intersect all parameter bounds and verify the endpoint candidates in the original model.

通用步骤

1. 先在有理数范围消元，同时保留全部整数与正性限制。

2. 求解剩余同余，并以一个整数参数写出所有解。

3. 对全部参数界取交集，再把端点候选代回原模型。

> Example. Find all positive integer pairs with $7x + 4y = 65$.

> 例题：求满足 $7x + 4y = 65$ 的全部正整数有序对。

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

Method 1 / 方法 1 [NT05-M01] (continued / 续)

Worked solution

1. Modulo 4, $3x \equiv 1$, hence $x = 3 + 4t$.

2. Substitution yields $y = 11 - 7t$. Positivity requires $t \geq 0$ and $t \leq 1$.

3. The pairs are $(3, 11), (7, 4)$, both of which satisfy the equation.

例题解析

1. 模 4 得 $3x \equiv 1$，故 $x = 3 + 4t$。

2. 代入得 $y = 11 - 7t$；正性要求 $0 \leq t \leq 1$。

3. 解为 $(3, 11), (7, 4)$，均满足原式。

An integer parameter still needs an exact feasible interval.

整数参数仍须有精确可行区间。

Practice / 配套练习: AIME-2001-II-Q02 (WB p. 657), AIME-2004-II-Q06 (WB p. 546), AIME-2006-I-Q09 (WB p. 474), AIME-2008-I-Q03 (WB p. 533), AIME-2009-I-Q14 (WB p. 555), AIME-2009-II-Q09 (WB p. 547), AIME-2012-I-Q04 (WB p. 11), AIME-2012-II-Q01 (WB p. 537), AIME-2013-II-Q07 (WB p. 549), AIME-2018-I-Q02 (WB p. 574), AIME-2020-I-Q03 (WB p. 577), AIME-2025-I-Q04 (WB p. 539), AIME-2026-II-Q05 (WB p. 732)

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

Method 2 / 方法 2 [NT05-M02]

Turn a square equation into parity-compatible factor pairs

把平方方程转为奇偶相容的因子对

Procedure

1. Complete a square or subtract two metric square equations to obtain $U^2 - V^2 = K$.

2. Factor $(U - V)(U + V) = K$. List signed or positive divisor pairs according to the original signs.

3. The two factors must have the same parity. Recover the original variables and check strict geometry, nonsquare exclusions and denominator conditions.

通用步骤

1. 配方，或相减两个度量平方方程，得到 $U^2 - V^2 = K$。

2. 分解 $(U - V)(U + V) = K$，按原符号列正因子对或带符号因子对。

3. 两个因子须奇偶相同。还原原变量，并检查严格几何、非平方排除与分母条件。

> Example. Find all integer pairs (u, v) with $u^2 - v^2 = 45$ and $u > v \geq 0$.

> 例题：求满足 $u^2 - v^2 = 45, u > v \geq 0$ 的全部整数对。

Worked solution

1. The positive factors $u - v, u + v$ multiply to 45 and are odd.

2. The increasing pairs are $(1, 45), (3, 15), (5, 9)$.

3. Halving their sums and differences gives $(23, 22), (9, 6), (7, 2)$.

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

Method 2 / 方法 2 [NT05-M02] (continued / 续)

例题解析

1. 正因子 $u-v, u+v$ 乘积为 45，且都是奇数。

2. 递增因子对为 $(1,45),(3,15),(5,9)$。

3. 对和与差分别除二，得 $(23,22),(9,6),(7,2)$。

The parity test is part of the inverse transformation, not an optional final check.

奇偶检验是逆变换的一部分，并非可有可无的末尾检查。

A sum of consecutive positive odd integers is m^2-k^2 with $m>k\geq 0$. Therefore its representations correspond to factor pairs of equal parity; the smaller factor gives the length. Only after this bijection is proved should a prescribed number of representations be converted into a divisor-count pattern (NT08).

连续正奇数之和为 m^2-k^2，其中 $m>k\geq 0$。所以表示方法对应奇偶相同的因子对，较小因子给出长度。先证明此双射，才可将指定表示数转为约数计数模式（NT08）。

Practice / 配套练习: AIME-2000-II-Q02 (WB p. 528), AIME-2003-I-Q07 (WB p. 543), AIME-2003-II-Q10 (WB p. 545), AIME-2005-I-Q04 (WB p. 532), AIME-2008-I-Q04 (WB p. 534), AIME-2008-II-Q15 (WB p. 554), AIME-2017-II-Q06 (WB p. 551), AIME-2006-II-Q13 (WB p. 610)

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

Method 3 / 方法 3 [NT05-M03]

Shift a rational or polynomial constraint into a fixed product

把分式或多项式限制平移为固定乘积

Procedure

1. Clear denominators within the original nonzero domain, then look for a constant product after adding or subtracting a fixed shift.

2. For an integer-coefficient polynomial, use divisibility of $P(a)-P(b)$ by $a-b$ or division by monic integer factors.

3. Enumerate all fixed-product pairs consistent with the signs, then reverse every substitution.

通用步骤

1. 在原非零定义域内清分母，再寻找加减固定平移后得到的常数乘积。

2. 整系数多项式可用 $a-b \mid P(a)-P(b)$，或除以首一整数因子。

3. 穷尽符合符号的固定乘积因子对，再逆转每个代换。

> Example. Find positive integer solutions to $1/x+1/y=1/6$ with $x \le y$.

> 例题：求 $1/x+1/y=1/6$ 且 $x \le y$ 的正整数解。

Worked solution

1. Clearing denominators gives $xy=6x+6y$, hence $(x-6)(y-6)=36$.

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

Method 3 / 方法 3 [NT05-M03] (continued / 续)

2. Both shifts are positive: each reciprocal is smaller than $1/6$. The ordered factor pairs with first no larger are $(1,36),(2,18),(3,12),(4,9),(6,6)$.

3. Adding six gives $(7,42),(8,24),(9,18),(10,15),(12,12)$.

例题解析

1. 清分母得 $xy=6x+6y$，故 $(x-6)(y-6)=36$。

2. 两平移均为正，因为每个倒数都小于 $1/6$。较小在前的因子对为 $(1,36),(2,18),(3,12),(4,9),(6,6)$。

3. 分别加六，得 $(7,42),(8,24),(9,18),(10,15),(12,12)$。

A fixed product gives a finite list only after signs and zero cases are controlled.

控制符号与零情形后，固定乘积才给出有限列表。

Practice / 配套练习: AIME-2000-I-Q02 (WB p. 349), AIME-2002-I-Q04 (WB p. 127), AIME-2003-II-Q01 (WB p. 530), AIME-2004-II-Q03 (WB p. 531), AIME-2005-II-Q13 (WB p. 553), AIME-2009-II-Q04 (WB p. 535), AIME-2015-II-Q08 (WB p. 550), AIME-2016-I-Q07 (WB p. 79), AIME-2021-I-Q05 (WB p. 538), AIME-2026-I-Q04 (WB p. 540)

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

Method 4 / 方法 4 [NT05-M04]

Convert rational radicals to integer square conditions

把有理根式转为整数平方条件

Procedure

1. Isolate a sum, difference or product of radicals and determine whether it must be rational.

2. If an integer has a rational square root, write it in lowest terms and prove the denominator is one.

3. Replace the radicals by integer parameters; keep any nonsquare or strict inequality condition in the resulting finite search.

通用步骤

1. 隔离根式的和、差或积，判断其是否必须为有理数。

2. 若整数的平方根为有理数，将其约至最简，并证明分母为一。

3. 用整数参数替代根式，并在所得有限搜索中保留非平方或严格不等式条件。

> Example. If $\sqrt{a}+\sqrt{b}=7$ for positive integers a, b, determine all unordered pairs.

> 例题：若正整数 a, b 满足 $\sqrt{a}+\sqrt{b}=7$，求全部无序对。

Worked solution

1. Squaring $\sqrt{b}=7-\sqrt{a}$ gives $\sqrt{a}=(49+a-b)/14$, which is rational.

2. A reduced r/s with $(r/s)^2=a$ has $s^2 \mid r^2$, forcing $s=1$. Thus both radicals are integers.

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

Method 4 / 方法 4 [NT05-M04] (continued / 续)

3. The positive integer root pairs summing to seven are $(1,6),(2,5),(3,4)$, giving $(a,b)=(1,36),(4,25),(9,16)$.

例题解析

1. 将 $\sqrt{b}=7-\sqrt{a}$ 平方，得 $\sqrt{a}=(49+a-b)/14$ 为有理数。

2. 若最简 r/s 满足 $(r/s)^2=a$，则 $s^2\mid r^2$ 迫使 $s=1$，故两个根式均为整数。

3. 和为七的正整数根对为 $(1,6),(2,5),(3,4)$，故 $(a,b)=(1,36),(4,25),(9,16)$。

A rational sum does not make each summand rational without an additional algebraic argument.

和为有理数并不自动使各项有理，必须补上代数论证。

Practice / 配套练习: AIME-2000-I-Q06 (WB p. 541), AIME-2003-II-Q10 (WB p. 545), AIME-2013-I-Q10 (WB p. 76)

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

Method 5 / 方法 5 [NT05-M05]

Certify a short integer search from algebraic or geometric bounds

由代数或几何界证明短整数搜索完整

Procedure

1. Obtain an equation with nonnegative terms, a discriminant square, or a triangle interval.

2. Order symmetric variables to avoid duplicates and derive the exact finite bounds for each variable.

3. Test every remaining integer case exactly, then reconstruct coefficients, geometry or probabilities to check attainability.

通用步骤

1. 得到非负项方程、平方判别式或三角形范围。

2. 给对称变量排序以避免重复，并推导每个变量的精确有限界。

3. 精确检查全部剩余整数情形，再重建系数、几何或概率以检验可达到性。

> Example. Find positive integers $a \le b \le c$ with $a^2 + b^2 + c^2 = 14$.

> 例题：求满足 $a^2 + b^2 + c^2 = 14$ 的正整数 $a \le b \le c$。

Worked solution

1. Ordering implies $3a^2 \le 14$, so $a = 1$ or 2.

2. For $a = 1$, $b^2 + c^2 = 13$ and $1 \le b \le 2$; only $b = 2, c = 3$ works.

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

Method 5 / 方法 5 [NT05-M05] (continued / 续)

3. For $a = 2$, $b \geq 2$ and $b^2 + c^2 = 10$, so $b = 2$ would leave six, not a square. The sole triple is $(1, 2, 3)$.

例题解析

1. 排序给出 $3a^2 \leq 14$，故 $a = 1$ 或 2。

2. $a = 1$ 时 $b^2 + c^2 = 13$ 且 $1 \leq b \leq 2$，仅 $b = 2, c = 3$ 可行。

3. $a = 2$ 时 $b \geq 2, b^2 + c^2 = 10$，但 $b = 2$ 留下六，非平方数。唯一三元组为 $(1, 2, 3)$。

Record why each bound is sufficient; a spreadsheet search alone does not prove completeness.

应写清每个界为何足够；电子表格搜索本身不能证明穷尽。

Practice / 配套练习: AIME-2000-I-Q05 (WB p. 719), AIME-2000-II-Q11 (WB p. 385), AIME-2002-II-Q03 (WB p. 529), AIME-2003-I-Q08 (WB p. 544), AIME-2005-II-Q05 (WB p. 101), AIME-2009-II-Q11 (WB p. 174), AIME-2010-II-Q12 (WB p. 278), AIME-2011-I-Q15 (WB p. 556), AIME-2011-II-Q15 (WB p. 176), AIME-2015-I-Q11 (WB p. 323), AIME-2015-II-Q06 (WB p. 48), AIME-2016-I-Q12 (WB p. 482), AIME-2016-II-Q05 (WB p. 222), AIME-2016-II-Q09 (WB p. 152), AIME-2018-II-Q03 (WB p. 575), AIME-2019-I-Q11 (WB p. 346), AIME-2020-II-Q12 (WB p. 409), AIME-2022-I-Q14 (WB p. 326), AIME-2026-II-Q08 (WB p. 318), AIME-2021-II-Q15 (WB p. 216)

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

Method 6 / 方法 6 [NT05-M06]

Prove a discrete minimum with a bound and an attaining construction

用下界与达到下界的构造证明离散最小值

Procedure

1. Compare any feasible object with a proposed optimum using integrality, factor order or a forced equality.

2. If two parameterizations collide, subtract their equations and use coprimality to bound the first possible collision.

3. Construct an object meeting the bound and verify every domain restriction.

通用步骤

1. 用整数性、因子大小或被迫等号，将任意可行对象与候选最优值比较。

2. 若两个参数表示重合，先相减，再用互质性确定首次可能重合的下界。

3. 构造达到界的对象，并检验全部定义域限制。

> Example. Prove that $4x + 7y = 25$ has at most one integer solution in the rectangle $0 \le x \le 6, 0 \le y \le 3$, and determine that solution.

> 例题：证明 $4x + 7y = 25$ 在矩形 $0 \le x \le 6, 0 \le y \le 3$ 中至多有一个整数解，并求出该解。

Worked solution

1. If $(x, y), (u, v)$ are two solutions, then $4(x - u) = 7(v - y)$. Coprimality gives $7 \mid x - u$.

2. But $|x - u| \le 6$, so $x = u$ and then $y = v$. This proves uniqueness from the width bound.

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

Method 6 / 方法 6 [NT05-M06] (continued / 续)

3. The pair $(1, 3)$ lies in the rectangle and satisfies $4 + 21 = 25$, so the upper bound of one solution is attained.

例题解析

1. 若 $(x, y), (u, v)$ 均为解，则 $4(x - u) = 7(v - y)$。由互质性得 $7 \mid x - u$。

2. 而 $|x - u| \leq 6$，所以 $x = u$，继而 $y = v$；由宽度界证明唯一性。

3. $(1, 3)$ 位于矩形内，且 $4 + 21 = 25$，所以一个解的上界可以达到。

An extremum needs both a universal bound and a valid witness.

最值须同时有普遍界与有效达到例。

Practice / 配套练习: AIME-2002-I-Q08 (WB p. 542), AIME-2004-II-Q09 (WB p. 144), AIME-2009-I-Q13 (WB p. 212), AIME-2010-I-Q05 (WB p. 536), AIME-2020-I-Q10 (WB p. 478)

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

Method 7 / 方法 7 [NT05-M07]

Analyze the gaps between consecutive powers

分析相邻幂之间的间隔

Procedure

1. Partition integers into blocks between consecutive squares or cubes.

2. Express the first available next power or the gap length exactly; identify the worst endpoint of a block.

3. Prove later blocks satisfy the requirement monotonically, then locate and verify the last failing block.

通用步骤

1. 按相邻平方数或立方数之间的区间给整数分块。

2. 精确表达下一个可用幂或间隔长度，并确定块中的最差端点。

3. 证明后续区块的条件按单调性成立，再定位并检验最后失败区块。

> Example. Find the least Q such that every integer $n \geq Q$ has a square in $(n, 2n]$.

> 例题：求最小 Q，使每个整数 $n \geq Q$ 的区间 $(n, 2n]$ 中都有平方数。

Worked solution

1. If $k^2 \leq n < (k+1)^2$, the next square is $(k+1)^2$ and the largest ratio to n occurs at $n = k^2$.

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

Method 7 / 方法 7 [NT05-M07] (continued / 续)

2. For $k \geq 3$, $(1+1/k)^2 \leq 16/9 < 2$, so all such blocks work. For $4 \leq n \leq 8$, only $n = 4$ fails because nine exceeds eight.

3. The integers $n = 5, 6, 7, 8$ are covered by nine. Thus the last failure is four and $Q = 5$.

例题解析

1. 若 $k^2 \leq n < (k+1)^2$，下一个平方数为 $(k+1)^2$，与 n 的比在 $n = k^2$ 时最大。

2. 当 $k \geq 3$，$(1+1/k)^2 \leq 16/9 < 2$，这些区块全可行。在 $4 \leq n \leq 8$ 中，仅 $n = 4$ 因九大于八而失败。

3. $n = 5, 6, 7, 8$ 均由九覆盖。因此最后失败值为四，$Q = 5$。

The ratio is tested at the block start, but the final failing integer may lie later in that block.

比值在块起点检查，但最后失败的整数可能在块内更后位置。

Practice / 配套练习: AIME-2013-II-Q06 (WB p. 548), AIME-2017-I-Q13 (WB p. 557)

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

Method 8 / 方法 8 [NT05-M08]

Minimize a denominator inside a narrow interval

在窄区间中最小化分母

Procedure

1. Convert a required decimal prefix to a half-open rational interval.

2. Compare with a nearby simple fraction a/b and form the nonzero integer $bm - an$.

3. Use the interval width to bound the denominator, test the first possible denominators, and confirm reduction and endpoint strictness.

通用步骤

1. 把指定小数前缀转为左闭右开的有理区间。

2. 与附近简单分数 a/b 比较，并形成非零整数 $bm - an$。

3. 用区间宽度给分母建立界，检查最先可能的分母，并核对既约性与端点严格性。

> Example. Find the least denominator of a reduced fraction strictly between $1/3$ and $2/5$.

> 例题：求严格位于 $1/3$ 与 $2/5$ 之间的既约分数的最小分母。

Worked solution

1. If $m/n > 1/3$, then $3m - n \geq 1$. If $m/n < 2/5$, then $2n - 5m \geq 1$.

2. Multiply the first inequality by five and the second by three; addition gives $n \geq 8$.

3. The fraction $3/8$ is reduced and lies strictly between the endpoints, so the minimum denominator is 8.

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

Method 8 / 方法 8 [NT05-M08] (continued / 续)

例题解析

1. 由 $m/n > 1/3$ 得 $3m - n \geq 1$；由 $m/n < 2/5$ 得 $2n - 5m \geq 1$。

2. 第一不等式乘五、第二乘三，相加得 $n \geq 8$。

3. $3/8$ 既约且严格位于两端之间，所以最小分母为 8。

If a digit string starts later, shifting the decimal and reducing can only decrease the denominator.

若数位串在较后位置出现，移动小数点后约分只会使分母不增。

Practice / 配套练习: AIME-2003-I-Q14 (WB p. 597)

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

Method 9 / 方法 9 [NT05-M09]

Use integer descent to force a rigid recurrence

用整数递降迫使递推具有刚性

Procedure

1. Clear two neighboring recurrence equations and subtract them to expose a relation between differences.

2. If a nonzero integer difference would shrink by a factor exceeding one forever, derive a contradiction.

3. Force that difference to zero, identify the resulting periodic family, and verify every member satisfies the original recurrence.

通用步骤

1. 清除相邻两式的分母并相减，找出差量之间的关系。

2. 若某非零整数差将永远按大于一的倍数缩小，则导出矛盾。

3. 迫使差为零，识别所得周期族，并验证每个成员满足原递推。

> Example. An infinite integer sequence satisfies $d_{n+1} = d_n/3$. Prove every term is zero.

> 例题：无限整数列满足 $d_{n+1} = d_n/3$，证明每项均为零。

Worked solution

1. If $d_1 \neq 0$, then $|d_n| = |d_1|/3^{n-1}$ remains positive.

2. For sufficiently large n this value is below one, impossible for a nonzero integer.

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

Method 9 / 方法 9 [NT05-M09] (continued / 续)

3. Therefore $d_1 = 0$, and the recurrence makes every subsequent term zero.

例题解析

1. 若 $d_1 \neq 0$，则 $|d_n| = |d_1|/3^{n-1}$ 始终为正。

2. 当 n 足够大，该值小于一，不可能是非零整数。

3. 故 $d_1 = 0$，由递推其后各项均为零。

Descent requires integrality and an infinite continuation; a finite real sequence gives no contradiction.

递降须有整数性与无限延续；有限实数列不会产生此矛盾。

Practice / 配套练习: AIME-2009-I-Q13 (WB p. 212)

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

Method 10 / 方法 10 [NT05-M10]

Project a nonnegative integer solution set onto the requested statistic

把非负整数解集投影到目标统计量

Procedure

1. Parametrize the admissible objects by nonnegative integer multiplicities.

2. Eliminate one variable and express the requested quantity as a linear function of the remaining ones.

3. For each bounded outer parameter, find the full integer interval of target values; take the union, not the sum of interval lengths.

通用步骤

1. 用非负整数重数参数化允许对象。

2. 消去一个变量，将目标量写为其余变量的线性式。

3. 固定每个有界外参数，求目标值的完整整数区间；最后取并集，不能直接累加区间长度。

> Example. Nonnegative integers satisfy $x+3y+5z=12$. How many distinct values can $x+y+z$ take?

> 例题：非负整数满足 $x+3y+5z=12$，$x+y+z$ 可取多少个不同值?

Worked solution

1. The target is $12-2y-4z=12-2v$, where $v=y+2z$.

2. For $z=0,1,2$, the possible v intervals are respectively $[0,4],[2,4],[4,4]$ on the integers.

Section 3.5 Diophantine equations & factor pairs

丢番图方程与因子配对

Method 10 / 方法 10 [NT05-M10] (continued / 续)

3. Their union is $0, 1, 2, 3, 4$, giving five distinct targets $12, 10, 8, 6, 4$.

例题解析

1. 目标为 $12 - 2y - 4z = 12 - 2v$，其中 $v = y + 2z$。

2. 当 $z = 0, 1, 2$，整数 v 区间分别为 $[0, 4], [2, 4], [4, 4]$。

3. 并集为 $0, 1, 2, 3, 4$，故五个不同目标值为 $12, 10, 8, 6, 4$。

Different representations may give the same target; the question may count values rather than representations.

不同表示可能产生同一目标；题目有时数的是值而非表示数。

Practice / 配套练习: AIME-2004-II-Q14 (WB p. 552)

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 1 / 方法 1 [NT06-M01]

Expand the numeral before manipulating it, and distinguish digit values from their place weights. Digit bounds, leading zeros and carry behavior belong in the model from the first line.

运算前先展开数码，区分数字值与位权。数位范围、首位零与进位行为从建模第一行就应明确。

Expand place values and enforce the common digit range

展开位值并施加共同数位范围

Procedure

1. Write a base-b numeral as $\sum d_i b^i$, with fractional positions using negative powers.

2. A digit shared between bases must be valid in every base in which it occurs; each leading digit is nonzero.

3. Eliminate variables, use a small congruence or a power bound, then test the remaining digits exactly.

4. When all digits are nonnegative, compare the value monotonically as the base changes.

通用步骤

1. 把 b 进制数写成 $\sum d_i b^i$，小数位使用负指数。

2. 跨进位制共用数字须在每个出现的进位制中有效，且每个首位非零。

3. 消元后使用小模数或幂的范围界，再精确检查剩余数字。

4. 所有数位非负时，可按底数变化单调比较数值。

> Example. The numeral $\overline{1d2}_b$ has value $2b^2$ for an integer base $b > 2$. Find all valid (b, d).

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 1 / 方法 1 [NT06-M01] (continued / 续)

> 例题：整数底数 $b > 2$ 下，数码 $\overline{1d2}_b$ 的值为 $2b^2$。求全部有效 (b, d)。

Worked solution

1. Place value gives $b^2 + db + 2 = 2b^2$, or $b(b - d) = 2$.

2. Because $0 \le d < b$, the integer $b - d$ is positive, so b must divide 2.

3. This contradicts $b > 2$. There are no valid digit/base pairs.

例题解析

1. 位值给出 $b^2 + db + 2 = 2b^2$，即 $b(b - d) = 2$。

2. 因 $0 \le d < b$，整数 $b - d$ 为正，故 b 须整除 2。

3. 这与 $b > 2$ 矛盾，故无有效数位底数对。

A decimal-looking symbol is a digit string until its base has been specified.

外观像十进制数的符号，在底数确定前只是数位串。

Practice / 配套练习: AIME-2001-I-Q08 (WB p. 585), AIME-2001-I-Q01 (WB p. 558), AIME-2004-I-Q01 (WB p. 563), AIME-2009-I-Q01 (WB p. 566), AIME-2017-I-Q05 (WB p. 572), AIME-2018-I-Q02 (WB p. 574), AIME-2018-II-Q03 (WB p. 575), AIME-2020-I-Q03 (WB p. 577), AIME-2022-I-Q02 (WB p. 579), AIME-2025-I-Q01 (WB p. 582), AIME-2026-II-Q04 (WB p. 584), AIME-2024-II-Q14 (WB p. 593)

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 2 / 方法 2 [NT06-M02]

Isolate a prefix or suffix and reduce digit changes modulo a divisor

隔离前缀或后缀，并对数位变化取模

Procedure

1. For a suffix of length k, write $N = b^k q + r$ with $0 \le r < b^k$.

2. Deletion or replacement changes a known place-value term; reduce the resulting equation or divisibility before enumerating.

3. For a square suffix, solve the congruence for its root modulo prime powers and order the roots, not the suffixes.

4. Check leading zeros, positivity of the remaining prefix and any boundary block separately.

通用步骤

1. 对长度为 k 的后缀，写 $N = b^k q + r$，其中 $0 \le r < b^k$。

2. 删位或替换数位会改变已知位值项；枚举之前先约化所得方程或整除式。

3. 平方数后缀题应按质数幂求根的同余，并给根排序，而非仅给后缀排序。

4. 单独检查首位零、剩余前缀正性及边界区块。

> Example. If $100q + 37$ is divisible by 7, find $q \pmod 7$. Can $100q + 12$ also be divisible by 7?

> 例题：若 $100q + 37$ 可被 7 整除，求 q 模 7 的剩余。$100q + 12$ 也能被 7 整除吗?

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 2 / 方法 2 [NT06-M02] (continued / 续)

Worked solution

1. Modulo seven, $100q + 37 \equiv 2q + 2$, so $q \equiv -1 \equiv 6$.

2. The two integers differ by 25, which is not divisible by seven.

3. Thus they cannot both be divisible by seven. In fact $100q + 12 \equiv 2 \cdot 6 + 5 \equiv 3 \pmod 7$.

例题解析

1. 模七有 $100q + 37 \equiv 2q + 2$，所以 $q \equiv -1 \equiv 6$。

2. 两整数之差为 25，不能被七整除。

3. 因此它们不能同时被七整除；实际 $100q + 12 \equiv 2 \cdot 6 + 5 \equiv 3 \pmod 7$。

A suffix may contain initial zeroes even though the whole numeral may not.

后缀允许开头为零，整个整数的首位则不允许。

Practice / 配套练习: AIME-2000-I-Q01 (WB p. 463), AIME-2006-I-Q03 (WB p. 564), AIME-2012-I-Q10 (WB p. 499), AIME-2014-I-Q08 (WB p. 500), AIME-2020-I-Q04 (WB p. 488), AIME-2024-II-Q07 (WB p. 506)

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 3 / 方法 3 [NT06-M03]

Use reversal and palindrome symmetry before counting digits

计数前先使用翻转与回文对称

Procedure

1. Expand a numeral and its reversal; paired places often cancel or acquire equal coefficients.

2. A palindrome is determined by its left half and, for odd length, its center digit.

3. Apply nonzero-leading and digit-sum parity restrictions, then count the independent positions.

4. For averages, average each free digit separately; for two bases, first bound the possible lengths.

通用步骤

1. 展开原数及其翻转，成对位值常会消去或得到相同系数。

2. 回文数由左半部分及奇数长度时的中央数字决定。

3. 施加首位非零与数位和奇偶限制，再统计独立位置。

4. 求平均时分别平均每个自由数字；跨进位制时先限制可能长度。

> Example. How many three-digit decimal palindromes are divisible by 3, and what is their mean?

> 例题：有多少个三位十进制回文数可被 3 整除？它们的平均数是多少？

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 3 / 方法 3 [NT06-M03] (continued / 续)

Worked solution

1. Write $N = 101a + 10b$, with $1 \le a \le 9, 0 \le b \le 9$. Divisibility requires $2a + b \equiv 0 \pmod 3$.

2. For $a \equiv 0$ there are four possible b; for the other two classes there are three. Each a class has three members, giving $3 \cdot 4 + 6 \cdot 3 = 30$.

3. Over these thirty choices, $\sum a = 3(1 + \cdots + 9) + (3 + 6 + 9) = 153$, while the three residue classes give $\sum b = 3(18 + 12 + 15) = 135$. The mean is $(101 \cdot 153 + 10 \cdot 135)/30 = 5601/10$.

例题解析

1. 写 $N = 101a + 10b$，其中 $1 \le a \le 9, 0 \le b \le 9$。整除要求 $2a + b \equiv 0 \pmod 3$。

2. $a \equiv 0$ 时 b 有四种，另两类各三种；每类 a 有三个值，故共 $3 \cdot 4 + 6 \cdot 3 = 30$ 个。

3. 这三十组选取中，$\sum a = 3(1 + \cdots + 9) + (3 + 6 + 9) = 153$，三类剩余给出 $\sum b = 3(18 + 12 + 15) = 135$。平均数为 $(101 \cdot 153 + 10 \cdot 135)/30 = 5601/10$。

> Do not impose distinct digits unless the question says so.

> 题目未要求时，不应额外施加数字互异。

Practice / 配套练习: AIME-2002-I-Q03 (WB p. 560), AIME-2002-II-Q01 (WB p. 561), AIME-2012-I-Q01 (WB p. 568), AIME-2021-II-Q01 (WB p. 794), AIME-2023-II-Q02 (WB p. 580), AIME-2026-I-Q02 (WB p. 583)

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 4 / 方法 4 [NT06-M04]

Make carries explicit in column sums and digit-sum changes

明确逐列进位与数位和变化

Procedure

1. Write one equation per column, introducing the carry entering and leaving it.

2. In base b, every carry reduces the naive digit sum by $b-1$; include carries created by incoming carries.

3. Translate fixed totals into constraints on column digit sums, then optimize or count their allowed digit assignments.

4. Verify the complete numeral at the end, including any new leading carry.

通用步骤

1. 每列写一式，设置进入与离开的进位。

2. b 进制中每次进位使原始数位和减少 $b-1$，并须计入由前列进位诱发的进位。

3. 把固定总数转为各列数位和限制，再优化或统计允许数位分配。

4. 末尾检验整个数值，包括新产生的最高位进位。

> Example. For binary integers of a fixed length of at least two bits, characterize when adding one leaves the number of ones unchanged.

> 例题：对位数固定且至少有两位的二进制整数，刻画何时加一后数字一的数目保持不变。

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 4 / 方法 4 [NT06-M04] (continued / 续)

Worked solution

1. If the number ends in t ones, adding one changes those t ones to zeroes and the preceding zero to one.

2. The net change in weight is $1-t$. Equality of weights requires exactly one trailing one.

3. Thus the numeral must end in 01 and become one ending in 10. An all-one string of length at least two has change $1-t<0$, so its overflow cannot qualify.

例题解析

1. 若末尾连续有 t 个一，加一会把它们变成零，并把此前的零变成一。

2. 一的数目净变化为 $1-t$，保持不变要求恰有一个末尾一。

3. 故原数须以 01 结尾，更新后以 10 结尾。至少两位的全一串满足 $1-t<0$，所以溢出不能合格。

Count carry events, not the number of columns whose original digit sum exceeds the base.

应统计实际进位事件，不能只数原数位和超过底数的列。

For $a+b=10^k$, first separate the case of zero units digits. Otherwise the units sum to ten and every higher pair of padded digits sums to nine. A zero in one higher position therefore forces nine in the complementary position, which helps make forbidden-zero cases disjoint. Leading padding zeroes are not digits of the original positive numeral.

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 4 / 方法 4 [NT06-M04] (continued / 续)

For coefficients $0 \le a_i < b^2$, split $a_i = bb_i + c_i$ into legal base-b digits. Then $\sum a_i b^i = bB + C$, with $B = \sum b_i b^i$ and $C = \sum c_i b^i$. Bound both B, C in their k-digit ranges before counting; conversely their padded digit expansions uniquely recover every coefficient. This supplies a reversible digit-stream encoding rather than only a carry count.

对 $a + b = 10^k$，先分出个位均为零的情形；否则个位之和为十，此后每对补齐位数的高位数字之和为九。某个高位为零便迫使互补位置为九，可据此证明某些禁零情形互斥。首部补齐的零不属于原正整数的数位。

若系数满足 $0 \le a_i < b^2$，将其拆为合法 b 进制数字 $a_i = bb_i + c_i$，则 $\sum a_i b^i = bB + C$，其中 $B = \sum b_i b^i, C = \sum c_i b^i$。计数前先给 B, C 都施加 k 位范围界；反过来，二者补齐后的数位展开能唯一恢复每个系数。这提供可逆的数位流编码，而非仅仅统计进位。

Practice / 配套练习: AIME-2006-II-Q07 (WB p. 586), AIME-2012-I-Q05 (WB p. 569), AIME-2015-I-Q08 (WB p. 588), AIME-2016-I-Q08 (WB p. 589), AIME-2019-I-Q01 (WB p. 576), AIME-2024-II-Q03 (WB p. 581), AIME-2010-I-Q10 (WB p. 712)

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 5 / 方法 5 [NT06-M05]

Count and unrank digit strings under an upper bound

统计并反向定位上界内的数位串

Procedure

1. Count shorter lengths first, fixing a nonzero leading digit unless zero padding is explicitly allowed.

2. For equal lengths, scan from the leading position and count all allowed smaller-prefix blocks before following the bound.

3. For a specified rank, subtract entire block counts until the desired rank lies in one block, then continue recursively.

4. Use binomial coefficients for a fixed digit weight and handle partial final blocks exactly.

通用步骤

1. 先统计较短长度；除非明确允许补零，首位须非零。

2. 等长时从高位扫描，先统计所有允许的较小前缀块，再沿上界前缀继续。

3. 给定排名时，逐块减去完整块数，直到目标位于某块，再递归处理。

4. 固定数位重数时用二项式系数，并精确处理最后不完整块。

> Example. List the fifth nonnegative binary string of length six with exactly two ones, when strings are ordered by value.

> 例题：六位二进制串（允许前补零）恰有两个一，按数值排序的第五个是什么？

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 5 / 方法 5 [NT06-M05] (continued / 续)

Worked solution

1. The first four are $000011, 000101, 000110, 001001$. A prefix block can be counted by choosing remaining one positions.

2. Among strings beginning 00, those beginning 000 contribute $\binom{3}{2} = 3$, and the next prefix 001 contributes $\binom{3}{1} = 3$.

3. Rank five is the second member of the latter block: $001010_2 = 10$.

例题解析

1. 前四个为 $000011, 000101, 000110, 001001$；每个前缀块的大小可由剩余一的位置选择数求出。

2. 在以 00 开头的串中，前缀 000 贡献 $\binom{3}{2} = 3$，随后前缀 001 贡献 $\binom{3}{1} = 3$。

3. 第五名是后块第二个：$001010_2 = 10$。

Numerical order equals lexicographic order only after the length is fixed.

只有固定长度后，数值顺序才等于字典序。

Practice / 配套练习: AIME-2003-I-Q13 (WB p. 591), AIME-2012-II-Q07 (WB p. 587), AIME-2017-II-Q04 (WB p. 573), AIME-2004-II-Q10 (WB p. 736)

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 6 / 方法 6 [NT06-M06]

Optimize a numeral by digit mass and lexicographic priority

按数位质量与字典序优先级优化整数

Procedure

1. Compare lengths first: every valid longer positive numeral exceeds every shorter one.

2. For a minimum with a prescribed digit sum, use the fewest positions and place as much mass as possible in low positions.

3. For a maximum with suffix constraints, maximize the earliest digit whose choice still permits a valid completion.

4. Use divisibility or geometric-progression constraints to certify that no rejected earlier prefix can work.

通用步骤

1. 先比较长度：每个有效的更长正整数都大于较短者。

2. 给定数位和求最小时，使用最少位数，并尽量把数位质量放在低位。

3. 带后缀限制求最大时，优先最大化仍允许合法补全的最早一位。

4. 用整除或等比限制证明被舍弃的更优前缀均不可能。

> Example. Find the smallest base-five numeral with digit sum 11.

> 例题：求数位和为 11 的最小五进制正整数。

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 6 / 方法 6 [NT06-M06] (continued / 续)

Worked solution

1. Two positions hold digit sum at most 8, so at least three positions are needed.

2. To minimize the leading digit, the two lower digits should total as much as possible, namely eight. The leading digit is therefore three.

3. The unique minimum is $344_5 = 99$; any larger length or earlier larger digit increases the value.

例题解析

1. 两位的数位和至多为 8，故至少需要三位。

2. 为最小化首位，低两位应合计最大值八，所以首位为三。

3. 唯一最小值为 $344_5 = 99$；更长位数或更大高位均会增大数值。

A greedy prefix choice needs a feasible completion, not just an attractive leading digit.

贪心选择前缀须有合法补全，不能只看首位大小。

Practice / 配套练习: AIME-2003-II-Q02 (WB p. 562), AIME-2009-I-Q01 (WB p. 566), AIME-2010-II-Q01 (WB p. 567), AIME-2015-II-Q03 (WB p. 571), AIME-2020-II-Q05 (WB p. 578)

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 7 / 方法 7 [NT06-M07]

Model local digit rules by a directed transition graph

用有向转移图建模局部数位规则

Procedure

1. Use digits or the necessary recent-digit memory as states, and draw every allowed next-digit transition.

2. For a largest unrestricted-length numeral, check whether a reachable directed cycle permits arbitrarily long strings.

3. For a fixed length, count paths by a recurrence or by a uniform predecessor count; enforce the initial nonzero rule.

通用步骤

1. 以数字或所需最近数位记忆为状态，画出每个允许后继转移。

2. 位数不受限求最大数时，先检查可达有向循环是否允许任意长串。

3. 固定位数时，以递推或统一前驱数统计路径，并施加首位非零。

> Example. Count four-digit binary strings with no adjacent ones, allowing initial zeros.

> 例题：统计无相邻一的四位二进制串，允许前补零。

Worked solution

1. Use the last digit as the state. A zero permits either next digit; a one permits only zero.

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 7 / 方法 7 [NT06-M07] (continued / 续)

2. If a_n is the total count, final zero contributes a_{n-1} and final 01 contributes a_{n-2}. Thus $a_n = a_{n-1} + a_{n-2}$.

3. With $a_0 = 1, a_1 = 2$, the counts are $a_2 = 3, a_3 = 5, a_4 = 8$.

例题解析

1. 以末位为状态；零允许后接零或一，一只允许后接零。

2. 设总数为 a_n，以零结尾贡献 a_{n-1}，以 01 结尾贡献 a_{n-2}，故 $a_n = a_{n-1} + a_{n-2}$。

3. 初值 $a_0 = 1, a_1 = 2$，得 $a_2 = 3, a_3 = 5, a_4 = 8$。

Counting an acyclic longest path differs from counting all paths of a fixed length.

无环图的最长路径与固定长度全部路径是不同计数问题。

Practice / 配套练习: AIME-2001-II-Q01 (WB p. 559), AIME-2007-II-Q06 (WB p. 671)

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 8 / 方法 8 [NT06-M08]

Prove uniqueness and carries in nonstandard positional systems

证明非标准位值系统的唯一性与进位

Procedure

1. Identify the legal digit range and the carry identity for the system.

2. Reduce modulo the smallest radix to determine the final digit, subtract it and divide to prove uniqueness inductively.

3. Use the carry rule to normalize signed digits or encode a combinatorial state, checking that supports do not overlap unexpectedly.

通用步骤

1. 确定系统的合法数位范围与进位恒等式。

2. 模最低进位基数确定末位，减去后相除，以归纳证明唯一性。

3. 用进位规则规范带符号数位或编码组合状态，并检查数位支撑不会意外重叠。

> Example. Write 8 in balanced ternary, with digits $-1, 0, 1$, and explain why the last digit is forced.

> 例题：把 8 写成数位为 $-1, 0, 1$ 的平衡三进制，并说明末位为何被迫确定。

Worked solution

1. Modulo three, $8 \equiv -1$, so its final balanced digit is -1.

2. Subtracting and dividing gives $(8+1)/3 = 3$. Its last digit is zero, followed by leading one.

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 8 / 方法 8 [NT06-M08] (continued / 续)

3. Thus $8 = 3^2 - 1$, with balanced digits $(1, 0, -1)$. Repeating the forced-residue argument proves uniqueness.

例题解析

1. 模三有 $8 \equiv -1$，故末位必为 -1。

2. 减去末位再相除得 $(8+1)/3 = 3$；它的末位为零，再往上为一。

3. 所以 $8 = 3^2 - 1$，平衡数位为 $(1, 0, -1)$；反复使用被迫剩余的论证可证明唯一性。

Uniqueness comes from legal digit ranges; an arbitrary signed expansion need not be unique.

唯一性来自合法数位范围，任意带符号展开未必唯一。

Factorial digits satisfy $0 \le f_i \le i$ because $(i+1)! = (i+1)i!$. The telescoping identity $a! - b! = \sum_{i=b}^{a-1} i\, i!$ produces a legal block of maximal digits; verify disjoint blocks before adding such expansions.

A set-count or prime-index state may encode each allowed element by a binary bit. Prove the update changes trailing one-bits to zero and the next zero to one; the claimed index then follows by binary uniqueness, not by guessing successive terms.

阶乘数位满足 $0 \le f_i \le i$，因为 $(i+1)! = (i+1)i!$。望远镜恒等式 $a! - b! = \sum_{i=b}^{a-1} i\, i!$ 产生一块合法最大数位；叠加展开之前须验证各块不相交。

集合计数或质数索引状态可用二进制位编码每个允许元素。须证明更新把末尾连续一变零、随后一个零变一；所求下标来自二进制唯一性，而非猜测逐项数值。

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 8 / 方法 8 [NT06-M08] (continued / 续)

Practice / 配套练习: AIME-2000-II-Q14 (WB p. 590), AIME-2008-II-Q04 (WB p. 565), AIME-2014-II-Q15 (WB p. 592), AIME-2024-II-Q06 (WB p. 642)

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 9 / 方法 9 [NT06-M09]

Count digit patterns with leading-zero and multiplicity corrections

按首位零与重复数位修正数位模式计数

Procedure

1. Choose a pattern invariant: digit set, common digit sum, alternating order or palindrome half.

2. Count each pattern before substituting actual digit values, separating whether zero is present.

3. Apply bounded compositions or combinations only after checking that no digit exceeds its base limit; subtract repeated descriptions exactly once.

通用步骤

1. 选择模式不变量：数字集合、共同数位和、交替大小或回文半串。

2. 代入实际数字值前先数模式，并区分是否含零。

3. 使用有界拆分或组合前，先核对数位不超过底数限制；重复描述只消除一次。

> Example. How many two-digit decimal integers have digit sum 9 and distinct digits?

> 例题：有多少个两位十进制整数的数位和为 9 且两位不同?

Worked solution

1. Write $10a+b$ with $1 \le a \le 9$ and $b = 9 - a$.

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 9 / 方法 9 [NT06-M09] (continued / 续)

2. Every such a gives a valid digit b in $0, \ldots, 8$. Equality $a = b$ would require $2a = 9$, impossible.

3. There are nine values. Zero is permitted in the units place, so 90 is included.

例题解析

1. 写为 $10a + b$，其中 $1 \le a \le 9, b = 9 - a$。

2. 每个 a 都给出 $0, \ldots, 8$ 中有效数字 b；若 $a = b$ 则 $2a = 9$，不可能。

3. 共有九个值；个位允许零，故包括 90。

Leading-zero restrictions act on positions; repeated-digit restrictions act on values.

首位零限制针对位置，重复数位限制针对数值。

Practice / 配套练习: AIME-2003-I-Q09 (WB p. 649), AIME-2004-I-Q06 (WB p. 625), AIME-2004-II-Q04 (WB p. 614), AIME-2004-II-Q14 (WB p. 552), AIME-2013-I-Q02 (WB p. 570), AIME-2014-II-Q04 (WB p. 594), AIME-2024-II-Q03 (WB p. 581), AIME-2026-I-Q02 (WB p. 583)

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 10 / 方法 10 [NT06-M10]

Count each digit uniformly by zero-padding a full block

补零成完整区块以均匀统计各数位

Procedure

1. Write all integers from 0 through $b^k - 1$ as strings of exactly k digits.

2. In any fixed position, each digit occurs exactly b^{k-1} times; sum a digit contribution position by position.

3. If leading zeroes have zero contribution they cause no change; otherwise correct them explicitly, and add or remove the interval endpoints.

通用步骤

1. 将 0 到 $b^k - 1$ 的整数全部写成恰有 k 位的数位串。

2. 每个固定位置上，各数字恰出现 b^{k-1} 次；逐位求数字贡献之和。

3. 若首补零贡献为零，则不改变结果；否则须明确修正，并增删区间端点。

> Example. Find the sum of the decimal digit sums of all integers from 0 through 999.

> 例题：求 0 到 999 各整数的十进制数位和之总和。

Worked solution

1. Pad every numeral to three digits; zero padding does not change its digit sum.

2. At each of three positions, every digit appears 100 times and the digits total 45.

3. The total is $3 \cdot 100 \cdot 45 = 13500$.

Section 3.6 Number bases, digits & palindromes

进位制、数位与回文

Method 10 / 方法 10 [NT06-M10] (continued / 续)

例题解析

1. 每个数补为三位，补零不会改变数位和。

2. 三个位置的每一个位置上，每个数字出现 100 次，而所有数字之和为 45。

3. 所以总和为 $3 \cdot 100 \cdot 45 = 13500$。

For reciprocal-digit sums, define the contribution of zero separately rather than divide by zero.

倒数数位和须单独定义零的贡献，不能除以零。

Practice / 配套练习: AIME-2006-II-Q14 (WB p. 481)

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

Method 1 / 方法 1 [NT07-M01]

A fraction problem is usually controlled by reduction, not by decimal approximation. Keep denominators exact, identify the cancellation gcd, and separate distinct values from different representations of the same value.

分数题通常由约分控制，而非小数近似控制。保留精确分母，确定约分公因数，并区分不同数值与同一数值的不同表示。

Convert a repeating block to an exact fraction

将循环节转为精确分数

Procedure

1. If a length-k block represents integer A in base b, then $0.\overline{A}_b = A/(b^k-1)$.

2. Leading zeroes in the block are legal; a shorter actual period may still be represented by the full block.

3. For sums over digit permutations, count how often each digit occupies each position before reducing fractions.

通用步骤

1. 若底数 b 下长度 k 的块表示整数 A，则 $0.\overline{A}_b = A/(b^k-1)$。

2. 循环块开头允许零，实际周期较短者仍可用完整循环块表示。

3. 对数位排列求和时，先统计每个数字在每个位置的出现次数，再约分。

> Example. Find the sum of $0.\overline{12}$, $0.\overline{21}$ and $0.\overline{03}$.

> 例题：求 $0.\overline{12}, 0.\overline{21}, 0.\overline{03}$ 之和。

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

Method 1 / 方法 1 [NT07-M01] (continued / 续)

Worked solution

1. The common denominator for a two-digit block is 99.

2. The numerators are $12, 21, 3$, including the leading-zero block 03.

3. The sum is $36/99 = 4/11$.

例题解析

1. 两位循环块的共同分母为 99。

2. 分子为 $12, 21, 3$，其中包括开头为零的循环块 03。

3. 和为 $36/99 = 4/11$。

The block length is fixed even when its integer value has fewer digits.

即使循环块的整数值位数较少，循环块长度仍固定。

Practice / 配套练习: AIME-2006-I-Q06 (WB p. 596), AIME-2014-II-Q04 (WB p. 594)

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

Method 2 / 方法 2 [NT07-M02]

Control reduction with a gcd and denominator prime structure

用最大公因数与分母质因数结构控制约分

Procedure

1. Write $r = a/b$ in lowest terms and compute explicitly the gcd introduced by a multiplier or common denominator.

2. After multiplying by c, the cancellation is $\gcd(c, b)$ because $\gcd(a, b) = 1$.

3. For an integrality condition, every prime-power in the reduced denominator must be supplied by the numerator; use valuation bounds to prove the first possible index.

4. With a fixed numerator-plus-denominator sum, replace their gcd by the gcd with that fixed sum.

通用步骤

1. 将 $r = a/b$ 写成既约形式，并明确计算乘数或公分母引入的最大公因数。

2. 乘以 c 后，约分公因数为 $\gcd(c, b)$，因为 $\gcd(a, b) = 1$。

3. 整数条件要求分子提供既约分母的每个质数幂，用赋值界证明最早可能下标。

4. 若分子分母之和固定，可将其公因数改写为与固定和的公因数。

> Example. A reduced fraction a/b becomes a fraction with denominator $b/6$ after multiplication by 18. Characterize b.

> 例题：既约分数 a/b 乘以 18 后，最简分母为 $b/6$。刻画 b。

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

Method 2 / 方法 2 [NT07-M02] (continued / 续)

Worked solution

1. The new denominator is $b/\gcd(18,b)$, so the condition is $\gcd(18,b)=6$.

2. Since $18=2\cdot 3^2$, this requires $v_2(b)\ge 1$ and $v_3(b)=1$; other primes are unrestricted.

3. Equivalently $b=6t$ with $3\nmid t$. Any a coprime to such a b realizes the stated cancellation.

例题解析

1. 新分母为 $b/\gcd(18,b)$，所以条件为 $\gcd(18,b)=6$。

2. 因 $18=2\cdot 3^2$，须 $v_2(b)\ge 1, v_3(b)=1$，其他质数不受限制。

3. 等价于 $b=6t$ 且 $3\nmid t$；任取与该 b 互质的 a 均实现所述约分。

Compute the cancellation gcd before comparing reduced numerators and denominators.

比较最简分子分母前，先算约分公因数。

Practice / 配套练习: AIME-2006-II-Q14 (WB p. 481), AIME-2014-I-Q03 (WB p. 603), AIME-2023-II-Q05 (WB p. 595)

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

Method 3 / 方法 3 [NT07-M03]

Characterize the possible reduced numerators using divisor denominators

用约数分母刻画可能的最简分子

Procedure

1. For fractions k/N with $1 \le k \le N$, every reduced denominator is a divisor D of N.

2. A positive numerator m occurs exactly when some such D satisfies $m \le D$ and $\gcd(m, D) = 1$.

3. Partition m by which primes of N divide it. In each class choose the largest divisor of N avoiding those primes, then count under that bound.

通用步骤

1. 对 $1 \le k \le N$ 的分数 k/N，每个最简分母都是 N 的约数 D。

2. 正分子 m 出现，当且仅当某个这样的 D 满足 $m \le D, \gcd(m, D) = 1$。

3. 按 N 的哪些质数整除 m 分类。每类选避开这些质数的最大约数作分母，再在该上界下计数。

> Example. Among $k/12$ for $1 \le k \le 12$, which positive reduced numerators occur?

> 例题：在 $1 \le k \le 12$ 的 $k/12$ 中，会出现哪些正的最简分子?

Worked solution

1. If m is coprime to 6, choose denominator 12; the allowed numerators are $1, 5, 7, 11$.

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

Method 3 / 方法 3 [NT07-M03] (continued / 续)

2. If $2 \mid m$ but $3 \nmid m$, the largest coprime divisor denominator is 3, giving only $m = 2$.

3. If $3 \mid m$ but $2 \nmid m$, denominator 4 permits only $m = 3$. A numerator divisible by both primes is too large for denominator one. Thus the set is $\{1, 2, 3, 5, 7, 11\}$.

例题解析

1. 若 m 与 6 互质，可取分母 12，得到 $1, 5, 7, 11$。

2. 若 $2 \mid m$ 而 $3 \nmid m$，最大互质约数分母为 3，只得到 $m = 2$。

3. 若 $3 \mid m$ 而 $2 \nmid m$，分母 4 只允许 $m = 3$。同时被两质数整除的分子对分母一而言太大。故集合为 $\{1, 2, 3, 5, 7, 11\}$。

Count distinct reduced numerators, not unreduced blocks or fraction representations.

应数不同最简分子，而非未约分循环块或分数表示。

Practice / 配套练习: AIME-2022-I-Q13 (WB p. 598)

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

Method 4 / 方法 4 [NT07-M04]

Use a decimal interval to bound a reduced denominator

用小数区间限制最简分母

Procedure

1. Turn a fixed length-k prefix A into $[A/10^k, (A+1)/10^k)$.

2. If the prefix begins later, shift the decimal and take the fractional part; its reduced denominator divides the original denominator.

3. Compare with a nearby simple fraction and exploit a nonzero integer cross-product difference to get a lower bound.

通用步骤

1. 将固定长度 k 的前缀 A 转为 $[A/10^k, (A+1)/10^k)$。

2. 若前缀在较后位置开始，移动小数点并取小数部分；其最简分母整除原分母。

3. 与附近简单分数比较，并利用非零整数交叉乘积差得到下界。

> Example. Prove no reduced fraction with denominator at most 10 lies in $[0.501, 0.510)$.

> 例题：证明分母不超过 10 的既约分数不可能位于 $[0.501, 0.510)$。

Worked solution

1. Such a fraction m/n is strictly above $1/2$, so $2m - n \geq 1$.

2. Its distance above $1/2$ is at least $1/(2n) \geq 1/20 = 0.05$.

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

Method 4 / 方法 4 [NT07-M04] (continued / 续)

3. But the stated interval requires that distance be below 0.010. This contradiction excludes every such denominator.

例题解析

1. 这样的分数 m/n 严格大于 $1/2$，所以 $2m - n \geq 1$。

2. 它与 $1/2$ 的距离至少为 $1/(2n) \geq 1/20 = 0.05$。

3. 而给定区间要求该距离小于 0.010，矛盾，故排除所有这些分母。

> The strict upper endpoint is essential when the next decimal prefix begins there.

> 右端点正是下一小数前缀开始处，严格性不可忽略。

Practice / 配套练习: AIME-2003-I-Q14 (WB p. 597)

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

Method 5 / 方法 5 [NT07-M05]

Locate the unique least valuation in a rational sum

定位有理和中唯一最低赋值项

Procedure

1. Reduce each summand and compare its exponent of a chosen prime in the denominator.

2. Clear the largest denominator power. If exactly one resulting numerator is a unit modulo the prime, the total cannot cancel that prime.

3. Compute the exact surviving denominator exponent; use a factorial valuation formula when needed.

通用步骤

1. 约分每项，并比较所选质数在分母中的指数。

2. 清除最大分母幂。若所得分子中恰有一项模该质数为单位，则总和不能约去该质数。

3. 算出保留的精确分母指数，必要时使用阶乘赋值公式。

> Example. Find the power of 2 in the reduced denominator of $1/2+3/8+5/32$.

> 例题：求 $1/2+3/8+5/32$ 的最简分母中 2 的指数。

Worked solution

1. The common denominator is 32, giving numerator $16+12+5$.

2. Only the last contribution is odd, so the total numerator 33 is odd.

3. No two-factor cancellation occurs: the reduced denominator remains $32=2^5$.

Section 3.7 Rational fractions & repeating decimals

有理分数与循环小数

Method 5 / 方法 5 [NT07-M05] (continued / 续)

例题解析

1. 共同分母为 32，分子为 $16+12+5$。

2. 只有最后一项贡献为奇数，故总分子 33 为奇数。

3. 不能约去任何因子二，所以最简分母仍为 $32=2^5$。

If two terms share the lowest valuation, cancellation may occur and needs another analysis.

若两项共享最低赋值，则可能相消，须另作分析。

Practice / 配套练习: AIME-2009-II-Q07 (WB p. 475)

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

Method 1 / 方法 1 [NT08-M01]

Divisor functions are products of local exponent choices. Learn the formulas together with their inverse patterns, parity exceptions, coprimality restrictions and exact tail bounds; these are what make the formulas useful in contest questions.

约数函数是局部指数选择的乘积。公式须与反向指数模式、奇偶例外、互质限制及精确尾项界一起掌握，才能在竞赛题中真正使用。

Count and sum divisors by independent prime exponents

以独立质指数统计约数及约数和

Procedure

1. Factor $N = \prod p^{e_p}$ and list each exponent choice $0, 1, \ldots, e_p$.

2. Use $\tau(N) = \prod(e_p + 1)$ and $\sigma(N) = \prod(1 + p + \cdots + p^{e_p})$.

3. If only certain exponents are allowed, replace each local factor by that permitted count or sum.

4. Only after the exact divisor count is known should it be inserted into a probability or a later equation.

通用步骤

1. 分解 $N = \prod p^{e_p}$，每个质数指数可选 $0, 1, \ldots, e_p$。

2. 使用 $\tau(N) = \prod(e_p + 1)$ 与 $\sigma(N) = \prod(1 + p + \cdots + p^{e_p})$。

3. 若只允许某些指数，将每个局部因子替换为允许指数的个数或幂和。

4. 明确精确约数数目之后，才代入概率或后续方程。

> Example. For $N = 2^4 3^2$, count its square divisors and find their sum.

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

Method 1 / 方法 1 [NT08-M01] (continued / 续)

> 例题：对 $N = 2^4 3^2$，求平方约数的个数及其总和。

Worked solution

1. Square exponents are $0, 2, 4$ for two and $0, 2$ for three, so there are $3 \cdot 2 = 6$ square divisors.

2. The sum factors as $(1 + 2^2 + 2^4)(1 + 3^2)$.

3. The value is $21 \cdot 10 = 210$.

例题解析

1. 二的平方指数为 $0, 2, 4$，三的为 $0, 2$，所以共有 $3 \cdot 2 = 6$ 个平方约数。

2. 其和分解为 $(1 + 2^2 + 2^4)(1 + 3^2)$。

3. 结果为 $21 \cdot 10 = 210$。

Restrictions belong on exponents; listing divisors is usually unnecessary.

限制应施加于指数，通常无需逐个列约数。

Practice / 配套练习: AIME-2005-I-Q02 (WB p. 600), AIME-2010-I-Q01 (WB p. 723), AIME-2011-I-Q07 (WB p. 498), AIME-2019-II-Q03 (WB p. 487), AIME-2023-I-Q04 (WB p. 473), AIME-2023-II-Q10 (WB p. 716), AIME-2026-I-Q06 (WB p. 124), AIME-2026-I-Q08 (WB p. 507)

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

Method 2 / 方法 2 [NT08-M02]

Invert a divisor count into prime-exponent patterns

将约数个数反解为质指数模式

Procedure

1. Write the required count as a product of factors $e_i + 1 > 1$.

2. List every multiplicative partition, then impose prime restrictions and numerical bounds.

3. To minimize an integer with fixed exponents, assign larger exponents to smaller primes; swapping proves this rule.

4. For constrained divisors of a large power, check that every recovered exponent lies in the original range.

通用步骤

1. 将指定约数个数写成若干 $e_i + 1 > 1$ 的乘积。

2. 列出全部乘法拆分，再施加质数限制与数值界。

3. 固定指数求最小整数时，将较大指数配较小质数；交换可证明此规则。

4. 对大幂的受限约数，核对每个还原指数均在原范围内。

> Example. Find the least positive integer with exactly eight positive divisors.

> 例题：求恰有八个正约数的最小正整数。

Worked solution

1. The exponent patterns are 7, $(3, 1)$ and $(1, 1, 1)$, from 8, $4 \cdot 2$ and $2 \cdot 2 \cdot 2$.

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

Method 2 / 方法 2 [NT08-M02] (continued / 续)

2. Their least values are $2^7 = 128$, $2^3 \cdot 3 = 24$ and $2 \cdot 3 \cdot 5 = 30$.

3. The minimum is 24, with divisor count $(3+1)(1+1) = 8$.

例题解析

1. 由 $8, 4 \cdot 2, 2 \cdot 2 \cdot 2$ 得指数模式 $7, (3,1), (1,1,1)$。

2. 各模式最小值为 $2^7 = 128, 2^3 \cdot 3 = 24, 2 \cdot 3 \cdot 5 = 30$。

3. 最小值为 24，其约数数为 $(3+1)(1+1) = 8$。

Include repeated exponents and the case of a single prime power.

须包含重复指数与只有一个质数幂的情形。

For consecutive integers with a small prescribed sum of divisor counts, classify the possible count pairs first. A count of three forces a prime square, five forces a prime fourth power, and four forces p^3 or pq with distinct primes p, q. This turns a long integer search into a bounded search around prime powers.

相邻整数的约数数目之和为小定值时，先分类可能的计数对。约数数为三迫使质数平方，为五迫使质数四次幂，为四迫使 p^3 或不同质数 p, q 的乘积 pq。这能把很长的整数搜索化为质数幂附近的有界搜索。

Practice / 配套练习: AIME-2000-II-Q04 (WB p. 599), AIME-2004-II-Q08 (WB p. 605), AIME-2005-I-Q03 (WB p. 601), AIME-2006-II-Q13 (WB p. 610), AIME-2019-I-Q09 (WB p. 606), AIME-2019-II-Q09 (WB p. 607)

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

Method 3 / 方法 3 [NT08-M03]

Characterize divisor counts of kth powers

刻画 k 次幂的约数个数

Procedure

1. If $a = \prod p_i^{e_i}$, then $\tau(a^k) = \prod(ke_i + 1)$.

2. Every factor is one modulo k, giving a necessary congruence on the divisor count.

3. To prove sufficiency, realize a count $N = ke + 1$ using a single prime power $a = p^e$, including $e = 0$.

通用步骤

1. 若 $a = \prod p_i^{e_i}$，则 $\tau(a^k) = \prod(ke_i + 1)$。

2. 每个因子都模 k 余一，从而得到约数个数的必要同余。

3. 证明充分性时，用单一质数幂 $a = p^e$ 实现个数 $N = ke + 1$，并包括 $e = 0$。

> Example. Which positive integers can be the number of divisors of an integer fourth power?

> 例题：哪些正整数可以是某整数四次幂的正约数个数?

Worked solution

1. Every such count is $\prod(4e_i + 1) \equiv 1 \pmod 4$.

2. Conversely, for $N = 4e + 1$, the fourth power of 2^e has exactly $4e + 1 = N$ divisors.

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

Method 3 / 方法 3 [NT08-M03] (continued / 续)

3. Thus exactly the positive integers congruent to one modulo four occur, including one from the integer one.

例题解析

1. 每个这样的个数均为 $\prod(4e_i+1)\equiv 1 \pmod 4$。

2. 反之，对 $N=4e+1$，2^e 的四次幂恰有 $4e+1=N$ 个约数。

3. 所以恰为模四余一的正整数，包括整数一产生的个数一。

> Necessity from a product congruence needs a separate realization for sufficiency.

> 乘积同余给出的必要性，须另给实现才能证明充分性。

Practice / 配套练习: AIME-2016-II-Q11 (WB p. 611)

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

Method 4 / 方法 4 [NT08-M04]

Count coprime classes with the totient and a symmetry pairing

用欧拉函数与对称配对统计互质类

Procedure

1. For $N = \prod p^e$, inclusion-exclusion gives $\varphi(N) = N \prod_{p \mid N} (1 - 1/p)$.

2. When an interval contains full residue periods, count those periods and then the boundary remainder.

3. Before halving a count by $a \leftrightarrow N - a$, prove the map has no fixed admissible element.

4. If some primes may divide the numerator, partition by those primes and apply the largest permitted coprime denominator bound.

通用步骤

1. 对 $N = \prod p^e$，容斥给出 $\varphi(N) = N \prod_{p \mid N} (1 - 1/p)$。

2. 区间含完整剩余周期时，先数完整周期，再处理边界余段。

3. 用 $a \leftrightarrow N - a$ 将个数减半前，证明没有合格不动点。

4. 若某些质数允许整除分子，按这些质数分类，并施加最大允许互质分母界。

> Example. Choose a residue uniformly from $0, \ldots, 29$. Given that it is odd, what is the probability it has a multiplicative inverse modulo 30?

> 例题：从 $0, \ldots, 29$ 等可能选一个剩余。已知它是奇数，其模 30 可逆的概率是多少？

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

Method 4 / 方法 4 [NT08-M04] (continued / 续)

Worked solution

1. Invertibility means coprimality with $30 = 2 \cdot 3 \cdot 5$.

2. There are $\varphi(30) = 30(1 - 1/2)(1 - 1/3)(1 - 1/5) = 8$ invertible residues, all odd.

3. The conditioning leaves fifteen odd residues, so the probability is $8/15$.

例题解析

1. 可逆等价于与 $30 = 2 \cdot 3 \cdot 5$ 互质。

2. 可逆剩余有 $\varphi(30) = 30(1 - 1/2)(1 - 1/3)(1 - 1/5) = 8$ 个，且全为奇数。

3. 条件下共有十五个奇数剩余，故概率为 $8/15$。

Totient counts all coprime residues, not automatically only half of an interval.

欧拉函数统计全部互质剩余，并非自动给出某区间的一半。

Practice / 配套练习: AIME-2004-I-Q08 (WB p. 604), AIME-2014-I-Q03 (WB p. 603), AIME-2022-I-Q13 (WB p. 598)

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

Method 5 / 方法 5 [NT08-M05]

Use divisor pairing to track parity through square blocks

用约数配对按平方区块追踪奇偶

Procedure

1. Pair every divisor d with n/d; an unpaired divisor exists exactly when n is square.

2. Therefore $\tau(n)$ is odd exactly at squares, and a cumulative divisor-count sum changes parity only there.

3. Count entire intervals between consecutive squares and then the final partial interval.

通用步骤

1. 把每个约数 d 与 n/d 配对；仅平方数有未配对约数。

2. 因此 $\tau(n)$ 恰在平方数时为奇数，累积约数计数和也仅在此改变奇偶。

3. 统计连续平方数之间的完整区间，再处理最后不完整区间。

> Example. Among $1 \le n \le 20$, for how many n is $\sum_{j=1}^{n} \tau(j)$ odd?

> 例题：在 $1 \le n \le 20$ 中，有多少个 n 使 $\sum_{j=1}^{n} \tau(j)$ 为奇数?

Worked solution

1. The parity is the parity of the number of squares at most n, namely $\lfloor \sqrt{n} \rfloor$.

2. Odd square-root floors occur on $1 \le n \le 3$ and $9 \le n \le 15$.

3. These contain $3 + 7 = 10$ integers.

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

Method 5 / 方法 5 [NT08-M05] (continued / 续)

例题解析

1. 其奇偶等于不超过 n 的平方数数目，即 $\lfloor\sqrt{n}\rfloor$ 的奇偶。

2. 平方根下取整为奇数的区间是 $1 \le n \le 3$ 与 $9 \le n \le 15$。

3. 共含 $3+7=10$ 个整数。

A square switches the cumulative parity at the square itself.

累积奇偶从平方数本身起切换。

Practice / 配套练习: AIME-2005-I-Q12 (WB p. 609)

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

Method 6 / 方法 6 [NT08-M06]

Intersect divisor sets with gcds and use multiplicativity carefully

以最大公因数求约数集交，并谨慎使用积性

Procedure

1. Divisors common to several integers are exactly divisors of their gcd.

2. Count unions by inclusion-exclusion, including the divisor one in every intersection.

3. For σ or τ of a product, multiply values only when the factors are coprime, or merge their prime exponents first.

通用步骤

1. 若干整数的公共约数恰为其最大公因数的约数。

2. 用容斥统计并集，并在每个交集中包括约数一。

3. 乘积的 σ 或 τ 只有在因子互质时才可相乘，否则须先合并质指数。

> Example. How many positive integers divide 18 or 24?

> 例题：有多少正整数整除 18 或 24?

Worked solution

1. $\tau(18) = (1+1)(2+1) = 6$ and $\tau(24) = (3+1)(1+1) = 8$.

2. The intersection is the divisor set of $\gcd(18, 24) = 6$, containing $\tau(6) = 4$ values.

3. The union count is $6 + 8 - 4 = 10$.

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

Method 6 / 方法 6 [NT08-M06] (continued / 续)

例题解析

1. $\tau(18)=(1+1)(2+1)=6, \tau(24)=(3+1)(1+1)=8$。

2. 交集为 $\gcd(18,24)=6$ 的约数集，有 $\tau(6)=4$ 个值。

3. 并集个数为 $6+8-4=10$。

The functions are multiplicative on coprime inputs, not on every product.

这些函数只对互质输入具有积性，并非对任意乘积都可相乘。

Practice / 配套练习: AIME-2005-II-Q04 (WB p. 602), AIME-2021-I-Q14 (WB p. 612)

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

Method 7 / 方法 7 [NT08-M07]

Regroup a positive series into divisor coefficients and bound its tail

把正项级数重组为约数系数，并估计尾项

Procedure

1. Expand each geometric denominator into a nonnegative series. Nonnegativity justifies regrouping by exponent.

2. An exponent n receives one contribution for each factorization $n = jk$, giving coefficient $\tau(n)$.

3. After scaling by a base power, separate the integral prefix from the tail and prove the tail is below the next integer before reading digits.

4. Use an elementary bound such as $\tau(n) \le n$ for distant terms, while calculating the first few tail coefficients exactly.

通用步骤

1. 把每个等比分母展开为非负级数；非负性允许按指数重组。

2. 指数 n 对每种 $n = jk$ 的分解得到一次贡献，因此系数为 $\tau(n)$。

3. 乘以底数幂后，将整数前缀与尾项分开；读取数位之前先证明尾项未达到下一整数。

4. 对远处项使用 $\tau(n) \le n$ 等简单界，对尾项开头若干系数则精确计算。

> Example. For $T = \sum_{k \ge 1} 1/(100^k - 1)$, identify the coefficient of 100^{-6} and prove $T < 1$.

> 例题：对 $T = \sum_{k \ge 1} 1/(100^k - 1)$，求 100^{-6} 的系数，并证明 $T < 1$。

Section 3.8 Divisor functions & totients

约数函数与欧拉函数

Method 7 / 方法 7 [NT08-M07] (continued / 续)

Worked solution

1. Expand $1/(100^k-1)=\sum_{j\ge1}100^{-kj}$. Exponent six arises from $k=1,2,3,6$, so its coefficient is four.

2. Thus $T=\sum_{n\ge1}\tau(n)100^{-n}\le\sum_{n\ge1}n100^{-n}$.

3. The last sum equals $(1/100)/(1-1/100)^2=100/9801<1$.

例题解析

1. 展开 $1/(100^k-1)=\sum_{j\ge1}100^{-kj}$。指数六来自 $k=1,2,3,6$，故系数为四。

2. 所以 $T=\sum_{n\ge1}\tau(n)100^{-n}\le\sum_{n\ge1}n100^{-n}$。

3. 最后一和为 $(1/100)/(1-1/100)^2=100/9801<1$。

Coefficients exceeding the base may carry; a series coefficient is not automatically a digit.

系数超过底数时会进位，级数系数并不自动等于数位。

Practice / 配套练习: AIME-2026-II-Q09 (WB p. 608)

Section 4.1 Restricted permutations & arrangements

受限排列

Method 1 / 方法 1 [CO01-M01]

Separate the choice of a type pattern from the assignment of distinct labels. Product rules need a fixed number of legal continuations; quotient rules need a proof of equal multiplicity. Restrictions often become simpler after choosing an extreme element, a run, or a cycle decomposition.

先分离类型模式与不同标签的安排。乘法原理须保证合法后继数确定，归并除法须证明每个对象被数同样多次。固定极值元素、连续段或循环分解后，限制常会简化。

Multiply sequential choices after identifying the state

明确状态后逐步相乘

Procedure

1. State what has already been chosen and what remains available.

2. Count legal choices at each step, checking whether the count depends on the earlier history.

3. Multiply only when these continuation counts are valid for every allowed history; sum disjoint process lengths or states separately.

通用步骤

1. 写清已选内容与尚可使用的内容。

2. 逐步计算合法选择数，并检验是否依赖此前具体历史。

3. 只有所有合法历史均具有所用后继数时才相乘；过程长度或状态不同则分类相加。

> Example. Choose an ordered three-station tour from five distinct stations without repeats. At each visited station choose one of two activities. How many tours with activities are there?

Section 4.1 Restricted permutations & arrangements

受限排列

Method 1 / 方法 1 [CO01-M01] (continued / 续)

> 例题：从五个不同站点选不重复的三站顺序游览，每站选两种活动之一，共有多少个带活动安排的行程?

Worked solution

1. Station choices are $5, 4, 3$, independent of which earlier stations were used.

2. Every station choice has 2^3 activity assignments.

3. The total is $5 \cdot 4 \cdot 3 \cdot 2^3 = 480$; the activity choice does not remove stations.

例题解析

1. 站点选择依次为五、四、三，与具体已选站点无关。

2. 每个站点顺序有 2^3 种活动安排。

3. 总数为 $5 \cdot 4 \cdot 3 \cdot 2^3 = 480$，活动选择不减少后续站点。

A condition with a uniform predecessor count may be easier to build right-to-left than left-to-right.

若前驱数恒定，从右向左构造可能比从左向右简单。

If a two-symbol local rule forbids only UD, every legal word is D^aU^b, so its length-n patterns number $n+1$. This structural split replaces an unnecessary long recurrence. In a substitution process, the number currently on the field may stay fixed while the unused incoming pool shrinks; distinguish these two counts.

若双符号局部规则只禁 UD，每个合法串均为 D^aU^b，长度 n 的模式数为 $n+1$，无须建立长递推。换人过程可能保持场上人数固定而未使用替补池缩小，须区分这两个选择数。

Section 4.1 Restricted permutations & arrangements

受限排列

Method 1 / 方法 1 [CO01-M01] (continued / 续)

Practice / 配套练习: AIME-2003-II-Q03 (WB p. 613), AIME-2005-I-Q05 (WB p. 615), AIME-2016-I-Q03 (WB p. 694), AIME-2019-I-Q04 (WB p. 620)

Section 4.1 Restricted permutations & arrangements

受限排列

Method 2 / 方法 2 [CO01-M02]

Choose positions when the internal order is forced

内部顺序被迫时只选位置

Procedure

1. Identify the element forced to be an extreme, such as the minimum at a valley.

2. Choose which labels go on each monotone side or type class.

3. Place each class in its forced order; for alternating inequalities, count admissible rank patterns before assigning values.

通用步骤

1. 找出被迫为极值的元素，例如谷底的最小值。

2. 选择各单调侧或类型类所用的标签。

3. 按被迫顺序放置各类；若为交错不等式，先数合法排名模式再赋实际值。

> Example. Merge three alphabetically ordered mathematics titles and three alphabetically ordered art titles into one shelf, preserving each list order. Count the shelves.

> 例题：将各三本已按字母顺序排列的数学书、艺术书合并成一排，分别保持原列表顺序，求排法数。

Worked solution

1. Choose the three of six shelf positions occupied by mathematics titles.

Section 4.1 Restricted permutations & arrangements

受限排列

Method 2 / 方法 2 [CO01-M02] (continued / 续)

2. The mathematics list must fill those slots in its given order; the art list then fills the other slots in its given order.

3. This is a bijection with three-subsets of six positions, giving $\binom{6}{3} = 20$.

例题解析

1. 选择六个位置中哪三个放数学书。

2. 数学书按原顺序填这些位置，艺术书按原顺序填其余位置，均无额外选择。

3. 所以与六位置的三元子集双射，共 $\binom{6}{3} = 20$ 种。

A forced monotone order contributes one arrangement, not a factorial.

被迫单调的顺序只贡献一种，不能再乘阶乘。

Practice / 配套练习: AIME-2004-I-Q06 (WB p. 625), AIME-2006-II-Q04 (WB p. 616), AIME-2017-I-Q11 (WB p. 796)

Section 4.1 Restricted permutations & arrangements

受限排列

Method 3 / 方法 3 [CO01-M03]

Count multiplicity patterns with multinomial weights

用多项式系数加权重数模式

Procedure

1. Solve the total-count and weighted-total equations for all possible multiplicity vectors.

2. For a vector $(n_1, \ldots, n_r)$, count its words by $N!/\prod n_i!$.

3. If only some symbols repeat, split by their used multiplicities before choosing and ordering the other symbols.

4. For unlabeled groups or unordered entries, remove exceptional repeated cases before dividing by their permutation multiplicity.

通用步骤

1. 解总数量与加权总量方程，列全可行重数向量。

2. 重数为 $(n_1, \ldots, n_r)$ 的串有 $N!/\prod n_i!$ 个。

3. 若只有部分符号可重复，先按实际使用重数分类，再选排其余符号。

4. 无标识组或无序项应先处理重复例外，再按排列重数归并。

> Example. Six positions contain three red, two blue, and one green marker, identical within a color. How many words have green before every blue?

> 例题：六个位置放三红、二蓝、一绿同色不可区分标记，要求绿标记在所有蓝标记之前，求串数。

Section 4.1 Restricted permutations & arrangements

受限排列

Method 3 / 方法 3 [CO01-M03] (continued / 续)

Worked solution

1. First choose the three positions occupied by nonred markers in $\binom{6}{3}$ ways.

2. In those positions the order must be green, blue, blue; the three red positions are also forced.

3. The answer is 20. Equivalently, among the $6!/(3!2!) = 60$ unrestricted words, green has three equiprobable ranks among the nonred markers.

例题解析

1. 先选三个非红位置，有 $\binom{6}{3}$ 种。

2. 这些位置只能依次为绿、蓝、蓝，红位置也全部确定。

3. 故答案为二十；亦可从无限制的 $6!/(3!2!) = 60$ 串中按绿色在非红中的三个等可能排名取一。

The same symbol-count vector may carry a different probability if individual outcome probabilities differ.

若各类单次概率不同，相同重数向量的概率还须乘相应幂权重。

Practice / 配套练习: AIME-2001-II-Q11 (WB p. 789), AIME-2004-II-Q04 (WB p. 614), AIME-2007-II-Q01 (WB p. 617), AIME-2008-I-Q09 (WB p. 786), AIME-2009-I-Q09 (WB p. 650), AIME-2016-II-Q08 (WB p. 476), AIME-2018-II-Q08 (WB p. 674), AIME-2018-II-Q15 (WB p. 655), AIME-2019-I-Q05 (WB p. 730), AIME-2021-I-Q04 (WB p. 648), AIME-2025-I-Q03 (WB p. 624)

Section 4.1 Restricted permutations & arrangements

受限排列

Method 4 / 方法 4 [CO01-M04]

Glue runs, place gaps, then fill labels

先粘连续段、放间隔，再填标签

Procedure

1. Turn each required consecutive group into a block, retaining its possible internal orders.

2. Place blocks in a line, a circle, or gaps of the other objects; check the endpoints and empty-gap rules.

3. For forbidden adjacency, count selected forced-adjacency blocks inside inclusion-exclusion.

4. Only after the color or group pattern is fixed, assign distinct members within each group.

通用步骤

1. 将须连续的每组粘成区块，并保留内部可排顺序。

2. 在线、圆或其他物体间隔中放区块，核对两端与空间隔规则。

3. 禁相邻时，在容斥内部计算所选强制相邻边产生的区块。

4. 先固定颜色或群体模式，再把不同成员填入各组位置。

> Example. Six distinct people A, B, C, D, E, F stand in a row. The first three named people must be consecutive, while D, E must be separated. Count the rows.

> 例题：六个不同的人 A, B, C, D, E, F 排成一行，A, B, C 必须连续，D, E 不得相邻，求排法数。

Section 4.1 Restricted permutations & arrangements

受限排列

Method 4 / 方法 4 [CO01-M04] (continued / 续)

Worked solution

1. Glue A, B, C into one internally ordered block. With D, E, F, there are $4! \cdot 3! = 144$ rows.

2. Among these, glue D, E too: three outer blocks have $3!$ orders, the first block has $3!$ internal orders, and D, E have two orientations, giving 72.

3. Subtract to obtain $144 - 72 = 72$. The two blocks use disjoint people, so their counts combine directly.

例题解析

1. 把 A, B, C 粘成可内部排列的区块，与 D, E, F 共四个单位，有 $4! \cdot 3! = 144$ 种。

2. 其中再把 D, E 粘成块，外部三块有 $3!$ 顺序，第一块内部 $3!$，二人方向两种，共七十二种。

3. 相减得七十二种，两块人员不交，因此其选择可相乘。

A circular block needs a specified anchor unless the rotation action has already been accounted for.

圆形区块须有指定锚点，除非已明确处理旋转归并。

Practice / 配套练习: AIME-2000-II-Q05 (WB p. 647), AIME-2009-I-Q10 (WB p. 651), AIME-2011-I-Q12 (WB p. 765), AIME-2011-II-Q07 (WB p. 652), AIME-2011-II-Q12 (WB p. 745), AIME-2014-II-Q09 (WB p. 663), AIME-2018-I-Q03 (WB p. 728), AIME-2019-II-Q05 (WB p. 621), AIME-2020-II-Q09 (WB p. 665), AIME-2025-II-Q10 (WB p. 653)

Section 4.1 Restricted permutations & arrangements

受限排列

Method 5 / 方法 5 [CO01-M05]

Decompose permutations into cycles and track symmetry

将置换分解为循环并追踪对称

Procedure

1. Translate an iterate, invariant-subset, or degree-two graph condition into allowed cycle lengths.

2. List the cycle-length partitions of the total size.

3. For a permutation with m_j cycles of length j, use $N!/\prod_j (j^{m_j} m_j!)$; undirected cycles have an extra reversal identification.

4. Divide a labeled circular arrangement by its rotation count only after proving there are no nontrivial rotational stabilizers.

通用步骤

1. 将迭代、不变子集或二度图条件转成允许循环长度。

2. 列出总大小的循环长度分拆。

3. 长度 j 循环有 m_j 个的置换数为 $N!/\prod_j (j^{m_j} m_j!)$；无向圈还须处理反向归并。

4. 有标识圆排列按旋转归并前，先证明没有非平凡旋转稳定子。

> Example. Count permutations of eight labels having two cycles of length three and one cycle of length two.

> 例题：求八个标签的置换中，恰有两个三循环及一个二循环的个数。

Section 4.1 Restricted permutations & arrangements

受限排列

Method 5 / 方法 5 [CO01-M05] (continued / 续)

Worked solution

1. Order all eight labels, giving $8!$ descriptions.

2. Each three-cycle has three choices of starting point, the two-cycle has two, and the two three-cycles can be interchanged in $2!$ ways.

3. The count is $8!/(3^2 \cdot 2 \cdot 2!) = 1120$. Cycles are directed permutations, so do not divide for reversal.

例题解析

1. 先排列八个标签，得到 $8!$ 个描述。

2. 每个三循环起点有三种，二循环有两种，两个三循环还可用 $2!$ 种方式交换。

3. 所以计数为 $8!/(3^2 \cdot 2 \cdot 2!) = 1120$。置换循环有方向，不能按反向再除二。

A map on a finite set that is onto is a permutation; an iterate equal to identity restricts every cycle length.

有限集到自身的满射是置换，某次迭代为恒等会限制每个循环的长度。

Practice / 配套练习: AIME-2011-I-Q05 (WB p. 618), AIME-2012-II-Q14 (WB p. 701), AIME-2014-II-Q13 (WB p. 747), AIME-2026-I-Q07 (WB p. 628)

Section 4.1 Restricted permutations & arrangements

受限排列

Method 6 / 方法 6 [CO01-M06]

Count category patterns before permuting within categories

先计类别模式，再排类内标签

Procedure

1. Replace actual labels by residues, low/high classes, parity classes, or repeated-position classes.

2. Derive the allowed category patterns from the original local or global conditions.

3. For each pattern multiply the independent within-class label permutations; check whether all patterns have the same weight.

通用步骤

1. 将实际标签替换为剩余类、高低类、奇偶类或重复位置类。

2. 由原局部整体条件推导允许的类别模式。

3. 对每个模式乘独立类内标签排列数，并检验各模式是否同权。

> Example. Use digits $0, 1, 2, 3, 4, 5$ once in six labeled slots. Slots one, three, five must contain even digits, and slot one cannot be zero. Count the assignments.

> 例题：将数位 $0, 1, 2, 3, 4, 5$ 各用一次填六个有标识位置，第一三五位须为偶数且首位不能为零，求填法数。

Worked solution

1. The category pattern is fixed: even, odd, even, odd, even, odd.

Section 4.1 Restricted permutations & arrangements

受限排列

Method 6 / 方法 6 [CO01-M06] (continued / 续)

2. The first even slot has two nonzero choices; the other two even digits can be permuted in $2!$ ways. The odd digits have $3!$ assignments.

3. The count is $2 \cdot 2! \cdot 3! = 24$. Zero is handled inside the even category.

例题解析

1. 类别模式已固定为偶、奇、偶、奇、偶、奇。

2. 第一偶数位有两个非零选择，其余偶数以 $2!$ 种排列；奇数以 $3!$ 种排列。

3. 共有 $2 \cdot 2! \cdot 3! = 24$ 种，零的限制在偶数类内部处理。

If two histories yield the same repeated-position pattern, count the intrinsic pattern once rather than choosing a repeated pair twice.

若两段构造给出同一重复位置模式，应按内在模式计一次，不能重复选择其见证。

For a regular even polygon, an inscribed rectangle is determined by two diameters. Group opposite vertex pairs by their color type; forbid two solid pairs of the same color rather than testing all rectangles separately. In a residue-constrained array, count residue or category fillings first and multiply the factorials of the actual labels in each class.

正偶数边形的内接矩形由两条直径确定。按对顶点对的颜色类型分类，禁止出现两对同色纯色对，比逐个矩形检查更简。剩余类限制点阵先计类别填法，再乘各类实际标签的阶乘。

Practice / 配套练习: AIME-2006-I-Q06 (WB p. 596), AIME-2011-I-Q05 (WB p. 618), AIME-2011-II-Q14 (WB p. 509), AIME-2012-I-Q15 (WB p. 510), AIME-2016-I-Q08 (WB p. 589), AIME-2017-I-Q11 (WB p. 796), AIME-2021-II-Q03 (WB p. 623), AIME-2022-I-Q09 (WB p. 742), AIME-2023-II-Q07 (WB p. 626), AIME-2025-I-Q05 (WB p. 495), AIME-2025-I-Q10 (WB p. 627), AIME-2026-I-Q09 (WB p. 763)

Section 4.1 Restricted permutations & arrangements

受限排列

Method 7 / 方法 7 [CO01-M07]

Count constrained injections with nested choices or matchings

计算嵌套选项与匹配限制的单射

Procedure

1. Model an assignment as an injection or a perfect matching between two labeled sets.

2. When allowed sets are nested, process the smallest allowed set first and subtract the previously used targets.

3. When one intersection or exceptional match controls feasibility, condition on whether it occurs and prove its representation is unique.

通用步骤

1. 把分配建模为两个有标识集合之间的单射或完美匹配。

2. 允许集合嵌套时，先处理最小集合，依次扣除已用目标。

3. 若一个交点或例外匹配决定可行性，按其是否出现分类，并证明表示唯一。

> Example. Three labeled tasks need distinct machines. Their allowed machine sets are $\{1,2\},\{1,2,3\},\{1,2,3,4,5\}$. Count assignments.

> 例题：三个有标识任务须用不同机器，允许机器集依次为 $\{1,2\},\{1,2,3\},\{1,2,3,4,5\}$，求分配数。

Worked solution

1. Assign the first task in two ways.

Section 4.1 Restricted permutations & arrangements

受限排列

Method 7 / 方法 7 [CO01-M07] (continued / 续)

2. Its machine lies in the second allowed set, so the second task has $3-1=2$ choices. Both used machines lie in the third allowed set, leaving $5-2=3$.

3. The count is $2\cdot 2\cdot 3=12$; nesting makes the continuation counts independent of the chosen machines.

例题解析

1. 第一任务有两种选择。

2. 其机器必在第二允许集中，所以第二任务有 $3-1=2$ 种；已用两机器都在第三集中，故第三有 $5-2=3$ 种。

3. 共有 $2\cdot 2\cdot 3=12$ 种，嵌套性保证后继数与具体选择无关。

A matching counted through a removable object is counted once only if that removable object is unique.

通过可删除物体计算匹配时，只有该物体唯一，表示才不会重复。

Unrestricted perfect matchings on $2n$ distinct items number $(2n-1)!!$: pair a fixed item first, then recurse. For a left-right matching, use $n!$. In a hand covering all rows and columns of a card grid with exactly one repeated row and column, removability is equivalent to presence of their intersection card; establish that it is uniquely removable before counting through the remaining matching.

二 n 个不同物体的无向完美匹配有 $(2n-1)!!$ 个：先为固定物体选搭档，再递推。左右两组之间的匹配则用 $n!$。若牌组覆盖所有行列，且恰一行与一列重复，则可删除性等价于它们交点牌出现；须证明该牌唯一可删，才能通过其余匹配计数。

Practice / 配套练习: AIME-2012-I-Q03 (WB p. 619), AIME-2016-II-Q13 (WB p. 779), AIME-2017-II-Q09 (WB p. 762), AIME-2025-I-Q07 (WB p. 743)

Section 4.1 Restricted permutations & arrangements

受限排列

Method 8 / 方法 8 [CO01-M08]

Insert an extreme element and prove deletion is the inverse

插入极值元素并证明删除为逆过程

Procedure

1. Remove the largest or smallest label and prove the remaining object stays legal.

2. Classify every slot in which reinserting that label preserves the condition.

3. If every smaller object has a fixed number of slots, form a recurrence; if multiple deletions witness the same object, correct their multiplicities.

通用步骤

1. 删除最大或最小标签，并证明剩余对象仍合法。

2. 分类重新插入时所有保持条件的位置。

3. 若每个较小对象具有相同位置数，就建递推；若同一对象有多个可删见证，则修正重数。

> Example. Count permutations of $1, \ldots, 5$ for which, for each k, the entries $1, \ldots, k$ occupy consecutive positions.

> 例题：求 $1, \ldots, 5$ 的排列数，使对每个 k，标签一至 k 都占据连续位置。

Worked solution

1. Begin with the one-entry row containing one. To preserve the condition, each new largest label must be attached to the left or right end.

Section 4.1 Restricted permutations & arrangements

受限排列

Method 8 / 方法 8 [CO01-M08] (continued / 续)

2. Deleting the largest label reverses this operation uniquely, so the construction neither misses nor repeats a permutation.

3. There are two choices at each of four insertions, giving $2^4 = 16$.

例题解析

1. 从只含一的行开始，为保持条件，每个新增最大标签只能附在左端或右端。

2. 删除最大标签唯一逆转此操作，所以构造完备且不重复。

3. 四次插入各有两种选择，共 $2^4 = 16$ 种。

Do not infer a recurrence from legal insertion alone; deletion must preserve legality and recover every object.

仅有合法插入还不足以得到递推，删除也须保持合法并覆盖每个对象。

For permutations with no proper prefix equal to an initial value set, split every unrestricted permutation at its earliest closed prefix. If c_k counts legal first blocks, then $n! = \sum_{k=1}^{n} c_k (n-k)!$. The word earliest is what gives a unique decomposition.

对不允许任何真前缀恰为初始值集合的排列，将任意排列按最早闭合前缀切开。若 c_k 计合法首块，则 $n! = \sum_{k=1}^{n} c_k (n-k)!$；“最早”保证分解唯一。

Practice / 配套练习: AIME-2015-II-Q10 (WB p. 673), AIME-2018-II-Q11 (WB p. 685), AIME-2020-I-Q05 (WB p. 622)

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 1 / 方法 1 [CO02-M01]

Subsets are unordered choices. Translate each condition into forced membership, a size split, or a complement before expanding formulas. Binomial identities become reliable when both sides count the same objects, while modular coefficient arguments require a clearly specified polynomial ring.

子集是无序选取。展开公式前，先把条件转成强制归属、大小分类或补集。二项式恒等式应由两边计算同一对象来保证，模系数论证则须明确使用的多项式环。

Condition a selection on its type counts

按类型数量分类选取

Procedure

1. Choose a size or type-count vector whose cases are disjoint and exhaustive.

2. Multiply the binomial choices from disjoint groups, including any forced or excluded elements.

3. Sum the valid vectors; when solving for an unknown size, cancel factorial ratios before testing its integer domain.

通用步骤

1. 选择互斥且完备的大小或类型数量向量。

2. 相乘各不交组的组合选法，并处理必选或禁选元素。

3. 对可行向量求和；未知总大小时先约去阶乘比例，再在整数定义域内求解。

> Example. Choose four books from six novels and four poetry books. Both genres must appear, and the number of novels must be even. Count the choices.

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 1 / 方法 1 [CO02-M01] (continued / 续)

> 例题：从六本小说、四本诗集中选四本，要求两类均出现且小说数量为偶数，求选法数。

Worked solution

1. The novel count can only be two: zero or four would omit a genre.

2. Choose two novels and two poetry books independently.

3. The count is $\binom{6}{2}\binom{4}{2} = 15 \cdot 6 = 90$. No ordering factor is needed.

例题解析

1. 小说数只能为二，零或四都使一类缺席。

2. 独立选择两本小说与两本诗集。

3. 共有 $\binom{6}{2}\binom{4}{2} = 15 \cdot 6 = 90$ 种，不需要排列因子。

Choose the sample space once: subsets, ordered draws, and assignments to labeled people have different denominators.

先固定样本空间：子集、有序抽取与向有标识人分配具有不同分母。

Canceling $\binom{n}{r}/\binom{n}{s}$ often leaves a short product of consecutive factors; prove its monotonicity on the legal range before declaring an integer solution unique. For boundary-point triangles, subtract triples lying on a common side from all triples, and justify that no other collinear triple exists.

约去 $\binom{n}{r}/\binom{n}{s}$ 的阶乘常剩连续因子的短乘积；须在合法范围证明单调后才可断言整数解唯一。对边界点三角形，从全部三元组扣除同边共线三元组，并证明没有其他共线三点。

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 1 / 方法 1 [CO02-M01] (continued / 续)

Practice / 配套练习: AIME-2000-II-Q03 (WB p. 720), AIME-2004-II-Q02 (WB p. 749), AIME-2005-I-Q03 (WB p. 601), AIME-2005-II-Q01 (WB p. 629), AIME-2006-II-Q04 (WB p. 616), AIME-2007-I-Q10 (WB p. 634), AIME-2009-I-Q11 (WB p. 254), AIME-2010-II-Q13 (WB p. 764), AIME-2011-I-Q10 (WB p. 739), AIME-2012-II-Q03 (WB p. 630), AIME-2013-I-Q06 (WB p. 740), AIME-2015-I-Q02 (WB p. 726), AIME-2017-I-Q01 (WB p. 631), AIME-2018-I-Q09 (WB p. 664), AIME-2018-I-Q10 (WB p. 715), AIME-2022-II-Q09 (WB p. 700), AIME-2023-I-Q01 (WB p. 731), AIME-2024-I-Q04 (WB p. 760), AIME-2025-I-Q05 (WB p. 495), AIME-2025-I-Q10 (WB p. 627)

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 2 / 方法 2 [CO02-M02]

Encode subsets by membership states and intrinsic levels

用归属状态与内在层次编码子集

Procedure

1. Describe each ground-set element by which sets contain it; translate global conditions into allowed states or fixed atoms.

2. Choose elements for any distinguished atoms and freely assign all remaining allowed states.

3. If the object is a function, choose intrinsic image or preimage levels first, then count maps between those levels.

4. Check that the encoding can be recovered from the object; otherwise different state descriptions may be duplicates.

通用步骤

1. 按每个基础元素属于哪些集合来编码，将整体条件转为允许状态或固定区域。

2. 为特殊区域选择元素，再自由分配其余允许状态。

3. 若对象为函数，先选其内在像集或原像层，再计层间映射。

4. 检验对象能否恢复编码，否则不同状态描述可能重复。

> Example. For three subsets of a five-element set, exactly two elements belong to all three, and each other element belongs to at most one. Count the ordered triples of subsets.

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 2 / 方法 2 [CO02-M02] (continued / 续)

例题：五元集合的三个子集中，恰有两个元素同时属于三者，其余每个元素至多属于一个子集，求有序子集三元组数。

Worked solution

1. Choose the common two-element atom in $\binom{5}{2} = 10$ ways.

2. Each of the other three elements has four states: only in the first, only in the second, only in the third, or in none.

3. Every triple uniquely recovers its common atom and states, so the count is $10 \cdot 4^3 = 640$.

例题解析

1. 选择三者公共的二元区域有 $\binom{5}{2} = 10$ 种。

2. 余下三个元素各有仅属第一、仅属第二、仅属第三、均不属四种状态。

3. 每个集合三元组唯一恢复公共区域与各状态，故为 $10 \cdot 4^3 = 640$。

The empty subset is one legal state unless it is explicitly excluded.

除非明确排除，空子集本身也是一个合法状态。

Exactly 2^{a-1} nonempty finite subsets of positive integers have maximum a: include a and choose freely below it. Therefore a sum of such counts can be decoded uniquely by binary expansion.

For $f \circ f$ constant with value a, the intrinsic level $S = f^{-1}(a)$ contains the fixed point a; points outside S must map into $S \setminus \{a\}$. For $f^2 = f^3$, use fixed points, their nonfixed direct preimages, and the remaining level. The levels must be intrinsic to avoid counting the same function more than once.

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 2 / 方法 2 [CO02-M02] (continued / 续)

最大元恰为 a 的有限非空正整数子集有 2^{a-1} 个：必选 a，其下任意选。因此此类数量之和可以按二进制展开唯一反解。

若 $f \circ f$ 恒为 a，内在层 $S = f^{-1}(a)$ 含不动点 a，S 外元素必须映入 $S \setminus \{a\}$。若 $f^2 = f^3$，则分为不动点、直接映入不动点的非不动元素以及剩余层。各层必须由函数本身唯一确定，以免重复计数。

Practice / 配套练习: AIME-2002-II-Q09 (WB p. 633), AIME-2010-I-Q07 (WB p. 636), AIME-2010-II-Q08 (WB p. 637), AIME-2013-II-Q11 (WB p. 215), AIME-2017-II-Q01 (WB p. 660), AIME-2018-II-Q10 (WB p. 207), AIME-2021-II-Q06 (WB p. 640), AIME-2024-II-Q06 (WB p. 642)

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 3 / 方法 3 [CO02-M03]

Collapse binomial sums by symmetry and convolution

用对称与卷积合并二项式和

Procedure

1. Rewrite every factor so the lower indices total a fixed selection size.

2. Interpret a convolution by choosing from disjoint groups: $\sum_k \binom{a}{k}\binom{b}{r-k} = \binom{a+b}{r}$.

3. Use complementary indices to obtain half-row or squared-row sums; retain any middle coefficient when the row length is even.

4. For signed or residue weights, use generating functions or an explicit parity filter rather than discarding signs.

通用步骤

1. 改写各因子，使下标之和成为固定选取数。

2. 把卷积解释为从不交组中选取：$\sum_k \binom{a}{k}\binom{b}{r-k} = \binom{a+b}{r}$。

3. 用互补下标处理半行和或平方和；行长为偶数时保留中间系数修正。

4. 有符号或剩余类权重时，使用生成函数或明确奇偶筛选，不能忽略符号。

> Example. From six red and six blue objects, choose six. How many choices contain an even number of red objects?

> 例题：从六红六蓝物体中选六个，含偶数个红物体的选法有多少？

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 3 / 方法 3 [CO02-M03] (continued / 续)

Worked solution

1. Let E, O count choices with even and odd red counts. Vandermonde gives $E + O = \binom{12}{6} = 924$.

2. The signed count $E - O$ is the coefficient of x^6 in $(1-x)^6(1+x)^6 = (1-x^2)^6$, namely $-\binom{6}{3} = -20$.

3. Adding the equations gives $E = (924 - 20)/2 = 452$.

例题解析

1. 令 E, O 分别计偶数与奇数个红物体，范德蒙德给出 $E + O = \binom{12}{6} = 924$。

2. 带符号差 $E - O$ 是 $(1-x)^6(1+x)^6 = (1-x^2)^6$ 的 x^6 系数，即 $-\binom{6}{3} = -20$。

3. 两式相加得 $E = (924 - 20)/2 = 452$。

A complement changes an index but does not change a binomial coefficient: $\binom{n}{k} = \binom{n}{n-k}$.

取补改变下标但不改变组合数：$\binom{n}{k} = \binom{n}{n-k}$。

Practice / 配套练习: AIME-2000-II-Q07 (WB p. 632), AIME-2003-I-Q13 (WB p. 591), AIME-2006-II-Q10 (WB p. 785), AIME-2007-I-Q10 (WB p. 634), AIME-2007-II-Q10 (WB p. 738), AIME-2008-II-Q12 (WB p. 654), AIME-2017-I-Q07 (WB p. 638), AIME-2018-I-Q12 (WB p. 748), AIME-2020-I-Q07 (WB p. 639), AIME-2022-I-Q12 (WB p. 668), AIME-2025-I-Q10 (WB p. 627)

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 4 / 方法 4 [CO02-M04]

Count weakly ordered indices and telescope binomial rows

计算弱次序下标并望远镜合并二项式行

Procedure

1. Translate inverse-branch choices or compressed positions into a weakly increasing index list.

2. Add successive offsets to turn weak inequalities into a strict subset selection.

3. When summing over the last index, use Pascal differences or the hockey-stick identity, with the starting index checked.

通用步骤

1. 把反函数分支或压缩位置转为弱递增下标串。

2. 依次加偏移量，将弱不等式转为严格子集选取。

3. 按最后下标求和时，用帕斯卡差分或曲棍球杆恒等式，并核对起始下标。

> Example. Count triples of days $2 \le i \le j \le k \le 7$, then count those whose last day is exactly seven.

> 例题：求 $2 \le i \le j \le k \le 7$ 的日期三元组数，并求末日恰为七的三元组数。

Worked solution

1. The map $(i, j, k) \mapsto (i, j+1, k+2)$ gives three distinct choices from $2, \ldots, 9$, so the total is $\binom{8}{3} = 56$.

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 4 / 方法 4 [CO02-M04] (continued / 续)

2. If $k = 7$, choose a weak pair $2 \le i \le j \le 7$, equivalent to two distinct elements from $2, \ldots, 8$.

3. This gives $\binom{7}{2} = 21$, also equal to $\binom{8}{3} - \binom{7}{3}$.

例题解析

1. 映射 $(i, j, k) \mapsto (i, j+1, k+2)$ 给出从二至九选三个不同数，总数为 $\binom{8}{3} = 56$。

2. 若 $k = 7$，只需选弱递增二元组 $2 \le i \le j \le 7$，等价于从二至八选两个不同数。

3. 故为 $\binom{7}{2} = 21$，也等于 $\binom{8}{3} - \binom{7}{3}$。

A half-open analytic branch can impose a strict upper endpoint even when the index inequality is weak.

解析分支若为半开区间，即使下标弱递增，也须保留严格上端点。

Practice / 配套练习: AIME-2009-II-Q06 (WB p. 635), AIME-2020-II-Q14 (WB p. 178), AIME-2022-II-Q10 (WB p. 641), AIME-2023-I-Q11 (WB p. 687)

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 5 / 方法 5 [CO02-M05]

Propagate binomial weights modulo a prime

在素数模下传递二项式权重

Procedure

1. Repeated adjacent addition produces Pascal weights; write the coefficient of each initial entry.

2. Use $(1+x)^p \equiv 1 + x^p \pmod{p}$ to reduce a large exponent.

3. For sums over exponents modulo m, work also modulo $x^m - 1$, where those coefficients combine.

4. Check which remaining factorial coefficients are nonzero modulo the prime before counting zero classes.

通用步骤

1. 反复相邻相加产生帕斯卡权重，写出每个初始数的系数。

2. 用 $(1+x)^p \equiv 1 + x^p \pmod{p}$ 降低大指数。

3. 若按指数模 m 求和，还须模去 $x^m - 1$，相应系数才会合并。

4. 计算哪些剩余阶乘系数在素数模下非零，再数零剩余类。

> Example. Find the coefficient of x^6 in $(1+x)^{13}$ modulo five.

> 例题：求 $(1+x)^{13}$ 的 x^6 系数模五的值。

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 5 / 方法 5 [CO02-M05] (continued / 续)

Worked solution

1. In characteristic five, $(1+x)^5 \equiv 1 + x^5$, so $(1+x)^{13} \equiv (1+x)^3(1+x^5)^2$.

2. To obtain degree six, take x from the first factor and x^5 from the second; no other exponents can combine to six.

3. The coefficient is $\binom{3}{1}\binom{2}{1} = 6 \equiv 1 \pmod 5$.

例题解析

1. 模五时 $(1+x)^5 \equiv 1 + x^5$，所以 $(1+x)^{13} \equiv (1+x)^3(1+x^5)^2$。

2. 次数六只能由第一因子的一次项与第二因子的五次项相乘。

3. 系数为 $\binom{3}{1}\binom{2}{1} = 6 \equiv 1 \pmod 5$。

Reducing exponents modulo $p-1$ applies in the quotient by $x^{p-1}-1$, not as an identity of ordinary polynomials.

指数模 $p-1$ 约化发生在模 $x^{p-1}-1$ 的商环中，不是普通多项式恒等式。

Practice / 配套练习: AIME-2007-II-Q13 (WB p. 643), AIME-2026-I-Q13 (WB p. 646)

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 6 / 方法 6 [CO02-M06]

Rank fixed-weight binary strings by prefix blocks

按前缀区块确定定重二进制串排名

Procedure

1. Fix the length and required number of ones, including a leading-digit rule if present.

2. At a possible one, count the smaller zero-prefix block by a binomial coefficient.

3. For ranking, add all skipped block sizes; for unranking, subtract a block only when the target rank lies beyond it.

4. For consecutive equal-weight integers, analyze the terminal carry chain before counting the shared prefix.

通用步骤

1. 固定长度与一的总数，并写明是否允许前导零。

2. 遇到可选的一位时，用组合数计算该位为零的较小前缀块。

3. 求排名时相加所有跳过区块；反解排名时，仅当目标越过区块才减去其大小。

4. 对相邻且同重量整数，先分析末尾进位链，再计算公共前缀。

> Example. Among length-six binary strings with three ones, including initial zeros and ordered as integers, what is the rank of 101010?

> 例题：六位二进制串恰有三个一，允许前导零，按整数大小排列。101010 排第几?

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 6 / 方法 6 [CO02-M06] (continued / 续)

Worked solution

1. Before its initial one are strings beginning zero, giving $\binom{5}{3} = 10$.

2. At the third bit, the smaller prefix 100 leaves two ones in three places, giving $\binom{3}{2} = 3$. At the fifth bit, prefix 10100 leaves one one in one place, giving one more.

3. There are $10 + 3 + 1 = 14$ smaller strings, so the rank is 15.

例题解析

1. 首位一之前是以零开头的串，有 $\binom{5}{3} = 10$ 个。

2. 第三位的一之前，较小前缀 100 余三位放两个一，有 $\binom{3}{2} = 3$ 个；第五位之前，前缀 10100 余一位放一个一，再有一个。

3. 较小串共 $10 + 3 + 1 = 14$ 个，排名为十五。

Rank one is the first object; count smaller objects and then add one.

排名从一开始：先计更小对象，再加一。

Practice / 配套练习: AIME-2012-I-Q05 (WB p. 569), AIME-2012-II-Q07 (WB p. 587)

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 7 / 方法 7 [CO02-M07]

Classify a forbidden-subset family before selecting members

先分类禁用子集族结构再选成员

Procedure

1. Find a distinguished member or minimum element that forces exclusions.

2. Convert pairwise restrictions into an auxiliary graph when members are edges or small subsets.

3. Prove the structural list is complete, then choose freely only within the surviving independent components.

通用步骤

1. 找出能迫使排除条件的特殊成员或最小元素。

2. 若成员为边或小子集，将两两限制转成辅助图。

3. 证明结构分类完备，再仅在剩余独立部分自由选取。

> Example. Prove that four distinct edges of a simple graph which pairwise intersect must all share one vertex.

> 例题：证明简单图中四条不同的边若两两相交，则必共用一个顶点。

Worked solution

1. Take two edges ab, ac. If every remaining edge contains a, the conclusion holds.

2. Any edge avoiding a must intersect both, so it is bc.

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 7 / 方法 7 [CO02-M07] (continued / 续)

3. An edge meeting all of ab, ac, bc can only be one of these three unless it contains a common vertex of all three, which does not exist. Thus a fourth distinct edge rules out the triangle case, leaving a star.

例题解析

1. 取两边 ab, ac，若其余边均含 a，结论成立。

2. 若某边不含 a，为了与这两边相交，只能是 bc。

3. 能同时交 ab, ac, bc 的边只能是这三条，因为三者没有公共顶点。所以第四条不同边排除了三角形情况，只剩星形。

Pairwise intersecting does not mean globally intersecting: a triangle is the essential exceptional case.

两两相交不等于全部有公共交点，三角形是关键例外。

For a pairwise-intersecting family of exactly half the subsets of a finite set, complementary pairs allow at most one member, hence exactly one from every pair. A chosen singleton forces all members to contain that element. Without a singleton, analyze the selected two-subsets as an intersecting edge family: empty, star subfamily, or triangle.

For product-free subsets, condition on the least chosen integer. Its products create explicit forbidden pairs or triples among the remaining integers. Count independent components only after verifying all product relations have been covered, including a repeated factor such as a^2.

两两相交子集族若恰含全部子集的一半，每对互补集至多选一个，因此每对恰选一个。若选单点集，全部成员都必须含该元素；没有单点集时，将选中的二元集视为两两相交边族，只可能为空、某星形的子族或三角形。

Section 4.2 Subsets & binomial identities

子集与二项式恒等式

Method 7 / 方法 7 [CO02-M07] (continued / 续)

对无乘积关系的子集，按最小所选整数分类，它与其他数相乘会产生明确的禁用二元或三元组合。只有确认覆盖全部乘积关系，包括 a^2 这类重复因子后，才可按独立分量相乘。

Practice / 配套练习: AIME-2017-I-Q12 (WB p. 644), AIME-2023-II-Q11 (WB p. 645)

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

Method 1 / 方法 1 [CO03-M01]

Replace an arrangement by its ordered part sizes, multiplicities, or gaps. Before applying stars and bars, decide which parts are labeled, whether zero is allowed, and whether upper bounds are active. Generating functions record these same choices algebraically.

将排列转成有序部件大小、重数或间隔。使用隔板法前，须判断部件是否有标识、能否为零以及上界是否起作用。生成函数是同一选择过程的代数记录。

Normalize lower bounds and use stars and bars

平移下界并使用隔板法

Procedure

1. Introduce one variable for each labeled part and translate any lower bounds to zero.

2. For r nonnegative parts totaling N, use $\binom{N+r-1}{r-1}$; for positive parts, subtract one first.

3. For a multiset, interpret parts as multiplicities; for prime factors, distribute each prime exponent independently.

通用步骤

1. 为每个有标识部件设变量，将下界平移为零。

2. 把 N 分成 r 个非负部分有 $\binom{N+r-1}{r-1}$ 种；正部分先各减一。

3. 多重集用重数作为部分，素因子分配则对每种指数独立拆分。

> Example. Four identical tokens go into five labeled boxes, with exactly two boxes nonempty. Count the distributions.

> 例题：四枚相同棋子放入五个有标识盒子，恰有两个非空，求分配数。

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

Method 1 / 方法 1 [CO03-M01] (continued / 续)

Worked solution

1. Choose the two nonempty boxes in $\binom{5}{2} = 10$ ways.

2. If their positive counts are u, v, then $u + v = 4$ has $\binom{3}{1} = 3$ solutions.

3. The choice of boxes and counts uniquely gives the distribution, so the answer is 30.

例题解析

1. 选两个非空盒子有 $\binom{5}{2} = 10$ 种。

2. 两个正重数满足 $u + v = 4$，有 $\binom{3}{1} = 3$ 组。

3. 盒子与重数唯一决定分配，所以共有三十种。

A sorted list with repetitions corresponds to multiplicities, not to an arbitrary ordering of the parts.

可重复的有序非降列表对应重数，不能再任意排列部分。

The number of nondecreasing k-tuples drawn from r values is $\binom{r+k-1}{k}$, because their multiplicities are nonnegative and total k. The number of all positive compositions of N with an arbitrary number of parts is 2^{N-1}, choosing cuts among its $N-1$ internal gaps.

从 r 种值中选非降 k 元组共有 $\binom{r+k-1}{k}$ 种，因为重数非负且和为 k。将 N 分成任意数量正部分的所有有序分拆共有 2^{N-1} 种，等价于选择内部 $N-1$ 个间隔是否切开。

Practice / 配套练习: AIME-2000-II-Q05 (WB p. 647), AIME-2001-I-Q06 (WB p. 733), AIME-2004-II-Q08 (WB p. 605), AIME-2006-II-Q08 (WB p. 711), AIME-2007-II-Q02 (WB p. 484), AIME-2011-II-Q06 (WB p. 714), AIME-2013-II-Q12 (WB p. 56), AIME-2020-I-Q09 (WB p. 741), AIME-2024-II-Q03 (WB p. 581), AIME-2026-I-Q02 (WB p. 583)

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

Method 2 / 方法 2 [CO03-M02]

Enforce upper bounds by coefficients or clipped intervals

用系数或截断区间处理上界

Procedure

1. Encode a variable taking values in A by $\sum_{a\in A} x^a$; multiply for independent labeled variables.

2. Extract the target coefficient, or fix some variables and intersect the remaining integer interval with every bound.

3. If upper-bound violations are easier to count, shift a violating variable down by its first forbidden value and apply inclusion-exclusion.

4. When congruences force residues, substitute those residues first; the remaining equation may become ordinary stars and bars.

通用步骤

1. 变量可取集合 A 时，用 $\sum_{a\in A} x^a$ 编码；独立有标识变量对应相乘。

2. 提取目标系数，或固定部分变量并把其余整数区间与所有边界相交。

3. 若越界容易计数，将越界变量减去首次禁止值，再容斥。

4. 若同余迫使变量落在指定剩余类，先代入剩余类，余下方程常可化为隔板法。

> Example. How many triples $0 \le x, y, z \le 3$ satisfy $x + y + z = 7$?

> 例题：满足 $0 \le x, y, z \le 3$ 且 $x + y + z = 7$ 的整数三元组有多少个?

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

Method 2 / 方法 2 [CO03-M02] (continued / 续)

Worked solution

1. The coefficient is $[t^7](1+t+t^2+t^3)^3$. Without the upper bounds there are $\binom{9}{2}=36$ triples.

2. For $x \geq 4$, shifting x down by four gives $\binom{5}{2}=10$ triples, and likewise for the other two variables.

3. Two violations would sum to at least eight and cannot occur. Therefore the answer is $36-3\cdot 10=6$.

例题解析

1. 所求为 $[t^7](1+t+t^2+t^3)^3$。不加上界有 $\binom{9}{2}=36$ 组。

2. 若 $x \geq 4$，将它减四后有 $\binom{5}{2}=10$ 组，其余两变量同理。

3. 两变量同时越界至少和为八，不可能，所以共有 $36-3\cdot 10=6$ 组。

A coefficient of a reciprocal product counts solutions only while numerator terms above the target degree are irrelevant.

倒数乘积的系数可以计解，但须确认分子的其他项次数均高于目标次数。

For $b>0$ and $L \leq ax+by \leq U$ and fixed x, the allowed y lie between $\lceil (L-ax)/b \rceil$ and $\lfloor (U-ax)/b \rfloor$, further clipped by its own bounds. Use zero if the lower endpoint exceeds the upper. For products of geometric series, a coefficient of $\prod_i (1-x^{w_i})^{-1}$ counts nonnegative solutions of $\sum_i w_i u_i = N$.

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

Method 2 / 方法 2 [CO03-M02] (continued / 续)

对固定 x，若 $b>0$ 且 $L \le ax+by \le U$，则 y 的上下界分别为 $\lceil (L-ax)/b \rceil$ 与 $\lfloor (U-ax)/b \rfloor$，还须与自身范围取交；下界大于上界时贡献零。几何级数乘积 $\prod_i (1-x^{w_i})^{-1}$ 的 x^N 系数计算非负方程 $\sum_i w_i u_i = N$ 的解数。

Practice / 配套练习: AIME-2003-I-Q09 (WB p. 649), AIME-2009-I-Q09 (WB p. 650), AIME-2013-I-Q15 (WB p. 513), AIME-2018-I-Q01 (WB p. 35), AIME-2020-I-Q11 (WB p. 58), AIME-2022-II-Q13 (WB p. 656)

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

Method 3 / 方法 3 [CO03-M03]

Describe rows and paths by runs and gaps

用连续段与间隔描述行和路径

Procedure

1. Replace each maximal block by its positive length and record its type.

2. Separate choices of block order, block lengths, and the gaps that receive them.

3. Treat the two ends and a circular closing gap explicitly; insert distinct labels only after the type pattern is counted.

通用步骤

1. 把每个极大同类段替换为其正长度并记录类型。

2. 分开计算段顺序、段长度与放置这些段的间隔。

3. 明确处理两端与圆形闭合间隔；先计类型模式，再放不同标签。

> Example. Count binary words of length nine with four ones, where the ones form exactly two runs, of lengths one and three.

> 例题：长度九的二进制串有四个一，且它们恰形成长度一与三的两个连续段，求串数。

Worked solution

1. The five zeros create six gaps, including the ends. The two one-runs must occupy different gaps.

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

Method 3 / 方法 3 [CO03-M03] (continued / 续)

2. Choose their gaps in $\binom{6}{2}$ ways, then assign which gap gets the length-three run in two ways.

3. The count is $2\binom{6}{2} = 30$; the separating zero is already built into choosing different gaps.

例题解析

1. 五个零产生含两端在内的六个间隔，两个一段须放在不同间隔。

2. 选间隔有 $\binom{6}{2}$ 种，再用两种方式指定哪个间隔放长度三的段。

3. 共有 $2\binom{6}{2} = 30$ 种，不同间隔已自动保证两段间有零。

The number of turns in a path is one less than its number of nonempty directional runs.

路径转弯次数比非空方向段数少一。

For j occupied runs using s cells and all run lengths one or two, the number of doubletons is $s - j$ and their ordered positions can be chosen in $\binom{j}{s-j}$ ways. If e empty cells supply the separators, placing runs in $e + 1$ distinct gaps contributes $\binom{e+1}{j}$.

Run lengths in $\{1, \ldots, h\}$ have generating function $x + \cdots + x^h$. With the initial symbol fixed, binary words with bounded runs correspond to positive compositions with those allowed part sizes.

若 j 个占用段共含 s 格，且段长仅一或二，则二元段有 $s - j$ 个，其顺序位置有 $\binom{j}{s-j}$ 种。若 e 个空格提供分隔，将各段放入 $e + 1$ 个不同间隔另有 $\binom{e+1}{j}$ 种。

段长限于 $\{1, \ldots, h\}$ 时，单段生成函数为 $x + \cdots + x^h$。固定首字母后，段长有限的二元串与允许这些部分大小的正有序分拆双射。

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

Method 3 / 方法 3 [CO03-M03] (continued / 续)

Practice / 配套练习: AIME-2008-II-Q12 (WB p. 654), AIME-2009-I-Q10 (WB p. 651), AIME-2011-II-Q07 (WB p. 652), AIME-2015-II-Q12 (WB p. 682), AIME-2019-II-Q05 (WB p. 621), AIME-2024-I-Q06 (WB p. 675), AIME-2025-II-Q10 (WB p. 653)

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

Method 4 / 方法 4 [CO03-M04]

Separate part-size patterns from label assignments

分离部分大小模式与标签分配

Procedure

1. List all feasible multiplicity or component-size patterns using the total constraints.

2. For labeled groups of sizes n_i, assign distinct objects in $N!/\prod_i n_i!$ ways.

3. If some groups are interchangeable, divide by factorials of repeated group multiplicities; never divide for groups already distinguished by size or label.

通用步骤

1. 按总量限制列全可行重数或部件大小模式。

2. 向大小为 n_i 的有标识组分配不同物体，有 $N!/\prod_i n_i!$ 种。

3. 若某些组可互换，按重复大小组的重数阶乘归并；已由大小或标签区分的组不能再相除。

> Example. Split six distinct samples into unlabeled groups of sizes one, two, and three. Compare with splitting them into three unlabeled pairs.

> 例题：将六个不同样本分成无标识的一元、二元、三元组；再与分成三个无标识二元组比较，各有多少种?

Worked solution

1. The sizes one, two, and three distinguish the groups, so the first count is $6!/(1!2!3!) = 60$.

2. For three pairs, first label the pairs: $6!/(2!)^3 = 90$.

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

Method 4 / 方法 4 [CO03-M04] (continued / 续)

3. Every unlabeled partition has $3!$ labelings of its equal-sized pairs, so the second answer is $90/6 = 15$.

例题解析

1. 一、二、三的大小已区分各组，第一种为 $6!/(1!2!3!) = 60$。

2. 三个二元组先加标签，计数为 $6!/(2!)^3 = 90$。

3. 每个无标识分拆有 $3!$ 种等大小组标签，所以第二种为 $90/6 = 15$。

> A multinomial count is a weight attached to a size pattern, not the number of size patterns.

> 多项式系数是每种大小模式的权重，不是大小模式数。

Practice / 配套练习: AIME-2012-II-Q14 (WB p. 701), AIME-2016-II-Q08 (WB p. 476), AIME-2018-II-Q08 (WB p. 674), AIME-2021-I-Q04 (WB p. 648), AIME-2025-I-Q03 (WB p. 624)

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

Method 5 / 方法 5 [CO03-M05]

Use exchangeable gap coordinates for order statistics

用可交换间隔求次序统计量

Procedure

1. Write a chosen subset increasingly and record all unchosen gaps, including both ends.

2. Show this is a bijection to weak compositions of the fixed unchosen total.

3. Use coordinate exchangeability to average the gaps, then express the requested statistic in them.

通用步骤

1. 把选中子集递增排列，记录未选元素的所有间隔，包括两端。

2. 证明其与固定未选总数的弱有序分拆双射。

3. 利用间隔坐标可交换性求均值，再用间隔表示目标统计量。

> Example. Choose three integers uniformly from $1, \ldots, 9$. What is the expected difference between the largest and smallest?

> 例题：从一至九中均匀选三个整数，最大数与最小数之差的期望是多少?

Worked solution

1. The six unchosen integers occupy four gap coordinates, each with the same expected value $6/4 = 3/2$.

2. The range is two chosen-to-chosen increments plus the two internal gaps: $2 + g_1 + g_2$.

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

Method 5 / 方法 5 [CO03-M05] (continued / 续)

3. Its expected value is $2+2(3/2)=5$. The two exterior gaps do not enter the range.

例题解析

1. 六个未选整数分布在四个间隔坐标，均值均为 $6/4=3/2$。

2. 最大最小之差为两个已选相邻步长加两个内部间隔，即 $2+g_1+g_2$。

3. 期望为 $2+2(3/2)=5$，两端外部间隔不计入极差。

Exchangeability follows from the uniform bijection to all gap vectors; do not assume unrelated gaps are independent.

可交换性来自均匀双射到全部间隔向量，不能假设这些间隔相互独立。

Practice / 配套练习: AIME-2015-I-Q12 (WB p. 777)

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

Method 6 / 方法 6 [CO03-M06]

Translate function bounds into all interval sums

把函数约束转成所有区间步长和

Procedure

1. Use consecutive increments $d_i = f(i) - f(i-1)$ as variables.

2. Convert every pairwise distance condition into a bound on each consecutive sum of increments.

3. Use the total sum to bound exceptional negative increments, then prove any proposed local rule suffices for every longer interval before counting its orders.

通用步骤

1. 以相邻增量 $d_i = f(i) - f(i-1)$ 为变量。

2. 把每对输入的距离条件转为每段连续增量和的界。

3. 利用总和限制例外负增量，再证明所设局部规则对全部较长区间也充分，之后才排列计数。

> Example. Start at zero with increments $(3, -1, 3)$, giving values $(0, 3, 2, 5)$. Verify $|i-j| \le |f(i) - f(j)| \le 3|i-j|$ for all indices, and explain why replacing the middle step by -2 fails.

> 例题：从零出发取增量 $(3, -1, 3)$，得到 $(0, 3, 2, 5)$。检验所有下标满足 $|i-j| \le |f(i) - f(j)| \le 3|i-j|$，并解释中间步改为负二为何失败。

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

Method 6 / 方法 6 [CO03-M06] (continued / 续)

Worked solution

1. Every single increment has absolute value between one and three. The two length-two interval sums are both two, and the length-three sum is five.

2. These absolute sums lie respectively in $[2, 6]$ and $[3, 9]$, proving every pairwise inequality.

3. After replacing -1 by -2, an adjacent two-step sum becomes one, below its required lower bound two. Bounds on individual steps alone are insufficient.

例题解析

1. 单步绝对值均在一与三之间；两段长度二的和均为二，长度三的和为五。

2. 这些绝对和分别位于 $[2, 6]$ 与 $[3, 9]$，所以全部成对不等式成立。

3. 若中间改为负二，某两步和为一，低于应有下界二；仅检验单步界不充分。

Local restrictions may suggest a count, but their sufficiency for nonadjacent input pairs must be proved.

局部限制可提示计数，但必须证明其对非相邻输入对也充分。

If all other increments lie in $\{1, 2, 3\}$ and the only negative increment is -1, each existing neighbor must be three for the length-two lower bound. These conditions suffice: an interval containing -1 alone has absolute sum one; an interval extending to one side begins with sum two over two steps, and every added positive step preserves sum at least length. Intervals extending both ways have an even larger initial margin. The upper bound follows by the triangle inequality.

Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

Method 6 / 方法 6 [CO03-M06] (continued / 续)

若其他增量均属于 $\{1,2,3\}$ 且唯一负增量为负一，长度二下界要求其每个现有邻居均为三。这也充分：仅含负一步的区间绝对和为一；向一侧延伸的区间从两步和为二开始，每加一步正增量都保持总和至少为长度；向两侧延伸有更大初始余量。上界由三角不等式得到。

Practice / 配套练习: AIME-2018-II-Q15 (WB p. 655)

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

Method 1 / 方法 1 [CO04-M01]

A construction may count witnesses rather than objects. Define the witness, compute its multiplicity, and inspect intersections before applying inclusion-exclusion. For extrema, an incidence bound is only half a solution: exhibit an arrangement satisfying every original condition.

一种构造可能数到的是见证而非对象。须定义见证、计算其重数，并在容斥前检查交集。对最值，关联计数界只是解答的一半，还须构造满足原全部条件的配置。

Use exact event intersections and the complement

精确计算事件交集与补集

Procedure

1. Choose an ambient set of equally counted objects and define the forbidden or required events.

2. Describe every intersection in structural terms, such as a common divisor, fixed cells, or a restricted image set.

3. Add single events, subtract pairs, add triples, and continue; include the empty or full object where appropriate.

4. For an exactly-one condition, account for intersection multiplicity rather than using a union count unchanged.

通用步骤

1. 选择统一计数的全集，并定义禁用或要求的事件。

2. 用结构描述每个交集，例如公约数、固定格或限制像集。

3. 单事件相加、两两相减、三者相加，依次进行，并妥善处理空对象或全对象。

4. 恰一事件发生时须调整交集重数，不能直接使用并集数。

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

Method 1 / 方法 1 [CO04-M01] (continued / 续)

> Example. Among integers one through sixty, how many are divisible by exactly one of four and nine?

> 例题：一至六十的整数中，有多少个恰能被四、九之一整除?

Worked solution

1. There are $\lfloor 60/4 \rfloor = 15$ multiples of four and $\lfloor 60/9 \rfloor = 6$ multiples of nine.

2. The intersection is the multiples of $\operatorname{lcm}(4, 9) = 36$, of which there is one.

3. For exactly one event, subtract that intersection twice: $15 + 6 - 2 = 19$.

例题解析

1. 四的倍数有 $\lfloor 60/4 \rfloor = 15$ 个，九的倍数有六个。

2. 交集是最小公倍数三十六的倍数，只有一个。

3. 恰一事件须减去交集两次，得到 $15 + 6 - 2 = 19$。

Intersections of divisor sets are divisors of a gcd; intersections of multiple sets are multiples of an lcm.

约数集合的交集是最大公因数的约数集，倍数集合的交集是最小公倍数的倍数集。

Practice / 配套练习: AIME-2001-II-Q09 (WB p. 735), AIME-2002-I-Q01 (WB p. 721), AIME-2002-II-Q09 (WB p. 633), AIME-2005-II-Q04 (WB p. 602), AIME-2006-II-Q07 (WB p. 586), AIME-2007-II-Q10 (WB p. 738), AIME-2008-II-Q12 (WB p. 654), AIME-2009-II-Q09 (WB p. 547), AIME-2013-II-Q09 (WB p. 672), AIME-2014-I-Q11 (WB p. 792), AIME-2014-I-Q12 (WB p. 746), AIME-2015-I-Q09 (WB p. 150), AIME-2016-II-Q11 (WB p. 611), AIME-2017-II-Q01 (WB p. 660), AIME-2017-II-Q11 (WB p. 702), AIME-2021-II-Q06 (WB p. 640), AIME-2022-I-Q13 (WB p. 598), AIME-2024-II-Q11 (WB p. 61), AIME-2025-I-Q04 (WB p. 539), AIME-2025-II-Q07 (WB p. 492)

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

Method 2 / 方法 2 [CO04-M02]

Classify compatible overlapping patterns

分类相容的重叠模式

Procedure

1. Specify each local bad pattern, including its direction and any surrounding separator.

2. For selected patterns, prove whether they can coexist and how many positions or labels their union fixes.

3. Glue overlapping adjacency runs or count the free remaining cells, then apply inclusion-exclusion.

4. Handle patterns that wrap around a circle and the all-selected exceptional object separately.

通用步骤

1. 明确每个局部坏模式，包括方向与周围分隔。

2. 对选中的模式，证明能否共存以及并集固定多少位置或标签。

3. 把重叠相邻关系粘成连续段，或计算剩余自由格，再容斥。

4. 单独处理绕圆闭合的模式与全部选中的例外对象。

> Example. How many permutations of A, B, C, D contain neither directed adjacent block AB nor directed adjacent block BC?

> 例题：字母 A, B, C, D 的排列中，有多少个既不含有向相邻块 AB，也不含 BC?

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

Method 2 / 方法 2 [CO04-M02] (continued / 续)

Worked solution

1. The ambient count is $4! = 24$. Each forbidden block, glued as one unit, gives $3! = 6$ permutations.

2. Both blocks can coexist only as ABC, which is one block together with D, giving two permutations.

3. Inclusion-exclusion gives $24 - 6 - 6 + 2 = 14$. The overlap is a single ordered run, not two freely orientable blocks.

例题解析

1. 全集有 $4! = 24$ 个排列，每个禁块粘成一单位后有 $3! = 6$ 种。

2. 两块共存只能形成 ABC，与 D 共两个单位，给出两种排列。

3. 容斥得 $24 - 6 - 6 + 2 = 14$。重叠是一个定向连续段，不是两个可自由转向的区块。

Overlapping witnesses may force an extra occurrence; prove that your multiplicity correction covers it.

重叠见证可能迫使额外出现，须证明重数修正覆盖全部情况。

If selected undirected forbidden edges on a line form r nonempty runs and contain k edges, they collapse to $N - k$ blocks and contribute $2^r(N - k)!$. A circular long run can be marked by its preceding unselected position; this excludes the all-selected object, which must be restored separately.

For fixed-sum pair constraints on a small selected set, model each pair as an edge. Shared endpoints form path witnesses, while disjoint pairs form matching witnesses. List possible witness multiplicities before adding their counts; a set with two witnesses must lose one copy, not two.

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

Method 2 / 方法 2 [CO04-M02] (continued / 续)

直线上选中的无向禁邻边若共 k 条且形成 r 个非空连续段，会粘成 $N-k$ 个区块，贡献 $2^r(N-k)!$。圆上的长段可由它前一个未选位置标记；这会排除全选对象，须另外补回。

对固定和二元组限制，把二元组视为边。共端点形成路径见证，不交边形成匹配见证。相加前列出可能见证重数；有两个见证的对象应扣去一份重复，而非扣去两份。

Practice / 配套练习: AIME-2001-II-Q09 (WB p. 735), AIME-2010-II-Q11 (WB p. 667), AIME-2011-II-Q12 (WB p. 745), AIME-2014-II-Q09 (WB p. 663), AIME-2018-I-Q09 (WB p. 664), AIME-2020-II-Q09 (WB p. 665), AIME-2022-I-Q06 (WB p. 666)

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

Method 3 / 方法 3 [CO04-M03]

Mark an incidence to sum weights or powers

标记关联以求权重或幂次总和

Procedure

1. Replace a weight by the number of marks that produce it.

2. Count pairs or tuples consisting of an object and its marks in the more convenient order.

3. Separate coincident marks from distinct marks when taking a square or higher moment.

4. For geometric incidences, count underlying pairs rather than assuming one pair per visible point.

通用步骤

1. 把权重替换为产生该权重的标记数。

2. 按更方便的顺序计算对象与标记组成的二元或多元组。

3. 平方或更高次矩须区分标记相同与不同。

4. 几何关联应计底层对象对，不能默认每个可见交点只对应一对。

> Example. Compute $\sum_{A\subseteq\{1,\ldots,5\}} |A|^2$ without listing subsets.

> 例题：不逐个列子集，计算 $\sum_{A\subseteq\{1,\ldots,5\}} |A|^2$。

Worked solution

1. For each subset, $|A|^2$ counts ordered marked pairs $(u, v) \in A^2$.

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

Method 3 / 方法 3 [CO04-M03] (continued / 续)

2. When $u = v$, choose the mark in five ways and the other four elements freely, giving $5 \cdot 2^4 = 80$.

3. When $u \neq v$, choose $5 \cdot 4$ ordered marks and the other three elements freely, giving160. Total 240.

例题解析

1. 对每个子集，$|A|^2$ 计算有序标记对 $(u, v) \in A^2$。

2. 若两标记相同，先选五种标记，其余四元素自由选，贡献八十。

3. 若不同，选 $5 \cdot 4$ 个有序标记，其余三元素自由选，贡献一百六十，总和二百四十。

A point where k lines concur consumes $\binom{k}{2}$ line pairs.

有 k 条直线共点时，该点占用 $\binom{k}{2}$ 个直线对。

For an incidence-weighted subset sum, fix one marked element first and count the remaining choices. For a sum of valuations in a product, count one contribution from each multiple of p, another from each multiple of p^2, and so on. These are the same object-mark counting principle.

子集的关联加权和可先固定一个标记元素，再计其余选择。乘积的赋值总和则可对每个 p 的倍数计一次、每个 p^2 的倍数再计一次，依此类推；本质都是对象与标记的双计数。

Practice / 配套练习: AIME-2003-I-Q03 (WB p. 658), AIME-2010-II-Q03 (WB p. 468), AIME-2022-I-Q12 (WB p. 668), AIME-2023-I-Q03 (WB p. 661), AIME-2024-II-Q01 (WB p. 662)

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

Method 4 / 方法 4 [CO04-M04]

Correct nonuniform witness multiplicities

修正不均匀的见证重数

Procedure

1. Determine how many witnesses each finished object has.

2. Partition into generic and exceptional multiplicity classes.

3. Convert incidence counts to object counts using those exact multiplicities, verifying that all exceptions have been listed.

通用步骤

1. 确定每个完成对象有几个见证。

2. 按一般重数与例外重数分类。

3. 用精确重数把关联数转为对象数，并证明例外已经列全。

> Example. Choose three elements from $\{1,2,3,4,5,6\}$ and require at least one even element. A construction first marks an even selected element. Why can its count not simply be used as the answer?

> 例题：从一至六中选三数，要求至少一个偶数。若构造先标记一个被选偶数，为什么不能直接把构造数当答案?

Worked solution

1. With exactly $r = 1, 2, 3$ even elements, the subset counts are $\binom{3}{r}\binom{3}{3-r} = 9, 9, 1$.

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

Method 4 / 方法 4 [CO04-M04] (continued / 续)

2. Marking an even element counts each such subset r times, so the incidence total is $9 + 18 + 3 = 30$.

3. The desired object count is $9 + 9 + 1 = 19$, also $\binom{6}{3} - 1$. The overcount is nonuniform.

例题解析

1. 恰有 $r = 1, 2, 3$ 个偶数时，子集数为 $\binom{3}{r}\binom{3}{3-r} = 9, 9, 1$。

2. 标记一个偶数使每个子集被数 r 次，关联总数为 $9 + 18 + 3 = 30$。

3. 真正对象数为 $9 + 9 + 1 = 19$，也即 $\binom{6}{3} - 1$；重复倍数并不统一。

Three equal sides make an equilateral triangle have three possible apex witnesses; a generic isosceles triangle has one.

等边三角形有三个顶角见证，一般等腰三角形只有一个。

Practice / 配套练习: AIME-2002-I-Q05 (WB p. 690), AIME-2004-I-Q03 (WB p. 691), AIME-2017-II-Q13 (WB p. 703), AIME-2017-II-Q14 (WB p. 704), AIME-2020-I-Q05 (WB p. 622), AIME-2024-II-Q11 (WB p. 61)

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

Method 5 / 方法 5 [CO04-M05]

Double count a capacity bound and construct equality

双计数容量界并构造等号

Procedure

1. Count the same total across two families of layers, vertices, or groups.

2. Combine the resulting identities with integrality or a capacity bound.

3. Determine the equality requirements, then build one configuration satisfying every layer or group condition.

通用步骤

1. 从两族层、顶点或群体分别计算同一总量。

2. 结合所得恒等式与整数性或容量界。

3. 找出等号要求，再构造同时满足每层每组条件的配置。

> Example. A zero-one matrix has three rows and five columns. Each row has two ones, and each column has at most two. What is the least possible number of nonzero columns?

> 例题：三行五列零一矩阵每行有两个一，每列至多两个一，非零列最少有多少列？

Worked solution

1. Row counting gives six ones in total. With at most two per nonzero column, at least three columns are needed.

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

Method 5 / 方法 5 [CO04-M05] (continued / 续)

2. Use rows $11000, 10100, 01100$. Each row has two ones and the first three columns each have two.

3. Thus three nonzero columns attain the lower bound; the unused columns do not violate any condition.

例题解析

1. 按行计共有六个一，非零列各至多两个，所以至少三列。

2. 取三行为 $11000, 10100, 01100$，各行两个一，前三列各两个一。

3. 故三列达到下界，其余空列不违反条件。

An integer dimension satisfying the counting equations may still need an explicit color or incidence construction.

满足计数方程的整数尺寸仍可能需要显式颜色或关联构造。

Practice / 配套练习: AIME-2001-II-Q02 (WB p. 657), AIME-2016-II-Q04 (WB p. 659)

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

Method 6 / 方法 6 [CO04-M06]

Count cyclic overlaps before analyzing exact exchange

先计循环重叠再分析完全交换

Procedure

1. For a subset B of a cyclic group, count ordered pairs of distinct selected points by their displacement.

2. Use $\sum_{k\neq 0} |B \cap (B+k)| = |B|(|B|-1)$ to force a zero overlap when the average is below one.

3. At half the positions, zero overlap means the shift exchanges the set with its complement; analyze each rotation cycle and its parity.

通用步骤

1. 对循环群子集 B，按位移计算不同所选点的有序对。

2. 利用 $\sum_{k\neq 0} |B \cap (B+k)| = |B|(|B|-1)$，平均值小于一时必有零重叠。

3. 若恰选一半位置，零重叠意味着平移交换集合与补集；再分析各旋转轨道及其奇偶性。

> Example. Three positions are selected on a labeled ten-cycle. Prove that at least three nonidentity rotations make the selected set disjoint from its image, and show this bound can be sharp.

> 例题：十个有标识循环位置中选三个，证明至少三个非恒等旋转使所选集与其像不交，并说明此界可达到。

Section 4.4 Inclusion-exclusion & double counting

容斥与双计数

Method 6 / 方法 6 [CO04-M06] (continued / 续)

Worked solution

1. The nine nonzero displacement overlaps total $3 \cdot 2 = 6$. Therefore at most six rotations have a positive overlap, leaving at least three zero-overlap rotations.

2. For $B = \{0, 1, 3\}$ modulo ten, its six directed nonzero differences are $1, 2, 3, 7, 8, 9$, all distinct.

3. Exactly rotations by $4, 5, 6$ have zero overlap, so the lower bound three is sharp.

例题解析

1. 九个非零位移的重叠总数为 $3 \cdot 2 = 6$，至多六个旋转有正重叠，所以至少三个无重叠。

2. 取模十子集 $B = \{0, 1, 3\}$，六个非零有向差为 $1, 2, 3, 7, 8, 9$，彼此不同。

3. 恰有位移四、五、六无重叠，因此三个的下界可达到。

Complement exchange requires alternating membership on every cycle, so every cycle must have even length.

交换集合与补集要求每个轨道上成员交替出现，故每个轨道长度必须为偶数。

For a half-size subset, a shift k partitions the cycle into $\gcd(n, k)$ cycles of length $n/\gcd(n, k)$. If that length is even, complement-exchanging colorings number $2^{\gcd(n,k)}$. These families for different shifts may overlap, so do not add them without checking intersections.

选半数位置时，位移 k 将循环分成 $\gcd(n, k)$ 个长度 $n/\gcd(n, k)$ 的轨道。长度为偶数时，交换补集的染色有 $2^{\gcd(n,k)}$ 种。但不同位移对应的染色族可能重叠，不能未经交集检验直接相加。

Practice / 配套练习: AIME-2024-I-Q11 (WB p. 669)

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 1 / 方法 1 [CO05-M01]

A recurrence is an exhaustive decomposition into disjoint smaller cases. Store exactly the information needed to determine legal continuations, state the empty and boundary cases, and explain why the update counts each object once. For a computed count, preserve the finite recurrence and an independently checkable certificate.

递推须把对象完备地分为互不重叠的较小情形。状态只保留决定合法后继所需的信息，明确空状态与边界，并解释每个对象为何恰计一次。依赖计算的计数应保留有限递推与可独立核验的证书。

Remember the suffix or last move needed by the rule

记住规则所需的后缀或末步

Procedure

1. Choose states recording the shortest suffix, last direction, used subset, or edge type that determines all legal extensions.

2. For each state list every legal next symbol or move and its new state.

3. Initialize the empty object explicitly, then add counts from disjoint predecessor states.

4. Sum only the terminal states satisfying the full endpoint or closure condition.

通用步骤

1. 状态记录决定后继所需的最短后缀、末方向、已用子集或末棱类型。

2. 对每态列出全部合法下一符号或动作及其新状态。

3. 明确空对象初值，再从互斥前驱状态相加。

4. 最后只汇总满足完整终点或闭合条件的终态。

> Example. Count binary words of length five avoiding the consecutive pattern 101.

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 1 / 方法 1 [CO05-M01] (continued / 续)

> 例题：求不含连续模式 101 的五位二进制串个数。

Worked solution

1. Use states $0, 1, 2$ for the longest suffix matching a proper prefix of 101: empty, 1, and 10.

2. The updates are $(a, b, c) \mapsto (a + c, a + b, b)$; from state 10, appending one is forbidden. Begin at $(1, 0, 0)$.

3. Lengths one through five give $(1, 1, 0), (1, 2, 1), (2, 3, 2), (4, 5, 3), (7, 9, 5)$. The answer is 21.

例题解析

1. 用状态零、一、二记录与 101 真前缀相同的最长后缀：空、一、十。

2. 更新为 $(a, b, c) \mapsto (a + c, a + b, b)$，状态十后不能再接一。初值 $(1, 0, 0)$。

3. 一至五位状态依次为 $(1, 1, 0), (1, 2, 1), (2, 3, 2), (4, 5, 3), (7, 9, 5)$，总数二十一。

A nonbacktracking walk must remember its previous edge; the current vertex alone is insufficient.

非回退游走须记住前一条棱，仅当前顶点不足以决定后继。

For a Hamilton path, use $H(S, v)$, with visited set S and last vertex v; append only unused adjacent vertices. A final cycle count needs the additional edge back to the start. For a cycle coloring, compare proper row colorings with equal versus unequal end colors; identifying equal ends yields $R_n + R_{n-1} = q(q-1)^{n-1}$.

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 1 / 方法 1 [CO05-M01] (continued / 续)

哈密顿路径可用已访集合与末点状态 $H(S, v)$，只添加未访问相邻点；若计圈，还须检验最后能接回起点。环染色可按一行合法染色的两端相同或不同分类，合并同色端点得到 $R_n + R_{n-1} = q(q-1)^{n-1}$。

Practice / 配套练习: AIME-2001-I-Q14 (WB p. 677), AIME-2001-I-Q15 (WB p. 744), AIME-2005-I-Q13 (WB p. 679), AIME-2007-II-Q06 (WB p. 671), AIME-2008-I-Q11 (WB p. 681), AIME-2013-II-Q09 (WB p. 672), AIME-2015-II-Q12 (WB p. 682), AIME-2016-II-Q12 (WB p. 683), AIME-2021-II-Q08 (WB p. 788)

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 2 / 方法 2 [CO05-M02]

Extend acyclic paths by their accumulated totals

按累计总量扩展无环路径

Procedure

1. Use coordinates, total weight, or another quantity that strictly increases at every step.

2. Write a predecessor or forward-extension recurrence over that quantity, storing any running-value or turn information needed.

3. For a fixed multiset of moves, check the result independently with a multinomial interleaving count.

4. If turns are prescribed, replace maximal directional runs by positive compositions before summing.

通用步骤

1. 使用坐标、总重量等每步严格增加的量。

2. 按该量写前驱或向前扩展递推，并记录所需运行值或转弯信息。

3. 固定动作重数时，可用多项式交错计数独立核验。

4. 若规定转弯数，先把极大方向段转成正有序分拆，再求和。

> Example. A path uses steps $(1,0),(0,1),(2,1)$. Count paths from $(0,0)$ to $(3,2)$.

> 例题：路径允许步长 $(1,0),(0,1),(2,1)$，求从 $(0,0)$ 到 $(3,2)$ 的路径数。

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 2 / 方法 2 [CO05-M02] (continued / 续)

Worked solution

1. Let $F(x,y)=F(x-1,y)+F(x,y-1)+F(x-2,y-1)$, with $F(0,0)=1$ and out-of-range states zero.

2. Alternatively, without a long step there are three horizontal and two vertical moves, giving $\binom{5}{2}=10$.

3. With one long step, one move of each of the three types remains, giving $3!=6$; two long steps exceed the horizontal total. Thus $F(3,2)=16$.

例题解析

1. 令 $F(x,y)=F(x-1,y)+F(x,y-1)+F(x-2,y-1)$，初值 $F(0,0)=1$，范围外状态为零。

2. 独立核验：无长步时有三横两竖，排列数为 $\binom{5}{2}=10$。

3. 恰一长步时三类动作各一次，有 $3!=6$ 种；两长步横总量过大。因此共有十六条。

A recurrence ordered by total weight remains finite even when its running value can be negative.

即使运行值可为负，只要按严格增加的总重量排序，递推仍然有限。

For a nonstandard arithmetic operation, track both ordinary accumulated total s and running result v. A move of positive weight t advances to total $s+t$, so no cycle is possible in the dynamic program. A parity invariant may simplify the transition, but does not replace counting every admissible weight choice.

非标准运算可同时记录普通累计总量 s 与运行结果 v。新增正重量 t 后总量变为 $s+t$，故动态规划不会循环。奇偶不变量可简化转移，但不能替代对每个合法重量选择的计数。

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 2 / 方法 2 [CO05-M02] (continued / 续)

Practice / 配套练习: AIME-2005-I-Q13 (WB p. 679), AIME-2007-I-Q06 (WB p. 670), AIME-2018-II-Q08 (WB p. 674), AIME-2024-I-Q06 (WB p. 675), AIME-2026-II-Q14 (WB p. 689)

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 3 / 方法 3 [CO05-M03]

Insert and delete an extreme object to obtain a product recurrence

插删极值对象得到乘法递推

Procedure

1. Identify where the largest label must occur in a legal object.

2. Show deleting it preserves every local condition.

3. Count its legal reinsertion slots in every smaller object and use the resulting recurrence with its smallest valid initial case.

通用步骤

1. 确定合法对象中最大标签可出现的位置。

2. 证明删去它仍保持每个局部条件。

3. 对每个较小对象计算重新插入的合法位置数，结合最小有效初值递推。

> Example. An increasing rooted binary tree is required to be a single path and has labels $1, 2, 3, 4, 5$, increasing away from the root. Count its left-right shapes.

> 例题：递增二叉树只允许形成一条从根出发的路径，标签一至五沿路径递增，求左右子节点形状数。

Worked solution

1. The labels along the path are forced to be $1, 2, 3, 4, 5$.

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 3 / 方法 3 [CO05-M03] (continued / 续)

2. Insert the next largest label as the left or right child of the current final leaf. Deleting that largest leaf uniquely reverses this operation.

3. Four insertions each have two choices, so there are $2^4 = 16$ shapes.

例题解析

1. 路径上标签顺序被迫为一、二、三、四、五。

2. 每次把新增最大标签接为当前末叶的左或右子节点，删除最大叶可唯一逆转。

3. 四次插入各两种选择，共十六种形状。

If the deletion joins two former neighbors, check their new adjacency explicitly.

若删除后两位旧邻居相接，须明确检验新相邻关系。

For a permutation whose downward jumps are at most two, the new maximum may go at the end or immediately before either of the next two largest labels. Deleting it is safe because the newly adjacent predecessor is at most the old maximum minus one and its successor at least the old maximum minus two. This proves the three-slot recurrence rather than merely guessing it.

若排列每次下降至多二，新增最大值只能放末尾，或紧靠在接下来两个最大标签之前。删除它后，新的前项至多为旧最大值减一、后项至少为旧最大值减二，所以新邻接仍合法，由此严格得到三位置递推。

Practice / 配套练习: AIME-2006-I-Q11 (WB p. 680), AIME-2015-II-Q10 (WB p. 673)

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 4 / 方法 4 [CO05-M04]

Split at the earliest closed prefix or unique defect

按最早闭合前缀或唯一缺陷分解

Procedure

1. Choose a distinguished split that every object has uniquely.

2. Count the first component and the independent remainder.

3. Write a convolution recurrence, subtracting earlier component sizes when solving for the indecomposable count.

4. For a unique adjacent defect, forbid its neighboring extra defects before multiplying the two side counts.

通用步骤

1. 选择每个对象唯一具有的特殊分界。

2. 分别计算第一部件与独立余部。

3. 写卷积递推，求不可分计数时扣除更早部件大小。

4. 唯一相邻缺陷出现后，先禁止它两旁新增缺陷，再相乘左右计数。

> Example. Let $c_1, \ldots, c_5 = 1, 1, 3, 13, 71$ count indecomposable permutations, where no proper prefix is an initial value set. How many permutations of six labels have exactly two such consecutive components?

> 例题：不可分排列指没有真前缀恰为初始值集合，已知一至五项计数为 $1, 1, 3, 13, 71$。六标签排列中，恰有两个这样的连续部件者有多少？

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 4 / 方法 4 [CO05-M04] (continued / 续)

Worked solution

1. The first component has a unique size k, from one to five. Its values must be $1, \ldots, k$ and the second uses the others.

2. Each size contributes $c_k c_{6-k}$, with the second component standardized without changing its order.

3. The total is $2(1 \cdot 71) + 2(1 \cdot 13) + 3^2 = 177$. The earliest closed prefix makes the split unique.

例题解析

1. 第一部件大小 k 唯一，范围一至五，其值必须为一至 k，第二部件使用其余值。

2. 每个大小贡献 $c_k c_{6-k}$，第二部件标准化不改变相对顺序。

3. 总数为 $2(1 \cdot 71) + 2(1 \cdot 13) + 3^2 = 177$。最早闭合前缀保证分界唯一。

The word earliest or unique is a proof device that prevents a convolution from overcounting.

“最早”或“唯一”是防止卷积重复计数的关键。

An arbitrary permutation splits at its earliest closed prefix, giving $n! = \sum_{k=1}^{n} c_k (n-k)!$. For independent sets in a line, the choose-or-omit last position recurrence is $T_n = T_{n-1} + T_{n-2}$, with $T_0 = 1, T_1 = 2$. A prescribed unique adjacent pair leaves two independent no-adjacency blocks after excluding the two positions immediately beside it.

任意排列按最早闭合前缀分解，得到 $n! = \sum_{k=1}^{n} c_k (n-k)!$。直线上的不相邻子集按末位选或不选满足 $T_n = T_{n-1} + T_{n-2}$，初值 $T_0 = 1, T_1 = 2$。指定唯一相邻对后，排除其两侧紧邻位置，余下为两个独立不相邻区块。

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 4 / 方法 4 [CO05-M04] (continued / 续)

Practice / 配套练习: AIME-2018-II-Q11 (WB p. 685), AIME-2023-I-Q11 (WB p. 687)

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 5 / 方法 5 [CO05-M05]

Reverse a deterministic process into exact first-hit levels

反转确定性过程得到首次命中层

Procedure

1. Define whether the hitting index counts terms or moves.

2. List every predecessor of a state from the original forward rule, including exceptional absorbing states.

3. Start at the target and generate one reverse level at a time, proving predecessor classes do not create duplicates or earlier hits.

4. For a queue or stack, reverse the full state or track original positions rather than guessing final labels.

通用步骤

1. 明确命中下标计算项数还是移动次数。

2. 从原正向规则列全每态前驱，包括特殊吸收态。

3. 从目标逐层反向生成，证明各前驱类不重复也不提前命中。

4. 队列或堆叠应逆转完整状态或追踪原位置，不能猜测最终标签。

> Example. Define $f(1)=1$; for even n, let $f(n)=n/2$; for odd $n>1$, let $f(n)=n-1$. Which starts reach one for the first time on their fourth term?

> 例题：定义 $f(1)=1$，偶数除以二，奇数且大于一时减一。哪些起点在第四项首次到一？

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 5 / 方法 5 [CO05-M05] (continued / 续)

Worked solution

1. The first level is $T_1 = \{1\}$ and the second is $T_2 = \{2\}$; the absorbing predecessor one must be excluded.

2. An even target z has predecessors $2z$ and $z+1$; an odd target greater than one has only $2z$. Thus $T_3 = \{3, 4\}$.

3. Reversing once more gives $T_4 = \{6, 8, 5\}$. Direct trajectories verify all three first-hit lengths.

例题解析

1. 初层为 $T_1 = \{1\}$、次层为 $T_2 = \{2\}$，须排除吸收前驱一。

2. 偶目标 z 有前驱 $2z, z+1$，大于一的奇目标只有 $2z$，所以第三层为 $\{3, 4\}$。

3. 再反向一步得到 $T_4 = \{6, 8, 5\}$，逐一正向检查均在第四项首次到一。

Different reverse branches can merge only if the forward map sends one source to two outputs, which a deterministic map cannot do.

同一反向层不同目标的前驱不会重复，因为确定性映射不可能把一个起点送到两个输出。

Practice / 配套练习: AIME-2000-I-Q15 (WB p. 676), AIME-2004-I-Q15 (WB p. 678)

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 6 / 方法 6 [CO05-M06]

Normalize divisor chains to smaller integer states

将整除链归一到较小整数状态

Procedure

1. Choose an initial term or a final divisor so the remainder has a smaller integer parameter.

2. Divide out the common scale and translate strict growth into the new first-term restriction.

3. Sum only legal divisors, give the one-term case separately, and memoize by the strictly decreasing parameter.

通用步骤

1. 选择首项或末端约数，使余部具有较小整数参数。

2. 除去公共比例，把严格增长转成新首项限制。

3. 仅对合法约数求和，单独给单项情况，并按严格减小的参数缓存。

> Example. Count chains $1 = a_0 < a_1 < \cdots < a_t = 12$ in which every term divides the next.

> 例题：求从一严格递增到十二、且每项整除下一项的链数。

Worked solution

1. Let $D(n)$ count such chains ending at n, with $D(1) = 1$. The penultimate term is a unique proper divisor d of n, so $D(n) = \sum_{d \mid n, d < n} D(d)$.

2. This gives $D(2) = D(3) = 1, D(4) = 2, D(6) = 3$.

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 6 / 方法 6 [CO05-M06] (continued / 续)

3. Therefore $D(12) = D(1) + D(2) + D(3) + D(4) + D(6) = 8$. Each chain is counted by its unique penultimate term.

例题解析

1. 令 $D(n)$ 计终点为 n 的链，$D(1) = 1$。倒数第二项是唯一真约数 d，故 $D(n) = \sum_{d \mid n, d < n} D(d)$。

2. 依次得 $D(2) = D(3) = 1, D(4) = 2, D(6) = 3$。

3. 所以 $D(12) = 1 + 1 + 1 + 2 + 3 = 8$，每链按唯一倒数第二项计一次。

Dividing the remaining terms also changes their sum; subtract the removed normalized first term.

将余项除以首项会改变总和，还须扣除已移去的归一首项。

For a strictly increasing positive divisor chain of total s and first term at least two, define $g(s)$. The one-term chain contributes one. If its first term is $d \mid s$, $2 \le d < s$, dividing later terms by d leaves sum $s/d - 1$ and first term at least two. Hence $g(s) = 1 + \sum_{2 \le d < s, d \mid s} g(s/d - 1)$, with impossible small states zero.

对总和 s、首项至少二的严格递增正整数整除链，令计数为 $g(s)$。单项链贡献一；若首项为 $d \mid s$ 且 $2 \le d < s$，后续项除以 d 后总和为 $s/d - 1$，首项仍至少二。因此 $g(s) = 1 + \sum_{2 \le d < s, d \mid s} g(s/d - 1)$，不可行小状态设零。

Practice / 配套练习: AIME-2019-II-Q12 (WB p. 686)

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 7 / 方法 7 [CO05-M07]

Separate active and absorbing states in a weighted recurrence

在加权递推中区分活动与吸收状态

Procedure

1. Choose states with the same future transition law; verify any symmetry-based merging.

2. For counts, sum legal predecessor counts; for probabilities, multiply by transition probabilities.

3. Remove paths as soon as they hit the absorbing event, so later steps cannot count them again.

4. For expected remaining time, use $E = 1 + \sum pE_{next}$ with zero at absorbing states.

通用步骤

1. 选择未来转移规律相同的状态，按对称合并前须验证。

2. 计路径时相加合法前驱数，计概率时乘转移概率。

3. 一旦命中吸收事件便移除路径，避免以后再次计入。

4. 剩余期望用 $E = 1 + \sum pE_{next}$，吸收态取零。

> Example. Flip a fair coin until two consecutive heads appear. Find the expected number of flips.

> 例题：反复掷公平硬币，首次连续两次正面时停止，求掷币次数的期望。

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 7 / 方法 7 [CO05-M07] (continued / 续)

Worked solution

1. Let E_0 be the expectation with no terminal head and E_1 with one terminal head.

2. First-step equations are $E_0 = 1 + (E_0 + E_1)/2$ and $E_1 = 1 + E_0/2$; the second head reaches an absorbing state with zero remaining time.

3. Solving gives $E_0 = 6$. The time already spent in the next flip supplies the leading one in each equation.

例题解析

1. 令 E_0 为末尾无正面时的期望，E_1 为末尾恰有一个正面时的期望。

2. 首次步骤方程为 $E_0 = 1 + (E_0 + E_1)/2$、$E_1 = 1 + E_0/2$，再出正面进入剩余时间零的吸收态。

3. 解得 $E_0 = 6$。每式开头的一是下一次掷币已经花费的时间。

Merged states must have identical transition totals into every merged class, not merely look symmetric in a sketch.

合并状态须对每个合并类别具有相同转移总量，不能只凭图上看似对称。

For a prefix success boundary, retain only states with hit count $j \le \lfloor pn \rfloor$ at each required prefix length; a final total condition is an additional terminal filter. For predicting the next color of an unchanged deck, the prediction does not affect later information or transitions, so choosing the currently more numerous color maximizes the immediate success probability and is globally optimal.

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 7 / 方法 7 [CO05-M07] (continued / 续)

对前缀成功率边界，在每个要求的前缀长度只保留命中数 $j \le \lfloor pn \rfloor$ 的状态，最终命中总数另作终态筛选。预测剩余牌颜色时，预测本身不改变以后的信息与转移，因此每次猜当前数量较多的颜色最大化即时成功率，也给出整体最优策略。

Practice / 配套练习: AIME-2002-II-Q12 (WB p. 790), AIME-2018-I-Q14 (WB p. 684), AIME-2021-I-Q12 (WB p. 781)

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 8 / 方法 8 [CO05-M08]

Count exact covers with the first uncovered element

用首个未覆盖元素计算精确覆盖

Procedure

1. Represent every allowed piece by its set of occupied cells.

2. Choose the first uncovered cell in a fixed order and branch only over legal pieces containing it and disjoint from the covered set.

3. Stop with count one exactly when every cell is covered and the prescribed number of pieces is used.

4. Save the candidate list, recurrence, and resulting cover list or checksums so a computed answer can be independently reproduced.

通用步骤

1. 用占用单元集合表示每个允许部件。

2. 按固定顺序取首个未覆盖格，只对含该格且不与已覆盖集相交的部件分支。

3. 仅在全部单元覆盖且使用指定部件数时计一。

4. 保存候选表、递推与最终覆盖列表或校验值，使计算结果可独立复现。

> Example. Use three dominoes to cover a two-row, three-column board. Count the tilings by branching at the upper-left cell.

> 例题：用三块骨牌覆盖两行三列棋盘，按左上首格分支计算铺法。

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 8 / 方法 8 [CO05-M08] (continued / 续)

Worked solution

1. If its domino is vertical, the remaining two-by-two square has two covers: both vertical or both horizontal.

2. If it is horizontal, the lower-left cell must also belong to a horizontal domino; the last column is then vertical, giving one cover.

3. The branches are disjoint because the piece covering the chosen cell is unique. Total $2+1=3$.

例题解析

1. 若首格骨牌竖放，余下二乘二方块有全竖或全横两种。

2. 若首格横放，左下也须横放，末列被迫竖放，只有一种。

3. 各分支按覆盖首格的唯一部件区分，互不重复，共三种。

Pruning is valid only when its condition is proved necessary; the unpruned pivot recurrence already counts each partition once.

剪枝条件必须证明为必要条件；即使不剪枝，首格递推本身已保证每个分拆恰计一次。

In symbols, $C(U,j)=\sum_L C(U\cup L,j-1)$ over candidate pieces containing the first cell outside U and avoiding U. Set $C(U,0)=1$ for full coverage and zero otherwise. The 10×10 five-loop problem has a companion certificate listing all 2025 candidate boundary masks, all 83 accepted covers, and the exact search totals; no unproved geometric slab classification is needed for the count.

Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

Method 8 / 方法 8 [CO05-M08] (continued / 续)

符号上，$C(U,j)=\sum_L C(U\cup L,j-1)$，其中 L 含首个不在 U 中的单元且不交 U。$j=0$ 时仅全部覆盖取一。10×10 棋盘的五环题配有证书，记录全部二千零二十五个候选边界掩码、八十三个有效覆盖及精确搜索总量，计数不依赖未经证明的条块分类。

Practice / 配套练习: AIME-2026-I-Q15 (WB p. 688)

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 1 / 方法 1 [CO06-M01]

Draw the correct graph before counting. Vertices may represent cells, faces, subsets, or states; edges may represent adjacency, compatibility, or an incidence constraint. Geometric counts also need a proof that every construction is legal and that its witness multiplicity is known.

计数前先建立正确的图。顶点可以代表单元格、面、子集或状态，边可以代表相邻、相容或关联限制。几何计数还须证明每个构造合法，并明确见证重数。

Translate adjacency into a finite graph

把相邻关系转成有限图

Procedure

1. Specify exactly what vertices and edges represent, and whether the edges are directed.

2. Use coordinates, parity, a dual graph, or an explicit edge list to prove the representation preserves every legal move.

3. Count graph objects or states, retaining visited-set memory or reachability constraints when required.

通用步骤

1. 明确顶点与边各代表什么，以及边是否有方向。

2. 用坐标、奇偶、对偶图或明确边表，证明图表示保留全部合法动作。

3. 再计算图对象或状态，必要时保留已访问集合或可达性条件。

> Example. A counter moves between side-adjacent cells of a two-by-three board. Can it return to its start after an odd number of moves?

> 例题：棋子在两行三列棋盘的共边相邻格之间移动，能否在奇数步后回到起点？

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 1 / 方法 1 [CO06-M01] (continued / 续)

Worked solution

1. Color each cell by the parity of its row plus column. Every graph edge joins opposite colors.

2. Each move flips the color, so after an odd number the counter is on the other color.

3. It cannot be at the starting cell. This is a bipartite-graph obstruction independent of the chosen route.

例题解析

1. 按行号加列号的奇偶给格子两色染色，每条图边连接异色格。

2. 每步翻转颜色，奇数步后必位于另一色。

3. 因此不能回起点，这是二部图障碍，与具体路线无关。

For face-adjacency problems, the dual graph often turns a solid into a small familiar graph.

面的相邻问题常可通过对偶图，把立体变成小型常见图。

An octahedron has the cube as its face-adjacency dual. Label cube vertices by binary triples: adjacent vertices differ in one bit, so avoiding face adjacency means Hamming distance two or three. For a digit rule, use digits as vertices and allowed consecutive digit blocks as directed edges; a directed cycle may produce unbounded lengths, while a finite acyclic graph has a longest path.

八面体的面相邻对偶图是立方体。用三位二进制标顶点，相邻恰差一位，因此避开共边面意味着相差两位或三位。数位规则可把数位作顶点、允许相邻数位块作有向边；有向环可能产生无限长度，无环有限图则有最长路径。

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 1 / 方法 1 [CO06-M01] (continued / 续)

Practice / 配套练习: AIME-2001-I-Q15 (WB p. 744), AIME-2001-II-Q01 (WB p. 559), AIME-2012-I-Q11 (WB p. 799), AIME-2018-I-Q14 (WB p. 684)

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 2 / 方法 2 [CO06-M02]

Count geometric witnesses and certify their multiplicity

计几何见证并证明其重数

Procedure

1. Choose an elementary witness: a vertex tuple, an apex, a pair of lines, or a pair of monotone branches.

2. Classify all degenerate or exceptional configurations and determine their witness multiplicity.

3. Subtract invalid cases or correct multiplicities only after proving every remaining witness produces one valid object.

4. For intersections of curves, prove existence and uniqueness within each counted branch pair; monotonicity of both curves alone does not suffice.

通用步骤

1. 选择基本见证：顶点元组、顶角、一对直线或一对单调分支。

2. 分类全部退化与例外构型，并确定见证重数。

3. 证明剩余每个见证均产生合法对象之后，再扣除无效情形或修正重数。

4. 曲线交点按分支对计数时，须证明每对中的存在性与唯一性，两曲线各自单调并不足够。

> Example. Four vertices of a square and its center are marked. How many nondegenerate triangles use three marked points?

> 例题：标记正方形四顶点与中心，从中选三个点能组成多少个非退化三角形？

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 2 / 方法 2 [CO06-M02] (continued / 续)

Worked solution

1. All triples number $\binom{5}{3} = 10$.

2. Exactly two triples are collinear: the center with either opposite vertex pair. No other line contains three marked points.

3. Subtract these two invalid witnesses to obtain eight triangles.

例题解析

1. 全部三元组有 $\binom{5}{3} = 10$ 个。

2. 恰有两个共线三元组，即中心与任一对对顶点；没有其他含三个标记点的直线。

3. 扣除两个退化见证，得到八个三角形。

A count of branch rectangles requires a local intersection certificate; never infer one crossing from a picture.

分支矩形的计数需要局部交点证书，不能凭图猜定一个交点。

An isosceles triangle has one apex witness except for an equilateral triangle, which has three. For a regular n-gon, apex counting gives $n\lfloor (n-1)/2 \rfloor$ incidences; subtract two extra copies of each equilateral triangle. For pairwise-intersecting edge families, two edges ab, ac force every edge either through a or equal to bc, leaving star families and the triangle exception.

For a curve-branch pair, use endpoint signs for existence. Opposite orientations can give direct uniqueness; equal orientations need a further derivative or convexity argument. A sufficient crossing certificate is that the difference of the two graphs has the same nonzero derivative sign at every zero: two such crossings would force an intervening zero of the opposite orientation. The full trigonometric-fold certificate is supplied with its companion solution.

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 2 / 方法 2 [CO06-M02] (continued / 续)

一般等腰三角形只有一个顶角见证，等边三角形有三个。正 n 边形按顶角计数有 $n\lfloor(n-1)/2\rfloor$ 次关联，应对每个等边三角形扣掉额外两份。两两相交边族中，已有 ab, ac 后，其他边要么经过 a，要么为 bc，只剩星形与三角形例外。

每对曲线分支可用端点符号证明存在；方向相反时可直接唯一，方向相同时还须导数或凸性。一个充分的穿越证书是两图差在每个零点的导数均具有相同非零符号；若有两个同向穿越，中间必有反向零点。完整三角折叠证书见配套解析。

Practice / 配套练习: AIME-2002-I-Q05 (WB p. 690), AIME-2003-I-Q06 (WB p. 445), AIME-2004-I-Q03 (WB p. 691), AIME-2010-II-Q11 (WB p. 667), AIME-2015-II-Q05 (WB p. 727), AIME-2017-I-Q01 (WB p. 631), AIME-2017-II-Q13 (WB p. 703), AIME-2018-I-Q07 (WB p. 699), AIME-2022-II-Q05 (WB p. 695), AIME-2023-I-Q03 (WB p. 661), AIME-2023-II-Q11 (WB p. 645), AIME-2024-I-Q12 (WB p. 179)

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 3 / 方法 3 [CO06-M03]

Count planar regions by Euler or incremental crossings

用欧拉公式或逐段交点计平面区域

Procedure

1. Replace each intersection by a graph vertex and split every segment there.

2. Use $V - E + F = 1 + C$ with the correct component and exterior-face counts, or add one segment at a time.

3. In a disk partitioned by full chords, a new chord with no overlapping old edge and r distinct interior crossings adds $r + 1$ regions. Coincident crossings count as one point; free-ended segments need a separate Euler calculation.

通用步骤

1. 将每个交点作为图顶点，并把线段在交点处分边。

2. 用 $V - E + F = 1 + C$ 并正确处理连通分量与外部面，或逐条添加线段。

3. 圆盘已由完整弦分割时，新增弦若不与旧边重合，且有 r 个不同内部交点，就新增 $r + 1$ 个区域。重合交点只计一个点；端点自由的线段须另用欧拉公式计算。

> Example. Three distinct diameters of a disk pass through its center. How many disk regions do they form, and why does a generic pair-crossing formula overcount?

> 例题：圆盘内三条不同直径都过圆心，形成多少个区域？为什么一般位置的线段对交点公式会多数？

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 3 / 方法 3 [CO06-M03] (continued / 续)

Worked solution

1. The first diameter creates two regions. The second crosses it at one point and adds two, giving four.

2. The third meets both old diameters at the same center point, so it also adds two, giving six.

3. Using three pairwise crossings in the generic formula $1+3+3$ gives seven because the last two crossings are not distinct.

例题解析

1. 第一条直径得到两个区域，第二条有一个交点，新增两个，得到四个。

2. 第三条与旧直径都在同一圆心相交，仍只新增两个，最终六个。

3. 把三对相交代入一般公式 $1+3+3$ 会得七，因为第三条上的两个交点并不不同。

Probability-one general position is valid only after excluding the finitely described degeneracies.

只有排除明确描述的退化事件后，才能使用概率一的一般位置。

For complete bipartite segments between m, n ordered points on parallel lines with no triple interior crossing, $X=\binom{m}{2}\binom{n}{2}$ crossings and Euler give $mn+X-1$ bounded faces. For a random chord arrangement, linearity of expectation applies to crossing indicators; endpoint interleaving determines whether two chords cross.

平行两线上分别有 m, n 个有序点，连全两组之间线段且无三线内部共点，则交点 $X=\binom{m}{2}\binom{n}{2}$，欧拉公式给有界面数 $mn+X-1$。随机弦安排可对相交指示变量使用期望线性性，两弦是否相交由端点是否交错决定。

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 3 / 方法 3 [CO06-M03] (continued / 续)

Practice / 配套练习: AIME-2022-II-Q09 (WB p. 700), AIME-2025-I-Q13 (WB p. 782)

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 4 / 方法 4 [CO06-M04]

Normalize lattice directions and count crossed cells

归一格点方向并计算穿越单元

Procedure

1. Divide an integer direction vector by the gcd of its coordinates to obtain the primitive step.

2. Use coordinate spans to bound possible directions when a line contains a prescribed number of grid points.

3. Normalize a line by its first boundary point or least coordinate to avoid counting it from several points.

4. For a segment joining lattice points, count crossed unit-cell interiors with the gcd correction.

通用步骤

1. 将整数方向向量各坐标除以最大公因数，得到本原步长。

2. 规定直线含多少格点时，利用坐标跨度限制方向。

3. 按首个边界点或最小坐标归一每条线，避免从多个点重复计。

4. 格点连线穿越单位格内部时，使用最大公因数修正。

> Example. Find all lattice points on the segment from $(0,0)$ to $(6,4)$, and count the unit-square interiors crossed by the segment.

> 例题：求从 $(0,0)$ 到 $(6,4)$ 线段上的全部整点，并计算线段穿过多少个单位正方形内部。

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 4 / 方法 4 [CO06-M04] (continued / 续)

Worked solution

1. The coordinate gcd is two, so the primitive step is $(3, 2)$. The lattice points are $(0, 0)$, $(3, 2), (6, 4)$.

2. A segment with positive integer spans a, b crosses $a + b - \gcd(a, b)$ unit-square interiors: simultaneous grid-line crossings at internal lattice points merge two changes.

3. The count here is $6 + 4 - 2 = 8$. Using the nonprimitive step would miss the interior lattice point.

例题解析

1. 坐标最大公因数为二，本原步长为 $(3, 2)$，整点为 $(0, 0), (3, 2), (6, 4)$。

2. 横纵正整数跨度为 a, b 的线段穿过 $a + b - \gcd(a, b)$ 个单位格内部；内部整点同时穿两条格线，将两次变化合并。

3. 本题为 $6 + 4 - 2 = 8$；用非本原方向会漏掉中间整点。

A line and a directed traversal are different objects; identify opposite primitive directions when counting lines.

直线与有向行进不同，计直线时须归并相反本原方向。

Practice / 配套练习: AIME-2007-II-Q05 (WB p. 692), AIME-2017-II-Q14 (WB p. 704)

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 5 / 方法 5 [CO06-M05]

Decompose regular graphs into cycles and gcd components

把规则图分解为循环与最大公因数组件

Procedure

1. Use degree two or a fixed modular step to identify disjoint cycle components.

2. For a step k modulo n, there are $\gcd(n,k)$ cycles, each of length $n/\gcd(n,k)$.

3. Apply the local cycle count, then multiply across independent components; remember diameter or single-edge degeneracies.

4. If the labels of components are being assigned, correct for indistinguishable components of the same size.

通用步骤

1. 通过度为二或固定模步长识别互不相交的循环分量。

2. 模 n 步长 k 产生 $\gcd(n,k)$ 个长度 $n/\gcd(n,k)$ 的循环。

3. 使用局部圈计数，再按独立分量相乘，并处理直径或单边退化。

4. 若需给分量分配标签，还须按相同大小分量的可交换性修正。

> Example. A graph is the disjoint union of a four-cycle and a six-cycle. Count its perfect matchings.

> 例题：一个图是四循环与六循环的不交并，求其完美匹配数。

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 5 / 方法 5 [CO06-M05] (continued / 续)

Worked solution

1. In an even cycle, choosing which of the two edges at a fixed vertex is matched forces every remaining edge alternately.

2. Each component therefore has exactly two perfect matchings.

3. The components are disjoint, so the total is $2 \cdot 2 = 4$. An odd-cycle component would make the total zero.

例题解析

1. 偶循环中，选定某顶点两条邻边中哪一条属于匹配后，其余边被迫交替确定。

2. 因此每个分量恰有两种完美匹配。

3. 不交分量独立，总数为四；若含奇循环分量，则总数为零。

A permutation cycle is oriented; an undirected graph cycle identifies its two orientations.

置换循环有方向，无向图循环则归并正反两方向。

An undirected cycle on k fixed labels has $(k-1)!/2$ realizations for $k \geq 3$. A proper q-coloring of an n-cycle has $(q-1)^n + (-1)^n(q-1)$ choices, obtainable by the row-versus-cycle recurrence. A primitive regular-star step requires $\gcd(n,k) = 1$; complementing the step reverses traversal, and step one must be excluded if crossings are required.

$k \geq 3$ 个固定标签的无向圈有 $(k-1)!/2$ 种。n 循环的合法 q 色染色数为 $(q-1)^n + (-1)^n(q-1)$，可由一行与环的关系递推得到。规则星形的本原步长须与 n 互素；互补步长反转方向，要求交叉时还须排除步长一。

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 5 / 方法 5 [CO06-M05] (continued / 续)

Practice / 配套练习: AIME-2004-I-Q08 (WB p. 604), AIME-2012-II-Q14 (WB p. 701), AIME-2014-II-Q13 (WB p. 747), AIME-2016-II-Q12 (WB p. 683), AIME-2024-I-Q11 (WB p. 669), AIME-2025-II-Q11 (WB p. 707)

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 6 / 方法 6 [CO06-M06]

Use layers and distance levels to bound simple paths

用层次与距离等级限制简单路径

Procedure

1. Identify a quantity that changes monotonically along the path, such as height.

2. Decompose the path into one visit to each layer and count its simple horizontal pieces.

3. If edge lengths must increase strictly, list all possible lengths to bound the edge count.

4. For an extremal path, enforce every required length in reverse order and exhibit all surviving constructions.

通用步骤

1. 找出沿路径单调变化的量，例如高度。

2. 把路径分成对各层的一次访问，计算各层内简单横向部分。

3. 若步长严格增加，列全可用长度以限制边数。

4. 最大路径须反向依次满足每个必要长度，并展示所有幸存构造。

> Example. A graph has top T, bottom B, and one middle triangular cycle. T joins every middle vertex, and every middle vertex joins B. Count simple paths from T to B with no return to an earlier level.

> 例题：图由顶点 T、底点 B 与中间三角循环组成，顶点与底点各连接全部中层点。求不返回较高层的简单 T 到 B 路径数。

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 6 / 方法 6 [CO06-M06] (continued / 续)

Worked solution

1. Choose the first middle vertex in three ways.

2. From it, a simple path within the triangle has one zero-edge option and two options each for one or two edges, totaling five.

3. It then goes to B uniquely, giving $3 \cdot 5 = 15$ paths. Turning back around the triangle would revisit a vertex and is excluded.

例题解析

1. 进入中层的首点有三种选择。

2. 从该点沿三角形的简单路径有零步一种、一步和两步各两种，共五种。

3. 之后唯一走到底点，所以有十五条，沿循环折返会重访顶点，已排除。

A bound from the number of distance levels needs an explicit path to prove that it is achievable.

由距离等级数得到的上界，还须有具体路径证明可达到。

Practice / 配套练习: AIME-2008-II-Q10 (WB p. 697), AIME-2016-I-Q03 (WB p. 694)

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 7 / 方法 7 [CO06-M07]

Normalize graph equilibrium by degree and apply a maximum principle

按度数归一平衡并使用最大值原理

Procedure

1. Write the fixed transfer equation $c_v = \sum_{u \sim v} c_u / d_u$.

2. Divide by d_v so c_v / d_v is the average of its neighbors corresponding ratios.

3. At a maximal ratio, every neighboring ratio must equal it; connectedness propagates equality.

4. Use the total amount to recover the common constant and all vertex allocations.

通用步骤

1. 写固定转移方程 $c_v = \sum_{u \sim v} c_u / d_u$。

2. 除以 d_v，使 c_v / d_v 成为邻居相应比值的平均。

3. 最大比值处所有邻居都须同值，连通性将等值传遍全图。

4. 用总量求公共常数及每个顶点的分配。

> Example. Three vertices form a path $A - B - C$. At every simultaneous step each sends its entire amount equally to its neighbors. A nonnegative total of forty is unchanged at every vertex. Find the amounts.

> 例题：三个顶点组成路径 $A - B - C$，每步同时把全部数量平分给邻居。非负总量为四十且每点数量保持不变，求各点数量。

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 7 / 方法 7 [CO06-M07] (continued / 续)

Worked solution

1. The degrees are one, two, one. Degree-normalized ratios must be constant by the connected-graph maximum principle.

2. Therefore the amounts are proportional to $1:2:1$.

3. The common factor is $40/4=10$, so the amounts are $(10,20,10)$. Direct transfer verifies the fixed state.

例题解析

1. 度数为一、二、一，连通图最大值原理使按度归一的比值恒定。

2. 故数量比为一比二比一。

3. 公共系数为十，数量为 $(10,20,10)$，直接转移也可核验。

The averaging identity uses an undirected graph; a directed transfer graph has a different stationary law.

此平均关系使用无向图，有向转移图具有不同的平衡规律。

Practice / 配套练习: AIME-2012-I-Q07 (WB p. 698)

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 8 / 方法 8 [CO06-M08]

Analyze tournament strong components before counting orientations

先分析竞赛图强分量再计方向

Procedure

1. A tournament has exactly one directed edge between each vertex pair.

2. Collapse strongly connected components; their directions form a total order.

3. Use small component sizes to determine the possible source or sink obstruction, then count its event intersections.

通用步骤

1. 竞赛图每对顶点间恰有一条有向边。

2. 将强连通分量缩点，各分量之间形成全序方向。

3. 利用小分量大小确定源点或汇点障碍，再计算相关事件交集。

> Example. Start with a directed triangle cycle and add a fourth vertex with one edge to each old vertex. How many choices of its three edge directions make the whole graph strongly connected?

> 例题：从一个有向三角循环出发，加第四顶点并与每个旧顶点各连一条有向边。三个方向有多少种选择使全图强连通？

Worked solution

1. The old cycle already allows every old vertex to reach every other.

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 8 / 方法 8 [CO06-M08] (continued / 续)

2. The new vertex must have at least one outgoing and one incoming edge. This condition also suffices, because the old cycle connects the entry and exit points.

3. Of the $2^3 = 8$ orientations, exclude all-outgoing and all-incoming, leaving six.

例题解析

1. 旧循环中各点已能互相到达。

2. 新顶点须至少有一条出边与一条入边；这也充分，因为旧循环连接进出点。

3. 八种方向中排除全出与全入，剩六种。

A nonstrong tournament need not have a global source or sink for arbitrary vertex counts.

一般大小的非强连通竞赛图未必有全局源点或汇点。

A nontrivial strongly connected tournament has at least three vertices. On five vertices, a nonstrong tournament cannot have both its first and last strong components nontrivial, since they would require at least six vertices. Thus its first or last component is a singleton, giving a global source or sink. This is why the source/sink inclusion-exclusion is complete specifically in the five-vertex problem.

非平凡强连通竞赛图至少有三个顶点。五顶点非强竞赛图的首尾强分量不可能都非平凡，否则至少需六点，因此首分量或末分量必为单点，形成全局源点或汇点。这才保证五顶点题中的源汇容斥完备。

Practice / 配套练习: AIME-2017-II-Q11 (WB p. 702)

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 9 / 方法 9 [CO06-M09]

Use circular order to characterize strict separation

用圆周顺序刻画严格分隔

Procedure

1. Restrict a separating circle to the common circumcircle of the points.

2. Two distinct circles meet at at most two points, so membership can change at most twice around the cyclic order.

3. For sufficiency, strictly separate a consecutive run by a line in the adjoining gaps, then approximate the finite half-plane choice by a sufficiently large circle.

通用步骤

1. 将分隔圆限制到所有点的公共外接圆上。

2. 两个不同圆至多相交两点，所以沿循环顺序成员身份至多改变两次。

3. 证明充分性时，在相邻空隙中用直线严格分隔连续段，再用足够大的圆在有限点集上近似该半平面。

> Example. Can a circle contain one opposite pair of a square strictly inside while placing the other opposite pair strictly outside?

> 例题：能否用一个圆，使正方形一对对顶点严格在圆内、另一对严格在圆外？

Worked solution

1. Along the square circumcircle the required membership pattern alternates inside, outside, inside, outside.

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 9 / 方法 9 [CO06-M09] (continued / 续)

2. This requires at least four boundary changes, but two distinct circles have at most two intersection points.

3. The circles cannot coincide under strict separation, so the configuration is impossible.

例题解析

1. 沿正方形外接圆，要求成员状态交替为内、外、内、外。

2. 这至少需要四次边界变化，而两个不同圆至多有两个交点。

3. 严格分隔也不允许两圆重合，因此不可能。

Strict separation leaves positive margins on a finite point set; those margins justify the large-circle approximation.

有限点集的严格分隔具有正余量，这一余量保证大圆近似有效。

Practice / 配套练习: AIME-2014-I-Q05 (WB p. 693)

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 10 / 方法 10 [CO06-M10]

Count chord-supported rectangles by perpendicular offsets

按垂直弦的偏移量计算弦支撑矩形

Procedure

1. Group chords by direction and pair perpendicular direction classes.

2. Represent each chord by its signed distance from the circumcenter.

3. Choose two offsets in each class; all four corners lie on their finite chords exactly when the largest squared offsets sum to at most the radius squared.

4. Count offset-pair multiplicities, taking care not to count the same perpendicular class pair twice.

通用步骤

1. 将弦按方向分组，再配对互相垂直的方向类。

2. 用弦到外心的带符号距离表示每条弦。

3. 各类选两个偏移量，四角都在有限弦上当且仅当两类最大偏移绝对值的平方和不超过半径平方。

4. 按偏移对重数计数，避免同一垂直方向对被数两次。

> Example. In a unit disk, two vertical chords have offsets $\pm 3/5$. What is the greatest symmetric horizontal offset $\pm t$ that forms a rectangle with corners on all four chord segments?

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 10 / 方法 10 [CO06-M10] (continued / 续)

> 例题：单位圆盘中两条竖弦偏移量为 $\pm 3/5$，两横弦偏移量为 $\pm t$。矩形四角均落在这些有限弦上时，t 最大为多少?

Worked solution

1. The four corners are $(\pm 3/5, \pm t)$.

2. Every corner lies in the disk exactly when $9/25 + t^2 \leq 1$. This also ensures it lies on each relevant chord segment.

3. Thus $t \leq 4/5$, attained when all four corners are on the circumcircle.

例题解析

1. 四角为 $(\pm 3/5, \pm t)$。

2. 它们在圆盘内恰要求 $9/25 + t^2 \leq 1$，这同时保证位于各自有限弦段上。

3. 所以最大 $t = 4/5$，等号时四角都在圆周。

Perpendicular supporting lines alone do not guarantee their intersections lie on the available chord segments.

支撑直线互相垂直，并不保证交点落在可用弦段上。

Practice / 配套练习: AIME-2024-II-Q15 (WB p. 706)

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 11 / 方法 11 [CO06-M11]

Recover a Hamiltonian rule from one incoming edge

由唯一入边恢复哈密顿转移规则

Procedure

1. Encode each allowed outgoing choice by a binary rule at the state.

2. Impose one incoming edge at every state; compare the two possible predecessors.

3. Propagate the resulting equality to reduce the rule to diagonal classes.

4. Compute the displacement after one rule period and use a gcd condition to distinguish one full cycle from several short cycles.

通用步骤

1. 用每个状态的二元规则编码出边选择。

2. 每态要求恰一入边，比较两种可能前驱。

3. 传播所得等式，将规则压缩到对角类别。

4. 计算一个规则周期后的位移，用最大公因数区分一个大循环与多个短循环。

> Example. On $\mathbb{Z}_3^2$, repeat commands A, A, B, where A increases the first coordinate and B the second. Show the resulting walk from $(0,0)$ visits all nine states before returning.

> 例题：在 $\mathbb{Z}_3^2$ 上重复指令 A, A, B，其中 A 增第一坐标，B 增第二坐标。证明从原点出发先遍历九态再返回。

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 11 / 方法 11 [CO06-M11] (continued / 续)

Worked solution

1. Each command increases the coordinate sum by one modulo three, so states within a three-step block have different diagonal classes.

2. One block changes coordinates by $(2,1) = (2,-2)$ modulo three. This translation has order three because $\gcd(2,3) = 1$.

3. The three diagonal classes and three distinct block translations give nine distinct states, followed by the return.

例题解析

1. 每条指令使坐标和模三增加一，同一三步块的状态位于不同对角类。

2. 一块位移为 $(2,1) = (2,-2)$，因 $\gcd(2,3) = 1$，此平移的阶为三。

3. 三个对角类配三个不同块平移，得到九个不同状态后返回。

A locally bijective successor rule gives a union of cycles; the gcd test is what proves there is one Hamiltonian cycle.

局部一入一出的规则只保证若干循环，最大公因数检验才证明只有一个哈密顿大循环。

For states $(a,b) \in \mathbb{Z}_n^2$, let $h(a,b) = 1$ choose the first-coordinate move and zero the second. One incoming edge gives $h(a-1,b) = h(a,b-1)$, so h depends only on $a+b$. If the resulting length-n rule has c first-coordinate moves, each period translates by $(c,-c)$, and the walk covers all n^2 states exactly when $\gcd(c,n) = 1$.

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 11 / 方法 11 [CO06-M11] (continued / 续)

对模 n 二维状态，以 $h(a,b)=1$ 选第一坐标步、零选第二坐标步。唯一入边给出 $h(a-1,b)=h(a,b-1)$，故规则只依赖 $a+b$。长度 n 规则若含 c 个第一坐标步，每周期平移 $(c,-c)$，遍历全部 n^2 状态的充要条件为 $\gcd(c,n)=1$。

Practice / 配套练习: AIME-2023-I-Q14 (WB p. 705)

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 12 / 方法 12 [CO06-M12]

Use finite incidence geometry to count signed configurations

用有限关联几何计算符号构型

Procedure

1. Recognize the small hypergraph and prove its point-line incidences.

2. Condition on zero positions; only hyperedges avoiding zeros contribute to a product sum.

3. Translate the remaining signs into line-product signs, or into degrees of a smaller auxiliary graph.

4. Count every incidence type disjointly and verify the small total by exhaustive enumeration when available.

通用步骤

1. 识别小型超图并证明点线关联规律。

2. 按零位置分类，只有避开零的超边对乘积和有贡献。

3. 把余下正负号转成线乘积符号，或转成更小辅助图的度数。

4. 按不交关联类型求和，可用穷举独立核验小规模总量。

> Example. The Fano plane has seven points and seven three-point lines, with three lines through each point and one line through each point pair. Give every point sign $+1$ except one point of sign -1. What is the sum of the seven line products? What changes with two negative points?

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 12 / 方法 12 [CO06-M12] (continued / 续)

> 例题：法诺平面有七点七条三点线，每点过三线、每对点恰共一线。仅一个点为负一，其余为正一，七条线的符号乘积和为多少？若有两个负点呢？

Worked solution

1. One negative point lies on exactly three lines, making three products negative and four positive. The sum is one.

2. Two negative points share one line, whose product is positive. Their other two lines each are distinct, giving four negative products.

3. There are then three positive products and four negative ones, so the sum is negative one. The shared line must not be counted as negative twice.

例题解析

1. 一个负点所在三条线乘积负，其余四条正，和为一。

2. 两个负点共用的一条线乘积反为正，各自其余两线不同，共四条负乘积线。

3. 于是三正四负，和为负一；公共线不能被当作两条负线。

Finite geometry is useful only after verifying the claimed incidence pattern, not merely recognizing its name.

须验证所声称的关联结构，不能只认出有限几何名称。

The cyclic triples $\{i, i+1, i+3\}$ modulo seven form the Fano plane: every pair occurs in one triple. With one zero point, four lines survive. Each of the six remaining points lies on exactly two surviving lines, identifying it with an edge of K_4. A choice of negative points becomes a graph edge set; a line product is negative exactly when that vertex degree is odd.

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 12 / 方法 12 [CO06-M12] (continued / 续)

With three negative signs among those six points, exactly two of the four products must be negative to make their sum zero modulo three. Three-edge subsets of K_4 with exactly two odd-degree vertices are precisely its spanning three-edge paths, numbering $4!/2 = 12$. With two zero points, the two surviving lines meet once, and a minority sign changes only one line product exactly when it is outside their intersection. These lemmas justify the nontrivial zero strata in the full enumeration.

模七的循环三元组 $\{i, i+1, i+3\}$ 构成法诺平面，每对点恰出现一次。一个零点后剩四条线，每个其余点属于其中两线，故对应 K_4 的一条边。选择负点就成为选择边，某线乘积为负恰对应其顶点度为奇数。

这六点中若三个为负，要使四个乘积之和模三为零，必须恰有两个负乘积。K_4 三边子集中恰有两个奇度顶点的正是覆盖四点的三边路径，有 $4!/2 = 12$ 个。两个零点时剩两线且相交一次，少数符号只有放在非交点位置才仅改变一个线乘积。这些引理支撑完整枚举中的非平凡零分层。

Practice / 配套练习: AIME-2026-II-Q15 (WB p. 708)

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 13 / 方法 13 [CO06-M13]

Condition on shared edges and multiply boundary completions

按公共边分类并相乘边界补全

Procedure

1. Choose the colors of all edges shared by multiple local regions.

2. For each region determine the number of remaining boundary choices compatible with its fixed internal edges.

3. Multiply only when those remaining boundary edge sets are disjoint, then sum over shared-edge patterns.

通用步骤

1. 先选择被多个局部区域共用边的颜色。

2. 对每个区域按已定内部边计算剩余边界补全数。

3. 只有剩余边界集合互不重叠时才相乘，再对公共边模式求和。

> Example. Two unit squares share one edge. Color their seven distinct edges red or blue so each square has exactly one blue edge. Count the colorings.

> 例题：两个单位正方形共用一条边，将七条不同边染红蓝，使每个正方形恰有一条蓝边，求染色数。

Worked solution

1. If the shared edge is blue, all six outside edges are forced red, giving one coloring.

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 13 / 方法 13 [CO06-M13] (continued / 续)

2. If the shared edge is red, each square chooses one of its three outside edges to be blue; these edge sets are disjoint.

3. The count is $1 + 3 \cdot 3 = 10$. Conditioning on the shared edge is what makes the completions independent.

例题解析

1. 公共边蓝时，六条外边均被迫红，只有一种。

2. 公共边红时，每个正方形各从自己的三条外边选一条蓝，两组边互不相交。

3. 共有 $1 + 3 \cdot 3 = 10$ 种，按公共边分类后，补全过程才独立。

Regions sharing unassigned edges do not have independent completion counts.

仍共用未定边的区域，其补全数不能独立相乘。

Practice / 配套练习: AIME-2025-II-Q03 (WB p. 696)

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 14 / 方法 14 [CO06-M14]

Bound graph edge differences by weighted deviations

用加权偏差限制图边差总和

Procedure

1. Orient every edge from its lower label to its higher label and collect the coefficient of each vertex value.

2. The coefficient sum is zero and its absolute value is at most the vertex degree. Subtract a common center from every label.

3. Use rearrangement to pair the largest absolute deviations with the largest degrees.

4. Prove equality by a compatible low-high bipartition and the required degree allocations; a numerical bound alone is insufficient.

通用步骤

1. 将边从较小标签定向到较大标签，收集每个顶点值的系数。

2. 系数和为零，绝对值不超过度数，因此可给所有标签减同一中心。

3. 用重排把最大偏差绝对值配最大度数。

4. 以相容的低高二部分配及指定度数分配证明等号，只有数值上界并不够。

> Example. Place labels one through four on the four vertices of a path. Maximize the sum of absolute differences on its three edges.

> 例题：将一至四放在四顶点路径上，求三条边绝对差之和的最大值。

Section 4.6 Graphs & geometric counting

图论与几何计数

Method 14 / 方法 14 [CO06-M14] (continued / 续)

Worked solution

1. Center labels at $5/2$. Their absolute deviations are $3/2, 3/2, 1/2, 1/2$; the path degrees are two, two, one, one.

2. The degree-weighted bound is $2(3/2 + 3/2) + (1/2 + 1/2) = 7$.

3. The order $2, 4, 1, 3$ has differences $2, 3, 2$, attaining seven. Its interior vertices receive the largest deviations and every edge joins low to high.

例题解析

1. 将标签减去 $5/2$，偏差绝对值为 $3/2, 3/2, 1/2, 1/2$，度数为二、二、一、一。

2. 加权上界为 $2(3/2 + 3/2) + (1/2 + 1/2) = 7$。

3. 排列 $2, 4, 1, 3$ 的边差为二、三、二，达到七；内部点占最大偏差，每条边均连接低值与高值。

The degree multiset alone gives a bound; the graph must permit the equality sign pattern and value allocation.

度数多重集只能给界，还须图结构允许等号符号模式与数值分配。

Practice / 配套练习: AIME-2026-I-Q11 (WB p. 809)

Section 4.7 Bijections & structural counting

双射与结构计数

Method 1 / 方法 1 [CO07-M01]

A structural count starts by changing what one object means. Write both the encoding and its inverse, and track fixed points before identifying symmetric descriptions. A useful bijection must preserve every boundary, maximality, and noncollision condition.

结构计数从改变对象的描述方式开始。必须写出编码与逆过程，并在按对称性归并之前处理不动点。有效双射须保留所有边界、极大性与不碰撞条件。

Write an index map for a shuffle

写出洗牌的位置映射

Procedure

1. Choose one indexing direction and keep it throughout.

2. Express the destination of each source interval as an affine function of its old index.

3. Solve fixed-point or target-position equations, then check the source interval and the other required conditions.

通用步骤

1. 统一使用一种位置编号方向。

2. 对每段来源位置，把新位置写成旧下标的仿射式。

3. 解不动点或目标位置方程，再检验来源区间与其他条件。

> Example. A strip has cards $1, \ldots, 12$. Reverse each consecutive block of four cards. Which cards retain their positions?

> 例题：纸条上依次放牌 $1, \ldots, 12$，把每个连续四张区块内部倒序。哪些牌保留原位置？

Section 4.7 Bijections & structural counting

双射与结构计数

Method 1 / 方法 1 [CO07-M01] (continued / 续)

Worked solution

1. In block $j = 0, 1, 2$, write the old position as $4j + t$, $1 \le t \le 4$.

2. The new position is $4j + 5 - t$. A fixed position would require $t = 5 - t$.

3. This gives $t = 5/2$, not an integer. Therefore no card stays fixed, even though every block remains in place.

例题解析

1. 第 $j = 0, 1, 2$ 块中，旧位置为 $4j + t$，$1 \le t \le 4$。

2. 新位置为 $4j + 5 - t$，不动条件要求 $t = 5 - t$。

3. 得到 $t = 5/2$ 非整数，所以没有牌保持原位置，尽管每块整体位置不变。

A map written from the bottom cannot be substituted into an equation indexed from the top.

从下方编号的映射不能直接代入从上方编号的方程。

Practice / 配套练习: AIME-2005-II-Q06 (WB p. 710)

Section 4.7 Bijections & structural counting

双射与结构计数

Method 2 / 方法 2 [CO07-M02]

Compress mandatory gaps

压缩必须保留的间隔

Procedure

1. Sort the selected positions and write their minimum separation.

2. Subtract the accumulated compulsory gaps to obtain an ordinary increasing selection.

3. Construct the inverse by adding those gaps back and verify both endpoint bounds.

通用步骤

1. 将选中位置排序并写明最小间距。

2. 减去累计强制间隔，化为普通递增选取。

3. 加回间隔写出逆过程，并检验两端界。

> Example. Three event days $a < b < c$ are chosen from days one through twelve. The second must be at least two days after the first, and the third at least four days after the second. Count the schedules.

> 例题：从第一至第十二天中选三个活动日 $a < b < c$，第二次须比第一次至少晚两天，第三次比第二次至少晚四天，求日程数。

Worked solution

1. Set $(u, v, w) = (a, b-1, c-4)$. The unequal compulsory gaps become $1 \le u < v < w \le 8$.

Section 4.7 Bijections & structural counting

双射与结构计数

Method 2 / 方法 2 [CO07-M02] (continued / 续)

2. Conversely, every such triple gives $(a, b, c) = (u, v + 1, w + 4)$ with gaps at least two and four.

3. The map is reversible, so the answer is $\binom{8}{3} = 56$.

例题解析

1. 令 $(u, v, w) = (a, b - 1, c - 4)$，不同的强制间隔统一变为 $1 \le u < v < w \le 8$。

2. 反之，任意此类三元组都给出 $(a, b, c) = (u, v + 1, w + 4)$，两段间距分别至少为二和四。

3. 映射可逆，故答案为 $\binom{8}{3} = 56$。

Subtract only the compulsory extra gaps; the strict ordering already accounts for a gap of one.

只减去额外强制间隔；严格递增本身已包含一单位间距。

Practice / 配套练习: AIME-2009-II-Q06 (WB p. 635)

Section 4.7 Bijections & structural counting

双射与结构计数

Method 3 / 方法 3 [CO07-M03]

Pair objects by an involution, including fixed points

用对合配对并计入不动点

Procedure

1. Define a transformation that is its own inverse.

2. Prove it exchanges the two classes being compared, or identifies the two descriptions of one object.

3. Count the fixed objects separately; only nonfixed objects come in pairs.

通用步骤

1. 定义一个连续作用两次回到原对象的变换。

2. 证明它交换要比较的两类，或交换同一对象的两种描述。

3. 单独计算不动对象；只有非不动对象能两两配对。

> Example. A designer chooses two subsets of a four-element icon library, allowing the two subsets to coincide. The two display panels are interchangeable. How many designs are there?

> 例题：设计者为两个可互换面板各选一个四元素图标库的子集，允许两子集相同。共有多少种设计?

Worked solution

1. There are $16^2 = 256$ ordered pairs of subsets.

Section 4.7 Bijections & structural counting

双射与结构计数

Method 3 / 方法 3 [CO07-M03] (continued / 续)

2. Swapping the panels is an involution. Its fixed pairs have the same subset on both panels, giving 16 fixed designs.

3. The number of unordered designs is $(256+16)/2=136$. Equal-panel designs must not be divided by two.

例题解析

1. 有序子集对共有 $16^2=256$ 个。

2. 交换面板是对合，不动对要求两面板选同一子集，共十六个。

3. 无序设计数为 $(256+16)/2=136$，相同面板设计不能除以二。

For a symmetry exchanging strict inequalities, remove the equality class before halving.

对交换严格不等式的对称，先扣除等号类再除二。

Practice / 配套练习: AIME-2010-II-Q10 (WB p. 713), AIME-2011-II-Q06 (WB p. 714)

Section 4.7 Bijections & structural counting

双射与结构计数

Method 4 / 方法 4 [CO07-M04]

Use symmetry orbits before counting colors

先看对称轨道再计颜色

Procedure

1. Write the permutation of positions induced by the required symmetry.

2. An invariant coloring is constant on each orbit; impose any prescribed color totals using orbit sizes.

3. When counting up to all symmetries, describe orbit representatives or average fixed-coloring counts; divide by the group size only for a free action.

通用步骤

1. 写出指定对称对位置产生的置换。

2. 不变染色在每个轨道上必须同色，按轨道大小施加颜色总量限制。

3. 按全部对称归并时，描述轨道代表或平均不动染色数；只有自由作用才能直接除群的大小。

> Example. A wheel has a center and eight labeled rim positions. How many red-blue colorings with exactly three red positions are unchanged by a half-turn?

> 例题：一个轮形有中心与八个有标识轮缘位置。恰有三个位置为红色的红蓝染色中，有多少个在半转后不变？

Worked solution

1. The half-turn fixes the center and splits the rim into four pairs.

Section 4.7 Bijections & structural counting

双射与结构计数

Method 4 / 方法 4 [CO07-M04] (continued / 续)

2. Three red positions require a red center and exactly one red pair; a blue center would leave an impossible odd red count on pairs.

3. Choose the red pair in four ways. The answer is 4.

例题解析

1. 半转固定中心，将轮缘分为四个二元轨道。

2. 红色总数为三，必须中心红色、恰一个轨道红色；中心为蓝时二元轨道无法给出奇数个红位。

3. 红轨道有四种选择，答案为四。

A prescribed invariant coloring and a coloring counted up to symmetry are different counting questions.

要求染色在某对称下不变，与把对称染色视为同类，是两种不同计数。

Practice / 配套练习: AIME-2006-II-Q08 (WB p. 711), AIME-2013-I-Q04 (WB p. 725)

Section 4.7 Bijections & structural counting

双射与结构计数

Method 5 / 方法 5 [CO07-M05]

Encode a constrained path by independent local choices

用独立局部选择编码受限路径

Procedure

1. Choose a short code for each available local move.

2. Prove each code word determines one legal complete object, including any forced intermediate moves.

3. Translate the endpoint or count condition into a condition on the code, and count only code words meeting it.

通用步骤

1. 给每个可选局部动作指定简短编码。

2. 证明每个编码串都唯一决定合法完整对象，包括被迫的中间动作。

3. 将终点或数量条件转成编码条件，再计算符合条件的串。

> Example. Two lamps begin off. Each of four commands either toggles both lamps or swaps their states. Count command histories ending with both lamps off.

> 例题：两盏灯起初均关闭。四次指令每次可同时翻转两灯，或交换两灯状态。求最后两灯均关闭的指令历史数。

Worked solution

1. Use T for toggle and S for swap. The lamps always have equal states, so S keeps them unchanged and T reverses them.

Section 4.7 Bijections & structural counting

双射与结构计数

Method 5 / 方法 5 [CO07-M05] (continued / 续)

2. The code word uniquely determines the state history, and conversely each step changes both lamps or leaves both unchanged, recovering its code.

3. The endpoint condition is an even number of T commands. Hence the count is $\binom{4}{0} + \binom{4}{2} + \binom{4}{4} = 8$.

例题解析

1. 用 T 表示翻转、S 表示交换。两灯始终同态，所以 S 保持状态，T 同时改变状态。

2. 编码串唯一决定状态历史，反之每步要么均变要么均不变，可恢复指令。

3. 终态条件是 T 数为偶数，故共有 $\binom{4}{0} + \binom{4}{2} + \binom{4}{4} = 8$ 种。

A code must remember enough state to recover orientation; counting only unordered column types needs a proof of forced orientation.

编码必须保留恢复方向所需的状态；只计无序列类型时须证明方向可被唯一恢复。

For a two-row proper three-color strip, if a column is (a, b) and the third color is c, the next column can only be $(b, a), (b, c), (c, a)$. Each unordered color pair occurs once in this list, so a sequence of unordered pairs plus the first orientation uniquely reconstructs the strip.

For a two-ring walk, a code that determines the ending ring can also turn angular displacement into a difference of code counts. Verify the cancellation of opposite spoke moves before applying a residue condition.

两行三色相邻异色条中，某列为 (a, b)、第三色为 c 时，下一列仅有 $(b, a), (b, c), (c, a)$。三种无序色对各出现一次，所以无序色对序列加首列方向即可唯一重构整条。

Section 4.7 Bijections & structural counting

双射与结构计数

Method 5 / 方法 5 [CO07-M05] (continued / 续)

双环游走中，若编码决定每步终止在哪个环，可将角位移化为两类编码数之差；应用同余条件前须证明向内向外辐条动作抵消。

Practice / 配套练习: AIME-2018-I-Q10 (WB p. 715), AIME-2023-II-Q10 (WB p. 716), AIME-2026-II-Q02 (WB p. 709)

Section 4.7 Bijections & structural counting

双射与结构计数

Method 6 / 方法 6 [CO07-M06]

Split overlapping place values into digit streams

将重叠位权拆成数位流

Procedure

1. Separate every bounded coefficient into ordinary base digits.

2. Regroup equal shifts into independent digit streams.

3. Solve for one stream and show the remaining stream has legal digits and bounds, giving an inverse.

通用步骤

1. 把每个有界系数拆成普通进位制数位。

2. 将相同位移重新归并成独立数位流。

3. 解出其中一流，证明另一流满足数位与范围限制，从而得到逆映射。

> Example. Count integer pairs $0 \le a, b \le 24$ satisfying $a + 5b = 48$.

> 例题：求 $0 \le a, b \le 24$ 且 $a + 5b = 48$ 的整数对数。

Worked solution

1. Write $a = 5u + v$ with digits $0 \le u, v \le 4$. Modulo five the equation gives $v = 3$.

2. Then $u + b = 9$, so $u = 0, 1, 2, 3, 4$ uniquely determines $b = 9 - u$.

3. All five recovered pairs satisfy the bounds, so the answer is 5. This explicitly accounts for the overlapping place value carried by a.

Section 4.7 Bijections & structural counting

双射与结构计数

Method 6 / 方法 6 [CO07-M06] (continued / 续)

例题解析

1. 写 $a = 5u + v$，其中 u, v 均为零至四的数位。模五得到 $v = 3$。

2. 于是 $u + b = 9$，每个 $u = 0, 1, 2, 3, 4$ 唯一给出 $b = 9 - u$。

3. 五个逆向恢复的数对均满足范围，故答案为五。这一编码明确处理了系数 a 内重叠的位权。

Padding a stream with leading zero digits is allowed only when it matches the original coefficient bounds.

只有符合原系数界时，才可给数位流补前导零。

Practice / 配套练习: AIME-2010-I-Q10 (WB p. 712)

Section 4.7 Bijections & structural counting

双射与结构计数

Method 7 / 方法 7 [CO07-M07]

Recover a maximal placement from its row and column labels

由行列标签恢复极大放置

Procedure

1. Use maximality to exclude an empty structural part that could receive another object.

2. Label every occupied row and column by its forced type.

3. Prove every compatible intersection must be filled and every incompatible one empty; count label assignments whose supports satisfy the nonempty conditions.

通用步骤

1. 利用极大性排除仍可加入物体的空结构部分。

2. 用被迫的类型给每个已占用行列标号。

3. 证明每个相容交点必须填、每个不相容交点必须空，再计算满足非空支持条件的标号方案。

> Example. A monochromatic-row-and-column token board is maximal. Its row colors are red, red, blue, and its column colors are red, blue, blue, green. Can such a placement exist? If the green column is changed to red, reconstruct the placement.

> 例题：一个每行每列各自单色的棋子放置为极大放置。行颜色为红、红、蓝，列颜色为红、蓝、蓝、绿。此放置存在吗？若把绿列改红，重构放置。

Worked solution

1. A maximal placement cannot have an empty column, but no row can supply a green token. The green column makes the given labels impossible.

Section 4.7 Bijections & structural counting

双射与结构计数

Method 7 / 方法 7 [CO07-M07] (continued / 续)

2. After replacing green by red, both row and column color supports are exactly red and blue. Fill precisely the equal-color intersections.

3. The two red rows each meet two red columns, and the blue row meets two blue columns. The unique reconstructed placement has $2 \cdot 2 + 1 \cdot 2 = 6$ tokens and is maximal.

例题解析

1. 极大放置不能有空列，但没有行能提供绿棋子，所以原来的绿列标签不可能实现。

2. 绿列改红后，行列的颜色集合都恰为红蓝两色，填满且仅填同行列同色的交点。

3. 两条红行各与两条红列相交，蓝行与两条蓝列相交，唯一重构有 $2 \cdot 2 + 1 \cdot 2 = 6$ 枚棋子且确为极大。

> Maximal means that no object can be added; it does not mean that the number already placed is greatest.

> 极大指不能再加物体，不表示已放物体数最多。

Practice / 配套练习: AIME-2024-II-Q09 (WB p. 717)

Section 4.7 Bijections & structural counting

双射与结构计数

Method 8 / 方法 8 [CO07-M08]

Factor independent collision chains, then optimize their lengths

分解独立碰撞链并优化链长

Procedure

1. Determine which local patterns force a move or make every matching impossible.

2. Join blocks only when they can compete for the same target; prove distinct chains are independent.

3. Count each chain by its allowed monotone choice pattern, then factor the target count to minimize construction cost.

通用步骤

1. 判断哪些局部结构迫使动作、哪些使配对完全不可能。

2. 仅将可能争夺同一目标的区块连成链，并证明不同链独立。

3. 按允许的单调选择模式计算每条链，再分解目标计数以最小化构造代价。

> Example. Place isolated marked integers so consecutive marks in a chain differ by two. Each mark shifts left or right by one, and target integers must be distinct. How many target sets arise from chains of lengths two and four, placed far apart?

> 例题：若干标记整数构成间隔为二的链，每个标记向左或右移一，目标整数不得重复。相距很远的两条链长度分别为二和四，共能得到多少个目标集合？

Section 4.7 Bijections & structural counting

双射与结构计数

Method 8 / 方法 8 [CO07-M08] (continued / 续)

Worked solution

1. Inside one chain, a right shift followed by a left shift would target the same missing integer and is forbidden.

2. Therefore the choices are an initial block of left shifts followed by right shifts. A chain of k marks has $k+1$ distinct target sets.

3. The separated chains do not compete, giving $(2+1)(4+1)=15$ target sets.

例题解析

1. 同一链中，向右后紧接向左会争夺同一个缺失整数，所以禁止。

2. 故动作只能先若干个向左、再向右；长度 k 的链产生 $k+1$ 个不同目标集合。

3. 分离的两链不争目标，总数为 $(2+1)(4+1)=15$。

Independence is a statement about disjoint target choices, not merely disjoint starting sets.

独立性要求目标选择互不冲突，仅起始集合不相交还不够。

For disjoint sets paired by differences one, a consecutive block of three source integers is impossible: the middle source has both possible targets still in the source set. A two-block forces its members outward. Chains are formed by singleton or two-block components separated by one missing integer. A chain with no two-block has $k+1$ images, one with one two-block has one image, and one with two two-blocks has none, because the forced directions would create a right-left collision.

If an independent singleton chain contributing factor q costs $q-1$ source elements, replacing a composite factor gh by separate factors costs $g+h-2<gh-1$ for $g,h>1$. Therefore the least cost for a prescribed product is the sum of prime-factor costs, counted with multiplicity; separate chains realize the bound.

Section 4.7 Bijections & structural counting

双射与结构计数

Method 8 / 方法 8 [CO07-M08] (continued / 续)

对不交且配对差为一的集合，连续三个源整数不可能，因为中间数的两个目标都仍在源集合中。连续二元块的两个数被迫向外移。由间隔一个缺失整数的单元块或二元块形成链：无二元块的链有 $k+1$ 个像，恰一二元块的链只有一个像，两个二元块的链无像，因强制方向最终产生右后左碰撞。

若产生因子 q 的独立单元链耗费 $q-1$ 个源元素，将合因子 gh 分成两链只耗费 $g+h-2<gh-1$。故指定乘积的最小代价是按重数相加的各素因子代价，分离各链即可达到。

Practice / 配套练习: AIME-2026-II-Q13 (WB p. 718)

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

Method 1 / 方法 1 [PS01-M01]

Start by proving what is equally likely. Labeled samples, color patterns, multisets and stopped paths are different sample spaces; use the representation that makes the event easiest to count while preserving its weights.

首先证明什么结果等可能。有标签样本、颜色模式、多重集与停时路径属于不同样本空间；应选择最便于统计事件且保留权重的表示。

Choose equally likely labeled samples before counting colors

统计颜色前先选择等可能的有标签样本

Procedure

1. Decide whether the elementary outcome is an ordered draw, an unordered subset, a permutation, or a full deal.

2. For k draws without replacement from N distinct objects, use $\binom{N}{k}$ unordered outcomes or $N(N-1)\cdots(N-k+1)$ ordered outcomes consistently.

3. Count favorable group multiplicities with products of binomial coefficients; color words with different multiplicities need not have equal probability.

4. Divide favorable by total outcomes only after proving the chosen elementary outcomes are uniform.

通用步骤

1. 确定基本结果是有序抽取、无序子集、排列还是完整分配。

2. 从 N 个不同物品不放回取 k 个时，一致使用 $\binom{N}{k}$ 个无序结果，或 $N(N-1)\cdots(N-k+1)$ 个有序结果。

3. 用二项式系数之积统计有利类别重数；重数不同的颜色串未必等概率。

4. 证明所选基本结果均匀后，才能以有利数除以总数。

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

Method 1 / 方法 1 [PS01-M01] (continued / 续)

> Example. An urn contains four red and three blue distinct balls. Find the chance that a uniformly chosen three-ball subset has exactly two red balls.

> 例题：袋中有四个红球和三个蓝球，球均可区分。均匀选择三球子集，恰有两红的概率是多少?

Worked solution

1. There are $\binom{7}{3} = 35$ equally likely subsets.

2. Choose two of the four red balls and one of the three blue balls: $\binom{4}{2}\binom{3}{1} = 18$.

3. The probability is $18/35$.

例题解析

1. 共有 $\binom{7}{3} = 35$ 个等可能子集。

2. 从四红中选二、三蓝中选一，有 $\binom{4}{2}\binom{3}{1} = 18$ 种。

3. 概率为 $18/35$。

Different color words can correspond to different numbers of labeled outcomes.

不同颜色串可能对应不同数量的有标签结果。

Practice / 配套练习: AIME-2000-II-Q03 (WB p. 720), AIME-2005-II-Q02 (WB p. 722), AIME-2010-I-Q01 (WB p. 723), AIME-2013-I-Q06 (WB p. 740), AIME-2015-I-Q02 (WB p. 726), AIME-2015-I-Q05 (WB p. 753), AIME-2018-I-Q03 (WB p. 728), AIME-2026-II-Q05 (WB p. 732)

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

Method 2 / 方法 2 [PS01-M02]

Count discrete geometric pairs through edges, gaps or position sets

用边、间隔或位置集合统计离散几何配对

Procedure

1. Represent the sampled geometric objects by labeled vertices or position subsets.

2. Count admissible pairs by graph edges or by separation, preserving ordered versus unordered sampling.

3. For a cyclic configuration, identify a unique anchor or a family of disjoint opposite-position pairs to avoid overcounting.

4. Use exact endpoint conventions for open semicircles and strict triangle inequalities.

通用步骤

1. 用有标签顶点或位置子集表示被抽取的几何对象。

2. 按图的边或间隔统计允许配对，并保持有序或无序抽样一致。

3. 循环构型须确定唯一锚点，或使用互不相交的对位配对，避免重复计数。

4. 对开半圆与严格三角不等式使用准确端点约定。

> Example. A graph on vertices $1, 2, 3, 4$ has edges $12, 23, 31, 34$. Choose two distinct vertices uniformly. What is the probability they are joined by an edge?

> 例题：顶点为 $1, 2, 3, 4$ 的图具有边 $12, 23, 31, 34$。均匀选两个不同顶点，它们有边相连的概率是多少?

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

Method 2 / 方法 2 [PS01-M02] (continued / 续)

Worked solution

1. The unordered sample space has $\binom{4}{2} = 6$ pairs.

2. The four listed edges are four distinct favorable pairs.

3. The probability is $4/6 = 2/3$. Ordered sampling doubles both counts and gives the same result.

例题解析

1. 无序样本空间有 $\binom{4}{2} = 6$ 对。

2. 所列四条边是四个不同的有利配对。

3. 概率为 $4/6 = 2/3$；有序抽样会同时使分子分母加倍，结果相同。

A discrete geometry question uses counts of choices; lengths alone do not weight the sample space.

离散几何题使用选择个数；长度本身不会给样本空间加权。

An obtuse inscribed triangle has its vertices in an open semicircle, with a unique first clockwise vertex. For an even polygon the antipodal endpoint is excluded, removing right triangles; for an odd polygon no vertex is exactly antipodal. Derive the two half-circle counts separately.

钝角内接三角形的顶点位于一个开半圆中，且有唯一顺时针首顶点。偶数边形须排除对径端点以去掉直角三角形；奇数边形没有恰好对径的顶点。应分别推导两种半圆计数。

Practice / 配套练习: AIME-2010-II-Q04 (WB p. 724), AIME-2011-I-Q10 (WB p. 739), AIME-2015-II-Q05 (WB p. 727), AIME-2019-I-Q02 (WB p. 729), AIME-2023-I-Q01 (WB p. 731)

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

Method 3 / 方法 3 [PS01-M03]

Use product coordinates and inclusion-exclusion on event unions

在事件并集中使用乘积坐标与容斥

Procedure

1. Encode each independent elementary choice locally, such as membership bits or cell colors.

2. Count an intersection by the union of all forced conditions; shared objects are forced once, not once per event.

3. Apply inclusion-exclusion for overlapping good or bad events, and use a complement when simpler.

4. For a uniform random subset, every membership bit is independent with probability one half; conditioning on size changes that model.

通用步骤

1. 将每个独立基本选择作局部编码，例如成员位或格子颜色。

2. 交集须按全部被迫条件之并计数；共用对象只被固定一次，不能按事件重复固定。

3. 对重叠好事件或坏事件使用容斥；补集较简单时先数补集。

4. 均匀随机子集的成员位独立且各为二分之一；若附加大小条件，模型便改变。

> Example. Choose two independent uniform subsets A, B of a four-element set. Find $\Pr(|A \cap B| = 1)$.

> 例题：从四元素集合的所有子集中独立均匀选 A, B，求 $\Pr(|A \cap B| = 1)$。

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

Method 3 / 方法 3 [PS01-M03] (continued / 续)

Worked solution

1. At each element the four membership pairs are equally likely, and only the pair $(1,1)$ enters the intersection.

2. The four elements act independently, so exactly one intersection element has probability $\binom{4}{1}(1/4)(3/4)^3$.

3. This simplifies to $27/64$.

例题解析

1. 每个元素的四种成员位对等可能，仅 $(1,1)$ 进入交集。

2. 四元素彼此独立，恰一个交集元素的概率为 $\binom{4}{1}(1/4)(3/4)^3$。

3. 化简为 $27/64$。

Inclusion-exclusion concerns overlapping events, even when the underlying random choices are independent.

即使基本随机选择独立，由它们构成的事件仍可能重叠而需要容斥。

Practice / 配套练习: AIME-2001-II-Q09 (WB p. 735), AIME-2002-I-Q01 (WB p. 721), AIME-2007-II-Q10 (WB p. 738), AIME-2011-II-Q12 (WB p. 745), AIME-2014-I-Q12 (WB p. 746), AIME-2025-II-Q07 (WB p. 492), AIME-2024-I-Q11 (WB p. 669)

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

Method 4 / 方法 4 [PS01-M04]

Compress outcomes into weighted residue or exponent classes

将结果压缩为带权剩余类或指数类

Procedure

1. Choose a class label that exactly determines the event, such as parity, a prime-exponent vector or a residue sum.

2. Attach the correct weight to each class: the number or total probability of its elementary outcomes.

3. Count permitted class patterns, restoring multinomial ordering and class weights; remove repeated-object samples if distinctness is required.

4. For independent uniform exponent coordinates, count nondecreasing exponent tuples by combinations with repetition.

通用步骤

1. 选择能准确决定事件的类别标签，例如奇偶、质指数向量或剩余和。

2. 为每类附上正确权重，即其基本结果数量或总概率。

3. 统计允许类别模式，并恢复多项式排列权重与类权重；若要求对象不同，剔除重复抽到同一对象的样本。

4. 指数坐标独立均匀时，用可重复组合统计非降指数元组。

> Example. Two fair dice are rolled. Find the probability that their faces have the same parity but are unequal.

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

Method 4 / 方法 4 [PS01-M04] (continued / 续)

> 例题：投两枚公平骰子，求点数奇偶相同但数值不同的概率。

Worked solution

1. There are 36 equally likely ordered outcomes. Each parity class has three faces.

2. Within either class, the first face has three choices and the distinct second face two choices, giving $2 \cdot 3 \cdot 2 = 12$ favorable outcomes.

3. The probability is $12/36 = 1/3$.

例题解析

1. 共有 36 个等可能有序结果，每个奇偶类有三个点数。

2. 每类首点数有三种，不同的第二点数有两种，有利结果为 $2 \cdot 3 \cdot 2 = 12$。

3. 概率为 $12/36 = 1/3$。

Compressing outcomes does not make the compressed classes equally likely.

压缩结果并不会使各压缩类别自动等概率。

For a square product, encode the parity of every prime exponent and require their coordinatewise sum to vanish modulo two. A face or number with zero parity vector may have more than one elementary representative and must retain that multiplicity.

For k equally likely exponent values, a nondecreasing r-tuple has $\binom{k+r-1}{r}$ possibilities out of k^r. Do not divide by $r!$, since tuples with repeated exponents have different orbit sizes.

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

Method 4 / 方法 4 [PS01-M04] (continued / 续)

乘积为平方数时，编码每个质指数的奇偶，并要求逐坐标和模二为零。零奇偶向量可能对应不止一个点数或数字，必须保留其重数。

指数有 k 个等可能值时，非降 r 元组有 $\binom{k+r-1}{r}$ 个，总数为 k^r。不能直接除以 $r!$，因为含重复指数的元组轨道大小不同。

Practice / 配套练习: AIME-2000-I-Q05 (WB p. 719), AIME-2001-I-Q06 (WB p. 733), AIME-2001-I-Q10 (WB p. 734), AIME-2004-II-Q10 (WB p. 736), AIME-2016-I-Q02 (WB p. 755), AIME-2018-I-Q12 (WB p. 748), AIME-2019-II-Q04 (WB p. 470), AIME-2020-I-Q09 (WB p. 741)

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

Method 5 / 方法 5 [PS01-M05]

Count invariant colorings and independent orientation types

统计不变涂色与独立朝向类型

Procedure

1. For a fixed symmetry, partition positions into its permutation orbits.

2. An invariant coloring is constant on every orbit; if a color total is fixed, sum the orbit sizes rather than choose colors independently.

3. For many independent oriented objects, classify positions by how many outward constraints they impose and multiply the correctly weighted success probabilities.

4. For existence of some symmetry, count the union of successful symmetry families and resolve their intersections.

通用步骤

1. 对固定对称，将位置分成其置换轨道。

2. 不变涂色在每个轨道内恒定；若某颜色总数固定，须按轨道大小求和，不能独立选色。

3. 对许多独立定向物体，按位置施加的外露限制分类，再乘正确加权的成功概率。

4. 若要求存在某个对称，须数各成功对称族的并，并处理其交集。

> Example. Color the vertices of a square independently red or blue with equal probabilities. What is the probability of invariance under the reflection fixing one diagonal?

> 例题：独立以相等概率把正方形顶点涂红或蓝，求涂色在固定某条对角线的反射下不变的概率。

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

Method 5 / 方法 5 [PS01-M05] (continued / 续)

Worked solution

1. The reflection fixes two vertices and exchanges the other two, giving orbit sizes $1, 1, 2$.

2. Each of three orbits can choose either color, so there are $2^3 = 8$ invariant colorings.

3. All $2^4 = 16$ vertex colorings are equally likely, so the probability is $1/2$.

例题解析

1. 该反射固定两个顶点，交换另两个，轨道大小为 $1, 1, 2$。

2. 三个轨道各可任选两色，故不变涂色有 $2^3 = 8$ 种。

3. 全部 $2^4 = 16$ 个顶点涂色等可能，故概率为 $1/2$。

Fixed color totals make the sample space a subset layer rather than independent bits.

固定颜色总数时，样本空间是固定大小子集层，而非独立成员位。

Practice / 配套练习: AIME-2005-I-Q09 (WB p. 737), AIME-2013-I-Q04 (WB p. 725), AIME-2024-I-Q11 (WB p. 669)

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

Method 6 / 方法 6 [PS01-M06]

Weight stopped paths by length and count constrained paths by state

按长度加权停时路径，并按状态统计受限路径

Procedure

1. Describe a terminal path including its stopping step; paths of different lengths generally have different probabilities.

2. For a fixed step multiplicity, use a multinomial count and multiply by that path's step probabilities.

3. For distinct-vertex or closing constraints, use a state recording the visited subset and last vertex, then test the closing edge.

4. Divide by a uniform total only when all retained elementary paths are equally likely.

通用步骤

1. 描述终止路径时包括停止步骤；不同长度路径通常具有不同概率。

2. 固定步型重数后，用多项式系数计数并乘该路径逐步概率。

3. 对顶点互异或闭合限制，使用记录已访问子集及末顶点的状态，再检验闭合边。

4. 只有保留的基本路径均等概率时，才能除以统一总数。

> Example. Toss a fair coin until the first head or three tosses, whichever occurs first. What is the probability that the terminal sequence contains more tails than heads?

> 例题：投公平硬币，直到首次正面或已投三次即停止。求终止串中反面多于正面的概率。

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

Method 6 / 方法 6 [PS01-M06] (continued / 续)

Worked solution

1. The terminal paths are H, TH, TTH, TTT, with probabilities $1/2, 1/4, 1/8, 1/8$.

2. Only TTH and TTT have more tails than heads.

3. Add their probabilities to get $1/4$. Counting two favorable paths out of four would incorrectly treat unequal paths as uniform.

例题解析

1. 终止路径为 H, TH, TTH, TTT，概率分别为 $1/2, 1/4, 1/8, 1/8$。

2. 只有 TTH, TTT 的反面数多于正面数。

3. 相加得 $1/4$；按四条路径中两条有利计数会错误地把不等概率路径看成均匀。

The stopping rule is part of the sample-space definition.

停止规则本身就是样本空间定义的一部分。

Practice / 配套练习: AIME-2001-I-Q15 (WB p. 744), AIME-2019-I-Q05 (WB p. 730)

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

Method 7 / 方法 7 [PS01-M07]

Model pairings and ownership constraints as permutations or matchings

把配对与归属限制建模为置换或匹配

Procedure

1. For a left-right bijection use a permutation; for arbitrary unordered pairings use a perfect matching with $(2n-1)!!$ outcomes.

2. Translate a closed collection of owners into a union of permutation cycles, or a sorted pair condition into a restriction on smaller endpoints.

3. Count allowable cycle types or pairing cases, correcting indistinguishable cycles and case intersections.

4. For repeated-color rows, use a uniform labeled model or prove every color pattern has the same number of labeled realizations.

通用步骤

1. 左右两类一一配对使用置换；任意无序配对使用完美匹配，共 $(2n-1)!!$ 种。

2. 将归属封闭子集转为置换循环之并，或将排序配对条件转为较小端点限制。

3. 统计允许循环型或配对情形，并修正不可区分循环与分类交集。

4. 对重复颜色行，使用均匀有标签模型，或证明每个颜色模式具有同数目的有标签实现。

> Example. Choose a uniform permutation of four labels. Find the probability that it has exactly one fixed point.

Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

Method 7 / 方法 7 [PS01-M07] (continued / 续)

例题：均匀选择四个标签的一个置换，求恰有一个不动点的概率。

Worked solution

1. Choose the fixed label in four ways.

2. The remaining three labels must have no fixed point, so they form one of the two three-cycles.

3. There are 8 favorable permutations out of $4! = 24$, giving probability $1/3$.

例题解析

1. 固定标签有四种选择。

2. 其余三个标签不能固定，因此构成两个三循环之一。

3. 有利置换为八个，全部为 $4! = 24$ 个，故概率为 $1/3$。

A pairing count and a permutation count use different elementary outcomes.

匹配计数与置换计数使用不同的基本结果。

Practice / 配套练习: AIME-2014-II-Q13 (WB p. 747), AIME-2022-I-Q09 (WB p. 742), AIME-2025-I-Q07 (WB p. 743)

Section 5.2 Conditional probability & independence

条件概率与独立性

Method 1 / 方法 1 [PS02-M01]

Information changes the probability space. Keep prior weights, evidence likelihoods, conditional cases and independence assumptions explicit; the formulas are short once the correct restricted or state-based model is chosen.

信息会改变概率空间。明确写出先验权重、证据似然、条件分类与独立假设；选对受限模型或状态模型后，公式本身通常很短。

Renormalize the sample space after information is given

给定信息后重新归一样本空间

Procedure

1. Name the conditioning event B and require $P(B) > 0$.

2. Use $P(A \mid B) = P(A \cap B)/P(B)$; in a uniform model count only outcomes surviving B.

3. Express structural information through gaps, degrees, repeated positions or disjoint Venn regions before counting.

4. Check the restricted outcomes remain equally likely; do not inherit independence without proof.

通用步骤

1. 命名条件事件 B，并要求 $P(B) > 0$。

2. 使用 $P(A \mid B) = P(A \cap B)/P(B)$；均匀模型中只统计条件 B 下保留的结果。

3. 计数前先用间隙、度数、重复位置或互斥 Venn 区域表达结构信息。

4. 检查受限结果仍等概率，不能未经证明沿用独立性。

Section 5.2 Conditional probability & independence

条件概率与独立性

Method 1 / 方法 1 [PS02-M01] (continued / 续)

> Example. Two fair dice are rolled. Given their sum is even, find the chance that both are odd.

> 例题：投两枚公平骰子，已知和为偶数，求两枚都为奇数的概率。

Worked solution

1. Even sum means both odd or both even. Each class has $3 \cdot 3 = 9$ ordered outcomes.

2. The restricted sample space therefore has eighteen equally likely outcomes, with nine favorable.

3. The conditional probability is $1/2$. The oddness events are dependent after conditioning, since knowing one determines the other.

例题解析

1. 和为偶数表示同奇或同偶，每类均有 $3 \cdot 3 = 9$ 个有序结果。

2. 条件样本空间有十八个等可能结果，其中九个有利。

3. 条件概率为 $1/2$；条件下两个奇偶事件相关，因为知道一个就确定另一个。

Conditional denominators count the information event, not the original whole space.

条件概率的分母统计已知信息事件，而非原整个空间。

Section 5.2 Conditional probability & independence

条件概率与独立性

Method 1 / 方法 1 [PS02-M01] (continued / 续)

For a uniformly chosen bipartite edge set covering all vertices with one extra edge, the repeated degree-two row and column control whether an edge is removable. Prove uniqueness of the removable edge before using a matching-plus-extra-edge count; otherwise that construction can overcount.

Visibility of a sticker attached at time j means that no later roll uses its face. Convert each visibility statement into forbidden equalities of roll positions, then count the allowed sequences under all conditions simultaneously.

For conditioned runs in a random lineup, place blocks in the gaps around the other group and enumerate every ordered composition of the block sizes. Cancel common internal-label factors only after verifying they are shared by all retained patterns.

均匀选取的二部边集若覆盖全部顶点且只多一条边，重复的二度行与列决定是否有边可删。用“匹配加一条边”计数前，须证明可删边唯一，否则该构造会重复计数。

时刻 j 贴上的贴纸最终可见，表示后续投掷都未使用其所在面。把每个可见声明转为投掷位置之间被禁止的相等关系，再同时满足全部条件计数。

随机队列的条件连续段问题，可将各块放入另一组周围的间隙，并枚举全部有序块长拆分。只有确认保留模式共享相同类内标签因子后，才能将它们约去。

Practice / 配套练习: AIME-2010-II-Q13 (WB p. 764), AIME-2011-I-Q12 (WB p. 765), AIME-2014-II-Q02 (WB p. 752), AIME-2015-II-Q02 (WB p. 754), AIME-2017-II-Q09 (WB p. 762), AIME-2024-I-Q04 (WB p. 760), AIME-2026-I-Q09 (WB p. 763)

Section 5.2 Conditional probability & independence

条件概率与独立性

Method 2 / 方法 2 [PS02-M02]

Update a hidden model with Bayes before predicting again

再次预测前先用贝叶斯更新隐藏模型

Procedure

1. List the possible hidden models H_i and their prior probabilities.

2. Multiply each prior by the likelihood of the observed evidence and normalize the resulting weights.

3. Average the next-event probability using these posterior weights; conditional independence given a model need not survive averaging over models.

通用步骤

1. 列出可能隐藏模型 H_i 及先验概率。

2. 将每个先验乘观测证据的似然，再归一所得权重。

3. 以后验权重平均下一事件概率；给定模型的条件独立，未必在对模型平均后仍成立。

> Example. Choose equally between a fair coin and an always-heads coin, then observe one head. Are the next two tosses independent given that observation?

> 例题：等可能选择公平硬币或总为正面的硬币，观察到一次正面后，接下来两次投掷在此条件下独立吗?

Worked solution

1. The evidence likelihoods are $1/2$ and one, so posterior probabilities are fair $1/3$, always-heads $2/3$.

Section 5.2 Conditional probability & independence

条件概率与独立性

Method 2 / 方法 2 [PS02-M02] (continued / 续)

2. Each future toss is heads with probability $(1/3)(1/2) + (2/3) = 5/6$.

3. Two future heads have probability $(1/3)(1/4) + (2/3) = 3/4$.

4. Since $3/4 \neq (5/6)^2$, the future tosses are dependent under the observed-information condition.

例题解析

1. 证据似然分别为 $1/2$ 与一，故后验为公平币 $1/3$、总正面币 $2/3$。

2. 每次未来投掷为正面的概率为 $(1/3)(1/2) + (2/3) = 5/6$。

3. 未来两次均正面的概率为 $(1/3)(1/4) + (2/3) = 3/4$。

4. 因 $3/4 \neq (5/6)^2$，在已观察信息条件下，两次未来投掷相关。

Evidence changes the model weights even when the physical die or coin remains unchanged.

即使实际骰子或硬币不变，证据仍会改变各模型权重。

Practice / 配套练习: AIME-2014-II-Q06 (WB p. 761)

Section 5.2 Conditional probability & independence

条件概率与独立性

Method 3 / 方法 3 [PS02-M03]

Condition on a complete first-stage partition

按完整的第一阶段分类取条件

Procedure

1. Choose disjoint cases H_i whose union is the whole space, such as opponents, bracket pairings or geometric regions.

2. Compute $P(A \mid H_i)$ within each case, using the probabilities that hold there.

3. Combine by $P(A) = \sum_i P(H_i)P(A \mid H_i)$. This also gives an equation for unknown conditional rates.

4. If coordinates are claimed independent after conditioning, identify the product region or independent coordinate variables that justify it.

通用步骤

1. 选择互斥且并为全空间的情形 H_i，例如对手、赛程配对或几何区域。

2. 在各情形内计算 $P(A \mid H_i)$，使用该情形实际成立的概率。

3. 以 $P(A) = \sum_i P(H_i)P(A \mid H_i)$ 合并，也可由此求未知条件概率。

4. 若声称取条件后坐标独立，须指出相应乘积区域或独立坐标变量。

> Example. A mode is selected: mode A with probability $1/4$, otherwise mode B. Success rates are $1/2$ and $1/3$. Find the overall success probability and the posterior chance of mode A after success.

Section 5.2 Conditional probability & independence

条件概率与独立性

Method 3 / 方法 3 [PS02-M03] (continued / 续)

例题：先选模式：概率 $1/4$ 选 A，否则选 B；成功率分别为 $1/2, 1/3$。求总体成功率及成功后为模式 A 的概率。

Worked solution

1. The disjoint success contributions are $(1/4)(1/2) = 1/8$ and $(3/4)(1/3) = 1/4$.

2. The total success probability is $1/8 + 1/4 = 3/8$.

3. Of that success mass, mode A contributes $1/8$, so its posterior probability is $(1/8)/(3/8) = 1/3$.

例题解析

1. 两种互斥成功贡献为 $(1/4)(1/2) = 1/8$ 与 $(3/4)(1/3) = 1/4$。

2. 总成功率为 $1/8 + 1/4 = 3/8$。

3. 成功质量中模式 A 贡献 $1/8$，故其后验概率为 $(1/8)/(3/8) = 1/3$。

Use the probability of reaching a branch as well as its conditional success rate.

既要乘分支内成功率，也要乘到达该分支的概率。

Practice / 配套练习: AIME-2016-II-Q02 (WB p. 756), AIME-2017-II-Q02 (WB p. 757), AIME-2022-II-Q02 (WB p. 759), AIME-2023-II-Q06 (WB p. 772)

Section 5.2 Conditional probability & independence

条件概率与独立性

Method 4 / 方法 4 [PS02-M04]

Normalize independent weights and convolve their distributions

归一独立权重并卷积其分布

Procedure

1. Turn proportional weights into probabilities by dividing by their total.

2. For independent variables, multiply probabilities of elementary coordinate outcomes and sum disjoint ways to reach the target.

3. A probability-generating polynomial encodes a nonnegative integer count; multiply polynomials for independent sums.

4. Matching two independent identical distributions gives $\sum p_k^2$; transformed walk coordinates require proof that their joint one-step signs are uniform.

通用步骤

1. 将成比例权重除以总权重，化为概率。

2. 独立变量的基本坐标结果概率相乘，再对达到目标的互斥方式求和。

3. 概率生成多项式编码非负整数计数；独立和对应多项式相乘。

4. 两个独立同分布计数相等的概率为 $\sum p_k^2$；变换后的随机游走坐标则须证明每步联合符号均匀。

> Example. Independent coins have head probabilities $1/2$ and $1/3$. Find the probability of exactly one head using a generating polynomial.

> 例题：两枚独立硬币正面率为 $1/2, 1/3$，以生成多项式求恰一次正面的概率。

Section 5.2 Conditional probability & independence

条件概率与独立性

Method 4 / 方法 4 [PS02-M04] (continued / 续)

Worked solution

1. The two head-count polynomials are $(1+t)/2$ and $(2+t)/3$.

2. Their product is $(2+3t+t^2)/6$.

3. The coefficient of t is $3/6 = 1/2$, agreeing with $(1/2)(2/3) + (1/2)(1/3)$.

例题解析

1. 两枚硬币的正面计数多项式为 $(1+t)/2, (2+t)/3$。

2. 乘积为 $(2+3t+t^2)/6$。

3. t 的系数为 $3/6 = 1/2$，也等于 $(1/2)(2/3) + (1/2)(1/3)$。

Independence is a property to prove or a stated assumption, not a consequence of naming two variables.

独立性须有已知条件或证明，并非命名两个变量后便自动成立。

Practice / 配套练习: AIME-2006-II-Q05 (WB p. 750), AIME-2010-I-Q04 (WB p. 784), AIME-2014-I-Q02 (WB p. 751), AIME-2014-I-Q11 (WB p. 792), AIME-2016-I-Q02 (WB p. 755), AIME-2023-II-Q06 (WB p. 772)

Section 5.2 Conditional probability & independence

条件概率与独立性

Method 5 / 方法 5 [PS02-M05]

Update the remaining population or use an exchangeable full deal

更新剩余总体，或使用可交换的完整分配

Procedure

1. After each draw, record the remaining multiplicities; without replacement changes later probabilities.

2. Multiply the next-step conditional probabilities along a path and add disjoint draw patterns.

3. If chronological cases are cumbersome, deal all labeled objects in advance and prove permuting batch labels preserves uniformity.

4. Track the number of complete groups left after a mismatch, rather than assume each unobserved pair is equally intact.

通用步骤

1. 每次抽取后记录剩余重数，不放回会改变后续概率。

2. 沿路径乘下一步条件概率，再对互斥抽取模式求和。

3. 若按时间分类繁琐，可先完整分配所有有标签物品，并证明批次标签置换保持均匀性。

4. 不匹配后须追踪剩余完整组数，不能假设所有未见配对仍完整。

> Example. An urn has two red and two blue balls. Draw two without replacement. Show that the events first-red and second-red are not independent.

> 例题：袋中有二红二蓝，不放回抽两球。证明首次红与第二次红两个事件不独立。

Section 5.2 Conditional probability & independence

条件概率与独立性

Method 5 / 方法 5 [PS02-M05] (continued / 续)

Worked solution

1. The first-red probability is $1/2$, and by exchangeability so is the marginal second-red probability.

2. Given the first is red, only one red remains among three balls, so $P(R_2 \mid R_1) = 1/3$.

3. This differs from the marginal $1/2$; equivalently $P(R_1 \cap R_2) = 1/6 \neq 1/4$.

例题解析

1. 首次红概率为 $1/2$，由可交换性，第二次红的边缘概率也为 $1/2$。

2. 已知首次红，剩余三球仅一红，故 $P(R_2 \mid R_1) = 1/3$。

3. 它不同于边缘概率 $1/2$；等价地 $P(R_1 \cap R_2) = 1/6 \neq 1/4$。

Exchangeability of batch labels does not mean independence of the batches.

批次标签可交换，并不意味着各批次独立。

Practice / 配套练习: AIME-2004-II-Q02 (WB p. 749), AIME-2015-I-Q05 (WB p. 753)

Section 5.2 Conditional probability & independence

条件概率与独立性

Method 6 / 方法 6 [PS02-M06]

Condition on the minimal memory state and track parity if needed

按最小记忆状态取条件，必要时追踪奇偶

Procedure

1. Choose the shortest memory that determines the next transition probabilities, such as the previous winner or a matched pattern prefix.

2. For a fixed finite event, multiply transition probabilities along each allowed state path, counting switches rather than assuming independent outcomes.

3. For a stopping parity, let $u_s = E((-1)^R)$ for remaining time R; each next step changes sign, and immediate stopping contributes -1.

4. Solve the first-step equations and convert u to odd probability $(1-u)/2$; establish almost-sure stopping.

通用步骤

1. 选择决定下一步转移概率的最短记忆，例如前次胜者或已匹配模式前缀。

2. 固定有限事件沿各允许状态路径乘转移概率，并统计切换次数，不能假设结果独立。

3. 停时奇偶可设 $u_s = E((-1)^R)$，其中 R 为剩余步数；每前进一步变号，立即停止贡献 -1。

4. 解首步方程，并以 $(1-u)/2$ 转为奇数概率；同时证明几乎必然停止。

> Example. Toss a fair coin until two consecutive heads. Find the probability that the stopping time is odd.

> 例题：投公平硬币，直到连续两次正面，求停止次数为奇数的概率。

Section 5.2 Conditional probability & independence

条件概率与独立性

Method 6 / 方法 6 [PS02-M06] (continued / 续)

Worked solution

1. Use states 0 (no trailing head) and 1 (one trailing head). Let u_0, u_1 be signed remaining-time expectations.

2. First-step conditioning gives $u_0 = -(u_0 + u_1)/2$ and $u_1 = -(u_0 + 1)/2$.

3. Thus $u_1 = -3u_0$ and $u_0 = 1/5$, so the odd probability is $(1 - 1/5)/2 = 2/5$.

4. Each disjoint two-toss block has probability $1/4$ of being HH, so the chance of avoiding a stop forever is zero.

例题解析

1. 取状态 0（无末尾正面）与 1（末尾一个正面），设 u_0, u_1 为带符号剩余时间期望。

2. 首步取条件得 $u_0 = -(u_0 + u_1)/2, u_1 = -(u_0 + 1)/2$。

3. 故 $u_1 = -3u_0, u_0 = 1/5$，奇数概率为 $(1 - 1/5)/2 = 2/5$。

4. 每个互不相交的两投块有 $1/4$ 概率为 HH，故永远不停止的概率为零。

A failed prefix may retain a shorter matching suffix; recalculate the state rather than always resetting to zero.

前缀失败后可能保留较短匹配后缀，应重新确定状态，不能总是归零。

Practice / 配套练习: AIME-2018-II-Q13 (WB p. 780), AIME-2021-I-Q01 (WB p. 758)

Section 5.3 Geometric probability

几何概率

Method 1 / 方法 1 [PS03-M01]

Continuous uniform probability is a ratio of measures in the original sampling coordinates. Draw the feasible region first, translate the event exactly, and justify all conditioning or coordinate changes before using symmetry.

连续均匀概率是原抽样坐标中的度量比。先画可行区域，再精确转化事件；使用对称之前，须证明条件化或坐标变换成立。

Measure the valid parameter intervals after algebraic reduction

代数约化后度量有效参数区间

Procedure

1. Keep the exact uniform parameter and its interval; a transformed variable is not automatically uniform.

2. Translate the event into algebraic or trigonometric inequalities, discriminants, or floor-constant intervals.

3. Find every favorable subinterval with the original domains restored, then divide its total length by the sample length.

4. Isolated endpoints have zero probability in a continuous uniform interval, but the interval interiors must be correct.

通用步骤

1. 保留实际均匀参数及其区间，变换后的变量不自动均匀。

2. 将事件转为代数或三角不等式、判别式条件，或取整恒定的区间。

3. 恢复原定义域后找出所有有利子区间，再以总长度除以样本长度。

4. 连续均匀区间中孤立端点概率为零，但区间内部必须判断正确。

Section 5.3 Geometric probability

几何概率

Method 1 / 方法 1 [PS03-M01] (continued / 续)

> Example. Choose θ uniformly from $[0, \pi/2]$. Find $P(\sin\theta \geq \cos 2\theta)$.

> 例题：从 $[0, \pi/2]$ 均匀选择 θ，求 $P(\sin\theta \geq \cos 2\theta)$。

Worked solution

1. Put $s = \sin\theta \in [0, 1]$ only to solve the inequality: $s \geq 1 - 2s^2$.

2. Factor $(2s - 1)(s + 1) \geq 0$, so $s \geq 1/2$ and $\theta \in [\pi/6, \pi/2]$.

3. The probability is the angle-length ratio $(\pi/2 - \pi/6)/(\pi/2) = 2/3$, rather than a length ratio in s.

例题解析

1. 仅为解不等式而令 $s = \sin\theta \in [0, 1]$，得 $s \geq 1 - 2s^2$。

2. 分解得 $(2s - 1)(s + 1) \geq 0$，故 $s \geq 1/2, \theta \in [\pi/6, \pi/2]$。

3. 概率为角度长度比 $(\pi/2 - \pi/6)/(\pi/2) = 2/3$，而非 s 区间的长度比。

Uniform angle does not mean uniform sine or uniform chord length.

角度均匀不代表正弦值均匀或弦长均匀。

If a floor expression is constant on $[n, n + 1)$, first check whether the other side can have the required integer value. For each surviving integer n, solve the exact monotone threshold where the floor changes, and intersect it with that unit interval.

Section 5.3 Geometric probability

几何概率

Method 1 / 方法 1 [PS03-M01] (continued / 续)

若某取整表达式在 $[n, n+1)$ 恒定，先检查另一侧是否能取所需整数值。对每个存活整数 n，求取整改变时的精确单调门槛，并与该单位区间相交。

Practice / 配套练习: AIME-2003-I-Q11 (WB p. 773), AIME-2011-II-Q15 (WB p. 176), AIME-2018-II-Q06 (WB p. 50)

Section 5.3 Geometric probability

几何概率

Method 2 / 方法 2 [PS03-M02]

Clip the correct feasible region by half-planes and distance strips

以半平面与距离带裁切正确可行区域

Procedure

1. Identify the actual sampling region; a moving disk center must stay its radius away from every container side.

2. Expand squared-distance comparisons to linear half-planes, or use distance to a line $|ax+by+c|/\sqrt{a^2+b^2}$.

3. When dividing by a coordinate difference to test slope, split its positive and negative cases before changing inequality direction.

4. Find the clipped polygon vertices, calculate its area, and divide by the area of the feasible sampling region.

通用步骤

1. 确定实际抽样区域；移动圆盘的圆心须距容器每边至少其半径。

2. 展开距离平方比较得到线性半平面，或使用点线距离 $|ax+by+c|/\sqrt{a^2+b^2}$。

3. 以坐标差相除判断斜率时，先按其正负分类，再决定不等号方向。

4. 求裁切多边形顶点、计算面积，并除以可行抽样区域面积。

> Example. Choose a point uniformly from $[0,3]\times[0,2]$. Find the probability that $x\ge 1/2$ and $x+y\le 2$.

> 例题：从 $[0,3]\times[0,2]$ 按面积均匀选点，求 $x\ge 1/2$ 且 $x+y\le 2$ 的概率。

Section 5.3 Geometric probability

几何概率

Method 2 / 方法 2 [PS03-M02] (continued / 续)

Worked solution

1. The rectangle has area six. The two inequalities clip out a right triangle.

2. Its vertices are $(1/2, 0), (2, 0), (1/2, 3/2)$, so its area is $\frac{1}{2}(3/2)^2 = 9/8$.

3. The probability is $(9/8)/6 = 3/16$.

例题解析

1. 矩形面积为六，两不等式裁出一个直角三角形。

2. 三顶点为 $(1/2, 0), (2, 0), (1/2, 3/2)$，面积为 $\frac{1}{2}(3/2)^2 = 9/8$。

3. 概率为 $(9/8)/6 = 3/16$。

Normalize by where the point or center was actually sampled.

分母须为点或圆心实际被抽取的区域。

Practice / 配套练习: AIME-2004-I-Q10 (WB p. 769), AIME-2010-II-Q02 (WB p. 766), AIME-2017-II-Q03 (WB p. 767), AIME-2020-II-Q02 (WB p. 768)

Section 5.3 Geometric probability

几何概率

Method 3 / 方法 3 [PS03-M03]

Use the parameter square for two independent uniform choices

对两个独立均匀选择使用参数正方形

Procedure

1. The two original parameters form a uniform rectangle because they are independent.

2. Use geometry to translate a chord or distance condition into a condition on their sum or difference.

3. Draw the resulting diagonal half-planes or band and use a complementary triangle count where possible.

4. Keep any angle range needed for the monotonicity of sine or another inverse transformation.

通用步骤

1. 两个原参数独立，因此形成均匀参数矩形。

2. 用几何将弦长或距离条件转为两参数和差的条件。

3. 画出所得对角半平面或带状区域，可能时用补三角形求面积。

4. 保留保证正弦或其他逆变换单调的角度范围。

> Example. Independent U, V are uniform on $[0,1]$. Find $P(U+V \geq 1,\ U-V \leq 1/2)$.

> 例题：U, V 独立且均匀分布于 $[0,1]$，求 $P(U+V \geq 1,\ U-V \leq 1/2)$。

Section 5.3 Geometric probability

几何概率

Method 3 / 方法 3 [PS03-M03] (continued / 续)

Worked solution

1. The joint sample square has area one. The condition $U+V \geq 1$ leaves a triangle of area $1/2$.

2. Within it the forbidden region $U-V>1/2$ is the triangle with vertices $(3/4,1/4),(1,0),(1,1/2)$, of area $1/16$.

3. The required probability is $1/2-1/16=7/16$.

例题解析

1. 联合样本正方形面积为一；条件 $U+V \geq 1$ 留下面积 $1/2$ 的三角形。

2. 其中禁用区 $U-V>1/2$ 是顶点为 $(3/4,1/4),(1,0),(1,1/2)$ 的三角形，面积为 $1/16$。

3. 所求概率为 $1/2-1/16=7/16$。

The natural probability coordinates may be angles even when the picture is a circle.

即使图形为圆，天然概率坐标仍可能是角度。

Practice / 配套练习: AIME-2017-I-Q08 (WB p. 771)

Section 5.3 Geometric probability

几何概率

Method 4 / 方法 4 [PS03-M04]

Translate chord events into cyclic endpoint order and arc masses

把弦事件转为端点循环次序与弧质量

Procedure

1. For endpoints uniform on a circle, an arc is selected with probability equal to its fraction of circumference.

2. A chord misses a convex inscribed polygon exactly when both endpoints lie on one vertex gap arc, apart from boundaries.

3. Two chords cross internally exactly when their four distinct endpoints alternate cyclically.

4. Under arc or quadrant conditioning, enumerate arc-pair cases first, then use symmetry of endpoint order within any shared arc.

通用步骤

1. 端点在圆周均匀时，选到某弧的概率等于该弧占周长的比例。

2. 除边界外，弦不交凸内接多边形，当且仅当两端点落在同一顶点间隙弧。

3. 两弦内部相交当且仅当四个互异端点在圆周上交替出现。

4. 附加弧或象限条件时，先枚举弧配对情形，再使用共用弧内端点次序的对称性。

> Example. A fixed chord cuts the circumference into arcs of fractions $1/3$ and $2/3$. Another chord has two independent uniform endpoints. Find its probability of crossing the fixed chord internally.

Section 5.3 Geometric probability

几何概率

Method 4 / 方法 4 [PS03-M04] (continued / 续)

例题：固定弦把圆周分成比例为 $1/3, 2/3$ 的两弧。另一条弦的两端点独立且圆周均匀，求它与固定弦内部相交的概率。

Worked solution

1. Crossing requires one endpoint on each of the two open arcs.

2. The two ordered endpoint cases have probabilities $(1/3)(2/3)$ and $(2/3)(1/3)$.

3. The total is $4/9$. Coincident endpoints have probability zero.

例题解析

1. 相交要求两个端点各在一条开弧上。

2. 两种有序端点情形概率为 $(1/3)(2/3)$ 与 $(2/3)(1/3)$。

3. 合计为 $4/9$，端点重合概率为零。

Specify the chord-generation rule; uniform perimeter endpoints is a particular model.

须明确弦的生成规则；圆周端点均匀是一种特定模型。

For four equal arcs and chords conditioned to use two different arcs, each chord has six equiprobable unordered arc pairs. For two chords, among the 36 ordered pair choices, two disjoint alternating choices force crossing, four disjoint nonalternating choices forbid it, and the remaining thirty have crossing probability $1/2$ by reversal of order in a shared arc. The resulting crossing probability is $(2 + 30/2)/36 = 17/36$.

Section 5.3 Geometric probability

几何概率

Method 4 / 方法 4 [PS03-M04] (continued / 续)

圆周分为四条等弧，弦端点条件为落在不同弧时，每弦有六种等可能无序弧对。两弦共有 36 种有序弧对选择：两种互不共弧且交替的选择必相交，四种互不共弧且不交替的选择必不相交，其余三十种由共用弧内端点顺序反转而各有 $1/2$ 概率相交。故交叉概率为 $(2+30/2)/36=17/36$。

Practice / 配套练习: AIME-2017-I-Q06 (WB p. 770), AIME-2025-I-Q13 (WB p. 782)

Section 5.3 Geometric probability

几何概率

Method 5 / 方法 5 [PS03-M05]

Condition on geometric pieces with their area weights

按几何分区及其面积权重取条件

Procedure

1. Partition a nonrectangular region into disjoint pieces of known areas; their selection probabilities are proportional to area.

2. For two independent points, ordered piece choices have product probabilities.

3. Within a rectangular piece, coordinates are independent uniform variables; justify any further coordinate independence after conditioning.

4. Compute the event inside each feasible piece pair and combine by total probability, often using the complement.

通用步骤

1. 将非矩形区域分为已知面积的互不相交部分，各部分被选概率与面积成正比。

2. 两个独立点的有序分区选择，其概率为两分区概率之积。

3. 矩形分区内坐标为独立均匀变量；若继续取条件，须证明后续坐标独立性。

4. 在每个可行分区对内求事件概率，再以全概率合并，通常可先求补事件。

> Example. A region consists of two disjoint rectangles of areas one and three. Choose two independent uniform points in their union. Find the chance that they occupy different rectangles.

Section 5.3 Geometric probability

几何概率

Method 5 / 方法 5 [PS03-M05] (continued / 续)

> 例题：某区域由面积一与三的两个互不相交矩形组成。在其并集中独立均匀选两点，求两点分属不同矩形的概率。

Worked solution

1. The piece probabilities are $1/4$ and $3/4$, rather than one half each.

2. The two disjoint ordered cases are first-small/second-large and first-large/second-small.

3. Their probabilities sum to $2(1/4)(3/4) = 3/8$.

例题解析

1. 两分区概率为 $1/4, 3/4$，而非各二分之一。

2. 两种互斥有序情形为先小后大、先大后小。

3. 相加得 $2(1/4)(3/4) = 3/8$。

A region made of equal-area pieces gives uniform piece labels; unequal pieces require weights.

等面积分区才给出均匀分区标签；面积不等必须加权。

Practice / 配套练习: AIME-2023-II-Q06 (WB p. 772)

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 1 / 方法 1 [PS04-M01]

An expectation is a weighted total, not necessarily a typical outcome. Use symmetry for exchangeable gaps, tails for integer scores, and complete states for stopping processes. A strategy is locally optimal only when its choice does not change the future state distribution.

期望是加权总量，不一定是典型结果。可交换间隙用对称性，整数分数用尾概率，停止过程用完整状态。只有决策不影响未来状态分布时，逐步贪心才自动最优。

Average order statistics through exchangeable gaps

用可交换间隙求顺序统计量均值

Procedure

1. Write the selected positions increasingly and record all unselected gaps, including both end gaps.

2. Prove a bijection to weak compositions of the unselected total; uniform subsets then give exchangeable gaps.

3. Each gap has the same mean, and their means sum to the fixed total.

4. Express the desired ordered position as its rank plus the preceding gaps.

通用步骤

1. 将所选位置递增排列，记录所有未选间隙，含首尾间隙。

2. 证明它们与未选总量的非负整数组成双射；均匀子集于是给出可交换间隙。

3. 各间隙均值相同，且均值总和等于固定未选总量。

4. 第几小的位置等于其序号加上它前面的所有间隙。

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 1 / 方法 1 [PS04-M01] (continued / 续)

> Example. Choose four positions uniformly from 1 through 9. Find the expected second-smallest position.

> 例题：从一至九中等可能选四个位置，求第二小位置的期望。

Worked solution

1. The five unchosen positions form five nonnegative gaps $g_0, \ldots, g_4$ with sum five. Each weak composition occurs once.

2. Exchangeability gives $E(g_i) = 1$ for every gap.

3. The second chosen position is $2 + g_0 + g_1$, so its expectation is 4.

例题解析

1. 五个未选位置形成五个非负间隙 $g_0, \ldots, g_4$，总和五，每种非负组成恰对应一个子集。

2. 由可交换性，每个间隙的期望为一。

3. 第二个所选位置为 $2 + g_0 + g_1$，故期望为四。

The gaps are exchangeable, not independent; equal means need only symmetry.

间隙可交换但并不独立；求相同均值只需对称性。

Practice / 配套练习: AIME-2015-I-Q12 (WB p. 777)

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 2 / 方法 2 [PS04-M02]

Sum tail probabilities for an integer score

对整数分数累加尾概率

Procedure

1. For a nonnegative integer score X, use $E(X)=\sum_{j\ge 1}\Pr(X\ge j)$.

2. If a lower bound L is automatic, start with L and sum only tails above it.

3. Translate each tail event into simultaneous restrictions and count it in the original uniform sample space.

4. For nested allowed sets, assign the smallest set first and subtract already-used choices.

通用步骤

1. 非负整数分数满足 $E(X)=\sum_{j\ge 1}\Pr(X\ge j)$。

2. 若自动有下界 L，先记 L，只累加更高的尾概率。

3. 把每个尾事件转为同时满足的限制，并在原等可能样本空间中计数。

4. 允许集合嵌套时，先分配最小集合，再减去已使用的选择。

> Example. Roll two independent fair dice. Find the expected larger result.

> 例题：独立投两枚公平骰子，求较大点数的期望。

Worked solution

1. Let X be the maximum. For $1\le j\le 6$, $\Pr(X\ge j)=1-((j-1)/6)^2$.

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 2 / 方法 2 [PS04-M02] (continued / 续)

2. Therefore $E(X) = 6 - (0^2 + 1^2 + 2^2 + 3^2 + 4^2 + 5^2)/36$.

3. The square sum is 55, so the expected maximum is $161/36$.

例题解析

1. 令 X 为最大点数，则 $1 \le j \le 6$ 时，$\Pr(X \ge j) = 1 - ((j-1)/6)^2$。

2. 所以 $E(X) = 6 - (0^2 + 1^2 + 2^2 + 3^2 + 4^2 + 5^2)/36$。

3. 平方和为 55，故最大值期望为 $161/36$。

A minimum exceeding a threshold means every component passes it, which often makes tails easier than the full distribution.

最小值超过门槛等价于所有分量都过关，故尾概率往往比完整分布好数。

The identity is pointwise: $X = \sum_{j \ge 1} \mathbf{1}_{X \ge j}$. For a rook assignment with nested row-allowed sets $S_1 \subseteq \cdots \subseteq S_r$, earlier choices all belong to the next set, so row i has exactly $|S_i| - (i-1)$ available columns. Complete any unrestricted rows by a factorial.

尾和恒等式来自逐点等式 $X = \sum_{j \ge 1} \mathbf{1}_{X \ge j}$。车的行列分配若允许集合满足 $S_1 \subseteq \cdots \subseteq S_r$，之前所选列均属于下一集合，故第 i 行恰有 $|S_i| - (i-1)$ 个可用列；余下无限制行用阶乘排列。

Practice / 配套练习: AIME-2016-II-Q13 (WB p. 779)

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 3 / 方法 3 [PS04-M03]

Write first-step equations with the correct boundary reward

按边界奖励写首步方程

Procedure

1. Choose states that determine every future transition, including holding and reflecting behavior.

2. A hitting probability satisfies $p_s = \sum_t P_{st} p_t$ with boundary values zero or one.

3. An expected step count satisfies $T_s = 1 + \sum_t P_{st} T_t$ with zero at absorption; keep any self-transition term.

4. Solve the finite system and justify absorption or finite expectation using a uniform positive chance of success in bounded blocks.

通用步骤

1. 选择能决定所有未来转移的状态，包括原地停留及反射行为。

2. 命中概率满足 $p_s = \sum_t P_{st} p_t$，边界值为零或一。

3. 期望步数满足 $T_s = 1 + \sum_t P_{st} T_t$，吸收状态为零；原地转移项不能漏。

4. 解有限方程组，并用有限步块中统一的正成功概率证明吸收或期望有限。

> Example. A walker on $0, 1, 2, 3$ moves right with probability $2/3$ and left with $1/3$, stopping at either endpoint. Starting at one, find the probability of stopping at three and the expected number of moves.

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 3 / 方法 3 [PS04-M03] (continued / 续)

> 例题：行者在 $0,1,2,3$ 上移动，向右概率 $2/3$、向左 $1/3$，到任一端点就停止。从一出发，求停在三的概率及期望步数。

Worked solution

1. The hitting probabilities obey $p_1 = 2p_2/3$ and $p_2 = 2/3 + p_1/3$. Thus $p_1 = 4/7, p_2 = 6/7$.

2. The expected times obey $T_1 = 1 + 2T_2/3$ and $T_2 = 1 + T_1/3$. Solving gives $T_1 = 15/7, T_2 = 12/7$.

3. From either interior state, two consecutive right moves guarantee absorption with probability at least $4/9$. Thus the expected stopping time is finite, validating these equations.

例题解析

1. 命中概率满足 $p_1 = 2p_2/3, p_2 = 2/3 + p_1/3$，解得 $p_1 = 4/7, p_2 = 6/7$。

2. 期望满足 $T_1 = 1 + 2T_2/3, T_2 = 1 + T_1/3$，解得 $T_1 = 15/7, T_2 = 12/7$。

3. 从任一内点连续两次右行都保证已吸收，概率至少 $4/9$，故停止期望有限，以上方程有效。

Probability equations have boundary values; time equations have a unit cost per step. Do not interchange them.

概率方程靠边界值，时间方程每步另加一，不能混用。

For a height walk with equally likely east, west, north, south, the two horizontal holding moves give $T_{j+1} - 2T_j + T_{j-1} = -4$. At a reflecting boundary where only east, west, north are available, $T_0 = 1 + (2T_0 + T_1)/3$, hence $T_0 - T_1 = 3$. First differences reduce the interior equation to an arithmetic sequence.

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 3 / 方法 3 [PS04-M03] (continued / 续)

For a strictly increasing walk, a target can be visited at most once. Partitioning paths by the last step gives a visit-probability recurrence without an expected-time unit cost.

若高度行走等可能向东西南北，两种水平停留使 $T_{j+1} - 2T_j + T_{j-1} = -4$。反射边界若只可向东西北，则 $T_0 = 1 + (2T_0 + T_1)/3$，故 $T_0 - T_1 = 3$。对期望作一阶差分，内点方程便化为等差数列。

严格向前的行走至多访问目标一次，因此可按最后一步分类得到访问概率递推；这里不能额外加期望时间的单位代价。

Practice / 配套练习: AIME-2016-I-Q13 (WB p. 778), AIME-2019-II-Q02 (WB p. 774), AIME-2021-I-Q12 (WB p. 781)

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 4 / 方法 4 [PS04-M04]

Aggregate symmetric states only after checking transition totals

核对转移总量后才能合并对称状态

Procedure

1. Replace absolute positions by relative gaps or another symmetry invariant.

2. Check that every microscopic state in one proposed class has the same probabilities of moving to every class.

3. Enumerate all simultaneous elementary moves, including collisions and any memory restrictions.

4. Use the resulting class transition table in the first-step equations.

通用步骤

1. 以相对间隙等对称不变量代替绝对位置。

2. 核对同一拟定类别中的每个细状态，转向各类别的概率均相同。

3. 枚举同时发生的全部基本移动，包括碰撞和历史限制。

4. 把所得类别转移表代入首步方程。

> Example. Two walkers start at opposite vertices of a square. Each second both independently choose a neighboring vertex with equal probabilities. They stop on their first simultaneous meeting. Find the expected stopping time.

> 例题：两名行者从正方形对顶点出发，每秒同时、独立等可能选一个相邻顶点，到首次同时相遇时停止。求停止时间期望。

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 4 / 方法 4 [PS04-M04] (continued / 续)

Worked solution

1. Before meeting they remain opposite: equal clockwise choices keep them opposite, while opposite choices send them to the same vertex.

2. Of the four equally likely simultaneous choices, two meet and two return to the same opposite-state class.

3. For expected time T, $T = 1 + T/2$, so $T = 2$. The geometric survival probability 2^{-n} also proves finite expectation.

例题解析

1. 未相遇时两人仍在对顶点；同向转动仍对顶，反向选择则在同一顶点相遇。

2. 四种等可能同时选择中，两种相遇，两种返回相同的对顶状态类。

3. 期望 T 满足 $T = 1 + T/2$，故为两秒；存活概率 2^{-n} 也证明期望有限。

Equal-looking states can have different transition counts; symmetry must preserve the process, not only the drawing.

外观相同的状态可能转移数不同；所用对称必须保持过程，而不只是图形。

Practice / 配套练习: AIME-2021-I-Q12 (WB p. 781)

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 5 / 方法 5 [PS04-M05]

Use a signed generating value for stopping-time parity

用带符号生成值求停止次数奇偶

Procedure

1. Track the longest suffix that matches a prefix of the target pattern.

2. Let $u_s = E((-1)^R)$ for remaining rolls. Every additional roll contributes a factor -1, and a completed pattern has remaining value one.

3. Write the first-step equations, including fallback to shorter matched prefixes.

4. Recover $\Pr(R \text{ odd}) = (1 - u_s)/2$ after proving almost-sure stopping.

通用步骤

1. 以“当前结尾能匹配目标前缀的最长长度”为状态。

2. 令 $u_s = E((-1)^R)$，每增加一步乘负一，已完成状态的余下次数为零、对应值为一。

3. 写首步方程，并包含退回较短匹配前缀的情况。

4. 证明几乎必然停止后，用 $\Pr(R\text{为奇数}) = (1 - u_s)/2$ 求概率。

> Example. Toss a fair coin until the first consecutive pattern HT. Find the probability the total toss count is odd.

> 例题：投公平硬币，首次连续出现 HT 时停止，求总次数为奇数的概率。

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 5 / 方法 5 [PS04-M05] (continued / 续)

Worked solution

1. Use state zero for no trailing H, and state one for a trailing H. Let the corresponding signed expectations be u_0, u_1.

2. First-step equations are $u_0 = -(u_0 + u_1)/2$ and $u_1 = -(u_1 + 1)/2$. Thus $u_1 = -1/3, u_0 = 1/9$.

3. The desired probability is $(1 - 1/9)/2 = 4/9$. Disjoint two-toss blocks have a fixed $1/4$ chance of HT, so stopping is almost sure.

例题解析

1. 用零状态表示末尾不为 H，一状态表示末尾为 H，带符号期望分别为 u_0, u_1。

2. 首步方程为 $u_0 = -(u_0 + u_1)/2, u_1 = -(u_1 + 1)/2$，解得 $u_1 = -1/3, u_0 = 1/9$。

3. 所求概率为 $(1 - 1/9)/2 = 4/9$。互不重叠的两次块各有 $1/4$ 概率出现 HT，故几乎必然停止。

Immediate completion still uses one roll, so its contribution has a minus sign.

立即完成仍消耗一步，因此在方程中的贡献带负号。

Practice / 配套练习: AIME-2018-II-Q13 (WB p. 780)

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 6 / 方法 6 [PS04-M06]

Optimize a prediction when choices do not affect future states

决策不影响未来状态时优化即时预测

Procedure

1. Prove that the current guess or action does not alter the next-state distribution.

2. Maximize the conditional immediate expected reward at each state.

3. Add that reward to the transition-weighted future value; keep terminal states exact.

4. Use a two-parameter recurrence when the remaining counts determine the state.

通用步骤

1. 先证明当前猜测或行动不会改变下一状态分布。

2. 在每个状态最大化当前条件期望奖励。

3. 将该奖励加上按转移概率加权的未来价值，并保留准确边界状态。

4. 若剩余数量决定状态，就用双参数递推。

> Example. A shuffled deck has two red cards and one black card. Before each reveal, a correct red guess earns two points and a correct black guess one point. Find the maximum expected total reward.

> 例题：随机洗匀两红一黑牌，每次翻牌前猜色，猜中红色得两分、黑色得一分。求最大期望总得分。

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 6 / 方法 6 [PS04-M06] (continued / 续)

Worked solution

1. At state (r, b), choose red when $2r \geq b$. The guess does not affect which card is revealed.

2. Let F be optimal expected reward. Then $F(1,0) = 2, F(0,1) = 1, F(2,0) = 4$, and $F(1,1) = 1 + (1+2)/2 = 5/2$.

3. At $(2,1)$, the immediate expectation is $4/3$, so $F(2,1) = 4/3 + (2/3)(5/2) + (1/3)4 = 13/3$.

例题解析

1. 状态 (r, b) 下，若 $2r \geq b$ 就猜红；猜测并不改变翻出的牌。

2. 令 F 为最优期望，边界 $F(1,0) = 2, F(0,1) = 1, F(2,0) = 4$，且 $F(1,1) = 1 + (1+2)/2 = 5/2$。

3. 在 $(2,1)$ 中，即时奖励期望为 $4/3$，故 $F(2,1) = 4/3 + (2/3)(5/2) + (1/3)4 = 13/3$。

Majority color is optimal for equal correct-guess rewards; unequal rewards change the threshold.

只有猜中两色奖励相同时才直接选多数色；奖励不等会改变阈值。

Practice / 配套练习: AIME-2023-I-Q06 (WB p. 775)

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 7 / 方法 7 [PS04-M07]

Count regions by linearity of crossing indicators

用交叉指示量的线性性数区域

Procedure

1. When adding a segment with distinct interior crossings, its new regions equal one plus its number of crossings.

2. Express the total as a fixed segment contribution plus one indicator for each crossing pair.

3. Compute each pair probability by endpoint interleaving under the actual conditioned endpoint distribution.

4. Use linearity; pairwise crossing indicators need not be independent. Treat prescribed multiple crossings separately.

通用步骤

1. 加入线段时，若内部交点互异，新增区域数等于交点数加一。

2. 将总量写为固定线段贡献加上每对交叉的指示量。

3. 根据真实的条件端点分布，用端点交错求各对交叉概率。

4. 使用期望线性性，无须交叉事件独立；预设的多线共点须另行处理。

> Example. A disk has two fixed perpendicular diameters and three independent chords, each formed by two uniform independent perimeter endpoints. Find the expected number of regions.

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 7 / 方法 7 [PS04-M07] (continued / 续)

> 例题：圆盘内有两条固定垂直直径，再独立画三条弦，每弦两端独立均匀取自圆周，求期望区域数。

Worked solution

1. There are five segments. Apart from probability-zero coincidences, regions equal $1+5+$ the number of crossing pairs. The fixed diameters contribute one crossing.

2. Each random chord crosses a given diameter with probability $1/2$, since its endpoints must lie in opposite semicircles. Two independent random chords cross with probability $1/3$: for four cyclically ordered endpoints, exactly one of the three pairings interleaves.

3. Thus the expectation is $6+1+3\cdot 2\cdot(1/2)+\binom{3}{2}(1/3)=11$.

例题解析

1. 共有五条线段。除零概率重合外，区域数等于 $1+5+$交叉线段对数；固定直径贡献一个交点。

2. 随机弦与一条直径相交概率为 $1/2$，因为两端须在不同半圆；两随机弦相交概率为 $1/3$，四个圆周顺序端点的三种配对中恰一种交错。

3. 所以期望为 $6+1+3\cdot 2\cdot(1/2)+\binom{3}{2}(1/3)=11$。

Conditioning endpoints to different quadrants changes the crossing law; do not reuse the unrestricted one-third value.

若端点被限制在不同象限，相交概率会改变，不能沿用无条件的三分之一。

Practice / 配套练习: AIME-2025-I-Q13 (WB p. 782)

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 8 / 方法 8 [PS04-M08]

Stop at the first third-category event and sum completed prefixes

在第三类首次出现时停止并累加已完成前缀

Procedure

1. Partition trials into two desired categories and a stopping category.

2. Condition on the full pre-stop word, including its order and the final stopping event.

3. Count or weight words that complete both desired quotas; handle overlap of deficient-category events.

4. Sum the resulting geometric or arithmetic-geometric series and check the starting index.

通用步骤

1. 把试验分为两种目标类别及一种停止类别。

2. 按停止前的完整词分类，保留顺序并计入最后的停止事件。

3. 对达到两种配额的词计数或加权；缺额事件可能有交集。

4. 累加等比或等差乘等比级数，并核对起始下标。

> Example. Independent trials produce A, B, C with probabilities $1/2, 1/4, 1/4$. What is the chance both A and B appear before the first C?

> 例题：独立试验出现 A, B, C 的概率为 $1/2, 1/4, 1/4$。求首次出现 C 之前，A, B 都已出现的概率。

Section 5.4 Expected values & stopping processes

期望与停止过程

Method 8 / 方法 8 [PS04-M08] (continued / 续)

Worked solution

1. No A before the first C means a word B^kC, with total probability $\sum_{k\ge0}(1/4)^{k+1}=1/3$.

2. No B similarly has probability $\sum_{k\ge0}(1/2)^k(1/4)=1/2$. Their intersection is an immediate C, with probability $1/4$.

3. By inclusion-exclusion, success is $1-1/3-1/2+1/4=5/12$. The final stopping-event weight is included in both series.

例题解析

1. 首次 C 之前无 A，对应词 B^kC，总概率为 $\sum_{k\ge0}(1/4)^{k+1}=1/3$。

2. 无 B 的概率同理为 $\sum_{k\ge0}(1/2)^k(1/4)=1/2$，两者交集为立即出现 C，概率 $1/4$。

3. 容斥得成功概率 $1-1/3-1/2+1/4=5/12$；两级数都计入了末次停止事件的权重。

For two quotas of two, deficient binary counts are disjoint only once the prefix length is at least four.

两类配额各为二时，只有前缀长至少四，两个缺额二元词类才互斥。

For $|x|<1$, $\sum_{k\ge0}x^k=1/(1-x)$ and $\sum_{k\ge0}(k+1)x^k=1/(1-x)^2$. Subtract the finitely many terms before the valid quota length. The second identity follows by multiplying two geometric series, so differentiation is optional.

当 $|x|<1$，有 $\sum_{k\ge0}x^k=1/(1-x)$ 及 $\sum_{k\ge0}(k+1)x^k=1/(1-x)^2$。减去达到配额前的有限项即可；第二式也可由两个等比级数相乘得到，无须使用求导。

Practice / 配套练习: AIME-2026-II-Q07 (WB p. 776)

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 1 / 方法 1 [PS05-M01]

Distinguish a probability mass from an outcome count. Independent trials give binomial, multinomial, geometric, or state-dependent laws; the right law depends on whether order, stopping, or memory matters.

必须区分点概率与结果数。独立试验可形成二项、多项、几何或状态分布；使用哪种分布，取决于顺序、停止规则及历史记忆。

Pair symmetric score outcomes and count the balance cases

配对对称得分并单独计算平衡情形

Procedure

1. Find a probability-preserving involution exchanging the two strict score comparisons.

2. Count the fixed balance event with the correct multinomial weights.

3. Subtract the balance probability from one and divide by two.

4. Check that the two compared categories have identical laws before using this symmetry.

通用步骤

1. 寻找保持概率且交换两个严格得分比较的对合。

2. 用正确的多项权重计算平衡事件。

3. 从一减去平衡概率，再除以二。

4. 使用该对称前，确认两个被比较类别的规律相同。

> Example. Three independent increments equal $1, -1, 0$ with probabilities $1/4, 1/4, 1/2$. Find the probability their sum is positive.

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 1 / 方法 1 [PS05-M01] (continued / 续)

> 例题：三个独立增量各以概率 $1/4, 1/4, 1/2$ 取 $1, -1, 0$，求总和为正的概率。

Worked solution

1. Negating all increments exchanges positive and negative totals and preserves their probabilities.

2. A zero total is either three zeros, probability $1/8$, or one of each symbol, probability $6(1/4)^2(1/2) = 3/16$. Its total probability is $5/16$.

3. Thus the positive-total probability is $(1 - 5/16)/2 = 11/32$. Counting the three symbols as equiprobable would be incorrect.

例题解析

1. 全部增量反号，会交换正和与负和，且概率不变。

2. 总和为零只能是三个零，概率 $1/8$，或三种符号各一个，概率 $6(1/4)^2(1/2) = 3/16$，合为 $5/16$。

3. 因此正和概率为 $(1 - 5/16)/2 = 11/32$，不能把三个符号当成等概率。

A tie must be removed before a symmetry argument halves the sample space.

利用对称性把样本空间平分之前，必须先扣去平局。

Practice / 配套练习: AIME-2001-II-Q11 (WB p. 789)

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 2 / 方法 2 [PS05-M02]

Compare binomial scores by complementing one count

用一个计数的补数比较二项得分

Procedure

1. Write the score inequality in terms of independent remaining-success counts.

2. For fair trials, replace one count Y from m trials by $m-Y$, which has the same binomial law.

3. Use convolution or Vandermonde to turn the comparison into a single binomial tail.

4. Only when the original two laws match may exchange symmetry split the strict comparisons equally.

通用步骤

1. 将得分比较写成两个独立剩余成功次数的不等式。

2. 公平试验中，将 m 次中的 Y 替换为同分布的 $m-Y$。

3. 用卷积或范德蒙德恒等式，把比较转为一个二项尾事件。

4. 只有原来两种分布相同时，交换对称性才能平分严格大小事件。

> Example. Independently let X count heads in three fair flips and Y heads in two fair flips. Find $\Pr(X \geq Y)$.

> 例题：独立地，X 为三次公平投币正面数，Y 为两次正面数，求 $\Pr(X \geq Y)$。

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 2 / 方法 2 [PS05-M02] (continued / 续)

Worked solution

1. The variable $2-Y$ is an independent two-flip head count. Thus $Z=X+(2-Y)$ is a five-flip head count.

2. The event $X \geq Y$ is exactly $Z \geq 2$.

3. Its probability is $1-(\binom{5}{0}+\binom{5}{1})/2^5=26/32=13/16$. The unequal trial counts prevent directly halving the non-tie event.

例题解析

1. $2-Y$ 仍为独立的两次公平投币正面数，所以 $Z=X+(2-Y)$ 为五次正面数。

2. 事件 $X \geq Y$ 等价于 $Z \geq 2$。

3. 概率为 $1-(\binom{5}{0}+\binom{5}{1})/2^5=13/16$。两边试验次数不同，不能直接把非平局事件平分。

The convolution identity $\sum_j \binom{n}{j}\binom{m}{j}=\binom{n+m}{n}$ counts equal scores.

卷积恒等式 $\sum_j \binom{n}{j}\binom{m}{j}=\binom{n+m}{n}$ 可计算相等得分。

Practice / 配套练习: AIME-2006-II-Q10 (WB p. 785)

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 3 / 方法 3 [PS05-M03]

Enumerate multinomial count vectors under a weighted total

按加权总量枚举多项计数向量

Procedure

1. Let n_i count occurrences of each outcome value. Write both $\sum n_i = n$ and the weighted-sum equation.

2. Bound and list all nonnegative integer count vectors satisfying the equations.

3. A vector has $n!/\prod n_i!$ orders and probability $(n!/\prod n_i!)\prod p_i^{n_i}$.

4. Sum the disjoint vector probabilities; a common denominator is valid only when the elementary outcomes really are equiprobable.

通用步骤

1. 令 n_i 为各结果出现次数，同时写出总次数及加权总和方程。

2. 给出界并列全满足条件的非负整数计数向量。

3. 一个向量有 $n!/\prod n_i!$ 种顺序，概率为其乘 $\prod p_i^{n_i}$。

4. 相加互斥向量的概率；只有基本结果确实等可能，才能统一按结果数作分母。

> Example. Four independent draws take values $0, 1, 3$ with probabilities $1/2, 1/3, 1/6$. Find the probability their sum is three.

> 例题：四次独立抽取，各以概率 $1/2, 1/3, 1/6$ 取值 $0, 1, 3$，求总和为三的概率。

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 3 / 方法 3 [PS05-M03] (continued / 续)

Worked solution

1. If a three occurs, the other values are all zero, giving counts $(3,0,1)$ and probability $4(1/2)^3(1/6)=1/12$.

2. Without a three, there must be three ones and one zero, giving probability $4(1/3)^3(1/2)=2/27$.

3. The cases are disjoint, so the total is $1/12+2/27=17/108$.

例题解析

1. 若出现三，其余都为零，次数向量为 $(3,0,1)$，概率 $4(1/2)^3(1/6)=1/12$。

2. 若不出现三，就需三个一和一个零，概率为 $4(1/3)^3(1/2)=2/27$。

3. 两类互斥，总概率为 $1/12+2/27=17/108$。

A fixed total can arise from several count vectors; one convenient vector is not an exhaustive list.

固定总和可能对应多个计数向量，找到一个方便向量并不等于穷尽。

Practice / 配套练习: AIME-2008-I-Q09 (WB p. 786)

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 4 / 方法 4 [PS05-M04]

Cancel binomial probabilities to recover success odds

约去二项概率以恢复成功赔率

Procedure

1. Write both exact-success probabilities with their binomial coefficients.

2. Divide by positive common factors, preserving any unequal combinatorial coefficient.

3. Solve for the positive odds $p/(1-p)$ and restore $0 < p < 1$.

4. Check the original probability ratio rather than only the powered equation.

通用步骤

1. 写出两个准确成功次数的二项概率及组合系数。

2. 除去正的公共因子，但保留不相等的组合系数。

3. 求正赔率 $p/(1-p)$，并恢复 $0 < p < 1$。

4. 核对原概率比，而不只核对取幂后的方程。

> Example. In four independent flips of a coin with head probability $0 < p < 1$, zero heads and exactly two heads have equal probabilities. Find p.

> 例题：独立投一枚正面概率满足 $0 < p < 1$ 的硬币四次，零次正面与恰两次正面的概率相等，求 p。

Worked solution

1. The equality is $(1-p)^4 = 6p^2(1-p)^2$. A valid nondegenerate coin has $0 < p < 1$.

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 4 / 方法 4 [PS05-M04] (continued / 续)

2. Cancel $(1-p)^2$ and take the positive square root: $1-p=\sqrt{6}p$.

3. Thus $p=1/(1+\sqrt{6})=(\sqrt{6}-1)/5$, which lies in $(0,1)$ and satisfies the original equality.

例题解析

1. 方程为 $(1-p)^4=6p^2(1-p)^2$，非退化硬币满足 $0<p<1$。

2. 约去 $(1-p)^2$ 并取正平方根，得 $1-p=\sqrt{6}p$。

3. 故 $p=(\sqrt{6}-1)/5$，位于零与一之间，代回原式成立。

Equal binomial coefficients are a special cancellation, not a general rule.

两边组合系数相等只是特殊情况，不能普遍略去。

Practice / 配套练习: AIME-2009-I-Q03 (WB p. 783)

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 5 / 方法 5 [PS05-M05]

Count allowed prefixes before attaching trial weights

先数合法前缀再附加试验权重

Procedure

1. Let $D(n, j)$ count allowed prefixes of length n with j successes.

2. Update by success or failure, discarding each state that violates the prefix boundary at that time.

3. If all final words have the same success total, multiply their count by the common Bernoulli word weight.

4. Keep the terminal rule separate when the boundary applies only to proper prefixes.

通用步骤

1. 令 $D(n, j)$ 为长 n、含 j 次成功的合法前缀数。

2. 按成功或失败转移，每一步都丢弃当时越过前缀边界的状态。

3. 若终态成功次数相同，各词权重相同，可用词数乘共同伯努利权重。

4. 若边界只限制真前缀，须单独处理末步规则。

> Example. Six independent trials succeed with probability $1/3$. Find the probability of exactly two successes with no prefix having more successes than failures.

> 例题：六次独立试验成功概率为 $1/3$，求恰两次成功且任一前缀中成功数不多于失败数的概率。

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 5 / 方法 5 [PS05-M05] (continued / 续)

Worked solution

1. Use $D(0,0)=1$ and $D(n,j)=D(n-1,j)+D(n-1,j-1)$ only when $2j \le n$.

2. The rows through $n=6$, restricted to relevant successes, end with $D(4,2)=2, D(5,2)=5, D(6,2)=9$; the last follows $D(5,1)+D(5,2)=4+5$.

3. Each accepted word has weight $(1/3)^2(2/3)^4$. Thus the probability is $9 \cdot 16/729 = 16/81$.

例题解析

1. 取 $D(0,0)=1$，仅保留 $2j \le n$ 的状态，用递推 $D(n,j)=D(n-1,j)+D(n-1,j-1)$。

2. 逐行计算得 $D(4,2)=2, D(5,2)=5, D(6,2)=9$，末项来自 $D(5,1)+D(5,2)=4+5$。

3. 每个合法词权重为 $(1/3)^2(2/3)^4$，所以概率为 $9 \cdot 16/729 = 16/81$。

A correct final success fraction does not guarantee that every earlier prefix was legal.

最终成功比例正确，不代表每个先前前缀都合法。

Practice / 配套练习: AIME-2002-II-Q12 (WB p. 790)

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 6 / 方法 6 [PS05-M06]

Match independent stopping counts by summing mass products

累加点概率乘积以匹配独立停止次数

Procedure

1. Write each complete stopping-count mass, such as $p(1-p)^{k-1}$.

2. For equal counts sum $\sum_k p_k q_k$; for an offset sum the shifted products over their common support.

3. Use a geometric series for infinite support and include both offset directions separately.

4. Do not double one offset when the two distributions differ.

通用步骤

1. 写出完整停止次数的点概率，如 $p(1-p)^{k-1}$。

2. 次数相等时求 $\sum_k p_k q_k$，有偏移时按共同支持集累加移位乘积。

3. 无穷支持用等比级数，两个偏移方向要分别计入。

4. 若两种分布不同，不能把某一方向直接乘二。

> Example. Two independent processes stop on their first successes, with per-trial success probabilities $1/2$ and $1/3$. Find the chance their stopping counts differ by at most one.

> 例题：两个独立过程分别以每次成功概率 $1/2$、$1/3$ 等待首次成功，求停止次数相差不超过一的概率。

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 6 / 方法 6 [PS05-M06] (continued / 续)

Worked solution

1. Equal counts have probability $\sum_{k\ge1}(1/2)^k(1/3)(2/3)^{k-1}=(1/6)/(1-1/3)=1/4$.

2. The first process stopping one later contributes $(1/2)(1/4)=1/8$; the second stopping one later contributes $(2/3)(1/4)=1/6$.

3. These are disjoint, totaling $1/4+1/8+1/6=13/24$. Their unequal offset weights show why symmetry cannot be assumed.

例题解析

1. 次数相等概率为 $\sum_{k\ge1}(1/2)^k(1/3)(2/3)^{k-1}=1/4$。

2. 第一个过程晚一次的概率为 $(1/2)(1/4)=1/8$，第二个晚一次为 $(2/3)(1/4)=1/6$。

3. 三类互斥，总概率为 $1/4+1/8+1/6=13/24$。两方向权重不同，不能假设对称。

Equality of two independent counts is a sum of products of masses, not a product of their total probabilities.

两个独立次数相等应累加各点概率的乘积，而不是乘各自总概率。

Practice / 配套练习: AIME-2009-II-Q08 (WB p. 787)

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 7 / 方法 7 [PS05-M07]

Multiply probability-generating polynomials for unequal trials

用概率生成多项式处理不同成功率

Procedure

1. A Bernoulli trial with success probability p contributes $(1-p)+pt$.

2. Multiply these factors for independent trials, retaining exact coefficients.

3. The coefficient of t^j is the full probability mass for j successes.

4. For independent repetitions, match masses by a sum of squared coefficients or convolve the two different laws.

通用步骤

1. 成功概率 p 的伯努利试验对应 $(1-p)+pt$。

2. 独立试验对应因子的乘积，保留精确系数。

3. t^j 的系数就是成功 j 次的完整点概率。

4. 独立重复时，以系数平方和求同次数概率；两种不同分布则作对应卷积。

> Example. Three independent coins have head probabilities $1/2, 1/3, 2/3$. Find the probability of exactly two heads.

> 例题：三枚独立硬币的正面概率为 $1/2, 1/3, 2/3$，求恰两次正面的概率。

Worked solution

1. The generating polynomial is $((1+t)/2)((2+t)/3)((1+2t)/3)$.

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 7 / 方法 7 [PS05-M07] (continued / 续)

2. Expanding gives $(2+7t+7t^2+2t^3)/18$; its coefficients sum to one.

3. The coefficient of t^2 is $7/18$, the desired probability. Equal-probability binomial formulas would not apply to these three coins.

例题解析

1. 生成多项式为 $((1+t)/2)((2+t)/3)((1+2t)/3)$。

2. 展开为 $(2+7t+7t^2+2t^3)/18$，系数和为一。

3. t^2 系数为 $7/18$，即所求概率；三枚硬币成功率不同，不能套同参数二项公式。

Check normalization at $t=1$ before using any coefficient as a probability.

把系数当作概率之前，先代入 $t=1$ 检查总和为一。

Practice / 配套练习: AIME-2010-I-Q04 (WB p. 784)

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 8 / 方法 8 [PS05-M08]

Propagate a walk distribution with enough memory

以足够历史信息传播行走分布

Procedure

1. Choose a state recording all information needed for the next move; a no-backtracking rule needs the previous edge or equivalent type.

2. Group states only when their class transition totals match.

3. Use either probabilities or path counts consistently; if using counts, prove that every complete path has the same weight.

4. Initialize after any exceptional first move and iterate to the requested horizon.

通用步骤

1. 状态须记录决定下一步的全部信息；禁止立即返回时，要记上一条边或等价类型。

2. 只有各类别转移总量相同时才能合并状态。

3. 统一使用概率或路径数；若数路径，必须证明完整路径等权。

4. 首步若规则特殊，应在首步后设初值，再迭代到所求时刻。

> Example. A walker on the complete graph with four vertices starts at A. Its first move is uniform among three neighbors; later moves are uniform among neighbors other than the previous vertex. Find its return probability after four moves.

> 例题：行者在四顶点完全图上从 A 出发，首步等可能去三个邻点，之后等可能去除上一顶点外的邻点。求四步后回到 A 的概率。

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 8 / 方法 8 [PS05-M08] (continued / 续)

Worked solution

1. After two moves the path is A, B, C with all three distinct; symmetry makes the actual names irrelevant.

2. The third step goes to A or the fourth vertex D, each with probability $1/2$. From A at step three, step four cannot remain at A.

3. From D at step three, excluding C leaves A and B as equiprobable choices. Thus the four-step return probability is $(1/2)(1/2) = 1/4$.

例题解析

1. 两步后的路径为互异三点 A, B, C，由对称性，具体标号无关。

2. 第三步等可能到 A 或第四点 D；若第三步到 A，第四步不可能留在 A。

3. 若第三步到 D，排除上一点 C 后，可等可能去 A 或 B。所以四步返回概率为 $1/4$。

Being at the same vertex does not imply the same future law when the incoming edge matters.

若来边影响可走方向，到达同一顶点并不代表未来规律相同。

Practice / 配套练习: AIME-2003-II-Q13 (WB p. 791), AIME-2021-II-Q08 (WB p. 788)

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 9 / 方法 9 [PS05-M09]

Test independence after a coordinate transformation

坐标变换后重新检验独立性

Procedure

1. List the transformed increment pair for every elementary move.

2. Independence requires joint probabilities to equal products of marginal probabilities; a symmetric drawing alone is insufficient.

3. If every sign pair occurs with probability $1/4$, the two coordinate walks are independent fair sign walks.

4. Translate the target event exactly, then use binomial tails and any required union correction.

通用步骤

1. 逐一列出每个基本移动对应的变换后增量对。

2. 独立要求联合概率等于边缘概率乘积，仅有图形对称还不够。

3. 若四种符号对各以概率 $1/4$ 出现，两坐标才是独立的公平正负行走。

4. 准确转换目标事件，再用二项尾概率及必要的并集修正。

> Example. One axis step chooses east or west with probability $1/3$ each, north or south with $1/6$ each. Are $U = x + y$ and $V = x - y$ independent?

> 例题：一步在东西方向各以概率 $1/3$、南北各以 $1/6$ 移动。$U = x + y, V = x - y$ 是否独立?

Section 5.5 Distributions & repeated trials

分布与重复试验

Method 9 / 方法 9 [PS05-M09] (continued / 续)

Worked solution

1. The four transformed pairs for east, west, north, south are $(1,1),(-1,-1),(1,-1),(-1,1)$.

2. Both marginals satisfy $\Pr(U=1)=\Pr(V=1)=1/2$. But $\Pr(U=1,V=1)=1/3$.

3. Since $1/3 \neq (1/2)(1/2)$, the transformed variables are not independent. Equal positive and negative marginals do not certify independence.

例题解析

1. 东西南北对应的变换对分别为 $(1,1),(-1,-1),(1,-1),(-1,1)$。

2. 边缘概率均有 $\Pr(U=1)=\Pr(V=1)=1/2$，但联合概率 $\Pr(U=1,V=1)=1/3$。

3. 由于 $1/3 \neq 1/4$，两变量不独立。边缘分布的正负对称并不能证明独立。

Under uniform axis moves the sign-pair table is uniform; changing move weights can destroy the factorization.

等可能轴向移动给出均匀符号对；改变移动权重可能破坏分解。

Practice / 配套练习: AIME-2014-I-Q11 (WB p. 792)

Section 5.6 Data & descriptive statistics

数据与描述统计

Method 1 / 方法 1 [PS06-M01]

Means preserve total sums; medians preserve order thresholds; a unique mode constrains multiplicities. Choose the matching representation before performing arithmetic or counting.

平均数对应总和，中位数对应有序位置，唯一众数对应重数限制。先选对表示，再计算或计数。

Convert changed means to changed totals

把平均数变化转为总量变化

Procedure

1. For an unknown sample size n and mean M, write its sum as nM.

2. Account for every added or removed observation in both the sum and the sample size.

3. Subtract the mean equations to eliminate a shared unknown, then verify a positive integer size and the original observations.

通用步骤

1. 若样本量为 n、平均数为 M，先把总和写成 nM。

2. 每次增删观测值，都同时修改总和及样本量。

3. 将平均数方程相减，消去公共未知量，再验证正整数样本量及原观测条件。

> Example. A data set has mean 18. Adding observations 10 and 30 raises its mean to 18.4. How many original observations were there?

> 例题：一组数据平均数为 18，加入 $10, 30$ 后，平均数变为 18.4。原来有多少个观测值?

Section 5.6 Data & descriptive statistics

数据与描述统计

Method 1 / 方法 1 [PS06-M01] (continued / 续)

Worked solution

1. For original size n, the new sum is $18n + 40$ and the new size is $n + 2$.

2. Thus $18n + 40 = (92/5)(n + 2)$, or $(2/5)n = 16/5$, giving $n = 8$.

3. The original sum is 144; the new sum 184 divided by ten is 18.4, verifying the result.

例题解析

1. 设原样本量为 n，新总和为 $18n + 40$，新样本量为 $n + 2$。

2. 所以 $18n + 40 = (92/5)(n + 2)$，解得 $n = 8$。

3. 原总和为 144，新总和 184 除以十恰为 18.4，验证无误。

Adding a value to a set may leave the size unchanged if it was already present; determine the intended set or list convention.

若对象为集合，加入已存在的值不会增加元素数；须先分清集合与列表的约定。

Practice / 配套练习: AIME-2001-I-Q02 (WB p. 793)

Section 5.6 Data & descriptive statistics

数据与描述统计

Method 2 / 方法 2 [PS06-M02]

Average positional contributions by symmetry and linearity

用对称性与线性性逐位求平均

Procedure

1. Write each data value as a weighted sum of digits or position indicators.

2. Find the mean at each position from its actual allowed choices or symmetry; positions need not be independent for linearity.

3. Add the weighted means, checking that the enumeration weights every allowed object equally.

通用步骤

1. 将每个数写成数字或位置指示量的加权和。

2. 根据各位置真实的允许取值或对称性求平均；线性性不要求各位置独立。

3. 加总加权平均，并检查所枚举对象是否确实等权。

> Example. Find the mean integer represented by all five-bit binary strings with exactly three ones, allowing leading zeros.

> 例题：求所有恰有三个一的五位二进制串所代表整数的平均值，允许前导零。

Worked solution

1. There are $\binom{5}{3} = 10$ strings. Each bit position occurs in $\binom{4}{2} = 6$ of them.

Section 5.6 Data & descriptive statistics

数据与描述统计

Method 2 / 方法 2 [PS06-M02] (continued / 续)

2. Thus the average indicator at every position is $6/10 = 3/5$. The five positional weights sum to $1 + 2 + 4 + 8 + 16 = 31$.

3. The mean is therefore $(3/5)31 = 93/5$. The bit indicators are dependent, but linearity still applies.

例题解析

1. 共有 $\binom{5}{3} = 10$ 个串，每个位置恰在 $\binom{4}{2} = 6$ 个串中为一。

2. 所以每位指示量平均为 $3/5$，五个位置权重和为 31。

3. 整数平均值为 $(3/5)31 = 93/5$。各位虽不独立，平均数的线性性仍然成立。

Repeated digit appearances share a parameter; include its full positional weight.

重复出现的同一数字共用参数，必须计入它的全部位权。

Practice / 配套练习: AIME-2021-II-Q01 (WB p. 794)

Section 5.6 Data & descriptive statistics

数据与描述统计

Method 3 / 方法 3 [PS06-M03]

Count nested medians through threshold patterns

按阈值模式计数嵌套中位数

Procedure

1. Replace values by categories below, equal to, or above the target median.

2. Translate each inner median condition into counts in these categories, then impose the outer median condition.

3. Count category patterns first; restore distinct labels within each category afterward.

4. Check whether row order, within-row order, and identical symbols create different multiplicities.

通用步骤

1. 按低于、等于或高于目标中位数，把数值替换为类别符号。

2. 将内层中位数条件转为类别个数，再加入外层中位数条件。

3. 先数类别模式，再恢复每类内部的不同标号。

4. 检查行顺序、行内顺序及相同符号分别造成的重数。

> Example. Choose four cells uniformly from a 3×3 grid and write ones there, zeros elsewhere. Take the median of each row, then the median of those three values. What is the probability the final value is one?

> 例题：从 3×3 方格中等可能选四格填一，其余填零。先求各行中位数，再求三个行中位数的中位数。最终为一的概率是多少?

Section 5.6 Data & descriptive statistics

数据与描述统计

Method 3 / 方法 3 [PS06-M03] (continued / 续)

Worked solution

1. A row has median one exactly when it contains at least two ones. The final median is one exactly when at least two rows qualify.

2. There are only four ones, so the row counts must be $2, 2, 0$. Choose the empty row in three ways and the two selected positions of each other row in three ways each.

3. There are 27 favorable choices out of $\binom{9}{4} = 126$. The probability is $27/126 = 3/14$.

例题解析

1. 一行至少有两个一时，其中位数为一；最终中位数为一，当且仅当至少两行满足此条件。

2. 总共只有四个一，所以行内个数只能为 $2, 2, 0$。空行三种，每个非空行选两个位置又各三种。

3. 有利选择为 27，总选择为 $\binom{9}{4} = 126$，概率为 $3/14$。

A median of medians is determined by row thresholds; it need not equal the median of all entries.

中位数的中位数取决于各行的阈值，不能直接当作全体数据的中位数。

Practice / 配套练习: AIME-2017-I-Q11 (WB p. 796)

Section 5.6 Data & descriptive statistics

数据与描述统计

Method 4 / 方法 4 [PS06-M04]

Use unique mode and an absent integer median to bound multiplicities

用唯一众数及缺失的整数中位数约束重数

Procedure

1. Sort the list. An integer median absent from the list requires an even length and two distinct middle entries.

2. Use uniqueness of the mode to bound every other multiplicity and the number of entries.

3. Enumerate the permitted positions of the repeated mode, enforcing middle-entry parity and the original size or sum bound.

4. Check all shorter and longer feasible cases before declaring uniqueness.

通用步骤

1. 先排序。若整数中位数不在列表中，长度必须为偶数，且中间两项不同。

2. 由众数唯一，限制其他值的重数以及项数。

3. 枚举重复众数的位置，同时使用中间两项的奇偶条件及原数量或总和界。

4. 宣称唯一之前，必须排除所有较短或较长的可行情形。

> Example. Classify all sorted four-entry integer lists using values 1 through 6, with unique mode 4 and an integer median absent from the list.

> 例题：分类所有使用 1 至 6 的整数、含四项的有序非降列表，要求唯一众数为 4，且中位数是未出现在列表中的整数。

Section 5.6 Data & descriptive statistics

数据与描述统计

Method 4 / 方法 4 [PS06-M04] (continued / 续)

Worked solution

1. Three or four copies of 4 would make the median 4. Thus there are exactly two copies, and the other two entries $a < b$ are distinct and different from 4.

2. If they straddle 4, the middle entries are both 4. If both exceed 4, they must be $5, 6$, with noninteger median $9/2$.

3. If both lie below 4, the median is $(b + 4)/2$, so b must be even. The only choice is $b = 2, a = 1$. Therefore the unique list is $1, 2, 4, 4$, with absent median 3.

例题解析

1. 若有三个或四个 4，中位数就是 4，不合要求。因此恰有两个 4，另两项 $a < b$ 互异且均不为 4。

2. 若两项分处 4 两侧，中间两项均为 4；若都大于 4，只能为 $5, 6$，中位数为非整数 $9/2$。

3. 若都小于 4，中位数为 $(b + 4)/2$，故 b 为偶数，只能取 $b = 2, a = 1$。唯一列表为 $1, 2, 4, 4$，其中位数 3 未出现。

Unique mode means a strictly greater frequency, so tied repetitions must be excluded.

唯一众数的频数须严格大于其他频数，不能保留并列重复值。

Practice / 配套练习: AIME-2024-II-Q02 (WB p. 795)

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 1 / 方法 1 [LO01-M01]

An invariant proves a necessary restriction; a construction or a legal strategy proves sufficiency. When a state changes by folding, cancellation, or a custom operation, record the information that survives that change.

不变量证明必要限制，构造或合法策略证明充分性。面对折叠、约分或自定义运算，应记录在变化中仍有效的信息。

Prove losing residues by induction on legal moves

按合法移动归纳证明必败余数

Procedure

1. Define a losing position as one whose every legal move is winning; a winning position needs one legal move to a losing state.

2. Guess a residue pattern only after computing enough small positions.

3. Prove both directions of the residue rule, checking that the proposed move is available at small pile sizes.

4. Count or apply the resulting strategy, with zero treated as a losing terminal state under normal play.

通用步骤

1. 必败状态的所有合法后继都必胜；必胜状态只需存在一个合法必败后继。

2. 先算足够多的小状态，再猜测余数模式。

3. 证明余数规则的两个方向，并检查小堆中所提移动是否合法。

4. 由规则计数或执行策略；正常取尽获胜规则下，零为必败终态。

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 1 / 方法 1 [LO01-M01] (continued / 续)

> Example. Players alternately remove $1, 3$, or 4 tokens, with the last removal winning. Classify the losing pile sizes.

> 例题：两人轮流取走一、三或四枚棋子，取尽者胜，分类所有必败堆大小。

Worked solution

1. Direct computation for 0 through 6 gives losing sizes $0, 2$ and all other sizes winning.

2. From residues $0, 2 \pmod 7$, subtracting a legal $1, 3, 4$ always leads to residues $1, 3, 4, 5, 6$, which are outside the proposed losing class.

3. From residues $1, 3, 4, 5, 6$, respectively subtract $1, 1, 4, 3, 4$ to reach residue 0 or 2. These moves are legal at the least positive representatives. Induction proves exactly $n \equiv 0, 2 \pmod 7$ are losing.

例题解析

1. 直接计算零至六，必败大小为零、二，其余必胜。

2. 从模七余零或二的状态，合法减去一、三、四后，都到余一、三、四、五、六，不在拟定必败类中。

3. 从余一、三、四、五、六分别减一、一、四、三、四，会到余零或二，且在最小正代表处也合法。归纳得必败状态恰为 $n \equiv 0, 2 \pmod 7$。

A repeating numerical table suggests a strategy; only the two legal-move implications prove it.

数值表重复只提示策略，必须证明合法移动的两个方向才算完成。

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 1 / 方法 1 [LO01-M01] (continued / 续)

Practice / 配套练习: AIME-2024-I-Q03 (WB p. 797)

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 2 / 方法 2 [LO01-M02]

Propagate forced colors, then construct below the threshold

传播被迫颜色并在门槛以下构造

Procedure

1. Assume a forbidden monochromatic relation is absent and fix one initial color by symmetry.

2. Propagate forced opposite colors through the relation.

3. Find an element whose two possible colors both cause a contradiction.

4. For a sharp threshold, provide an explicit coloring below it and check every same-color relation.

通用步骤

1. 假设不存在禁止的同色关系，并利用颜色对称固定起始颜色。

2. 沿关系传播被迫的相反颜色。

3. 找出一个无论染哪色都矛盾的元素。

4. 若要求精确门槛，必须给出门槛以下的明确染色并核对全部同色关系。

> Example. Find the least m such that every two-coloring of $\{1,\dots,m\}$ has same-colored a,b,c with $a+b=c$, allowing $a=b$.

> 例题：求最小 m，使 $\{1,\dots,m\}$ 的任意两色染色都存在同色 a,b,c 满足 $a+b=c$，允许 $a=b$。

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 2 / 方法 2 [LO01-M02] (continued / 续)

Worked solution

1. At $m = 5$, color one red. Avoiding $1 + 1 = 2$ forces two blue; avoiding $2 + 2 = 4$ forces four red; avoiding $1 + 4 = 5$ forces five blue.

2. If three is red, $1 + 3 = 4$ is monochromatic; if blue, $2 + 3 = 5$ is monochromatic. Hence five always forces the relation.

3. For $m = 4$, color $\{1, 4\}$ red and $\{2, 3\}$ blue. A direct check of sums within each color finds none in that same color. Restricting this coloring covers all smaller m, so the least is five.

例题解析

1. 当 $m = 5$，不妨一为红。由 $1 + 1 = 2$，二须蓝；由 $2 + 2 = 4$，四须红；由 $1 + 4 = 5$，五须蓝。

2. 若三红，$1 + 3 = 4$ 同色；若三蓝，$2 + 3 = 5$ 同色，因此五必然产生目标关系。

3. 当 $m = 4$，把一、四染红，二、三染蓝，逐一检查同色和均不落在本色中；更小范围可限制该染色。因此最小值为五。

A contradiction proves an upper bound; an explicit avoiding coloring proves the matching lower bound.

矛盾只证明上界，明确的避免染色才能证明相应下界。

Practice / 配套练习: AIME-2010-I-Q12 (WB p. 798)

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 3 / 方法 3 [LO01-M03]

Pair a modular invariant with a forward reachability certificate

将同余不变量与正向可达构造配对

Procedure

1. Find linear combinations of coordinates preserved modulo small integers by every allowed move.

2. Treat these as necessary conditions only.

3. Build nonnegative combinations of legal moves that generate every point satisfying the conditions; integer combinations with negative move counts are insufficient.

4. When counting, transform the region and retain any parity compatibility between new coordinates.

通用步骤

1. 寻找每种合法移动都保持的坐标线性组合及小模数。

2. 先把它们视为必要条件。

3. 以合法移动的非负次数组合构造全部满足条件的点；允许负次数的整数线性组合还不够。

4. 计数时转换区域，并保留新坐标间的奇偶兼容条件。

> Example. From the origin, allowed moves are $(2,0),(0,2),(-1,-1)$. Characterize all reachable integer points.

> 例题：从原点出发，可走 $(2,0),(0,2),(-1,-1)$，刻画全部可达整数点。

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 3 / 方法 3 [LO01-M03] (continued / 续)

Worked solution

1. Each move preserves evenness of $x-y$, so reachable coordinates must have the same parity.

2. For any such target (x,y), choose an integer $c \geq \max(0,-x,-y)$ with the common parity of x,y. Then $a=(x+c)/2, b=(y+c)/2$ are nonnegative integers.

3. The legal sequence using the moves a,b,c times reaches $(2a-c,2b-c)=(x,y)$. Thus equal coordinate parity is both necessary and sufficient.

例题解析

1. 每步均保持 $x-y$ 为偶数，所以可达点两坐标须同奇偶。

2. 对任一这样的目标，选与两坐标同奇偶且 $c \geq \max(0,-x,-y)$ 的整数，令 $a=(x+c)/2, b=(y+c)/2$，均为非负整数。

3. 三种移动分别走 a,b,c 次，就到达 $(2a-c,2b-c)=(x,y)$，故同奇偶既必要又充分。

A lattice generated algebraically by moves can be larger than the forward-reachable semigroup; prove the nonnegative construction.

移动所生成的整数格可能大于正向可达半群，必须证明非负次数构造。

Practice / 配套练习: AIME-2012-I-Q11 (WB p. 799)

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 4 / 方法 4 [LO01-M04]

Search for a conserved algebraic expression before iterating

迭代前先寻找守恒代数式

Procedure

1. Write the transformation on a complete tuple and compare the same expression before and after it.

2. Clear denominators or factor the difference to prove exact invariance.

3. Use the invariant to reduce a nonlinear recurrence to a linear relation or a fixed algebraic curve.

4. Restore positivity and monotonic bounds before extracting floors or extrema.

通用步骤

1. 把变换写在完整元组上，比较变换前后的同一个表达式。

2. 清分母或将差因式分解，证明精确不变。

3. 利用不变量把非线性递推化为线性关系或固定代数曲线。

4. 求整数部分或最值前，恢复正性及单调界。

> Example. Show that replacing c by $3ab-c$ preserves $a^2+b^2+c^2-3abc$. Apply it to $(1,1,1)$.

> 例题：证明将 c 换成 $3ab-c$ 保持 $a^2+b^2+c^2-3abc$ 不变，并应用于 $(1,1,1)$。

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 4 / 方法 4 [LO01-M04] (continued / 续)

Worked solution

1. Write $c' = 3ab - c$. Then $(c')^2 - 3abc' = 9a^2b^2 - 6abc + c^2 - 9a^2b^2 + 3abc = c^2 - 3abc$.

2. The other terms $a^2 + b^2$ are unchanged, proving the invariant.

3. The initial value is zero. The replacement gives $(1, 1, 2)$, and $1 + 1 + 4 - 6 = 0$ verifies the same algebraic curve.

例题解析

1. 令 $c' = 3ab - c$，展开得 $(c')^2 - 3abc' = c^2 - 3abc$。

2. 其余两项 $a^2 + b^2$ 不变，所以表达式守恒。

3. 起始值为零，变换得到 $(1, 1, 2)$，且 $1 + 1 + 4 - 6 = 0$，仍位于同一代数曲面。

An invariant is useful only after its domain and any needed inequality bounds are checked.

只有核对定义域及所需不等式界后，不变量才足以支持结论。

If $ac = b^2 + C$ and $bd = c^2 + C$, then $c(a + c) = b(b + d)$, hence $(a + c)/b = (b + d)/c$ for nonzero b, c. Thus the adjacent ratio $K = (a_{n-1} + a_{n+1})/a_n$ is constant and the recurrence becomes $a_{n+1} = Ka_n - a_{n-1}$. The original identity also gives $a_{n+1}^2 + a_n^2 = Ka_na_{n+1} - C$.

若 $ac = b^2 + C, bd = c^2 + C$，则 $c(a + c) = b(b + d)$，非零 b, c 下有 $(a + c)/b = (b + d)/c$。因此相邻比值 $K = (a_{n-1} + a_{n+1})/a_n$ 为常数，递推化为 $a_{n+1} = Ka_n - a_{n-1}$；原式还给出 $a_{n+1}^2 + a_n^2 = Ka_na_{n+1} - C$。

Practice / 配套练习: AIME-2007-I-Q14 (WB p. 158)

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 5 / 方法 5 [LO01-M05]

Track an invariant through reduction of rational pairs

在分数约分中跟踪不变量

Procedure

1. Use the reduced numerator and denominator as the state, because the next rule depends on them.

2. Find a linear determinant invariant before reduction; after dividing by a gcd, that invariant divides by the same factor.

3. For the shift $(m, n) \mapsto (m + r, n + r + 1)$, put $d = n - m, I = (r + 1)m - rn$. The cancellation factor is $\gcd(d + 1, I)$.

4. Use the new difference and invariant together to prove whether a target shape can recur.

通用步骤

1. 将既约分子分母作为状态，因为下一步规则依赖它们。

2. 先找约分前的线性行列式不变量；除以最大公因数后，它也除以同一因子。

3. 若平移为 $(m, n) \mapsto (m + r, n + r + 1)$，令 $d = n - m, I = (r + 1)m - rn$，约分因子为 $\gcd(d + 1, I)$。

4. 同时跟踪新差值与不变量，以证明某个目标形式能否再次出现。

> Example. Start with $5/7$. Repeatedly add two to its reduced numerator and three to its reduced denominator, then reduce. Can a term equal $t/(t + 1)$ for a positive integer t?

> 例题：从 $5/7$ 出发，反复给既约分子加二、分母加三，再约分，能否出现正整数 t 的 $t/(t + 1)$?

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 5 / 方法 5 [LO01-M05] (continued / 续)

Worked solution

1. Here $I = 3m - 2n$ starts at $15 - 14 = 1$ and is unchanged before reduction.

2. The cancellation factor is $\gcd(d+1, 1) = 1$, so no reduction ever occurs and d increases by one each time.

3. Initially $d = 2$, hence always $d \geq 2$. A reduced fraction $t/(t+1)$ has difference one, so it can never occur.

例题解析

1. 此处 $I = 3m - 2n$ 初值为一，未约分前保持不变。

2. 约分因子为 $\gcd(d+1, 1) = 1$，所以永不约分，差 d 每步增加一。

3. 初始差为二，始终至少二，而 $t/(t+1)$ 已既约且差为一，因此不可能出现。

An unreduced update and a reduced update are different dynamical systems.

未约分与约分后的递推是不同的动力系统。

Practice / 配套练习: AIME-2021-I-Q10 (WB p. 491)

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 6 / 方法 6 [LO01-M06]

Track both position and layer orientation in repeated folds

反复折叠时同时跟踪位置与层次反向

Procedure

1. Record current strip length L, stack thickness H, horizontal unit position p, and bottom-up layer b.

2. A square in the retained half keeps its state. A right-half square folded onto the left changes to $p' = L + 1 - p, b' = 2H + 1 - b$.

3. After the fold, halve L and double H, then repeat.

4. For an entire stack, append the reversed moving-layer list above the retained-layer list.

通用步骤

1. 记录当前条长 L、厚度 H、水平位置 p 及自下而上层号 b。

2. 保留半区的方格状态不变；右半向左盖上时，$p' = L + 1 - p, b' = 2H + 1 - b$。

3. 每次折后 L 减半、H 加倍，再重复。

4. 若跟踪整叠顺序，应把移动侧层表反向后接在保留侧层表之上。

> Example. An eight-square strip is labeled 1 through 8. Fold the right half over the left three times. Give the final stack from bottom to top.

> 例题：八格纸条从左到右标一至八，连续三次将右半盖到左半，求最终自下而上层序。

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 6 / 方法 6 [LO01-M06] (continued / 续)

Worked solution

1. After the first fold, the four vertical stacks from left to right are $[1,8]$, $[2,7]$, $[3,6]$, $[4,5]$.

2. After the next fold, they are $[1,8,5,4]$ and $[2,7,6,3]$, reversing each moved stack.

3. The final fold gives $[1,8,5,4,3,6,7,2]$. Reversing horizontal order alone would miss the reversed layer orientation.

例题解析

1. 第一折后，从左到右四叠为 $[1,8]$, $[2,7]$, $[3,6]$, $[4,5]$。

2. 第二折时反向移动层表，得到 $[1,8,5,4]$ 及 $[2,7,6,3]$。

3. 最后得到 $[1,8,5,4,3,6,7,2]$。只反转水平位置而不反转层次会出错。

Use bottom-up or top-down layers consistently; changing conventions mid-process reverses the answer.

必须始终统一用自下而上或自上而下层号，途中换约定会反转答案。

Practice / 配套练习: AIME-2004-II-Q15 (WB p. 390)

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 7 / 方法 7 [LO01-M07]

Use ordinary prefix parity to control a custom operation

用普通前缀奇偶控制自定义运算

Procedure

1. Compare the operation modulo two; addition and subtraction have the same parity effect.

2. Prove that the running-value parity equals the ordinary prefix-sum parity.

3. Mark exactly which terms are subtracted; the final value equals the ordinary sum minus twice their total.

4. For counting, keep both total size and running value as states and extend by every allowed positive part.

通用步骤

1. 先模二比较运算，加与减对奇偶的影响相同。

2. 证明当前运算值与普通前缀和同奇偶。

3. 标出恰被减去的项，最终值等于普通总和减去这些项总和的两倍。

4. 计数时同时保存总量及当前值，并用每个允许正部件作扩展。

> Example. Define $a \circ b = a - b$ if a is odd and b even, and $a + b$ otherwise. Evaluate the left-associated composition $2 \circ 1 \circ 4 \circ 3$ using prefix parity.

> 例题：定义 $a \circ b$：若 a 奇且 b 偶则取 $a - b$，否则取 $a + b$。用前缀奇偶求左结合的 $2 \circ 1 \circ 4 \circ 3$。

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

Method 7 / 方法 7 [LO01-M07] (continued / 续)

Worked solution

1. The ordinary prefixes before the later terms are $2, 3, 7$, with parities even, odd, odd.

2. Only the even term four following the odd prefix three is subtracted. The other terms are added.

3. Thus the result is $(2 + 1 + 4 + 3) - 2 \cdot 4 = 2$. Direct left evaluation $2 \circ 1 = 3, 3 \circ 4 = -1, -1 \circ 3 = 2$ confirms it.

例题解析

1. 后三项之前的普通前缀分别为二、三、七，奇偶为偶、奇、奇。

2. 只有跟在奇前缀三后的偶数四被减去，其余均相加。

3. 所以结果为 $10 - 2 \cdot 4 = 2$；直接左结合计算依次为三、负一、二，吻合。

The operation is left-associated and generally not associative; parentheses are part of the state rule.

运算按左结合进行，一般不满足结合律，括号顺序属于状态规则的一部分。

Practice / 配套练习: AIME-2026-II-Q14 (WB p. 689)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 1 / 方法 1 [LO02-M01]

A discrete optimum is a bound plus a witness. Use integrality, exchange moves, residue capacities and resource counts to narrow the search, then construct an arrangement that meets every local condition.

离散最优值由界与见证共同组成。用整数性、交换操作、余数容量及资源总量缩小搜索，再构造满足全部局部条件的配置。

Bound digit positions, then optimize lexicographically

先界定数位再按字典序优化

Procedure

1. Write every arithmetic digit condition as a weighted place-value equation.

2. Bound the total contribution of lower positions to rule out impossible high positions.

3. Among the remaining strings, optimize the leading digit first while preserving feasible remaining digit mass.

4. Use exchange of digit mass toward cheaper positions to prove a minimum, or the reverse for a maximum.

通用步骤

1. 把数字算术条件写成位权方程。

2. 用低位总贡献的界排除过高数位。

3. 在剩余字符串中先优化首位，同时保留余下数字总量的可行性。

4. 通过把数字总量移向低成本位置的交换证明最小，求最大则反向。

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 1 / 方法 1 [LO2-M01] (continued / 续)

> Example. Find the smallest four-digit decimal number whose digits are at most four and sum to ten.

> 例题：求各位数字均不超过四、数字和为十的最小四位十进制数。

Worked solution

1. The leading digit is at least one; choosing one is feasible because the other positions can hold up to twelve.

2. After choosing one, the last two positions hold at most eight, so the hundreds digit must be at least one. Choose it as one.

3. The remaining mass eight forces the final digits four and four. The number is 1144; each earlier digit was proved minimal while completion remained possible.

例题解析

1. 首位至少一，取一可行，因为余三位容量十二。

2. 首位取一后，末两位容量仅八，因此百位至少一，取一。

3. 剩余数字和八迫使末两位均四，得到 1144；每个高位均在可完成条件下取了最小值。

A leading-digit choice is valid only if the remaining positions can complete the weighted condition.

只有余下数位能完成位权条件，首位选择才有效。

Practice / 配套练习: AIME-2001-I-Q08 (WB p. 585), AIME-2020-II-Q05 (WB p. 578)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 2 / 方法 2 [LO02-M02]

Turn a forbidden pattern into forced sorted growth

将禁止模式转为被迫的有序增长

Procedure

1. Sort the positive entries and test the closest preceding entries to a proposed largest member.

2. The absence of a local forbidden pattern gives a recurrence lower bound.

3. Iterate the bound from the smallest permitted initial entries.

4. Check that the equality sequence avoids every nonlocal forbidden pattern too.

通用步骤

1. 将正数排序，对某个最大成员先测试最接近的前两项。

2. 局部禁止模式不存在会给出递推下界。

3. 从最小允许初项迭代该界。

4. 检查取等序列也能避免所有非局部禁止模式。

> Example. Prove that a sorted positive list contains a nondegenerate triangle if and only if some three consecutive entries form one.

> 例题：证明递增正数列表含有非退化三角形，当且仅当某三个连续成员可作三角形。

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 2 / 方法 2 [LO02-M02] (continued / 续)

Worked solution

1. One direction is immediate: a consecutive triple forming a triangle is also a triple in the whole list.

2. Otherwise every consecutive triple obeys $a_{j-2} + a_{j-1} \leq a_j$.

3. For arbitrary $i < k < j$, $a_i + a_k \leq a_{j-2} + a_{j-1} \leq a_j$, so no arbitrary triple forms a triangle. This proves the converse and the exhaustive local test.

例题解析

1. 连续三项若能作三角形，自然给出所需三项。

2. 否则每个连续三项均有前两项之和不超过第三项。

3. 任意更早两项之和也不超过最大项前的两项之和，故同样不能作三角形，完成反向证明。

A locally defined extremal sequence needs a proof that it also avoids nonconsecutive patterns.

局部定义的极值序列还须证明能避免非连续模式。

Practice / 配套练习: AIME-2001-II-Q05 (WB p. 800)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 3 / 方法 3 [LO02-M03]

Combine a common residue class with a distinctness capacity

结合公共余数类与互异容量

Procedure

1. Translate every deletion or divisibility condition into a common congruence for all entries.

2. Use the difference of the largest and smallest entries to restrict the modulus.

3. Distinct entries in one residue class require minimum successive gaps equal to that modulus.

4. Construct an initial arithmetic block plus the prescribed endpoint and verify every deletion condition.

通用步骤

1. 把每次删除或整除条件化为所有成员的公共同余。

2. 由最大最小差限制模数。

3. 同余类内互异成员的连续间距至少为该模数。

4. 用起始等差块加规定端点构造，并验证每次删除。

> Example. For positive integers $m < M$ and $n \geq 2$, characterize when an n-element set with minimum m, maximum M has integer mean after any one deletion.

> 例题：给定正整数 $m < M, n \geq 2$，刻画何时存在最小值 m、最大值 M 的 n 元集，删除任一成员后的平均数均为整数。

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 3 / 方法 3 [LO02-M03] (continued / 续)

Worked solution

1. Put $d = n - 1$ and let the sum be S. Every deletion gives $S - x \equiv 0 \pmod{d}$, so all entries are congruent and $d \mid M - m$.

2. There are d positive gaps of at least d, so $M - m \geq d^2$.

3. Conversely, if both conditions hold, take $m, m + d, \ldots, m + (d - 1)d, M$. They are distinct, and all entries are congruent to m; the total is also $m \pmod{d}$. Thus every deletion sum is divisible by d.

例题解析

1. 令 $d = n - 1$，总和为 S。每次删除给出 $S - x \equiv 0 \pmod{d}$，故所有成员同余，且 $d \mid M - m$。

2. 共有 d 个正间距，每个至少 d，所以 $M - m \geq d^2$。

3. 反之，若两条件都满足，取起始 d 项等差块及端点 M，全部互异且同余；总和也与 m 同余，故任删一项后被 d 整除。

Prove both the divisor restriction and the spacing capacity; either alone is too weak.

约数限制与间距容量两者都要证明，单用一者不够。

Practice / 配套练习: AIME-2002-I-Q14 (WB p. 508)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 4 / 方法 4 [LO02-M04]

Round per-group bounds before summing integer totals

先对各组整数界取整再求和

Procedure

1. Convert percentage or ratio constraints to exact inequalities with positive denominators.

2. For strict upper bounds on integers use $x < r \Rightarrow x \le \lceil r \rceil - 1$; for lower bounds use ceilings.

3. Add the local integer bounds and compare with the global total.

4. Distribute any remaining excess explicitly and verify every group separately.

通用步骤

1. 将百分比或比例限制化为正分母的精确不等式。

2. 整数严格上界用 $x < r \Rightarrow x \le \lceil r \rceil - 1$，下界用向上取整。

3. 相加各局部整数界，再与全局总量比较。

4. 明确分配剩余量，并逐组验证。

> Example. Three bags have capacities $5, 11, 16$. Each contains a positive integer number of marked items, strictly less than five-eighths of its capacity. Maximize the total marked count.

> 例题：三个袋子容量为五、十一、十六，标记物数量均为正整数且严格小于各容量的八分之五，求标记总数最大值。

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 4 / 方法 4 [LO02-M04] (continued / 续)

Worked solution

1. The three integer upper bounds are $\lceil 25/8 \rceil - 1 = 3$, $\lceil 55/8 \rceil - 1 = 6$, and $10 - 1 = 9$.

2. Thus at most eighteen items are marked, stronger than rounding only the combined capacity bound.

3. Counts $(3, 6, 9)$ satisfy all strict per-bag inequalities and attain eighteen.

例题解析

1. 三组整数上界分别为三、六、九。

2. 所以总数至多十八，比只对合并容量取整更强。

3. 取标记数三、六、九，逐组均严格符合比例，故十八可达。

A strict ratio bound can lose a full integer when its unrounded bound is itself integral.

当未取整上界本身为整数时，严格比例还须再减一。

Practice / 配套练习: AIME-2003-II-Q12 (WB p. 806), AIME-2004-I-Q05 (WB p. 801)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 5 / 方法 5 [LO02-M05]

Use unit exchanges to prove no gaps in attainable sums

用单位交换证明可达和无缺口

Procedure

1. Compute the smallest and largest possible sums.

2. At every nonmaximal configuration find a legal exchange increasing the sum by exactly one.

3. The sum increases within a finite set, so the exchanges eventually reach the maximum.

4. Every intermediate integer has then been attained; count the full interval.

通用步骤

1. 先算最小及最大可能总和。

2. 在每个非最大配置中，找出合法且恰增加一的交换。

3. 总和在有限集合中严格增加，故最终到达最大值。

4. 于是每个中间整数都达到，可直接数完整区间。

> Example. Prove the sums of k-element subsets of $\{1, \ldots, N\}$ fill an integer interval, and find its size.

> 例题：证明一至 N 的 k 元子集和填满一个整数区间，并求区间大小。

Worked solution

1. The endpoint sums are $k(k+1)/2$ and $k(2N-k+1)/2$.

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 5 / 方法 5 [LO02-M05] (continued / 续)

2. Unless the set is the terminal block, some chosen $j < N$ has unchosen successor $j + 1$. Replacing it legally raises the sum by one.

3. Following such exchanges from the initial block reaches the terminal block and visits every integer between. The number of sums is $k(N - k) + 1$.

例题解析

1. 两端和分别为最小前块及最大末块之和。

2. 若还不是末块，必有被选 j 的后继 $j + 1$ 未选，替换后合法且和增加一。

3. 从前块不断交换到末块，中间整数无遗漏，故总数为 $k(N - k) + 1$。

Knowing the endpoint sums is insufficient without the no-gap exchange proof.

只知道两端和还不够，必须证明中间无缺口。

Practice / 配套练习: AIME-2006-I-Q02 (WB p. 802)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 6 / 方法 6 [LO02-M06]

Telescope to a quantized terminal state and construct it

望远镜化到离散终态并构造

Procedure

1. Square an absolute recurrence to remove the arbitrary sign.

2. Sum the resulting differences so the objective depends only on a terminal square.

3. Prove which terminal parity and size classes are attainable.

4. Choose the best allowed terminal value and give a complete legal sign sequence reaching it.

通用步骤

1. 将绝对值递推平方，消去任意符号。

2. 累加差式，使目标只依赖终态平方。

3. 证明终态允许的奇偶及大小类。

4. 选择最优允许终值，并给出完整合法符号序列。

> Example. Let $u_0 = 0, |u_j| = |u_{j-1} + 1|$. Find all possible values of $u_0 + u_1 + u_2 + u_3$.

> 例题：设 $u_0 = 0, |u_j| = |u_{j-1} + 1|$，求前四项和的全部可能值。

Worked solution

1. Squaring and summing through step four gives $u_4^2 = 4 + 2\sum_{j=0}^{3} u_j$.

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 6 / 方法 6 [LO02-M06] (continued / 续)

2. Induction on the recurrence gives terminal values $u_4 \in \{-4, -2, 0, 2, 4\}$, all attained: each stage gives every integer of its parity between its two extreme values.

3. The terminal squares are $16, 4, 0$, so the sums are exactly $6, 0, -2$. The reachability induction supplies every listed value.

例题解析

1. 平方后累加到第四步，得 $u_4^2 = 4 + 2\sum_{j=0}^{3} u_j$。

2. 递推归纳得第四步可达负四、负二、零、二、四：每步均填满其两端之间的同奇偶整数。

3. 终态平方为十六、四、零，故和恰为六、零、负二，每个值均由可达性归纳保证。

A nearest-square bound becomes an optimum only when the terminal state can actually occur.

最近平方给出的界，只有终态确实可达时才成为最优值。

Practice / 配套练习: AIME-2006-I-Q15 (WB p. 156)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 7 / 方法 7 [LO02-M07]

Check the nearest admissible integers of a concave product

比较凹乘积顶点附近的允许整数

Procedure

1. Eliminate one count using the linear resource equation.

2. Complete the square in the remaining concave quadratic.

3. Solve its integrality congruence and test the nearest admissible points on both sides of the vertex.

4. Convert the maximizing counts back to a valid geometric or combinatorial arrangement.

通用步骤

1. 用线性资源方程消去一个计数。

2. 对剩下的凹二次式配方。

3. 解整数性同余，并检查顶点两侧最近的允许点。

4. 将最优计数恢复为合法几何或组合配置。

> Example. Positive integers $x+y=12$ satisfy $x \equiv 1 \pmod 3$. Maximize xy.

> 例题：正整数 $x+y=12$ 且 $x \equiv 1 \pmod 3$，求 xy 最大值。

Worked solution

1. The product is $36-(x-6)^2$.

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 7 / 方法 7 [LO02-M07] (continued / 续)

2. Allowed x are $1, 4, 7, 10$. The closest to six is seven, at distance one; four is distance two.

3. Thus $(x, y) = (7, 5)$ attains the maximum thirty-five.

例题解析

1. 乘积可配方为 $36 - (x - 6)^2$。

2. 允许的一、四、七、十中，七距顶点最近。

3. 取七、五得最大乘积三十五。

Rounding the real vertex without enforcing the residue class can give an impossible count.

若不执行余数限制就将实顶点取整，可能得到不可能的计数。

Practice / 配套练习: AIME-2007-II-Q08 (WB p. 803)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 8 / 方法 8 [LO02-M08]

Distinguish a finite-window count from a steady rate

区分有限时间窗计数与稳定速率

Procedure

1. Translate the spacing rule to a lower bound on time between successive events.

2. m events in a closed window require only $m-1$ inter-event gaps.

3. Use strict gap inequalities to rule out the next integer count.

4. Choose an initial phase and explicit legal gaps that attain the claimed count.

通用步骤

1. 将间距规则转为连续事件之间的时间下界。

2. 闭时间窗内的 m 个事件只需 $m-1$ 个间隔。

3. 用严格间隔不等式排除下一个整数计数。

4. 选择初始相位及明确合法间隔，达到所称数量。

> Example. Events are separated by strictly more than two minutes. What is the largest number in the closed ten-minute interval $[0,10]$?

> 例题：相邻事件间隔严格大于两分钟，在闭区间零至十分钟内最多可有几个?

Worked solution

1. Six events would require five gaps whose total is strictly greater than ten, impossible.

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 8 / 方法 8 [LO02-M08] (continued / 续)

2. Five events at times $0, 9/4, 9/2, 27/4, 9$ have gaps $9/4 > 2$ and fit in the window.

3. Thus five is the maximum. An initial event is counted even though it does not require an earlier gap inside the window.

例题解析

1. 六个事件需五个间隔，总长严格大于十，矛盾。

2. 取 $0, 9/4, 9/2, 27/4, 9$ 分钟这五个时刻，间隔均为 $9/4 > 2$。

3. 因此最多五个，初始事件不需要窗内更早的一个间隔。

State whether endpoints and a freely chosen initial phase are included before converting flow to a count.

把流率变成数量前，先说明端点及自由初相是否计入。

Practice / 配套练习: AIME-2008-I-Q12 (WB p. 173)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 9 / 方法 9 [LO02-M09]

Bound a growing path by its distinct distance levels

用不同距离层数限制递增长路径

Procedure

1. List every possible positive pair distance or squared distance.

2. A strictly increasing path uses each level at most once, giving a point-count bound of one more than the number of levels.

3. If the bound is reached, every level must be used in order; backtrack from the largest level to force predecessors.

4. Check distinct vertices and construct or count all surviving paths.

通用步骤

1. 列出所有可能正距离或距离平方。

2. 严格递增路径每层至多一次，故顶点数至多距离层数加一。

3. 若取等，必须按顺序使用全部距离层，可从最大层倒推前驱。

4. 检查顶点互异，构造或计数全部剩余路径。

> Example. Among the four vertices of a 1×2 rectangle, what is the longest path visiting distinct vertices with strictly increasing successive distances?

> 例题：在一乘二矩形的四个顶点中，以不同顶点构成路径且相邻距离严格增加，最多经过几个点?

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 9 / 方法 9 [LO02-M09] (continued / 续)

Worked solution

1. The distance levels are $1, 2, \sqrt{5}$, suggesting an upper bound of four points.

2. A four-point path would need distances $1, 2, \sqrt{5}$ in order. The first two are perpendicular rectangle sides, after which the unused fourth vertex is distance one from the current vertex, not $\sqrt{5}$. Thus four is impossible.

3. Three points are possible by a length-one side followed by a diagonal. Hence the true maximum is three; the level bound alone was not attained.

例题解析

1. 三种距离是一、二、根号五，形式上顶点数至多四。

2. 若走四点，距离须依次一、二、根号五；前两段为相邻垂直边，此时唯一未用顶点距当前位置却是一，不能完成。

3. 先走短边再走对角线可走三点，所以准确最大为三，距离层数界并不自动可达。

A level-count upper bound is only a candidate until a path realizes its entire ordering.

只有有路径实现完整距离顺序，层数界才可达到。

Practice / 配套练习: AIME-2008-II-Q10 (WB p. 697)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 10 / 方法 10 [LO02-M10]

Count one resource in two ways and build a compatible block

从两种角度计同一资源并构造相容块

Procedure

1. Write the same total as a sum over each of two families of groups.

2. Use distinctness or fixed per-group counts to give lower and upper bounds.

3. Solve the resulting divisibility or size obstruction.

4. Build repeated local blocks or disjoint parity classes to satisfy every local rule, not only the total equation.

通用步骤

1. 按两类分组分别表示同一个总量。

2. 由互异性或固定组内数量给出上下界。

3. 解所得整除或大小障碍。

4. 用重复局部块或不相交奇偶类构造，使全部局部规则成立，而不只满足总量方程。

> Example. A nonempty zero-one matrix has two ones in every row and three ones in every column. Find the least possible number of entries.

> 例题：非空零一矩阵每行两个一、每列三个一，求总格数最小值。

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 10 / 方法 10 [LO02-M10] (continued / 续)

Worked solution

1. With r rows and c columns, counting ones gives $2r = 3c$, hence $r = 3t, c = 2t$ for positive integer t.

2. Therefore the entry count is $6t^2 \geq 6$.

3. The 3×2 all-one matrix meets both local rules and attains six entries, proving the minimum.

例题解析

1. 按行列计一的总数，得 $2r = 3c$，所以行列数为 $3t, 2t$。

2. 总格数为 $6t^2 \geq 6$。

3. 三乘二全一矩阵满足各行各列要求并达到六，故最小值为六。

Equality of global counts does not guarantee a compatible local arrangement; exhibit one.

全局数量相等不保证局部配置相容，必须给出见证。

For k disjoint pairs from $\{1, \ldots, N\}$, the used-label sum is at least $k(2k+1)$. If pair sums are distinct and at most N, their sum is at most $k(2N-k+1)/2$. Comparing the same total gives $5k \leq 2N-1$. A sharp construction must separately ensure disjoint labels and distinct sums, often by giving one block even sums and another odd sums.

从一至 N 选 k 个不交元素对，已用标号总和至少 $k(2k+1)$。若对和互异且不超过 N，其总和至多 $k(2N-k+1)/2$，比较同一总量得 $5k \leq 2N-1$。取等构造还须分别保证标号不重、对和互异，常可用一块给偶和、另一块给奇和。

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 10 / 方法 10 [LO02-M10] (continued / 续)

Practice / 配套练习: AIME-2009-II-Q12 (WB p. 807), AIME-2016-II-Q04 (WB p. 659)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 11 / 方法 11 [LO02-M11]

Build product-free color bands with endpoint checks

以端点核查构造无同色积色带

Procedure

1. Choose multiplicative scale thresholds so products of the low band jump to the other color.

2. Put a far high band in the first color only when multiplying it by any allowed factor leaves the whole domain.

3. Check products within the other color land outside that color.

4. Include every endpoint and permit repeated factors when the original relation allows them.

通用步骤

1. 选择乘法尺度门槛，使低色带乘积跳到另一颜色。

2. 只有高色带乘任一允许因子都超出全域时，才可把它放回第一色。

3. 检查另一色内部的积不会仍落在本色。

4. 逐一核查端点，并保留原关系允许的重复因子。

> Example. For any integer $a \geq 2$, construct a two-coloring of $\{a, \ldots, a^5 - 1\}$ with no monochromatic product $xy = z$.

> 例题：对任意整数 $a \geq 2$，构造从 a 至 $a^5 - 1$ 的两色染色，使无同色乘法关系。

Worked solution

1. Color $[a, a^2 - 1] \cup [a^4, a^5 - 1]$ red and $[a^2, a^4 - 1]$ blue.

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 11 / 方法 11 [LO02-M11] (continued / 续)

2. Two low red factors multiply into the blue band; a product involving a high red factor is at least a^5 and leaves the domain.

3. Two blue factors have product at least a^4, so any in-domain product is red. Thus every same-color product is excluded, including equal factors.

例题解析

1. 将整数带 $[a, a^2-1] \cup [a^4, a^5-1]$ 染红，将 $[a^2, a^4-1]$ 染蓝。

2. 两低红因子的积落在蓝带；含高红因子的积至少 a^5，已出域。

3. 两蓝因子的积至少 a^4，若仍在域内则为红，因此同色积关系全部排除，包括重复因子。

The high band is safe only because its product with the smallest allowed factor is already outside the domain.

高色带安全的关键，是它乘最小允许因子也已越界。

Practice / 配套练习: AIME-2010-I-Q12 (WB p. 798)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 12 / 方法 12 [LO02-M12]

Saturate a run-count bound before counting compositions

先使连续段界取满再数正整数组成

Procedure

1. Bound the number of color changes or runs by the scarcer symbol count.

2. Translate any equal-pair condition into a bound on total adjacent pairs.

3. At equality, identify all forced singleton runs and endpoint colors.

4. Count the remaining positive run lengths by compositions, restoring labels only if the objects are distinct.

通用步骤

1. 用稀少符号数量限制颜色变化数或连续段数。

2. 将相邻对数量关系转为总相邻对界。

3. 取等时找出所有被迫的单元素段及端点颜色。

4. 用正整数组成数剩余段长，只有对象不同才恢复标号。

> Example. A binary word has three ones and seven zeros. Maximize its color changes and count the maximizing words.

> 例题：一个二进制词有三个一、七个零，求最多变色次数及达到最大值的词数。

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 12 / 方法 12 [LO02-M12] (continued / 续)

Worked solution

1. Three one-runs allow at most six changes. To reach six, all ones are singletons and there are four positive zero-runs, including both endpoints.

2. Write their lengths as $z_1 + z_2 + z_3 + z_4 = 7$ with each positive.

3. There are $\binom{6}{3} = 20$ such compositions, each giving exactly six changes. Thus the maximum is six and the count twenty.

例题解析

1. 三个一段至多产生六次变色，取等须三个一均单独，且四个零段均非空，包含两端。

2. 四个正零段长之和为七。

3. 隔板法得 $\binom{6}{3} = 20$ 种，每种均恰有六次变色，因此最大次数为六，取等词数为二十。

Counting before analyzing equality can include run patterns that do not attain the bound.

尚未分析取等就计数，会混入未达到上界的段型。

Practice / 配套练习: AIME-2011-II-Q07 (WB p. 652)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 13 / 方法 13 [LO02-M13]

Optimize uncoupled choices term by term

逐项优化互不耦合的选择

Procedure

1. Express the total objective as a sum of contributions, one from each choice.

2. Prove that choosing one option imposes no restriction on the others.

3. Take the maximum contribution independently for each term.

4. Give the actual full selection attaining all local maxima simultaneously.

通用步骤

1. 将总目标拆为每次选择的一项贡献。

2. 证明一个选项不限制其他选择。

3. 每项独立取最大贡献。

4. 给出同时实现全部局部最大值的完整选择。

> Example. For each of $1-2i, -3+4i, 5-i$, choose the number or its conjugate. Maximize the imaginary part of their sum.

> 例题：对 $1-2i, -3+4i, 5-i$ 各选原数或共轭，求总和虚部最大值。

Worked solution

1. Each conjugation can change only the sign of its own imaginary part.

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 13 / 方法 13 [LO02-M13] (continued / 续)

2. The independent maxima are two, four, one, summing to seven.

3. Choose $1+2i, -3+4i, 5+i$. Their sum is $3+7i$, attaining the bound.

例题解析

1. 每次共轭只改变自身虚部符号。

2. 三个独立最大虚部分别为二、四、一，总和七。

3. 选取 $1+2i, -3+4i, 5+i$，总和为 $3+7i$，确取等。

Local maximization fails when a shared constraint couples the choices.

若存在耦合选择的共同限制，局部最优不一定能同时实现。

Practice / 配套练习: AIME-2011-II-Q08 (WB p. 72)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 14 / 方法 14 [LO02-M14]

Find a first omitted block using consecutive square gaps

用连续平方间隙寻找首个空块

Procedure

1. Use square gaps $2j+1$ to exclude all blocks too short to fit between consecutive squares.

2. For the remaining aligned blocks, track the quotient $\lfloor j^2/B \rfloor$.

3. A missing block occurs when consecutive square quotients skip its index.

4. Exhibit the two bounding consecutive squares and prove every earlier block had a square.

通用步骤

1. 用平方间距排除无法完整藏在相邻平方间的早期块。

2. 对剩下的对齐块，跟踪平方除以块长的整数商。

3. 当相邻平方的块下标跳过某个整数，就出现空块。

4. 给出夹住该块的连续平方，并证明此前各块都含平方。

> Example. For positive integer B and nonnegative integer N, give an exact test for whether the integer block $BN, \ldots, BN+B-1$ contains a square.

> 例题：设 B 为正整数、N 为非负整数，给出整数块 BN 至 $BN+B-1$ 是否含平方数的精确判断式。

Worked solution

1. The least square at least BN is q^2 with $q=\lceil \sqrt{BN} \rceil$.

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 14 / 方法 14 [LO02-M14] (continued / 续)

2. If $q^2 \le BN + B - 1$, that square belongs to the block.

3. If it is larger, every subsequent square is even larger and no earlier square reaches BN. Thus the block is empty exactly when $\lceil \sqrt{BN} \rceil^2 > BN + B - 1$.

例题解析

1. 不小于块起点的最小平方为 q^2，其中 $q = \lceil \sqrt{BN} \rceil$。

2. 若它不超过块终点，则块内有该平方。

3. 否则更大平方都越界、更小平方都不及起点，因此空块的充要条件正是该平方严格超过终点。

A large square gap alone does not guarantee that an aligned whole block fits inside it.

平方间距大，并不保证某个对齐整块恰好放在其中。

Practice / 配套练习: AIME-2013-II-Q06 (WB p. 548)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 15 / 方法 15 [LO02-M15]

Use competing integer bounds to force a small candidate class

用相竞争的整数界迫使小候选类

Procedure

1. Find one feasible candidate to establish a numerical benchmark.

2. Use a global denominator or resource bound to exclude all objectives with a larger integer numerator or quotient class.

3. Apply the remaining local bound to enumerate only potentially improving cases.

4. Check every surviving case and retain the original witness if none improves it.

通用步骤

1. 先找一个可行候选作为数值基准。

2. 用全局分母或资源界，排除整数分子或商类别较大的情况。

3. 用另一局部界，只枚举可能改进的少量情形。

4. 检查所有保留候选，若无改进则原见证最优。

> Example. Positive integers a, b satisfy $b \leq 30$ and $a/b < 1/17$. Determine their possible form.

> 例题：正整数 a, b 满足 $b \leq 30$ 且 $a/b < 1/17$，求它们的全部可能形式。

Worked solution

1. If $a \geq 2$, then $a/b \geq 2/30 = 1/15 > 1/17$, impossible.

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 15 / 方法 15 [LO02-M15] (continued / 续)

2. Thus $a = 1$, and the strict comparison gives $b > 17$.

3. The possibilities are exactly $(a, b) = (1, b)$ for $18 \le b \le 30$, each of which satisfies both conditions.

例题解析

1. 若分子至少二，分数至少十五分之一，比十七分之一还大，矛盾。

2. 所以分子为一，严格比较要求分母大于十七。

3. 因此恰为分子一、分母十八至三十，且全部满足条件。

A good candidate converts a huge search into the small class capable of beating it.

一个好候选能把巨大搜索缩成仅有可能胜过它的小类。

For a remainder of n by $k \le n/2$, the quotient is at least two, so the remainder is bounded by both $k - 1$ and $n - 2k$. Split the divisor range near their intersection, then exhibit a divisor realizing the better bound and check its actual quotient; a divisor of n has remainder zero regardless of a loose upper bound.

用 $k \le n/2$ 除 n 时商至少二，所以余数同时不超过 $k - 1$ 与 $n - 2k$。在两界交会附近分割除数区间，再给出达到较好界的除数并检查真实商；整除时余数为零，不能把宽松上界当作实际余数。

Practice / 配套练习: AIME-2013-II-Q14 (WB p. 514), AIME-2022-I-Q07 (WB p. 804)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 16 / 方法 16 [LO02-M16]

Use least representable residues to certify a largest gap

用最小可表示余数值核实最大缺口

Procedure

1. First check that the denominations have gcd one; otherwise infinitely many amounts are missing. Choose a denomination d and find the least representable amount M_r in every residue class modulo d.

2. Adding copies of d then reaches every larger amount in that residue.

3. The last possible gap in residue r is $M_r - d$; maximize these values. If that maximum is negative, no nonnegative amount is missing.

4. Prove every residue minimum is minimal and construct its representation.

通用步骤

1. 先检验所有面额的最大公因数为一，否则有无穷多个金额无法表示。选一种面额 d 作模数，在每个余数类找最小可表示金额 M_r。

2. 再加该面额，就能到达同余类中全部更大金额。

3. 余数类 r 的最后可能缺口为 $M_r - d$，再对各类取最大；若最大值仍为负，则所有非负金额都已可表示。

4. 必须证明各最小额真正最小，并给出表示。

> Example. With unlimited stamps of values four and seven, find the largest nonrepresentable positive amount.

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 16 / 方法 16 [LO02-M16] (continued / 续)

> 例题：四与七面额邮票无限，求最大不可表示正金额。

Worked solution

1. Modulo four, zero through three have least represented values $0, 21, 14, 7$: respectively use zero, three, two, one seven-stamps.

2. Adding four-stamps reaches every larger value in the same residue; fewer seven-stamps cannot reach that residue at a lower value.

3. The largest gaps are $-4, 17, 10, 3$, so the largest positive gap is seventeen.

例题解析

1. 模四的零至三余数最小表示额分别为零、二十一、十四、七。

2. 再加四即可覆盖同余类后续所有值；更少七票不能以更小金额得到该余数。

3. 各类最大缺口为负四、十七、十、三，所以最大正缺口为十七。

One large residue minimum is not enough; all other residue classes must also obey the claimed largest gap.

只有一个余数最小额很大还不够，其他所有类也须符合所称最大缺口。

Practice / 配套练习: AIME-2019-II-Q14 (WB p. 515)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 17 / 方法 17 [LO02-M17]

Prove the no-gap subset-sum coverage criterion

证明子集和无缺口覆盖准则

Procedure

1. Sort the available positive coin or stamp values and keep the covered interval $[0, S]$.

2. The next value v extends coverage without a gap exactly when $v \leq S + 1$.

3. Then the old interval and its shifted copy overlap or touch, giving all values through $S + v$.

4. Use mandatory denomination counts to prove a lower bound, then exhibit a collection satisfying the criterion and the target total.

通用步骤

1. 将正面额排序，维护已覆盖区间 $[0, S]$。

2. 下一面额 $v \leq S + 1$ 时，且仅此时不会产生新缺口。

3. 原区间与平移区间相接或重叠，覆盖扩展至 $S + v$。

4. 以必须保留的面额数量证明下界，再给出满足准则及目标总量的具体集合。

> Example. Show that stamps $1, 1, 3, 6$ make every amount from zero to eleven, and explain why replacing three by four creates a gap.

> 例题：证明邮票一、一、三、六可拼出零至十一的所有金额，并说明将三改为四为何产生缺口。

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 17 / 方法 17 [LO02-M17] (continued / 续)

Worked solution

1. The two ones cover $[0, 2]$. Adding three gives the shifted interval $[3, 5]$, contiguous with it, hence coverage $[0, 5]$.

2. Adding six gives $[6, 11]$, so the full interval through eleven is covered.

3. If three is replaced by four, after the ones the next value exceeds $2 + 1$. The amount three cannot be formed, since all remaining stamps are at least four.

例题解析

1. 两张一覆盖零至二，加入三的平移区间为三至五，与原区间相接。

2. 再加入六，平移部分覆盖六至十一，因此全覆盖。

3. 若三改四，两张一后下一票超过三，金额三无法拼出，因为其他票至少四。

The greedy interval criterion proves all intermediate sums, not merely that the total is large enough.

区间准则证明所有中间和都可达，而不仅是总面额足够。

For denominations $1 < b < c$ with positive counts x, y, z, coverage through the entire collection total is equivalent to $x \geq b - 1$ and $x + yb \geq c - 1$. For a smaller target T, process sorted stamps while the covered endpoint $S < T$: each next stamp must satisfy $v \leq S + 1$; stop once $S \geq T$. Thus a denomination threshold is necessary only if coverage has not already reached T before that denomination is used. This qualification applies to both b and c, as well as requiring total value at least T.

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 17 / 方法 17 [LO02-M17] (continued / 续)

三种面额 $1 < b < c$、张数 x, y, z 均正时，覆盖直到全体总额的每个值，充要条件为 $x \geq b - 1, x + yb \geq c - 1$。若仅要求覆盖至较小目标 T，按面额顺序处理邮票：当已覆盖端点 $S < T$ 时，下一票必须满足 $v \leq S + 1$；一旦 $S \geq T$ 即可停止。因此，只有在使用该面额之前尚未覆盖至 T，其门槛才是必要条件；这个限定同时适用于 b、c，并仍要求总额至少为 T。

Practice / 配套练习: AIME-2022-II-Q14 (WB p. 808)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 18 / 方法 18 [LO02-M18]

Use improving coin exchanges to reduce every optimum to a few cases

用改进换币把所有最优解缩成少量情形

Procedure

1. Find local equal-value exchanges that strictly reduce coin count.

2. Every optimum must avoid those patterns, bounding the lower-denomination remainder.

3. This bounds the possible number of largest coins to a short list near the greedy count.

4. Compare every normalized case; greedy is optimal only if none improves it.

通用步骤

1. 寻找同值但严格减少枚数的局部换币。

2. 最优解不能含这些模式，从而限制小面额剩余总值。

3. 由此把大面额枚数限制为贪心数附近的少数候选。

4. 比较全部归一候选，只有无改进时贪心才最优。

> Example. Using coins one, four, six, find the minimum number for amount fourteen and compare with greedy.

> 例题：用一、四、六面额拼十四，求最少枚数，并与贪心比较。

Worked solution

1. An optimum has at most three pennies and two fours, since four pennies exchange for one four and three fours for two sixes.

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 18 / 方法 18 [LO02-M18] (continued / 续)

2. The non-six remainder is at most eleven, so only two or one sixes need comparison. With two sixes, two pennies give four coins; with one six, two fours give three coins.

3. Thus $6+4+4$ uses the minimum three coins, beating greedy $6+6+1+1$.

例题解析

1. 最优解至多三枚一和两枚四，否则可同值换成更少硬币。

2. 非六剩余至多十一，所以只需比较两枚六或一枚六：前者还需两枚一，共四枚；后者还需两枚四，共三枚。

3. 因此六加四加四为最少三枚，比贪心的四枚更好。

> A normalization proof must cover every optimum, not just a rival example.

> 归一论证必须覆盖所有最优解，不能只给一个竞争例子。

Practice / 配套练习: AIME-2025-II-Q08 (WB p. 805)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 19 / 方法 19 [LO02-M19]

Check degree-weight bounds against graph sign compatibility

检查度数加权界与图符号构造是否相容

Procedure

1. Use a common center to bound each edge difference by the deviations at its two ends.

2. Rearrange deviations against degrees to obtain an upper bound.

3. Equality needs every edge to join compatible low/high labels; test the capacities of those sign classes.

4. If the bound cannot be attained, refine the cases rather than report it as the maximum.

通用步骤

1. 以公共中心将每条边差界为两端偏差之和。

2. 把偏差与度数重排，得到上界。

3. 取等要求每边连接相容高低标号，须检查两符号类的容量。

4. 若不能取等，就细分情形，而不能将上界报成最大值。

> Example. Label a three-leaf star with one through four. Find the greatest sum of edge differences and explain why a naive degree bound can be loose.

> 例题：将一至四标在三叶星图上，求边差总和最大值，并说明简单度数界为何可能偏松。

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 19 / 方法 19 [LO02-M19] (continued / 续)

Worked solution

1. At center value $5/2$, deviations are $3/2, 3/2, 1/2, 1/2$. Assigning the largest to degree three gives upper bound seven.

2. If the central vertex has label $c = 1, 2, 3, 4$, the actual three-edge sums are respectively six, four, four, six.

3. Thus the maximum is six. The degree bound seven cannot attain its required edge sign pattern; a bipartite graph alone does not guarantee the needed label capacities.

例题解析

1. 以二分之五为中心，偏差为二分之三、二分之三、二分之一、二分之一，重排度数给出上界七。

2. 但中心顶点分别标一、二、三、四时，真实边差和分别为六、四、四、六。

3. 所以最大为六，七的重排取等符号无法实现；仅有二分图结构还不足以保证标号容量。

An explicit checkerboard construction must allocate each degree class on both colors correctly.

明确棋盘构造还须把每种度数的位置在两色中正确分配。

Practice / 配套练习: AIME-2026-I-Q11 (WB p. 809)

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 20 / 方法 20 [LO02-M20]

Split composite counting factors to minimize additive cost

拆分复合计数因子以降低加法成本

Procedure

1. Prove the combinatorial count factors independently as a product of integers $d_i \geq 2$.

2. Express the resource cost of each factor, often $d_i - 1$.

3. If a factor splits as gh, compare the split and unsplit costs exactly.

4. Use prime factorization for the lower bound and construct independent components realizing every prime factor.

通用步骤

1. 先证明组合计数独立分解为若干至少二的整数因子。

2. 表示每因子的资源成本，常为因子减一。

3. 若因子为 gh，精确比较拆分及未拆成本。

4. 以素因数分解给下界，再构造分别实现各素因子的独立部件。

> Example. For $M \geq 2$, prove the minimum of $\sum_i (d_i - 1)$ over all factorizations $M = \prod_i d_i$, $d_i \geq 2$, is $\sum_p v_p(M)(p-1)$.

> 例题：对 $M \geq 2$，证明所有分解 $M = \prod_i d_i, d_i \geq 2$ 中，成本 $\sum_i (d_i - 1)$ 的最小值为 $\sum_p v_p(M)(p-1)$。

Section 6.2 Discrete extrema & constructions

离散最值与构造

Method 20 / 方法 20 [LO02-M20] (continued / 续)

Worked solution

1. If a factor gh is composite with $g, h \geq 2$, its cost is $gh - 1$. The split cost is $g + h - 2$.

2. The saving is $(g-1)(h-1) > 0$, so no minimum can keep a composite factor.

3. All factors at a minimum are therefore primes. Unique factorization fixes their multiplicities and gives the claimed cost, which the prime-factor list itself attains.

例题解析

1. 复合因子 gh 未拆成本为 $gh - 1$，拆分成本为 $g + h - 2$。

2. 节省量为 $(g-1)(h-1) > 0$，所以最优分解不能保留复合因子。

3. 最优因子全为素数，由唯一分解确定重数，得到公式，素因子列表本身即取等。

The product factorization needs a bijection or unique matching proof before costs may be optimized independently.

独立优化成本之前，计数乘积必须先有双射或唯一匹配证明。

Practice / 配套练习: AIME-2026-II-Q13 (WB p. 718)

Section 6.3 Constraint deductions & logical cases

约束推理与逻辑分类

Method 1 / 方法 1 [LO03-M01]

A logical case split must preserve every condition and exhaust every possible model. Separate ordinary feasibility, ambiguity after partial information, and public knowledge: they remove possibilities in different ways.

逻辑分类必须保留所有条件，并覆盖全部可能模型。普通可行性、部分信息造成的歧义及公开知识更新，会以不同方式排除可能性，应分别处理。

Propagate the tightest local constraint before branching

先传播最紧的局部限制再分类

Procedure

1. Choose the smallest uncovered position, a forced repeated value, or a variable involved in many restrictions.

2. For each possible branch, immediately propagate all forced inclusions and exclusions.

3. Count only after proving that every remaining constraint has been satisfied or explicitly retained.

4. Use size bounds to close the search: k distinct positive integers sum to at least $k(k+1)/2$.

通用步骤

1. 选择最小未覆盖位置、被迫重复的值，或参与限制最多的变量。

2. 每个分支一旦确定，立刻传播所有必选和禁选条件。

3. 只有证明其余限制均已满足或仍被明确保留后，才能计数。

4. 用数量界封闭搜索：k 个不同正整数之和至少为 $k(k+1)/2$。

Section 6.3 Constraint deductions & logical cases

约束推理与逻辑分类

Method 1 / 方法 1 [LO03-M01] (continued / 续)

> Example. A set of chosen tasks is selected from A, B, C, D, E. Choosing A forbids B, C, and choosing D requires C. How many task sets are possible?

> 例题：从任务 A, B, C, D, E 中选择子集。选择 A 就不能选 B, C，选择 D 则必须选 C。共有多少种可行任务集?

Worked solution

1. If A is chosen, B, C are excluded, hence D is excluded too. Only E is free, giving two sets.

2. If A is absent, B, E are free. The allowed (C, D) states are $(0,0), (1,0), (1,1)$, giving $2 \cdot 2 \cdot 3 = 12$ sets.

3. The two branches are disjoint and exhaustive, so the total is 14. The implication was enforced before the independent choices were multiplied.

例题解析

1. 若选 A，则 B, C 不选，由此 D 也不能选，只剩 E 自由，共两种。

2. 若不选 A，则 B, E 自由，(C, D) 只能取 $(0,0), (1,0), (1,1)$，共 $2 \cdot 2 \cdot 3 = 12$ 种。

3. 两类不相交且覆盖全部可能，共 14 种。先执行蕴含条件，才可把独立选择相乘。

A short list is exhaustive only after an explicit bound or forced first case has been proved.

只有先证明界或被迫的首个分类，短候选表才称得上穷尽。

Section 6.3 Constraint deductions & logical cases

约束推理与逻辑分类

Method 1 / 方法 1 [LO03-M01] (continued / 续)

For a progression $r, r+d, r+2d, \ldots$ to cover an uncovered index q, its step must divide $q-r$ and $q \geq r$. To forbid overlap of two progressions, use their congruence compatibility modulo the gcd of their steps.

For an absent integer median, the list is even and its two middle entries are distinct with the same parity. A unique mode has strictly greater multiplicity than every other value; these constraints combine with the distinct-positive sum bound.

若数列 $r, r+d, r+2d, \ldots$ 要覆盖未覆盖位置 q，必须有 $d \mid (q-r)$ 且 $q \geq r$。判断两等差数列是否重叠，可检查它们的同余条件在公差最大公因数下是否相容。

若中位数为未出现的整数，则列表长为偶数，中间两项不同且同奇偶；唯一众数的频数严格超过每个其他值。这些限制应与不同正数的最小和界合用。

Practice / 配套练习: AIME-2002-I-Q09 (WB p. 496), AIME-2017-I-Q12 (WB p. 644), AIME-2024-II-Q02 (WB p. 795)

Section 6.3 Constraint deductions & logical cases

约束推理与逻辑分类

Method 2 / 方法 2 [LO03-M02]

Group complete models by the information that is known

按已知信息把完整模型分组

Procedure

1. Enumerate the complete feasible models under a proved finite bound.

2. Compute the reported summary for each model, and group models having the same summary.

3. Translate “still two possibilities” into a group-size condition, then compute the requested quantity from the surviving group.

4. Distinguish distinct models from sign, order, or labeling copies that describe the same model.

通用步骤

1. 在已证明的有限范围内，列出完整可行模型。

2. 计算每个模型的已知摘要，按摘要相同分组。

3. 将“仍有两种可能”转为组大小条件，再由保留组求目标量。

4. 区分真正不同的模型与符号、顺序或标号变化造成的重复。

> Example. Three distinct members of $\{1, 2, 3, 4, 5, 6\}$ are chosen. Only their product is announced. Which products leave exactly two possible chosen triples?

> 例题：从 $\{1, 2, 3, 4, 5, 6\}$ 中选三个不同数，只公布乘积。哪些乘积仍对应恰好两种所选三元集？

Section 6.3 Constraint deductions & logical cases

约束推理与逻辑分类

Method 2 / 方法 2 [LO03-M02] (continued / 续)

Worked solution

1. There are $\binom{6}{3} = 20$ complete models. Compare products for these unordered triples rather than their six orderings.

2. The repeated-product groups are $12 : \{1, 2, 6\}, \{1, 3, 4\}$; $24 : \{1, 4, 6\}, \{2, 3, 4\}$; $30 : \{1, 5, 6\}, \{2, 3, 5\}$; and $60 : \{2, 5, 6\}, \{3, 4, 5\}$.

3. The other twelve products are $6, 8, 10, 15, 18, 20, 36, 40, 48, 72, 90, 120$, each occurring once. Thus exactly $12, 24, 30, 60$ have the required ambiguity.

例题解析

1. 共有 $\binom{6}{3} = 20$ 个完整模型，应对无序三元集的乘积分组，而非重复计其六种顺序。

2. 重复组为 $12 : \{1, 2, 6\}, \{1, 3, 4\}$；$24 : \{1, 4, 6\}, \{2, 3, 4\}$；$30 : \{1, 5, 6\}, \{2, 3, 5\}$；$60 : \{2, 5, 6\}, \{3, 4, 5\}$。

3. 其余十二个乘积为 $6, 8, 10, 15, 18, 20, 36, 40, 48, 72, 90, 120$，各出现一次。因此所求恰为 $12, 24, 30, 60$。

An ambiguity condition refers to the entire feasible class, not merely two examples you happened to find.

歧义条件针对完整可行类，不能只找到两个例子就断言恰有两个。

Practice / 配套练习: AIME-2015-II-Q06 (WB p. 48)

Section 6.3 Constraint deductions & logical cases

约束推理与逻辑分类

Method 3 / 方法 3 [LO03-M03]

Pair complementary sums of four unknown numbers

将四个未知数的两两和配成互补对

Procedure

1. For four numbers with total T, partition the six pair sums into three complementary pairs each totaling T.

2. Use the sum of all six entries, $3T$, and any known largest or smallest sums to restrict the pairing.

3. For each surviving pairing, solve three linear pair equations to recover the four numbers.

4. Check distinctness and reproduce all six pair sums to prove attainment.

通用步骤

1. 四个数总和为 T 时，六个两两和可配成三对，每对之和均为 T。

2. 利用六和总和 $3T$，以及已知最大、最小和来限制配对。

3. 对每种保留配对，解三个线性两两和方程，恢复四个数。

4. 检查数互异并重算全部六和，证明该配对可实现。

> Example. Four distinct real numbers have pair sums $4, 7, 8, 9, 10, 13$. Find all possible sorted quadruples.

> 例题：四个不同实数的六个两两和为 $4, 7, 8, 9, 10, 13$。求所有从小到大的四元组。

Section 6.3 Constraint deductions & logical cases

约束推理与逻辑分类

Method 3 / 方法 3 [LO03-M03] (continued / 续)

Worked solution

1. The total of the four numbers is $T = (4+7+8+9+10+13)/3 = 17$. Complementary pairs are $(4,13),(7,10),(8,9)$.

2. For $a<b<c<d$, the smallest sums are $a+b=4, a+c=7$, and $c+d=13$. The remaining ambiguity is $a+d=8, b+c=9$ or the reverse.

3. The first gives $(a,b,c,d)=(1,3,6,7)$; the second gives $(3/2,5/2,11/2,15/2)$. Both are distinct sorted quadruples and reproduce exactly the six given sums, so the list is complete.

例题解析

1. 四数和为 $T = (4+7+8+9+10+13)/3 = 17$，互补对为 $(4,13),(7,10),(8,9)$。

2. 若 $a<b<c<d$，则最小两和为 $a+b=4, a+c=7$，最大和为 $c+d=13$。余下仅两种：$a+d=8, b+c=9$，或交换八与九。

3. 第一种给出 $(1,3,6,7)$，第二种给出 $(3/2,5/2,11/2,15/2)$。两组均严格递增，且重算得到全部六和，因此已穷尽。

Equal complementary totals are necessary; explicit reconstruction proves sufficiency.

互补对总和相等只是必要条件，恢复具体数值才能证明充分。

Practice / 配套练习: AIME-2017-II-Q05 (WB p. 18)

Section 6.3 Constraint deductions & logical cases

约束推理与逻辑分类

Method 4 / 方法 4 [LO03-M04]

Update possible worlds after a public statement of ignorance

公开“不知道”后更新可能世界

Procedure

1. List all worlds allowed by the initial public information.

2. For each possible private observation, count how many initial worlds contain it; a unique world would make its observer know.

3. A simultaneous public statement that everyone is ignorant removes each world containing any initially unique observation.

4. Recompute observation classes among the surviving worlds and check the later statement, using the correct simultaneous round.

通用步骤

1. 先列出初始公开信息允许的全部可能世界。

2. 对每种私人观察，数它属于多少个初始世界；若只属于一个，观察者就能确定。

3. 若所有人同时公开表示不知道，就排除含有任何初始唯一观察的世界。

4. 在保留世界中重新分组私人观察，检查下一轮陈述，且必须遵守同时发言的轮次。

> Example. A teacher chooses one pair from $\{a,b\},\{b,c\},\{c,d\}$ and gives its two distinct symbols privately to two perfectly logical pupils. Both simultaneously say they cannot identify the pair. Which pair was chosen, and can both identify it afterward?

Section 6.3 Constraint deductions & logical cases

约束推理与逻辑分类

Method 4 / 方法 4 [LO03-M04] (continued / 续)

> 例题：老师从 $\{a,b\},\{b,c\},\{c,d\}$ 中选择一对，把两个不同符号分别私下给两位完全理性的学生。两人同时表示不能确定所选对。选的是哪一对？听完之后两人能否确定？

Worked solution

1. Initially a occurs only in $\{a,b\}$ and d only in $\{c,d\}$, so either symbol would allow its pupil to know immediately.

2. Unanimous initial ignorance therefore excludes both pairs containing a or d, leaving only $\{b,c\}$.

3. After the public update, either private symbol b or c belongs to that sole surviving world. Both pupils can now identify the pair.

例题解析

1. 初始时，a 仅属于 $\{a,b\}$，d 仅属于 $\{c,d\}$，拿到它们的人会立即知道。

2. 两人都表示初始不知道，就排除了含 a 或 d 的两对，只剩 $\{b,c\}$。

3. 公开更新后，无论私人符号为 b 还是 c，都只对应这个保留世界，所以两人此时都能确定。

Analyze all initial replies against the same initial world set; processing simultaneous replies sequentially changes the puzzle.

同一轮初始回答必须都针对同一个初始世界集；把同时回答逐个更新会改变题意。

Practice / 配套练习: AIME-2021-II-Q11 (WB p. 810)